Corporate Finance Casebook and Companion

Cases, Modeling Guides, Problem Sets and Instructor Materials

Volume 3 of a Three-Volume Textbook for MBA Students

About This Volume

Volume 3 accompanies Volumes 1 and 2 with full-length cases and teaching notes, executed modeling guides in Excel and Python, graded problem sets with full solutions, and instructor materials including an examination bank. All companies, figures and cases are fictional and illustrative unless stated; they are for teaching only and are not investment, legal or tax advice. Figures were computed by the authors' working and should be verified before use in professional decisions; legal and regulatory descriptions are general and should be checked against current rules.

Contents

Part I — Cases in Valuation and Investment

Chapter 1: Case: Helios Cell, a Battery Gigafactory Decision 5

Chapter 2: Case: Norwind Air, a Fleet Renewal Decision 24

Chapter 3: Case: Lumen Therapeutics, Valuing a Biotech Pipeline 41

Chapter 4: Case: Harbor & Pine, a Retail Turnaround 62

Chapter 5: Case: Cirrus Ledger, Valuing a SaaS Company 81

Chapter 6: Case: Ironvale Copper, a Mine Expansion under Commodity Risk 103

Part II — Cases in Financing

Chapter 7: Case: Calder & Sons, a Family Firm Considers an IPO 126

Chapter 8: Case: Meridian Power, Capital Structure for a Regulated Utility 147

Chapter 9: Case: Quantra Systems, Dividend or Buyback? 166

Chapter 10: Case: Andara Foods, an Emerging-Market Bond Issue and FX Risk 186

Chapter 11: Case: Nimbus Robotics, a Series B Term Sheet Negotiation 206

Part III — Cases in Transactions and Restructuring

Chapter 12: Case: Tri-County and Lakeshore, a Regional Bank Merger 229

Chapter 13: Case: Brightwell Instruments, a Hostile Bid and Its Defense 247

Chapter 14: Case: Oakhaven Brands, a Consumer-Brand LBO 266

Chapter 15: Case: Rollan Industrial, a Distressed Manufacturer 283

Chapter 16: Case: Zephyr Components Buys into a Fictional Emerging Market 307

Chapter 17: Case: Sterling Allied, Break Up the Conglomerate? 325

Part IV — Modeling Companion

Chapter 18: Excel Modeling Standards and Building an Integrated Three-Statement and DCF Model 349

Chapter 19: Python for Corporate Finance: NPV, IRR, Bonds and Monte Carlo 378

Chapter 20: Building LBO and Merger Models 403

Chapter 21: Options, Risk and Hedging Models in Python 425

Part V — Practice and Instructor Materials

Chapter 22: Comprehensive Problem Sets with Full Solutions I: Foundations and Valuation 452

Chapter 23: Comprehensive Problem Sets with Full Solutions II: Financing, Derivatives and Transactions 469

Chapter 24: Instructor Manual and Examination Bank 492

Part I — Cases in Valuation and Investment

Chapter 1: Case: Helios Cell, a Battery Gigafactory Decision

Illustrative case. Helios Cell Technologies, its people, its lenders, the Province of Ardmore and every number in this chapter are fictional and were constructed for teaching. Units: $M means millions of US dollars; GWh means gigawatt-hours of cell capacity. Because one GWh is one million kWh, output in GWh multiplied by a price in $/kWh gives revenue in $M directly. The case draws on Volume 1, Chapters 6 to 8, 11 and 14, and on Volume 2 for adjusted present value and options.

The Situation

On 4 November 2026, Dev Raman, chief financial officer of Helios Cell Technologies Inc., opened a folder labelled “Project Meridian” and found at the top the third revision of a term sheet from the Province of Ardmore. The Province would support a new lithium-ion cell plant at Port Alder, a deep-water industrial site on its northern coast, with a capital grant, a production credit, a property-tax abatement and a low-interest development loan. The offer expired on 31 January 2027. The Helios board would meet on 15 December 2026, and Raman had to bring it a recommendation.

Helios, headquartered in Marlow Bay, makes prismatic lithium-ion cells for stationary energy storage (battery containers for utilities and large commercial sites) and for commercial electric vehicles such as buses, delivery vans and port equipment. Its two plants, Brightwater (8 GWh a year, eight years old) and Corrin Valley (5 GWh, four years old), ran at high utilisation through 2026. The company expected revenue of $3,210M and EBITDA of $545M for 2026, and its shares traded at $20.00, an equity value of about $6,100M on 305 million shares.

Meridian would be larger than anything Helios had attempted. The engineering study called for a 15 GWh plant (Phase 1) costing $1,500M, spent $900M in the first year of construction and $600M in the second, with the first cells off the line in the third year. Phase 2 was a further 15 GWh, taking the site to 30 GWh, at $1,300M if started three years after Phase 1 or $1,248M (a 4% discount for ordering together) if committed with Phase 1. Helios did not have to decide Phase 2 now, but it did have to decide whether to reserve the land, power connection and permits for it, and whether to build it at once.

The people

Marguerite Okonkwo-Hale, the chief executive, was an engineer and a believer in scale: “cell manufacturing is a learning business, and whoever makes the most cells learns fastest.” She wanted to approve all 30 GWh in December. Raman was more cautious: every earlier Helios plant had run over budget and late, and the first plant’s cell price had fallen by about a third in five years. Lena Bergström, strategy chief, had negotiated the term sheet and favoured Phase 1 with an option on Phase 2. Tomasz Wierzba, manufacturing vice president, owned the learning-curve assumptions. Priya Anand, the treasurer, owned the funding plan and covenant. Dr. Nasrin Kaveh, audit committee chair, would oppose any plan that endangered the covenant.

The decision

The board had to answer three connected questions: whether to commit to Phase 1 on the terms offered; whether to build Phase 2 now or hold it as an option to be exercised in three years; and what conditions to attach. Beneath them lay a fourth, which Raman felt he owed the directors: how much of the project’s apparent value came from subsidies, and how secure were they?

“Year 0” is the date of commitment, January 2027. Later cash flows occur at year end. Construction ends in Year 1 (2028) and first production comes in Year 2 (2029).

Industry Context

Demand. Cell demand comes from the shift of road vehicles to electric drive, now spreading from cars to buses and trucks, and from grid storage that absorbs solar and wind output. Both are growing fast and both are policy-sensitive. The consultants Helios hired (a fictional firm, Calder & Voss) gave low, central and high views, and the gap between low and high exceeded the central growth rate.

Price. Cell prices have fallen for a decade as raw-material prices dropped, processes improved and capacity arrived. Declines come in steps when a capacity surplus coincides with falling metal prices. Margin is the gap between a falling price and a falling cost, and survivors are those whose costs fall faster. Brightwater earned a high margin in its third year and a modest one in its eighth.

Costs and learning. A cell plant has three cost layers. Materials (active materials, electrolyte, separators, foils) are about half the early selling price and largely pass through from commodity markets. Conversion costs (labour, energy, consumables, scrap) fall as operators learn. Fixed cash costs (supervision, engineering, insurance) do not vary with volume, which is why utilisation matters so much. The manufacturing literature describes conversion cost with an experience curve: each doubling of cumulative output cuts unit cost by a fixed fraction. A learning rate of 85% means cost after a doubling is 85% of cost before. The curve runs on cumulative output across the site, so Phase 2 would inherit Phase 1’s experience.

Policy support. Governments competing for plants offer grants, production credits, tax abatements and subsidised loans. Their value is large relative to the plant’s margin and they carry political risk that private contracts do not: a credit is legislated, and legislatures change course. The Ardmore credit statute is subject to review in 2031.

Capital intensity. Phase 1 costs $100M per GWh of annual capacity. A plant takes two years to build and three to four to ramp, so cash flow dips for years before a long plateau. That shape makes NPV very sensitive to the discount rate, the speed of ramp-up and the price path in the plateau years. Helios’s edge is a licence for a high-density chemistry, an anchor-customer letter and the Province’s goodwill; its weakness is scale, since 15 GWh is small by global standards.

Financial Data

The tables below are the complete data pack that Dev Raman’s team assembled for the board. All amounts are in $M unless stated. Treat management’s assumptions as given, with the caveats in the next section.

Table 1: Helios consolidated income statement

$M, year ended 31 December	FY2024A	FY2025A	FY2026E
Revenue	2,640	2,930	3,210
Cash operating costs	(2,230)	(2,455)	(2,665)
EBITDA	410	475	545
Depreciation and amortisation	(176)	(192)	(210)
EBIT	234	283	335
Net interest expense	(47)	(49)	(52)
Profit before tax	187	234	283
Income tax	(47)	(58)	(71)
Net income	140	176	212
EBITDA margin	15.5%	16.2%	17.0%

Table 2: Helios consolidated balance sheet

$M, at 31 December	FY2025A	FY2026E
Cash and equivalents	360	410
Trade receivables	470	520
Inventories	440	480
Property, plant and equipment, net	2,240	2,350
Other non-current assets	600	640
Total assets	**4,110**	**4,400**
Trade payables	560	610
Interest-bearing debt	930	930
Other liabilities	450	478
Shareholders' equity	2,170	2,382
Total liabilities and equity	**4,110**	**4,400**
Net debt (debt less cash)	570	520

Helios's bank facilities carry one financial covenant: net debt must not exceed 3.75 times trailing twelve-month EBITDA, tested every six months. The company has 305 million shares at $20.00. Its senior unsecured debt trades at a yield of about 7.0%.

Table 3: Project Meridian, Phase 1 capital budget and timing

Item	Year 0	Year 1	Total
Buildings, process equipment and installation	900	600	1,500
Land at Port Alder (non-depreciable, resaleable at cost)	40	–	40
Total cash outlay before subsidies	940	600	1,540
Provincial capital grant (12% of plant cost, paid in Year 1)	–	(180)	(180)
Net outlay	940	420	1,360

The plant is depreciated for tax purposes on a straight-line basis over ten years starting in Year 2. At the end of Year 15 the plant is assumed to be sold for 8% of its original cost and the land for its purchase price. Net working capital is 12% of revenue and is recovered in full at the end of Year 15. The corporate tax rate is 25%. Meridian will be held in a ring-fenced subsidiary, as the Province requires, so its losses cannot be set against Helios's other profits. They can be carried forward without expiry and can offset up to 80% of taxable income in any later year.

Table 4: Management’s Phase 1 operating forecast

Year	Phase 1 utilisation	Cell price ($/kWh)	Materials ($/kWh)	Fixed cash opex ($M)	Phase 1 output (GWh)	Cumulative output (GWh)
2	35%	97.0	49.0	110.0	5.2	5.2
3	60%	94.1	47.0	112.2	9.0	14.2
4	78%	91.3	45.2	114.4	11.7	26.0
5	85%	88.5	43.4	116.7	12.8	38.7
6	85%	85.9	41.6	119.1	12.8	51.5
7	85%	83.3	40.0	121.4	12.8	64.2
8	85%	80.8	38.4	123.9	12.8	77.0
9	85%	78.4	36.8	126.4	12.8	89.7
10	85%	76.0	35.3	128.9	12.8	102.5
11	85%	73.7	33.9	131.5	12.8	115.2
12	85%	71.5	32.6	134.1	12.8	128.0
13	85%	69.4	31.3	136.8	12.8	140.7
14	85%	67.3	30.0	139.5	12.8	153.4
15	85%	65.3	28.8	142.3	12.8	166.2

Cell price starts at $97.0 per kWh in Year 2 and falls by 3% a year; materials start at $49.0 per kWh and fall by 4% a year as the supply chain matures. Fixed cash operating costs start at $110M in Year 2 and rise 2% a year. Selling, general and administrative cost is 3% of revenue. Conversion cost follows an 85% learning curve with a cost of $36 per kWh for the first GWh of cumulative output at the site. In any year the average conversion cost is the average of the curve over that year’s output, so a year’s cost falls as cumulative output grows through the year (the details are the subject of Question 3).

Table 5: Comparable companies and market data

Company (illustrative)	Levered equity beta	Debt/equity (market)	Marginal tax rate
Aurelia Cells	1.69	0.45	21%
Kestrel Power Systems	1.51	0.30	25%
Nordvik Energy Storage	1.34	0.20	22%
Orrin Electrochem	1.61	0.55	24%
Tamsin Materials Group	1.09	0.25	26%

Market input	Value
Risk-free rate (long government bond)	4.0%
Equity market risk premium	5.5%
Helios pre-tax cost of debt for a project of this risk	7.0%
Target debt/(debt + equity) for the project	30%
Corporate tax rate	25%

Debt betas may be assumed to be zero. The hurdle rate Helios used in the past was 10%, which nobody could trace to a calculation.

Table 6: Provincial support and financing terms

Instrument	Terms
Capital grant	12% of Phase 1 plant cost (not land), paid in Year 1, subject to construction milestones
Production credit	$10 per kWh of cells sold from the plant, for the first eight years of production (Years 2 to 9); Helios may sell the credit to a third party at 92 cents on the dollar, and this is how it is valued in the base case
Property-tax abatement	$8M a year reduction in property tax for the first ten years of production (Years 2 to 11)
Development loan	$500M, drawn in Year 1; interest 2.5% a year paid annually in Years 2 to 11; principal repaid in a single payment at the end of Year 11
Conditions	Start construction by 31 March 2027; employ at least 900 people at the site by Year 3; no change of control during the credit period

Table 7: Management's Phase 1 funding plan

Source	$M
Cash on balance sheet	300
State development loan	500
Capital grant (received year 1)	180
New bank term loan	350
New equity (private placement)	210
Total sources	1,540

Total uses are $1,540M (plant $1,500M plus land $40M), which equals total sources. The loan and grant are conditional on the provincial package.

Table 8: Scenario definitions (Calder & Voss and Helios management)

Driver	Downside	Base	Upside
Initial price ($/kWh)	91.2	97.0	102.8
Annual price decline	4.0%	3.0%	2.5%
Steady-state utilisation	78.2%	85.0%	89.3%
Learning rate	88%	85%	82%
Probability	25%	50%	25%

Table 9: Phase 2 option inputs

Item	Value
Phase 2 capacity	15 GWh
Phase 2 capex if started at the end of Year 3	1,300
Discount on Phase 2 capex if committed with Phase 1	4%
Phase 2 ramp-up utilisation (Years 1, 2, 3 and later after start)	50%, 75%, 85%
Price realised on Phase 2 output relative to Phase 1 price	15% lower (sold on spot contracts)
Additional fixed opex as share of Phase 1 fixed opex	60%
Volatility of project value (consultants' range 25% to 45%; point estimate)	35%
Risk-free rate for option valuation (continuously compounded)	4.0%

Complications and Exhibits

Real decisions arrive with untidy data. The exhibits below, assembled by Raman's staff in late October, raise problems students must decide how to handle. None has a single correct answer; the Teaching Note gives the model treatment.

Exhibit A: Costs that may or may not be relevant

Helios has already paid a consultant $22M for site studies and permits. Head office allocates $14M a year to the Meridian unit in proportion to revenue, but head-office headcount would not change. The land will be bought for $40M; the Province will repurchase it at cost at any time, which management takes to mean Year 15 recovery at cost. The marketing director thinks Meridian might take $18M a year of Brightwater's contribution in Years 2 to 6, but Brightwater is full and sales could be replaced from a waiting list.

Exhibit B: Learning data from Brightwater

Wierzba supplied the average conversion cost at Brightwater during its first four years. Meridian's newer technology has a lower first-GWh cost, but he believes the rate of learning is comparable. Management's 85% was rounded down from a first look at these data.

Cumulative output at Brightwater (GWh)	Average conversion cost ($/kWh)
1.5	41.0
3.0	35.6
6.0	30.2
12.0	26.9
24.0	22.6

Exhibit C: Tax counsel's open questions

Counsel could not say whether the production credit is taxable income when received ("probably not", 60% confidence), or whether the capital grant reduces the depreciable basis of the plant ("yes" under general principles, but the statute is silent). The base case assumes the credit is not taxable and the grant does reduce basis. The CFO wants to know how much this matters.

Exhibit D: Policy risk

The credit statute is reviewed in 2031, Year 4 of the project, and the opposition has promised to "review the cost-effectiveness" of the credit. Bergström gives a 30% chance that the credit is repealed or sharply curtailed for production from Year 5 onward. The grant, abatement and loan are contractual and not subject to review. Students should model repeal as loss of the credit for

all production from Year 5 on, with probability 30%, while recognising that no one can know it.

Exhibit E: Anchor customer

Northgate Coachworks, a bus maker, has signed a non-binding letter of intent to buy 4 GWh a year for five years at a price no lower than $90 per kWh, with no penalty to either side. Sales treats it as validation. The base case ignores it.

Exhibit F: The covenant

Net debt after the Phase 1 funding plan would exceed today's level while plant EBITDA is negative in Year 2. Anand has asked the banks for a construction holiday on the covenant but has no written commitment.

Exhibit G: Volatility and rivals

For the Phase 2 option the consultants offered volatility of 25% (long-run pack-price volatility), 35% (listed manufacturers' equity volatility, adjusted for leverage) and 45% (spot cell prices in the last shortage). Raman chose 35%. Rival plants announced in the region explain why Phase 2 output is assumed to sell at spot at a discount.

Assignment Questions

Answer all ten questions, defend your treatment of ambiguous items and show calculations. Use the Question 1 discount rate (rounded to one decimal) throughout.

Question 1: Cost of capital. Unlever each comparable's beta, average the results, relever at the target structure, and estimate the project's cost of equity and WACC. Explain why Helios's own beta and capital structure are the wrong inputs and why the untraced 10% hurdle should be retired.

Question 2: Base case without subsidies. Build Phase 1 free cash flows for Years 0 to 15 with no grant, credit or abatement, including land and the tax-loss carry-forward. Say which Exhibit A items you include and why. Compute NPV, IRR, modified IRR, payback, discounted payback and profitability index, and say what they imply for the unsubsidised business.

Question 3: Learning curve. (a) Estimate the learning rate from Exhibit B by log-log regression and judge its precision against management's 85%. (b) Derive the average conversion cost for Years 2 and 3 from an 85% curve with a $36 first-GWh cost. (c) Tabulate NPV, with and without subsidies, for learning rates from 80% to 100%. (d) Why might a sponsor overstate the rate, and how should the board use the sensitivity?

Question 4: Subsidies. (a) Value the grant, credit and abatement separately and together; explain why the parts do not sum to the whole. (b) Compute NPV if the credit is taxable and, separately, if the grant does not reduce depreciable basis. (c) Compute NPV if the credit is lost from Year 5 on, the expected NPV at 30% probability, and the break-even probability. (d) What share of the project's value comes from subsidies?

Question 5: Tax. (a) Show the loss schedule for Years 2 to 7. (b) Compare NPV under 5-, 10- and 15-year depreciation and immediate expensing, and explain why expensing helps so little. (c) How much would NPV change without the 80% limit on loss use?

Question 6: Sensitivity and scenarios. (a) Prepare a tornado for eight inputs: initial price, price decline, utilisation, learning rate, Phase 1 capex, materials cost, fixed cost and discount rate. (b) Find break-even values for initial price and utilisation. (c) Compute NPV in each Table 8 scenario, the probability-weighted NPV and the standard deviation.

Question 7: The Phase 2 option. (a) Compute the incremental cash flows and static NPV of Phase 2 committed at the end of Year 3. (b) Value the option by Black–Scholes and by a three-step binomial lattice. (c) Show sensitivity to volatility. (d) Explain how an option can be valuable when static NPV is negative.

Question 8: Phase or build all now? (a) Compute the NPV of building both phases now. (b) Compare it with Phase 1 plus the option and find the volatility at which they are equal. (c) Compute pro forma leverage under both plans against the 3.75x covenant. (d) Compare the plans under four price outcomes and recommend.

Question 9: Financing and APV. (a) Value the $500M development loan by Myers's method, discounting after-tax payments at the 7% market rate. (b) Compute base-case NPV at the unlevered cost of capital, add the loan and report the APV. (c) Should the loan count in the investment decision?

Question 10: Recommendation. (a) Compute the probability-weighted NPV across scenarios including the 30% repeal risk. (b) Say how Helios should use the anchor-customer letter. (c) Draft the board recommendation: decision, Phase 2 structure, conditions precedent, and three actions for the next sixty days.

Teaching Note and Full Solutions

Case synopsis

Helios Cell is a capital-budgeting case in which the arithmetic is straightforward and the judgement is not. A cell plant has a deep early cash dip, a long plateau, a falling price, an experience curve that rewards volume, and a subsidy package worth more than the project's unsubsidised value. The questions are how to value each feature, how to separate subsidy value from operating value, and whether to pay for flexibility by phasing. It works best after Volume 1, Chapters 6 to 8. All results below come from one Python model of the cash flows, checked by hand on selected years. They use year-end flows, a 10.2% discount rate (Question 1) and the base-case inputs; students working with rounded intermediate numbers may differ by a unit or two.

Learning objectives

Students should be able to (1) estimate a project cost of capital by unlevering and relevering comparable betas; (2) build incremental after-tax cash flows including working capital, land, salvage and a loss carry-forward; (3) model a learning curve on cumulative output; (4) separate and value subsidies and price political risk; (5) run sensitivity, break-even and scenario analysis; (6) value a staged-investment option by Black-Scholes and a lattice and weigh it against the cost of delay; (7) use APV for a subsidised loan; and (8) write a board recommendation with conditions.

Suggested class plan

Two 90-minute sessions. Preparation: Questions 1 to 3.

Time	Activity	Questions
S1, 0 to 10	Cold poll: commit, delay or decline	–
10 to 30	Discount rate; cash flow construction; relevant costs	1, 2
30 to 50	Learning curve and its sensitivity	3
50 to 75	Subsidies, tax and repeal risk	4, 5
75 to 90	Tornado and scenarios at the whiteboard	6
S2, 0 to 30	Phase 2 option: static NPV, Black-Scholes, lattice	7
30 to 55	Phase versus all now; covenant	8
55 to 70	Financing and APV	9
70 to 90	Recommendation; revisit the opening poll	10

Question 1 solution: Cost of capital

Step 1: unlever each beta. With zero debt beta, the asset beta is the equity beta divided by one plus (1 minus tax rate) times debt/equity.

Company	Levered beta	1 + (1 − t) × D/E	Unlevered beta
Aurelia Cells	1.69	1 + 0.79 × 0.45 = 1.3555	1.2468
Kestrel Power Systems	1.51	1 + 0.75 × 0.30 = 1.2250	1.2327
Nordvik Energy Storage	1.34	1 + 0.78 × 0.20 = 1.1560	1.1592
Orrin Electrochem	1.61	1 + 0.76 × 0.55 = 1.4180	1.1354
Tamsin Materials Group	1.09	1 + 0.74 × 0.25 = 1.1850	0.9198

The simple average is 1.14 and the median is 1.16. Tamsin is a materials supplier and a reader might drop it (average 1.19; Alternative Answers).

Step 2: relever. The target debt/(debt + equity) is 30%, so debt/equity is 0.30 / 0.70 = 0.4286.

```
βL = βU × [1 + (1 − t) × D/E]
   = 1.1388 × [1 + 0.75 × 0.4286]
   = 1.1388 × 1.3214
   = 1.505
```

Step 3: cost of equity (CAPM).

```
ke = rf + βL × ERP = 4.0% + 1.505 × 5.5% = 12.28%
```

Step 4: WACC.

```
WACC = E/V × ke + D/V × kd × (1 − t)
     = 0.70 × 12.28% + 0.30 × 7.0% × 0.75
     = 8.59% + 1.58%
     = 10.17%, rounded to 10.2%
```

The equity part contributes 8.59 points and the after-tax debt part 1.57 points. For use in Question 9, the unlevered cost of capital is 4.0% + 1.1388 × 5.5% = 10.26%.

Why not Helios's own inputs? Helios is about 13% debt-financed, and its beta reflects mature plants with contracted volumes. Meridian is a greenfield single-site plant in a falling-price market. The discount rate depends on the risk of the project's cash flows, not on the sponsor's name (Volume 1, Chapter 11), and 30% is a project-appropriate structure. The 10% hurdle has no traceable derivation. It is close here, but an untraced hurdle gives no guidance when risk differs across projects, and each added point of discount rate costs about $90M of NPV (Question 6).

Question 2 solution: Base-case cash flows without subsidies

Relevant versus irrelevant costs. Only incremental cash flows count.

Item	Treatment	Reason
$22M site studies already paid	Excluded	Sunk; paid whatever the decision
$14M allocated head-office cost	Excluded	Not incremental: headcount does not change
$18M Brightwater cannibalisation	Excluded	Brightwater is full and the sales can be replaced from the waiting list; if the waiting list were empty the item would be a real opportunity cost
$40M land	Included (Year 0 outflow, Year 15 recovery)	Real cash; recoverable at cost
Working capital at 12% of revenue	Included	Real cash tied up
Salvage 8% of plant cost	Included, taxed as a gain	Book value is zero after ten years

Year-by-year mechanics. For each year:

```
Revenue            = output (GWh) × price ($/kWh)                    [$M]
EBITDA             = revenue − materials − conversion − fixed − SG&A
Taxable income     = EBITDA − tax depreciation (+ gain on sale in Year 15)
Tax                = 25% × (taxable income − NOL used)
NOL used           = min(NOL balance, 80% × taxable income)
FCF                = EBITDA − tax − capex − ΔNWC + salvage and land recovery
```

Worked example, Year 6 (no subsidies). Output is 15 × 85% = 12.75 GWh and price is 97.0 × 0.97^4 = $85.87 per kWh, so revenue is 12.75 × 85.87 = $1,094.9M. Materials are 12.75 × $41.62 = $530.6M. The conversion cost (Question 3) is $14.75 per kWh, so $188.1M. Fixed opex is $119.1M and SG&A is 3% × 1,094.9 = $32.8M. EBITDA is 1,094.9 − 530.6 − 188.1 − 119.1 − 32.8 = $224.2M. Tax depreciation is 1,500 / 10 = $150.0M, so taxable income before losses is $74.2M. The loss balance at the end of Year 5 is $162M, and 80% × 74.2 = $59.4M can be used, leaving taxable income of $14.8M and tax of $3.7M. Working capital falls from $135.4M to $131.4M because revenue falls, releasing $4.1M. Free cash flow is 224.2 − 3.7 + 4.1 = $224.6M.

Full schedule.

Year	Volume (GWh)	Price ($/kWh)	Revenue	EBITDA	Tax dep.	Taxable income	NOL balance	Cash tax	FCF
0	0.0	–	0	0	0	0	0	0	(940)
1	0.0	–	0	0	0	0	0	0	(600)
2	5.2	97.0	509	(41)	150	(191)	191	0	(102)
3	9.0	94.1	847	94	150	(56)	247	0	53
4	11.7	91.3	1,068	184	150	34	220	2	155
5	12.8	88.5	1,129	222	150	72	162	4	211
6	12.8	85.9	1,095	224	150	74	103	4	225
7	12.8	83.3	1,062	222	150	72	46	4	222
8	12.8	80.8	1,030	217	150	67	0	5	216
9	12.8	78.4	999	211	150	61	0	15	199
10	12.8	76.0	969	203	150	53	0	13	194
11	12.8	73.7	940	195	150	45	0	11	187
12	12.8	71.5	912	186	0	186	0	47	143
13	12.8	69.4	885	177	0	177	0	44	136
14	12.8	67.3	858	168	0	168	0	42	129
15	12.8	65.3	832	158	0	278	0	69	351

Year 0 shows 940 (900 plant + 40 land). In Year 15 the FCF includes the $120M salvage (taxed in full, because book value is zero), the $40M land recovery and the $99.9M release of working capital.

Results at 10.2%.

Measure	Result
NPV	−$537M
IRR	4.60%
Modified IRR (finance and reinvest at 10.2%)	7.16%
Payback (undiscounted)	10.89 years after Year 0
Discounted payback	not recovered within 15 years
Profitability index (PV of Years 2 to 15 flows / PV of Years 0 and 1 outlays)	0.64

The PV of Year 0 and Year 1 outlays is 940 + 600 / 1.102 = $1,484.5M, and the PV of all later flows is $947.2M, so the NPV is 947.2 − 1,484.5 = −$537.3M.

Interpretation. On its own, the plant earns an IRR of 4.6%, well below the 10.2% cost of capital. A plant that returns less than its cost of capital destroys value. Payback of 10.9 years is meaningless because discounted payback never occurs: EBITDA peaks at $224M on a $1,540M outlay, a 14.5% pre-tax cash yield, while price falls 3% a year. The conclusion is that **Meridian is not an economically viable unsubsidised business at these prices and costs**, so the case must turn on the subsidies, the learning curve and the options, which is exactly what the next questions examine.

Question 3 solution: The learning curve

(a) Estimating the learning rate. The experience-curve model is cost = a × Q^b, so log(cost) = log(a) + b × log(Q). Regressing the five points of Exhibit B:

Q (GWh)	ln Q	Cost ($/kWh)	ln cost
1.5	0.4055	41.0	3.7136
3.0	1.0986	35.6	3.5722
6.0	1.7918	30.2	3.4078
12.0	2.4849	26.9	3.2921
24.0	3.1781	22.6	3.1179

The means are ln Q = 1.7918 and ln cost = 3.4208. The sum of squares of ln Q about its mean is 4.8045 and the cross-product sum is −1.0199, so the slope is b = −1.0199 / 4.8045 = −0.2123. The learning rate is 2^b = 2^(−0.2123) = **86.3%**, meaning each doubling of cumulative output reduces cost by 13.7%. The intercept gives a cost of $44.75 per kWh at Q = 1 GWh for Brightwater. R^2 is 0.997. The standard error of the slope is 0.0065, so with three degrees of freedom (t = 3.18) the 95% confidence interval for b is −0.2329 to −0.1917, which corresponds to learning rates of 85.1% to 87.6%. Management's 85% is just outside the favourable end of that interval. The four successive doubling ratios (86.8%, 84.8%, 89.1% and 84.0%) show how noisy a single estimate is, and with only five observations from one plant the interval is probably too narrow.

(b) Deriving the average conversion cost. The exponent for an 85% curve is b = ln(0.85) / ln(2) = -0.2345. If the cost of the Q-th GWh is 36 × Q^b, the total conversion cost of the output between cumulative volumes Q_1 and Q_2 is the integral of 36 × Q^b from Q_1 to Q_2, which equals 36 × (Q_2^{b+1} − Q_1^{b+1}) / (b + 1), with b + 1 = 0.7655. The average cost per kWh is that total divided by the output Q_2 − Q_1.

```
Year 2: Q1 = 0, Q2 = 5.25 GWh
  5.25^0.7655 = 3.5588
  total = 36 × 3.5588 / 0.7655 = 167.4  ($M)
  average = 167.4 / 5.25 = $31.88 per kWh

Year 3: Q1 = 5.25, Q2 = 14.25 GWh
  14.25^0.7655 = 7.6434
  total = 36 × (7.6434 − 3.5588) / 0.7655 = 192.1  ($M)
  average = 192.1 / 9.00 = $21.34 per kWh
```

The average cost falls from $31.88 in Year 2 to $21.34 in Year 3 and $14.75 in Year 6, as cumulative output rises from 5.3 GWh to 51.5 GWh. A common error is to apply the curve to each year's output separately, which resets cost every year. The correct unit of account is cumulative output over the plant's life, which is also why Phase 2 inherits Phase 1's learning.

(c) NPV by learning rate (project discount rate 10.2%).

Learning rate	Exponent b	Conv. cost yr 2	Yr 6	Yr 15	NPV no subsidy	NPV with subsidy
80%	0.32	31.1	10.6	7.0	(315)	334
85%	0.23	31.9	14.8	11.0	(537)	115
90%	0.15	33.0	20.2	16.6	(851)	(186)
95%	0.07	34.4	27.2	24.7	(1,342)	(660)
100%	0.00	36.0	36.0	36.0	(1,987)	(1,305)

At the base rate of 85%, NPV with subsidies is $115M. Each one-point worsening of the learning rate removes about $52M of NPV. At the regression estimate of 86.3% the NPV is $45M, and at 87.1% the NPV is about $1M, so the subsidised project breaks even at a learning rate of roughly 87%. Without subsidies, even an 80% learning rate leaves NPV at −$315M.

(d) Overstatement and use. Sponsors own their assumptions, and a faster learning rate is the cheapest way to improve the numbers. The estimate comes from one plant, an earlier technology and a period when yields improved for other reasons; a change of chemistry would reset the curve. The board should treat 85% as the optimistic end of a range of 85% to 90% and tie management incentives to yield and conversion-cost milestones at fixed cumulative volumes.

Question 4 solution: Subsidies

(a) Value of each instrument (base model with one instrument switched on).

Case	NPV ($M)	Increment vs none
No subsidies	(537)	–
Capital grant only (12%)	(395)	143
Production credit only ($10/kWh, 92% monetised)	(63)	475
Property-tax abatement only	(502)	35
Sum of individual effects		653
All three together	115	652
Interaction (all - sum)		(1)

The grant adds $143M, the production credit $475M (equal to its present value of $475M, since it is untaxed) and the abatement $35M (less than the $44M pre-tax present value because the saved property tax is deductible). Together they add $652M. The $1M interaction arises because the grant lowers depreciation and the abatement raises taxable income, which changes how fast losses are used.

(b) Tax ambiguities. If the credit is taxable, NPV falls from $115M to $2M, a loss of $113M: one line of tax law separates value creation from break-even. If the grant does not reduce depreciable basis, NPV rises to $136M. The board should require tax counsel's resolution of the credit question, ideally an advance ruling, before approval.

(c) Repeal risk. If the credit is lost from Year 5, NPV is −$185M, $300M lower. The expected NPV at 30% is

```
E[NPV] = 0.70 × 114.7 + 0.30 × (−185.3) = 80.3 − 55.6 = $24.7M
```

The break-even probability is 114.7 / 300.0 = 38%, not far above Bergström's unverifiable 30%. The risk bites when the plant is operating and Helios has least flexibility.

(d) Share from subsidies. Unsubsidised NPV is −$537M and subsidised NPV +$115M, so subsidies contribute $652M and **all of the positive NPV comes from government support**; 73% of that support (475 / 652) is the production credit, which is subject to review. The board should know how much value rests on a statute.

Question 5 solution: Tax effects

(a) Loss schedule (base case, with subsidies).

Year	Taxable income before NOL	NOL used	NOL balance end	Cash tax
2	(165)	0	165	0
3	(30)	0	195	0
4	60	48	147	3
5	98	78	69	5
6	100	69	0	8
7	98	0	0	24

Losses of $165M and $30M build a balance of $195M. In Year 4 only 80% of the $60M taxable income ($48M) can be offset, and the balance is exhausted in Year 6, so cash tax in Years 4 to 6 is $3M, $5M and $8M instead of $15M, $25M and $25M.

(b) Depreciation alternatives.

Tax depreciation	NPV with subsidies	Change vs base	Peak NOL balance
Straight-line 5 years	122	8	597.1
Straight-line 10 years (base)	115	0	194.9
Straight-line 15 years	88	(27)	120.6
Immediate expensing (1 year)	112	(2)	1,352.6

Five-year depreciation adds only $8M and 15-year costs $27M. Early deductions create losses, not savings, and the 80% rule makes losses slow to use. Immediate expensing builds a loss balance of $1,353M and is worth less than five-year depreciation. **A deduction is worth the tax rate only when there is income to absorb it.**

(c) No 80% limit. NPV would be $115M, a change of about $1M, because losses are used within three years of profit. The ring-fence matters more: if Meridian's losses could be set against Helios's other profits, NPV would be $124M, $10M higher.

Question 6 solution: Sensitivity and scenarios

(a) Tornado (with subsidies; base NPV $115M; one input varied at a time).

Variable	Adverse case	NPV	Favourable case	NPV	Swing
Learning rate	90%	(186)	80%	334	520
Annual price decline	4%	(115)	2%	358	473
Initial price level (p0)	-5%	(118)	+5%	339	457
Steady-state utilisation	76.5%	(91)	93.5% (cap 100%)	319	410
Materials cost level	+5%	0	-5%	228	228
Phase 1 capex	+10%	4	-10%	224	220
Discount rate	+1 pt	24	-1 pt	215	191
Fixed opex	+15%	21	-15%	208	187

Learning, price path and utilisation have swings above $400M. Discount rate, capex and fixed cost, which management argues over most, matter least.

(b) Break-even values. NPV is zero at an initial price of $94.6 per kWh, only 2.5% below the $97.0 base, and at steady-state utilisation of 80.2% against 85% in the base. Without subsidies the price must be 11.9% above the base.

(c) Scenarios.

Scenario	Probability	NPV no subsidy	NPV with subsidy	IRR with subsidy
Upside	25%	105	776	17.7%
Base	50%	(537)	115	11.5%
Downside	25%	(1,409)	(765)	-1.9%

The probability-weighted NPV is $60M (0.25 × 776 + 0.50 × 115 + 0.25 × (−765)) with a standard deviation of $548M. **The expected value is small relative to its dispersion**: the board is betting on price, learning and utilisation. It can manage the last two through milestones and hiring but the first only through offtake contracts.

Question 7 solution: The Phase 2 option

(a) Incremental cash flows. Cash flows with Phase 2 (capex at the end of Year 3, output from Year 4) minus cash flows without it, including shared learning, the 15% price discount, extra fixed cost and tax effects.

Year	Incremental volume	Incr. EBITDA	Incr. cash tax	Incr. capex	Incr. FCF	Discount factor	PV
3	0.0	0	0	1,300	(1,300)	0.7472	(971)
4	7.5	35	(3)	0	37	0.6781	25
5	11.2	113	(1)	0	185	0.6153	114
6	12.8	149	(2)	0	259	0.5584	144
7	12.8	155	(11)	0	287	0.5067	146
8	12.8	157	7	0	271	0.4598	125
9	12.8	156	7	0	270	0.4172	113
10	12.8	154	6	0	151	0.3786	57
11	12.8	151	5	0	149	0.3436	51
12	12.8	147	4	0	145	0.3118	45
13	12.8	142	3	0	142	0.2829	40
14	12.8	137	34	0	106	0.2567	27
15	12.8	132	59	0	265	0.2330	62

The PV of the post-capex flows is **$949M** and the PV of the $1,300M capex at 10.2% is $971M, so static NPV is −$23M.

(b) Valuation. The option is a European call on an asset worth S = $949M today with strike X = $1,300M in three years, r = 4% and σ = 35%. PV(X) = 1,300 × e^(−0.12) = $1,153M, and σ√T = 0.6062.

```
d1 = [ln(948.8 / 1,153.0) + 0.5 × 0.35² × 3] / 0.6062 = (−0.1948 + 0.1838) / 0.6062 = −0.0185
d2 = d1 − 0.6062 = −0.6247        N(d1) = 0.4926     N(d2) = 0.2661
C  = 948.8 × 0.4926 − 1,153.0 × 0.2661 = 467.4 − 306.8 = $160.6M
```

Lattice. With Δt = 1 year, u = e^0.35 = 1.4191, d = 1/u = 0.7047 and q = (e^0.04 − d)/(u − d) = 0.4705.

Year	Project value S (top to bottom)	Option value
0	949	145
1	1,346 / 669	310 / 9
2	1,911 / 949 / 471	662 / 21 / 0
3	2,711 / 1,346 / 669 / 332	1,411 / 46 / 0 / 0

Year 3 payoffs, top to bottom, are 1,411 / 46 / 0 / 0; discounting at e^(−0.04) with probabilities q and 1 − q gives $145M. 200 steps give $161M, matching the formula. The plant is built only in the two best states.

(c) Volatility.

Volatility	Option value ($M)
25%	96
30%	128
35%	161
40%	193
45%	226

(d) Interpretation. Static NPV of −$23M says building in Year 3 loses money at expected prices, but Helios need not build. If prices are high it builds and captures the upside; if low it walks away and loses only the holding cost. The loss is truncated at zero, and the $161M is the value of that truncation, so Phase 1 with the option is worth $115M + $161M = $275M.

Question 8 solution: Phase it or build it all now?

(a) All now. With Phase 2 capex of $1,248M at the end of Year 1 and output from Year 2, NPV is **$302M** and IRR 12.13%, $187M above Phase 1 alone.

(b) Comparison. Phase 1 plus the option is worth $275M (formula) or $260M (three-step lattice), so building now is worth $27M to $42M more. The cost of waiting (two years of margin, the lost 4% discount, delayed learning) nearly matches the value of flexibility; the two are equal at a volatility of **39%**, inside the 25% to 45% range.

(c) Covenant. After Phase 1, net debt is 520 + 500 + 350 + 300 = $1,670M, or 3.06 times EBITDA of $545M. Building Phase 2 now adds $1,248M, giving $2,918M or **5.35 times**. The covenant allows 3.75 × 545 = $2,044M, so Phase 2 could be funded only with about $874M of new equity (1,248 − 374), 14% of market value.

Case	Pro forma net debt ($M)	Existing EBITDA ($M)	Net debt / EBITDA	Covenant
Today	520	545	0.95	3.75x
Phase 1 financed per sources table	1,670	545	3.06	3.75x
Build all 30 GWh now (extra debt-funded capex $1,248M)	2,918	545	5.35	3.75x

(d) Price outcomes (with subsidies; “phased” builds Phase 2 only if it adds value).

Price scenario	Phase 1 only	Phase 1 + Phase 2 at yr 3	Phased (best choice)	Build all now	Phased minus all-now
Base	115	92	115	302	(187)
Price -15%	(665)	(1,244)	(665)	(1,214)	549
Price -10%	(372)	(758)	(372)	(656)	284
Price +10%	562	832	832	1,154	(322)

If prices are 10% to 15% low, phasing is worth $284M to $549M more; if 10% high, building now is worth $322M more. With near-symmetric payoffs, the covenant breach decides: **build Phase 1 and hold Phase 2 as an option**. The answer changes if lenders give covenant relief or non-recourse project debt, or if rival capacity is clearly delayed. Repeal risk also favours phasing.

Question 9 solution: Financing and APV

(a) The development loan. Myers’s method values a loan as proceeds minus the PV of after-tax payments discounted at the pre-tax market rate (7%). Interest after tax is 500 × 2.5% × 0.75 = $9.375M a year in Years 2 to 11; principal is repaid at the end of Year 11.

```
PV of proceeds (Year 1)      = 500 / 1.07                  = 467.3
PV of after-tax interest     = 9.375 × annuity, Years 2-11 =  61.5
PV of principal              = 500 / 1.07^11               = 237.5
NPV of loan                  = 467.3 − 61.5 − 237.5        = $168.3M
```

A market-rate loan would be worth $57M (tax shield only), so $111M is the subsidy in the 4.5-point rate gap.

(b) APV.

Component	$M
Base-case NPV at WACC 10.2% (Question 4)	115
Base-case NPV at unlevered cost of capital 10.26%	109
+ NPV of state loan (Myers, discounted at 7%)	168
APV	277

The unlevered NPV is slightly below the WACC figure because the unlevered rate (10.26%) exceeds the WACC (10.17%), which embeds a debt tax shield. The APV is **$277M**.

(c) Should it count? Yes in principle, since the loan exists only because Meridian is built. But the unlevered NPV is already positive, so the loan is not what makes the project acceptable and must not rescue a negative one. The conditions (build start by 31 March 2027, 900 jobs, no change of control) carry costs and default risk, and the loan counts once only, not also as cheap debt in the WACC.

Question 10 solution: Recommendation

(a) Scenarios with repeal risk.

Scenario	NPV credit intact	NPV credit repealed after year 4
Upside	776	461
Base	115	(185)
Downside	(765)	(1,041)

The probability-weighted NPV is $60M with the credit intact and −$238M with it repealed, so with a 30% chance of repeal it is $0.70 \times 60.1 + 0.30 \times (-237.7) =$ **−$29M**, about zero. As an operating investment Meridian is a coin flip. Its value lies in what sits outside the operating DCF: the Phase 2 option (+$161M) and the subsidised loan (+$168M), together $329M, provided both are secured.

(b) Anchor customer. The letter is non-binding, so its $90 floor is not a hedge. The base price reaches $91 in Year 4, so the floor bites only after that. Helios should convert it to a take-or-pay contract with a floor: 4 GWh is 31% of the 12.75 GWh plateau output, which would remove part of the price risk the tornado ranked second. Its value is the risk it removes, not the average price.

Sensitivity discussion

Value is concentrated in a few drivers: a 2.5% price shortfall or a learning rate of 87% erases the subsidised NPV, and a 30% repeal probability brings expected value to about zero. The drivers are also correlated: an oversupply that cuts price tends to coincide with weak utilisation and slower learning, so the downside scenario is more plausible than one-at-a-time swings suggest. And flexibility is worth most near the strike: Phase 2 is almost at the money, so its value responds strongly to volatility, the one input nobody can observe.

Common student errors

1. **Applying the learning curve to each year's output** instead of cumulative output, which resets cost every year (Question 3).
2. **Using Helios's capital structure or beta,** or leaving the 10% hurdle in place (Question 1).
3. **Including sunk, allocated or cannibalisation costs** (Exhibit A), or omitting land and working capital recovery.

4. **Ignoring the tax-loss mechanics,** either giving the project an immediate tax refund or forgetting that salvage is taxed in full.
5. **Treating subsidies as certain,** or as additive without checking taxability; presenting the subsidised NPV without the unsubsidised one.
6. **Reporting payback as evidence of value** when discounted payback never occurs.
7. **Computing the option on the static NPV (−23)** instead of the PV of operating flows (949), or **counting the loan twice.**
8. **Choosing between phasing and building now on NPV alone,** ignoring the covenant.

Alternative defensible answers

The model solution recommends Phase 1 with an option, but a well-argued case can be made for other answers. **Build all 30 GWh now** is defensible on pure value (it exceeds phasing by $27M to $42M at 35% volatility and by more if volatility is below 39%), but only if the covenant can be amended and about $874M of equity found. **Decline** is defensible if the board regards the 30% repeal probability as too low, since at 38% the expected NPV is negative, or if it will not rely on a learning rate that sits at the edge of the confidence interval. **Use a different discount rate**: a student who removes Tamsin from the comparables obtains an asset beta near 1.19 and a WACC near 10.4%, which lowers NPV by about $24M but does not change the ranking of alternatives. A student who treats the credit as taxable (the 40% case in counsel's opinion) finds a project worth about $2M, so the decision becomes one of negotiation, not valuation. Reward a clear decision rule, correct mechanics and honest treatment of uncertainty, not agreement with any single figure.

Board recommendation memo

To: The Board of Directors, Helios Cell Technologies Inc. **From:** Dev Raman, Chief Financial Officer **Date:** 1 December 2026
Re: Project Meridian, Port Alder: recommendation to proceed with Phase 1 on conditions

Recommendation. Approve Phase 1 (15 GWh, $1,500M plus $40M of land) subject to the five conditions below. Do not approve Phase 2 now; reserve the land, power connection and permits for it and review the decision at the end of Year 3.

Basis. At a project discount rate of 10.2% (comparable-company WACC), Phase 1 has an NPV of $115M and an IRR of 11.5% with the Province's support. **Without it the NPV is −$537M and the IRR 4.6%**, so all of the value, and more, is subsidy: $652M in total, of which $475M is the production credit that is reviewed in 2031. If the credit is lost from Year 5 NPV falls to −$185M, and the expected NPV with a 30% chance of repeal is about −$29M across scenarios. The project loses its subsidised value with a 2.5% price shortfall, a learning rate of 87% or steady-state utilisation of 80%. Against that, two items outside the operating forecast are worth more: the option to build Phase 2 (about $161M at 35% volatility) and the development loan ($168M against a 7% market rate). Building all 30 GWh now is worth about $27M to $42M more than phasing, but would take net debt to 5.35 times EBITDA against a 3.75 covenant.

Conditions precedent. 1. A binding provincial agreement that makes the production credit contractual for eight years or compensates Helios for repeal. 2. Tax counsel's written opinion, or an advance ruling, that the credit is not taxable income. 3. Written covenant relief from the banks (a construction holiday or project-level debt outside the covenant), and a signed funding plan including the equity placement. 4. A take-or-pay offtake contract with Northgate Coachworks for at least 4 GWh a year with a price floor. 5. Learning and ramp milestones (conversion cost, yield and utilisation at fixed cumulative volumes) reported quarterly and linked to management incentives, with a stage-gate review before the second construction payment.

Next sixty days. Obtain the Class 3 cost estimate and a fixed-price contract for at least 70% of plant cost to address overrun risk. Complete the covenant negotiation. Secure the Phase 2 land and permit reservation at no more than the holding cost assumed here.

Risks not captured. Construction delay and overrun, chemistry change, rival capacity and the political cycle. The structure buys a position in a business that may be worth building, partly paid for by the Province.

Chapter 2: Case: Norwind Air, a Fleet Renewal Decision

Illustrative case. Norwind Air Group, its people, Aerovance Industries, Lakeshore Aircraft Leasing and every number in this chapter are fictional and were constructed for teaching. Amounts are in $M (millions of US dollars) unless stated. The case draws on Volume 1, Chapters 7, 8, 11, 14, 19 and 20, and on Volume 2 for options and lease analysis. Norwind reports under IFRS, so IFRS 16 applies to leases.

The Situation

Tobias Renner, chief financial officer of Norwind Air Group, had a decision on his desk with a hard date attached. Aerovance Industries, the manufacturer of the NX-200 narrowbody, would hold twelve delivery slots for Norwind in 2027 only if Norwind signed by 15 December 2026. The board would meet on 17 November 2026. Renner had to tell it whether to replace twelve ageing aircraft with twelve NX-200s, how to pay for them, whether to hedge the fuel the new fleet would burn, and how much of the order to commit now.

Norwind is a regional and short-haul carrier based at Skarholm, flying 54 aircraft to about 70 destinations. It carries leisure and business passengers, with a heavy summer peak, and its traffic moves with the business cycle. Its fleet is described in Table 1. The twelve Series 700 aircraft, an average of 19 years old, seat 174 passengers, burn 2.95 tonnes of fuel per block hour and need two expensive heavy maintenance checks, in Years 3 and 9. The NX-200 seats 186 and burns 2.35 tonnes per block hour, 20% less. Norwind would sell the old aircraft to a freight-conversion buyer for $5.0M each.

Table 1: Norwind fleet at 31 December 2026

Type	Number	Average age (years)	Seats	Ownership
Series 700 narrowbody (to be replaced)	12	19	174	Owned, mostly depreciated
Series 800 narrowbody	28	8	180	16 owned, 12 leased
Regional turboprop	14	11	72	Owned
Total	**54**			

The people

Ingrid Halvorsen, the chief executive, wanted the whole order: “A modern fleet is the only fuel hedge that cannot be cancelled.” Renner was less sure. Norwind had been close to its leverage covenant in the last downturn, and he distrusted a deal that looked good only if demand stayed strong. Captain Amara Osei, head of fleet planning, valued the commonality of one type across the short-haul network. Wei Lindqvist, the treasurer, managed fuel hedging and the banks, and had been asked to design a hedging policy for the new fleet. Professor Marcus Eddington, chair of the risk committee, asked a question that framed the case: “If we wait, what do we lose? If we go now, what do we risk?”

The decision

The board faced five linked questions: (1) is the renewal worth doing at all; (2) should Norwind buy the aircraft with secured debt or lease them from Lakeshore, which had offered a 12-year operating lease; (3) how much fuel price risk should be hedged, and with what; (4) given that Norwind’s demand is cyclical and its covenant is tight, how much of the order should be

committed now; and (5) should Norwind use Aerovance's contractual rights to defer six deliveries and to buy six more aircraft at a fixed price?

"Year 0" is delivery, March 2027. Later cash flows occur at year end, and the analysis horizon is twelve years.

Industry Context

Cyclicality. Airline earnings move with GDP, corporate travel budgets and leisure confidence. Capacity is added in booms and cannot be removed quickly in busts, because aircraft are bought or leased on long contracts. The result is a sequence of profitable and unprofitable years, and fleet decisions made in a boom are tested in the next bust. Many airlines have gone out of business with modern fleets and heavy leverage.

Fuel and carbon. Fuel is typically the largest or second-largest cost, about 15% of Norwind's revenue. Airlines recover some fuel price changes through fares, but with a lag and incompletely. New-generation aircraft use roughly a fifth less fuel per seat. Carbon regulation adds a second price on every tonne burned: Norwind's routes fall under an emissions trading scheme, and the carbon price is expected to rise.

Aircraft values and leasing. Aircraft are long-lived, mobile, standardised assets with active resale markets, which makes them easy to finance with secured debt and attractive to lessors. A lessor earns its return from rentals and from the residual value of the aircraft; a lessee that leases passes the residual risk to the lessor. Residual values depend on the model's popularity, the age of the aircraft at disposal and the state of the cycle, and appraisers' estimates are wide.

Manufacturers' contracts. Manufacturers allocate delivery slots years ahead and often sell flexibility: rights to defer deliveries for a fee, to convert orders from one model to another, or to buy additional aircraft at a pre-agreed price. These rights are options and have value, though their prices rarely reflect it explicitly.

IFRS 16. Since 2019 lessees under IFRS recognise most leases on the balance sheet as a right-of-use asset and a lease liability. EBITDA no longer bears the rent, which is replaced by depreciation of the asset and interest on the liability. Operating leases therefore no longer hide leverage, but they still shift residual-value risk to the lessor, and covenant definitions determine how they count.

Financial Data

Table 2: Norwind consolidated income statement (IFRS)

$M, year ended 31 December	FY2024A	FY2025A	FY2026E
Revenue	2,210	2,395	2,480
Fuel and carbon	(345)	(366)	(372)
Staff	(540)	(575)	(590)
Maintenance	(210)	(222)	(230)
Airport and handling	(395)	(425)	(440)
Other operating costs	(300)	(321)	(328)
EBITDA	420	486	520
Depreciation (including right-of-use assets)	(214)	(231)	(240)
EBIT	206	255	280
Net interest (including lease interest)	(70)	(72)	(74)
Profit before tax	136	183	206
Tax at 24%	(33)	(44)	(49)
Net income	103	139	157

Table 3: Norwind consolidated balance sheet (IFRS)

$M, at 31 December	FY2025A	FY2026E
Cash	380	410
Receivables and other current assets	260	270
Spare parts	90	95
Aircraft and equipment (owned)	1,520	1,580
Right-of-use assets	540	520
Other non-current assets	210	225
Total assets	**3,000**	**3,100**
Air traffic liability (tickets sold in advance)	330	350
Trade payables	290	300
Borrowings	1,010	980
Lease liabilities	560	540
Other liabilities	230	240
Equity	580	690
Total liabilities and equity	**3,000**	**3,100**
Net debt including leases	1,190	1,110

Norwind has 200 million shares trading at $9.50. It also has an undrawn revolving credit facility of $300M at 7.5%. Its lending covenant requires net debt including lease liabilities to stay at or below 3.5 times EBITDA, tested annually. At the end of 2026 the ratio is 1,110 / 520 = 2.13 times. In the last downturn EBITDA fell to 76% of its normal level.

Table 4: Capital budgeting inputs per aircraft

Item	Value
NX-200 net price (list $60.0M less 13.3% discount)	$52.0M
Pre-delivery payments (PDPs)	20% of price, in four equal instalments 24, 18, 12 and 6 months before delivery
Sale of each Series 700 aircraft (book value $2.0M)	$5.0M, taxable gain
Block hours per year	3,300
Fuel burn, Series 700 / NX-200 (tonnes per block hour)	2.95 / 2.35
Jet fuel price Year 0 / growth	$820 per tonne / 2% a year
Carbon intensity / carbon price Year 0 / growth	3.16 t CO2 per tonne fuel / $65 per t CO2 / 7% a year
Maintenance cost, Series 700 (Year 0 prices)	$3.1M a year, rising 3% a year
Maintenance cost, NX-200 (Year 0 prices, then rising 3% a year)	$1.0M in Years 1 to 6; $2.0M in Years 7 to 12
Avoided heavy checks on Series 700 (Year 0 prices, rising 3% a year)	$8.0M in Year 3 and in Year 9
Extra-seat contribution (12 more seats, 100% of expected demand index)	$0.90M a year, rising 2% a year
Tax depreciation (NX-200)	Straight line over 15 years to 10% residual
Expected resale value at Year 12	52% of price ($27.04M); taxed on gain over book value
Tax rate	24%

Operating benefits are measured relative to keeping the Series 700 aircraft in service, and all tax effects arise in Norwind's profitable group, so there is no loss carry-forward issue.

Table 5: Financing terms

Item	Terms
Secured aircraft debt	6.5% a year pre-tax; the market rate on Norwind's own secured borrowing, also its incremental borrowing rate under IFRS 16
Unsecured debt	7.2%
Operating lease offered by Lakeshore	12-year term, 0.80% of the $52.0M price per month ($0.416M a month, $4.99M a year, treated as paid annually in arrears); lessor retains residual; return at end of term
Target capital structure for fleet decisions	40% net debt (including leases) / total capital
Market inputs	Risk-free rate 4.2%; equity risk premium 5.5%

Table 6: Airline comparables

Airline (illustrative)	Levered equity beta	Net debt incl. leases / equity	Marginal tax rate
Corvid Airways	1.88	1.10	22%
Skyreach Air	1.64	0.80	25%
Halcyon Airlines	1.40	0.60	20%
Brightfeather Air	1.98	1.30	26%
Oriel Airlines	1.53	0.70	24%

Debt betas may be taken as zero.

Table 7: The demand cycle

Demand moves among three states each year. The utilisation index multiplies block hours (and so fuel savings); the yield index multiplies the extra-seat contribution.

State	Utilisation index	Yield index	Norwind EBITDA before renewal ($M)
Boom	1.08	1.35	not modelled
Normal	1.00	1.00	520
Bust	0.82	0.45	395

Annual transition probabilities (the cycle starts in the Normal state):

From to	Boom	Normal	Bust
Boom	0.50	0.40	0.10
Normal	0.20	0.60	0.20
Bust	0.10	0.40	0.50

Table 8: Fuel exposure and hedging data

Norwind burns 450,000 tonnes of fuel a year (a bill of $369M at $820 per tonne). Fares recover part of fuel increases with a lag. The table gives the last ten years of annual changes in the fuel price and in EBITDA before hedge results.

Year	Change in fuel price (%)	Change in EBITDA before hedges ($M)
Y-9	+2.9	+37
Y-8	+38.7	-91
Y-7	+35.1	-61
Y-6	-11.8	+23
Y-5	-6.0	+3
Y-4	-12.2	+41
Y-3	+17.4	-17
Y-2	+0.5	-7
Y-1	+22.2	-55
Y0	-47.9	+129

Hedging would be done on a rolling 12-month basis with either a fixed-price swap at $820 per tonne or a zero-cost collar (buy a call at $900, sell a put at $740). Swaps and collars are subject to margin calls when the market moves against Norwind.

Table 9: Aerovance options

Item	Terms
Firm order	Up to 12 aircraft, delivered Year 0
Deferral right on 6 aircraft	Delivery may be deferred by 1 or 2 years on payment of a fee of $0.8M per aircraft at the time of deferral; price escalates 4% a year
Purchase rights	Six additional aircraft at a fixed price of $52.0M (net of old aircraft sale, $47.72M), exercisable at any time within 3 years, for a non-refundable deposit of $1.0M per right
Volatility of project value (consultants' estimate)	25%
Risk-free rate (continuously compounded)	4.2%
Annual cash-flow yield forgone by waiting (after-tax operating benefits / project value)	10%

Complications and Exhibits

Exhibit A: Residual value is uncertain

Three appraisers gave 12-year resale values for the NX-200 of 35%, 52% and 60% of price. Norwind's base case uses 52%. A lessee that leases does not bear this risk; a buyer does.

Exhibit B: Carbon price

The $65 per tonne starting price and 7% growth are a planning assumption. Policy could soften (carbon price near zero) or tighten. Renner wants to know how much of the case rests on it.

Exhibit C: How rents are really paid

Lakeshore's monthly rent is payable monthly in advance. The model treats rent as an annual payment in arrears, which understates the cost slightly, and for IFRS 16 the lease liability is the present value of payments at Norwind's incremental borrowing rate of 6.5%.

Exhibit D: Margin calls and liquidity

A 60% swap that is 30% out of the money loses about $66M a year in mark-to-market value, payable in cash if the counterparty requires collateral. Norwind has cash of $410M and an undrawn revolver of $300M.

Exhibit E: Pre-delivery payments

Aerovance requires 20% of the price in four instalments before delivery. Norwind would fund PDPs on its revolver at 7.5%. Under a lease, Lakeshore would pay them, with an arrangement fee not modelled.

Exhibit F: Items excluded

Crew training for the new type, spares provisioning and the commonality benefits of a single type are not quantified; Osei believes they roughly offset. The 12 Series 700 aircraft would otherwise be parked, not sold, in a bust, which is a further reason to prefer keeping Aerovance deliveries flexible.

Assignment Questions

Answer all ten questions. Use the data given, defend any treatment of ambiguous items and show every calculation. Work per aircraft, then scale to the fleet of twelve. Use the discount rate from Question 1 (rounded to one decimal) for operating cash flows.

Question 1: Cost of capital. Unlever the comparable airlines' betas, average them, relever at the 40% target structure and compute the WACC. Explain why the WACC, not the cost of debt, discounts operating cash flows.

Question 2: Operating NPV of renewal. Compute the after-tax incremental cash flows of replacing one Series 700 with one NX-200 (assuming purchase), using the expected demand cycle. Report NPV per aircraft and for twelve, IRR, payback and discounted payback, and decompose the value into its sources.

Question 3: Lease or buy. Compute the net advantage to leasing (NAL) per aircraft, discounting the financing flows at the after-tax cost of secured debt. Find the lessor's implicit rate, the break-even rent and the break-even residual value. Why does the lessee's view of residual value matter?

Question 4: IFRS 16 and the covenant. Compute the lease liability at inception, the first three years of its schedule, and the pro forma net debt/EBITDA under no renewal, buying 12, leasing 12 and buying 6 and leasing 6, in a normal year and a bust year. Which plans comply with the 3.5x covenant?

Question 5: Fuel hedging. (a) Regress the EBITDA changes on fuel price changes in Table 8 to find the minimum-variance hedge ratio, with a confidence interval. (b) Compare the standard deviation of EBITDA at hedge ratios of 0%, 60% and 100%. (c) Compare swap and collar outcomes across fuel prices. (d) Test the covenant at fuel +50% and in a bust with fuel +50%, and discuss the liquidity risk.

Question 6: Fuel and carbon as project risks. Tabulate NPV for fuel price levels from 70% to 150% of base, and for carbon prices of zero to 1.5 times base. Find the break-even fuel price. In what sense is the NX-200 a natural hedge, and what does this imply for the hedging policy?

Question 7: Cyclical demand. Compute NPV per aircraft if the demand state is permanently Boom, Normal or Bust. Find the state probabilities by year and the stationary distribution, and interpret a simulation of 20,000 cycles.

Question 8: Option to defer. Value Aerovance's right to defer six deliveries using a two-step binomial lattice with the Table 9 inputs, allowing early exercise. Compare the result with immediate purchase and show sensitivity to volatility and the deferral fee.

Question 9: Purchase rights and PDPs. (a) Value the six purchase rights with the Black–Scholes formula adjusted for the cash-flow yield forgone, and compare with the $1.0M deposit. (b) Compute the carrying cost of the PDPs for twelve aircraft funded on the revolver.

Question 10: Recommendation. Combine the analyses into a recommended structure: how many aircraft to commit now, how to finance them, what hedge policy to adopt, and which options to exercise or buy. Draft the board memo.

Teaching Note and Full Solutions

Case synopsis

Norwind Air looks like a lease-versus-buy case and is really a case about risk and flexibility. The operating renewal has a thin positive NPV. The financing choice is close and turns on the residual value. Fuel is a risk that the new aircraft partly hedges and that treasury can hedge further. Demand is cyclical and the covenant is tight, so the binding question is how much to commit now. Aerovance's contractual rights are options worth more than they cost. The case suits a module on capital budgeting, leasing and risk management after Volume 1, Chapters 7, 8, 14, 19 and 20. All figures come from one Python model, checked by hand on selected years; year-end flows, the 9.5% WACC from Question 1 and Table 4 inputs are used throughout, and results are per aircraft unless stated.

Learning objectives

Students should be able to (1) compute an airline's cost of capital from comparables; (2) build incremental after-tax cash flows for fleet replacement and decompose NPV; (3) analyse lease versus buy by the net advantage to leasing and explain the role of residual value; (4) apply IFRS 16 and test a leverage covenant under normal and bust conditions; (5) estimate a minimum-variance fuel hedge ratio and compare swaps and collars including liquidity risk; (6) treat fuel and carbon as project risks; (7) model a cyclical demand process; (8) value deferral and purchase options; and (9) write a recommendation that combines value, risk and constraints.

Suggested class plan

Two 90-minute sessions. Preparation: Questions 1 and 2.

Time	Activity	Questions
S1, 0 to 10	Cold poll: all 12 now, none, or something between	–
10 to 30	Operating NPV and where value comes from	1, 2
30 to 55	Lease versus buy; residual value	3
55 to 90	IFRS 16, covenant and the bust test	4
S2, 0 to 30	Fuel exposure, hedge ratio, swap versus collar, margin	5, 6
30 to 50	Demand cycle and simulation	7
50 to 75	Deferral and purchase options	8, 9
75 to 90	Recommendation; revisit the poll	10

Question 1 solution: Cost of capital

Unlevering. With zero debt beta, $\beta U = \beta L / [1 + (1 - t) \times D/E]$.

Airline (illustrative)	Levered beta	Net debt incl. leases / equity	Tax rate	1 + (1 − t) × D/E	Unlevered beta
Corvid Airways	1.88	1.10	22%	1.8580	1.0118
Skyreach Air	1.64	0.80	25%	1.6000	1.0250
Halcyon Airlines	1.40	0.60	20%	1.4800	0.9459
Brightfeather Air	1.98	1.30	26%	1.9620	1.0092
Oriel Airlines	1.53	0.70	24%	1.5320	0.9987

The average unlevered beta is 0.998. The target net debt ratio of 40% gives D/E = 0.40 / 0.60 = 0.6667.

```
βL  = 0.998 × [1 + 0.76 × 0.6667] = 0.998 × 1.5067 = 1.504
ke  = 4.2% + 1.504 × 5.5% = 12.47%
kd  = 6.5% × (1 − 0.24) = 4.94%
WACC = 0.60 × 12.47% + 0.40 × 4.94% = 7.48% + 1.98% = 9.46%, rounded to 9.5%
```

Why the WACC. The renewal's operating cash flows (fuel, maintenance and seat contributions) carry the business risk of an airline, which falls on both lenders and shareholders, so they are discounted at the blended WACC, not at the cost of debt. The financing decision is a separate matter and uses the after-tax cost of debt (Question 3), because lease payments are contractual and nearly as safe as debt service (Volume 1, Chapter 14).

Question 2 solution: Operating NPV of renewal

Year 0. Norwind pays $52.00M, sells the old aircraft for $5.00M and pays tax on the $3.00M gain (24% × 3.00 = $0.72M): −52.00 + 5.00 − 0.72 = **−$47.72M**. Sunk or irrelevant costs (the old aircraft's book value, past depreciation) do not enter, and the old aircraft's sale price is a relevant inflow.

Worked Year 1. Fuel saved per year is (2.95 − 2.35) × 3,300 = 1,980 tonnes. Year 1 fuel is 820 × 1.02 = $836.4 per tonne and carbon is 3.16 × 65 × 1.07 = $219.8 per tonne, so the saving is 1,980 × 1,056.2 = $2.091M at normal utilisation, or $2.05M with the expected utilisation index of 0.98. Maintenance saving is 3.1 × 1.03 − 1.0 × 1.03 = $2.16M. The extra-seat contribution is 0.90 × 1.02 × 0.9654 (the expected utilisation-times-yield index) = $0.89M. Incremental EBITDA is $5.10M. Tax depreciation is 52 × 0.90 / 15 = $3.12M a year, so tax is 0.24 × (5.10 − 3.12) = $0.47M and the Year 1 cash flow is **$4.62M**.

Year 12 sale. Book value is 52 − 12 × 3.12 = $14.56M. The expected sale price is 0.52 × 52 = $27.04M, the gain is $12.48M, tax is $3.00M and net proceeds are $24.04M.

Year	Fuel & carbon saving	Maintenance saving	Avoided heavy check	Extra-seat contribution	Incr. EBITDA	Cash tax	After-tax cash flow
0	–	–	–	–	–	–	(47.72)
1	2.05	2.16	–	0.89	5.10	0.47	4.62
2	2.10	2.23	–	0.90	5.23	0.51	4.72
3	2.17	2.29	8.74	0.91	14.12	2.64	11.48
4	2.23	2.36	–	0.93	5.53	0.58	4.95
5	2.30	2.43	–	0.95	5.69	0.62	5.07
6	2.38	2.51	–	0.97	5.86	0.66	5.20
7	2.46	1.35	–	0.99	4.80	0.40	4.39
8	2.54	1.39	–	1.01	4.94	0.44	4.50
9	2.62	1.44	10.44	1.03	15.52	2.98	12.55
10	2.71	1.48	–	1.05	5.24	0.51	4.73
11	2.80	1.52	–	1.07	5.40	0.55	4.85
12	2.90	1.57	–	1.09	5.56	0.59	4.98
12 sale	–	–	–	–	–	3.00	24.04

Results at 9.5%.

Measure	Per aircraft	Twelve aircraft
NPV	$2.38M	$28.5M
IRR	10.33%	
Payback (years)	8.22	
Discounted payback (years)	11.76	

Decomposition of PV (after tax, per aircraft).

Source	PV ($M)
Fuel and carbon savings	12.51
Maintenance savings	10.59
Avoided heavy checks	8.57
Extra-seat contribution	5.11
Depreciation tax shield	5.23
After-tax resale value	8.09
Total PV of inflows	**50.10**
Net outlay (Year 0)	(47.72)
NPV	**2.38**

The renewal clears its cost of capital by only 0.8 percentage points of IRR. Fuel and carbon contribute about a quarter of the PV, and maintenance and avoided heavy checks together nearly 40%. **Nearly 40% of the PV rests on avoiding maintenance on the old aircraft,** which is why the assumption that the old aircraft would otherwise be kept in service matters; if Norwind would park them in a bust, avoided maintenance is overstated in that state.

Question 3 solution: Lease or buy

Method. Leasing is a financing alternative to buying, so its cash flows are discounted at the after-tax cost of secured debt, 6.5% × 0.76 = 4.94%. Annuity factor for 12 years is 8.8933 and the 12-year discount factor is 0.5607. The annual rent is 0.80% × 12 × 52.0 = $4.99M, or $3.79M after tax.

Item (per aircraft, $M)	Amount
Purchase price avoided (Year 0)	52.00
PV of after-tax rent (12 years at 4.94%)	(33.74)
PV of depreciation tax shield lost	(6.66)
PV of after-tax residual value lost	(13.48)
Net advantage to leasing (NAL)	−1.88

Each aircraft costs $1.88M more to lease than to buy, **−$22.6M for twelve**, on the base residual of 52%. Operating NPV plus NAL (the lease path) is 2.38 − 1.88 = $0.50M per aircraft.

Implicit rate and break-evens. The rate at which the lessor's flows (−52.0, rent of 4.99 for 12 years, and a 27.04 residual) have zero NPV is **6.90%**, above Norwind's 6.5% cost of debt; leasing is a higher-priced source of finance if the lessee expects the residual to hold. The break-even annual rent is $4.71M ($0.393M a month, 0.755% of price), compared with the offered $4.99M (0.80%). The break-even residual is 43.5% of price.

Residual value at Year 12 (% of price)	Residual ($M)	NAL per aircraft	NAL fleet of 12
35%	18.20	1.89	22.6
40%	20.80	0.78	9.3
46%	23.92	−0.55	−6.6
52%	27.04	−1.88	−22.6
60%	31.20	−3.65	−43.8

Residual matters. Leasing becomes attractive if the residual falls below 43.5%. The three appraisals averaged 49%, giving a NAL of −$1.22M, still negative. The buyer holds the residual-value risk and the upside; a lessee holds none of it and the lease contains an implicit put on the aircraft's value. Students should note that the NAL uses expected residual and ignores the value of that put, so the true benefit of leasing is higher than the table suggests. They should also note two other effects: leasing preserves debt capacity (Question 4) and avoids the PDPs (Question 9), and Exhibit C's monthly-in-advance payment makes true rent costlier than modelled, which strengthens the case for buying. A defensible answer buys when the board has confidence in the residual and needs the ownership, and leases when covenant headroom is scarce.

Question 4 solution: IFRS 16 and the covenant

Lease liability at inception. The present value of 12 rent payments of $4.992M at 6.5% is 4.992 × 8.1536 = **$40.73M per aircraft, $489M for twelve**; the right-of-use asset is the same at inception. Schedule for the first three years:

Year	Opening liability	Interest 6.5%	Rent paid	Closing liability	ROU depreciation
1	40.73	2.65	4.99	38.38	3.39
2	38.38	2.49	4.99	35.89	3.39
3	35.89	2.33	4.99	33.23	3.39

Year 1 expense under IFRS 16 is interest $2.65M plus depreciation $3.39M = $6.04M, against a cash rent of $4.99M: expenses are front-loaded. EBITDA excludes the rent, so it is $4.99M per aircraft higher than under the old rules.

Leverage. Pro forma net debt includes the old aircraft's $60M sale proceeds. EBITDA adds the renewal's operating benefit: $61.2M in a normal year and $50.6M in a bust. The base EBITDA is $520M normal and $395M in a bust.

Plan	Pro forma net debt incl. leases	EBITDA, normal year	Net debt/EBITDA	EBITDA, bust year	Net debt/EBITDA
No renewal (today)	1,110	520	2.13x	395	2.81x
Buy 12 (debt-funded)	1,674	581	2.88x	446	3.76x
Lease 12	1,539	581	2.65x	446	3.45x
Buy 6, lease 6	1,606	581	2.76x	446	3.60x

Interpretation. In a normal year every plan complies (2.65x to 2.88x). In a bust, buying all twelve breaches the 3.5x covenant (3.76x) and a six-and-six split also breaches (3.60x); only leasing all twelve complies (3.45x), and with just 0.05x of headroom. Leasing produces lower measured leverage because the lease liability (489) is lower than the debt needed to buy (about $564M net of sale proceeds): the lessor keeps the residual and the lease liability excludes it. **Compliance in the bust state, not NPV, is the binding constraint on the size of the order.** With no renewal, a bust leaves Norwind at 2.81x. The maximum net debt at 3.5x in a bust is $1,560M, below the $1,606M of the six-and-six plan.

Question 5 solution: Fuel hedging

(a) Hedge ratio. If a fraction $(1 - \theta)$ of a fuel price change is not recovered through fares, the EBITDA change is $-(1 - \theta) \times$ fuel bill $\times$ (fuel price change), and the minimum-variance hedge covers that fraction of consumption. Regress ΔEBITDA on ΔFuel (as fractions): mean fuel change 3.89%, mean EBITDA change $0.20M, $\Sigma(x - \bar{x})^2 = 0.6001$, $\Sigma(x - \bar{x})(y - \bar{y}) = -140.5228$.

```
slope = −140.52 / 0.6001 = −$234.2M per 100% change (−$2.34M per 1%)
h*    = 234.2 / 369 = 0.635, about 63% of consumption (≈ 285,600 tonnes)
R²    = 0.92, standard error of slope = $23.7M
95% interval for h* (t = 2.306, 8 d.f.) = 49% to 78%
```

The implied pass-through through fares is about 36%. The interval is wide because there are ten observations.

(b) Risk reduction. The standard deviation of the annual EBITDA change is $\sqrt{[((\text{slope} + h \times 369)^2 \times \sigma F^2) + \sigma\varepsilon^2]}$, with σF = 25.8% and a residual standard deviation of $18.3M:

Hedge ratio	Std dev of EBITDA change ($M)
0%	63.2
60%	18.6
63% (minimum variance)	18.3
100%	39.4

A 60% hedge removes about 70% of the volatility, and a 100% hedge is worse than 60%, since it makes Norwind net long fuel. The remaining $18M is demand noise that a fuel hedge cannot touch.

(c) Swap versus collar (60% of consumption; the collar buys a call at $900 and sells a put at $740, at zero cost; no-hedge EBITDA assumes 38% pass-through and base $520M).

Fuel price ($/t)	Change	EBITDA no hedge	EBITDA, 100% swap	EBITDA, 60% swap	EBITDA, 60% collar
574	-30%	589	478	522	544
656	-20%	566	492	521	543
738	-10%	543	506	521	542
820	+0%	520	520	520	520
902	+10%	497	534	519	498
984	+20%	474	548	519	497
1,066	+30%	451	562	518	496
1,230	+50%	406	590	516	495

The swap flattens EBITDA but gives up the benefit of falling prices (−30% gives 522 against 589 unhedged). The collar keeps most of the benefit down to $740, at the cost of exposure above $900. A 100% swap makes EBITDA fall as prices fall. A swap or collar is a hedge of EBITDA, not a source of value, since fair-priced derivatives have zero expected value (Volume 1, Chapter 19); the benefit is lower distress cost and covenant protection.

(d) Covenant and liquidity. For the buy-12 plan in a normal year with fuel +50%:

Hedge policy	EBITDA at fuel +50% (buy plan, normal demand)	Net debt/EBITDA
No hedge	467	3.59x
60% swap	577	2.90x
100% swap	651	2.57x

Without hedging, a fuel spike alone takes leverage to 3.59x, a breach. In the worst combination, a bust with fuel +50%:

Hedge policy	EBITDA in bust with fuel +50%	Buy 12	Lease 12
No hedge	331	5.05x	4.65x
60% swap	442	3.79x	3.48x

Even a 60% hedge leaves buy-12 in breach (3.79x), while lease-12 hedged just complies (3.48x). The cost of hedging is margin liquidity: a 60% swap with fuel down 30% loses $-66.4M of mark-to-market in a year, which Norwind could need to post as cash, against cash of $410M and a $300M undrawn revolver. Collars lose less on the downside, and a policy that hedges 50% to

65% of net consumption, uses collars for part, and agrees collateral thresholds with counterparties is defensible.

Question 6 solution: Fuel and carbon as project risks

Fuel price level vs base	Year-0 price ($/t)	NPV per aircraft	NPV, 12 aircraft	IRR
70%	574	−0.44	−5.3	9.3%
85%	697	0.97	11.6	9.8%
100%	820	2.38	28.5	10.3%
115%	943	3.78	45.4	10.8%
130%	1,066	5.19	62.3	11.3%
150%	1,230	7.07	84.8	11.9%

Each 10% change in fuel price moves NPV by $0.94M per aircraft ($11.3M for twelve). The break-even fuel price is **74.7% of base, about $612 per tonne**. A fall of a quarter in fuel, plausible after a spike, erases the operating NPV.

Carbon price Year 0 ($/t CO2)	NPV per aircraft	NPV, 12 aircraft
0	−0.75	−9.0
32	0.81	9.8
65	2.38	28.5
98	3.94	47.3

If carbon pricing is abandoned, NPV is −$0.75M per aircraft (−$9.0M for twelve). The case therefore rests on a policy variable as much as on a commodity.

Natural hedge. The renewal's benefit rises when fuel rises: it reduces Norwind's exposure by 12 × 1,980 = 23,760 tonnes a year, 5.3% of consumption. The firm as a whole is hurt by high fuel; the renewal is worth more in exactly that state. Hedging policy should therefore apply the hedge ratio to post-renewal net consumption (426,240 tonnes, so about 268,531 tonnes at 63%) and not to the pre-renewal figure. The cost of this logic is that low fuel prices hurt twice, on the renewal's NPV and on the swap, while the airline's base operations gain; the board should view the three together.

Question 7 solution: Cyclical demand

NPV by state (the state fixed for twelve years):

Demand state (every year)	Utilisation index	Yield index	NPV per aircraft	NPV, 12 aircraft	IRR
Boom	1.08	1.35	6.38	76.6	11.7%
Normal	1.00	1.00	2.91	34.9	10.5%
Bust	0.82	0.45	−2.76	−33.1	8.5%
Markov expectation (start Normal)	–	–	2.38	28.5	10.3%

The Bust state turns the project into a loser (−$2.76M per aircraft, IRR 8.5%) and the Boom state gives $6.38M. The deviations from Normal are asymmetric: a boom adds $3.47M and a bust removes $5.67M, because the yield index falls further below one in a bust (0.45) than it rises in a boom (1.35).

State probabilities. From the Normal start, $P(t) = P(0) \times P^t$:

Year	P(Boom)	P(Normal)	P(Bust)	E[utilisation]
1	20.0%	60.0%	20.0%	0.980
2	24.0%	52.0%	24.0%	0.976
3	24.8%	50.4%	24.8%	0.975
6	25.0%	50.0%	25.0%	0.975
12	25.0%	50.0%	25.0%	0.975

The stationary distribution solves $\pi = \pi P$ and is 25% / 50% / 25% (Boom / Normal / Bust). The chain mixes quickly; by Year 3 the probabilities are within half a point of stationary. Expected utilisation settles at 0.975, below 1.00 because the bust (0.82) outweighs the boom (1.08) in the expectation. The cycle costs \$0.54M of NPV per aircraft relative to a permanently Normal economy (2.91 against 2.38).

Simulation. Over 20,000 simulated 12-year paths, mean NPV is \$2.39M per aircraft (matching the analytic 2.38), the standard deviation is \$1.38M, the 5th percentile is \$0.01M and the 95th \$4.53M; the probability of a negative NPV is 5%. Fleet-wide, mean \$28.7M and standard deviation \$16.5M. **NPV is a poor measure of this risk:** the cycle averages out over twelve years and the project looks safe on a lifetime basis. The risk is the bust year in which leverage breaches, which is why the covenant, not the NPV, governed Question 4.

Question 8 solution: Option to defer six deliveries

Setup. S = PV of operating benefits of a delivered aircraft = \$50.10M; K = net outlay = 52.00 − 5.00 + 0.72 = \$47.72M; immediate NPV = \$2.38M. With σ = 25%, Δt = 1 year, risk-free rate 4.2% and yield forgone δ = 10%:

```
u = e^0.25 = 1.2840     d = 1/u = 0.7788
q = [e^(0.042 − 0.10) − d] / (u − d) = (0.9436 − 0.7788) / 0.5052 = 0.3263
Strike at year t = 47.72 × 1.04^t + 0.8 (t ≥ 1)
```

Early exercise is allowed, and because δ exceeds the risk-free rate, waiting costs cash flow, so exercise at an in-the-money node can be optimal.

Year	Node	Project value S	Strike incl. escalation and fee	Exercise value	Option value	Best action
0	start	50.10	47.72	2.38	4.35	wait
1	u	64.33	50.43	13.90	13.90	exercise
1	d	39.02	50.43	0.00	0.00	wait
2	uu	82.60	52.41	30.18	30.18	exercise
2	ud	50.10	52.41	0.00	0.00	lapse
2	dd	30.39	52.41	0.00	0.00	lapse

At Year 2: uu 30.18; others lapse. At the up-node in Year 1, exercising yields 13.90, while waiting is worth e^(−0.042) × 0.3263 × 30.18 = 9.44, so exercise. At the down-node, the project is out of the money and waiting costs nothing. At the start, immediate exercise gives 2.38 and waiting gives e^(−0.042) × 0.3263 × 13.90 = **\$4.35M**. The deferral right is worth \$1.97M more per aircraft than buying now, **\$11.8M for six aircraft** (\$26.1M against \$14.3M).

Volatility drives the value:

Volatility	Option value per aircraft	Premium over immediate NPV	Six aircraft
15%	2.38	0.00	14.3
20%	3.20	0.82	19.2
25%	4.35	1.97	26.1
30%	5.49	3.11	32.9
35%	6.63	4.25	39.8

At about 16% volatility the right is worth no more than immediate exercise; below that, buy now. The fee matters little:

Deferral fee per aircraft ($M)	Option value per aircraft	Six aircraft
0.00	4.60	27.6
0.80	4.35	26.1
1.50	4.13	24.8
2.40	3.85	23.1
3.50	3.50	21.0

Interpretation and limits. A positive-NPV project can still be better deferred when uncertainty is high and the cost of waiting is moderate. The result depends on δ: waiting forgoes $5M a year of benefits per aircraft, and at 10% the leakage is large, yet the option is still worth $2M extra. The model ignores that aircraft not delivered must be replaced by costly maintenance on the old fleet, which is already inside δ, and that Aerovance's slots may disappear.

Question 9 solution: Purchase rights and PDPs

(a) Black–Scholes with a yield. C = S e^(−δT) N(d1) − K e^(−rT) N(d2), with S = 50.10, K = 47.72, T = 3, r = 4.2%, δ = 10%, σ = 25%.

```
S × e^(−0.30)   = 37.11        K × e^(−0.126) = 42.07
d1 = [ln(50.10 / 47.72) + (0.042 − 0.10 + 0.5 × 0.0625) × 3] / (0.25 × √3) = -0.0731
d2 = d1 − 0.4330 = -0.5061     N(d1) = 0.4709     N(d2) = 0.3064
C = 17.48 − 12.89 = $4.58M per right
```

Six rights are worth **$27.5M** against deposits of $6.0M, a net gain of $21.5M.

Volatility	Right value per aircraft ($M)	Six rights ($M)
15%	2.08	12.5
20%	3.31	19.9
25%	4.58	27.5
30%	5.87	35.2
35%	7.15	42.9

With δ = 6% the right is worth $7.08M, so the value is sensitive to the yield assumption. The rights are cheap because the manufacturer values slot flexibility, and they should be bought only if Norwind may grow beyond a replacement programme, since aircraft purchased under them add capacity, not replace old aircraft, and the operating benefits assumed in S would differ.

(b) PDP carrying cost. Each 5% instalment on twelve aircraft is 0.05 × 624 = $31.2M. Interest at 7.5% for 2.0, 1.5, 1.0 and 0.5 years is 31.2 × 0.075 × 5.0 = **$11.7M** pre-tax, $8.9M after tax; total PDPs are $124.8M. Including the carrying cost reduces the fleet NPV from $28.5M to $19.6M, and under a lease the lessor funds PDPs, but its fee is not modelled.

Question 10 solution: Recommendation

Comparing structures (per the base model; leverage is net debt including leases over EBITDA in a bust year; the covenant is 3.5x).

Structure	Value ($M)	Bust-year leverage	Comment
No renewal	0	2.81x	Keeps maintenance burden and fuel exposure
Buy all 12 now	28.5	3.76x	Breaches covenant in a bust
Lease all 12 now	5.9 (NPV plus NAL)	3.45x	Passes with 0.05x of headroom; costs $22.6M of NAL
Buy 6 now, defer 6 with option	40.3 (6 × 2.38 plus 6 × 4.35)	3.31x	Passes; option value captured
Lease 6 now, defer 6 with option	6 × (2.38 − 1.88) plus 26.1 = 29.1	3.15x	Passes; lower value than buying

Recommended structure. Buy the first six aircraft, defer the other six by one year using the Aerovance right, and decide in Year 1 whether to exercise. This is worth about $40.3M against $28.5M for all twelve now, an improvement of $11.8M, and keeps bust-year leverage at 3.31x. If a bust seems likely when the second batch is due, lease it, accepting the NAL cost in return for covenant headroom. The Year 1 decision rule should combine value and constraint: exercise if the demand state is not Bust and projected bust-year leverage after delivery is no more than about 3.2x.

Hedging. Hedge 55% to 65% of next-12-month post-renewal net consumption, about 270,000 tonnes at 63%, using a mix of swaps and collars with collateral thresholds, so that the margin outflow in a 30% fuel fall stays within the revolver. Re-estimate the hedge ratio annually.

Options. Do not buy the purchase rights as part of the renewal. They are worth about $27M against $6M of deposits, but only if Norwind grows beyond replacement, a question for the strategy review.

Sensitivity discussion

Operating NPV is thin: it disappears if fuel is 25% lower, if the carbon price is zero, or if the maintenance saving is overstated. Financing value is small and sign-flips with residual value (the break-even is 43.5% against 52%). Option values are large relative to NPV because the base project is close to the money and volatility is 25%. And the covenant constraint, not the NPV, decides the size of the commitment, so a plausible change to the covenant definition or to the bust-state EBITDA changes the answer more than any change in discount rate.

Common student errors

1. **Discounting operating cash flows at the cost of debt,** or discounting the lease-versus-buy flows at the WACC (Questions 1 and 3).
2. **Omitting the tax on the sale of the old aircraft,** or treating its book value as a cost.
3. **Counting fuel savings at the base fuel price only,** forgetting the carbon price, or leaving out utilisation.
4. **Comparing the lease rate with the coupon on debt** instead of computing the NAL.
5. **Ignoring the residual value** or treating it as certain.
6. **Treating IFRS 16 as irrelevant to leverage,** or computing the covenant with the pre-renewal EBITDA.
7. **Assuming a fuel swap adds value,** or hedging 100% of consumption, which overshoots the minimum-variance ratio.
8. **Valuing the deferral as a European call** without the yield forgone, which gives a value of zero or below.
9. **Using NPV standard deviation as the measure of cyclical risk** and concluding the project is safe.

Alternative defensible answers

The model recommendation (buy six, defer six) is not unique. **Lease all twelve** is defensible if the board wants maximum covenant headroom and treats the residual put as valuable: the NAL cost of $22.6M is the price of insurance. **Buy all twelve with equity support** is defensible if Norwind can raise equity or negotiate a covenant at 4.0x. **Do nothing** is defensible if the board doubts the carbon price, since NPV is negative without it. A student who treats deferral as costless, or who uses a lower yield forgone, will find an even larger deferral value and may recommend deferring all twelve; this is sound if slots are available later. The hedge ratio could reasonably be set anywhere in the interval of 49% to 78%. Assessment should reward a clear decision rule, correct mechanics and an explicit treatment of the covenant.

Board recommendation memo

To: The Board of Directors, Norwind Air Group **From:** Tobias Renner, Chief Financial Officer **Date:** 10 November 2026 **Re:** NX-200 fleet renewal: commit to six aircraft now, defer six, hedge 60% of fuel

Recommendation. Commit to six NX-200 aircraft for delivery in March 2027, financed by secured debt, and exercise Aerovance's right to defer six further deliveries by one year. Adopt a fuel hedging policy covering 55% to 65% of the next twelve months' post-renewal fuel burn. Do not buy the purchase rights now.

Basis. At a WACC of 9.5%, replacing a Series 700 with an NX-200 has an NPV of $2.38M per aircraft ($28.5M for twelve) and an IRR of 10.33%. The margin is thin: the value disappears if fuel falls to about $612 a tonne or the carbon price is abandoned. Buying is worth about $1.9M per aircraft more than leasing at the offered rent, if the aircraft resell at 52% of price after 12 years; the lease wins only if the resale value falls below 43.5%. Buying all twelve would take net debt including leases to 3.76 times EBITDA in a bust year, above the 3.5x covenant. Six aircraft keep the ratio at 3.31 times in a bust. The deferral right is worth $4.35M per aircraft against $2.38M for buying now, so deferring six is worth about $11.8M more than taking them immediately.

Hedging. Fuel is the main source of volatility not related to demand. A 63% hedge minimises the volatility of EBITDA. We propose a mixture of swaps and collars and collateral thresholds that keep margin calls within the undrawn revolver.

Conditions and risks. The recommendation depends on a residual value near 50%, a continuing carbon price, and a covenant that remains at 3.5x including leases. Management will return to the board in about twelve months with a Year 1 decision on the deferred six: exercise if demand is not in a bust and projected bust-year leverage after delivery is below about 3.2x, and lease them if a bust seems likely. The board should recognise that this plan is a deliberate trade of some value for protection against a bust, not a bet that no bust will occur.

Chapter 3: Case: Lumen Therapeutics, Valuing a Biotech Pipeline

Type: Case. Prerequisites: Volume 1, Chapters 6 (NPV), 8 (uncertainty and real options), 10 (CAPM) and 11 (WACC); Volume 2, the chapters on advanced valuation. Lumen Therapeutics, its people, its programs and every figure in this chapter are fictional and illustrative. No real company, trial result, regulatory statistic or market datum is intended or reproduced.

The Situation

It is Wednesday 4 November 2026. In the glass-walled conference room on the third floor of a converted laboratory building, Dr. Priya Raman, chief executive officer of Lumen Therapeutics, Inc., has spread three printed spreadsheets across the table and is not looking at any of them. She is looking at a calendar. On 17 November the board of directors meets. On 18 November, at 17:00, the exclusivity window that Corvane Pharma granted to Lumen expires, and with it the term sheet on the table: a regional licensing and co-funding deal for Lumen's lead asset, LUM-101. A second bidder, Vantex Biopharma, has put forward a competing structure that expires on the same day. The board must tell management what to do.

Lumen is a clinical-stage company listed on a US exchange. It was founded nine years ago by Dr. Raman and a medicinal chemist, Dr. Anil Sorensen, on the back of a license from the University of Calder, and it has never sold a product. Today it has three clinical programs and an early discovery platform:

- **LUM-101**, an oral, biomarker-selected inhibitor for a genetically defined subtype of second-line lung cancer. It is in a pivotal Phase III trial that has just begun enrolling its final cohort. Topline data are expected in December 2028.
- **LUM-205**, an injectable antibody for moderate-to-severe ulcerative colitis. It has finished a small, encouraging Phase IIa study and is about to start the larger Phase IIb trial that will decide its fate. Topline data are expected in December 2027.
- **LUM-310**, an oral small molecule for a rare pediatric metabolic disorder. It is in Phase I.
- A **discovery platform** that screens for protein-degradation compounds and has produced no clinical candidate yet.

Lumen has 58.4 million shares outstanding, and 62.0 million on a diluted basis once in-the-money options and restricted stock units are counted. The shares closed at $14.50 on 3 November, giving a market capitalization of $899.0 million on the diluted count. The company holds $412.0 million in cash and marketable securities and owes $75.0 million on a senior term loan. It burned $142.9 million of cash in 2026 and will burn more in 2027 and 2028, when all three programs are funded and Phase III is at full enrollment. The arithmetic of that burn is the first thing the chief financial officer, Marcus Feldt, wants the board to see, because the term loan contains a minimum-liquidity covenant of $100 million. On the current plan, with no new money, Lumen breaches that covenant in the third quarter of 2028, some months before the Phase III result it is waiting for.

The people around the table matter, because each of them wants something different from the answer.

Dr. Priya Raman (CEO) wants to keep LUM-101 whole, believing oral convenience and biomarker selection will make it a franchise drug.

Marcus Feldt (CFO), a former equity analyst, wants a funded company, a covenant out of play, and a defensible valuation. He has asked for a risk-adjusted NPV of every program and an honest comparison of three ways to fund 2028: an equity raise, the Corvane deal or the Vantex deal.

Dr. Helen Ostrow (chief medical officer) owns the probability estimates and argues that biomarker enrichment justifies a LUM-101 Phase III probability above the oncology benchmark. The board's science adviser thinks she is anchoring on a small Phase II cohort.

Tomas Beringer (head of business development) negotiated both term sheets and worries about what a partner with a competing asset will prioritize.

Ellen Okafor-Lang (board chair) is a former pharmaceutical executive. She will not accept a recommendation that treats an expected value as a forecast, and she has said so in advance: "Tell me what each choice is worth in the world where the trial works, and in the world where it does not."

Deadline and decision. The board must decide by 17 November whether to (a) accept the Corvane term sheet, perhaps after a counter-proposal, (b) accept the Vantex alternative, or (c) decline both and raise equity in the market, retaining all rights. Whichever it chooses, it must also decide whether to continue funding all three programs through their next gates, and what valuation to present publicly. You have been asked, as a member of Marcus Feldt's finance team, to prepare the analysis.

Industry Context

Drug development is a sequence of experiments that mostly fail. A compound enters first-in-human testing (Phase I) to establish safety and dosing. If it is tolerated, it moves into Phase II, where a first test of efficacy is run in a modest number of patients. Programs that show a signal enter Phase III, the large pivotal trials regulators use to decide on approval. A successful Phase III is followed by a regulatory filing, a review period and, if all goes well, approval and launch. At each gate the sponsor learns something, and the sponsor can stop. That structure, a series of staged commitments with a termination right at every gate, is what makes the economics of a pipeline different from the economics of a factory.

Staged cost, back-loaded revenue. Development spending rises sharply with each phase: Phase I costs tens of millions, Phase III costs hundreds. Revenue arrives only after approval, often six to ten years after the first dose, and then follows a characteristic life cycle: a slow launch ramp as physicians, payers and patients adopt the drug, a plateau at peak sales, and a steep decline when patents and exclusivity expire and generic or biosimilar competitors enter. A pipeline asset is therefore a long, thin stream of negative cash flows followed, with some probability, by a very large and then fading positive one.

Probability of success is the dominant driver. Practitioners estimate the probability that a program clears each gate, often by therapeutic area, and multiply the cash flows that depend on clearing a gate by the cumulative probability of reaching it. The result is a risk-adjusted net present value, written rNPV. Estimates of phase-transition probabilities differ by area. In this case we use an illustrative benchmark panel (see Financial Data) in which oncology Phase III success is lower than immunology or rare-disease Phase III success, and Phase II is the steepest gate everywhere. These benchmarks are fictional teaching inputs. In practice analysts calibrate to the company's own history and to published industry studies, and they should expect the answer to be sensitive to the choice.

Two ways to price risk. An asset can be valued by discounting success-case cash flows at a high venture rate that implicitly absorbs failure risk, or by probability-weighting cash flows and discounting at a rate that reflects only systematic risk. Volume 1, Chapter 8 argued for explicitness: technical failure of one compound is diversifiable and belongs in the cash flows. A 25 percent rate combined with explicit probabilities counts the same risk twice.

Partnering is the industry's financing and risk-sharing technology. Small companies license regional or global rights in exchange for an upfront payment (certain), milestones (contingent), royalties (contingent and long-dated) and sometimes a share of development cost. The seller gives up upside in the success state in return for cash and risk transfer in every state.

Whether that is attractive depends on the probability of success, on the cost of external capital, and on whether the alternative is a dilutive raise at a depressed price. Upstream academic licensors usually take a share of sublicense income, and the wording can matter.

Market valuation of clinical-stage companies. Clinical-stage equities often trade far from the rNPV of their pipelines. The gap reflects more conservative probabilities, higher discount rates and anticipated dilution when a company must raise money before its data arrive. A good analyst asks what probability or rate the price implies. Under both US GAAP and IFRS research costs are expensed, so the balance sheet shows almost nothing for the pipeline and the whole valuation must be forward-looking.

Financial Data

All amounts are in US$ millions unless stated. The valuation date is 31 December 2026. Year 0 is calendar 2026, Year 1 is 2027, and so on. Cash flows are treated as arriving at the end of each year, and the transaction values in the deal terms are treated as received at Year 0 or at the end of the year in which the triggering event occurs.

Table 3.1: Lumen Therapeutics, income statement extract

US$ millions	FY2024A	FY2025A	FY2026E
Collaboration and grant revenue	14.2	17.5	12.0
Research and development	(118.4)	(139.6)	(151.2)
General and administrative	(24.1)	(26.3)	(28.0)
Operating loss	(128.3)	(148.4)	(167.2)
Interest expense	(6.9)	(7.1)	(7.1)
Interest income	11.6	15.9	18.2
Net loss	(123.6)	(139.6)	(156.1)
Weighted basic shares (millions)	56.1	57.2	58.0
Loss per share (US$)	(2.20)	(2.44)	(2.69)
Memo: stock-based compensation included above	11.8	12.9	14.0

Collaboration revenue is a government grant that ends in 2026. FY2026E combines nine months of actual results with a three-month estimate.

Table 3.2: Lumen Therapeutics, balance sheet extract

US$ millions	31 Dec 2025A	31 Dec 2026E
Cash and marketable securities	554.9	412.0
Prepaid and other current assets	8.8	9.5
Property and equipment, net	20.5	21.0
Right-of-use asset	15.8	14.2
Other non-current assets	6.0	6.0
Total assets	**606.0**	**462.7**
Accounts payable and accrued expenses	36.1	38.5
Deferred revenue	8.0	6.0
Senior term loan	75.0	75.0
Lease liability	17.2	15.6
Total liabilities	**136.3**	**135.1**
Paid-in capital	1,366.0	1,380.0
Accumulated deficit	(896.3)	(1,052.4)
Total equity	**469.7**	**327.6**
Total liabilities and equity	**606.0**	**462.7**

The term loan bears 9.5 percent interest paid in cash, matures in December 2030 as a single payment, and carries a covenant requiring at least $100.0 million of unrestricted cash and marketable securities at all times. Operating cash flow in 2026 is estimated at an outflow of $138.6 million and capital expenditure at $4.3 million, which reconciles the $142.9 million fall in cash. The company has a large accumulated tax loss carry-forward that this chapter ignores except where noted.

Table 3.3: Market data and capital structure (3 November 2026)

Item	Value
Share price (US$)	14.50
52-week range (US$)	9.80 to 21.30
Basic shares outstanding (millions)	58.4
Dilutive options and RSUs, treasury-stock method (millions)	3.6
Diluted shares (millions)	62.0
Market capitalization	899.0
Cash and marketable securities	412.0
Senior term loan	75.0
Net cash	337.0
Enterprise value (market capitalization less net cash)	562.0
Equity beta (against broad market)	1.25
Risk-free rate	4.3%
Equity risk premium	5.5%
Statutory tax rate	21.0%

The CFO’s team computes a CAPM cost of equity of 4.3% + 1.25 x 5.5% = 11.2 percent. Because Lumen has net cash and its cash flows are overwhelmingly equity-funded, it uses 11 percent as the discount rate on risk-adjusted program cash flows. Sector practitioners the team consulted use rates from 10 to 12 percent for rNPV models of this kind.

Table 3.4: Pipeline overview and scheduled development spending (US$ millions, spent only if the program reaches the stage)

Program	Stage and timing	Spend by year	Stage total	Stage success probability
LUM-101	Phase III, 2027-2028	2027: 90; 2028: 75	165	55%
LUM-101	Filing, review and pre-launch, 2029	2029: 50	50	90%
LUM-205	Phase IIb, 2027	2027: 55	55	34%
LUM-205	Phase III, 2028-2030	80; 100; 60	240	58%
LUM-205	Filing, review and pre-launch, 2031	2031: 55	55	90%
LUM-310	Phase I, 2027-2028	22; 10	32	66%
LUM-310	Phase II, 2029-2030	30; 32	62	42%
LUM-310	Phase III, 2031-2033	45; 40; 35	120	65%
LUM-310	Filing, review and pre-launch, 2034	2034: 25	25	92%

Each stage’s success probability is the probability of clearing that gate, given that the program entered it. A stage’s spending is committed only if every earlier gate was cleared. Approval and launch occur at the end of the filing year, so first sales are in 2030 for LUM-101 (Year 4), 2032 for LUM-205 (Year 6) and 2035 for LUM-310 (Year 9).

Table 3.5: Illustrative benchmark panel for stage success probabilities (fictional teaching data)

Therapeutic area	Phase I to II	Phase II to III	Phase III to filing	Filing to approval
Oncology, biomarker-selected	60%	30%	50%	88%
Immunology	62%	34%	58%	90%
Rare metabolic disease	66%	42%	65%	92%

Lumen’s inputs equal the panel for LUM-205 and LUM-310. For LUM-101, Dr. Ostrow uses 55 percent and 90 percent rather than the panel’s 50 percent and 88 percent, citing the biomarker enrichment. LUM-101 has already passed Phase II, so only the Phase III and filing values are used.

Table 3.6: Commercial assumptions by program

Assumption	LUM-101	LUM-205	LUM-310
Peak annual net sales (US$ millions)	1,400	1,300	620
First sales year (calendar)	2030	2032	2035
Launch ramp, percent of peak in launch years 1 to 8	5, 15, 32, 54, 74, 88, 97, 100	4, 12, 28, 48, 68, 84, 95, 100	12, 35, 65, 88, 100
First year of generic or biosimilar entry	2041	2044	2047
Sales after entry, percent of peak, years 1 to 4	45, 22, 12, 7	70, 48, 34, 24	60, 35, 20
Cost of goods sold plus licensor royalty, percent of sales	14%	18%	12%
Lifecycle R&D, percent of sales	4%	5%	3%
Sales and marketing, launch years 1-3	45% of sales	40%	30%
Sales and marketing, launch years 4-5	32%	30%	22%
Sales and marketing, launch year 6 onward	24%	22%	20%
Sales and marketing floor in first four launch years (US$ millions)	70	55	30

Sales at peak persist until the generic entry year, then fall as shown, and are treated as zero after the listed years (no terminal value). Sales and marketing in any year is the greater of the percentage of sales and the floor, where the floor applies only in the first four launch years and only while sales are positive. The 21 percent tax rate applies to operating profit and to development spending alike. The licensor royalty to the University of Calder is a 3 percent sales royalty that is included in the cost-of-goods line for LUM-101.

Table 3.7: Management’s unrisked commercial forecast (net sales and operating profit before tax, US$ millions)

Operating profit equals sales times (1 - COGS and royalty % - lifecycle R&D %) less sales and marketing. It excludes development spending in Table 3.4 and assumes the drug is approved. Year 0 is 2026.

Year	Calendar	LUM-101 sales	LUM-101 op. profit	LUM-205 sales	LUM-205 op. profit	LUM-310 sales	LUM-310 op. profit
4	2030	70.0	(12.6)	-	-	-	-
5	2031	210.0	77.7	-	-	-	-
6	2032	448.0	165.8	52.0	(15.0)	-	-
7	2033	756.0	378.0	156.0	57.7	-	-
8	2034	1,036.0	518.0	364.0	134.7	-	-
9	2035	1,232.0	714.6	624.0	293.3	74.4	33.2
10	2036	1,358.0	787.6	884.0	415.5	217.0	119.3
11	2037	1,400.0	812.0	1,092.0	600.6	403.0	221.7
12	2038	1,400.0	812.0	1,235.0	679.2	545.6	343.7
13	2039	1,400.0	812.0	1,300.0	715.0	620.0	390.6
14	2040	1,400.0	812.0	1,300.0	715.0	620.0	403.0
15	2041	630.0	365.4	1,300.0	715.0	620.0	403.0
16	2042	308.0	178.6	1,300.0	715.0	620.0	403.0
17	2043	168.0	97.4	1,300.0	715.0	620.0	403.0
18	2044	98.0	56.8	910.0	500.5	620.0	403.0
19	2045	-	-	624.0	343.2	620.0	403.0
20	2046	-	-	442.0	243.1	620.0	403.0
21	2047	-	-	312.0	171.6	372.0	241.8
22	2048	-	-	-	-	217.0	141.0
23	2049	-	-	-	-	124.0	80.6

Table 3.8: Comparable clinical-stage companies (fictional, 3 November 2026, US$ millions)

Measure (US$ millions)	Enterprise value
Kestrel Oncology (Phase III; market cap 1,240, net cash 310)	930
Brightwell Therapeutics (Phase III; 760, 285)	475
Norvale Immunology (Phase II; 540, 190)	350
Tessaly Rare Disease (Phase II; 410, 120)	290
Piedmont Biologics (Phase III; 1,020, 150)	870
Peer mean / median	583 / 475
Lumen Therapeutics (Phase III; 899, 337)	**562**

Table 3.9: Base cash plan with no new financing (US$ millions)

	2027	2028
Opening cash	412.0	226.4
Research and development (all three programs funded)	(167.0)	(165.0)
General and administrative (treated as cash)	(28.0)	(28.0)
Interest on term loan (9.5% of 75.0)	(7.1)	(7.1)
Interest income (4.0% of opening cash)	16.5	9.1
Closing cash	**226.4**	**35.3**

Table 3.10: Competing term sheets for the ex-US rights to LUM-101

Term	Corvane Pharma (Deal A)	Vantex Biopharma (Deal B)
Territory	All markets outside the United States, 40% of global LUM-101 sales	Same
Upfront payment at signing (before 31 December 2026)	$150.0M	$260.0M
Share of remaining Phase III cost reimbursed by partner	35% of 2027 and 2028 spending	None
Milestone on positive Phase III topline	$60.0M	$40.0M
Milestone on ex-US marketing authorization	$50.0M	$30.0M
Sales milestones (paid once, in first year the threshold is met)	$100.0M at ex-US annual sales of $300M; $150.0M at $500M	$75.0M at ex-US annual sales of $400M
Royalty on ex-US net sales	16% on first $250M; 20% on $250M-$450M; 24% above $450M	12% on first $300M; 15% on $300M-$500M; 18% above $500M
Royalty step-down	Rates halve from the first year of generic entry	Same
Partner manufactures ex-US supply	Yes	Yes
Exclusivity expires	18 November 2026, 17:00	18 November 2026, 17:00

For the valuation in this chapter, ex-US sales equal 40 percent of the global sales in Table 3.7 for LUM-101, ex-US approval follows US approval with certainty (as in the base program model), the ex-US authorization milestone is paid in 2030 (Year 4, the first sales year), the Phase III milestone is paid at the end of 2028 (Year 2) and the upfront payment is received at Year 0. Under either deal, Lumen keeps the United States, which is 60 percent of global sales, and bears the University of Calder obligations described in the next section.

Complications and Exhibits

Exhibit 1: The University of Calder sublicense clause. Lumen’s license entitles the University to (a) 3 percent of net sales of any licensed product sold by Lumen or its sublicensees and (b) 18 percent of “Sublicense Income,” defined as all consideration received from a sublicensee other than payments for research and development at cost and royalties on net sales subject to clause (a). Counsel’s first reading is that the 18 percent applies to upfronts and milestones, and that royalties Lumen receives are covered by the 3 percent sales royalty, which Lumen must fund. A second reading, hinted at by the University, is that royalties received are Sublicense Income subject to 18 percent in place of the 3 percent. A third, aggressive reading would also apply 18 percent to cost reimbursements. The base case uses the first reading.

Exhibit 2: The LUM-205 Phase IIa data. The Phase IIa study enrolled 38 patients; week-12 clinical remission was 41 percent on drug and 23 percent on placebo ($p = 0.07$ on the original primary endpoint). The analysis plan was amended before unblinding to add a stricter second definition on which the difference was larger. The science adviser calls it "encouraging but exactly the sort of signal that shrinks at scale." The 34 percent Phase IIb probability is the benchmark value, not an upgrade.

Exhibit 3: A competing readout. A rival compound against the same genetic subtype reads out a pivotal trial in the second half of 2027. A positive, clearly superior result would cut LUM-101 peak sales to about $1,000 million; a negative result would lift them to $1,800 million. The $1,400 million base sits between and is not formally probability-weighted.

Exhibit 4: Pricing policy. In the fictional jurisdiction in which Lumen sells, small-molecule drugs become eligible for government price negotiation nine years after approval, and biologics thirteen years after approval. Management's estimate is that this would reduce net price on LUM-101 by 15 percent from 2039, the ninth year after its 2030 launch, until generic entry. The base forecast in Table 3.7 does not include this effect. LUM-205 is a biologic and LUM-310 is protected by orphan-product exclusivity, so neither is assumed to be affected.

Exhibit 5: A potential priority review voucher. If LUM-310 is approved, management believes there is a 40 percent chance that Lumen is awarded a transferable priority review voucher for rare pediatric disease. Vouchers have been sold privately; management assumes a sale price of $100 million at the end of 2034, taxable at 21 percent. The base case excludes it.

Exhibit 6: Corvane's competing asset. Corvane owns a Phase II compound aimed at the same subtype. The term sheet requires only "commercially reasonable efforts" and contains no non-compete, minimum-sales or reversion right. Mr. Beringer believes a reversion right is worth more than a point of royalty but has not priced it.

Exhibit 7: Tax losses. Lumen has about $780 million of accumulated tax losses. The models here ignore their timing benefit and apply a 21 percent rate to operating profit, with a 21 percent shield on development spending, as if losses in one program offset profits in another.

Exhibit 8: Equity market access. The company's bankers advise that an underwritten follow-on of $250 million could be priced at $12.00 per share, a discount of about 17 percent to the last close, with gross underwriting spread and costs of 5.5 percent. They caution that a larger raise, or a raise priced after the market learns of a covenant problem, would price lower.

Exhibit 9: Structure of the deal cases. Under either deal, ex-US sales are 40 percent of global LUM-101 sales and are booked by the partner. Lumen keeps the 60 percent US business, with the same cost percentages as Table 3.6 but a first-four-years sales and marketing floor of $55 million instead of $70 million, and continues to owe the University 3 percent of ex-US net sales.

Assignment Questions

Use the 11 percent discount rate, end-of-year timing and a 21 percent tax rate throughout unless a question says otherwise. State every assumption you add.

1. **Probabilities and expected spending.** For each of LUM-101, LUM-205 and LUM-310, compute the cumulative probability of reaching approval and the probability of entering each stage. Compute the expected development spending of each program (stage spending times the probability the stage is entered) and compare it with the scheduled spending in Table 3.4. Explain why the expected figure is the right one for an rNPV.
2. **rNPV of LUM-101.** Using Tables 3.4 to 3.7, build the year-by-year risk-adjusted after-tax cash flow of LUM-101 and its rNPV at 11 percent. Show how the operating profit in Table 3.7 is built up from the commercial assumptions in Table 3.6 for at least the first four sales years. Report the present value of development cost and of commercial cash flow separately.

3. **rNPV of LUM-205 and LUM-310.** Repeat Question 2 for the other two programs. Explain why the rNPV of LUM-205 is positive even though its probability of reaching approval is below 18 percent, and why the rNPV of LUM-310 is much smaller.
4. **Sum of the parts and the market.** Value Lumen's equity per diluted share. Treat the discovery platform as worth $45.0 million, and capitalize corporate G&A of $28.0 million per year for ten years after tax at 11 percent as a deduction. Compare the result with the $14.50 share price. Find (a) the uniform scaling of all stage probabilities, (b) the Phase III success probability for LUM-101 alone, and (c) the discount rate, each of which would reproduce the market price.
5. **Discount rate versus probability.** Compute the unrisked NPV of each program at 11 percent. Then find the single discount rate which, applied to the unrisked flows, reproduces each rNPV. Compute each program's rNPV at 15 percent and 20 percent. Explain what goes wrong if a 20 percent rate is combined with explicit probabilities, and argue whether any argument for a rate above 11 percent is defensible.
6. **Sensitivity and outcomes.** For LUM-101, compute the change in value per share from (a) Phase III success probability of 40 and 70 percent, (b) peak sales of $1,000 million and $1,800 million, (c) discount rates of 13 and 9 percent, (d) a one-year launch delay, (e) the pricing-policy haircut of Exhibit 4, and (f) using the benchmark panel's Phase III and filing probabilities. Find the break-even Phase III probability at which LUM-101's rNPV is zero. Then compute LUM-101's value and Lumen's equity value per share in each of the three outcomes: Phase III fails, Phase III succeeds but the filing is rejected, and approval.
7. **The value of staging.** Compute each program's rNPV if the company committed to all scheduled development spending regardless of earlier results. The difference is the value of the option to stop at each gate. Explain why it is large for LUM-205 and LUM-310 and small for LUM-101, and what that implies about the order in which to cut programs if cash runs short.
8. **Valuing Deal A.** Value Lumen's US-only LUM-101 business, then value Corvane's term sheet (upfront, cost share, milestones and royalties, net of the University's share) on a risk-adjusted after-tax basis. Compare the total with retaining global rights. Compute the upfront payment at which Deal A would be value-neutral against retention. Quantify how the second and third readings of the University's clause in Exhibit 1 change the answer.
9. **Funding choices.** Compare three ways of funding 2027 and 2028: (I) the $250 million equity raise of Exhibit 8, (II) Deal A, and (III) Deal B. For each, produce the closing cash balances for 2027 and 2028, test the covenant, and compute equity value per share on the sum-of-the-parts basis. Compare the cash each deal pays in the state where Phase III fails with the cash it pays in the state where LUM-101 is approved. State which option you would choose and why.
10. **Board recommendation.** Prepare a one-page recommendation to the board for 17 November. It should state which funding path you recommend, what you would try to renegotiate (at least three specific terms, with values), what you would tell the market about valuation, which program you would cut first if cash became tight, and which three facts, if they changed, would reverse your recommendation.

Teaching Note and Full Solutions

Learning Objectives

After this case, students should be able to:

1. Convert stage success probabilities into cumulative probabilities and probability-weighted development spending, and build an rNPV and sum-of-the-parts equity value.

2. Explain why technical risk belongs in cash flows and why a high venture rate on top of explicit probabilities double counts, and reverse-engineer what a market price implies.
3. Value a licensing deal (upfront, cost share, milestones, royalties, upstream share) against retention, and recognize when financing risk drives the decision.
4. Use state-contingent analysis to communicate an expected value that no one will ever observe.

Suggested Class Plan (120 minutes)

Minutes	Segment	Activity
0-10	Opening	Cold-call the CFO's question and show the covenant timeline (Table 3.9).
10-25	Probabilities	Questions 1 and 5 in pairs.
25-55	rNPV build	Question 2 in three groups, one per program; reconcile to $698.0M, $139.8M, $52.6M.
55-80	Parts and sensitivity	Questions 3, 4 and 6; poll: is $14.50 cheap? Ask Ms. Okafor-Lang's question.
80-105	Staging and the deals	Questions 7 to 9, split by role (CEO, CFO, BD, chair).
105-120	Decision	Question 10: vote, then compare with the memo below.

Solutions

All figures were recomputed in Python from Tables 3.4 to 3.10 and are rounded to one decimal place (probabilities to four places) for display. Small differences in the last digit can arise if you round intermediate values.

Question 1: Probabilities and expected spending

Cumulative probability is the product of stage probabilities. The probability of entering a stage is the product of the stage probabilities before it.

- LUM-101: approval = 0.55 x 0.90 = 0.4950.
- LUM-205: probability of entering Phase III = 0.34; of entering filing = 0.34 x 0.58 = 0.1972; approval = 0.1972 x 0.90 = 0.1775 (0.17748).
- LUM-310: probability of entering Phase II = 0.66; Phase III = 0.66 x 0.42 = 0.2772; filing = 0.2772 x 0.65 = 0.1802; approval = 0.1802 x 0.92 = 0.1658 (0.16577).

Table 3.11: Expected development spending (US$ millions)

Program	Stage	Scheduled spend	Probability of entering	Expected spend
LUM-101	Phase III	165.0	1.0000	165.0
LUM-101	Filing and pre-launch	50.0	0.5500	27.5
	LUM-101 total	**215.0**		**192.5**
LUM-205	Phase IIb	55.0	1.0000	55.0
LUM-205	Phase III	240.0	0.3400	81.6
LUM-205	Filing and pre-launch	55.0	0.1972	10.8
	LUM-205 total	**350.0**		**147.4**
LUM-310	Phase I	32.0	1.0000	32.0
LUM-310	Phase II	62.0	0.6600	40.9
LUM-310	Phase III	120.0	0.2772	33.3
LUM-310	Filing and pre-launch	25.0	0.1802	4.5
	LUM-310 total	**239.0**		**110.7**
	All programs	**804.0**		**450.6**

Scheduled spending of $804.0 million overstates the expected outlay by 78 percent, because most of it is committed only in the states where earlier gates are cleared. The expected figure is correct for rNPV because cash flows are weighted by the probability that they occur. The other way of putting it: the sponsor’s decision to stop at a failed gate is built into the model, so spending after a failure is not a cost.

Question 2: rNPV of LUM-101

Building operating profit. Sales in launch year n equal the ramp percentage times peak sales of $1,400.0 million. Operating profit equals sales x (1 - 0.14 - 0.04) less sales and marketing, which is the greater of the percentage of sales and the $70.0 million floor.

- 2030 (launch year 1): sales 5% x 1,400 = 70.0. COGS and royalty 14% = 9.8. Lifecycle R&D 4% = 2.8. Sales and marketing = max(45% x 70.0 = 31.5; 70.0) = 70.0. Operating profit = 70.0 - 9.8 - 2.8 - 70.0 = (12.6).
- 2031: sales 15% x 1,400 = 210.0. COGS 29.4, R&D 8.4. Sales and marketing = max(94.5; 70.0) = 94.5. Operating profit = 210.0 - 29.4 - 8.4 - 94.5 = 77.7.
- 2032: sales 32% x 1,400 = 448.0. COGS 62.7, R&D 17.9. Sales and marketing 45% = 201.6. Operating profit = 448.0 - 62.7 - 17.9 - 201.6 = 165.8.
- 2033 (launch year 4): sales 54% x 1,400 = 756.0. COGS 105.8, R&D 30.2. Sales and marketing 32% = 241.9. Operating profit = 756.0 - 105.8 - 30.2 - 241.9 = 378.0.
- From 2039 (launch year 10) onward, sales and marketing is 24 percent, so at the 2040 plateau operating profit is 1,400 x (1 - 0.14 - 0.04 - 0.24) = 812.0. After generic entry in 2041, sales are 45 percent of peak (630.0), then 22, 12 and 7 percent (308.0, 168.0, 98.0), so operating profit is 365.4, 178.6, 97.4 and 56.8.

Risk adjustment and discounting. Development costs in 2027 and 2028 are certain to be incurred (the program is already in Phase III), so they carry probability 1. The 2029 filing and pre-launch cost of $50.0 million is incurred only if Phase III succeeds, so its expected value is 0.55 x 50.0 = 27.5. Operating profits are multiplied by the cumulative approval probability, 0.495. Everything is then taxed at 21 percent (multiplied by 0.79) and discounted at 1.11 to the power of the year.

Worked examples. Year 1: -90.0 x 0.79 = -71.1; x 0.9009 = -64.1. Year 5: 77.7 x 0.495 = 38.5; x 0.79 = 30.4; x 0.5935 = 18.0.

Table 3.12: LUM-101 rNPV (US$ millions)

Year	Cal.	Unrisked op. profit	x 0.495	Risked dev. cost	Risked pre-tax	After-tax (x 0.79)	DF 11%	PV
1	2027	-	-	(90.0)	(90.0)	(71.1)	0.9009	(64.1)
2	2028	-	-	(75.0)	(75.0)	(59.2)	0.8116	(48.1)
3	2029	-	-	(27.5)	(27.5)	(21.7)	0.7312	(15.9)
4	2030	(12.6)	(6.2)	-	(6.2)	(4.9)	0.6587	(3.2)
5	2031	77.7	38.5	-	38.5	30.4	0.5935	18.0
6	2032	165.8	82.1	-	82.1	64.8	0.5346	34.7
7	2033	378.0	187.1	-	187.1	147.8	0.4817	71.2
8	2034	518.0	256.4	-	256.4	202.6	0.4339	87.9
9	2035	714.6	353.7	-	353.7	279.4	0.3909	109.2
10	2036	787.6	389.9	-	389.9	308.0	0.3522	108.5
11	2037	812.0	401.9	-	401.9	317.5	0.3173	100.7
12	2038	812.0	401.9	-	401.9	317.5	0.2858	90.8
13	2039	812.0	401.9	-	401.9	317.5	0.2575	81.8
14	2040	812.0	401.9	-	401.9	317.5	0.2320	73.7
15	2041	365.4	180.9	-	180.9	142.9	0.2090	29.9
16	2042	178.6	88.4	-	88.4	69.9	0.1883	13.2
17	2043	97.4	48.2	-	48.2	38.1	0.1696	6.5
18	2044	56.8	28.1	-	28.1	22.2	0.1528	3.4
Total								**698.0**

The present value of risk-adjusted after-tax development cost is -128.0 (64.1 + 48.1 + 15.9). The present value of risk-adjusted commercial cash flow is 826.1. The **rNPV of LUM-101 is $698.0 million.** As a cross-check, the unrisked NPV (success case with all development costs certain) is $1,527.8 million, which is the approval-state value used in Question 6.

Question 3: rNPV of LUM-205 and LUM-310

Operating profits are taken from Table 3.7. For LUM-205 they are multiplied by 0.17748; for LUM-310 by 0.16577. Development costs are probability-weighted as in Table 3.11 and fall in the years shown in Table 3.4. The tables below risk-adjust development cost year by year and discount the commercial flows of Table 3.7 as a block (each year’s operating profit x 0.79 x discount factor), then weight the block by the cumulative approval probability. This is algebraically identical to the year-by-year layout used for LUM-101.

Table 3.13: LUM-205 rNPV (US$ millions)

Stage	Year	Cal.	Scheduled cost	Prob. of entering	Risked cost	After-tax	DF 11%	PV
Phase II	1	2027	55.0	1.0000	(55.0)	(43.5)	0.9009	(39.1)
Phase III	2	2028	80.0	0.3400	(27.2)	(21.5)	0.8116	(17.4)
Phase III	3	2029	100.0	0.3400	(34.0)	(26.9)	0.7312	(19.6)
Phase III	4	2030	60.0	0.3400	(20.4)	(16.1)	0.6587	(10.6)
Reg+prelaunch	5	2031	55.0	0.1972	(10.8)	(8.6)	0.5935	(5.1)
PV of risked development cost								**-91.9**
PV of unrisked commercial cash flow (Table 3.7 x 0.79 x DF)								1,305.4
Commercial PV x cumulative PoS 0.17748								231.7
rNPV								**139.8**

rNPV of LUM-205 = -91.9 + 231.7 = $139.8 million.

Table 3.14: LUM-310 rNPV (US$ millions)

Stage	Year	Cal.	Scheduled cost	Prob. of entering	Risked cost	After-tax	DF 11%	PV
Phase I	1	2027	22.0	1.0000	(22.0)	(17.4)	0.9009	(15.7)
Phase I	2	2028	10.0	1.0000	(10.0)	(7.9)	0.8116	(6.4)
Phase II	3	2029	30.0	0.6600	(19.8)	(15.6)	0.7312	(11.4)
Phase II	4	2030	32.0	0.6600	(21.1)	(16.7)	0.6587	(11.0)
Phase III	5	2031	45.0	0.2772	(12.5)	(9.9)	0.5935	(5.8)
Phase III	6	2032	40.0	0.2772	(11.1)	(8.8)	0.5346	(4.7)
Phase III	7	2033	35.0	0.2772	(9.7)	(7.7)	0.4817	(3.7)
Reg+prelaunch	8	2034	25.0	0.1802	(4.5)	(3.6)	0.4339	(1.5)
PV of risked development cost								**-60.3**
PV of unrisked commercial cash flow (Table 3.7 x 0.79 x DF)								680.8
Commercial PV x cumulative PoS 0.16577								112.8
rNPV								**52.6**

rNPV of LUM-310 = -60.3 + 112.8 = $52.6 million.

Interpretation. LUM-205 is worth $139.8 million despite a 17.7 percent approval probability because its success case is large (commercial PV $1,305.4 million) and risk-adjusted development cost is only $91.9 million rather than $350.0 million. LUM-310's success case is half the size, starts nine years out (discount factors of 0.39 or less) and must clear four gates.

Question 4: Sum of the parts and the market

Corporate G&A: $28.0 million x 0.79 = $22.12 million a year; the ten-year annuity factor at 11 percent is (1 - 1.11^-10) / 0.11 = 5.8892, so the present value is 22.12 x 5.8892 = $130.3 million.

Table 3.15: Sum-of-the-parts equity value

Component	US$ millions	Per diluted share (US$)
LUM-101 rNPV	698.0	11.26
LUM-205 rNPV	139.8	2.25
LUM-310 rNPV	52.6	0.85
Discovery platform	45.0	0.73
Corporate G&A (after tax, present value)	(130.3)	(2.10)
Net operating value	**805.1**	**12.99**
Cash and marketable securities	412.0	6.65
Term loan	(75.0)	(1.21)
Equity value	**1,142.1**	**18.42**
Market capitalization at $14.50	899.0	14.50
Premium of intrinsic value over price	243.1	27.0%

Market enterprise value of $562.0 million is 69.8 percent of the model's net operating value of $805.1 million. Three ways to read the market's implied view:

- **(a) Uniform scaling.** Multiplying every stage probability by 0.901 (so LUM-101's Phase III becomes 49.6 percent and its filing 81.1 percent) lowers equity value to $899.0 million.
- **(b) LUM-101 Phase III alone.** Holding all else fixed, a Phase III probability of 38.5 percent for LUM-101 reproduces the price.
- **(c) Discount rate.** A discount rate of 13.6 percent applied to all program cash flows and to the G&A deduction reproduces the price. Equity values per share at other rates are $22.48 at 9 percent, $20.32 at 10, $18.42 at 11, $16.75 at 12 and $15.28 at 13.

None of the three is obviously unreasonable, which is the point. The market is pricing a mix of lower probabilities, a higher rate and financing dilution risk that the model has not yet charged for. The financing charge is quantified in Question 9.

Question 5: Discount rate versus probability

Table 3.16: Discount rate and probability (US$ millions)

Measure	LUM-101	LUM-205	LUM-310
Unrisked NPV at 11% (success case, all costs certain)	1,527.8	1,100.2	564.8
rNPV at 11%	698.0	139.8	52.6
Single discount rate on unrisked flows that equals the rNPV	18.9%	24.9%	25.0%
rNPV at 15%	451.2	60.4	13.1
rNPV at 20%	259.2	6.6	(10.9)

The venture-style rate that replicates the explicit-probability answer is not one number: it is about 19 percent for a Phase III asset and 25 percent for earlier-stage assets, because the risk that a rate must mimic differs by stage and the rate does not capture the fact that development costs are avoided on failure. Applying a flat 20 percent to unrisked flows would misvalue all three programs, and applying a 20 percent rate together with explicit probabilities, as in the last row, charges for technical failure twice and values LUM-310 below zero.

Is any argument for a rate above 11 percent defensible? Yes, but about systematic risk, not technical risk: financing-risk, estimation-error or payer-correlation premia support 12 to 13 percent, and Question 4(c) shows the market implies 13.6 percent. The rate should not price whether the trial works.

Question 6: Sensitivity and outcomes

Table 3.17: LUM-101 sensitivity (US$ millions, per diluted share)

Case	LUM-101 rNPV	Change in rNPV	Change per share (US$)
Base	698.0		
Phase III probability 40%	477.1	(220.9)	(3.56)
Phase III probability 70%	919.0	221.0	3.56
Peak sales $1,000M	456.3	(241.7)	(3.90)
Peak sales $1,800M	939.2	241.2	3.89
Discount rate 13%	561.2	(136.8)	(2.21)
Discount rate 9%	869.2	171.2	2.76
Launch delayed one year (patent expiry unchanged)	555.1	(142.9)	(2.31)
Pricing-policy haircut (Exhibit 4: 15% price cut on 2039 and 2040 sales)	674.7	(23.3)	(0.38)
Benchmark probabilities (50% and 88%)	607.7	(90.3)	(1.46)

Each ten points of Phase III probability moves LUM-101 by $147.4 million, or $2.38 per share, because the change equals 0.10 x (0.90 x 1,668.8 - 28.9), where 1,668.8 is the unrisked present value of LUM-101 commercial cash flow and 28.9 is the present value of the after-tax filing cost. The break-even Phase III probability, at which LUM-101's rNPV is zero, is 7.6 percent. Running the pivotal trial is therefore justified at any plausible probability; what is uncertain is the value of the franchise, not whether to continue.

Three outcomes. Equity value outside LUM-101 is LUM-205 139.8 + LUM-310 52.6 + platform 45.0 - G&A 130.3 + cash 412.0 - loan 75.0 = 444.1.

Table 3.18: Outcomes for LUM-101 (US$ millions, per share)

Outcome	Probability	LUM-101 value	Equity value	Per diluted share (US$)
Phase III fails (development costs of $90M and $75M only)	45.0%	(112.1)	332.0	5.35
Phase III succeeds, filing rejected (adds $50M filing cost)	5.5%	(141.0)	303.1	4.89
Approved (unrisked NPV)	49.5%	1,527.8	1,971.9	31.80
Probability-weighted	**100%**	**698.0**	**1,142.1**	**18.42**

The $18.42 expected value is not a price the stock will ever trade at on the day after the data: it is $5.35 with 45 percent probability and $31.80 with 49.5 percent. That is the answer to Ms. Okafor-Lang's question, and the reason a financing plan must survive the failure state.

Question 7: The value of staging

Table 3.19: Value of the option to stop (US$ millions)

Program	rNPV with gating	rNPV if all scheduled cost were spent regardless	Option value
LUM-101	698.0	685.0	13.0
LUM-205	139.8	26.5	113.3
LUM-310	52.6	(3.1)	55.7

Without the right to stop, LUM-310 would be worth less than zero and LUM-205 only a fifth of its value, because LUM-205 has $240.0 million of Phase III spending behind a 34 percent gate. For LUM-101 the only gated spending is the $50.0 million filing stage. If cash runs short, compare rNPV with spending through the next gate: LUM-101 698.0 / 165.0 = 4.2; LUM-205 139.8 / 55.0 = 2.5; LUM-310 52.6 / 32.0 = 1.6. Slow LUM-310 and the platform first; slowing LUM-205 forfeits the valuable 2027 readout.

Question 8: Valuing Deal A

US-only business. Sales are 60 percent of the base case (peak $840.0 million, with every ramp percentage applied to that figure). COGS and royalty, lifecycle R&D and sales and marketing use the same percentages as before; the floor is $55.0 million. Development cost is unchanged. The rNPV of the US-only business is **$364.3 million**, so retained ex-US rights are worth 698.0 - 364.3 = $333.8 million.

The deal. The University takes 18 percent of the upfront payment and milestones, so Lumen keeps 82 percent. Cost reimbursements are not shared. Royalties are net of the University's 3 percent of ex-US net sales. Royalty step-downs halve the rates from 2041.

- Upfront: 150.0 x 0.82 = 123.0 at Year 0; after tax 97.2.
- Cost share: 35% x 90.0 = 31.5 (2027) and 35% x 75.0 = 26.25 (2028), both certain. After tax and discounting: 31.5 x 0.79 x 0.9009 = 22.4; 26.25 x 0.79 x 0.8116 = 16.8; total 39.2.
- Phase III milestone: 60.0 x 0.82 x 0.55 = 27.06 at 2028; x 0.79 x 0.8116 = 17.4.
- Authorization milestone: 50.0 x 0.82 x 0.495 = 20.30 at 2030; x 0.79 x 0.6587 = 10.6.
- Sales milestones: $100M is earned in 2033 (ex-US sales of $302.4M exceed $300M): 100 x 0.82 x 0.495 = 40.59; x 0.79 x 0.4817 = 15.4. $150M is earned in 2036 (sales of $543.2M exceed $500M): 150 x 0.82 x 0.495 = 60.89; x 0.79 x 0.3522 = 16.9. Total 32.4.
- Royalties (Table 3.20): 89.9 present value.

Table 3.20: Deal A royalty stream (US$ millions)

Year	Ex-US sales	Gross royalty	After step-down	University 3%	Net to Lumen	x 0.495	PV after tax
2030	28.0	4.5	4.5	0.8	3.6	1.8	0.9
2031-2032	263.2	42.1	42.1	7.9	34.2	16.9	7.4
2033	302.4	50.5	50.5	9.1	41.4	20.5	7.8
2034	414.4	72.9	72.9	12.4	60.4	29.9	10.3
2035	492.8	90.3	90.3	14.8	75.5	37.4	11.5
2036	543.2	102.4	102.4	16.3	86.1	42.6	11.9
2037	560.0	106.4	106.4	16.8	89.6	44.4	11.1
2038-2040 (each year)	560.0	106.4	106.4	16.8	89.6	44.4	10.0, 9.0, 8.1
2041	252.0	40.4	20.2	7.6	12.6	6.3	1.0
2042-2044	229.6	36.8	18.4	6.9	11.5	5.7	0.8

At peak, gross royalty is 16% x 250 + 20% x 200 + 24% x 110 = 40.0 + 40.0 + 26.4 = 106.4, an effective 19.0 percent of $560.0 million.

Total. Deal items are worth 97.2 + 39.2 + 17.4 + 10.6 + 32.4 + 89.9 = **$286.6 million**. Lumen's value under Deal A is 364.3 + 286.6 = $650.9 million, against $698.0 million from retention: **Deal A costs $47.2 million of rNPV ($0.76 per share).** The upfront payment at which Deal A matches retention is $222.8 million, because each extra $10 million of upfront adds $6.5 million (10 x 0.82 x 0.79).

The University's clause. Under the second reading (18 percent of royalties received, instead of 3 percent of ex-US sales), net royalties are 82 percent of gross; the deal total is $285.6 million, a reduction of $1.0 million, because 18 percent of the effective 19 percent royalty is close to the 3 percent sales royalty it replaces at this royalty level. Under the third reading, which also takes 18 percent of the cost reimbursement, the cost share loses 0.18 x 39.2 = $7.1 million. The clause matters mostly through the upfront ($27.0 million pre-tax to the University) and is not a reason to reject Deal A.

Question 9: Funding choices

Table 3.21: Funding options

Option	Closing cash 2027	Closing cash 2028	Covenant ($100M)	Equity value per share (US$)
No action	226.4	35.3	Breach in Q3 2028	18.42 on paper; not achievable
I: $250M raise at $12.00, 5.5% costs	472.1	290.8	Met	16.64
II: Deal A	385.8	227.3 (276.5 if Phase III milestone is paid)	Met	17.66
III: Deal B	448.1	265.9	Met	17.18

Cash paths use the Table 3.9 assumptions. Option I: net proceeds 250 x 0.945 = 236.25, opening cash 648.25, 2027 closing = 648.25 - 167.0 - 28.0 - 7.1 + 0.04 x 648.25 = 472.1. Option II: Lumen receives 123.0 upfront and reimbursements of 31.5 and 26.25. Option III: 260 x 0.82 = 213.2 upfront.

Per-share values. Option I issues 250 / 12.00 = 20.83 million shares. Post-money equity value is 1,142.1 + 236.25 = 1,378.4 million over 82.83 million shares = $16.64, a dilution of $1.78 per original share, or $110.4 million of value transferred to new investors. Option II: (1,142.1 - 47.2) / 62.0 = $17.66. Option III: Deal B total is 364.3 + 256.9 = 621.2, a loss of $76.9 million against retention: (1,142.1 - 76.9) / 62.0 = $17.18. The right comparison is therefore not $18.42, which the covenant makes infeasible, but $16.64: the deal is worth $1.02 per share more than the raise.

State-contingent payoffs. After-tax present values of deal payments by outcome:

Outcome	Probability	Deal A	Deal B	Value of retained ex-US rights
Phase III fails	45.0%	136.4	168.4	0
Succeeds, filing rejected	5.5%	168.0	189.5	0
Approved	49.5%	436.3	344.8	674.3

Deal B pays $32.0 million more than Deal A if the trial fails; Deal A pays $91.5 million more if the drug is approved. At a 50-50 chance, A is better; B is better only if the board believes the Phase III probability is below about 29 percent (solving 0.285 as the Phase III probability x at which -32.0(1 - x) - 21.5(0.1x) + 91.5(0.9x) = 0). Deal A gives up $238.0 million of retained value in the approval state in return for $136.4 million of certain value in the failure state.

Choice. Deal A, after negotiation, in preference to the raise or Deal B. It leaves covenant headroom, costs $0.76 per share in rNPV against a $1.78 dilution cost of the raise, and keeps the US franchise.

Question 10: Board recommendation

A strong answer recommends Deal A after renegotiation (see the memo below), quantifies each ask, states the valuation range, names the program to slow, and lists reversal triggers. Values per ask: each $10 million of upfront is worth $6.5 million; one point on every royalty tier $6.0 million; raising cost share from 35 to 50 percent $16.8 million; a Corvane non-compete or reversion right is unpriced and should be treated as insurance against the competing asset in Exhibit 6.

Sensitivity Discussion

Value per share is most exposed to peak sales and Phase III probability (about $3.6 to $3.9 per share each for the ranges tested), then to the discount rate and launch timing ($2.2 to $2.8). The pricing-policy haircut costs only $0.38 per share because it applies to two plateau years. The benchmark probabilities cost $1.46. The market price of $14.50 sits inside the range created

by simply moving two inputs to reasonable alternatives, so the stock is not obviously mispriced. The deal decision, by contrast, is insensitive to the University's clause and sensitive to the Phase III probability only below about 29 percent.

Common Student Errors

- Discounting success-case flows at a venture rate and then also multiplying by probabilities.
- Charging scheduled rather than probability-weighted development spending, which understates rNPV.
- Applying the cumulative approval probability to development costs that occur before the gate.
- Presenting the $18.42 expected value as a forecast of the share price, rather than as a weighted average of $5.35 and $31.80.
- Comparing the deal with $18.42 rather than with the feasible alternative, the dilutive raise.
- Treating the upfront as fully retained, forgetting that 18 percent goes to the University.
- Omitting the licensor's 3 percent on ex-US sales after the rights are licensed, or applying the royalty step-down to milestones.
- Forgetting to deduct corporate G&A and the loan in the sum of the parts.

Alternative Defensible Answers

A class may reasonably use the benchmark probabilities (LUM-101 falls to $607.7 million and equity to $16.96 per share), a rate of 10 or 12 percent, mid-year discounting, or explicit use of the tax-loss carry-forward, which raises every value. A risk-averse board may prefer Deal B for its larger certain cash ($213.2 million against $180.8 million for Deal A before tax) if it believes the Phase III probability is below about 29 percent. A board that believes the stock is cheap might raise a smaller amount and keep global rights; even at a no-discount issue price of $14.50 the raise yields $17.39 per share, below Deal A's $17.66, so the case for retention rests on a belief that rNPV is materially understated. A CEO may reasonably choose retention with a staged raise after the 2027 LUM-205 readout, but must accept a covenant waiver conversation.

Board Recommendation Memo

To: Board of Directors, Lumen Therapeutics, Inc. **From:** Finance. **Date:** 5 November 2026. **Re:** LUM-101 ex-US licensing and 2028 funding (illustrative).

Recommendation. Accept the Corvane structure (Deal A), conditional on the three improvements below, and do not raise equity now. Continue all three programs through their next gates.

Why. On a risk-adjusted basis Lumen is worth $1,142.1 million, or $18.42 per share (range $15.28 to $20.32 at discount rates of 13 to 10 percent), but that value is bimodal: $31.80 if LUM-101 is approved, about $5 if Phase III fails (probability 50.5 percent including filing rejection). With no action Lumen breaches its $100 million covenant in Q3 2028, before the Phase III result. An equity raise at $12.00 transfers about $110 million of value to new investors. Deal A costs $47.2 million of rNPV against retention, about $0.76 per share, but delivers certain after-tax value of $136.4 million in the failure state, keeps cash at $227.3 million at the end of 2028, and leaves the United States, 60 percent of sales, with Lumen. It is worth $1.02 per share more than the raise and $0.48 more than Deal B.

Negotiate. (1) Upfront to $190 million or more: worth about $25.9 million. (2) Cost share to 50 percent: worth $16.8 million. (3) A non-compete or reversion right over Corvane's competing Phase II asset. Secure the University's written confirmation of its reading of Sublicense Income before signing.

Message to the market. Publish the rNPV range with the outcome distribution, not a point estimate.

What would change our view. A Phase III probability below 29 percent (favoring Deal B); a Corvane refusal of any reversion or diligence protection; an equity window that prices at or above $14.50 with low costs and a belief that rNPV is understated; or a negative rival readout that lifts peak sales toward $1,800 million, which raises the cost of giving away ex-US rights.

If cash tightens. Slow LUM-310 first, then the platform; protect the 2027 LUM-205 readout.

Chapter 4: Case: Harbor & Pine, a Retail Turnaround

Type: Case. Prerequisites: Volume 1, Chapters 2 (statements and ratios), 11 (WACC), 15 (capital structure), 21 (enterprise valuation) and 23 (LBO, private equity and restructuring); Volume 2, the chapters on advanced valuation and restructuring. Harbor & Pine Holdings, Granite Ridge Partners, the peer companies, their people and every figure in this chapter are fictional and illustrative. No real company, transaction or statistic is intended or reproduced.

The Situation

On Monday 2 November 2026, at 7:40 in the morning, the board secretary of Harbor & Pine Holdings, Inc. received a letter by courier and by email. It came from Granite Ridge Partners, a private equity sponsor, and it proposed to acquire every share of Harbor & Pine for **$9.40 per share in cash**, a premium of 38.2 percent to the unaffected closing price of $6.80 on 30 October. The letter expires on Friday 20 November 2026 at 17:00. It is conditional on due diligence and on committed financing of $400 million of new term debt and a $60 million asset-based revolver, which the sponsor says it holds in writing. It does not mention what the sponsor intends to do with the company's real estate.

Harbor & Pine is a specialty retailer of home furnishings, garden and outdoor-living products. It operates 214 stores in 31 states, 192 of them leased and 22 owned, plus a leased headquarters, a company-owned distribution center and an e-commerce site that now accounts for 18 percent of sales. Founded 61 years ago by a family nursery business, it listed 19 years ago and has been run since by a succession of merchants. Net sales peaked at $1.81 billion in fiscal 2024. Since then comparable-store sales have fallen three years running, closures have outpaced openings, the stock has fallen by roughly 60 percent from its high, and the company has funded a share repurchase and a dividend with borrowings.

The people involved:

Dana Whitcombe (chief executive officer) joined seven months ago from a national home-goods chain. She has a turnaround plan, "Anchor 2030", which closes 28 weak stores over two years, remodels 60 more, rebuilds the e-commerce platform and takes $55 million of cost out of the business. She believes the stock is worth well above $9.40 if she is given three years.

Raj Venkataraman (chief financial officer) has been with the company for nine years. He prepared the normalized earnings schedule that the committee is using and is uneasy about three of its adjustments.

Margaret Oyelaran (lead independent director) chairs the special committee formed on 2 November to evaluate the proposal. A former retail chief financial officer, she asks one question of every number: "Is that before rent or after rent?"

Tom Ferreira (head of real estate) knows that the owned stores and the distribution center were appraised in the spring at $354.0 million against a book value of $238.0 million, and that the headquarters lease and 41 other leases expire within three years.

Calloway Value Fund, which holds 9.6 percent of the shares, has written to the board that it will not support a sale below $11.00.

The decision. The special committee must decide by 20 November whether to (a) accept the proposal, (b) reject it and continue with Anchor 2030, or (c) engage with Granite Ridge and counter at a higher price. It needs a defensible view of what Harbor & Pine is worth. That means normalizing a messy earnings year, putting leased and owned properties on a comparable footing

with peers that report under a different lease standard, valuing four turnaround outcomes rather than one, and understanding what a sponsor can afford to pay. You are an analyst at the committee's financial adviser.

Industry Context

Specialty home and outdoor retail. Demand for furniture, décor and garden products follows housing turnover, household formation and consumer confidence, and is discretionary and seasonal. Merchants compete on assortment and price against large-format and online rivals, and stores serve as showrooms and fulfillment points for the online channel. Store productivity (sales per square foot), markdown discipline and inventory turns drive the gross margin; occupancy, labor and marketing absorb most of it. A fleet of aging stores in weaker locations produces a long tail of stores that earn less than their rent and a recurring program of closures.

Leases are quasi-debt. A retailer with 200 stores signs 200 long-term obligations. Rent is a contractual fixed charge that does not flex when sales fall. Closing a store does not end the obligation: the landlord must be bought out, or the lease subleased, often at a discount to remaining rent. Credit analysts therefore treat lease obligations as debt. Under US GAAP (ASC 842) an operating lease creates a right-of-use asset and a lease liability on the balance sheet, but the income statement still shows a single straight-line lease cost within operating expenses, so EBITDA is stated after rent. Under IFRS 16 nearly all leases are treated like financed purchases: the income statement shows depreciation of the right-of-use asset and interest on the liability, so EBITDA is stated before most rent and EBIT is higher. Two otherwise identical retailers report very different EBITDA and EV/EBITDA depending on the standard. Valuation work must put them on the same footing, either by adding rent back and lease liabilities to enterprise value (EBITDAR and lease-adjusted EV), or by treating rent as an operating cost for everyone.

Rules of thumb. Practitioners sometimes capitalize annual rent at a multiple (a figure of eight times is common) to estimate lease debt. The balance-sheet liability, which discounts the committed payments at the lessee's borrowing rate, is more precise for the committed term but ignores renewals and variable rent. The two measures can differ by tens of percent.

Turnarounds are distributions, not points. A retailer in decline can recover, stall or fail. Value depends on how many weak stores close and at what cost, whether remodels lift sales, whether the margin rebuilds, and whether lenders and landlords remain patient. Practitioners model discrete scenarios, attach probabilities and test how sensitive the weighted value is to those probabilities, because the single "management case" is rarely the expected case.

Real estate and the financing toolkit. Retailers that own property can sell it to an investor and lease it back, raising cash at a capitalization rate that often implies a far higher multiple than the stock market pays for the retailer's earnings. A sale-leaseback converts owned property into a lease obligation, raises rent and lease-adjusted debt, and triggers tax on any gain. Sponsors acquiring retailers often underwrite a sale-leaseback as part of the financing.

Take-privates. A sponsor values a target by what return it earns on its equity, given the debt it can raise, the cash flows it can extract and the price at which it can exit. A bid is therefore bounded by the sponsor's required return, and the committee can estimate that bound.

Financial Data

All amounts are US$ millions unless stated. Harbor & Pine reports under US GAAP. Its fiscal year ends on the Saturday nearest 31 January and is named for the calendar year in which it starts: fiscal 2026 ("FY2026") ends 30 January 2027 and has 53 weeks. FY2026E combines three quarters of actual results with an estimate for the fourth. The valuation date is the end of FY2026, and cash flows are discounted from the end of each fiscal year.

Table 4.1: Harbor & Pine, income statement extract

US$ millions	FY2024A	FY2025A	FY2026E (53 weeks)
Net sales	1,812.4	1,724.1	1,653.0
Cost of sales, including buying and occupancy (excluding D&A)	(1,199.7)	(1,155.2)	(1,125.0)
Gross profit	612.7	568.9	528.0
SG&A (excluding D&A)	(395.4)	(396.5)	(392.0)
Store closure and impairment charges	(5.1)	(7.9)	(14.2)
Other income (FY2026E: gain on sale-leaseback of one store)	0.5	0.8	9.0
EBITDA (reported)	**212.7**	**165.3**	**130.8**
Depreciation and amortization	(48.6)	(50.2)	(51.5)
EBIT	164.1	115.1	79.3
Net interest expense	(27.2)	(29.4)	(31.0)
Pre-tax income	136.9	85.7	48.3
Income tax at 25%	(34.2)	(21.4)	(12.1)
Net income	102.7	64.3	36.2
Memo: stock-based compensation inside SG&A	9.4	10.2	11.0

Table 4.2: Harbor & Pine, balance sheet extract

US$ millions	31 Jan 2026 (FY2025A)	30 Jan 2027 (FY2026E)
Cash	48.0	62.5
Inventories	372.0	402.0
Other current assets	36.5	38.0
Property and equipment, net (includes owned real estate with book value of 238.0)	582.8	548.0
Operating lease right-of-use assets	1,003.0	962.0
Goodwill and intangibles	112.0	112.0
Other assets	22.0	22.5
Total assets	**2,176.3**	**2,147.0**
Accounts payable and accrued expenses	252.0	265.0
Current operating lease liabilities	124.5	122.8
Current portion of term loan	8.0	8.0
Other current liabilities	41.0	40.0
Term loan, non-current	265.0	257.0
ABL revolver	30.0	55.0
Non-current operating lease liabilities	993.5	960.3
Finance lease liabilities	19.5	18.0
Other long-term liabilities	40.8	38.0
Total liabilities	**1,774.3**	**1,764.1**
Paid-in capital	509.0	520.0
Retained earnings (deficit)	(107.0)	(137.1)
Total equity	**402.0**	**382.9**
Total liabilities and equity	**2,176.3**	**2,147.0**

Equity fell in FY2026 although the company earned $36.2 million, because it paid dividends of $11.6 million and repurchased $54.7 million of stock in the first quarter, before the new chief executive arrived. The repurchase was funded by drawing on the revolver.

Table 4.3: Operating data

Item	FY2024A	FY2025A	FY2026E
Stores at year end	226	220	214
Comparable-store sales	(2.1%)	(5.0%)	(3.8%)
E-commerce share of sales	14%	16%	18%
Inventories at year end	355.0	372.0	402.0
Inventory days (inventories / cost of sales x 365)	108	118	130
Capital expenditure	58.0	44.0	21.5
Operating lease cost, fixed	177.0	181.0	184.0
Operating lease cost, variable (not in lease liability)	12.0	12.0	12.0
Sales in the 53rd week of FY2026			31.2

Table 4.4: Lease disclosures (US GAAP operating leases, 30 January 2027)

Item	Value
Weighted-average discount rate	6.2%
Weighted-average remaining term	6.8 years
Undiscounted payments, FY2027 / FY2028 / FY2029 / FY2030 / FY2031	190.0 / 178.0 / 165.0 / 150.0 / 135.0
Undiscounted payments, FY2032 to FY2036	630.0 (126.0 a year)
Total undiscounted payments	1,448.0
Operating lease liability on balance sheet (current plus non-current)	1,083.1
Average annual rent escalator	2.0%

Table 4.5: Capital structure, market data and the proposal

Item	Value
Term loan B (SOFR plus spread, about 9.0%), matures 15 September 2028	265.0
ABL revolver (about 7.5%), $120.0M commitment, borrowing-base limited, springing fixed-charge coverage covenant of 1.10x	55.0 drawn
Finance leases (about 6.0%)	18.0
Total financial debt (including finance leases)	338.0
Cash	62.5
Net financial debt (excluding operating leases)	275.5
Basic shares (millions) / dilutive securities (millions) / diluted shares (millions)	58.0 / 1.8 / 59.8
Unaffected share price, 30 October 2026 (52-week range $5.90 to $11.40)	$6.80
Market capitalization at $6.80	406.6
Granite Ridge proposal per share	$9.40
Equity value at $9.40 (59.8 million diluted shares)	562.1
Enterprise value at $9.40 (excluding operating leases)	837.6
Termination fee requested by sponsor	3.5% of equity value
Committed new debt	$400.0M term loan at about 10.5%, $60.0M ABL
Risk-free rate / equity risk premium	4.3% / 5.5%
Equity beta (observed against broad market)	1.45
Tax rate	25%

Table 4.6: Comparable specialty-retail companies (fictional, US$ millions)

Company	Standard	Market cap	Net financial debt	Lease liabilities	EBITDA as reported	Fixed operating lease cost in EBITDA
Cedar & Loom Home	US GAAP	1,480	210	1,720	262	302
Tidewater Living	US GAAP	760	395	980	128	171
Willow Row Stores	US GAAP	310	285	840	74	148
Nordhaven Home	IFRS 16	1,150	140	1,260	395	none (lease cost below EBITDA)
Alder Outdoor Group	IFRS 16	640	260	710	235	none

Table 4.7: Management’s EBITDA normalization, as submitted to the committee (FY2026E)

Item	US$ millions
Reported EBITDA	130.8
Add: inventory write-down	18.0
Add: legal settlement	6.5
Add: CEO transition costs	4.1
Add: store closure and impairment charges	14.2
Add: stock-based compensation	11.0
Less: gain on sale-leaseback	(9.0)
Management's adjusted EBITDA	**175.6**

Table 4.8: Turnaround scenarios (five-year assumptions, FY2027 to FY2031)

Assumption	Upside	Plan (Anchor 2030)	Muddle-through	Distress
Probability	20%	40%	25%	15%
Comparable-store sales growth	3.0%, 3.5%, 3.0%, 2.5%, 2.5%	0.5%, 2.0%, 2.5%, 2.0%, 2.0%	(2.0%), (1.0%), 0.0%, 0.5%, 0.5%	(5.0%), (4.0%), (3.0%), (2.0%), (1.0%)
Store closures by year	12, 10, 0, 0, 0	14, 14, 0, 0, 0	10, 8, 0, 0, 0	30, 25, 0, 0, 0
Small-format openings by year	0, 0, 4, 6, 6	0, 0, 0, 4, 4	none	none
EBITDAR margin (percent of sales)	20.5, 21.5, 22.5, 23.0, 23.0	19.8, 20.8, 21.8, 22.2, 22.2	19.0, 19.3, 19.6, 19.6, 19.6	17.5, 17.0, 17.0, 17.5, 17.5
Stores remodeled per year (FY2027 to FY2029), at $1.4M each	24	20	10	0
IT and e-commerce capex ($M), FY2027, FY2028, then each later year	18, 18, 8	18, 18, 8	9, 9, 6	4 a year
Cash cost per closed store (lease buyout and severance), $M	2.3	2.3	2.3	3.0
Exit multiple of FY2031 EBITDA (after rent)	6.25x	5.75x	5.25x	4.25x

Modeling conventions for all scenarios.

- The FY2026 base is the normalized 52-week year: sales of $1,621.8 million (reported $1,653.0 million less the $31.2 million 53rd week), and EBITDAR of $322.9 million, 19.9 percent of sales. Scenario EBITDAR margins apply to each year's sales.
- Sales in year t equal prior-year sales x (1 + comp growth), less closures x $5.0 million x 70 percent (closed stores average $5.0 million of sales and 30 percent transfers to nearby stores and online), plus openings x $3.0 million.
- Fixed rent in FY2026 is $184.0 million: $171.2 million on the 192 leased stores and $12.8 million for headquarters and distribution. Closing a leased store saves $1.00 million of rent (the weakest stores carry high rents), a new store adds $0.60 million, and all rent rises 2.0 percent a year. Closures are assumed to happen at the start of the fiscal year. The 22 owned stores pay no rent and no market rent is charged for them.
- EBITDA equals EBITDAR less fixed rent. Depreciation is 3.1 percent of sales. Closure costs are expensed and tax-deductible. Maintenance capex is 1.8 percent of sales; remodel and IT capex are added. Working capital is 12.0 percent of

sales (so a sales decline releases cash). Tax is 25 percent of positive EBIT.

- Enterprise value is the present value of free cash flow to the firm at the WACC plus a terminal value at the FY2031 exit multiple, discounted five years. Equity value is enterprise value less net financial debt of $275.5 million. Per-share values use 59.8 million shares and are floored at zero because equity is limited liability. Operating leases are not deducted again, because rent is already in EBITDA.

Complications and Exhibits

Exhibit 1: Management's normalization. Mr. Venkataraman is uneasy about three items in Table 4.7. (a) Stock-based compensation is a real cost of retaining staff. (b) Store closures have cost $5.1 million, $7.9 million and $14.2 million in the last three years, so some closure cost recurs; he judges $6.0 million a year to be normal. (c) The inventory write-down of $18.0 million exceeds a normal year, which he puts at $5.0 million. Table 4.7 also omits items that cut the other way: $3.5 million of vendor rebates that were pulled forward from FY2027, the extra 53rd week (which contributed about $2.2 million of EBITDA in a low-volume January week), $4.0 million of annual rent step-ups on 18 stores that renew at market in FY2027, and $5.0 million of marketing spend deferred in the second half that must be restored.

Exhibit 2: Leases and closures. Forty-one leases, including the headquarters, expire within three years. The 28 stores in the Anchor 2030 closure list average $5.0 million of sales and $1.00 million of annual rent, with 4.3 years remaining. A landlord buyout averages $1.9 million a store, and severance and liquidation add $0.4 million, for $2.3 million per closure ($3.0 million in distress, where landlords have less reason to negotiate). Under US GAAP the lease liability for a closed store stays on the balance sheet until the lease is exited.

Exhibit 3: Owned real estate. The 22 owned stores and the distribution center were appraised in spring 2026 at $354.0 million against a book value of $238.0 million. A sale-leaseback of the whole portfolio would be priced at a 7.0 percent capitalization rate, producing initial rent of $24.8 million (rising 2.0 percent a year) on a 20-year lease, transaction costs of $3.0 million and tax at 25 percent on the $116.0 million gain. The management projections charge no rent on these properties. Granite Ridge's letter does not mention a sale-leaseback.

Exhibit 4: Debt terms. The $265.0 million term loan matures on 15 September 2028, within 22 months. A change of control would trigger repayment. The revolver's springing fixed-charge coverage covenant of 1.10x applies when availability falls below 12.5 percent of the borrowing base. Fixed-charge coverage here is defined as EBITDAR divided by fixed rent plus cash interest.

Exhibit 5: Inventory. Inventory days have risen from 108 to 130 while sales fell. Merchandise older than twelve months totals $31.0 million, reserved at 25 percent. The auditors have suggested 40 percent. The additional $4.65 million exposure (31.0 x 15 percent) is not in any normalization above.

Exhibit 6: Peer set. Willow Row Stores is small and highly leveraged, and some analysts would exclude it. Nordhaven Home and Alder Outdoor Group report under IFRS 16, so their EBITDA is before most rent. Raw EV/EBITDA multiples taken straight from a data terminal mix the two standards and are not comparable.

Exhibit 7: Trading and shareholders. Third-quarter comparable-store sales were (3.1 percent), better than the (4.4 percent) of the first half. Ms. Whitcombe attributes this to new assortments; Mr. Venkataraman notes that part reflects customers pulling forward outdoor-furniture purchases ahead of a vendor price increase. Calloway Value Fund has said it will not support a sale below $11.00 and has asked for a market check. The committee's fairness opinion will be based on the work you do now.

Assignment Questions

Use the 10.0 percent WACC, 25 percent tax rate, and the conventions in Table 4.8 unless a question says otherwise.

1. **Normalize earnings.** Build the committee's normalized FY2026 EBITDA from the reported $130.8 million, classifying each adjustment as non-recurring, timing, forward-looking or judgment, and stating whether you add back stock compensation. Compare with management's $175.6 million and state a defensible range. Compute normalized and reported EBITDAR.
2. **Lease-adjusted metrics.** Verify the $1,083.1 million lease liability from Table 4.4. Compare it with capitalizing fixed rent at eight times. Compute lease-adjusted net debt, leverage on EBITDAR, and fixed-charge coverage. Compute EV/EBITDA and lease-adjusted EV/EBITDAR at $6.80 and at $9.40. Restate normalized EBITDA, EBIT and net debt to EBITDA as if Harbor & Pine reported under IFRS 16.
3. **Peer multiples on a common basis.** Compute EV/EBITDAR for each peer using lease-adjusted EV. Show what happens if raw EV/EBITDA is used. Apply the peer range to Harbor & Pine and derive an implied value per share. Explain which multiple you would choose and why.
4. **Cost of capital.** Compute the cost of equity and the WACC on a basis that excludes operating leases from debt and on a basis that includes them. State which cash flows each WACC must discount, and quantify the error from discounting after-rent cash flows at the lease-inclusive WACC.
5. **The Plan scenario.** Build the FY2027 to FY2031 projection for Anchor 2030 and value it. Show every line.
6. **Four scenarios and the bid.** Value the Upside, Muddle-through and Distress scenarios, compute the probability-weighted value per share, and compare it with $6.80 and $9.40. Show the implied perpetual growth rates in the terminal values.
7. **Sensitivity.** Test the weighted value against WACC, exit multiples, EBITDAR margins and comparable-store sales. Find the Plan probability at which the weighted value equals $9.40. Discuss what the sensitivities say about the bid.
8. **Real estate.** Evaluate a sale-leaseback of the owned properties: its effect on value in each scenario, on lease-adjusted leverage and fixed-charge coverage, and the pitfall in the base valuation.
9. **The sponsor's view.** Build Granite Ridge's sources and uses and its returns by scenario at $9.40 with and without a sale-leaseback. Estimate the price at which the sponsor earns a 20 percent return. What does this say about how much room the committee has?
10. **Recommendation.** Prepare a one-page memo to the special committee: accept, reject or counter; the price range you would defend; the three facts you would verify in diligence; and what you would say to Calloway.

Teaching Note and Full Solutions

Learning Objectives

1. Normalize a distorted earnings year by classifying adjustments (non-recurring, timing, forward-looking, judgment) and defend a range rather than a point.
2. Convert operating leases into lease-adjusted debt, EBITDAR and leverage, and restate US GAAP results as if under IFRS 16.
3. Compare peers reporting under different lease standards on a common basis, and keep numerators and denominators consistent (EV with leases over EBITDAR; EV without leases over EBITDA).
4. Value a turnaround with explicit scenarios, probabilities, and an equity floor, and test how much of the value sits in the terminal value.
5. Recognize hidden real-estate value and how a sale-leaseback changes value, leverage and coverage.

6. Estimate a sponsor's ability to pay from its required return, and use it to frame a negotiation.

Suggested Class Plan (120 minutes)

Minutes	Segment	Activity
0-10	Opening	Read the letter aloud; ask each student for a first-impression price. Record the votes.
10-30	Normalization	Question 1 in pairs; compare ranges; debate stock compensation and closure costs.
30-50	Leases	Questions 2 to 4: liability, adjusted leverage, IFRS 16 restatement, peer multiples, WACC consistency.
50-75	Scenarios	Questions 5 and 6: the Plan projection on a projected spreadsheet, then the other three scenarios in groups.
75-90	Sensitivity and real estate	Questions 7 and 8.
90-105	The sponsor	Question 9: role-play Granite Ridge and the committee.
105-120	Decision	Question 10: re-vote and compare with the memo below.

Solutions

All figures were recomputed in Python and are rounded for display; last-digit differences can arise from rounding of intermediate steps.

Question 1: Normalized earnings

Table 4.9: Normalized FY2026 EBITDA (US$ millions)

Item	Amount	Classification	Reasoning
Reported EBITDA	130.8		
Inventory write-down above normal (18.0 - 5.0)	13.0	Non-recurring	A normal $5.0M markdown level stays in the run-rate.
Legal settlement	6.5	Non-recurring	One-time.
CEO transition costs	4.1	Non-recurring	One-time.
Store closure and impairment above normal (14.2 - 6.0)	8.2	Judgment	Closures recur every year; only the excess is added back.
Gain on sale-leaseback	(9.0)	Non-recurring	Remove the gain.
Vendor rebate timing	(3.5)	Timing	Belongs to FY2027.
53rd week	(2.2)	Calendar	Not in a 52-week run-rate.
Rent step-ups on 18 renewals	(4.0)	Forward-looking	Will be paid from FY2027.
Marketing deferral to be restored	(5.0)	Timing	Run-rate marketing is higher.
Stock-based compensation	0.0	Judgment	Not added back: a real cost of paying staff.
Normalized EBITDA	**138.9**		8.6% of normalized sales of 1,621.8

Additions total 13.0 + 6.5 + 4.1 + 8.2 = 31.8; deductions total 9.0 + 3.5 + 2.2 + 4.0 + 5.0 = 23.7; net +8.1; 130.8 + 8.1 = 138.9.

Reconciliation to management's $175.6 million (gap 36.7): stock compensation 11.0, full closure add-back instead of the excess 6.0, full inventory add-back instead of the excess 5.0, and the four omitted deductions 14.7 (3.5 + 2.2 + 4.0 + 5.0). 11.0 + 6.0 + 5.0 + 14.7 = 36.7.

Range. Low: treat closures as recurring at their three-year average of $9.1M (add back 5.1, not 8.2, a difference of 3.1) and reserve the extra $4.65M of aged inventory: 138.9 - 3.1 - 4.65 = $131.2M. High: if the closure program is judged entirely one-time, add back the $6.0M normal level too: 138.9 + 6.0 = $144.9M. Adding back stock compensation (155.9) is outside the defensible range because it reverses a real cost. **Defensible range $131M to $145M, central $138.9M.**

EBITDAR. Normalized: 138.9 + 184.0 (fixed lease cost) = **$322.9M**, 19.9% of 1,621.8. Reported: 130.8 + 184.0 = $314.8M. Variable rent of $12.0M stays in cost because it is not in the lease liability.

Question 2: Lease-adjusted metrics

Table 4.10: Lease liability check

Fiscal year	Undiscounted payment	Discount factor at 6.2%	Present value
FY2027	190.0	0.9416	178.9
FY2028	178.0	0.8866	157.8
FY2029	165.0	0.8349	137.8
FY2030	150.0	0.7861	117.9
FY2031	135.0	0.7402	99.9
FY2032-FY2036 (126.0 a year)	630.0		390.8
Total	**1,448.0**		**1,083.1**

The computed liability of $1,083.1M reconciles to the balance sheet (122.8 + 960.3). The eight-times rent shortcut gives 8 x 184.0 = $1,472.0M (36 percent above the liability), or 8 x 196.0 = $1,568.0M (45 percent above) if variable rent is included. The shortcut overstates because remaining committed terms average only 6.8 years and the discount rate is 6.2 percent; neither measure captures renewals, so neither is the going-concern obligation.

Table 4.11: Lease-adjusted metrics

Metric	Reported basis	Normalized basis
Net financial debt / EBITDA	275.5 / 130.8 = 2.11x	275.5 / 138.9 = 1.98x
Lease-adjusted net debt: 338.0 + 1,083.1 - 62.5	1,358.6	1,358.6
Lease-adjusted net debt / EBITDAR	1,358.6 / 314.8 = 4.32x	1,358.6 / 322.9 = 4.21x
Fixed-charge coverage: EBITDAR / (184.0 + 31.0)	314.8 / 215.0 = 1.46x	322.9 / 215.0 = 1.50x
EV / EBITDA at $6.80 (EV 682.1)	5.22x	4.91x
EV / EBITDA at $9.40 (EV 837.6)	6.40x	6.03x
Lease-adjusted EV at $6.80 (682.1 + 1,083.1 = 1,765.2) / EBITDAR	5.61x	5.47x
Lease-adjusted EV at $9.40 (837.6 + 1,083.1 = 1,920.7) / EBITDAR	6.10x	5.95x

Leverage looks modest at 2x until leases are counted, and then it is above 4x. Fixed-charge coverage of 1.50x leaves 36 percent headroom before the 1.10x covenant.

IFRS 16 restatement (normalized). EBITDA becomes EBITDA plus fixed rent: 138.9 + 184.0 = $322.9M. Lease interest is 6.2% x 1,083.1 = 67.2, and right-of-use depreciation is 184.0 - 67.2 = 116.8 (a simplification that sets total lease expense equal to the straight-line cost). EBIT = 322.9 - 51.5 - 116.8 = **$154.6M**, against $87.4M under US GAAP (138.9 - 51.5), the difference being the lease interest of 67.2. Interest expense rises from 31.0 to 98.2. Net debt including leases is 1,358.6, or 4.21x EBITDA, equal to the EBITDAR figure above. Pre-tax income is the same in this simplified year; only the presentation (and the timing over the lease life) changes.

Question 3: Peer multiples on a common basis

Lease-adjusted EV is market capitalization plus net financial debt plus lease liabilities. EBITDAR for US GAAP peers is EBITDA plus fixed operating lease cost; for IFRS 16 peers, reported EBITDA is already before most rent.

Table 4.12: Peer multiples

Company	Lease-adjusted EV	EBITDAR	EV / EBITDAR	Raw EV / EBITDA (EV excluding leases)	Lease-adjusted net debt / EBITDAR
Cedar & Loom Home	3,410	564	6.05x	6.45x	3.42x
Tidewater Living	2,135	299	7.14x	9.02x	4.60x
Willow Row Stores	1,435	222	6.46x	8.04x	5.07x
Nordhaven Home	2,550	395	6.46x	3.27x	3.54x
Alder Outdoor Group	1,610	235	6.85x	3.83x	4.13x
Mean / median			6.59x / 6.46x	6.12x / 6.45x	

On a common basis the peers sit in a tight 6.05x to 7.14x band. Raw multiples run from 3.3x to 9.0x, a spread that comes entirely from accounting: the IFRS peers' EBITDA excludes rent while their EV excludes leases, so the ratio is understated. The raw median of 6.45x happens to equal the adjusted median, a coincidence that should not reassure anyone.

Implied value per share for Harbor & Pine (EBITDAR 322.9; less lease liability 1,083.1 and net financial debt 275.5; 59.8 million shares):

EV / EBITDAR	Enterprise value	Equity value	Per share
5.00x	1,614.5	255.9	4.28
5.50x	1,775.9	417.3	6.98
5.90x	1,905.1	546.5	9.14
5.95x (the bid)	1,921.3	562.7	9.41
6.05x (peer minimum)	1,953.5	594.9	9.95
6.30x	2,034.3	675.7	11.30
6.46x (peer median)	2,085.9	727.3	12.16
6.59x (peer mean)	2,127.9	769.3	12.86
7.14x (peer maximum)	2,305.5	946.9	15.83

Each 0.1x of multiple is worth about $0.54 a share, because lease and debt claims are large relative to equity. Harbor & Pine's falling comps, inventory build and 2028 refinancing justify a 5 to 9 percent discount to the median: 5.9x to 6.3x, or **$9.14 to $11.30**. The bid sits at the bottom of that range, and the market price of $6.80 equates to 5.47x.

Question 4: Cost of capital

Cost of equity = 4.3% + 1.45 x 5.5% = 12.275%. After-tax cost of term debt = 9.0% x (1 - 0.25) = 6.75%.

Excluding operating leases from debt. Weights use the unaffected market value of equity (406.6) and net financial debt (275.5), total 682.1: equity 59.6 percent, debt 40.4 percent. WACC = 0.596 x 12.275% + 0.404 x 6.75% = 7.32% + 2.73% = **10.04%, rounded to 10.0%**.

Including operating leases as debt. Total capital is 406.6 + 275.5 + 1,083.1 = 1,765.2: equity 23.0 percent, financial debt 15.6 percent, leases 61.4 percent. After-tax lease cost = 6.2% x 0.75 = 4.65%. WACC = 0.230 x 12.275% + 0.156 x 6.75% + 0.614 x 4.65% = 2.83% + 1.05% + 2.85% = **6.73%**.

The two numbers are both correct for different cash flows. The 10.0 percent WACC discounts cash flows after rent, which belong to financial debt and equity only. The 6.7 percent WACC discounts cash flows before rent (EBITDAR-based), and then requires explicit investment in replacement leases and subtraction of lease liabilities. Mixing them is the common error. If the after-rent Plan cash flows and terminal value are discounted at 6.73%, enterprise value becomes 1,117.7 instead of 972.2 and the value per share becomes (1,117.7 - 275.5) / 59.8 = **$14.08 instead of $11.65**, an overstatement of $2.43 (21 percent). This chapter uses the after-rent approach throughout.

Question 5: The Plan scenario

Worked FY2027. Sales = 1,621.8 x 1.005 = 1,629.9, less 14 closures x 5.0 x 0.70 = 49.0, equals 1,580.9. Fixed rent = (171.2 - 14 x 1.00 + 12.8) x 1.02 = 170.0 x 1.02 = 173.4. EBITDAR = 19.8% x 1,580.9 = 313.0; EBITDA = 313.0 - 173.4 = 139.6. D&A = 3.1% x 1,580.9 = 49.0. Closure costs = 14 x 2.3 = 32.2. EBIT = 139.6 - 49.0 - 32.2 = 58.4; tax = 25% x 58.4 = 14.6. Capex = 1.8% x 1,580.9 (28.5) + 20 remodels x 1.4 (28.0) + IT 18.0 = 74.5. Working capital: 12% x (1,580.9 - 1,621.8) = -4.9, a release of cash. Free cash flow = 139.6 - 14.6 - 32.2 - 74.5 + 4.9 = 23.2 (23.3 unrounded).

Table 4.13: Plan scenario projection (US$ millions)

US$ millions	FY2027	FY2028	FY2029	FY2030	FY2031
Net sales	1,580.9	1,563.5	1,602.6	1,646.7	1,691.6
Stores at year end	200	186	186	190	194
EBITDAR	313.0	325.2	349.4	365.6	375.5
Fixed rent	(173.4)	(162.3)	(165.5)	(171.5)	(177.5)
EBITDA	139.6	162.9	183.8	194.1	198.0
Closure costs	(32.2)	(32.2)	0.0	0.0	0.0
D&A	(49.0)	(48.5)	(49.7)	(51.0)	(52.4)
EBIT (after closure costs)	58.4	82.2	134.1	143.1	145.6
Taxes at 25%	(14.6)	(20.6)	(33.5)	(35.8)	(36.4)
Capex	(74.5)	(74.1)	(64.8)	(37.6)	(38.4)
Investment in working capital	4.9	2.1	(4.7)	(5.3)	(5.4)
Free cash flow to firm	**23.3**	**38.1**	**80.7**	**115.4**	**117.8**
Discount factor at 10.0%	0.9091	0.8264	0.7513	0.6830	0.6209
Present value	21.2	31.5	60.7	78.8	73.1

Present value of five-year free cash flow: 21.2 + 31.5 + 60.7 + 78.8 + 73.1 = **265.3**. Terminal value = 5.75 x FY2031 EBITDA of 198.0 = 1,138.5; discounted five years at 0.6209 gives **706.9**. Enterprise value = 265.3 + 706.9 = **972.2**. Equity value = 972.2 - 275.5 = 696.7, or **$11.65 per share** on 59.8 million shares. Terminal value is 73 percent of enterprise value.

Question 6: Four scenarios and the bid

Table 4.14: Scenario values

Scenario	Prob.	FY2031 sales	FY2031 EBITDA	Exit multiple	PV of FCF	PV of terminal value	Enterprise value	Equity value	Per share (raw)	Per share (floored at 0)
Upside	20%	1,835.1	232.6	6.25x	317.4	902.7	1,220.1	944.6	15.80	15.80
Plan	40%	1,691.6	198.0	5.75x	265.3	706.9	972.2	696.7	11.65	11.65
Muddle	25%	1,526.0	115.8	5.25x	197.3	377.5	574.9	299.4	5.01	5.01
Distress	15%	1,214.7	70.2	4.25x	83.5	185.1	268.6	(6.9)	-0.12	0.00
Probability-weighted	100%						816.9			**9.07**

The probability-weighted value is 0.20 x 15.80 + 0.40 x 11.65 + 0.25 x 5.01 + 0.15 x 0.00 = 3.16 + 4.66 + 1.25 + 0 = **$9.07 per share** (weighted enterprise value 816.9). The Distress scenario's raw equity value is negative (6.9 million), so shareholders receive nothing and the lenders bear the shortfall; ignoring the floor gives $9.05, almost no difference.

The bid of $9.40 is 3.6 percent above the weighted value, 38 percent above the unaffected price, 19 percent below the Plan value and 41 percent below the Upside. Equity is worth more than $9.40 in the Upside and Plan scenarios (combined probability 60 percent) and less in the other two (40 percent).

Implied terminal growth. The exit multiples imply perpetual growth after FY2031 that can be backed out as g = (TV x WACC - FCF) / (TV + FCF), using a terminal-year free cash flow of EBITDA less tax on EBIT less capex with no working-capital change:

Scenario	Terminal value	Terminal-year FCF	Implied perpetual growth	Terminal value share of EV
Upside	1,453.8	147.6	(0.1%)	74%
Plan	1,138.5	123.2	(0.7%)	73%
Muddle-through	608.0	65.2	(0.7%)	66%
Distress	298.2	36.2	(1.9%)	69%

The multiples are conservative: they assume a mature, slowly shrinking business even in the Plan. A growth-rate terminal value of 1 to 2 percent would raise every scenario, which is one reason the committee should not treat $9.07 as an upper bound.

Question 7: Sensitivity

Table 4.15: Probability-weighted value per share

Change	Weighted value	Change vs. $9.07
WACC 9.0% / 9.5% / 10.0% / 10.5% / 11.0%	9.64 / 9.34 / 9.07 / 8.80 / 8.55	+0.57 / +0.27 / 0 / (0.27) / (0.52)
Exit multiples all shifted (1.0x) / (0.5x) / 0 / +0.5x / +1.0x	7.46 / 8.27 / 9.07 / 9.91 / 10.77	(1.61) / (0.80) / 0 / +0.84 / +1.70
EBITDAR margin all years shifted (100bp) / (50bp) / 0 / +50bp / +100bp	7.57 / 8.32 / 9.07 / 9.90 / 10.74	(1.50) / (0.75) / 0 / +0.83 / +1.67
Plan scenario value alone at the same margin shifts	9.87 / 10.76 / 11.65 / 12.54 / 13.43	
Comparable-store sales (1 point) / +1 point a year	7.91 / 10.36	(1.16) / +1.29
Probabilities 20/40/15/25 (more Distress)	8.57	(0.50)
Probabilities 30/40/20/10 (more Upside)	10.40	+1.33
No Distress (others rescaled)	10.67	+1.60

The Plan probability at which the weighted value equals $9.40 is **47.7 percent**, with the other three scenarios scaled pro rata to 17.4, 21.8 and 13.1 percent. Alternatively, exit multiples about 0.2x higher, or EBITDAR margins about 20 basis points higher in every year of every scenario, or a WACC of about 9.4 percent, bring the weighted value to the bid. In other words, the bid is inside the noise of the valuation: nothing in the sensitivities says $9.40 is clearly cheap or clearly generous. The valuation is most sensitive to comparable-store sales and margins, which are what the turnaround must actually deliver, and to the terminal multiple, which is a market judgment.

Question 8: Real estate and a sale-leaseback

Proceeds. Appraised value 354.0 less tax on the 116.0 gain (25 percent, 29.0) is 325.0, less transaction costs of 3.0, or 322.0 of cash. Initial rent is 7.0% x 354.0 = 24.8. The 20-year lease with 2.0 percent escalators, discounted at 6.2 percent, creates a lease liability of **326.8**. Cash repays the term loan (265.0) and the revolver (55.0), saving interest of 265.0 x 9.0% + 55.0 x 7.5% =

28.0.

Value effect (Plan scenario). After-tax proceeds net of costs of +322.0; present value of the after-tax rent charged in FY2027 to FY2031 of -74.5; present value of the terminal value lost (5.75 x FY2031 rent of 24.8 x 1.02^5 = 27.4, so 157.3, discounted at 0.6209) of -97.7. Net uplift is 322.0 - 74.5 - 97.7 = **$149.8M, or $2.51 per share**. The mechanism: the base valuation capitalizes the property's rent-free earnings at an exit multiple of 5.75x, which values the real estate at about 157 million in FY2031, whereas an investor will pay 14.3 times rent (1 / 7.0%), and today's appraisal is 354.0.

Table 4.16: Value per share with and without a sale-leaseback

Scenario	Base	With sale-leaseback	Uplift
Upside	15.80	18.16	2.36
Plan	11.65	14.16	2.51
Muddle-through	5.01	7.65	2.65
Distress	0.00 (raw (0.12))	2.63	2.63 (raw 2.75)
Probability-weighted	**9.07**	**11.60**	**2.53**

Credit effect. After the transaction, lease-adjusted net debt is 18.0 (finance leases) + 1,083.1 + 326.8 - 64.5 (cash after repaying 320.0 of debt) = 1,363.4, or 4.22x EBITDAR against 4.21x before. Fixed-charge coverage is 322.9 / (184.0 + 24.8 + 3.0) = 1.52x against 1.50x. A sale-leaseback does not de-lever a retailer once leases count as debt; it swaps 9 percent debt for a 7 percent lease. Its benefit is that the market pays more for the property than for the earnings it supports, not that risk falls. Because the simple multiples method hides this value, the pitfall is to value the operating business on EBITDA that includes rent-free real estate, then forget the real estate. A better method values the operating company after market rent on all properties and adds the property portfolio at appraised value, net of tax.

Question 9: The sponsor's view

Sources and uses at $9.40. Uses: equity purchase 562.1, repayment of term loan and revolver 320.0, fees 30.0, total 912.1. Sources: new term loan 400.0, balance-sheet cash 22.5 (leaving 40.0), sponsor equity 489.6. With a sale-leaseback: uses rise by 3.0 to 915.1; sources add 325.0 of after-tax proceeds, and sponsor equity falls to **167.6**. At close, without a sale-leaseback, lease-adjusted net debt is (400.0 + 18.0 + 1,083.1 - 40.0) = 1,461.1, or 4.52x EBITDAR, and fixed-charge coverage is 322.9 / (184.0 + 42.0 + 1.1) = 1.42x.

Mechanics. Each year, interest of 10.5 percent on the opening term loan (plus 1.1 on finance leases) is paid, tax is 25 percent of positive pre-tax income, and remaining free cash flow repays the loan (or draws it if negative). At exit in FY2031, enterprise value is the scenario's exit multiple times EBITDA (less the new rent, in the sale-leaseback case); net debt includes finance leases less 40.0 of cash. Equity is floored at zero.

Table 4.17: Sponsor returns at $9.40 (five years)

Scenario	Probability	No sale-leaseback: exit equity	Multiple	IRR	With sale-leaseback: exit equity	Multiple	IRR
Upside	20%	1,383.7	2.83x	23.1%	1,097.6	6.55x	45.6%
Plan	40%	986.9	2.02x	15.0%	711.1	4.24x	33.5%
Muddle-through	25%	347.1	0.71x	(6.6%)	74.0	0.44x	(15.1%)
Distress	15%	0.0	0.00x	(100%)	0.0	0.00x	(100%)
Probability-weighted multiple			**1.55x**	**9.1%**		**3.12x**	**25.5%**

Sponsor equity is 489.6 without and 167.6 with the sale-leaseback; multiples are exit equity divided by equity invested, and IRR is multiple^(1/5) - 1.

What the sponsor can pay for a 20 percent IRR. Without a sale-leaseback, $7.84 on the Plan scenario and $6.31 on the probability-weighted multiple: the bid of $9.40 cannot be justified on operations alone. With a sale-leaseback, **$11.38** on the Plan and **$10.11** on the weighted multiple. The sponsor is evidently counting on the real estate (and probably on deeper cost cuts), and so it can afford to pay more than $9.40. The committee's room is the band of roughly $10.1 to $11.4, and the Calloway demand of $11.00 sits inside it.

Practical caution. The sponsor's returns are highly sensitive to the exit multiple and to the Distress probability, and a 15 percent chance of total loss limits how much the sponsor will concede. The committee should expect a final price near the lower end of the band unless a competing bidder appears.

Question 10: Recommendation

See the memo below. A strong answer rejects $9.40 as inadequate without refusing to engage, anchors on three independent value indications, uses the sponsor's ability to pay as leverage, and treats the real estate as the key unvalued asset.

Sensitivity Discussion

Table 4.18: Ranking of drivers (probability-weighted value per share, base $9.07)

Driver and range tested	Low	High	Swing
Exit multiples (1.0x lower / 1.0x higher)	7.46	10.77	3.31
EBITDAR margin (100bp lower / higher, every year)	7.57	10.74	3.17
Sale-leaseback (excluded / included)	9.07	11.60	2.53
Comparable-store sales (1 point lower / higher a year)	7.91	10.36	2.45
Scenario probabilities (more Distress / more Upside)	8.57	10.40	1.83
WACC (11.0% / 9.0%)	8.55	9.64	1.09

Four lessons. First, the largest swings come from operating outcomes and the terminal multiple, and the terminal value alone carries about 70 percent of enterprise value; the discount rate matters least, so debating 9.5 versus 10.5 percent wastes the committee's time. Second, the real estate decision is as large as any operating driver and is entirely within the board's control. Third, the bid sits within one standard swing of the weighted value, so a fairness opinion must be framed as a range. Fourth, probabilities are the least observable inputs, and the break-even Plan probability of 47.7 percent (against a 40 percent assumption) gives a more useful test than any point estimate: ask the committee whether it believes the Plan is more likely than not to work.

Teaching Tips

Ask half the class to argue as the CEO, using the Plan, and half as the sponsor, using the Distress and Muddle scenarios; then reveal the sale-leaseback returns and ask who is better informed. Insist that every multiple on the board names its numerator and denominator ("EV with leases over EBITDAR" or "EV without leases over EBITDA"). Close by asking whether Ms. Oyelaran's question, "before rent or after rent?", has been answered for every number on the page.

Common Student Errors

- Using raw EV/EBITDA from IFRS 16 and US GAAP peers without lease adjustment, and taking a tight raw median as confirmation.
- Adding back stock compensation and all closure and inventory charges, which reproduces management's $175.6M and inflates every multiple-based value.
- Forgetting to subtract lease liabilities when using a lease-adjusted EV, or subtracting them when valuing after-rent cash flows (a double count).
- Discounting after-rent cash flows at a lease-inclusive WACC, producing an apparent $14.08.
- Treating the Plan as the expected case, and ignoring that equity cannot be negative in the Distress case.
- Valuing the business on EBITDA that includes rent-free owned stores, and never adding the property back.
- Presenting a sale-leaseback as de-leveraging.
- Forgetting that the 53rd week, vendor rebates and rent step-ups cut earnings.

Alternative Defensible Answers

A class may reasonably (a) exclude Willow Row from the peer set (the median becomes about 6.7x on four peers, a higher value); (b) use a WACC of 9 or 11 percent; (c) probability-weight differently, for example giving Plan 30 percent (the Plan probability at which the bid equals weighted value is 47.7 percent, so a lower Plan probability argues for accepting the bid); (d) use a Gordon-growth terminal value of 1 percent, which raises values; (e) accept the bid because the committee is risk-averse about a 15 percent chance of equity loss and a September 2028 refinancing; or (f) reject outright and pursue the Plan with a sale-leaseback executed by management, which gives a weighted value of $11.60 before execution risk. The weakest answers ignore the real estate, or accept management's EBITDA without challenge.

Special Committee Recommendation Memo

To: Special Committee of the Board, Harbor & Pine Holdings, Inc. **From:** Financial adviser. **Date:** 6 November 2026. **Re:** Granite Ridge proposal of $9.40 per share (illustrative).

Recommendation. Do not accept $9.40. Engage with Granite Ridge, counter at $11.25, and authorize a 30-day market check. We would recommend a transaction at $10.75 or above if no better proposal emerges.

Why. Four indications of value, on normalized EBITDA of $138.9 million (range $131M to $145M, not management's $175.6M): lease-adjusted peer multiples at 5.9x to 6.3x EBITDAR give $9.14 to $11.30; the probability-weighted turnaround value is $9.07 before real estate and $11.60 with a sale-leaseback; the Plan alone is $11.65 and the Upside $15.80; and the sponsor's own returns allow a price of $10.1 to $11.4 if it monetizes the owned property. The bid equals the bottom of the comps range and the probability-weighted value of an operation that has no real estate. It is a 38 percent premium to an unaffected price that equates to 5.5x EBITDAR, but it is not a full price for an asset whose owned property is worth $354.0 million.

Risks to our view. Distress carries a 15 percent probability and wipes out equity; the term loan matures in September 2028; aged inventory may require a further $4.65 million; and 41 leases expire within three years. These support a negotiated outcome rather than a fight.

Diligence. Verify (1) the property appraisal and a third-party sale-leaseback bid at or below a 7 percent capitalization rate; (2) inventory reserves and fourth-quarter trading; (3) landlord buyout costs for the 28 closure stores and the renewal terms of the 41 expiring leases; and obtain Granite Ridge's committed financing letter.

Message to Calloway. The committee shares the view that $9.40 is inadequate, is testing the market, and will rely on evidence, not on any shareholder's number; $11.00 lies inside the range the committee's analysis considers achievable.

Chapter 5: Case: Cirrus Ledger, Valuing a SaaS Company

Cirrus Ledger, Tessara Systems, Kessler Dunmore & Co., every peer company, and every transaction in this chapter are fictional and illustrative. All numbers were constructed for teaching and are not statistics about any real company, market, or deal. The chapter draws on Volume 1, Chapter 5 (equity valuation), Chapter 11 (WACC), Chapter 18 (options), Chapter 21 (enterprise valuation), and Chapter 22 (M&A), and on Volume 2's treatment of valuation under uncertainty.

The Situation

On the morning of Wednesday, 4 November 2026, Marisol Quade, chief executive of Cirrus Ledger, Inc., opened a letter that she had been half expecting for a year. It came from Walter Osei, chief executive of Tessara Systems, a much larger enterprise software group, and it proposed to acquire every outstanding share of Cirrus Ledger for $66.50 in cash. The letter was short, courteous, and firm about timing. The offer would lapse at 5:00 p.m. on Friday, 20 November 2026 unless the Cirrus board entered into exclusive discussions before then. It also made clear that Tessara would be content to "explore other paths" if the board declined, a phrase that the company's general counsel read as a hint that Tessara might go directly to shareholders.

Cirrus Ledger sells cloud software that automates accounts payable, invoice capture, and the monthly financial close for finance teams at mid-sized companies, generally those with revenue between $200 million and $3 billion. The company was founded in 2014 by Marisol Quade and by Devraj Anand Rao, who is now chief financial officer, after both had spent years watching finance staff reconcile supplier invoices in spreadsheets. It went public in fiscal 2021, and its fiscal year ends on 30 June. At the end of fiscal 2026 it had 2,474 customers, ending annual recurring revenue (ARR) of $589.3 million, revenue for the year of $559.5 million, and 1,740 employees. It had never reported a GAAP profit. It had, however, produced $134.9 million of free cash flow in fiscal 2026, a fact that the company's investor presentations placed on the first slide.

The offer is 18.8 percent above the undisturbed closing price of $56.00 on Friday, 30 October 2026. Tessara describes it as a "full and fair premium reflecting the strategic fit of the two businesses." Cirrus's stock had traded between $41 and $63 over the preceding twelve months, so the offer is above the top of that range, but only modestly, and several large holders had said publicly in the summer that they saw the shares as worth "well north of $70" if the company delivered its plan. The founders and directors together own 11.4 percent of the shares. Two long-only funds, Larkspur Growth and the Eastmarch Pension Trust, own 9.8 and 7.1 percent respectively and have already called the chair.

The board's independent directors have formed a special committee chaired by Teodora Brandt, a former chief financial officer of an industrial distributor. She has asked her advisers at Kessler Dunmore & Co. for a valuation that the committee can defend to shareholders and, if it comes to it, to a court. She has also asked Dev Rao and his finance team to prepare the standalone management plan, and she has been blunt about her concern. "Every software valuation I have seen this year starts with 'adjusted' something," she told the committee. "I want to know what a share of this company is worth when we count every cost, including the shares we hand to our own employees, and then I want to know how far the offer sits from that number."

You are a member of the Kessler Dunmore team. By the committee's meeting on Thursday, 19 November, you must deliver three things. The first is a diagnosis of the quality of the business, meaning its cohort economics, its unit economics, and its rule-of-40 profile on several definitions. The second is a valuation range built from a discounted cash flow model and from market multiples, with an explicit treatment of stock-based compensation. The third is a recommendation: accept, reject, or counter, and if counter, at what price and with what reasoning. The committee is not asking for a fairness opinion, which the bank will issue separately. It is asking for judgment about value, and about how much weight each method deserves.

Three people will react to your work. Dev Rao will defend the management plan, and he is right that it was prepared carefully, but he is also the person whose compensation depends on it. Teodora Brandt will push on stock-based compensation and on the terminal value. A third director, Anselm Okafor, a retired software executive and a major option holder, will argue that the plan understates the company’s potential and that the board should reject any offer below $75. Your task is to give each of them a number they can interrogate and a reason they can accept or reject.

Industry Context

Software sold as a subscription, usually called software as a service (SaaS), has become the most heavily studied business model in equity valuation. The customer pays a recurring fee for access rather than a large upfront license fee. The vendor books revenue ratably over the contract term but often bills in advance, so cash arrives before revenue is recognized and the balance sheet carries a liability called deferred revenue. For a growing vendor, deferred revenue is a source of cash: each new dollar of annual contract billed up front is collected before it is earned. This is one reason that free cash flow at a growing SaaS company frequently exceeds net income by a wide margin, and it is the reason that valuation shortcuts built on earnings multiples fail for the sector.

The economics of the model rest on a few ideas that every chapter reader should hold in mind. A customer is expensive to win. The vendor spends on sales and marketing in the year of acquisition, and the spending is expensed immediately, so the income statement shows a loss on a customer that will be profitable later. If the customer stays and buys more, the vendor earns that investment back several times over. The central analytic question is therefore whether customers stay, whether they expand, and how much it costs to acquire the next one. Investors summarize the first two ideas with net revenue retention (NRR), the share of last year’s recurring revenue from existing customers that remains this year after churn, downgrades, and upsell, and with gross revenue retention (GRR), which excludes upsell. They summarize the third idea with customer acquisition cost (CAC), the payback period on that cost, and the ratio of lifetime value to CAC.

Because the model front-loads cost and back-loads cash, a growing SaaS company usually looks poor on GAAP earnings and strong on cash flow. Management teams respond by reporting “non-GAAP” operating income, which excludes stock-based compensation (SBC), and by reporting free cash flow before any adjustment for the cost of share issuance. Both measures are legitimate descriptions of cash, but neither shows the full economic cost of employing people. SBC is a real cost. The company pays engineers partly in shares, and the shares have a market value that would otherwise have to be paid in cash or would dilute existing owners. In the late 2010s and early 2020s, SBC ran at 15 to 30 percent of revenue at many growth-stage software companies, so the choice of how to treat it moves valuations by tens of percent. This chapter makes that choice explicit and measures its effect.

Investors also use a screening heuristic known as the rule of 40. It says that a healthy software company’s revenue growth rate plus its profit margin should sum to at least 40 percent. The rule trades growth against profitability on the theory that a business can buy growth with margin or margin with growth. The heuristic is useful because it is simple and because empirical work on public software companies has found that it correlates with revenue multiples. It is dangerous because “profit margin” has no fixed definition. Free cash flow margin, non-GAAP operating margin, and GAAP operating margin give different answers for the same company, and the gap between them is almost exactly the amount of SBC and capitalized costs. Part of the work in this case is to compute the rule of 40 on several definitions and to see which one correlates with value.

The market backdrop in late 2026 is one of selective discipline. Public software multiples compressed in the years after the pandemic-era peak and then recovered unevenly. Investors now pay for durable growth with demonstrated cash conversion and penalize growth that depends on heavy discounting or rising acquisition costs. Corporate buyers and private equity sponsors have been active in mid-capitalization software, and take-private premiums have clustered in the low-to-mid 30s percent. A

strategic acquirer such as Tessara can often pay more than a sponsor because it can remove duplicated sales, marketing, and administrative cost and can sell Cirrus's products into its own customer base. Tessara has not disclosed its synergy estimate. The committee must decide how much of that synergy it is willing to hand to its own shareholders.

Finally, a note on method. A discounted cash flow valuation of a company that is growing at 30 percent is a long-duration asset. Most of its value sits in cash flows that arrive ten years from now and later, and small changes in the discount rate, the terminal growth rate, or the long-run margin change the answer substantially. Multiples are easier to apply, but they import the market's current mood. A careful analyst uses both, understands why they disagree, and tells the client how much of the gap is judgment and how much is arithmetic. That is what the committee has asked you to do.

Financial Data

Tables 1 and 2 show the cohort data that the finance team assembled. Customers are grouped by the fiscal year in which they first signed, and the table shows each group's ARR at each fiscal year-end. ARR is the annualized value of subscription contracts in force at the date, excluding professional services. The "pre-FY2023" row pools every customer acquired before 1 July 2022. The FY2023 cohort is therefore the oldest cohort that is reported separately.

Table 1. Ending ARR by customer cohort ($ millions, fiscal years ending 30 June)

Cohort	FY2022	FY2023	FY2024	FY2025	FY2026
Pre-FY2023 customers (pooled)	175.0	192.5	209.8	226.6	242.5
FY2023 cohort	–	55.0	66.0	75.9	84.2
FY2024 cohort	–	–	66.0	77.2	86.5
FY2025 cohort	–	–	–	78.0	88.1
FY2026 cohort	–	–	–	–	88.0
Total ending ARR	**175.0**	**247.5**	**341.8**	**457.7**	**589.3**

Table 2. Customer counts, landing ARR per new customer and gross dollar retention

Fiscal year	Opening customers	New customers	Closing customers	Landing ARR per new customer ($000)	Gross dollar retention
FY2023	1,180	380	1,477	144.7	93%
FY2024	1,477	420	1,794	157.1	92%
FY2025	1,794	455	2,123	171.4	91%
FY2026	2,123	500	2,474	176.0	90%

The cohort table and the customer table should be read together. Landing ARR per new customer has risen from about $145,000 to $176,000, so Cirrus is selling larger initial deals. At the same time, gross dollar retention has slipped one point a year, from 93 percent to 90 percent. The finance team reports that customer counts are net of the roughly 7 percent of customers who leave each year, and that churn is concentrated in customers with annual contract values below $60,000.

Tables 3 through 5 give the financial statements. Revenue is recognized ratably, and subscription revenue equals the average of opening and closing ARR in each year, rounded. Professional services, mostly implementation work, run at thin margins. The income statement is on a GAAP basis, so stock-based compensation is inside every expense line. The finance team estimates that about one-third of SBC sits in sales and marketing, about 41 percent in research and development, 18 percent in general and administrative, and the balance in cost of revenue. The memo lines at the bottom of Table 3 are provided for convenience.

Table 3. Income statement (GAAP, $ millions)

	FY2024	FY2025	FY2026
Subscription revenue	294.7	399.8	523.5
Professional services revenue	24.0	30.5	36.0
Total revenue	318.7	430.3	559.5
Cost of revenue	92.4	118.3	148.3
Gross profit	226.3	312.0	411.2
Sales and marketing	141.8	182.9	229.4
Research and development	86.0	109.7	134.3
General and administrative	43.0	53.8	64.3
Operating income (loss)	(44.5)	(34.4)	(16.8)
Interest income	11.8	13.9	15.6
Interest expense	2.0	2.0	2.0
Income taxes	2.4	3.1	4.0
Net income (loss)	(37.1)	(25.6)	(7.2)
Memo: stock-based compensation (included above)	73.3	90.4	109.1
Memo: SBC in sales and marketing	24.4	30.1	36.4
Memo: non-GAAP operating income (excludes SBC)	28.8	56.0	92.3

Table 4. Balance sheet extract at 30 June ($ millions)

	FY2025	FY2026
Cash and marketable securities	372.0	489.8
Accounts receivable	86.1	114.7
Prepaid and other current assets	28.0	36.5
Property and capitalised software, net	61.0	64.3
Goodwill and acquired intangibles	48.0	48.0
Other non-current assets	30.0	30.0
Total assets	**625.1**	**783.3**
Accounts payable and accrued liabilities	62.0	79.0
Deferred revenue	192.2	247.5
Convertible senior notes, carrying value	171.0	172.1
Other non-current liabilities	25.0	25.0
Total liabilities	**450.2**	**523.6**
Additional paid-in capital	484.9	576.9
Accumulated deficit	(310.0)	(317.2)
Total stockholders’ equity	**174.9**	**259.7**
Total liabilities and equity	**625.1**	**783.3**

Table 5. FY2026 cash flow statement extract ($ millions)

	FY2026
Net income (loss)	(7.2)
Stock-based compensation	109.1
Depreciation and amortisation (incl. capitalised software)	16.8
Amortisation of debt issuance costs	1.1
Increase in accounts receivable	(28.6)
Increase in prepaid and other current assets	(8.5)
Increase in accounts payable and accruals	17.0
Increase in deferred revenue	55.3
Cash flow from operations	**155.0**
Purchases of property and capitalised software	(20.1)
Free cash flow (company definition)	**134.9**
Proceeds from option exercises	7.4
Taxes paid on net share settlement of RSUs	(24.5)
Net change in cash and marketable securities	**117.8**

Table 6. Free cash flow and growth history ($ millions)

	FY2024	FY2025	FY2026
Revenue	318.7	430.3	559.5
Free cash flow (company definition)	47.8	90.4	134.9
Free cash flow margin	15.0%	21.0%	24.1%
Revenue growth	38.4%	35.0%	30.0%

Deferred revenue rose from $192.2 million to $247.5 million as annual-in-advance billing grew, which contributed $55.3 million to cash flow from operations. That inflow is larger than the entire improvement in FY2026 free cash flow over FY2025. Receivables absorbed $28.6 million. The “taxes paid on net share settlement” line is cash that Cirrus pays to tax authorities on behalf of employees when restricted stock units vest. The company withholds shares rather than selling them, so the cash leaves the company. It does not appear in free cash flow as the company defines it, because it is classified as a financing item.

Table 7 gives the capital structure and market data on which the committee’s advisers have settled. The convertible notes carry a 0.5 percent coupon, mature in 2029, and convert into 1.902 million shares at $92.00. A change-of-control event gives holders a put at par, plus a make-whole premium that the company’s counsel estimates at no more than $4 million at the offer price. For the exercises that follow, treat the notes as debt at face value of $175.0 million unless the value per share exceeds the conversion price.

Table 7. Capital structure and market data

Item	Value
Basic common shares outstanding (millions)	101.4
Employee stock options outstanding (millions)	8.4
Weighted-average exercise price of options	$21.50
Unvested restricted stock units (millions)	6.2
Convertible senior notes, face value	$175.0 million
Conversion price of notes	$92.00 (1.902 million shares)
Coupon on notes	0.50%
Undisturbed share price (close, 30 October 2026)	$56.00
Tessara indicative offer, cash	$66.50 per share
Implied premium to undisturbed price	18.8%
Risk-free rate (10-year government)	4.2%
Equity risk premium used by the board’s advisers	4.5%
Pre-tax cost of straight debt for a company of this profile	6.2%
Statutory/cash tax rate on taxable income	23%
Federal and state net operating losses (30 June 2026)	$260.0 million

Management’s plan appears in Tables 8 and 9. Dev Rao’s team prepared the first five years in detail. The extension to fiscal 2036 comes from the team’s long-range model (Exhibit C), which assumes that growth fades and that the company reaches the margin profile of today’s most profitable mature software firms. “Non-GAAP operating margin” excludes SBC and nothing else. Capital expenditure is 3.5 percent of revenue and depreciation and amortization 3.0 percent. Because customers pay in advance, each dollar of incremental revenue releases about 15 cents of working-capital cash. Cirrus has $260.0 million of net operating losses that it can use against taxable income, and the cash tax rate on taxable income beyond the losses is 23 percent.

Table 8. Management plan, FY2027 to FY2031

	FY2027	FY2028	FY2029	FY2030	FY2031
Revenue growth	27.0%	25.0%	23.0%	21.0%	19.0%
Non-GAAP operating margin (excludes SBC)	19.5%	22.5%	26.0%	29.0%	31.5%
SBC as % of revenue	18.0%	16.5%	15.0%	14.0%	12.5%
Capital expenditure, % of revenue	3.5%	3.5%	3.5%	3.5%	3.5%
Depreciation and amortisation, % of revenue	3.0%	3.0%	3.0%	3.0%	3.0%
Deferred-revenue/working-capital inflow, % of incremental revenue	15%	15%	15%	15%	15%

Table 9. Management long-range extension, FY2032 to FY2036 (Exhibit C)

	FY2032	FY2033	FY2034	FY2035	FY2036
Revenue growth	17.0%	15.0%	13.0%	11.5%	10.0%
Non-GAAP operating margin	34.0%	36.0%	38.0%	39.5%	41.0%
SBC as % of revenue	11.5%	10.5%	10.0%	9.5%	9.0%

Table 10 shows six public peers that Kessler Dunmore screened as the closest comparables. None is an exact match. Halcyon Cloud and Meridian Stack are the closest in product, while Bramble Analytics grows fastest but with very little cash flow, and Tollgate Software is a slower, more profitable incumbent. Figures are next-twelve-month (NTM) consensus-style estimates as of 30 October 2026, and enterprise value is market capitalization plus debt less cash, taking dilution into account. The “FCF margin” column is free cash flow before any deduction for SBC, the way each company reports it. Table 11 lists four recent transactions, and Table 12 gives the beta data for the cost-of-capital estimate.

Table 10. Public comparable companies (fictional, illustrative; all figures next-twelve-months unless stated)

Company	Revenue growth	FCF margin (pre-SBC)	SBC / revenue	Rule of 40	Rule of 40 less SBC	Net revenue retention	EV / NTM revenue
Halcyon Cloud	33%	23%	21%	56	35	118%	11.2x
Quillon Systems	24%	27%	14%	51	37	114%	9.6x
Meridian Stack	29%	18%	22%	47	25	110%	9.1x
Bramble Analytics	38%	5%	27%	43	16	121%	8.9x
Tollgate Software	14%	29%	12%	43	31	101%	6.2x
Verdant Ops	21%	15%	17%	36	19	104%	6.0x
Mean	26.5%	19.5%	18.8%	46.0	27.2	111.3%	8.5x
Median	26.5%	20.5%	19.0%	45.0	28.0	112.0%	9.0x
Cirrus Ledger (NTM, management plan)	27%	22%	18%	49	31	110%	n/a

Table 11. Precedent software take-privates and acquisitions (fictional, illustrative)

Transaction	Announced	EV / NTM revenue	Premium to undisturbed	Buyer type
Aldermoor / Norwood Ledger	Eleven months ago	8.1x	31%	Strategic
Pinegate Capital / Heron Systems	Seven months ago	9.8x	38%	Sponsor
Corvane Group / Sable Cloud	Fourteen months ago	8.6x	24%	Strategic
Marlowe Partners / Linden Works	Nine months ago	10.4x	44%	Sponsor
Mean		9.2x	34.3%	
Median		9.2x	34.5%	

Table 12. Peer equity betas (fictional, illustrative; two-year weekly returns against a broad index)

Company	Levered beta	Debt / equity (market value)
Halcyon Cloud	1.20	3%
Quillon Systems	1.05	8%
Meridian Stack	1.25	2%
Bramble Analytics	1.35	0%
Tollgate Software	0.95	15%
Verdant Ops	1.15	5%

Complications and Exhibits

The data in the previous section are clean. The following exhibits are not, and the committee expects the advisers to say how each should change their view or why it should not.

Exhibit A: First-quarter flash. The first quarter of fiscal 2027 closed on 30 September 2026. Preliminary figures, not yet audited or reviewed, show net new ARR of $28.7 million, against $30.5 million in the first quarter of FY2026, a decline of about 6 percent, while revenue grew 29 percent year on year. Management attributes the shortfall to two large deals that slipped into the second quarter and to a deliberate decision to stop discounting below a floor price in the small-customer segment. The sales organization has an interest in the first explanation and the finance team in the second. The committee has no way to verify either, and the full-year plan in Table 8 has not been revised.

Exhibit B: Customer concentration. Cirrus's largest customer, Ostrava Freight Holdings, represents 6.4 percent of ending ARR, about $37.7 million, and its contract renews on 31 March 2027. The next nine customers together represent 11.1 percent. Ostrava's procurement team told Cirrus's account director in September that it is "evaluating alternatives, including Tessara's module." The management plan assumes Ostrava renews at flat pricing. Tessara's letter does not mention Ostrava, but the coincidence has not gone unnoticed.

Exhibit C: Long-range model. The extension in Table 9 comes from a management model built for the board's strategy offsite in March 2026 and not updated since. It assumes that Cirrus enters adjacent segments (treasury and procurement), which would lift growth in the middle years, and that sales and marketing falls to 22 percent of revenue by fiscal 2036. The finance team describes it as "ambitious but internally consistent." No peer in Table 10 currently has a non-GAAP operating margin above 33 percent while still growing above 10 percent, although Tollgate Software, growing at 14 percent, is at 31 percent.

Exhibit D: Non-recurring SBC. Fiscal 2026 SBC of $109.1 million includes $14.2 million of accelerated vesting for a former chief revenue officer who left in January, and a one-time grant of $6.0 million to retain the product leadership team. The finance team argues that "run-rate SBC" is therefore $88.9 million, or 15.9 percent of revenue. Counter-evidence: SBC as a percentage of revenue has fallen in each of the past two years without these adjustments, and the plan already assumes that it continues to fall. A new grant cycle in February 2027 will be priced at the then-current stock price, which will be higher if the offer becomes public.

Exhibit E: Billing terms. In the second half of FY2026, Cirrus moved roughly 30 percent of its customer base from quarterly to annual billing in advance, offering a 3 percent discount for the change. The move pulled forward cash but did not change revenue recognition. The finance team estimates that it added about $17 million to deferred revenue and therefore to FY2026 free cash flow, and that the benefit is non-repeating. The discount reduces future subscription revenue by roughly $2.6 million per year.

Exhibit F: Net operating losses. The $260.0 million of losses in Table 7 is the book figure. Tax counsel notes that on a change of control, an annual limitation under the relevant tax provision would restrict use of the losses to a fraction of the equity value multiplied by a published rate. Tessara may therefore value the losses differently from the standalone company. For a standalone

valuation, the losses are available without limitation, but tax counsel cautions that a large equity issuance could trigger the same limitation.

Exhibit G: Alternative interest. Redcliff Partners, a private equity sponsor, has told Kessler Dunmore informally that it would be prepared to discuss a transaction at "$62 to $64 per share, subject to diligence and financing," which the sponsor's partner described as a price that works if the company's non-GAAP operating margin reaches 30 percent within six years and the sponsor can finance the purchase with debt equal to six times next-twelve-month free cash flow. The sponsor has not produced a financing commitment, and the indication is not in writing.

Exhibit H: Ambiguous retention data. Cirrus defines net revenue retention on a trailing-twelve-month basis and includes price increases. About 2.4 percentage points of FY2026 NRR came from a list-price increase of 4 percent applied to about 60 percent of the base in January 2026. Management has not planned a second increase. The finance team also notes that the oldest "pre-FY2023" group contains 140 customers that Cirrus acquired through a small acquisition in fiscal 2022, and that these customers have been migrating to the main platform with temporary price protection that expires in 2027.

Exhibit I: The peer data. The peer multiples in Table 10 are as of 30 October 2026, but two of the six peers reported earnings after that date and moved by more than 8 percent. Bramble Analytics, in particular, has since announced a restructuring that would lift its FCF margin to 15 percent within a year, which, if true, would raise its rule of 40 from 43 to 53 without any change in its multiple. The committee has not decided whether to update the table.

Assignment Questions

Answer all ten questions. Where a question calls for a judgment, state your assumptions and say which exhibits you used. Unless told otherwise, use the data in Tables 1 through 12 as given, value the company as of 30 June 2026 with mid-year discounting for the forecast period, and ignore the passage of time between the balance sheet date and the offer date. State all dollar amounts in millions except per-share values.

1. **Cohort analysis.** Using Table 1, compute ending ARR, net new ARR, new-customer ARR, and net revenue retention for each of FY2023 to FY2026. Compute NRR for the FY2023 to FY2026 cohorts at each age, and for the pooled group. What does the pattern of retention across cohorts and ages tell you about the durability of the growth rate? Adjust FY2026 NRR for the January price increase described in Exhibit H.

2. **Unit economics.** Using Tables 2 and 3, estimate for FY2024, FY2025, and FY2026 the fully loaded cash CAC per dollar of new-customer ARR (excluding SBC and allocating 62 percent of sales and marketing to new customers), the payback period in months at a 78 percent subscription gross margin, and the sales efficiency ("magic number"). Compute lifetime value per dollar of landed ARR over ten years at a 10 percent discount rate, once using the cohort NRR curves and once using gross retention only, and compare each to CAC. Which view is more defensible?

3. **Rule of 40.** Compute the rule of 40 for FY2024 to FY2026 using revenue growth plus (a) free cash flow margin, (b) non-GAAP operating margin, (c) GAAP operating margin, and (d) free cash flow margin less SBC as a percentage of revenue. Rank Cirrus against the peers in Table 10 on definitions (a) and (d). Explain which definition best explains the peers' multiples and why.

4. **Stock-based compensation.** Quantify the economic cost of SBC in FY2026. Compute the annual increase in basic shares, the market value of shares issued at the undisturbed price, and free cash flow after the cost of neutralizing dilution. Compute free cash flow after the RSU tax withholding. Evaluate the finance team's "run-rate SBC" adjustment in Exhibit D, and the effect of Exhibit E on the quality of FY2026 free cash flow.

5. **Cost of capital.** Unlever the peer betas in Table 12, compute the average, relever at Cirrus's market capital structure at the undisturbed price (use the treasury stock method for dilution), and compute the cost of equity and the weighted average cost of capital. State the WACC you adopt, rounded to two decimals in percent.
6. **Forecast.** Extend Table 8 into a base case that differs from management's plan. Your base case must be reasoned from the cohort data (Question 1), the peer data (Table 10), and the exhibits. Present revenue, non-GAAP operating margin, SBC, GAAP operating income, cash taxes with the net operating losses, and free cash flow to the firm for FY2027 to FY2036, with SBC treated as an operating expense.
7. **Discounted cash flow.** Compute the enterprise value of your base case with a Gordon growth terminal value, using mid-year discounting for the explicit forecast. Bridge to equity value and to value per diluted share, applying the treasury stock method to the options, counting the RSUs in full, and treating the convertible notes correctly. Repeat for management's plan. Weight a downside, base, and management case 20, 55, and 25 percent and compute a probability-weighted value.
8. **Terminal value and SBC treatment.** State the share of enterprise value in the terminal value and the implied terminal multiples. Cross-check with an exit multiple. Construct sensitivity tables for WACC against terminal growth and for terminal margin against terminal growth. Recompute the base case with SBC added back instead of expensed and with no adjustment for dilution, and explain the difference.
9. **Multiples.** Apply the peer multiples in Table 10 in four ways: mean EV/NTM revenue, regression on growth, regression on the rule of 40 (free cash flow definition), and regression on the rule of 40 less SBC. Compute the regression fit for each. Apply the precedent multiples in Table 11. Build a football field of value per share and reconcile the market approach with the DCF.
10. **Recommendation.** Using a reverse DCF, determine what the undisturbed price and the offer each imply about the plan. Assess the offer against all your valuation evidence, the premium evidence in Table 11, and Exhibits A, B, and G. Recommend to the special committee whether to accept, reject, or counter, with a price and a walk-away level, and draft a board-style memorandum.

Teaching Note and Full Solutions

Learning Objectives

After working through this case, students should be able to:

1. Build and interpret a cohort table, separating growth from new customers and growth from the installed base, and detect deterioration that blended metrics hide.
2. Compute CAC, payback, magic number, and lifetime value, and explain why lifetime value built on net retention above 100 percent needs to be treated with suspicion.
3. Compute the rule of 40 on several definitions and explain why the definition that excludes SBC is the weakest guide to value.
4. Quantify the economic cost of stock-based compensation through dilution, cash withholding, and the difference between added-back and expensed discounted cash flow, and avoid double counting between cash flows and share counts.
5. Estimate a cost of capital from unlevered peer betas, handling a small convertible and the treasury stock method correctly.
6. Build a ten-year free cash flow to the firm forecast for a loss-making company with net operating losses and deferred-revenue working capital.
7. Value equity per diluted share when dilution depends on the value itself, and when a convertible is out of the money.

8. Explain what share of value sits in the terminal value, test it against an exit multiple, and run meaningful sensitivities.
9. Apply regression-based multiples with honest attention to sample size, and reconcile intrinsic and relative value in a football field.
10. Use a reverse DCF to translate a price into the plan it implies, and use that to frame a recommendation on a takeover offer.

Suggested Class Plan (90 minutes)

Minutes	Activity	Notes
0-10	Opening: cold call on the offer. “Would you sell at $66.50?” Record the vote	Revisit the vote at the end; most classes move
10-25	Business quality: Questions 1 to 3 on the board. Cohort table, unit economics, rule of 40 definitions	Draw the cohort triangle; push on first-year NRR falling from 120 to 113
25-40	Stock-based compensation: Question 4, then the add-back versus expense comparison from Question 8	The single largest swing in value; ask students to defend either side
40-60	Cost of capital and DCF mechanics: Questions 5 to 7. One group presents base case, one presents management case	Insist on the equity bridge with the treasury stock method
60-72	Terminal value, sensitivities, multiples: Questions 8 and 9 and the football field	Ask what the regression with six points can and cannot tell us
72-85	Recommendation: Question 10. Reverse DCF; counter-offer debate with a student playing Anselm Okafor	Test the walk-away price
85-90	Close: revisit the opening vote	Discuss what moved

For a 75-minute session, assign Questions 1 to 4 as pre-work and begin at the 40-minute mark. For a two-session version, devote the second session to Questions 5 to 10 with a spreadsheet built live.

Solutions

All solutions were recomputed in Python from the data tables. Tables show rounded values; intermediate calculations used full precision, so a student working with rounded intermediate values may differ by a few tenths of a million or a few cents per share.

Solution 1: Cohort analysis

Ending ARR is the column total of Table 1: $175.0 million (FY2022), $247.5 million (FY2023), $341.8 million (FY2024), $457.7 million (FY2025), and $589.3 million (FY2026). Net new ARR is the annual change, and new-customer ARR is the cohort’s first-year figure. Net revenue retention in a year equals ARR at year-end from customers who existed at the start of the year, divided by opening ARR, which is (ending ARR minus new-customer ARR) divided by opening ARR.

	FY2023	FY2024	FY2025	FY2026
Opening ARR	175.0	247.5	341.8	457.7
New-customer ARR	55.0	66.0	78.0	88.0
ARR from existing customers at year-end	192.5	275.8	379.7	501.3
Net expansion from base (existing less opening)	17.5	28.3	37.9	43.6
Ending ARR	247.5	341.8	457.7	589.3
Net new ARR	72.5	94.3	115.9	131.6
ARR growth	41.4%	38.1%	33.9%	28.8%
New-customer share of net new ARR	75.9%	70.0%	67.3%	66.9%
Net revenue retention	**110.0%**	**111.4%**	**111.1%**	**109.5%**

Worked example for FY2026: existing-customer ARR is 589.3 minus 88.0, or 501.3; dividing by 457.7 gives 1.0953, so NRR is 109.5 percent.

Retention by cohort and age, computed as each year’s ARR over the prior year’s ARR for the same cohort:

Cohort	First renewal (age 1)	Age 2	Age 3	Age 4
Pre-FY2023 (pooled)	110.0% (FY23)	109.0% (FY24)	108.0% (FY25)	107.0% (FY26)
FY2023	120.0%	115.0%	110.9%	
FY2024	117.0%	112.0%		
FY2025	113.0%			

Three patterns emerge. First, every cohort’s retention declines with age, as expected, so blended NRR depends on the age mix of the base. The base is aging, which pushes blended NRR down mechanically. Second, and more worrying, each new cohort retains less at the same age than its predecessor: age-one NRR fell from 120 percent to 117 percent to 113 percent. Age-two NRR fell from 115 percent to 112 percent. That is a genuine deterioration in the quality of customers or in the success of expansion selling, and it is the real story behind the slowing growth. Third, gross dollar retention fell from 93 to 90 percent, so churn is part of the story, and gross expansion (NRR minus GRR) in FY2026 was 19.5 points.

The January price increase contributed about 2.4 points of FY2026 NRR (Exhibit H), so the like-for-like figure is 109.5 minus 2.4, or 107.1 percent. That is a 4.0-point fall from 111.1 percent in FY2025, if FY2025 had no comparable increase, and 107.1 percent is the better base for forecasting because management has not planned a second increase.

The implication for the plan is large. Management’s FY2027 revenue of $710.6 million, with about $40.0 million of services, requires subscription revenue of $670.6 million and therefore ending ARR of 2 times 670.6 minus 589.3, or $751.9 million, which is 27.6 percent growth and net new ARR of $162.6 million, 24 percent more than the $131.6 million added in FY2026. If NRR holds at 107.1 percent, the installed base contributes $631.1 million, and new-customer ARR must reach $120.8 million, 37 percent above FY2026’s $88.0 million. If NRR is 109.5 percent, new-customer ARR must still rise 21 percent, to $106.6 million. Either way, the plan requires a reacceleration that the cohort data do not support, and that is consistent with Exhibit A’s soft first quarter.

Solution 2: Unit economics

Cash sales and marketing spend is GAAP sales and marketing less the SBC inside it (one-third of total SBC, per the finance team). New-customer spend is 62 percent of that. CAC per dollar of new-customer ARR is new-customer spend divided by new-customer ARR. Payback in months is CAC per dollar divided by the gross margin of 78 percent, times 12. The magic number is net new ARR divided by cash sales and marketing spend.

	FY2024	FY2025	FY2026
GAAP sales and marketing	141.8	182.9	229.4
Less SBC in sales and marketing	24.4	30.1	36.4
Cash sales and marketing	117.4	152.8	193.0
New-customer spend (62%)	72.8	94.7	119.7
New-customer ARR	66.0	78.0	88.0
New customers	420	455	500
CAC per $1 of new ARR	**1.10**	**1.21**	**1.36**
CAC per new customer ($000)	173.0	208.0	239.0
Payback (months)	**17.0**	**18.7**	**20.9**
Net new ARR	94.3	115.9	131.6
Magic number (cash S&M)	**0.80**	**0.76**	**0.68**
Magic number (GAAP S&M)	0.67	0.63	0.57

Worked example for FY2026: 229.4 minus 36.4 is 193.0; 62 percent of 193.0 is 119.7; 119.7 divided by 88.0 is 1.36; 1.36 divided by 0.78 is 1.744 years, or 20.9 months. CAC per dollar has risen 23 percent in two years while landing ARR per customer rose only 12 percent ($157,100 to $176,000). Acquisition is getting more expensive faster than deals are getting larger. The magic number has fallen from 0.80 to 0.68, below the level of about 0.75 that practitioners treat as a threshold for adding sales capacity. The trend in all three measures points the same way as the cohort data.

Lifetime value per dollar of landed ARR is the present value of gross profit over ten years, discounting mid-year at 10 percent, with ARR in year t equal to prior-year ARR times that year’s retention. For cohort curves, the observed NRR path is used and then 108 percent in later years. The gross-only view uses 90 percent retention every year.

Basis	LTV per $1 landed ARR	LTV / CAC at FY2026 CAC of $1.36
FY2023 cohort curve (120%, 115%, 111%, then 108%)	8.08	5.9x
FY2024 cohort curve (117%, 112%, then 108%)	7.58	5.6x
FY2025 cohort curve (113%, then 108%)	7.14	5.3x
FY2026 cohort (assume 110%, then 108%)	6.97	5.1x
Gross retention only (90% every year)	3.54	2.6x
Cohort curve net of expansion selling cost (FY2026)	5.64	4.1x

The cohort-curve LTV counts the revenue from upsell but not the cost of the selling effort that produced it. The cost of expansion can be estimated: 38 percent of cash sales and marketing (the share not allocated to new customers), or $73.3 million, produced gross expansion of $89.4 million in FY2026 (net expansion of $43.6 million plus churn of 10 percent of opening ARR, $45.8 million), or $0.82 per dollar of expansion ARR. Deducting the discounted expansion cost from the FY2026 cohort curve gives net LTV of $5.64 per dollar, or 4.1 times CAC. The gross-only measure of 2.6 times is a floor, since it assumes no expansion at all, which contradicts the data.

Which is more defensible? The loaded cohort view of roughly 4 times, with a floor of 2.6 times, is a good business by the usual 3-times rule of thumb. The conclusion that is not defensible is the 5 to 6 times headline that comes from applying the old cohorts' retention to today's much higher CAC. The correct statement is that Cirrus's unit economics are still healthy but are trending down on three independent measures.

Solution 3: Rule of 40

Revenue growth is 38.4 percent (FY2024), 35.0 percent (FY2025), and 30.0 percent (FY2026). FY2023 revenue was $230.3 million. Margins are on revenue.

	FY2024	FY2025	FY2026
Revenue growth	38.4%	35.0%	30.0%
(a) FCF margin	15.0%	21.0%	24.1%
Rule of 40, FCF	**53.4**	**56.0**	**54.1**
(b) Non-GAAP operating margin	9.0%	13.0%	16.5%
Rule of 40, non-GAAP operating	47.4	48.0	46.5
(c) GAAP operating margin	(14.0%)	(8.0%)	(3.0%)
Rule of 40, GAAP operating	24.4	27.0	27.0
SBC / revenue	23.0%	21.0%	19.5%
(d) Rule of 40, FCF less SBC	**30.4**	**35.0**	**34.6**

On the headline definition, Cirrus clears the bar easily, at 54.1 in FY2026, and the score has been flat for three years because falling growth has been offset by rising cash flow margin. On the GAAP operating definition it is at 27.0 and has not improved since FY2025. On the SBC-adjusted cash definition it is at 34.6, below 40. The same company is a star or a laggard depending on the definition, which is the lesson. Note also that the flat headline score hides the composition: the company is trading growth for margin at roughly one for one, which would be fine if margin came from scale, but part of the margin came from deferred revenue (Exhibit E) that will not repeat.

Peer ranking, using NTM figures (Cirrus 27 percent growth, 22 percent FCF margin, 18 percent SBC):

Rank	Definition (a): growth + FCF margin	Score	Definition (d): less SBC	Score
1	Halcyon Cloud	56	Quillon Systems	37
2	Quillon Systems	51	Halcyon Cloud	35
3	**Cirrus Ledger**	**49**	**Cirrus Ledger (tied)**	**31**
3=			Tollgate Software	31
4	Meridian Stack	47	Meridian Stack	25
5	Bramble Analytics, Tollgate Software	43	Verdant Ops	19
6	Verdant Ops	36	Bramble Analytics	16

Which definition explains multiples? A regression of EV/NTM revenue on each measure (Solution 9) gives an R-squared of 0.81 for definition (a) and only 0.19 for definition (d), and 0.52 for growth alone. In this small sample the market's multiples track the headline rule of 40 and ignore SBC. The market is, in effect, paying for SBC-heavy companies as if SBC were free. That is an observation about how peers are priced, not a statement that SBC is costless. A valuation anchored only on peer multiples therefore inherits the market's blind spot, which is the reason the DCF in Questions 6 to 8 treats SBC as an expense.

Solution 4: Stock-based compensation

Basic shares rose from 98.2 million to 101.4 million in FY2026, an increase of 3.2 million shares, or 3.3 percent (FY2025 saw 3.8 percent growth from 94.6 million). At the undisturbed price of $56.00, the shares issued are worth 3.2 times 56.00, or $179.2 million, which is 164 percent of the $109.1 million SBC expense. The difference arises because grants were priced at lower stock prices on their grant dates; the expense records the grant-date value, while the dilution is measured at today's price.

Measure	FY2026	% of revenue
Free cash flow (company definition)	134.9	24.1%
Less taxes paid on net share settlement	(24.5)	
FCF after RSU tax withholding	**110.4**	**19.7%**
FCF less SBC expense	25.8	4.6%
FCF less cost of buying back 3.2 million shares at $56.00	(44.3)	(7.9%)

Free cash flow after the cash cost of holding the share count flat is negative. Cirrus's cash generation is therefore financed by its employees and shareholders, not by customers alone: if the company bought back shares to prevent dilution, it would consume more than its entire free cash flow. This does not make the business uneconomic, because the shares issued are claims on a growing business, but it shows that "$134.9 million of free cash flow" overstates what accrues to a continuing shareholder.

Exhibit D, run-rate SBC. Removing $14.2 million of accelerated vesting and the $6.0 million retention grant gives $88.9 million, or 15.9 percent of revenue. The adjustment is defensible for the accelerated vesting, because it will not recur, but weaker for the retention grant, because retention grants are a normal cost of running a software business. Even accepting the whole adjustment, FCF less SBC rises only from $25.8 million to $46.0 million (8.2 percent of revenue), and the plan already assumes SBC falls to 18.0 percent of revenue in FY2027. The correct use of the adjustment is to avoid extrapolating FY2026's level, not to exclude the cost.

Exhibit E, billing terms. About $17 million of FY2026 free cash flow came from the one-time move to annual billing. Free cash flow on a repeatable basis is $117.9 million (21.1 percent of revenue), the headline rule of 40 falls to 51.1, and the SBC-adjusted rule of 40 to 31.6. The 3 percent discount also removes about $2.6 million a year of subscription revenue permanently. This one-time benefit is not wrong, since the cash is real, but it should not be capitalized at a multiple. It is a pull-forward of future cash.

Solution 5: Cost of capital

Unlevered beta equals levered beta divided by (1 + (1 minus 23%) times D/E). For Halcyon Cloud, 1.20 divided by (1 + 0.77 times 0.03) is 1.173. The six unlevered betas are 1.173, 0.989, 1.231, 1.350, 0.852, and 1.107, with an average of 1.117 (median 1.140).

Cirrus's diluted share count at $56.00 is 101.4 basic plus 6.2 RSUs plus 8.4 times (1 minus 21.50/56.00), or 5.174 million net option shares, a total of 112.774 million. Market capitalization is $6,315.4 million. Debt of $175.0 million gives D/E of 2.77 percent. Relevered beta is 1.117 times (1 + 0.77 times 0.0277), or 1.141. The cost of equity is 4.2% + 1.141 times 4.5%, or 9.33 percent. The after-tax cost of debt is 6.2% times 0.77, or 4.77 percent, and the weight of debt is 175.0 divided by (6,315.4 + 175.0), or 2.70 percent. WACC is 0.973 times 9.334% + 0.027 times 4.774%, which is 9.21 percent. The coupon of 0.5 percent is irrelevant to the cost of debt, since the note's low coupon is paid for by the conversion option. Using the coupon would understate WACC by roughly 0.1 point. Because the market price is itself an input, there is mild circularity, which is immaterial here. **Adopt 9.21 percent.**

Solution 6: Base-case forecast

The base case adjusts management's plan in four ways, each tied to evidence. Growth starts at 26.0 percent rather than 27.0 because the cohort data (Solution 1) show that FY2027 requires 21 to 37 percent more new-customer ARR and because the first quarter was soft, and it fades more quickly to 9.0 percent in FY2036 rather than 10.0 percent. Non-GAAP margin reaches 37.0 percent in FY2036, not 41.0 percent, because no peer combines more than 33 percent with growth above 10 percent (Exhibit C), and the sales-and-marketing efficiency decline argues against the plan's decline to 22 percent of revenue. SBC falls to 9.5 percent, not 9.0 percent. Capital expenditure, depreciation, working capital, and the tax assumptions are management's. The downside case, used in Solution 7, models loss of Ostrava, continued deterioration in new-customer cohorts, and growth fading to 6 percent.

Each year: revenue equals prior revenue times (1 + growth); GAAP EBIT equals revenue times (non-GAAP margin minus SBC percentage); cash tax is 23 percent of EBIT after using the NOL; FCFF equals EBIT minus tax, plus depreciation (3.0 percent), minus capex (3.5 percent), plus 15 percent of incremental revenue. For FY2027: revenue is 559.5 times 1.26, or 705.0; EBIT is 705.0 times (19.0% minus 18.0%), or 7.0; no tax because the NOL absorbs it (NOL falls from 260.0 to 253.0); depreciation is 21.2, capex is 24.7, and the working-capital inflow is 0.15 times (705.0 minus 559.5), or 21.8. FCFF is 7.0 + 21.2 - 24.7 + 21.8 = 25.3.

Table S6. Base case free cash flow to the firm ($ millions)

Year	Growth	Revenue	Non-GAAP margin	SBC %	GAAP EBIT	Cash tax	FCFF	Discount factor	PV of FCFF
FY2027	26.0%	705.0	19.0%	18.0%	7.0	0.0	25.3	1.0450	24.3
FY2028	23.5%	870.6	21.5%	16.5%	43.5	0.0	64.0	1.1413	56.1
FY2029	21.0%	1,053.5	24.5%	15.0%	100.1	0.0	122.2	1.2464	98.1
FY2030	19.0%	1,253.6	27.0%	14.0%	163.0	12.3	174.4	1.3612	128.1
FY2031	17.0%	1,466.7	29.5%	13.0%	242.0	55.7	211.0	1.4866	141.9
FY2032	15.0%	1,686.8	31.5%	12.0%	328.9	75.7	277.8	1.6235	171.1
FY2033	13.0%	1,906.0	33.0%	11.0%	419.3	96.4	346.2	1.7730	195.3
FY2034	11.5%	2,125.2	34.5%	10.5%	510.1	117.3	415.0	1.9363	214.3
FY2035	10.0%	2,337.8	36.0%	10.0%	607.8	139.8	488.2	2.1146	230.9
FY2036	9.0%	2,548.2	37.0%	9.5%	700.7	161.2	558.4	2.3094	241.8

The NOL is exhausted in FY2030 (cash tax of 12.3 against EBIT of 163.0). GAAP EBIT margin rises from 1.0 percent in FY2027 to 27.5 percent in FY2036. FCFF as a percentage of revenue rises from 3.6 percent to 21.9 percent. Management's case (Table S7) is built the same way with its own inputs and reaches revenue of $2,933.5 million and FCFF of $748.1 million in FY2036, compared with $2,548.2 million and $558.4 million in the base case.

Table S7. Management case free cash flow to the firm ($ millions)

Year	Growth	Revenue	Non-GAAP margin	SBC %	GAAP EBIT	Cash tax	FCFF	Discount factor	PV of FCFF
FY2027	27.0%	710.6	19.5%	18.0%	10.7	0.0	29.8	1.0450	28.5
FY2028	25.0%	888.2	22.5%	16.5%	53.3	0.0	75.5	1.1413	66.2
FY2029	23.0%	1,092.5	26.0%	15.0%	120.2	0.0	145.4	1.2464	116.6
FY2030	21.0%	1,321.9	29.0%	14.0%	198.3	28.2	197.9	1.3612	145.4
FY2031	19.0%	1,573.1	31.5%	12.5%	298.9	68.7	260.0	1.4866	174.9
FY2032	17.0%	1,840.5	34.0%	11.5%	414.1	95.2	349.8	1.6235	215.5
FY2033	15.0%	2,116.6	36.0%	10.5%	539.7	124.1	446.4	1.7730	251.8
FY2034	13.0%	2,391.7	38.0%	10.0%	669.7	154.0	545.0	1.9363	281.5
FY2035	11.5%	2,666.8	39.5%	9.5%	800.0	184.0	644.0	2.1146	304.5
FY2036	10.0%	2,933.5	41.0%	9.0%	938.7	215.9	748.1	2.3094	324.0

Solution 7: Discounted cash flow

Flows arrive mid-year, so the discount factor for FY2027 is 1.0921 raised to the power 0.5, which is 1.0450, and for FY2036 is 1.0921 raised to the power 9.5, which is 2.3094. The terminal value uses the Gordon formula on the FY2037 cash flow, computed as follows for the base case. FY2037 revenue is 2,548.2 times 1.035, or 2,637.3. EBIT at the FY2036 margin of 27.5 percent is 725.3, and NOPAT at a 23 percent tax rate is 558.5. Depreciation is 79.1, capex is set at 1.10 times depreciation (87.0), because a growing firm must invest slightly more than it depreciates, and working-capital inflow is 0.15 times 89.1, or 13.4. Terminal FCFF is 558.5 + 79.1 - 87.0 + 13.4 = 563.9. Terminal value is 563.9 divided by (9.21% minus 3.5%), which is 9,876.1. Because the Gordon formula values flows one period before the first flow, and the first terminal flow (FY2037) arrives at time 10.5, the terminal value sits at time 9.5 and is discounted by 2.3094, giving a present value of 4,276.5. The sum of present values of explicit flows is 1,501.9, so the enterprise value is 5,778.4.

The bridge to equity adds cash of 489.8 and subtracts the notes at face value of 175.0 (the notes are out of the money since the value per share is below $92.00): equity value is 5,778.4 + 489.8 - 175.0 = 6,093.2. Per-share value requires iteration because the option dilution depends on the price. At $54.08 the net option shares are 8.4 times (1 minus 21.50/54.08), or 5.061 million. Diluted shares are 101.4 + 6.2 + 5.061 = 112.661 million, and 6,093.2 divided by 112.661 is $54.08, which is the fixed point.

Table S9. Scenario valuation summary ($ millions except per share; shares in millions)

Case	Weight	FY2036 revenue	PV of FY27-36 FCFF	PV of terminal value	Enterprise value	Equity value	Diluted shares	Value per share
Downside	20%	1,706.4	684.3	1,832.5	2,516.8	2,831.6	109.05	$25.97
Base	55%	2,548.2	1,501.9	4,276.5	5,778.4	6,093.2	112.66	$54.08
Management	25%	2,933.5	1,908.7	5,721.0	7,629.7	7,944.5	113.42	$70.04

The probability-weighted value is 0.20 times $25.97 plus 0.55 times $54.08 plus 0.25 times $70.04, or **$52.45 per share**. The management case yields $70.04, which is 5.3 percent above the offer, and the base case is 18.7 percent below it. Note the asymmetry: the downside is $28 below base but the management upside is only $16 above it, so the probability-weighted value sits below the base.

Solution 8: Terminal value and SBC treatment

In the base case, terminal value is 74.0 percent of enterprise value (4,276.5 of 5,778.4); in the management case 75.0 percent, and in the downside 72.8 percent. The Gordon terminal value of 9,876.1 is 3.88 times FY2036 revenue (4.05 times when restated to the end of FY2036) and 17.5 times FY2037 FCFF, with a terminal FCFF margin of 21.4 percent. These multiples are well below the current 8.4 times NTM revenue because the business is assumed to be mature by then, with growth of 3.5 percent. The exit-multiple cross-check discounts a multiple of FY2036 revenue by ten full years:

Table S12. Exit-multiple cross-check, base case (multiple of FY2036 revenue, discounted ten years)

Exit EV / revenue	PV of terminal value	Enterprise value	Value per share	Terminal value share of EV
3.0x	3,167.5	4,669.4	$44.52	67.8%
4.0x	4,223.4	5,725.3	$53.63	73.8%
5.0x	5,279.2	6,781.1	$62.73	77.9%
6.0x	6,335.1	7,837.0	$71.83	80.8%

An exit multiple of 4.0 times gives $53.63, nearly identical to the Gordon result of $54.08 and implies terminal growth of 3.4 percent. A 5.0 times exit multiple would require 4.5 percent perpetual growth, above any plausible long-run nominal growth rate for the economy, so the market-style multiples of 5 to 6 times revenue embed assumptions about permanent growth that the Gordon model makes explicit.

Table S10. Base-case value per share: WACC (rows) against terminal growth (columns)

WACC	2.5%	3.0%	3.5%	4.0%	4.5%
8.25%	$57.13	$61.22	$66.16	$72.27	$80.00
8.75%	$52.06	$55.37	$59.31	$64.07	$69.96
9.21%	$48.08	$50.84	$54.08	$57.95	$62.64
9.75%	$44.08	$46.34	$48.96	$52.04	$55.70
10.25%	$40.89	$42.79	$44.97	$47.50	$50.47

Table S11. Base-case value per share: terminal non-GAAP margin shift (rows) against terminal growth (columns) at WACC 9.21%

Margin shift	2.5%	3.0%	3.5%	4.0%	4.5%
-6 pts	$41.37	$43.55	$46.12	$49.18	$52.89
-3 pts	$44.73	$47.20	$50.10	$53.56	$57.76
+0 pts	$48.08	$50.84	$54.08	$57.95	$62.64
+3 pts	$51.44	$54.49	$58.07	$62.34	$67.51
+6 pts	$54.79	$58.13	$62.05	$66.72	$72.39

Each 50 basis points of WACC moves value by about $5 to $6 per share, and each 50 basis points of terminal growth by about $3 to $4. A three-point change in terminal margin moves value by about $4. Only the combination of a WACC of about 8.25 percent with terminal growth of 3.5 percent or more, or a 4.5 percent terminal growth rate with a 8.75 percent WACC, brings the base case to the offer.

Adding back SBC. If SBC is added back to free cash flow in every year and the terminal value, and the share count is not adjusted for future issuance:

Table S5. Effect of treating SBC as an expense versus adding it back ($ millions except per share)

Case	EV, SBC expensed	Per share	EV, SBC added back	Per share	PV of SBC	PV of SBC as % of added-back EV
Base	5,778.4	$54.08	8,883.3	$80.85	3,104.9	35.0%
Management	7,629.7	$70.04	10,965.8	$98.69	3,336.1	30.4%

The base case jumps from $54.08 to $80.85 per share, an increase of $26.77 or 49.5 percent, and in the management case from $70.04 to $98.69. The present value of the SBC stream is $3,104.9 million in the base case: $1,204.8 million from the explicit period and $1,900.0 million from the terminal value. This is 35 percent of the added-back enterprise value. The add-back is wrong unless the share count rises to reflect the shares that will be issued. Employees will receive shares worth this much; those shares are a claim on the same cash flows. If they are neither deducted as a cost nor counted as future dilution, the value is overstated by exactly this amount. The expense approach is correct and self-contained, while the add-back approach requires a forecast of future share issuance and its price. Note that the offer of $66.50 sits between the expensed values and the added-back values, so a buyer who ignores SBC could justify more than $66.50, and the committee should know that.

Solution 9: Multiples

NTM revenue is $710.6 million (management plan; using the base case's $705.0 million changes values by under 1 percent). Regression of EV/NTM revenue on each measure across the six peers:

Method	Fit	R-squared	Cirrus input	Multiple	EV	Per share
Mean multiple	n/a	n/a	n/a	8.50x	6,039.8	$56.34
Median multiple	n/a	n/a	n/a	9.00x	6,395.1	$59.40
Regression on growth	4.00 + 0.170 x growth	0.52	27	8.58x	6,100.1	$56.86
Regression on Rule of 40 (FCF)	-3.53 + 0.262 x score	0.81	49	9.28x	6,597.2	$61.14
Regression on Rule of 40 less SBC	5.72 + 0.103 x score	0.19	31	8.89x	6,319.0	$58.74
Growth-adjusted (mean EV/revenue per point of growth, 0.336)	n/a	n/a	27	9.07x	6,446.1	$59.84
Precedent mean	n/a	n/a	n/a	9.23x	6,555.0	$60.78
Precedent low to high	n/a	n/a	n/a	8.1x to 10.4x	5,755.6 to 7,389.9	$53.89 to $67.98

Equity per share from EV follows the same bridge as Solution 7 (plus cash, less $175.0 million of notes, with iterated treasury-method dilution). With six data points, the regression's apparent precision is deceptive: the standard error on a slope estimated from six observations is large, and a single outlier (Bramble Analytics) moves the R-squared of the SBC-adjusted measure materially. The regressions are best read as saying that peer multiples cluster around 8.5 to 9.3 times for a company with Cirrus's profile. The premium evidence conflicts with the multiple evidence. Applying the mean take-over premium of 34.3 percent to $56.00 gives $75.18, while the mean precedent multiple gives $60.78. The two reconcile once it is recognized that Cirrus already trades at 8.4 times NTM revenue, inside the precedents' range of 8.1 to 10.4 times, so the "premium" the market already pays is high. Premium evidence should therefore be given less weight than multiple evidence here.

Football field (value per share):

Method	Low	High	Midpoint or point
DCF, SBC expensed: downside to management	$25.97	$70.04	Base $54.08; weighted $52.45
DCF base, WACC 8.75% to 9.75%, growth 3.5%	$48.96	$59.31	$54.08
DCF base, SBC added back (not recommended)			$80.85
Public comparables (mean, regressions)	$56.34	$61.14	$58.74
Precedent transactions	$53.89	$67.98	$60.78
Undisturbed price			$56.00
52-week trading range	$41.00	$63.00	
Tessara offer			**$66.50**

The reconciliation: the DCF base case ($54.08) sits about 8 percent below the comparables' midpoint of $58.74 because the market paid for SBC-heavy growth as if SBC were free and because it values the company's growth optimistically; the DCF's probability-weighted value is lower still. The market approach is a measure of what other investors pay today, not of intrinsic value, and the committee should treat it as an upper cross-check on the base case, not as independent confirmation of the offer.

Solution 10: Recommendation

Reverse DCF. The undisturbed price of $56.00 implies a plan that sits 13 percent of the way from the base case to the management case, or in other words, the market is pricing close to the base case with a small upside tilt. At $66.50, the offer implies a plan 79 percent of the way to management's. Holding the base case constant, the offer requires a WACC of 8.23 percent (versus 9.21), or perpetual growth of 4.85 percent (versus 3.5). Holding management's case constant, the offer implies a WACC of 9.48 percent: Tessara is paying roughly the value of a plan that has essentially no execution haircut, discounted at a slightly higher rate than ours.

Implied multiples at $66.50: equity value $7,533.4 million; enterprise value $7,218.6 million; 10.2 times NTM revenue, 12.9 times FY2026 revenue, and 12.3 times ending ARR. At $56.00 they are 8.4, 10.7, and 10.2 times. The offer is above every peer multiple in Table 10 except Halcyon Cloud at 11.2 times, and above all but the highest precedent (10.4 times). The 18.8 percent premium is below the 34.3 percent mean premium, but the multiple evidence, as noted, says the premium evidence overstates what is fair. Tessara's potential synergy is unquantified; Exhibits A and B argue against walking away from an offer that exceeds the base case by 23 percent, and Exhibit G shows the nearest alternative bid is $62 to $64 and unfinanced.

Recommendation: **counter at $72.00 and authorize acceptance at not less than $68.50.** At $72.00, EV is $7,856.6 million, 11.1 times NTM revenue, a 28.6 percent premium, and just above the management-case DCF value of $70.04, which asks Tessara to pay for the whole plan and to share the synergies. At $68.50, EV is $7,450.6 million, 10.5 times NTM revenue (just above the highest precedent of 10.4 times), and a 22.3 percent premium. That floor is 27 percent above the base-case value, 31 percent above the probability-weighted value, and above the sponsor indication by $4.50 to $6.50. Additional protections: a go-shop period, a reverse termination fee, no financing condition, and treatment of the converts at par plus make-whole.

Sensitivity Discussion

Four variables dominate the valuation. The first is the discount rate: 9.21 percent is an estimate, and a 0.5-point change moves the base case by $5 to $6. The second is the terminal growth rate and, jointly, the terminal margin. The sensitivity tables show that the full plausible range, from $41 to $80 on the base case, spans both the undisturbed price and the offer. The third is the treatment of SBC, which moves value by about $27 per share and is a choice rather than an estimate. The fourth is the plan path: management versus base is worth $16 per share. The marginal analysis suggests the most defensible negotiating position is to concede that the base case is below the offer and to argue that Tessara is purchasing optionality on the management plan and on its own synergies.

Common Student Errors

1. Computing NRR from blended totals and missing the cohort-by-cohort deterioration. Students who report “NRR of about 110 percent, stable” have not read the cohort triangle.
2. Computing LTV with the textbook formula, margin divided by churn, or with net retention above 100 percent for ten years, producing LTV/CAC above 5 without noting that expansion is not free.
3. Using the rule of 40 on one definition and calling Cirrus “best in class.”
4. Adding SBC back in the DCF while counting only today’s options and RSUs in the share count, so that all future issuance is free.
5. Counting SBC as an expense and also increasing the share count for future grants. This double counts. Only existing options and RSUs should be added to the share count.
6. Applying the treasury stock method at the offer price in one place and at the DCF value in another, or counting all 8.4 million options as shares.
7. Treating the convertible as converted at a price below $92.00.
8. Discounting the Gordon terminal value at the full ten years while using mid-year discounting for the cash flows, which understates value by about 4.5 percent.
9. Forgetting the NOL, or applying the tax rate to a negative EBIT.
10. Using the average of the football field as “the answer,” or applying the take-over premium to a price that already contains a premium multiple.

Alternative Defensible Answers

Several different recommendations can be defended. A student who accepts management’s plan, perhaps because of the March offsite model or because the upside is large, can argue for rejecting $66.50 and countering above $75, relying on the management DCF of $70.04, the 34 percent mean premium, and the holders who see “north of $70.” Anselm Okafor’s position is a variant of this. A student who stresses the first-quarter flash, Ostrava, and the cohort deterioration can reasonably recommend accepting the offer now, on the grounds that a certain $66.50 is well above any risk-adjusted standalone value and that Tessara might withdraw. A third answer is to run a short, limited market check with Redcliff and others before responding. The grade should reward the quality of the reasoning: whether SBC is treated consistently, whether the terminal value is tested, and whether the recommendation follows from the evidence rather than from the first number computed.

Board-Style Recommendation Memorandum

To: Special Committee of the Board of Directors, Cirrus Ledger, Inc. **From:** Kessler Dunmore & Co. **Date:** 19 November 2026
Re: Response to Tessara Systems’ proposal of $66.50 per share in cash

Recommendation. We recommend that the committee decline the proposal as submitted, counter at $72.00 per share in cash, and authorize management and advisers to accept a price of not less than $68.50 per share on customary terms. We recommend that the committee not enter exclusivity until the price is agreed.

Basis. Treating stock-based compensation as the operating cost it is, we value Cirrus at $54.08 per share in our base case, $70.04 in management's plan, and $25.97 in a downside case, a probability-weighted $52.45. Public comparables and precedent transactions suggest $56 to $61, with a precedent range of $54 to $68. The offer is 18.8 percent above the undisturbed price and 23 percent above our base case, and at 10.2 times next-twelve-month revenue it is above every precedent but one. It is nearly equal to the management plan's value, which requires reacceleration of growth that the cohort data do not support.

Risks to standalone value. New-customer cohorts retain less than earlier cohorts (first-year net retention 120, 117, 113 percent); acquisition cost has risen 23 percent in two years; first-quarter net new ARR fell about 6 percent; and the largest customer, 6.4 percent of ARR, is evaluating alternatives. Free cash flow of $134.9 million includes about $17 million of non-repeating billing-term benefit and costs $179 million in dilution at today's price.

Why counter. A price of $72.00 values the entire management plan at our cost of capital and allocates some of Tessara's synergy to our shareholders. The 34 percent mean premium of recent transactions is not a good guide here, because Cirrus already trades at a multiple within the precedent range. The floor of $68.50 equals 10.5 times revenue, above the highest precedent, and is 27 percent above our base case.

Process protections. We recommend a go-shop period of at least 30 days, a reverse termination fee, no financing condition, treatment of the convertible notes at par plus make-whole, and confirmation of the committee's authority over any change in management arrangements. We recommend that the committee also invite a written proposal from Redcliff Partners. Our fairness opinion will be delivered separately and will rest on the same analysis.

Chapter 6: Case: Ironvale Copper, a Mine Expansion under Commodity Risk

Ironvale Copper Corp., its Karsten mine and Ridge deposit, Ledgerline Capital, Kestrel Royalty Partners, the lenders, and every peer company in this chapter are fictional and illustrative. All numbers were constructed for teaching and are not statistics about any real company, mine, or market. The chapter draws on Volume 1, Chapter 6 (NPV), Chapter 7 (capital budgeting), Chapter 8 (uncertainty and real options), Chapter 11 (WACC), Chapter 14 (debt), Chapter 18 (options), Chapter 19 (derivatives), and Chapter 20 (risk management).

The Situation

On Thursday, 10 December 2026, the board of Ironvale Copper Corp. will be asked to approve a final investment decision (FID) on the largest project in the company's history: a $1.9 billion expansion of its Karsten copper mine into the adjacent Ridge deposit. The vote is five weeks away. On the board's table will be a funding plan that includes a $500 million equity raise, a project finance facility from three international banks, and a hedging program that the banks have made a condition of their loan. The banks' credit approvals lapse on 18 December, a fixed-price engineering contract offered by the main contractor lapses on 31 January 2027, and the equity market window the company's brokers favour is the second half of January.

Ironvale operates Karsten, an open-pit mine and concentrator at 3,400 metres in a remote highland district, which produced 76,400 tonnes of copper in 2026 at a C1 cash cost of $2.15 per pound. The company was assembled in 2011 from two bankrupt mines by Halvard Pryce, its chief executive, who took the listed shell public in 2015. Imogen Tarrant, the chief financial officer, joined in 2019 after fourteen years in project finance. Lucinda Okonjo, a former mining engineer and now a professional director, chairs the board. Rafael Quintero, a retired lender who sat on the credit committees of two of the banks now in the syndicate, sits on the audit committee, and he has said openly that he distrusts "any mine model whose price deck is above the forward curve."

Ridge is a separate pit on the same property. The company has defined 200 million tonnes of reserves at 0.54 percent copper, which can feed an incremental 14 million tonnes a year of throughput for fourteen years once a new concentrator train, a larger tailings facility, and a power line are built. First production would come in 2030. The technical work is mature, but the economics are not obviously compelling. Net present value is positive on the consensus price deck and roughly zero on the banks' more conservative deck, and it turns negative on the price that the futures market will pay today for copper delivered in 2030. A twenty percent overrun on capital cost would erase most of the value on the consensus deck.

You are a vice-president in corporate development, reporting to Imogen Tarrant. She has asked you for a board paper that does five things: value the Ridge project under several price decks and show what drives the answer; value Ironvale as a whole on a net asset value (NAV) basis and compare the result with the market price; evaluate the funding and hedging structure that the banks require; value the managerial flexibility that the project carries, including the option to delay, to stage the commitment, and to extend the life of the mine; and recommend whether the board should approve the project on 10 December, delay, or reject it.

Three directors have formed views. Halvard Pryce wants to proceed on the consensus deck and argues that "a mine is a long-dated bet on copper that we have been right about before." Rafael Quintero will argue that the project should be tested against the forward curve and the lenders' deck, not the consensus. Lucinda Okonjo is more concerned about execution: Ironvale has

never built anything of this scale, and its last concentrator upgrade overran by 14 percent. The paper must give each of them a number they can interrogate, and it must say honestly how much of the project's value is analysis and how much is opinion about copper.

Industry Context

Copper is a metal whose price is set on an international exchange by the balance between mine supply, scrap, and demand for wiring, pipes, motors, and the equipment of electricity grids. A copper mine is a price-taker. It sells a concentrate, a powder that is about one-quarter copper, to a smelter at a price that is a formula based on the exchange price less treatment and refining charges (TC/RCs), a fee that the smelter keeps. The miner's economics therefore depend on three prices: the metal price it cannot control, the TC/RC that it negotiates annually, and the costs of digging, hauling, and grinding rock.

The industry quotes costs per pound of payable copper. C1 is the cash cost of producing and delivering payable metal, including mining, milling, site administration, freight, and TC/RCs, net of by-product credits. All-in sustaining cost adds sustaining capital, royalties, and corporate costs. The cost curve is the ranking of all the world's mines by cost; mines at the high end of the curve set the price in weak markets because they close when the price falls below their cash cost. Copper mines are in general long-lived and slow to build, with seven to ten years from discovery to production commonly quoted, and the supply response to a price signal therefore comes late. That mismatch makes the price volatile, mean-reverting over long horizons, and difficult to forecast. Annualized price volatility around 20 to 25 percent is common in textbooks; the effect of mean reversion is that volatility over a ten-year horizon is much lower than the short-run figure would imply.

Mining companies value their assets with a net asset value model. The analyst forecasts production, prices, costs, and capital for each mine over its life, discounts the after-tax cash flows at a rate that reflects the risk of that asset, sums across assets, adds the value of exploration and resources, subtracts net debt and corporate overhead, and divides by shares. The ratio of the share price to NAV per share (P/NAV) is the central relative-value metric; it is usually below one for companies with higher risk, higher leverage, or poor delivery records, and above one for companies with growth and good jurisdictions. A NAV depends on the price deck. Analysts publish a "consensus" deck, typically a near-term forecast that converges to a long-term real price. Banks use a more conservative "lenders' deck" to size debt. Companies often present a "management" or "planning" deck, which tends to be the most optimistic. The futures market provides yet another view: the forward curve, which is the price at which anyone can lock in delivery today. For a long-lived asset, a forward curve is observable for only a few years, and the remainder must be extrapolated.

Hedging is controversial in mining. Many investors buy mining shares for their exposure to copper and regard company-level hedging as a waste of money, since investors can hedge themselves. Corporate-finance theory (Volume 1, Chapter 20) agrees that hedging cannot add value in a frictionless world, and gives three reasons it might add value in practice: it reduces expected costs of financial distress, it supports debt capacity, and it protects investment plans from being starved of cash when prices fall. Project lenders add a fourth, practical reason: they will not lend against a geared project with a single commodity exposure unless a portion of the cash flows is fixed during the years when the debt is largest. The hedging program in this case is therefore partly a financing necessity, and the company must decide the form of hedge, the volume, the floor, and the price at which the cost of protection is acceptable.

Finally, mines carry real options. A mine owner can delay a project until prices are more favourable, scale production up or down, suspend operations in a bad market, or extend the life by converting resources to reserves. Real-options theory (Volume 1, Chapter 8) values these rights using lattices or simulation, and the central finding is that flexibility is most valuable where uncertainty is high and the project's value is close to its cost. It also finds, uncomfortably, that the models often say "wait" even

for projects that managers should do now, because they leave out the costs of delay: the loss of contractor slots, the expiry of financing commitments, the escalation of capital costs, and the competitive reaction. The case asks students to value flexibility and then to challenge the number.

Capital cost overruns are the other persistent hazard. Studies of large resource projects commonly find that actual capital cost exceeds the approved estimate, and the pattern is stronger for projects in remote locations and for first-time builders. A company that is approving a project at a thin margin of safety against cost has little room for error, and the board's job is to separate what the project would be worth if built to budget from what it will be worth given the record of delivery.

Financial Data

Tables 1 and 2 give Ironvale's recent financial performance and balance sheet. Revenue is production multiplied by the average exchange price; TC/RCs, freight, and mining and milling costs are captured in C1 cost of sales. Hedging results come from the collars described below. The company's fiscal year is the calendar year, and 2026 includes about two months of forecast.

Table 1. Ironvale Copper: income statement summary ($ millions unless stated; December year-end; 2026 is nine months actual plus three months forecast)

	2024A	2025A	2026E
Copper sold (thousand tonnes)	71.2	74.1	76.4
Copper sold (million pounds)	157.0	163.4	168.4
Average LME price ($/lb)	4.02	4.10	4.22
C1 cash cost incl. TC/RC ($/lb)	2.05	2.10	2.15
Revenue	**631.0**	**669.8**	**710.8**
Realised hedging result	(3.4)	(4.8)	(6.0)
Cost of sales (C1 x pounds)	(321.8)	(343.1)	(362.1)
Royalty (2.5% of revenue)	(15.8)	(16.7)	(17.8)
General and administrative	(31.0)	(33.0)	(35.0)
EBITDA	**259.0**	**272.2**	**289.9**
Depreciation and amortisation	(118.0)	(124.0)	(129.0)
EBIT	**141.0**	**148.2**	**160.9**
Interest expense	(62.0)	(65.0)	(63.0)
Pre-tax income	79.0	83.2	97.9
Income tax (27%)	(21.3)	(22.5)	(26.4)
Net income	**57.7**	**60.7**	**71.5**
Capital expenditure (2026 includes $28.0m Ridge studies)	62.0	71.0	88.0

Table 2. Balance sheet extract at 31 December ($ millions)

	2025A	2026E
Cash	355.0	387.5
Receivables	84.0	91.5
Inventories	96.0	101.0
Property, plant and equipment, net	2,430.0	2,389.0
Other assets	60.0	60.0
Total assets	**3,025.0**	**3,029.0**
Accounts payable and accruals	118.0	124.5
Senior notes (6.25%, due 2031)	600.0	600.0
Bank term loan	420.0	380.0
Closure provision	145.0	151.0
Deferred tax liability	310.0	322.0
Other liabilities	25.0	25.0
Total liabilities	**1,618.0**	**1,602.5**
Shareholders’ equity	1,407.0	1,426.5
Total liabilities and equity	**3,025.0**	**3,029.0**

EBITDA margins have been stable at 41 percent as rising prices have been offset by cost inflation of about 2.5 percent a year. Net debt of $592.5 million is 2.04 times 2026 EBITDA. The notes carry a 6.25 percent coupon and mature in 2031. The bank term loan amortizes at $40.0 million a year. The company pays a dividend of $0.20 per share.

Table 3 sets out the three price decks and the market forward curve. All decks are expressed in real 2026 dollars and converted to nominal at 2.5 percent a year, the rate at which the project model escalates costs. The futures market is liquid for about five years. Beyond 2031, the finance team’s practice has been to extrapolate the forward curve at 2 percent a year, a number that approximates the cost of carry less a convenience yield.

Table 3. Copper price assumptions ($ per pound; spot at 30 October 2026 was $4.38; cost and price inflation 2.5% a year)

Deck	2027	2029	2030	2031	2035	2040
Market forward curve (nominal, liquid to 2031)	4.30	4.34	4.36	4.38	n/a	n/a
Consensus deck, long-term $4.25 real (nominal)	4.36	4.58	4.69	4.81	5.31	6.01
Lenders’ deck, $3.85 real (nominal)	3.95	4.15	4.25	4.36	4.81	5.44
Management deck, $4.60 real (nominal)	4.71	4.95	5.08	5.20	5.74	6.50

Table 4 lists the Ridge project parameters that the engineering and finance teams agreed. All cash flows are nominal, the valuation date is 1 January 2027, and flows are discounted mid-year. Table 5 shows how the company proposes to fund the project and Table 8 the lenders’ indicative terms. Dividends would be suspended from 2027 through 2029.

Table 4. Ridge expansion: technical and economic parameters

Item	Assumption
Incremental ore processed	14.0 million tonnes a year at full rate (200 Mt of Ridge reserves)
Head grade / recovery	0.54% copper / 87%
Payable copper at full rate	65.8 thousand tonnes a year (145.0 million pounds)
Production schedule	2030 at 70%; 2031 to 2043 at 100%; 14 years
Mining, processing and site G&A	$13.80 per tonne ore (2026 dollars), 2030 at 85% of full-rate cost
TC/RC and freight	$0.38 per pound (2026 dollars)
Royalty	2.5% of revenue
Initial capital cost (nominal, incl. 10% contingency)	$1,900.0 million: 2027 $665.0m; 2028 $760.0m; 2029 $475.0m
Sustaining capital	$30 million a year (2026 dollars); $15 million in 2030
Working capital	8% of revenue
Closure cost	$140.0 million nominal: 70% in 2044, 30% in 2045
Tax	27%, group-level; depreciation straight line over 14 years from 2030; losses shelter Karsten income
Valuation date and convention	1 January 2027; mid-year discounting; nominal cash flows

Table 5. Proposed funding of the Ridge expansion ($ millions)

Item	Amount
Ridge initial capital	1,900.0
Interest during construction (capitalised)	102.7
Financing fees	28.0
Debt service reserve account	60.0
Total uses	**2,090.7**
Project finance facility (principal $850.0m plus capitalised interest)	952.7
Equity raise at $6.30 per share (79.4 million new shares)	500.0
Karsten free cash flow retained after interest, dividends and amortisation, 2027-2029	405.0
Drawdown of corporate cash (leaving $154.5 million)	233.0
Total sources	**2,090.7**

The Karsten assumptions in Table 10 are used for the NAV of the existing operation. Table 6 compares Ironvale with six listed peers on NAV multiples. Table 7 shows the existing hedge book, which is small. Table 9 gives market data and the inputs for cost of capital.

Table 10. Karsten (existing operation) life-of-mine assumptions for the NAV

Item	Assumption
Remaining life and production	20 years (2027 to 2046) at 78.0 thousand tonnes (171.96 million pounds) a year
C1 cash cost including TC/RC	$2.15 per pound (2026 dollars), escalating at 2.5%
Royalty	2.5% of revenue
Sustaining capital	$55.0 million a year (2026 dollars)
Tax depreciation	Remaining tax base $620.0 million, straight line over ten years
Closure cost	$95.0 million (2026 dollars), paid in 2046
Corporate overhead	$36.0 million a year (2026 dollars), after-tax value to be deducted in NAV
Resources and exploration	Valued at $110.0 million (in-situ resource multiple, peers)
Discount rate	Same as Ridge (to be computed)

Table 6. Listed copper producers (fictional, illustrative; consensus decks; NAV at 8% real-equivalent nominal rates)

Company	P/NAV	EV/EBITDA (2026E)	C1 cost ($/lb)	Production (kt Cu)	Net debt/EBITDA	Production growth CAGR to 2030
Halden Copper	1.10x	7.8x	1.95	210	1.2x	12%
Marrow Ridge Mining	0.88x	6.9x	2.40	150	1.9x	5%
Tamsin Resources	1.02x	8.1x	2.05	320	0.8x	9%
Westerly Metals	0.95x	7.2x	2.25	180	1.5x	15%
Brackenfell Mining	0.81x	6.1x	2.60	95	2.4x	0%
Lorne Valley Copper	1.14x	8.9x	1.80	260	0.5x	20%
Mean	0.98x	7.5x	2.18	203	1.4x	10%
Median	0.99x	7.5x	2.15	195	1.4x	10.5%
Ironvale at $6.55 (P/NAV on base case)	0.97x	8.0x	2.15	76	2.0x	0% (before Ridge)

Peer NAV multiples are computed by each company’s own advisers using consensus decks and their own discount rates, so differences in P/NAV partly reflect differences in method. The relationship between P/NAV and the other columns is loose: the highest-growth peer (Lorne Valley Copper) trades at the highest multiple, and the most leveraged, highest-cost peer (Brackenfell Mining) trades at the lowest.

Table 7. Existing Karsten hedge book (zero-cost collars; Ironvale buys puts and sells calls)

Delivery year	Volume (thousand tonnes)	Floor ($/lb)	Cap ($/lb)
2027	30.0	3.80	4.75
2028	20.0	3.80	4.80

Table 8. Project finance term sheet (indicative, from the lead arrangers’ letter of 27 October 2026)

Term	Indication
Facility	$850.0 million senior term loan plus capitalised interest, drawn pro rata 2027-2029
Pricing	Term SOFR (assumed 4.0%) plus 3.25%, 7.25% all-in
Repayment	Interest only in 2030; 12 equal principal instalments 2031 to 2042
Sizing test	Minimum DSCR of 1.35x on the lenders' deck ($3.85 real)
Lock-up / default DSCR	1.20x / 1.10x, tested annually on actual cash flow
Hedging condition precedent	At least 50% of Ridge production for 2030-2033 hedged before first drawdown, at price floors not below $3.80 nominal
Completion	Cost overrun facility of 10% of capital cost funded by sponsor equity commitment
Hedge options considered	(A) none; (B) zero-cost collars, floor $3.80; (C) fixed-price swaps at forward less a 1% bank margin

Table 9. Market and cost-of-capital data

Item	Value
Diluted shares outstanding (millions)	262.0
Share price (close 30 October 2026)	$6.55
Market capitalisation	$1,716.1 million
Net debt at 31 December 2026E	$592.5 million (debt $980.0m less cash $387.5m)
Enterprise value	$2,308.6 million
Dividend	$0.20 per share ($52.0 million a year)
Risk-free rate (10-year)	4.3%
Equity risk premium	5.0%
Peer-median asset beta	0.92
Target debt / total capital	35%
Pre-tax cost of debt (project finance)	7.25%
Corporate tax rate	27%
Copper price volatility (short-run, annual)	22% (mean-reverting; long-run effective about 18%)
Risk-free rate for derivatives pricing	4.0%

Complications and Exhibits

Exhibit A: Contractor offer and cost history. The main contractor's fixed-price offer covers the process plant and tailings facility, which is 62 percent of the $1,900.0 million, with the balance (mining fleet, power line, owner's costs, and a 10 percent contingency allowance of about $173 million) on a cost-reimbursable basis. The engineering estimate is described as "plus or minus 15 percent." Ironvale's own last major capital project, the 2021 concentrator debottlenecking, overran by 14 percent and finished seven months late. The fixed-price offer lapses on 31 January 2027, and the contractor has said that a new offer would be priced "at then-current market rates, which we expect to be higher."

Exhibit B: Stream offer. Kestrel Royalty Partners has offered $400.0 million upfront, paid in the same 35/40/25 proportions as capital spending, in exchange for 15 percent of Ridge's payable copper for the life of the mine. Kestrel would pay Ironvale 20 percent of the spot price for each pound delivered. Kestrel's proposal would replace $400 million of the $500 million equity

raise. Management describes it as “non-dilutive.” The tax advisers believe the forgone revenue would be deductible when delivered.

Exhibit C: Grade reconciliation. The Ridge reserve is 0.54 percent copper. A 12-hole infill program completed in October returned an average of 0.49 percent in the northeast corner of the pit, which represents about 15 percent of the first five years of ore. At Karsten, the actual grade has averaged 4 percent below the reserve model over the past six years, and management attributes this to dilution that the model does not capture. The reserve estimate has not been revised. A further 120 million tonnes of Ridge material at a similar grade is classified as resource, not reserve, because drilling is too wide-spaced to support a mine plan.

Exhibit D: Fiscal risk. The government has announced a consultation on a “windfall surcharge” of 20 percent of revenue from price above $4.50 per pound. The consultation paper does not say whether the threshold would be indexed to inflation. Legal advisers say that stabilization terms in Ironvale’s mining agreement protect it only against increases in the general royalty, not against new instruments. The proposal is not law and may never be.

Exhibit E: Lenders’ deck review. The banks revise their deck semi-annually. Their next review is in May 2027. One bank’s research team has published a long-term real price of $3.60, and the lead arranger has told the company that “our credit committee will not be surprised if the deck moves toward the lower end of the range.” The loan documents allow the banks to size the facility to the revised deck only before first drawdown.

Exhibit F: Permits. The water-use permit for the new tailings facility is expected in the second quarter of 2027. The board is being asked to approve FID before it is granted; the facility agreement makes the permit a condition precedent to drawdown. Management says that the probability of refusal is “very low” but that a six-month delay is possible.

Exhibit G: Equity placement. Brokers have indicated that a $500 million placement, 29 percent of the existing share count, would likely price at a discount of 3 to 8 percent to the prevailing price. The case data use $6.30, a 3.8 percent discount to $6.55. One broker added that the share price might rise on announcement of the project, or fall because of the dilution, and said it was “a coin toss.”

Exhibit H: Management incentives. Short-term bonuses at Ironvale are tied to production growth and to a project milestone in 2027 (“FID and first drawdown”), and long-term awards vest on production in 2030. No part of the incentive is tied to returns on capital or to NAV per share. Rafael Quintero has asked for the incentive terms to be tabled with the investment paper.

Exhibit I: Hedging history. Ironvale’s board has historically hedged only small amounts. In 2024 to 2026 the realised hedging result was a loss every year (Table 1), because the collars’ caps were below the market price. Some shareholders have written that they “do not want a gold-plated hedge program.” The lenders’ letter, however, treats the hedging condition as non-negotiable.

Exhibit J: Forward-curve and deck mismatch. The consensus deck of $4.25 in real terms is $4.70 per pound in nominal terms in 2030, compared with a forward price of $4.36 for the same year. The finance team notes that analysts’ long-term decks are conventionally flat in real terms while futures curves are conventionally flat to modestly sloped in nominal terms, and that neither is “right.”

Assignment Questions

Answer all ten questions. State your assumptions. Unless told otherwise, value Ridge as of 1 January 2027 with nominal cash flows discounted mid-year at the WACC you compute in Question 3, and use Tables 1 through 10 as given. Money is in US dollars, in millions unless stated.

1. **Position and performance.** Using Tables 1 and 2, compute for 2024 to 2026 the EBITDA margin, EBITDA per pound, net debt to EBITDA, interest cover, and free cash flow after interest and tax (use EBITDA less cash tax, interest, and capital expenditure, treating the working-capital and non-cash movements as given). Where does Ironvale sit against the peers in Table 6, and what does that imply for its capacity to fund Ridge?
2. **Price decks and the forward curve.** Convert each deck in Table 3 to nominal prices for every year from 2027 to 2045. Extend the forward curve as described. Compute the real-2026-dollar equivalent of the forward-based price in 2031 and 2040. State why the decks differ and which one a board should use to approve the project.
3. **Cost of capital.** Relever the peer asset beta to the target capital structure, compute the cost of equity, the after-tax cost of debt, and the WACC. State and defend the rate you use for Ridge and for Karsten, and explain why a higher rate could be used for Ridge during construction.
4. **Project valuation.** Build the Ridge cash flow model for the consensus deck from the Table 4 parameters. Present revenue, operating costs, EBITDA, tax, capital, working capital, and free cash flow for each year, and compute the NPV, IRR, profitability index, and payback.
5. **Scenarios and sensitivities.** Compute NPV and IRR under the stress, lenders', market-forward, consensus, management, and upside prices. Compute the break-even real price and the break-even capital cost. Build a table of NPV against capital overrun and price, and a one-at-a-time sensitivity (tornado). Quantify the effect of a one-year delay in first production (assume $60 million of additional owner's costs) and of the windfall surcharge in Exhibit D under both readings of the threshold.
6. **Net asset value.** Value Karsten at the three decks, subtract corporate overhead and net debt, add resources and the hedge book, and compute NAV per share, P/NAV at the current price, and the deck implied by the market price. Add Ridge and the equity raise and compute pro forma NAV per share and the accretion. Explain why the equity raise appears accretive on the lenders' deck and what that tells you.
7. **Financing.** Evaluate the funding plan in Table 5. Compute the number of shares issued and the dilution. Compare the stream in Exhibit B with equity and with the project finance facility by computing the stream's internal rate of return to Ironvale at the three decks. Recommend a financing mix.
8. **Hedging.** Price zero-cost collars with a $3.80 floor for 2027 to 2033 using Black's model for forwards. Compute the mark-to-market of the existing hedge book. Simulate the project's debt service cover for 2030 to 2033 under the lenders' deck and the consensus deck with no hedge, collars, and swaps on 50 percent of volumes. Estimate the expected cost of each hedge if the consensus deck is correct. State whether hedging creates value here and recommend a program.
9. **Optionality.** Value, using a binomial lattice, the option to delay FID by one, two, and three years. Value a staged FID with $180 million of early works. Value the option to extend the mine life by converting the 120 million tonne resource. Discuss the suspension option. State what costs of delay the lattice omits and whether you would act on the result.
10. **Recommendation.** Recommend whether the board should approve FID on 10 December, delay it, or reject it. State the conditions you would attach, the walk-away points, and the monitoring metrics. Draft a board-style memorandum of about 500 words.

Teaching Note and Full Solutions

Learning Objectives

After working through this case, students should be able to:

1. Assess a mining company's cost position, leverage, and capacity to fund a transformational project.
2. Distinguish price decks from forward curves, convert real decks to nominal prices, and explain why the choice of deck can change the sign of a project's NPV.
3. Build a nominal after-tax project model for a mine, with ramp-up, escalation, working capital, depreciation, closure, and a mid-year discounting convention.
4. Run scenario, break-even, grid, and tornado analyses, and identify which risks matter: price, grade, capital cost, or timing.
5. Build a net asset value for a producer, interpret P/NAV, and back out the price deck implied by the market.
6. Compare equity, project debt, and a stream as funding sources, and recognise that "non-dilutive" is not "free."
7. Price a zero-cost collar with Black's model, simulate debt service cover with and without hedging, and judge whether hedging creates value or only manages financing risk.
8. Value real options with a binomial lattice, interpret the results with humility about omitted costs of delay, and value a staged commitment.
9. Weigh quantitative output against execution risk, and write a conditional recommendation.

Suggested Class Plan (90 minutes)

Minutes	Activity	Notes
0-10	Cold open: poll the class. "Approve, delay, or reject?" Record the votes	Revisit at the end
10-20	Position and decks: Questions 1 and 2. Draw the nominal deck against the forward curve on the board	The gap between deck and forward is the hidden story
20-40	Project model and scenarios: Questions 3 to 5. Groups compute NPV on one deck each	Insist on mid-year discounting and the closure and working-capital release
40-55	NAV and financing: Questions 6 and 7	Why does the raise look accretive on the low deck?
55-70	Hedging: Question 8. Walk through a collar price and the DSCR table	Hedging is a financing tool here, not a source of value
70-82	Optionality: Question 9; ask students to attack the lattice	What is missing from the lattice?
82-90	Recommendation and revisit the poll: Question 10	Where did the class end up?

For a 75-minute version, assign Questions 1 to 3 as pre-work and give the Question 4 model to students as a template. A Python or Excel model of the Ridge cash flows should be available to students in the second hour.

Solutions

All solutions were recomputed in Python from the data tables. Rounding in the tables is to one decimal. A student's Excel model with rounded intermediate values may differ by a few tenths of a million dollars.

Solution 1: Position and performance

	2024A	2025A	2026E
EBITDA margin	41.0%	40.6%	40.8%
EBITDA per pound sold ($)	1.65	1.67	1.72
Net debt ($ millions)	n/a	665.0	592.5
Net debt / EBITDA	n/a	2.44x	2.04x
EBITDA / interest	4.18x	4.19x	4.60x
EBIT / interest	2.27x	2.28x	2.55x
Free cash flow after interest and tax	113.7	113.7	112.5

Worked example, 2026: EBITDA 289.9 divided by 168.4 million pounds is $1.72 per pound; net debt is 380.0 + 600.0 - 387.5 = 592.5, and 592.5/289.9 is 2.04 times; free cash flow is 289.9 - 26.4 (tax) - 63.0 (interest) - 88.0 (capex) = 112.5. Margins and per-pound EBITDA have been flat to slightly rising as higher prices have offset cost inflation. Free cash flow has been flat at about $113 million because capital expenditure has risen from $62 to $88 million, of which $28 million in 2026 is Ridge study cost.

Against the peers, Ironvale's C1 cost of $2.15 equals the peer median, its production of 76 thousand tonnes is far below the median of 195, it is more leveraged than the median (2.0 versus 1.4 times), it trades at a slightly higher EV/EBITDA (8.0 versus 7.5 times), and it has no growth in the pipeline. The central observation concerns scale. Ridge's $1,900 million is 82 percent of the company's enterprise value of $2,309 million, 17 times its annual free cash flow, and 3.2 times its market capitalisation of $1,716 million. Internal cash flow cannot fund it. Ironvale is a mid-sized, mid-cost producer undertaking a project that more than doubles its size. The same facts imply that Ridge matters to the share price in both directions, which is why the funding, hedging, and cost-overrun risks deserve more attention than the headline NPV.

Solution 2: Price decks and the forward curve

The nominal price for year t equals the real price times 1.025 raised to the power (t - 2026). For the consensus deck, 2030 is 4.25 times 1.025^4 = 4.69; 2040 is 4.25 times 1.025^14 = 6.00; 2045 is 6.79. For the lenders' deck, $3.85 real is $4.25 nominal in 2030 and $5.44 in 2040. The forward curve is 4.30, 4.32, 4.34, 4.36, and 4.38 for 2027 to 2031, then rises at 2 percent a year: 4.47 (2032), 4.56, 4.65, 4.74, 4.84, 4.93, 5.03, 5.13, and 5.23 (2040), reaching 5.78 in 2045. The real-2026 equivalent of the 2031 forward price is 4.38/1.025^5 = $3.87, and of the extrapolated 2040 price is 5.23/1.025^14 = $3.70.

Three conclusions follow. First, the lenders' deck is close to the forward curve in real terms ($3.85 versus $3.87 in 2031), so the lenders are, in effect, pricing at the market. Second, the consensus deck is 10.4 percent above the lenders' deck and about 10 percent above the forward curve in 2031, and the gap is larger (about 15 percent) by 2040, because the deck is flat in real terms while the extrapolated curve falls in real terms. Third, the decks differ because they answer different questions: the forward curve is a price at which cash can be locked in, an analyst's deck is an expectation about the price in a mean-reverting market that will, on average, eventually reflect the cost of marginal supply, and a lender's deck is a conservative case chosen to size debt. A board approving a $1.9 billion project should not rely on any one of them. A reasonable standard is that the project should be approved only if it earns its cost of capital on the market (forward) deck, or if the board is prepared to explain why it believes it knows the price of copper better than the market does. Rafael Quintero's position is therefore defensible: the case for approval should be tested at the forward-based NPV and at the lenders' deck. Management's position is defensible if the board accepts that approving a mine is a bet on mean reversion.

Solution 3: Cost of capital

The target debt to capital is 35 percent, so D/E is 0.35/0.65 = 0.5385. Relevered equity beta is 0.92 times (1 + (1 - 0.27) times 0.5385) = 0.92 times 1.3931 = 1.282. Cost of equity is 4.3% + 1.282 times 5.0% = 10.71 percent. The after-tax cost of debt is 7.25% times 0.73 = 5.29 percent. WACC is 0.65 times 10.71% + 0.35 times 5.29% = 6.96% + 1.85% = **8.81 percent, rounded to 8.8 percent**.

The same rate is used for Ridge and Karsten. Karsten is a producing mine in the same jurisdiction, so no operating-risk premium is warranted, and it is a good practice for a NAV to apply one rate consistently so that differences in value reflect cash flows. For Ridge, a premium of 0.5 to 1.0 point during construction would be defensible, because completion, ramp-up, and cost-overrun risks are not diversifiable on a single-mine, single-commodity project, and 9.8 percent produces an NPV of $167.0 million (Solution 5), still positive. Using the cost of the financing package (the 7.25 percent facility rate) would be wrong: debt is not a cheaper source of capital in itself, because it raises the cost of equity. Using the rate on the stream as the discount rate would equally be wrong.

Solution 4: Project valuation

For the consensus deck the annual calculation is as follows, shown for 2031. Price is 4.25 times 1.025^5 = $4.81. Copper sold is 65.8 kt times 2.20462 = 145.0 million pounds. Revenue is 145.0 times 4.81 = 697.2. Mining and processing costs are 13.80 times 14.0 million tonnes times 1.025^5 = 218.6; TC/RC and freight are 0.38 times 145.0 times 1.1314 = 62.3; the royalty is 2.5 percent of revenue, or 17.4. EBITDA is 697.2 - 218.6 - 62.3 - 17.4 = 398.9. Depreciation is 1,900.0/14 = 135.7, sustaining capital is 30 times 1.1314 = 33.9, and taxable income is 398.9 - 135.7 - 33.9 = 229.3, so tax is 27 percent, or 61.9. The working-capital increase is 8 percent times (697.2 - 476.2) = 17.7. Free cash flow is 398.9 - 61.9 - 33.9 - 17.7 = 285.4. The discount factor for 2031 (4.5 years) is 1.088^4.5 = 1.4616, which gives a present value of 195.2.

Table S1. Ridge base case: consensus deck ($4.25 real), nominal $ millions, WACC 8.8%

Year	Price ($/lb)	Copper sold (Mlb)	Revenue	Opex, TC/RC, royalty	EBITDA	Tax	Capex, sustaining, closure	Working capital	FCF	Discount factor	PV
2027	4.36	0.0	0.0	-0.0	0.0	-0.0	-665.0	-0.0	-665.0	1.0431	-637.5
2028	4.47	0.0	0.0	-0.0	0.0	-0.0	-760.0	-0.0	-760.0	1.1349	-669.7
2029	4.58	0.0	0.0	-0.0	0.0	-0.0	-475.0	-0.0	-475.0	1.2347	-384.7
2030	4.69	101.5	476.2	-235.7	240.4	-23.8	-16.6	-38.1	162.0	1.3434	120.6
2031	4.81	145.0	697.2	-298.4	398.9	-61.9	-33.9	-17.7	285.4	1.4616	195.2
2032	4.93	145.0	714.7	-305.8	408.9	-64.4	-34.8	-1.4	308.3	1.5902	193.9
2033	5.05	145.0	732.5	-313.5	419.1	-66.9	-35.7	-1.4	315.1	1.7302	182.1
2034	5.18	145.0	750.9	-321.3	429.6	-69.5	-36.6	-1.5	322.1	1.8824	171.1
2035	5.31	145.0	769.6	-329.3	440.3	-72.1	-37.5	-1.5	329.2	2.0481	160.7
2036	5.44	145.0	788.9	-337.6	451.3	-74.8	-38.4	-1.5	336.5	2.2283	151.0
2037	5.58	145.0	808.6	-346.0	462.6	-77.6	-39.4	-1.6	344.0	2.4244	141.9
2038	5.72	145.0	828.8	-354.7	474.1	-80.5	-40.3	-1.6	351.7	2.6377	133.3
2039	5.86	145.0	849.5	-363.5	486.0	-83.4	-41.4	-1.7	359.6	2.8699	125.3
2040	6.01	145.0	870.8	-372.6	498.1	-86.4	-42.4	-1.7	367.6	3.1224	117.7
2041	6.16	145.0	892.5	-381.9	510.6	-89.5	-43.4	-1.7	375.9	3.3972	110.7
2042	6.31	145.0	914.8	-391.5	523.4	-92.6	-44.5	-1.8	384.4	3.6961	104.0
2043	6.47	145.0	937.7	-401.3	536.4	-95.9	-45.6	-1.8	393.1	4.0214	97.8
2044	6.63	0.0	0.0	-0.0	0.0	26.5	-98.0	75.0	3.5	4.3753	0.8
2045	6.79	0.0	0.0	-0.0	0.0	11.3	-42.0	0.0	-30.7	4.7603	-6.4
NPV											**307.8**

The NPV is **$307.8 million**. The IRR is 11.17 percent against a WACC of 8.8 percent. The profitability index, the ratio of the present value of inflows after the start of construction to the present value of the capital cost, is (307.8 + 1,691.9)/1,691.9 = 1.18. Cumulative undiscounted cash flow turns positive in 2036, seven years after first production. The project earns a modest spread over its cost of capital. Note three modelling points. Tax is computed at the group level, so depreciation shelters Karsten income immediately; a stand-alone calculation with loss carry-forwards would lower NPV. The 2044 closure cost is deducted for tax, and the release of working capital in 2044 offsets part of it. Revenue in the ramp year, 2030, is $476.2 million, and the 15 percent reduction of operating cost in the ramp year is a simplification, since in practice costs are largely fixed in the first year.

Solution 5: Scenarios and sensitivities

Table S2. Ridge scenario results (WACC 8.8%; capital cost $1,900.0m)

Scenario	NPV	IRR	Profitability index	Cumulative cash payback year
Stress, $3.50 flat real	-289.5	6.30%	0.83	2038
Lenders' deck, $3.85 real	-10.8	8.71%	0.99	2037
Market forward to 2031, then +2% a year	-60.1	8.30%	0.96	2037
Consensus deck, $4.25 real	307.8	11.17%	1.18	2036
Management deck, $4.60 real	586.5	13.11%	1.35	2035
Upside, $5.00 real	905.1	15.16%	1.53	2035

The price at which the NPV is zero is **$3.86 per pound in real terms** (nominal $3.96 in 2027, rising 2.5 percent a year), which is 9.1 percent below the consensus deck and almost exactly the lenders' deck. The capital cost at which the NPV is zero is 1.211 times the estimate, or $2,301 million, a 21 percent overrun. Break-even operating cost is $17.61 per tonne, 28 percent above the estimate.

Table S3. Ridge NPV ($ millions): capital cost overrun (rows) against real copper price (columns)

Capex overrun	$3.50	$3.85	$4.25	$4.60	$5.00
+0% ($1,900m)	-289.5	-10.8	307.8	586.5	905.1
+10% ($2,090m)	-435.3	-156.6	161.9	440.7	759.2
+20% ($2,280m)	-581.2	-302.4	16.1	294.9	613.4
+30% ($2,470m)	-727.0	-448.3	-129.7	149.0	467.6

Table S4. One-at-a-time sensitivities around the base NPV of $307.8 million

Variable (downside / upside)	NPV, downside	NPV, upside	Change, downside	Change, upside
Copper price -/+10% ($3.825 / $4.675)	-30.7	646.2	-338.5	338.5
Head grade 0.49% / 0.59%	23.4	592.2	-284.4	284.4
Capital cost +/-10%	161.9	453.6	-145.8	145.8
Discount rate 9.8% / 7.8%	167.0	466.6	-140.8	158.8
Opex per tonne +/-10%	196.2	419.3	-111.6	111.6
Recovery 84% / 90%	201.9	413.7	-105.9	105.9
TC/RC $0.46 / $0.30	241.8	373.7	-65.9	65.9

The tornado shows that copper price and head grade are equally important, each moving NPV by about $285 to $340 million for a roughly ten percent change. Capital cost and discount rate follow at about $145 million. Grade is the underrated variable, because it is under the company's partial control: a 0.05 percentage-point shortfall (Exhibit C describes both a 0.49 percent infill result and a history of 4 percent reconciliation shortfalls) removes as much value as an 8 percent price fall. Because a head-grade shortfall reduces revenue without reducing the cost of mining, it is operationally leveraged.

A one-year delay in first production, with $60 million of additional owner's costs and operating flows shifted one year later, reduces NPV from $307.8 to $147.3 million, a loss of $160.5 million. The extra owner's costs account for about $45 million of the loss and the deferral of the operating flows for the rest. This is the figure to hold in mind when Exhibit F's six-month permit

slippage is discussed.

The windfall surcharge, at 20 percent of revenue from price above $4.50 per pound, is worth nothing on the consensus deck if the threshold is indexed to inflation, since the real price is $4.25. If the threshold is a fixed nominal $4.50, it applies in every producing year, because the consensus deck is above $4.50 in nominal terms from 2030, and it costs $114.4 million of present value after tax, cutting the NPV to $193.4 million. The ambiguity in the consultation paper is therefore worth up to $114 million, more than a ten percent capital cost overrun, and the board paper should mention it as a risk that is binary and political, not as an estimate to be averaged.

Solution 6: Net asset value

Karsten’s after-tax cash flow in each year is revenue (78.0 kt, or 171.96 million pounds, times the nominal price), less C1 cost (2.15 per pound escalated), less the 2.5 percent royalty, less sustaining capital of $55 million escalated, less tax at 27 percent of EBITDA after depreciation (the $620 million tax base over ten years) and sustaining capital, with closure of $95 million escalated in 2046. For 2027 on the consensus deck: price 4.25 times 1.025 = $4.356; revenue is 171.96 times 4.356 = 749.1; C1 cost is 2.15 times 1.025 times 171.96 = 378.9; royalty is 18.7; EBITDA is 351.5; sustaining capital is 56.4; depreciation is 62.0; tax is 0.27 times (351.5 - 62.0 - 56.4) = 62.9; free cash flow is 351.5 - 62.9 - 56.4 = 232.2. Discounting the 20 years at 8.8 percent mid-year gives the Karsten values below. Corporate overhead of $36.0 million (2026 dollars), after tax at 27 percent and escalated, has a present value of $310.7 million.

Table S5. Net asset value of Ironvale at three price decks ($ millions except per share; 262.0 million diluted shares)

Deck	Karsten NPV	Resources	Corporate G&A	Hedge book	Net debt	NAV before Ridge	NAV per share	P/NAV at $6.55	Ridge NPV	NAV per share with Ridge, before raise
Lenders’ deck ($3.85)	1,996.6	110.0	-310.7	-5.3	-592.5	1,198.1	$4.57	1.43x	-10.8	$4.53
Consensus deck ($4.25)	2,575.5	110.0	-310.7	-5.3	-592.5	1,777.0	$6.78	0.97x	307.8	$7.96
Management deck ($4.60)	3,081.9	110.0	-310.7	-5.3	-592.5	2,283.4	$8.72	0.75x	586.5	$10.95

The market price of $6.55 equals 0.97 times NAV on the consensus deck, 1.43 times on the lenders’ deck, and 0.75 times on the management deck. P/NAV of 1.00 requires a deck of $4.21 real, and the peer-median P/NAV of 0.99 requires $4.23. The market is therefore pricing Ironvale at roughly the consensus deck, in line with its peers, and has not given credit for a Ridge premium, nor discounted for it. A corollary: the market’s implied real price, $4.21, is above the forward-based real equivalent of $3.87 in 2031, so the share price already embeds a belief that copper will do better than the futures curve.

The equity raise issues 500.0/6.30 = 79.4 million shares, raising the count from 262.0 to 341.4 million. Pro forma equity value on the consensus deck is 1,777.0 (NAV before Ridge) + 307.8 (Ridge NPV) + 500.0 (cash raised) = 2,584.8, divided by 341.4 million shares, or $7.57 per share. This is a gain of 11.6 percent over $6.78. The decomposition is instructive: Ridge alone, before the raise, lifts NAV per share to $7.96, a 17.3 percent gain; issuing shares at $6.30, below the $6.78 NAV per share, dilutes this by 4.9 percent. On the lenders’ deck, Ridge destroys $10.8 million, and yet the raise appears accretive (NAV per share rises from $4.57 to $4.94, or 8.1 percent) because the company sells shares at $6.30 against an NAV of $4.57. That

accretion is created by selling overvalued paper and should not be credited to the project. If the market price is right and the lenders' deck wrong, it is a mirage; if the lenders' deck is right, it is a transfer from new to old shareholders and a reason for caution about the placement's price. The only fair way to judge Ridge is the NPV; the equity raise is a separate decision.

Solution 7: Financing

The raise of $500 million at $6.30 issues 79.4 million shares, 30.3 percent of the existing count and 23.3 percent of the pro forma count. At the 8 percent discount at the top of Exhibit G's range ($6.03), the issue would be 83.0 million shares, and at the market price of $6.55, 76.3 million. Total uses of $2,090.7 million include $102.7 million of capitalised interest, $28.0 million of fees, and a $60.0 million reserve. The balance sheet at the end of 2029, before Ridge earns anything, would carry bank debt of $260.0 million, notes of $600.0 million, and project debt of $952.7 million, a total of $1,812.7 million less cash of $181.5 million (including the reserve), so net debt of $1,631.2 million. Against Karsten's 2029 EBITDA after overhead of $330.4 million on the consensus deck, consolidated leverage peaks at 4.9 times; on the lenders' deck it is 6.3 times ($258.2 million). The project debt is non-recourse to the extent of the completion support, so the notes' covenants may not be tested against it, but rating agencies and equity investors will look through. This concentration of leverage at the moment of maximum technical risk is the most important financing risk in the case.

The stream. Kestrel pays $400 million in 35/40/25 proportions and receives 15 percent of payable copper (21.75 million pounds a year at full rate, 70 percent in 2030) and pays 20 percent of the spot price. Ironvale's cost of the stream is the forgone 80 percent of the revenue, 0.8 times price times volume. The internal rate of return that equates the upfront payments to those forgone flows is:

Deck	Pre-tax IRR	After-tax IRR	PV of the stream at 8.8%, after tax
Lenders' ($3.85)	14.5%	9.7%	-24.6
Consensus ($4.25)	16.1%	11.2%	-64.2
Management ($4.60)	17.4%	12.4%	-98.8

The stream is not cheap capital. Its after-tax cost at the consensus deck, 11.2 percent, exceeds the 10.7 percent cost of equity and is double the 5.3 percent after-tax cost of project debt. It is, however, equity-like in a useful way: the cost rises with price, so it transfers upside to Kestrel and gives Ironvale some protection on the downside (the cost falls to 9.7 percent on the lenders' deck). That is a form of hedge, though one with no floor and no cap. The claim that it is "non-dilutive" is accurate only in the sense that share count is unchanged; it dilutes the owners of the Ridge cash flows by 15 percent of the copper forever, and on the management deck costs $99 million in present value after tax. The recommendation is to proceed with the base plan (project finance of $850 million plus capitalised interest, equity of $500 million) and to hold the stream as a contingency, to be activated only if the equity window is shut in January or the placement price falls below about $5.50. A hybrid of $150 to 200 million of stream and a smaller equity issue is defensible if the board values the downside protection. The hybrid reduces dilution to 18 to 20 percent but surrenders more of the upside that the shareholders are being asked to underwrite.

Solution 8: Hedging

Black's model for a European put on a forward price F with strike K, volatility sigma, time T, and discount rate r gives P = e^(-rT) times [K N(-d2) - F N(-d1)], where d1 = [ln(F/K) + 0.5 sigma^2 T]/(sigma sqrt(T)) and d2 = d1 - sigma sqrt(T). For 2030: F = 4.36, K = 3.80, T = 3.5, sigma = 24 percent. Then d1 = [ln(1.1474) + 0.5 times 0.0576 times 3.5]/(0.24 times 1.8708) = (0.1375 + 0.1008)/0.4490 = 0.5307; d2 = 0.0817; N(-d1) = 0.2978; N(-d2) = 0.4674; and the discount factor is e^(-0.14) = 0.8694. The put is 0.8694 times (3.80 times 0.4674 - 4.36 times 0.2978) = 0.8694 times (1.7761 - 1.2984) = $0.4153 per pound. The zero-cost cap K_c solves the call price = 0.4153, which gives $5.255.

Table S6. Zero-cost collar pricing, floor $3.80 (Black’s model, volatility 24%, risk-free 4.0%)

Delivery year	Forward ($/lb)	Years T	Discount factor	Put value ($/lb)	Zero-cost cap ($/lb)
2027	4.30	0.5	0.9802	0.0913	4.928
2028	4.32	1.5	0.9418	0.2434	5.045
2029	4.34	2.5	0.9048	0.3430	5.151
2030	4.36	3.5	0.8694	0.4154	5.255
2031	4.38	4.5	0.8353	0.4701	5.359
2032	4.40	5.5	0.8025	0.5123	5.464
2033	4.42	6.5	0.7711	0.5450	5.572

The existing hedge book, with puts at $3.80 and calls at $4.75 (2027, 30 kt) and $4.80 (2028, 20 kt), is worth (put - call) times pounds: 2027, (0.0913 - 0.1294) times 66.1 million pounds = -2.5; 2028, (0.2434 - 0.3074) times 44.1 million pounds = -2.8; a total mark-to-market of **-$5.3 million**, which is the hedge-book figure in the NAV. The company has sold calls that are worth more than the puts it bought, because the caps were set below the zero-cost levels. Over 2024-2026 the realised hedge losses of $3.4, $4.8, and $6.0 million were a similar story.

The deterministic debt-service cover for Ridge is:

Table S8. Deterministic CFADS ($ millions) and DSCR, 2030 to 2033, no hedge

Real price	CFADS 2030	2031	2032	2033	DSCR 2030	2031	2032	2033
$3.85	133.7	240.3	260.6	266.2	1.94x	1.62x	1.83x	1.94x
$4.25	162.0	285.4	308.3	315.1	2.35x	1.92x	2.16x	2.30x
$4.60	186.7	324.8	350.1	357.9	2.70x	2.19x	2.45x	2.61x

Debt service is interest only in 2030 ($69.1 million on $952.7 million at 7.25 percent), then $148.5 million (2031), $142.7 million (2032), and $136.9 million (2033), consisting of interest plus an equal principal instalment of $79.4 million. On the lenders’ deck the minimum DSCR is 1.62 times in 2031, which passes the 1.35 times sizing test with room to spare. Deterministic cover looks comfortable, but the cash flows are highly leveraged to price: a 10 percent price fall reduces 2031 CFADS by about 17 percent.

The simulation uses a mean-reverting log price, calibrated so that short-run volatility is 22 percent and mean reversion is 0.40 a year, centred on each deck. CFADS is EBITDA less tax, sustaining capital, and working capital, with hedge payoffs added after tax.

Table S7. Simulated minimum DSCR, 2030 to 2033 (200,000 paths; mean-reverting price, 22% short-run volatility; PV of hedge payoffs after tax at 8.8%, $ millions)

Price deck (centre of distribution)	Program	Mean of minimum DSCR	5th percentile	10th percentile	P(min DSCR < 1.20x)	P(min DSCR < 1.00x)	Expected PV of hedge payoff
Lenders' deck	No hedge	1.24	0.10	0.35	47.3%	35.2%	0.0
Lenders' deck	Collars, floor $3.80, 50%	1.39	0.66	0.81	37.5%	20.2%	4.1
Lenders' deck	Swaps, 50%	1.50	0.90	1.04	18.3%	8.2%	-8.6
Consensus deck	No hedge	1.57	0.41	0.68	29.7%	20.0%	0.0
Consensus deck	Collars, floor $3.80, 50%	1.64	0.84	1.00	21.7%	9.9%	-24.1
Consensus deck	Swaps, 50%	1.66	1.08	1.23	8.8%	3.3%	-67.3

Three results matter. First, an unhedged, geared single-mine structure is fragile: even if the consensus deck is correct, there is a 20 percent chance that cover falls below 1.0 times at some point in the first four years and a 30 percent chance it falls below the 1.20 times lock-up. A prudent lender could not accept this without protection, which explains the condition. Second, swaps reduce the probability of a sub-1.0 outcome to 3 percent (consensus) or 8 percent (lenders), and collars to 10 or 20 percent. Collars leave more upside, but they leave the tail unprotected between the floor and the level at which cover fails. Third, the expected cost depends on whose deck is right. If the consensus deck is right, the collars give up $24 million of present value and the swaps $67 million, because the forward price is below the deck; if the lenders' deck is right, the collars are worth $4 million and the swaps cost $9 million. That cost is not a loss in the sense of a fair-value shortfall: at forward prices both hedges are worth about zero, so the cost is the price of disagreeing with the market.

Does hedging create value? In the Modigliani-Miller sense, not directly. The expected payoff of a zero-cost hedge at forward prices is zero. Value creation, if any, comes through (a) making the financing possible, since without it the project cannot be debt-funded at 50 percent gearing; (b) lowering the probability of financial distress during the years of maximum debt; and (c) protecting the investment plan. The recommendation is to adopt the lenders' requirement in its cheapest form: 50 percent of Ridge volumes for 2030 to 2033 hedged with zero-cost collars with a $3.80 nominal floor (caps of $5.26, $5.36, and so on), placed progressively rather than all on the day of financial close, to avoid locking in a single price. A swap is not recommended, because the protective benefit above that of the collar is smaller than its expected cost on the board's own deck. Note also that the $3.80 nominal floor is 13 percent below the 2030 forward and 11 percent below the lenders' 2030 nominal price of $4.25, so it works as insurance and not as a price view.

Solution 9: Optionality

The valuation uses a binomial lattice for the project's gross present value V, defined as the PV of post-construction cash flows at the WACC, with V0 = 307.8 + 1,691.9 = $1,999.7 million; the exercise price X is the PV of the capital spending, $1,691.9 million, escalating at 6 percent a year if FID is deferred. The volatility of V equals the long-run effective copper volatility of 18 percent times the elasticity of V to price, 1.69 (computed from NPV at $4.24 and $4.26), or 30.5 percent, so each annual step multiplies V by u = 1.356 or d = 0.737. The risk-neutral drift is the risk-free rate less the project's cash yield: V grows at 2.5 percent (inflation) in expectation, and its required return is 8.8 percent, so the cash yield forgone by waiting is 6.3 percent and the risk-neutral growth rate is 4.0% - 6.3% = -2.3 percent. The probability of an up-move is p = (0.977 - 0.737)/(1.356 - 0.737) = 0.387. Option values are discounted at 4.0 percent, and the option can be exercised at each year-end.

Right	Value ($ millions)	Gain over investing now ($307.8m)
Invest now	307.8	0.0
Defer up to one year	342.0	34.2
Defer up to two years	363.6	55.9
Defer up to three years	398.9	91.2
Defer up to three years, volatility 12%	307.8	0.0
Defer up to three years, volatility 26%	544.6	236.8
Defer up to one year at lenders' deck (NPV -10.8)	181.1	191.9
Defer up to one year at management deck (NPV 586.5)	586.5	0.0

For the one-year option, at t = 1 the project value is $2,711.9 million (up) or $1,474.5 million (down), against a strike of 1,691.9 times 1.06 = $1,793.4 million; the up state pays 918.5 and the down state is out of the money, so the option is worth 0.387 times 918.5/1.04 = $342.0 million, which exceeds the intrinsic value of $307.8 million by $34.2 million. The option to delay is worth something, but on the consensus deck it is worth only about 11 percent of the NPV for one year, because the project's NPV is a substantial fraction of V and the cost of waiting, 6.3 percent a year of forgone cash yield plus capital escalation, is high. The option value is sensitive to volatility: at 12 percent the project should be built now; at 26 percent, waiting three years is worth $237 million more. On the lenders' deck the project is just out of the money and the option to wait is worth $181 million, which is a measure of what flexibility offers a marginal project.

The lattice omits costs of delay that are material here. The fixed-price contract and the banks' commitments lapse, so a deferral must be re-priced (Exhibits A and E); Ironvale's brokers' equity window may close; bonuses are tied to FID (Exhibit H); the construction schedule and the pit sequence would slip; competitors could secure contractors or power. Any deferral cost over about $34 million (for a one-year delay) erases the option value on the consensus deck. Thus the lattice says that waiting is not clearly better than proceeding if the base case is credible, and that waiting is valuable only if the project is marginal and volatile. Students who answer "wait" on the strength of the lattice should be asked what they would pay to keep the contractor's price.

Staged FID. Spend $180 million now on early works and long-lead items (a part of the $1,900 million), with a 12-month gate, with the remaining capital cost fixed, and a salvage value of 25 percent of the early works if abandoned. The value is -180 + [p times max(V_u - (X0 - 180), 45) + (1 - p) times max(V_d - (X0 - 180), 45)]/1.04. Up: 2,711.9 - 1,511.9 = 1,200.0; down: 1,474.5 - 1,511.9 < 0, so the abandon value of 45 applies. The value is -180 + [0.387 times 1,200.0 + 0.613 times 45.0]/1.04 = $293.3 million, slightly below the value of investing now ($307.8 million). The staged option is valuable only in worse states of the world: at the lenders' deck it is worth $132.5 million (against -10.8 now), and at the management deck it costs $80 million relative to proceeding. So staging is insurance whose premium (about $14 million on the consensus deck) is small relative to the protection it gives against a bad outcome, which is a reasonable bargain if the board thinks the lenders' deck is plausible, but it is not a free lunch.

Resource conversion. The 120 million tonnes of resource could add eight to nine years of production after 2043 if it converts to reserve and $250 million of stripping capital is spent in 2042 and 2043. On the consensus deck, the present value of the extension today is about $548 million (the cash flows are far in the future, but the nominal margins are large), $446 million on the lenders' deck, and $638 million on the management deck. At a 35 percent probability of conversion, the expected value is $192 million on the consensus deck, less roughly $30 million of present-value drilling and study cost, about $162 million, equal to half of Ridge's NPV. This is the largest single source of optionality, and it is exercised, if at all, in 2040, after the project has been paid for. It is not in the NAV, the banks do not credit it, and the market probably does not either. It deserves a place in the memo as qualitative support, with the caveat that its value depends on prices fifteen years out and that it should not be added to the NPV in the base case.

Suspension. Ridge's cash cost, including TC/RC, is about $1.71 per pound in 2026 dollars, and Karsten's is $2.15, plus the royalty. Under the mean-reverting price model, the probability that the annual average price in 2030 falls below $2.30 (the point at which Karsten would lose cash) is below one percent, so the option to suspend is worth little and should be excluded from the valuation. Its value would rise only if costs were higher or the price much more volatile. The low-cost position of the new pit is itself a protection.

Solution 10: Recommendation

The evidence can be summarised in five numbers. Ridge has an NPV of $307.8 million on the consensus deck (IRR 11.2 percent), about zero on the lenders' deck (-$10.8 million), and negative on the market forward curve (-$60.1 million, IRR 8.3 percent). The project's break-even price is $3.86 real, and its break-even capital cost is $2,301 million, 21 percent above the estimate. NAV per share rises from $6.78 to $7.57 on the consensus deck after the raise. The probability of a covenant-level cash shortfall is 22 to 30 percent without full hedging if the consensus deck holds. And the company has a record of 14 percent overrun on its last project.

The project clears its hurdle on the consensus deck by a margin that is thinner than the uncertainty in the estimate: the NPV equals about 16 percent of the capital cost, a 10 percent overrun removes 47 percent of it, and a 9 percent shortfall in grade removes 92 percent. The market-forward NPV is negative by a small amount ($60 million is 3 percent of capital), and the deck-versus-forward gap is the single most important judgment. The forward curve is not a forecast of the price in 2035; copper is mean-reverting and the curve is flat in nominal terms, which understates the long-run price if the real price recovers. But a project that is valuable only if the board knows better than the market has not been shown to be good. The real-options analysis says delay has modest value ($34 million for one year) and the lattice omits the costs of delay, which here are large: the lapse of contract and finance terms, and the cost of a one-year delay in first production, about $160 million. Approving the project is therefore preferable to delaying or rejecting it, provided that the board is protected against the cases in which it is wrong.

Recommendation: approve FID on 10 December 2026, subject to conditions precedent, with delegated authority to withdraw if any condition fails. The conditions are:

1. A fixed-price or capped-price contract covering at least 80 percent of the capital cost (up from 62 percent), or a contingency increased to at least 15 percent of the estimate, so that the break-even overrun of 21 percent is not consumed by expected overrun.
2. Satisfactory completion of the water permit before first drawdown, as in the facility agreement, and a sponsor completion guarantee limited to the 10 percent cost-overrun facility.
3. The hedging program recommended in Solution 8: zero-cost collars with a $3.80 floor on 50 percent of 2030-2033 Ridge volumes, layered over the first six months after financial close.
4. An equity raise of $500 million, with a minimum price of $5.50 per share, and a fallback to a part-stream hybrid if the minimum cannot be met.
5. Resolution of the head-grade question: a revised reserve reconciliation, including the 12-hole infill program, before the first drawdown. If the revised grade is 0.49 percent, the NPV on the consensus deck falls to $23 million, and the board should treat that as a stop condition.
6. An incentive structure for the executive team amended so that part of the long-term award depends on return on capital or NAV per share, not just production (Exhibit H).

Walk-away points. The board should not proceed if the lenders' revised deck falls below $3.60 real, since the project's NPV is then about -$210 million and DSCR cover will be too thin; if the capital cost, after contracts are signed, exceeds $2,100 million (about a 10 percent overrun, at which NPV on the consensus deck is $162 million but on the lenders' deck is -$157 million); if

the revised grade is below 0.50 percent; or if the windfall surcharge becomes law with a fixed nominal threshold, which would cut NPV by $114 million on the consensus deck. **Monitoring metrics:** monthly spend against the control budget with a 5 percent trigger for board review, drilled grade against the model, the forward curve against the lenders' deck, hedge coverage, DSCR forecast against 1.35 times, and NAV per share.

Sensitivity Discussion

The valuation has four dominant sensitivities. Price matters most: the NPV falls by $338 million for a 10 percent fall in the real price, and the sign flips at $3.86. The deck-versus-forward gap is, in practice, a 10 percent price difference. Head grade is nearly as large as price, and it is the variable the company can most influence. Capital cost matters less than the other two in a symmetric sense, but its distribution is skewed: overruns are common and underruns rare, so the expected loss from capex is larger than the table suggests. The discount rate matters at about $140 to $160 million per point, so the choice between a development-stage rate and the corporate rate is worth about half the NPV. Timing risk (the one-year slip) costs $160 million, mostly from the deferral of cash flows. The windfall surcharge ambiguity is worth $114 million. Hedging and funding structure do not change NPV materially but determine whether the project can be financed and whether the company survives the downside, which is why they take priority in the recommendation.

Common Student Errors

1. Using the real price deck without escalating it, while escalating costs, or the reverse. This changes NPV by hundreds of millions.
2. Discounting nominal cash flows at a real rate, or the reverse.
3. Forgetting working capital, or forgetting its release at closure, or omitting closure costs.
4. Treating the 2026 C1 cost of $2.15 as the Ridge cost. Ridge's cost is $1.71 per pound in real terms.
5. Using end-year discounting for a mid-year deck and cost profile and then comparing with a mid-year model.
6. Comparing the NPV on the consensus deck with the market capitalisation, or adding the NPV to the share price. NPV is value added to NAV, not value per share, and must be divided by the diluted shares after the issue.
7. Crediting the project with the accretion that arises from selling shares above NAV.
8. Believing that a hedge creates value because it "reduces risk." Zero-cost collars at forward prices have zero expected value.
9. Pricing options on the spot price instead of the forward price, or using a call price that does not equal the put price in a zero-cost collar.
10. Reading the lattice's value of waiting as a recommendation to wait, without considering the costs of delay.
11. Treating the stream as free because it is "non-dilutive."
12. Forgetting the tax shield on depreciation in the project calculation, or taking tax on negative taxable income as zero when the group can use the loss.

Alternative Defensible Answers

The case has no unique answer. A student who weights the forward-based NPV heavily can defend deferral or rejection: the project has a negative NPV at market prices, the margin of safety on cost is thin, and the company has a poor delivery record. Such a student should still address the one-year delay cost of about $160 million and the lapse of the lenders' approvals. A student who trusts management's deck can recommend approval without conditions, on a $586 million NPV, and argue that hedging destroys $24 to $67 million of expected value; that student should be pressed on whether the project can be financed. A third defensible answer is the staged approach: authorize $180 million of early works and defer full FID to the second quarter of 2027, when the permit, the lenders' May deck review, and the grade reconciliation are known. The lattice values this at $293

million against $308 million for immediate approval on the base deck, and at $133 million against -$11 million on the lenders' deck, so it is a sound choice for a board that puts more weight on the downside than the consensus deck implies. The lapse of the contractor's fixed-price offer on 31 January weakens the case for staging. Credit should go to answers that identify which assumption their recommendation depends on and state the evidence that would change their mind.

Board-Style Recommendation Memorandum

To: Board of Directors, Ironvale Copper Corp. **From:** Corporate Development (reporting to the Chief Financial Officer) **Date:** 10 December 2026 **Re:** Final investment decision on the Ridge expansion

Recommendation. We recommend that the board approve the Ridge expansion at a capital cost of $1,900 million, subject to the six conditions below, and delegate authority to the chair and the chief financial officer to withdraw if any condition is not met before first drawdown.

Value. On the consensus price deck of $4.25 per pound (real), Ridge has a net present value of $308 million and an internal rate of return of 11.2 percent against a cost of capital of 8.8 percent. On the lenders' deck of $3.85 the NPV is approximately zero, and on the market forward curve it is minus $60 million. The break-even price is $3.86 and the break-even capital cost is $2,301 million. A one-year delay costs about $160 million. Our NAV for the company is $6.78 per share before Ridge and $7.57 after Ridge and the equity raise, against a share price of $6.55.

Why proceed. The project is worth building if the price is at or above the lenders' deck, it has low operating cost ($1.71 per pound), and the cost of delay is high. A real-options analysis puts the value of waiting at no more than $34 million for one year, which is less than the likely cost of re-pricing the contractor and the financing. The 120 million tonne resource adjacent to the pit offers further value that is not in our numbers.

Why conditions. The margin of safety is thin. A 21 percent capital overrun, or a head grade of 0.49 percent, would remove the project's value on our base deck. Ironvale's last project overran by 14 percent. The company would be levered at about 4.9 times EBITDA at the end of 2029. We therefore require: (1) fixed or capped pricing on at least 80 percent of capital cost; (2) the water permit and a limited sponsor completion guarantee; (3) zero-cost collars with a $3.80 floor on 50 percent of 2030 to 2033 volumes; (4) a $500 million equity raise at no less than $5.50 per share, with a hybrid stream fallback; (5) a revised grade reconciliation before first drawdown; and (6) a change to management incentives to include return on capital.

What would change our view. A lenders' deck below $3.60, a capital cost above $2,100 million after contracting, a reserve grade below 0.50 percent, or a windfall surcharge with a fixed nominal threshold would each cause us to recommend withdrawal. The board should receive a monthly report on spend, grade, hedge coverage, and forecast debt-service cover.

Part II — Cases in Financing

Chapter 7: Case: Calder & Sons, a Family Firm Considers an IPO

Illustrative case. Calder & Sons Building Supply, Inc., its owners, its advisers, its peers and all figures are fictional and were constructed for teaching. Nothing here describes a real company, and no figure should be read as market data. Dollar amounts are in millions unless stated.

The Situation

On the morning of Thursday, 4 November 2026, Margaret Calder opened the envelope that every chief executive of a family company eventually receives. It was a letter, three pages long, from the investment bank Lindqvist Rowe, and it said in carefully lawyered language what the bank's coverage officer had been saying over lunch for a year: the market for building-products companies was open, the public peers were trading at their highest multiples in three years, and Calder & Sons Building Supply, Inc. ("Calder") could probably complete an initial public offering ("IPO") in the first quarter of 2027 at an equity value "in the range of one billion dollars."

Margaret, 52, is Calder's chief executive and a third-generation owner. The company was founded in 1952 by her grandfather Walter Calder as a single lumberyard in the Mid-Lakes region. Today it designs and manufactures engineered millwork (doors, windows and stair systems) in two plants, and distributes those products and a broad range of third-party building materials through 41 branches. It employs about 2,900 people. It has never had an outside shareholder. Its debt is owed to a bank syndicate and to a small group of insurance companies that hold private notes. Its dividend policy, for as long as anyone can remember, has been "forty percent of earnings, paid quarterly, and never cut."

The ownership of the company's 50.0 million shares is spread over four holders who do not always agree.

- **Margaret Calder** holds 17.0 million shares (34%). She runs the business and wants capital to expand it.
- **Hugh Calder**, her father, 81, is chairman and holds 14.0 million shares (28%). He built the distribution network. He wants the family to keep control "for at least another generation," and he is also thinking about his estate, which is almost entirely Calder stock.
- **The Calder Family Trust** holds 11.0 million shares (22%) for the benefit of Margaret's children and Hugh's other grandchildren. The trustee is an independent corporate trustee, which votes the shares in consultation with Margaret.
- **The Eastbrook branch**, descendants of Walter's second son, hold 8.0 million shares (16%) through Nora and Tomas Calder-Hale. They are not involved in management. Both are in their sixties, both have said clearly that they want to convert part of their holding into cash, and Tomas has twice raised the idea of a sale of the whole company.

The board has seven members: Margaret, Hugh, Nora Calder-Hale, the chief financial officer Anil Varghese, and three independent directors, one of whom, Beatrice Okafor, a retired treasurer of a listed industrial company, is the lead independent director. At its meeting on Friday, 19 November 2026, the board must decide four things.

1. **Whether to authorise a confidential submission** of a registration statement so that an IPO could price in the first quarter of 2027, or to wait.
2. **What range of prices is defensible**, and how large the offering should be. Lindqvist Rowe has proposed $225 million in total: $150 million of new shares sold by the company (primary) and $75 million of existing shares sold by family members (secondary).

3. **Whether to adopt a dual-class structure**, with the family's shares carrying ten votes each and the shares sold to the public carrying one vote each, and if so with what protections for public investors.
4. **Whether an IPO is the right route at all**, given two alternatives that arrived in the last month: a proposal from a private equity firm to buy a minority stake, and an informal indication of interest from a strategic buyer.

There is a hard timetable behind the decision. Calder's most recent financial statements cover the period ended 30 September 2026. For purposes of this case, assume the registration rules treat interim statements as stale 135 days after the period end, which is 12 February 2027. A deal priced after that date needs audited FY2026 numbers, which the auditors expect to deliver in the first half of March. The bankers therefore describe two practical windows: a pricing in the first half of February on the September numbers (too tight, in the view of the company's counsel), or a pricing in the second half of March on audited year-end numbers. If the March window is missed, the next realistic window is after the second-quarter numbers are available, in September 2027.

Margaret's own view is that the company has outgrown private ownership. Calder's capital budget has been constrained by the family's reluctance to borrow more, and she sees acquisition targets among regional distributors that she cannot buy with the family's cash flow. Hugh's view is that public markets "reward quarter-to-quarter behavior and punish long-term thinking." The independent directors want a clear, quantified answer to three questions: what the company is worth, what the family gives up, and what is lost by waiting. Anil Varghese has been asked to prepare the analysis for the meeting. You are advising him.

Who decides what

It helps to be explicit about who holds which levers, because the case turns as much on governance as on valuation. The company's charter requires a simple majority of the board for an IPO authorisation, but the shareholders' agreement, which dates from 1998, requires holders of 75% of the shares to approve any change in share classes or any issuance that would reduce the family's aggregate ownership below 60%. No single holder has 75%, and any holder with more than 25% (Margaret at 34% or Hugh at 28%) can block alone. Margaret and Hugh together have 62%; adding the trust gives 84%, so the Eastbrook branch cannot block if the other three are aligned. In practice, the real constraint is not legal but personal: Hugh will not support a structure that lets the family lose control, and Nora and Tomas will not support a structure that keeps their capital locked up indefinitely.

Industry Context

Calder operates in the middle of the building-products value chain. Upstream are lumber mills, glass and hardware makers, and fabricators of engineered wood. Downstream are professional contractors and home builders, and, at the margin, retail customers. Calder's economics blend two models.

Distribution (about 71% of revenue) buys products from many suppliers and sells them, with delivery and credit, to professional contractors. Gross margins are in the low-to-mid twenties of revenue, working capital is heavy (receivables and inventory together exceed a quarter of sales), and capital expenditure is modest. Returns come from density: each additional branch added to a metropolitan area lowers delivery cost per order and increases the share of a contractor's purchases that Calder captures. Distribution is consolidating, and larger players pay substantial premiums for regional networks.

Manufacturing (about 29% of revenue) produces made-to-order millwork. Gross margins are higher, in the mid-thirties of revenue, but the business needs plants, automation and skilled labour, and it is more exposed to input prices. Calder's plants run at roughly 80% of capacity, and the company's capital plan includes automation of the stair line and the construction of a third plant within five years.

Demand for both lines follows residential repair and remodelling, new housing starts and commercial fit-out, all of which are sensitive to interest rates and consumer confidence. Revenue is seasonal: the second and third quarters are the strongest, and the first quarter is the weakest. Calder’s EBITDA margin has been remarkably stable through the recent cycle, which the management team attributes to its mix of repair work and new construction and to the price discipline of the distribution branches. Investors will test that claim.

The public peers. Six listed companies are the natural comparables, and each differs from Calder in ways that matter for valuation. Ashgrove and Corvane are larger and more diversified; Brightwater and Eastfield are mostly distributors, with lower margins and higher leverage; Delmar is the only manufacturer of similar scale; Fallowgate sells materials to commercial builders. All six are larger than Calder by market value except Delmar, which is slightly smaller. None has a dual-class structure, although Corvane has a controlling shareholder. Over the last year the peer group’s median multiple of enterprise value to next-twelve-months EBITDA has risen from about 8.1 times to about 9.0 times, as investors have become more confident that interest rates have peaked.

The IPO market. Three comparable companies completed IPOs during the last eighteen months (see the exhibits). The market clearly values scale, visible growth and a credible plan for using proceeds. It has been less forgiving of offerings that consist mostly of selling shareholders cashing out, and of governance structures that leave public holders with little voice. Investors in this sector tend to be long-only funds that compare a new issue to the existing peers on cash-flow yield, so the price at which the new shares can be sold is strongly anchored to where the peers trade on the day.

A note on dual-class structures. In a dual-class structure the company issues two kinds of common stock that differ in voting rights but share equally in dividends and liquidation proceeds. Founders and families use them to raise equity capital while keeping control. Critics argue that they separate cash-flow rights from control and weaken accountability. Public investors in some markets price this by paying less for the low-vote shares; the size of the discount varies with the circumstances and with the quality of safeguards such as sunset clauses, which convert high-vote shares to ordinary shares after a period or upon a trigger. Some index providers and proxy advisers apply special rules to multi-class companies; in a real transaction the current policies should be checked with counsel. Volume 1, Chapter 24 introduced these governance issues, and Volume 1, Chapter 13 covered the mechanics of an IPO, including underwriting spreads and underpricing. Volume 2’s treatment of valuation under uncertainty is also relevant when comparing offers made at different dates.

Financial Data

All figures are in millions of dollars except per-share data. Fiscal years end 31 December. The statements below were prepared by Anil Varghese’s team from management accounts; FY2025 is audited, the nine-month periods are reviewed but unaudited, and FY2026E is management’s estimate (nine months actual plus a three-month forecast). Tax is modelled at 25% of pre-tax income. Interest is modelled at a blended 6.5% on average debt.

Exhibit 1: Income statement, FY2023 to FY2025 (audited)

$ millions	FY2023	FY2024	FY2025
Revenue	612.4	668.9	721.3
Cost of goods sold	438.9	476.3	511.2
Gross profit	173.5	192.6	210.1
SG&A	84.6	92.1	100.6
EBITDA	88.9	100.5	109.5
D&A	15.9	17.4	19.0
EBIT	73.0	83.1	90.5
Interest expense	13.5	13.9	13.8
Pre-tax income	59.5	69.2	76.7
Income tax (25%)	14.9	17.3	19.2
Net income	44.6	51.9	57.5
EPS (50.0 million shares, $)	0.89	1.04	1.15
Gross margin (%)	28.3	28.8	29.1
EBITDA margin (%)	14.5	15.0	15.2
Dividends paid	17.8	20.8	23.0

Exhibit 2: Interim results and last twelve months (LTM)

$ millions	9M 2025	9M 2026	LTM Sep-2026
Revenue	540.2	585.8	766.9
EBITDA	81.1	85.7	114.1
D&A	14.2	15.2	20.0
Interest expense	10.4	10.2	13.6
Net income	42.4	45.2	60.3
Dividends paid	17.3	17.4	23.1

The LTM column equals FY2025 plus 9M 2026 less 9M 2025 for each line. Management’s estimate for the fourth quarter of 2026 is revenue of $195.4 million (against $181.1 million in the fourth quarter of 2025) and EBITDA of $29.9 million, giving FY2026E revenue of $781.2 million and EBITDA of $115.6 million.

Exhibit 3: Balance sheet extracts

$ millions	31 Dec 2025 (audited)	30 Sep 2026 (reviewed)
Cash	27.4	33.8
Accounts receivable	96.2	109.5
Inventory	118.7	126.0
Other current assets	9.1	9.6
Property, plant and equipment, net	214.6	223.0
Goodwill	38.0	38.0
Other non-current assets	12.4	12.9
Total assets	**516.4**	**552.8**
Accounts payable	71.3	79.8
Accrued liabilities	30.2	33.0
Debt (term loan and notes)	212.0	209.0
Other long-term liabilities	24.8	25.1
Total liabilities	**338.3**	**346.9**
Shareholders' equity	178.1	205.9
Total liabilities and equity	**516.4**	**552.8**
Net debt (debt less cash)	184.6	175.2

At 31 December 2026 management expects debt of $205.0 million (a $120.0 million term loan at a floating rate currently near 6.4%, and $85.0 million of fixed-rate 6.6% senior notes) and cash of $31.0 million, for net debt of $174.0 million, which is 1.5 times FY2026E EBITDA. The credit agreement requires net debt to EBITDA of no more than 3.0 times. The term loan can be prepaid at par; the senior notes carry a make-whole premium if repaid before maturity. There are 1.5 million employee options outstanding with an exercise price of $9.00.

Exhibit 4: Management forecast and free cash flow

$ millions	2026E	2027E	2028E	2029E	2030E
Revenue	781.2	835.8	890.2	943.6	990.8
Growth (%)	8.3	7.0	6.5	6.0	5.0
EBITDA	115.6	126.2	136.2	146.3	153.6
EBITDA margin (%)	14.8	15.1	15.3	15.5	15.5
D&A (2.6% of revenue)	20.3	21.7	23.1	24.5	25.8
EBIT	95.3	104.5	113.1	121.7	127.8
Tax on EBIT (25%)	23.8	26.1	28.3	30.4	32.0
NOPAT	71.5	78.4	84.8	91.3	95.9
Capital expenditure (3.2% of revenue)	25.0	26.7	28.5	30.2	31.7
Increase in net working capital (18% of added revenue)	10.8	9.8	9.8	9.6	8.5
Free cash flow to the firm	56.0	63.5	69.7	76.0	81.4

The forecast assumes no acquisitions and no new plant. For the terminal year (2031), assume revenue grows at the long-run rate g from the 2030 level, the EBITDA margin stays at 15.5%, D&A is 2.6% of revenue, capital expenditure falls to 3.0% of revenue (a normalised level for a business growing at g), and net working capital grows with revenue at 18% of the increase.

Exhibit 5: Cost of capital inputs

Input	Value	Comment
Risk-free rate	4.2%	Long-dated government yield, illustrative
Equity risk premium	5.0%	Analyst consensus, illustrative
Equity beta (relevered to target structure)	1.12	Peer asset betas relevered at 20% debt to value
Cost of equity	9.8%	4.2% + 1.12 × 5.0%
Pre-tax cost of debt	6.5%	Blended current borrowing cost
After-tax cost of debt	4.875%	6.5% × (1 − 25%)
Target debt to value	20%	Peer median is about 22%
WACC	8.8%	80% × 9.8% + 20% × 4.875% = 8.815%, rounded
Long-run growth rate g	2.5%	Range 2.0% to 3.0%
Cost of equity for timing comparisons	10.5%	Includes a size premium, used for discounting expected IPO proceeds

Exhibit 6: Public comparable companies (illustrative, as of 3 November 2026)

Company	Price ($)	Shares (m)	Net debt	NTM EBITDA	NTM net income	NTM EBITDA margin (%)	NTM growth (%)
Ashgrove Building Products	43.80	100.0	750.0	520.0	295.0	14.1	6.5
Brightwater Supply	30.10	62.0	610.0	330.0	148.0	12.4	5.0
Corvane Industries	61.50	45.0	540.0	290.0	160.0	16.2	7.8
Delmar Millwork	21.40	38.0	210.0	120.0	52.0	15.0	8.4
Eastfield Distribution	36.20	70.0	980.0	480.0	175.0	11.3	4.1
Fallowgate Materials	26.90	55.0	330.0	190.0	95.0	13.8	6.9

Market capitalisation is price times shares; enterprise value is market capitalisation plus net debt. NTM means the next twelve months. For Calder, use FY2027E as the NTM period for a valuation date at the end of 2026.

Exhibit 7: Recent comparable IPOs and peer multiple history (illustrative)

IPO	Months ago	Gross proceeds ($m)	Primary share of deal (%)	Pricing vs range	Discount to peer-implied value (%)	First-day return (%)	Return after 90 days (%)
Northvale Fixtures	14	310	60	Top of range	12	14	6
Pellham Tile	9	180	35	Below midpoint	9	4	-8
Harker Flooring	5	260	80	Midpoint	11	9	12

Peer median NTM EV/EBITDA	Q4 2025	Q1 2026	Q2 2026	Q3 2026	Early Nov 2026
Multiple (x)	8.1	8.4	8.7	9.0	9.0

Exhibit 8: Shareholders and the proposed offering

Holder	Shares (m)	Ownership (%)	Wants
Margaret Calder	17.0	34.0	Growth capital; control
Hugh Calder	14.0	28.0	Control for a generation; estate liquidity
Calder Family Trust	11.0	22.0	Long-term holding; diversification at the margin
Eastbrook branch (Nora and Tomas Calder-Hale)	8.0	16.0	Cash now
Total	**50.0**	**100.0**	

Term	Lindqvist Rowe proposal
Primary offering (gross)	$150.0 million
Secondary offering (gross)	$75.0 million
Underwriting spread	6.0% of gross proceeds, primary and secondary
Other offering expenses (paid by the company)	$5.5 million
Use of primary proceeds	Repay $110.0 million of the term loan; balance of net proceeds to fund automation and the first three acquisitions
Share structure	Existing shares become Class B (10 votes per share); shares sold to the public are Class A (1 vote per share)
Over-allotment option	15% of base deal, assumed unexercised in the base case
Pricing assumption for the base case	$17.50 per share
Sellers in the secondary	Eastbrook branch and Hugh Calder

Treat the proposed secondary as 3.0 million shares sold by the Eastbrook branch and the remaining secondary shares (the amount needed to reach $75.0 million at the offer price) sold by Hugh Calder.

Exhibit 9: Two alternatives received in October 2026

Alternative	Terms
Minority investment by a private equity firm (Redfield Partners)	$240 million for a minority stake at an entry price equal to 8.0 times FY2026E EBITDA of enterprise value, with a 1.0x liquidation preference, two board seats, consent rights over acquisitions, debt and dividends, and a right to force a sale or IPO after year five
Informal indication from a strategic buyer (Marlow Group)	All-cash purchase of 100% of the equity at an enterprise value of 10.0 times LTM EBITDA, subject to diligence, with management to remain for two years

Complications and Exhibits

The data above are the clean version. The following facts surfaced during the first two weeks of preparation, and a good analysis has to decide how far each of them matters.

1. “Adjusted” EBITDA. Management presents “adjusted EBITDA” to bankers. For FY2025 it adds back $3.1 million for a one-time enterprise-system implementation cost and $2.2 million of family-related expenses (vehicles, a corporate aircraft share and the salaries of two relatives who do not work in the business), giving $114.8 million against reported EBITDA of $109.5 million. The Exhibit 1 figures are reported, not adjusted, and the forecast in Exhibit 4 assumes the family expenses continue. Lindqvist Rowe’s valuation letter is based on adjusted figures. A public investor will accept the system add-back only if it is demonstrably one-off, will probably accept the family add-back only if the expenses actually stop, and will insist on deducting the costs of being a public company, which Anil estimates at $3.5 million a year (directors, audit and controls, listing and investor relations).

2. Related-party leases. Six branches are leased from a family partnership in which Hugh and Nora hold interests. An independent appraiser estimates that the rents are about $1.2 million a year above market. Investors will ask for the leases to be reset at the IPO. The reset would raise EBITDA by $1.2 million, but it would reduce the income of two selling shareholders.

3. The notes. The $85.0 million of senior notes carry a make-whole premium that Anil estimates at 4.5% of principal if prepaid, about $3.8 million. This is the reason the proposed use of proceeds repays only the term loan. The insurance companies holding the notes have said informally that they would welcome a lower-leverage Calder and would waive the premium if the notes were refinanced into a public bond later, but they have not put this in writing.

4. Management equity. Besides the 1.5 million existing options at $9.00, the compensation committee proposes a post-IPO equity plan of 2.4 million shares, about 4% of the post-IPO share count, to retain senior managers who will be asked to stay through a lock-up. The plan is not in the forecast cost base. Assume that, at today’s pricing, the annual expense is about $2.0 million after tax.

5. Lock-ups. The underwriters require all existing holders to agree not to sell for 180 days after pricing. The Eastbrook branch has asked whether it can sell a further tranche earlier if the stock trades above the offer price. Lindqvist Rowe says an early release clause usually weakens the book building because investors read it as a signal that the sellers are in a hurry.

6. The over-allotment option. The underwriters can sell up to 15% more shares than the base deal if demand is strong and then buy shares back in the market or exercise an option to buy additional shares from the company. If exercised in full at $17.50, the company would sell another 1.286 million shares ($22.5 million) and the family would hold fewer votes. The base case assumes it is not exercised.

7. Tax on sales. Assume the sellers pay tax at 20% on capital gains, with a tax basis of $1.00 per share, and that the sale proceeds are net of the 6.0% spread. Hugh’s advisers note that under the tax rules assumed for this case, shares held at death receive a new tax basis equal to the market value at that date, which would eliminate the gain for his heirs. This makes Hugh’s

secondary sale more costly in after-tax terms than it appears, and it is the reason he has suggested that the secondary be borne entirely by the Eastbrook branch.

8. The strategic indication. The Marlow Group indication is informal. Marlow has asked for sixty days of exclusivity and access to detailed branch-level data. Tomas Calder-Hale supports it. Margaret says that if Calder is sold she will retire, and she believes that three of the five senior managers would leave. The strategic proposal is expressed on an LTM basis; Marlow has not said whether it would assume the notes or require them to be repaid, and has not said who would bear the make-whole premium.

9. The private equity proposal. Redfield's price is expressed as an enterprise value of 8.0 times FY2026E EBITDA of $115.6 million. The firm says that the control rights are "standard." Margaret's counsel says that the consent right over acquisitions would stop the family from doing precisely the deals that motivate the transaction.

10. A data conflict. The Lindqvist Rowe letter refers to peer valuation "at about 9.5 times EBITDA." The median of the six peers in Exhibit 6 is lower than that, and the mean is lower still. The letter appears to use the highest multiples in the set, or EBITDA before certain peer-specific costs; the bank has not been asked to explain. Another discrepancy: the letter quotes a price range of $18 to $22 per share, which cannot be reproduced from the data in Exhibits 4 to 6 without ignoring an IPO discount.

11. Timing scenarios. The board wants to compare pricing in the second half of March 2027 (Option A) with pricing in September 2027 (Option B). Management has prepared the following assumptions. The fully diluted pre-IPO share count for valuation purposes is 51.5 million, which is 50.0 million shares plus the 1.5 million options treated as exercised with $13.5 million of exercise proceeds. The offer price in each scenario is

```
Offer price = [ (Multiple × NTM EBITDA − Net debt + 13.5) / 51.5 ] × (1 − 10%) × (1 − 3%)
```

where 10% is the IPO discount and 3% is the dual-class governance discount. If the window is closed and the deal cannot be priced as planned, the family achieves a downsized or private-placement alternative worth 80% of the price the scenario would have produced. Expected prices are discounted to today at the 10.5% cost of equity from Exhibit 5.

Exhibit 10: Timing scenarios

	Option A: price late March 2027	Option B: price September 2027
Time from today	4.5 months	10.5 months
NTM EBITDA	$126.2 million (FY2027E)	$131.2 million (average of FY2027E and FY2028E)
Net debt at pricing	$174.0 million	$151.5 million
Peer multiple scenarios (probability)	9.0x (50%), 8.0x (30%), 10.0x (20%)	9.0x (45%), 7.8x (35%), 10.2x (20%)
Probability the window is closed	8%	15%
Outcome if window closed	80% of scenario price	80% of scenario price

Exhibit 11: Proposed shareholder-protection terms for the dual-class structure (draft)

Provision	Lindqvist Rowe draft
Class B votes per share	10
Transfers	Class B converts to Class A on transfer outside the family group
Sunset	None
Coattail	None
Independent directors	Majority of the board, but nominated by the controlling holders
Equal treatment in a sale of the company	Not specified

Assignment Questions

Answer the following as Anil Varghese's adviser. Show every calculation. State assumptions where the case is silent. Unless told otherwise, use the offer price of $17.50, a fully diluted pre-IPO share count of 51.5 million (including the 1.5 million options at $9.00, exercise proceeds $13.5 million), net debt of $174.0 million at 31 December 2026, and FY2027E as the NTM period.

1. **Comparable-company valuation.** Compute market capitalisation, enterprise value, EV/NTM EBITDA and P/NTM earnings for each peer in Exhibit 6, and the first quartile, median and third quartile of each multiple. Calculate Calder's FY2027E net income on a pre-IPO basis, then derive the implied equity value and per-share value for Calder from each method at the three quartile points.
2. **Discounted cash flow.** Value Calder at 31 December 2026 using the free cash flows for 2027 to 2030 and a terminal value from the normalised 2031 cash flow in the notes to Exhibit 4, at the 8.8% WACC and 2.5% growth. Report enterprise value, equity value and value per fully diluted share, the percentage of enterprise value from the terminal value and the implied terminal EV/EBITDA multiple. Construct a sensitivity table of value per share to WACC (7.8% to 9.8%) and growth (1.5% to 3.5%). Verify that the 8.8% WACC follows from Exhibit 5.
3. **Valuation range.** Combine Questions 1 and 2 into a valuation range for Calder's equity (low, mid, high) using weights of 40% for EV/EBITDA, 20% for P/E and 40% for DCF, using the quartile range for the multiples and a WACC and growth range of 9.3% and 2.0% (low), 8.8% and 2.5% (mid), 8.3% and 3.0% (high) for the DCF. Show a football field in table form. Then discuss the adjustments in Complication 1 and the public-company costs and quantify the effect of the $3.5 million public-company cost on value at the peer median multiple.
4. **Offer price and range.** Using the blended values from Question 3, a 10% IPO discount and a 3% governance discount, derive the offer price implied at the low, mid and high values, and propose a filing range. Test the sensitivity of the mid-point price to IPO discounts of 10%, 12% and 15% and governance discounts of 0%, 3% and 5%. For the base case of $17.50, compute the primary and secondary shares, the post-IPO share count, market capitalisation, enterprise value and the implied EV/EBITDA and P/E multiples, and compare these with the peers.
5. **Dilution and the cost of going public.** Compute each holder's ownership before and after the IPO. Calculate pro forma FY2027E EPS on a diluted basis before and after the IPO, including the interest saved from repaying $110.0 million of the term loan, and the percentage dilution. Measure the value transferred to new investors by underpricing, using an intrinsic post-IPO value per share derived from the mid-point valuation. Compute the total direct cost of the offering and each seller's after-tax proceeds.

6. **Dual-class design.** Compute the family's voting power after the IPO under one vote per share, five votes per share and ten votes per share for Class B. Find the maximum number of shares the family could sell in total (counting the IPO secondary) before losing a majority of votes under each ratio. Estimate the value cost, to the family's retained stake, of governance discounts of 3%, 5% and 8%. Recommend a Class B ratio and a set of protections, and comment on Exhibit 11.
7. **Control over time.** The family expects to sell a further 8.0 million shares in follow-on offerings and the company may issue 6.0 million shares for acquisitions. Compute voting power and economic ownership under the three vote ratios for several combinations of follow-ons and issuance. Compute each holder's voting power at the IPO under ten-to-one voting. Discuss whether the family is one bloc, and what governance arrangements are needed among the family members themselves.
8. **Use of proceeds and capital structure.** Compute net proceeds from the primary offering, pro forma debt, net debt and net debt/EBITDA before and after the IPO. Is Calder under-levered after the IPO? Compute what a leveraged recapitalisation to 2.0 times and 2.5 times net debt to FY2026E EBITDA would raise, the interest cost and the interest coverage ratio, and compare the recapitalisation with the IPO as a source of family liquidity.
9. **Timing.** Using Exhibit 10, compute the offer price in each scenario for Options A and B, the probability-weighted expected price including the effect of the closed-window risk, and the present value at the 10.5% cost of equity. Which option is better? What multiple at Option B makes the two options equal in present value? Comment on the weaknesses of this analysis.
10. **Alternatives and recommendation.** Compare the IPO, the Redfield minority investment, the Marlow sale and a leveraged recapitalisation on price per share, family liquidity, control and execution risk, using the data in Exhibit 9 and your earlier answers. Write the recommendation that you would give the board on 19 November 2026 as a one-page memorandum.

Teaching Note and Full Solutions

Learning objectives

After working through this case, students should be able to:

1. Build a valuation range from trading multiples and a discounted cash flow, and reconcile the methods rather than average them blindly.
2. Translate a valuation into an offer price by applying an IPO discount and a governance discount, and explain what each discount represents.
3. Measure the dilution a public offering imposes: ownership dilution, earnings-per-share dilution and the value transferred by underpricing.
4. Analyse a dual-class structure quantitatively: voting power, the sell-down path before control is lost, and the cost in valuation of weaker governance.
5. Evaluate IPO timing under uncertainty using probability-weighted outcomes, discounting and a break-even multiple.
6. Compare an IPO with a minority sale, a sale of the company and a leveraged recapitalisation, and defend a recommendation to a board that includes divided owners.

Suggested class plan (150 minutes)

Segment	Minutes	Content
Opening poll and the family map	10	Who wants what? Cold-call: “Who controls Calder after the IPO?”
Valuation: comps and DCF (Q1 to Q3)	30	Students present the multiples, the DCF and the football field on the board; discuss why the three methods differ
Price and dilution (Q4 and Q5)	25	Offer price, discounts, EPS dilution, the cost of underpricing
Dual-class and control (Q6 and Q7)	30	Voting arithmetic, the sell-down path, intra-family governance; role-play Hugh and Tomas
Capital structure and timing (Q8 and Q9)	25	Under-leverage after the IPO; the value of waiting
Alternatives and decision (Q10)	20	Vote by show of hands before and after the debrief
Wrap-up	10	The memorandum, alternative defensible answers, common errors

A useful device is to ask for a vote at the start (“IPO in March, wait, or sell?”) and again at the end. In most classes the final vote is more divided than the first, because the numbers show that the family’s real problem is not valuation but the structure of its own ownership.

Solution to Question 1: Comparable-company valuation

Step 1: peer multiples. Market capitalisation is price times shares, enterprise value (EV) adds net debt, and the multiples divide by NTM EBITDA and NTM net income.

Company	Market cap	EV	EV/EBITDA (x)	P/E (x)	Net debt/EBITDA (x)
Ashgrove	4,380.0	5,130.0	9.87	14.85	1.44
Brightwater	1,866.2	2,476.2	7.50	12.61	1.85
Corvane	2,767.5	3,307.5	11.41	17.30	1.86
Delmar	813.2	1,023.2	8.53	15.64	1.75
Eastfield	2,534.0	3,514.0	7.32	14.48	2.04
Fallowgate	1,479.5	1,809.5	9.52	15.57	1.74

For example, Ashgrove: 43.80 × 100.0 = 4,380.0; 4,380.0 + 750.0 = 5,130.0; 5,130.0 / 520.0 = 9.87; 4,380.0 / 295.0 = 14.85.

Step 2: quartiles. Using linear interpolation (the same as Excel’s QUARTILE.INC), EV/EBITDA is 7.76x (first quartile), 9.03x (median) and 9.78x (third quartile); P/E is 14.57x, 15.21x and 15.62x. The mean EV/EBITDA is 9.02x, so the bank’s reference to “about 9.5 times” (Complication 10) is above the median, the mean and the third quartile. Only Ashgrove, Corvane and Fallowgate trade at or above 9.5x.

Step 3: Calder’s FY2027E net income, pre-IPO. EBIT is $104.5 million (Exhibit 4; unrounded 104.47). Interest on year-end debt of $205.0 million at 6.5% is $13.3 million. Pre-tax income is 104.47 − 13.33 = 91.14; tax at 25% is 22.79; net income is **$68.4 million**.

Step 4: implied values. The FY2027E EBITDA is $126.2 million; net debt is $174.0 million. The per-share value is (equity value + $13.5 million option proceeds) / 51.5 million.

Method	Multiple	Implied EV	Implied equity value	Per share ($)
EV/EBITDA, first quartile	7.76x	979.3	805.3	15.90
EV/EBITDA, median	9.03x	1,139.1	965.1	19.00
EV/EBITDA, third quartile	9.78x	1,234.4	1,060.4	20.85
P/E, first quartile	14.57x	n/a	996.2	19.61
P/E, median	15.21x	n/a	1,039.9	20.45
P/E, third quartile	15.62x	n/a	1,068.0	21.00

Interpretation. The P/E method gives a narrower and higher range than EV/EBITDA. The reason is that Calder converts more of its EBITDA into net income than the peers do: Calder's FY2027E net income is 54% of EBITDA, while the peers range from 37% to 57% with an average of 48%. An EBITDA multiple is capital-structure neutral; a P/E multiple embeds each peer's own leverage, depreciation intensity and tax position. A good answer says which of the two is more reliable for this comparison (EV/EBITDA, with P/E as a cross-check) rather than averaging them mechanically.

Solution to Question 2: Discounted cash flow

WACC check. Cost of equity = 4.2% + 1.12 × 5.0% = 9.8%. After-tax cost of debt = 6.5% × 0.75 = 4.875%. WACC = 0.80 × 9.8% + 0.20 × 4.875% = 7.84% + 0.975% = 8.815%, rounded to 8.8%.

Present values at 31 December 2026 (year-end discounting, first flow is 2027):

Year	FCF ($m)	Discount factor at 8.8%	PV ($m)
2027	63.5	0.9191	58.4
2028	69.7	0.8448	58.9
2029	76.0	0.7764	59.0
2030	81.4	0.7136	58.1
Sum of PVs			234.4

Terminal value. 2031 revenue = 990.8 × 1.025 = 1,015.6. EBITDA = 15.5% × 1,015.6 = 157.4. D&A = 2.6% = 26.4. EBIT = 131.0; NOPAT = 98.3. Capital expenditure = 3.0% = 30.5. Increase in working capital = 18% × (1,015.6 − 990.8) = 4.5. FCF_{2031} = 98.3 + 26.4 − 30.5 − 4.5 = **89.7**. Terminal value at end-2030 = 89.7 / (0.088 − 0.025) = **1,424.3**. Present value = 1,424.3 × 0.7136 = 1,016.5.

Enterprise value = 234.4 + 1,016.5 = **$1,250.8 million**. Equity value = 1,250.8 − 174.0 = **$1,076.8 million**. Per fully diluted share = (1,076.8 + 13.5) / 51.5 = **$21.17**. The terminal value is **81.3%** of enterprise value, and the implied terminal multiple is 1,424.3 / 153.6 = **9.3x** 2030 EBITDA, which is close to the peer median, a useful sanity check: the DCF is not relying on a multiple far above the market.

Sensitivity of value per share ($).

WACC \ g	1.5%	2.0%	2.5%	3.0%	3.5%
7.8%	22.22	23.89	25.88	28.29	31.26
8.3%	20.29	21.69	23.32	25.27	27.62
8.8%	18.63	19.81	21.17	22.77	24.67
9.3%	17.18	18.19	19.34	20.67	22.23
9.8%	15.91	16.77	17.75	18.88	20.18

A one-point change in WACC moves the value by roughly $3.5 to $4.0 per share in the middle of the grid, which is why the DCF carries only 40% of the weight in Question 3.

Solution to Question 3: Valuation range

DCF at the low and high cases. At 9.3% and 2.0%, EV is $1,097.1 million and equity value is $923.1 million. At 8.3% and 3.0%, EV is $1,461.9 million and equity value is $1,287.9 million.

Football field (equity value, $ millions, and per fully diluted share).

Method	Low	Mid	High	Weight
EV/EBITDA	805.3 ($15.90)	965.1 ($19.00)	1,060.4 ($20.85)	40%
P/E	996.2 ($19.61)	1,039.9 ($20.45)	1,068.0 ($21.00)	20%
DCF	923.1 ($18.19)	1,076.8 ($21.17)	1,287.9 ($25.27)	40%
Blended	**890.6 ($17.56)**	**1,024.7 ($20.16)**	**1,152.9 ($22.65)**	

Calculation of the mid blend: 0.40 × 965.1 + 0.20 × 1,039.9 + 0.40 × 1,076.8 = 386.0 + 208.0 + 430.7 = 1,024.7. The blended mid is an enterprise value of 1,198.7, or 10.4x FY2026E EBITDA, 9.5x FY2027E EBITDA and 15.0x FY2027E earnings. The Lindqvist Rowe letter's "$1 billion" equity value is therefore consistent with the mid blend but is a pre-discount figure: it is what the company is worth, not what the IPO will raise it at.

Adjustments. The public-company cost of $3.5 million a year reduces EBITDA, and at the peer median multiple of 9.03x it reduces enterprise value by 3.5 × 9.03 = **$31.6 million**, or $0.61 per share. The related-party lease reset adds $1.2 million of EBITDA (worth 1.2 × 9.03 = $10.8 million, or $0.21 per share). The family-expense add-back of $2.2 million should be accepted only if the spending actually ceases; if it does, it is worth 2.2 × 9.03 = $19.9 million ($0.39 per share). The enterprise-system add-back should be ignored in a forward valuation, since the forecast does not include that cost. A reasonable analyst nets these items: −3.5 + 1.2 + 2.2 = −$0.1 million, which is about zero. The forecast in Exhibit 4 as published is therefore a fair base, provided the family expenses stop and the leases are reset. If only the public-company cost is deducted, the mid value falls by about $0.61 per share before discounts.

Solution to Question 4: Offer price and range

Offer prices at each value. Offer price = pre-IPO value per share × (1 − 10%) × (1 − 3%) = value × 0.873.

Case	Pre-IPO value per share ($)	Offer price ($)
Low	17.56	15.33
Mid	20.16	17.60
High	22.65	19.77

Proposed filing range. A filing range is usually narrow, about 10% to 12% wide, around the expected pricing. Pricing is expected at or near the mid value of $17.60; the comps-only price (EV/EBITDA median, same discounts) is $16.53 and the high end of the blend is $19.77. A defensible range is **$16.50 to $18.50**, with a midpoint of $17.50. The Lindqvist Rowe range of $18 to $22 can only be reached by ignoring the discount (the undiscounted mid is $20.16) and is not supportable.

Sensitivity of the mid price ($).

IPO discount governance discount	0%	3%	5%
10%	18.14	17.60	17.24
12%	17.74	17.21	16.85
15%	17.14	16.62	16.28

Base case at $17.50. Primary shares = 150.0 / 17.50 = **8.571 million**. Secondary shares = 75.0 / 17.50 = **4.286 million**, of which 3.000 million come from the Eastbrook branch and 1.286 million from Hugh. Post-IPO basic shares outstanding = 50.0 + 8.571 = **58.571 million**. The 1.5 million options add 1.5 × (1 − 9.00/17.50) = 0.729 million under the treasury stock method, giving 59.300 million diluted shares. Market capitalisation = 17.50 × 59.300 = **$1,037.8 million**. Pro forma net debt = 174.0 − 135.5 = $38.5 million, so enterprise value = 1,037.8 + 38.5 = **$1,076.3 million**, which is 9.3x FY2026E EBITDA and **8.5x** FY2027E (NTM) EBITDA. The pro forma FY2027E diluted EPS is $1.243 (Question 5), so the price/earnings multiple is **14.1x**. Both are below the peer medians (9.03x and 15.21x), by about 6% on EV/EBITDA and 7% on P/E, which is the visible form of the IPO discount.

Solution to Question 5: Dilution and the cost of going public

Ownership. Shares sold in the secondary come out of the sellers' holdings; the primary shares are new. Post-IPO basic shares are 58.571 million.

Holder	Pre-IPO shares (m)	Pre-IPO %	Sold in IPO (m)	Post-IPO shares (m)	Economic %	Votes % at 10 to 1
Margaret Calder	17.000	34.0	0.000	17.000	29.0	36.2
Hugh Calder	14.000	28.0	1.286	12.714	21.7	27.1
Calder Family Trust	11.000	22.0	0.000	11.000	18.8	23.4
Eastbrook branch	8.000	16.0	3.000	5.000	8.5	10.6
Public (Class A)	0.000	0.0	n/a	12.857	21.9	2.7
Total	**50.000**	**100.0**	**4.286**	**58.571**	**100.0**	**100.0**

The family as a whole falls from 100% to 78.05% of the equity: 14.63 points of dilution come from the primary shares (8.571 / 58.571) and 7.32 points from the secondary (4.286 / 58.571).

EPS dilution. FY2027E EBIT is 104.47. Before the IPO, interest is 205.0 × 6.5% = 13.33; net income is (104.47 − 13.33) × 0.75 = **$68.36 million**. After the IPO, debt is 95.0 and interest is 6.18; net income is (104.47 − 6.18) × 0.75 = **$73.72 million**. The difference is 110.0 × 6.5% × 0.75 = $5.36 million after tax. Diluted share counts are 50.0 + 0.729 = 50.729 million before and 50.0 + 8.571 + 0.729 = 59.300 million after.

	Pre-IPO	Post-IPO
Net income ($m)	68.36	73.72
Diluted shares (m)	50.729	59.300
Diluted EPS ($)	1.348	1.243
Change		−7.7%

The dilution is modest because most of the proceeds retire debt. The finance view: EPS is diluted because the earnings yield on the new shares (1.243 / 17.50 = 7.1%) exceeds the after-tax cost of the debt retired (4.9%), so each dollar of equity raised to repay debt costs more in earnings than it saves. If the $25.5 million of unspent proceeds earns nothing in the interim, the figures above overstate the benefit slightly by omitting interest income and understate it by omitting the future returns on the acquisitions.

If the incremental public-company costs ($3.5 million before tax, $2.6 million after tax) and the new equity plan ($2.0 million after tax) are included, net income becomes 73.72 − 2.63 − 2.00 = $69.10 million and diluted EPS is **$1.165, a decline of 13.5%** against the $1.348 pre-IPO figure. The comparison against "status quo" is not like for like, since the status quo has no public-company costs, but it is the comparison an investor will make against the pre-IPO forecast.

Value transferred by underpricing. The intrinsic post-IPO value per share is (mid-point equity value + net primary proceeds + option proceeds) / post-IPO fully diluted shares = (1,024.7 + 135.5 + 13.5) / (51.5 + 8.571) = 1,173.7 / 60.071 = **$19.54**. The offer price of $17.50 is 10.4% below that figure. Value transferred to new investors = (19.54 − 17.50) × 12.857 million shares = **$26.2 million**. This is borne by all pre-IPO holders: the secondary sellers receive less than the shares are worth, and the primary shares dilute the continuing holders.

Direct costs. Spread on the primary offering = 6% × 150.0 = 9.0; on the secondary = 6% × 75.0 = 4.5; other expenses 5.5. Total **$19.0 million, or 8.4% of the $225.0 million gross proceeds**. Adding the underpricing, the all-in cost of going public is 19.0 + 26.2 = **$45.2 million, or 20.1% of gross proceeds**. Many students quote only the spread; the board should see the all-in figure.

Seller proceeds. With a basis of $1.00 per share and a 20% tax rate:

Seller	Shares sold (m)	Gross	After 6% spread	Tax	After-tax proceeds
Eastbrook branch	3.000	52.50	49.35	9.27	40.08
Hugh Calder	1.286	22.50	21.15	3.97	17.18

For example, Eastbrook's tax is 3.000 × (17.50 × 0.94 − 1.00) × 20% = 3.000 × 15.45 × 0.20 = $9.27 million. Eastbrook converts 37.5% of its holding into $40.1 million after tax and remains locked up on the other 5.0 million shares for 180 days. If Hugh's heirs would receive a stepped-up basis at his death, his $3.97 million of tax is avoidable by waiting, which is a strong argument for his not selling.

Solution to Question 6: Dual-class design

Voting power after the IPO. Family shares: 45.714 million (Class B). Public shares: 12.857 million (Class A).

Class B votes per share	Total votes (m)	Family votes (%)	Family economic ownership (%)
1	58.571	78.0	78.0
5	241.43	94.7	78.0
10	470.00	97.3	78.0

At the IPO, the family holds a clear majority of the votes in all three cases. A dual-class structure adds little to control in the first year. Its value is in the future.

Maximum sell-down. Let s be the total number of shares the family sells (the IPO secondary plus later sales) with the primary offering of 8.571 million fixed. Family votes exceed 50% as long as r × (50 − s) > (8.571 + s), where r is the Class B vote ratio.

Class B votes per share	Maximum total family sales (m)	Further sales beyond the IPO secondary (m)	Family economic stake at that point (%)
1	20.7	16.4	50.0
5	40.2	36.0	16.7
10	44.7	40.4	9.1

A simple rule of thumb: the family keeps voting control as long as its economic stake exceeds 1/(r + 1) of the shares, which is 50% for one vote, 16.7% for five votes and 9.1% for ten votes.

Cost of the governance discount. The family's 45.714 million retained shares are worth $829.4 million at an offer price of $18.14 (no governance discount), against $804.6 million at 3%, $788.0 million at 5% and $763.1 million at 8%. The cost is therefore **$24.8 million, $41.4 million and $66.3 million** respectively. Since the discount depends on the strength of the safeguards, the structure should be designed with the safeguards that minimise the discount.

Recommendation. Class B should carry five votes, not ten. Five votes keep control until the family's stake falls to 16.7%, well beyond the sell-down the family contemplates, and the structure is easier to explain and defend; ten votes buy a further 7.6 points of cushion that the family does not expect to need and invite a larger discount. Exhibit 11 should be strengthened with: a **time-based sunset** (conversion of Class B to Class A twelve years after the IPO, extendable by a vote of Class A holders); an **ownership-based sunset** (automatic conversion when Class B shares fall below 15% of total shares outstanding); **conversion on transfer** outside a defined family group (already in the draft); an **equal-treatment clause** so that Class A holders receive the same per-share consideration in any sale of the company; and a **nominating committee of independent directors**. Each of these costs the family little today and reduces the probability that investors demand a larger discount.

Solution to Question 7: Control over time

Follow-on scenarios. Class B shares are 45.714 million at the IPO; Class A shares are 12.857 million. Each row removes family shares (they convert to Class A when sold) and adds any newly issued Class A shares.

Family sells further (m)	New shares issued (m)	Family economic stake (%)	Family votes, 1 vote (%)	Family votes, 5 votes (%)	Family votes, 10 votes (%)
0	0	78.0	78.0	94.7	97.3
8.0	0	64.4	64.4	90.0	94.8
8.0	6.0	58.4	58.4	87.5	93.4
16.0	6.0	46.0	46.0	81.0	89.5
24.0	6.0	33.6	33.6	71.7	83.5
30.0	10.0	22.9	22.9	59.8	74.8

Under one vote per share, the family keeps control through the planned follow-on of 8.0 million shares and the acquisition issuance of 6.0 million (58.4%), but loses it if it sells 16.0 million and the company issues 6.0 million (46.0%). With five votes the family retains control even in the most aggressive scenario shown. The analysis shows that the structure is insurance for the second and third decades, not a necessity today.

Each holder at the IPO, ten votes per share. Margaret 36.2%, Hugh 27.1%, the Trust 23.4%, Eastbrook 10.6%, public 2.7%. (Under five votes: 35.2%, 26.3%, 22.8%, 10.4% and 5.3%.) **No single family holder has a majority.** Margaret and the Trust together have 59.6% of the votes; Margaret and Hugh together have 63.3%; Hugh and the Trust together have 50.5%; Eastbrook's 10.6% is not decisive in any pairing of the other three.

Is the family one bloc? Not legally. The dual-class structure gives control to the Class B holders as a group but says nothing about how they decide among themselves. The analysis therefore points to a **voting agreement or family governance charter** among the Class B holders covering: how the board nominees are chosen (for example by a majority of Class B votes), a right of first refusal when a family member sells, a rule on which Class B transfers are permitted, procedures to resolve a deadlock between Margaret and Hugh, and what happens when Hugh dies and his shares pass to several heirs. Without such an agreement a split among the family would hand the decisive vote to the independent directors or to the public holders, which is the outcome Hugh fears most.

Solution to Question 8: Use of proceeds and capital structure

Net primary proceeds = 150.0 − 9.0 (spread) − 5.5 (expenses) = **$135.5 million**. Of this, $110.0 million repays the term loan (leaving $10.0 million of term loan and $85.0 million of notes, or $95.0 million of debt) and $25.5 million is added to cash.

	Before IPO (end-2026E)	After IPO
Debt	205.0	95.0
Cash	31.0	56.5
Net debt	174.0	38.5
Net debt / FY2026E EBITDA	1.51x	0.33x
Peer median net debt / EBITDA	1.80x	1.80x

Is Calder under-levered? Yes, on paper. At 0.33x it sits far below the peer median of 1.80x and the 20% debt-to-value target in the WACC. But the $25.5 million is earmarked for automation and acquisitions; spending it, and using debt for further acquisitions, would take leverage back toward 1.0x to 1.5x. The company should be explicit about this in the prospectus, since a public investor who sees an unlevered balance sheet will ask why the company sold more equity than it needed.

Leveraged recapitalisation. Raising net debt to a multiple of FY2026E EBITDA (115.6) and holding cash at 31.0:

Target net debt/EBITDA	Net debt	Incremental borrowing	Gross debt	Interest at 7.0%	FY2027E EBITDA / interest	Covenant headroom (EBITDA fall before 3.0x)
2.0x	231.2	57.2	262.2	18.4	6.9x	33.3%
2.5x	289.0	115.0	320.0	22.4	5.6x	16.7%

A recapitalisation to 2.5x could fund a $115.0 million dividend, which exceeds the $75.0 million secondary and costs no underwriting spread, and the family keeps 100% of the votes. It would also increase interest expense by 22.4 − 13.3 = $9.1 million before tax and leave headroom of only 16.7% of EBITDA against the 3.0x covenant, in a cyclical business. It gives no growth capital, no acquisition currency and no price discovery, and the make-whole premium on the notes (about $3.8 million) would arise if the notes were refinanced. The recapitalisation is a real alternative for liquidity but not for growth.

Solution to Question 9: Timing

Offer prices. Offer price = [(multiple × EBITDA − net debt + 13.5) / 51.5] × 0.90 × 0.97.

Option	Multiple	Probability	Offer price ($)
A (March 2027)	9.0x	50%	16.53
A	8.0x	30%	14.39
A	10.0x	20%	18.67
B (September 2027)	9.0x	45%	17.68
B	7.8x	35%	15.01
B	10.2x	20%	20.35

Worked example, A at 9.0x: 9.0 × 126.2 = 1,135.8; less 174.0 = 961.8; plus 13.5 = 975.3; divided by 51.5 = 18.94; times 0.873 = 16.53.

Expected values. A: 0.5 × 16.53 + 0.3 × 14.39 + 0.2 × 18.67 = **$16.32**. Allowing for an 8% chance of a closed window with outcome worth 80%: 16.32 × (0.92 + 0.08 × 0.80) = 16.32 × 0.984 = **$16.06**. Present value over 0.375 years at 10.5%: 16.06 / 1.105^0.375 = **$15.47**.

B: 0.45 × 17.68 + 0.35 × 15.01 + 0.20 × 20.35 = **$17.28**. With a 15% closed-window risk: 17.28 × (0.85 + 0.15 × 0.80) = 17.28 × 0.97 = **$16.76**. Present value over 0.875 years: 16.76 / 1.105^0.875 = **$15.36**.

Conclusion. Option A is better by $0.11 per share (0.7%), a small margin. The break-even multiple at which Option B equals Option A in present value is **8.88x**, only 0.15x below today's 9.03x peer median. Waiting is therefore only worth it if one believes multiples will be at least roughly flat. The weight of risk (a longer exposure to a window that may close, and more time for a bad quarter at Calder) tilts the decision toward March. **Weaknesses:** the scenario probabilities are subjective; the 80% fallback is arbitrary; the cost of equity is applied to a single cash flow when the family's real alternative is to continue to own the shares; the analysis ignores the option value of using the window to complete one of the acquisitions first; and the multiples in the two options are not independent of each other (a bad March makes a bad September more likely). The right use of the calculation is to show that the case for waiting is weak, not to claim a precise answer.

Solution to Question 10: Alternatives and recommendation

	IPO (base)	Redfield minority	Marlow sale	Recapitalisation (2.5x)
Price per fully diluted share	$17.50 offer (intrinsic about $19.54 post-IPO)	$14.84 (EV 924.8 = 8.0 × 115.6; equity 750.8)	$19.02 (EV 1,141.0 = 10.0 × 114.1; equity 965.8 using September net debt of 175.2)	n/a (no sale)
Cash to family	$70.5 million net of spread (secondary); retained stake of 45.7 million shares worth $800.0 million at $17.50	Up to $240 million, depending on primary and secondary mix	All equity value, about $966 million before tax; $3.8 million make-whole if the notes are repaid	$115.0 million dividend
Family control	78% of shares; 94.7% of votes at five votes per share	Retained, subject to consent rights	Lost	100% retained
Growth capital	$135.5 million net	$240 million only if primary	None (Marlow's balance sheet)	None
Main risks	Market window; public-company cost and scrutiny	Liquidation preference, forced exit after year five, veto over acquisitions	Non-binding; Margaret and managers likely to leave; tax on full sale	Leverage in a cyclical business

Reading the table. Redfield's price per share is 15.2% below the IPO offer price and 26.4% below the mid intrinsic value of $20.16, before the value of its preferences; this is a financing priced for a private, illiquid, control-constrained minority and should be used as a negotiating reference, not as a competitor to the IPO. The Marlow indication of $19.02 is 8.7% above the IPO offer price and 5.7% below intrinsic value, and for a 100% sale the family would pay tax on the whole gain (at 20% with a $1.00 basis, $15.41 per share after tax). It is non-binding. The comparison of an IPO at $17.50 with a sale at $19.02 is not one of price alone: in the IPO the family keeps 78% of a company whose value is expected to rise (the intrinsic value after the IPO is $19.54) and defers tax on the retained stake; in the sale it receives a certain amount now and ends its role in the business. For a board whose chief executive says she will retire on a sale, and whose chairman wants control, the strategic route has the lowest value to the people who must decide it, but should be kept alive as a price check.

Sensitivity discussion

The offer price is most sensitive to the market multiple. One turn of EV/EBITDA on FY2027E EBITDA of $126.2 million is worth 126.2 / 51.5 = $2.45 per share before discounts and $2.14 after the 10% and 3% discounts, which is why Question 9 turns on a 0.15x gap. Each percentage point of IPO discount is worth about $0.20 per share (20.16 × 0.01 × 0.97), and the 3% governance discount is worth about $0.54 per share at the mid-point price before the IPO discount, or $24.8 million on the family's retained shares. The $3.5 million public-company cost is worth $0.61 per share before discounts. The DCF is sensitive mainly to WACC: moving from 8.8% to 9.3% lowers the value by $1.83 per share at 2.5% growth. Taken together, no plausible

single change moves the offer price outside a $15 to $20 band, but the combination of a weaker market (8.0x), a 15% IPO discount and a 5% governance discount produces a price of about $13.31, a fall of 24% from the base case. At that price the company would have to issue about 11.3 million shares instead of 8.6 million to raise the same $150 million, an increase of 31%.

Common student errors

1. **Averaging methods without weighting or reconciling them**, or using the peer maximum multiple, as the bank letter does.
2. **Applying the EV multiple to equity**, or forgetting to subtract net debt (and to use the pre-IPO net debt in a pre-money valuation).
3. **Ignoring the options** in the share count or counting all 1.5 million as new shares without exercise proceeds.
4. **Confusing the IPO discount with the dual-class discount**, or applying both to the wrong base.
5. **Treating the primary and secondary the same way** in the dilution analysis; only primary shares add to the share count and raise cash for the company.
6. **Computing votes as if Class B shares were sold at ten votes** to the public (shares sold by the family convert to Class A).
7. **Stopping at the spread** when estimating the cost of the IPO and omitting underpricing.
8. **Describing dual-class as a pure governance evil or pure benefit**, instead of quantifying the control it buys and its cost.
9. **Discounting the expected price at the WACC** in the timing analysis (the cash flow is to equity holders, so the cost of equity is appropriate).
10. **Mixing LTM and NTM** multiples between the Marlow offer and the peer multiples.

Alternative defensible answers

Single-class structure. The family holds 78% after the IPO and could sell 16.4 million more shares before losing a majority. A single class avoids the governance discount ($25 to $41 million), the proxy-adviser and index issues, and the symbolism. Its drawback is that the sell-down the Eastbrook branch wants, plus acquisitions, would put control at risk within a few years. A board that expects the family to stay above 50% can defensibly choose it.

Delay. A board that values the March window less than the opportunity to complete the first acquisitions before listing, and so present a larger, more diversified company, could choose September. The numbers do not strongly prefer either date.

Sale to Marlow. If Margaret were willing to leave and Hugh to give up control, a firm offer near 10.0 times LTM EBITDA is above the IPO price per share and certain. This is a rational choice for owners whose objective is wealth rather than continuity.

Smaller IPO. Eastbrook alone sells (3.0 million shares, $52.5 million) and Hugh does not sell, for tax reasons, producing a $202.5 million offering with a free float of 19.8% of shares. The risk is thin trading.

Board recommendation memorandum

To: Board of Directors, Calder & Sons Building Supply, Inc. **From:** Anil Varghese, Chief Financial Officer **Date:** 19 November 2026 **Re:** Proposed initial public offering

Recommendation. Authorise a confidential submission now, targeting pricing in the second half of March 2027, with the following terms: a primary offering of $150 million and a secondary of up to $75 million; an initial filing range of **$16.50 to $18.50 per share**; and a dual-class structure with Class B carrying **five votes** and the safeguards in this memorandum. Do not grant exclusivity to Marlow, but ask Marlow for a firm, written price.

Valuation. Our blended equity value is $1,025 million at the mid-point ($20.16 per fully diluted share), within a range of $891 million to $1,153 million. After a 10% IPO discount and a 3% governance discount the expected offer price is about $17.60. This is below the $18 to $22 range in the bank's letter, which does not reflect an IPO discount and uses a peer multiple above the group median.

What we give up. The offering dilutes existing holders' ownership from 100% to 78.0%, reduces FY2027E EPS by about 7.7% before public-company costs (13.5% after them), and costs $19.0 million in fees plus an estimated $26.2 million of underpricing, 20% of gross proceeds in total. The family retains 94.7% of the votes under the proposed structure and keeps control until its economic stake falls below 16.7%.

Why now. Peer multiples are at a three-year high and the calculation in the timing analysis shows a probability-weighted advantage of about $0.11 per share for March over September. The break-even September multiple is 8.88x, against 9.03x today.

Conditions. (a) A family voting agreement among Class B holders, signed before filing. (b) Class B sunsets after twelve years and below 15% of shares outstanding, with an equal-treatment clause and an independent nominating committee. (c) Leases with family entities reset to market. (d) Family expense add-backs only to the extent that the expenses actually cease. (e) The notes are left in place to avoid the make-whole premium. (f) Hugh to decide, on tax advice, whether to participate in the secondary.

Risks and fallback. If the market multiple falls below about 8.0x before pricing, the price would be near $14.40 and the board should reconsider size and timing. In that case, defer to September, or consider the recapitalisation as a bridge for the Eastbrook branch's liquidity needs. The Redfield proposal should not be pursued: its price per share is 15% below the base IPO price and its consent rights would block the acquisition strategy that motivates the transaction.

Chapter 8: Case: Meridian Power, Capital Structure for a Regulated Utility

Illustrative case. Meridian Power Company, its regulator, its rating agency, its peers and all figures are fictional and were constructed for teaching. Nothing here describes a real utility, commission or agency, and no figure should be read as market data. Dollar amounts are in millions unless stated.

The Situation

Samuel Okonkwo, chief financial officer of Meridian Power Company (“Meridian”), has a number on his whiteboard that he does not like: **$7.5 billion**. It is the capital expenditure the company intends to make between 2027 and 2031 to modernise its distribution grid, add transmission lines, build owned solar and storage capacity and convert an older coal plant to gas. It is also 72% of the $10.45 billion of rate base the company expects to have at the end of 2026. Meridian has never spent money at this pace, and the cash flow from operations (FFO) covers only about three-quarters of the cost before any dividend is paid, and about half of it after.

Meridian is an electric utility serving 1.3 million customers across the northern part of the fictional state of Platte. It owns generation, a transmission network and a distribution system, and it is regulated by the Platte Public Utilities Commission (“the Commission”). Its common stock (180.0 million shares, currently $42.00) is held mainly by income-oriented funds and a large base of retail shareholders who have received a dividend every year since 1957. Its senior unsecured debt is rated BBB by Keystone Ratings (“Keystone”), one notch above the lowest investment-grade level but two below a threshold that the treasurer regards as comfortable.

The people around the table are these.

- **Elena Marchetti**, chief executive, who inherited the capital plan from her predecessor and believes in it. She wants to protect the dividend and the rating.
- **Samuel Okonkwo**, chief financial officer, who must present the financing plan.
- **Priya Raman**, treasurer, who runs the debt programme and the relationship with Keystone, and who is certain that the rating agency’s published thresholds are binding in practice.
- **Walter Lindgren**, chair of the board’s finance committee and a former commissioner of another state. He knows how commissions think and warns that “a commission will not pay us to be careless about our balance sheet.”
- **Dr. Hannah Osei**, the company’s regulatory economist, who is preparing the cost-of-capital testimony for the rate case.

The finance committee meets on **Tuesday, 8 December 2026**. It must decide three things.

1. **The 2027 to 2031 financing plan**: how much of the funding gap to cover with debt, common equity, hybrid securities and retained cash, and in what sequence, subject to a rating constraint that has become binding.
2. **The return Meridian will request in its general rate case**, to be filed on 15 March 2027: the requested return on equity (“ROE”) and the equity ratio. The present authorised return is 9.6% on a 50% equity ratio, set in the last case.
3. **The dividend policy** to adopt for the plan period, given the tension between a 62% payout ratio that investors treat as sacrosanct and a balance sheet that is losing its cushion.

The decision has a clock behind it. The commission's statutory timetable gives it about eleven months from filing to issue an order, so new rates cannot take effect before early 2028. During the first year of the plan Meridian will spend $1.35 billion against rates that were set using a rate base that is already out of date. Keystone has told Priya privately that it will review Meridian's rating after the rate case is filed and that its downgrade threshold for BBB is a ratio of funds from operations ("FFO") to debt below 15% on a sustained basis.

Elena's own summary of the dilemma, in the company's board paper, is that regulation gives Meridian a promise (an opportunity to earn a stated return on every dollar it invests prudently) but not a guarantee, and that the promise is only worth what the capital markets believe it is worth. A utility that is seen as stretching its balance sheet pays more for money, and eventually its customers pay more as well.

How regulation shapes the question

Regulation of a utility turns the usual corporate-finance problem on its head. An unregulated company chooses a capital structure to maximise value given its operating risk. A regulated company is told, in effect, what it may earn: the commission sets a **revenue requirement** equal to operating costs, depreciation and taxes plus a **return on rate base**. Rate base is the net, depreciated cost of the plant that is used and useful in serving customers, less the deferred taxes that customers have already advanced. The allowed return is the weighted cost of the capital that the commission treats as financing that rate base: an authorised equity ratio times an allowed ROE, plus the debt weight times the embedded cost of debt. The company's actual financing mix does not set the revenue requirement. If Meridian finances more than the authorised proportion with debt it keeps the extra margin between the allowed return on equity and the cost of debt, which raises its earned ROE and, simultaneously, its risk. If the commission becomes concerned it can impute a hypothetical capital structure that is more heavily weighted to equity than the actual one.

This setting makes three features of the case especially important. First, **regulatory lag**: because rates are set on a historical rate base, a company in the middle of a heavy capital programme earns less than its allowed return until rates catch up. Second, **credit metrics, not value maximisation, are the binding constraint**: the rating thresholds that the agency applies to utilities decide how much debt can be used. Third, the **spread between the allowed return and the market cost of capital** determines whether the capital programme creates or destroys shareholder value, which is why the choice of the requested ROE matters as much as the financing mix.

Industry Context

Capital intensity. Regulated electric utilities invest continuously in networks whose economic life runs to decades. A utility's earnings growth is mostly rate base growth: if rate base grows at 7.5% a year and the allowed return stays constant, then earnings grow at about the same rate, provided the financing is done at the authorised structure and costs. This has made the utility sector a favourite of income investors, who want a stable and growing dividend, but it also makes the sector a heavy and regular issuer of both debt and equity.

The credit framework. Rating agencies assess utilities on business risk (regulatory support, the predictability of cost recovery, the fuel mix and the physical exposure to storms or other hazards) and financial risk, measured through cash-flow ratios. The most widely used is FFO divided by adjusted debt. Keystone's published guidance, which the case treats as illustrative, is summarised below.

Rating	FFO / debt	Debt / EBITDA	Interpretation for a utility of Meridian's business profile
A-	22% or more	3.8x or less	Strong; requires a materially lower payout or much more equity
BBB+	18% to 22%	3.8x to 4.3x	Comfortable investment grade
BBB	15% to 18%	4.3x to 5.0x	Meridian's current level; downgrade threshold at 15% sustained
BBB-	12% to 15%	5.0x to 5.8x	Lowest investment grade; commercial paper and swap markets become harder
BB+	below 12%	above 5.8x	Non-investment grade; many institutional buyers are excluded

Keystone considers a ratio sustained if it is below the threshold for two consecutive years on its own forecast. It treats a hybrid security with the required features (long maturity, deep subordination and the ability to defer interest) as 50% debt and 50% equity for the calculation of FFO to debt.

The regulatory environment. The Commission has a reputation for being constructive on cost recovery but sceptical on returns. In recent years it has awarded ROEs a little below the average of its peers. Its staff routinely recommend lower returns and lower equity ratios than utilities request, and commissions commonly settle somewhere between the staff's and the company's positions. A commission that sees a utility using a thin equity layer tends to respond by imputing a thicker one.

The capital markets. Utility equity is valued on dividend yield, earnings growth and the regulatory environment; the price-to-book ratio of a utility exceeds one if investors believe that the allowed return exceeds the market cost of equity. Utility investors generally dislike equity issuance at a discount to the market price but accept it if it is regular, small relative to market value and clearly linked to a funded rate base. The usual instruments are an at-the-market ("ATM") programme, in which shares are sold gradually into the market through an agent; forward sale agreements, in which the company fixes the price today and delivers shares later; block offerings; and a dividend reinvestment plan. On the debt side, first-mortgage bonds and senior notes are issued across the maturity curve, and junior subordinated hybrids supply equity credit at a higher coupon. Volume 1, Chapter 11 (WACC), Chapter 14 (debt) and Chapter 15 (capital structure) set out the relevant theory; Volume 1, Chapter 13 covers equity issuance.

Peers. Meridian is compared with six regional utilities; their authorised returns, equity ratios, credit metrics and valuation multiples are shown in the next section. The peers' credit metrics are strongly related to their equity ratios and ratings. Meridian is at the lower end of the group on equity ratio and in the middle on FFO to debt.

Financial Data

Fiscal years end 31 December. Figures for 2024 and 2025 are audited; 2026 is management's estimate (ten months actual plus two months forecast). The model conventions are stated with the tables. Tax is 24% of pre-tax income. Meridian pays dividends at a constant 62% payout ratio.

Exhibit 1: Income statement

$ millions	2024A	2025A	2026E
Revenue	3,880.0	4,010.0	4,150.0
Fuel and purchased power (passed through to customers)	1,300.0	1,330.0	1,360.0
Operations and maintenance	975.0	1,000.0	1,030.0
Property and other taxes, and other costs	515.0	530.0	552.0
EBITDA	1,090.0	1,150.0	1,208.0
Depreciation and amortisation	330.0	345.0	360.0
EBIT (operating income)	760.0	805.0	848.0
Interest expense	225.6	237.6	264.0
Pre-tax income	534.4	567.4	584.0
Income tax (24%)	128.3	136.2	140.2
Net income	406.1	431.2	443.9
Shares outstanding (millions)	180.0	180.0	180.0
EPS ($)	2.26	2.40	2.47
Dividends (62% payout)	251.8	267.3	275.2
DPS ($)	1.40	1.49	1.53

Interest for 2026E is calculated on the year-end debt balance of $5,500.0 million at the embedded cost of 4.8%, a simplification used throughout the case. Deferred income taxes (the difference between book and cash tax, arising mainly from accelerated tax depreciation) were $95.0 million in 2024, $100.0 million in 2025 and $120.0 million in 2026E.

Exhibit 2: Balance sheet extracts and capitalisation

$ millions	31 Dec 2025	31 Dec 2026E
Net utility plant	11,290.0	12,130.0
Cash	60.0	55.0
Receivables	480.0	505.0
Fuel and materials inventory	310.0	320.0
Regulatory assets	620.0	650.0
Other assets	340.0	350.0
Total assets	**13,100.0**	**14,010.0**
Accumulated deferred income taxes (ADIT)	1,560.0	1,680.0
Long-term debt	5,000.0	5,500.0
Accounts payable and accruals	770.0	805.0
Regulatory liabilities	410.0	420.0
Other long-term liabilities	579.0	655.0
Common equity	4,781.0	4,950.0
Total liabilities and equity	**13,100.0**	**14,010.0**
Rate base (net plant less ADIT)	9,730.0	10,450.0
Debt / (debt + equity)	51.1%	52.6%
Equity ratio, actual	48.9%	47.4%

Rate base for regulatory purposes is net utility plant less ADIT, and the case ignores working capital allowances. The company's debt has an embedded cost of 4.8%. Book equity is the base on which the earned ROE is measured.

Exhibit 3: Credit metrics

	2024A	2025A	2026E
EBITDA ($m)	1,090.0	1,150.0	1,208.0
Cash interest ($m)	225.6	237.6	264.0
Cash taxes (income tax less deferred) ($m)	33.3	36.2	20.2
FFO = EBITDA − interest − cash taxes ($m)	831.1	876.2	923.9
Year-end debt ($m)	4,700.0	5,000.0	5,500.0
FFO / debt (%)	17.7	17.5	16.8
Debt / EBITDA (x)	4.31	4.35	4.55
EBITDA / interest (x)	4.83	4.84	4.58
Earned ROE on year-end book equity (%)	8.8	9.0	9.0

The earned ROE is net income divided by year-end book equity: 2026E is 443.9 / 4,950.0 = 9.0%, against an authorised 9.6%. The shortfall is regulatory lag. Year-end equity for 2024 was $4,617.1 million.

Exhibit 4: Capital expenditure plan, 2027 to 2031

$ millions	2027	2028	2029	2030	2031	Total
Grid modernisation	520	560	560	500	460	2,600
Transmission	250	280	300	240	230	1,300
Renewables and storage	300	400	480	380	340	1,900
Gas conversion	130	160	180	190	140	800
Other (IT, fleet, facilities)	150	100	80	240	330	900
Total capex	**1,350**	**1,500**	**1,600**	**1,550**	**1,500**	**7,500**

Exhibit 5: Regulatory and forecasting conventions

Item	Convention
Allowed ROE (current order)	9.6%
Authorised equity ratio	50% (debt 50%)
Regulatory cost of debt	4.8% (Meridian's embedded cost)
Allowed after-tax return on rate base	0.50 × 9.6% + 0.50 × 4.8% = **7.20%**
Allowed pre-tax return on rate base	0.50 × 4.8% + 0.50 × 9.6% / (1 − 24%) = **8.716%**
Operating income (EBIT) in year t	8.716% × rate base at the end of year t−1 (one-year regulatory lag)
Depreciation in year t	3.5% of rate base at the end of year t−1
Deferred taxes in year t	10% of capex in year t
Rate base roll-forward	Rate base(t) = rate base(t−1) + capex(t) − depreciation(t) − deferred taxes(t)
Interest	4.8% on the opening balance of existing debt of $5,500.0m, plus the cost of new instruments (half a year on instruments issued in the year)
FFO	EBITDA − interest − (tax − deferred taxes)
Dividends	62% of net income
Equity issuance cost	2.5% of gross proceeds (ATM); issue price $42.00
New senior notes	5.3% pre-tax cost
Legacy debt maturities	Assumed refinanced at the embedded cost of 4.8% (simplification)

Applying these conventions gives the rate base path in the table below. The EBIT row uses the previous year's rate base, which is how the model represents regulatory lag.

$ millions	2026E	2027E	2028E	2029E	2030E	2031E
Rate base, end of year	10,450.0	11,299.2	12,253.8	13,264.9	14,195.6	15,048.8
Depreciation	360.0	365.8	395.5	428.9	464.3	496.8
Deferred taxes	120.0	135.0	150.0	160.0	155.0	150.0
EBIT = 8.716% × prior-year rate base	848.0	910.8	984.8	1,068.0	1,156.1	1,237.3
EBITDA	1,208.0	1,276.5	1,380.3	1,496.9	1,620.4	1,734.1

Exhibit 6: Peer utilities (illustrative)

Utility	Allowed ROE (%)	Authorised equity ratio (%)	FFO / debt (%)	Rating	Beta	Dividend yield (%)	P/E (x)	P/B (x)
Valley Electric	9.5	50	17.2	BBB	0.80	3.9	16.8	1.48
Prairie States Energy	9.8	52	18.4	BBB+	0.78	3.6	18.1	1.62
Harbor Light & Gas	9.3	48	15.6	BBB	0.88	4.2	15.5	1.35
Summit Power	9.7	50	16.4	BBB	0.84	4.0	16.9	1.50
Northgate Utilities	10.0	53	19.1	BBB+	0.74	3.5	19.0	1.70
Lakeshore Energy	9.4	49	15.1	BBB-	0.92	4.6	14.6	1.28
Meridian	**9.6**	**50**	**16.8**	**BBB**	**0.85**	**3.6**	**17.0**	**1.53**

Meridian’s P/E is $42.00 / $2.47 = 17.0x; its dividend yield is $1.53 / $42.00 = 3.6%; its price-to-book ratio is $42.00 × 180.0 / $4,950.0 = 1.53x.

Exhibit 7: Market inputs for the cost of capital

Input	Value
Risk-free rate	4.2%
Equity risk premium	5.0%
Meridian equity beta (regression)	0.85
Share price; shares	$42.00; 180.0 million
Market capitalisation	$7,560.0 million
Debt (book, assumed to equal market)	$5,500.0 million
Cost of new 10-year senior notes	5.3%
Tax rate	24%
Expected 2027 DPS growth (analyst consensus)	about 7%
Long-run dividend growth used in the dividend discount model	5.0%
Meridian’s yield on its own long-term debt, bond-yield-plus method	5.3% plus a premium of 3.8 percentage points (illustrative)

Exhibit 8: Menu of financing instruments

Instrument	Pre-tax cost / price	Keystone treatment	Issue cost	Comment
Senior notes (10-year)	5.3%	100% debt	0.7%	Deep market; supports rating only if metrics hold
Common equity through an ATM programme	$42.00 per share	100% equity	2.5%	Gradual; flexible; modest market impact
Equity forward sale	$42.00, fixed now	Equity when settled	3.0%	Locks in price; settlement in 12 months
Underwritten block of common equity	About 4% below market	100% equity	4.0%	Fast; heavier signal to the market
Junior subordinated hybrid	6.6%	50% debt, 50% equity	1.0%	Tax-deductible; higher coupon than senior debt
Dividend reinvestment plan	$42.00	100% equity	Nil	Small, depends on shareholders electing to reinvest

Exhibit 9: Rate case positions

	Current order	Company draft request	Staff expectation	Plausible range for the order
Allowed ROE	9.6%	10.25%	9.0%	9.0% to 10.25%
Authorised equity ratio	50%	52%	48%	48% to 52%
Test year	Historical (one-year lag)	Historical	Historical	Historical, or partly forward-looking

Complications and Exhibits

The model in the exhibits is deliberately clean. The following facts complicate it, and an adviser must decide which ones to build into the answer.

1. Keystone's adjusted debt. Priya has learned that Keystone does not use the reported debt. It adds $310.0 million for pension deficits and operating leases. Applying the adjustment to 2026E, the ratio of FFO to adjusted debt is 923.9 / (5,500.0 + 310.0) = 15.9%, not 16.8%. The published 15% downgrade threshold is stated in terms of the adjusted ratio. This means that Meridian has less than one percentage point of cushion today, not 1.8 points. Priya has not shared the adjustment with the rest of the finance committee because she believes the agency may soften it if Meridian files a credible plan. (Keystone has made no such statement.)

2. The rate case outcome is uncertain. The company's draft request is 10.25% on a 52% equity ratio. Staff will probably argue for 9.0% on 48%. The final order will be somewhere in between, and might also change the test year. Dr. Osei believes that the Commission would find a request of more than 10% hard to justify given the peer average, and that a request that was seen as aggressive could damage credibility on the other issues, including the recovery of the renewable investments. The model assumes the current 9.6% and 50% throughout; the true outcome affects every ratio.

3. Partly forward-looking rates. A bill before the legislature would allow utilities to use a partly forward-looking test year, so that the return on investments made during the rate-effective year is recognised in rates. If passed, the effect would be to bring revenues closer to the current year's rate base. You may treat the current model as having no forward-looking element (weight 0), and test the effect of a 50% and a 100% weight.

4. A renewable tracker. The company is negotiating a tracker that would allow it to recover the cost of owned solar and storage capacity as it is placed in service, without waiting for a rate case. The Commission has not agreed. The tracker would cover about half of the $1.9 billion renewable programme. It would have a similar (but smaller) effect to Complication 3.

5. The shareholder base. About half of the shareholders are retail investors who hold the stock for the dividend, and Meridian has not cut the dividend since 1957. A cut in the payout would reduce the DPS below the current $1.53 in 2027 for any payout below about 58% (given the 2027 earnings in the model) and would probably be read by income investors as a signal of distress, even if the earnings continue to grow. Equity issuance is also sensitive: some large holders have said that they will tolerate “modest, regular” issuance but not a surprise block.

6. The share price. The share price of $42.00 has been stable for six months. A fall of 15% would increase the number of shares required for any given amount of equity, and utility shares are sensitive to changes in long-term interest rates. The financing plan should therefore be tested at $36.00 and $48.00.

7. Interest rates. The model assumes new senior notes at 5.3%. A rise of one percentage point would raise the cost of new debt to 6.3% and reduce every coverage ratio. The company could hedge part of the 2027 issuance with forward-starting swaps, at a cost that you need not model.

8. Legacy maturities. A $600.0 million series of notes with a 3.6% coupon matures in 2028. The model assumes it is refinanced at the embedded cost of 4.8%, whereas the likely cost is 5.3%; the effect is an extra $3.0 million of annual interest before tax (600.0 × 0.5%). It is small but it is an example of why the embedded cost of debt will drift above 4.8%.

9. Storm costs. The Commission has permitted a regulatory asset of $650.0 million (including storm costs from last year) which earns no return until it is recovered. The model leaves this in the rate base; the company believes that the treatment of this asset may be contested in the case.

10. Data ambiguity. Exhibit 6 shows a Meridian P/B of 1.53x and an allowed ROE of 9.6%, yet the CAPM cost of equity (Exhibit 7) is below the allowed ROE, which means that investors believe the regulatory promise is worth more than the cost of capital. The dividend-discount estimate and the bond-yield-plus estimate give higher costs of equity than the CAPM. A student must decide which estimates to trust, and the case does not say.

Assignment Questions

Answer as Samuel Okonkwo’s adviser. Show every calculation and state assumptions where the case is silent. Use the conventions in Exhibit 5 and the instrument costs in Exhibit 8. Where a question refers to “the model”, use Exhibits 1 to 5 as given.

1. **Allowed return and regulatory lag.** Compute the allowed after-tax and pre-tax returns on rate base from Exhibit 5. Show that the 2026E EBIT of $848.0 million follows from the rate base at the end of 2025. Compute the earned ROE for 2026E, the gap to the allowed ROE and what the gap is worth in dollars of net income. Explain how an increase in leverage above the authorised structure would change the earned ROE.
2. **Cost of equity and the market WACC.** Estimate Meridian’s cost of equity by the CAPM, a dividend discount model and the bond-yield-plus method using Exhibit 7. Compute the market-value WACC. Compare it with the regulatory WACC of 7.20% and with the allowed ROE. Estimate the value created for each $1 of rate base added when the allowed return is 7.20% and the market WACC is the figure you computed, assuming a 30-year straight-line life. Use this to estimate the value of the $7.5 billion programme. What ROE would you recommend requesting in the rate case?

3. **Rating analysis.** Compute the 2026E ratios in Exhibit 3 and then compute the ratio of FFO to Keystone-adjusted debt. Determine how much additional debt Meridian could carry in 2026 before reaching the 15% threshold, on both reported and adjusted bases.
4. **The funding gap.** Build the model for 2027 to 2031 on the assumption that all external funding is raised as senior notes at 5.3%, and the dividend is 62% of net income. Report for each year the rate base, EBIT, interest, net income, FFO, dividends, external funding need, year-end debt, FFO to debt, debt to EBITDA and the equity ratio. In which year does the plan breach the downgrade threshold? What is the earned ROE in 2031?
5. **Equity required.** Using the same model, find the minimum common equity (ATM, at $42.00 with a 2.5% issuance cost) that must be issued each year to hold FFO to reported debt at 16.0% or higher. Report the total, the number of shares and the dilution of existing holders, and the EPS and DPS growth rates from 2026E to 2031E.
6. **Alternative plans.** Compare the equity plan of Question 5 with (a) a plan that uses $500 million of hybrid securities in 2027 and $400 million in 2029, with 50% equity credit, so that the adjusted FFO to debt remains at 16.0%; (b) a plan that lowers the payout ratio to 55%; and (c) a plan that moves the actual equity ratio to the authorised 50%. Summarise the ratios, the equity needed, EPS and DPS in 2031, and the risks of each.
7. **The rate case and regulatory lag.** For each combination of allowed ROE (9.0%, 9.6%, 10.25%) and equity ratio (48%, 50%, 52%), compute the pre-tax allowed return and the equity needed to hold 16.0%. Show also the effect of a partly forward-looking test year (weights 0.5 and 1.0), and the effect of Keystone's adjusted debt (Complication 1) on the equity needed.
8. **The cost of a downgrade.** Estimate the direct annual cost, before and after tax, of a downgrade that widens the spread on new debt by 45 basis points, against the after-tax earnings cost of issuing equity to avoid it. Which is larger? Explain why a regulated utility might still prefer to protect the rating.
9. **Execution.** Compare the ATM, a forward sale and a block for the equity programme. Show the effect on shares issued and 2031 EPS of an issue price of $36.00, $42.00 and $48.00, and the effect of a 100 basis point rise in the cost of new debt on FFO to debt.
10. **Recommendation.** Write the financing plan and board recommendation that you would give the finance committee on 8 December 2026: the financing mix by year, the requested ROE and equity ratio, the dividend policy and the contingencies.

Teaching Note and Full Solutions

Learning objectives

After this case, students should be able to:

1. Explain how an allowed return is built from an authorised capital structure and why the actual financing mix does not set the revenue requirement.
2. Estimate a utility's cost of equity by three methods, compute a market WACC, and compare it with the regulatory return to judge whether a capital programme creates value.
3. Treat credit-rating thresholds as the binding constraint on debt capacity, and build a multi-year financing model that tests FFO-to-debt and leverage ratios.
4. Size the external equity needed to protect a rating, and compare common equity, hybrids, a payout change and a thicker equity ratio.

5. Analyse the effect of the rate case outcome, regulatory lag and rating-agency adjustments on the financing plan.
6. Make and defend a financing and regulatory recommendation that balances the rating, the dividend, shareholder dilution and customer costs.

Suggested class plan (150 minutes)

Segment	Minutes	Content
Opening: what is different about a utility?	10	Cold call: “Who sets Meridian’s return?” Introduce rate base, allowed return and lag
Allowed return, cost of capital, value of the programme (Q1 and Q2)	30	Build the 7.20% and 8.716% returns; the three cost-of-equity methods; the value per $1 of rate base
Ratings and the funding gap (Q3 and Q4)	30	Debt capacity today; the all-debt model; year of breach
Equity and alternative plans (Q5 and Q6)	30	Equity needed; compare hybrid, payout cut and 50% equity ratio
Rate case and downgrade economics (Q7 and Q8)	25	Sensitivities; the cost of a downgrade against the cost of equity
Execution and decision (Q9 and Q10)	15	Instrument choice; the committee vote
Wrap-up	10	The memorandum, alternatives, common errors

Ask the class to vote before Question 5 on the equity they think Meridian needs (none, about $500 million, about $1 billion, about $1.5 billion). The distribution usually surprises students: the answer depends on a ratio target, not on a view about value.

Solution to Question 1: Allowed return and regulatory lag

Allowed returns. After-tax: 0.50 × 9.6% + 0.50 × 4.8% = 4.80% + 2.40% = **7.20%**. Pre-tax: debt interest is deductible, so only the equity return needs grossing up for tax: 0.50 × 4.8% + 0.50 × 9.6% / (1 − 0.24) = 2.400% + 6.316% = **8.716%**.

EBIT. 8.716% × 9,730.0 (end-2025 rate base) = **$848.0 million**, which equals Exhibit 1.

Earned ROE. Net income 443.9 / equity 4,950.0 = **8.97%** (9.0%), compared with 9.6% allowed: a shortfall of 0.63 percentage points, worth 9.6% × 4,950.0 − 443.9 = **$31.3 million** of net income, or 7.1% of earnings.

Decomposition. If rates were set on the end-2026 rate base, EBIT would be 8.716% × 10,450.0 = 910.8 and net income (910.8 − 264.0) × 0.76 = 491.6, an earned ROE of 9.93%. Two effects separate the 9.0% from 9.6%: lag costs 9.93% − 8.97% = **0.96 points**, while the company’s heavier use of debt than the authorised structure adds 9.93% − 9.60% = **0.33 points**. At the authorised structure (equity 5,225.0, debt 5,225.0, interest 250.8) the no-lag net income would be (910.8 − 250.8) × 0.76 = 501.6 and the ROE exactly 9.60%.

Leverage. Replacing equity with debt at a given rate base raises net income by the after-tax interest saved relative to the equity return it displaces. A utility that holds 52.6% debt against an authorised 50% earns a higher ROE on book equity, but its allowed revenue does not change: the extra margin is compensation for risk that the agency prices through the rating. This is the reason the financing plan, not the allowed return, becomes the binding choice.

Solution to Question 2: Cost of equity, market WACC and value of the programme

Cost of equity.

Method	Calculation	Result
CAPM	4.2% + 0.85 × 5.0%	**8.45%**
Dividend discount	D1 / P + g = (1.53 × 1.07) / 42.00 + 5.0% = 3.90% + 5.0%	**8.90%**
Bond-yield-plus	5.3% + 3.8%	**9.10%**
Average		**8.82%**

The peer average allowed ROE is 9.62%, so the allowed ROE of 9.6% exceeds even the highest estimate by 0.5 to 1.2 points. That excess is the source of the 1.53 price-to-book ratio.

Market WACC. E = 42.00 × 180.0 = 7,560.0; D = 5,500.0; V = 13,060.0; E/V = 57.9%, D/V = 42.1%. After-tax cost of new debt = 5.3% × 0.76 = 4.03%. With the CAPM cost of equity: 0.579 × 8.45% + 0.421 × 4.03% = 4.89% + 1.70% = **6.59%**. With the 8.82% average: **6.80%**.

Spread. The regulatory after-tax return of 7.20% exceeds the market WACC by 0.61 points (0.40 points using the average cost of equity). Applied to a rate base of $10.45 billion, that is about $64 million a year (or $42 million), a measure of the annual economic profit that the regulatory contract gives shareholders.

Value per $1 of rate base. A $1 asset with a 30-year straight-line life earns depreciation of 1/30 each year plus 7.20% on the undepreciated balance. The present value at the market WACC is:

Discount rate	PV per $1 of rate base, no lag	Value of $7.5 billion programme (NPV, $m)	PV with one-year lag in recovery	NPV with lag ($m)
6.59%	1.0526	+394.8	0.9876	−93.3
6.80%	1.0340	+255.0	0.9682	−238.8
7.20%	1.0000	0.0	0.9328	−503.7

Without lag, the programme creates $255 million to $395 million, equal to $1.42 to $2.19 per share (255.0 / 180.0 and 394.8 / 180.0). With a one-year delay in earning the return, which is how the model treats regulatory lag, the programme destroys value at these discount rates, because delaying a stream by a year costs about one year of the discount rate (6.6% to 6.8%), more than the 3.4% to 5.3% premium. In practice an allowance for funds used during construction (AFUDC) and trackers recover part of this carrying cost; the case ignores them, but they show why the structure of recovery matters at least as much as the allowed ROE.

Recommended ROE request. The evidence supports a cost of equity of 8.5% to 9.1% and a peer range of 9.3% to 10.0%. A request of 10.25% exceeds the highest peer award and risks credibility, as Dr. Osei says. A defensible request is **10.0% on a 50% equity ratio**, supported by the peer range, the heavy investment programme and the relevance of a premium for execution risk, together with requests for trackers or a partly forward-looking test year. Question 7 shows that this request, if granted, would reduce the equity required by about $190 million relative to the 9.6% case.

Solution to Question 3: Rating analysis

Ratios for 2026E. FFO = 1,208.0 − 264.0 − (140.2 − 120.0) = 923.9. FFO/debt = 923.9 / 5,500.0 = **16.8%**. Debt/EBITDA = 5,500.0 / 1,208.0 = **4.55x**. EBITDA/interest = 1,208.0 / 264.0 = **4.58x**. Adjusted for $310.0 million of pensions and leases: 923.9 / 5,810.0 = **15.9%**; adjusted debt/EBITDA = 4.81x.

Debt capacity. At a 15% threshold, maximum debt is 923.9 / 0.15 = 6,159.2. Reported basis: additional debt of 6,159.2 − 5,500.0 = **$659.2 million**. Adjusted basis: maximum reported debt is 6,159.2 − 310.0 = 5,849.2, giving additional debt of only **$349.2 million**. On the leverage test (5.0x EBITDA), maximum debt is 6,040.0, an additional $540.0 million. Since the 2027 funding need alone is about $668 million, Meridian has no spare capacity: the plan cannot be financed with debt alone.

Solution to Question 4: The funding gap (all-debt plan)

Each year: rate base(t) = rate base(t−1) + capex − depreciation − deferred tax; EBIT = 8.716% × rate base(t−1); interest = 264.0 + 5.3% × (cumulative new debt before the year + half of the year's new debt); tax is 24% of pre-tax income; FFO = EBITDA − interest − (tax − deferred tax); external need = capex + dividends − FFO. New debt equals the need; the interest on new debt is solved iteratively because it depends on the year's issuance.

$ millions	2027	2028	2029	2030	2031
Rate base (end of year)	11,299.2	12,253.8	13,264.9	14,195.6	15,048.8
EBIT	910.8	984.8	1,068.0	1,156.1	1,237.3
EBITDA	1,276.5	1,380.3	1,496.9	1,620.4	1,734.1
Interest	281.7	319.6	361.2	401.4	436.8
Net income	478.1	505.6	537.2	573.6	608.3
FFO	978.9	1,051.1	1,126.1	1,192.8	1,255.2
Capex	1,350.0	1,500.0	1,600.0	1,550.0	1,500.0
Dividends (62%)	296.4	313.5	333.1	355.6	377.2
External funding need	667.6	762.4	807.0	712.8	622.0
Year-end debt	6,167.6	6,930.0	7,737.0	8,449.7	9,071.7
FFO / debt (%)	15.9	15.2	14.6	14.1	13.8
Debt / EBITDA (x)	4.83	5.02	5.17	5.21	5.23
Equity ratio (%)	45.4	43.4	41.7	40.5	39.7
Earned ROE (%)	9.3	9.5	9.7	10.0	10.2

Example for 2027: rate base = 10,450.0 + 1,350.0 − 365.8 − 135.0 = 11,299.2 (depreciation = 3.5% × 10,450.0 = 365.8). EBIT = 8.716% × 10,450.0 = 910.8. Debt need 667.6 gives interest of 264.0 + 5.3% × 0.5 × 667.6 = 281.7. Pre-tax income 629.1; tax 151.0; net income 478.1; FFO = 1,276.5 − 281.7 − (151.0 − 135.0) = 978.9. Need = 1,350.0 + 296.4 − 978.9 = 667.6 (rounding).

Breach. FFO/debt falls below 15% in **2029** (14.6%), the first year in which it is below the threshold, and remains below it in 2030, so a Keystone downgrade would be triggered on its two-year test by 2030, and on the adjusted basis (15.1%, 14.5%, 14.0%, 13.6% and 13.4% for 2027 to 2031) the breach would begin in 2028. Debt to EBITDA exceeds 5.0x from 2028. The cumulative external debt would be $3.57 billion. **Earned ROE** rises to 10.2% in 2031: the cheap debt displaces equity, so the ROE on book equity exceeds the allowed 9.6% even though the allowed revenue is unchanged, which is the temptation of the all-debt plan, and the reason an agency downgrade and a hypothetical-structure order are real risks.

Solution to Question 5: Equity required

Each year, equity is chosen as the smallest amount (gross proceeds at $42.00) for which FFO divided by year-end reported debt is at least 16.0%. Net proceeds are 97.5% of gross. Equity reduces the new debt needed and the interest on it, so the calculation is solved year by year with the interest iteration used in Question 4.

$ millions	2027	2028	2029	2030	2031	Total
Gross equity issued	44.8	275.8	269.7	227.2	170.8	**988.3**
Net equity proceeds (97.5%)	43.7	268.9	263.0	221.5	166.5	963.6
New senior notes	623.5	490.7	537.1	480.5	441.6	2,573.5
Shares issued (millions, at $42.00)	1.07	6.57	6.42	5.41	4.07	23.53
Net income	479.0	512.8	555.3	601.8	644.9	
FFO	979.8	1,058.3	1,144.2	1,221.1	1,291.8	5,695.1
Dividends (62%)	297.0	317.9	344.3	373.1	399.8	1,732.2
Year-end debt	6,123.5	6,614.3	7,151.4	7,631.9	8,073.5	
FFO / debt (%)	16.0	16.0	16.0	16.0	16.0	
Debt / EBITDA (x)	4.80	4.79	4.78	4.71	4.66	
Equity ratio (%)	45.8	46.0	46.1	46.2	46.4	
Earned ROE (%)	9.3	9.1	9.1	9.2	9.2	
Shares outstanding (millions)	181.1	187.6	194.1	199.5	203.5	
EPS ($)	2.65	2.73	2.86	3.02	3.17	
DPS ($)	1.64	1.70	1.77	1.87	1.97	

Sources and uses, 2027 to 2031. Uses: capex 7,500.0 plus dividends 1,732.2 = 9,232.2. Sources: FFO 5,695.1 plus new notes 2,573.5 plus net equity 963.6 = 9,232.2.

Dilution. Meridian issues 23.5 million shares, 13.1% of the 180.0 million existing shares (203.5 million at the end), at a total cost of $24.7 million in issuance fees. EPS grows from $2.47 to $3.17, a compound rate of **5.1%** a year, against 6.5% in the all-debt plan ($3.38), a difference of 6.2% in 2031 EPS. DPS grows from $1.53 to $1.97, also 5.1% a year, so the dividend is maintained and grows. Book value per share is not diluted, since the issue price of $42.00 exceeds the $27.50 book value per share.

The shape of the programme. The equity is back-loaded: only $45 million is needed in 2027 because the starting ratio (16.8%) provides a cushion, but the programme must then run at about $225 million to $275 million a year. This pattern should be explained to investors as “regular, modest” issuance of roughly 3% to 3.5% of market capitalisation a year.

Solution to Question 6: Alternative financing plans

	All debt	A. Common equity (16.0%)	B. Hybrid plus equity	C. Payout 55%	D. 50% equity ratio
Gross common equity ($m)	0	988.3	693.7	787.7	1,515.6
Hybrid ($m)	0	0	900.0	0	0
New senior notes ($m)	3,571.7	2,573.5	1,988.9	2,573.5	2,024.3
Lowest FFO / debt, reported (%)	13.8 (2031)	16.0	15.0 (2029)	16.0	17.5
Lowest FFO / debt, 50% credit for hybrid (%)	13.8	16.0	16.0	16.0	17.5
2031 debt / EBITDA (x)	5.23	4.66	4.84	4.66	4.34
2031 equity ratio (%)	39.7	46.4	44.3	46.4	50.0
2031 shares (millions)	180.0	203.5	196.5	198.8	216.1
2031 EPS ($)	3.38	3.17	3.17	3.24	3.09
2031 DPS ($)	2.10	1.97	1.97	1.78	1.91
2027 DPS ($)	1.65	1.64	1.64	1.46	1.57

Plan B (hybrids). The hybrid securities of $500 million in 2027 and $400 million in 2029 at 6.6% reduce the common equity needed by $294.6 million, to $693.7 million. Because the hybrid coupon (6.6%) is above senior debt (5.3%) but below the 8.8% average cost of equity, the plan is roughly EPS-neutral relative to Plan A (2031 EPS $3.17 in both) but with 7.0 million fewer new shares. Its risk is that it depends on Keystone's 50% equity credit: reported FFO to debt falls to 15.0% in 2029, and interest coverage is lower (EBITDA/interest 4.15x against 4.44x in Plan A in 2029). It also adds $900 million of long-dated, deeply subordinated paper.

Plan C (payout cut). Lowering the payout to 55% reduces the equity need to $787.7 million (a saving of $200.6 million) and lifts 2031 EPS to $3.24, but 2027 DPS of $1.46 is 4.4% below the 2026 level of $1.53, the first cut since 1957, and 2031 DPS is $1.78, 9% below Plan A's. For a stock priced on its yield (3.6%) this would probably cost more in market value than the equity it avoids.

Plan D (50% equity ratio). Raising the equity ratio to the authorised 50% needs $1,515.6 million of equity (216.1 million shares at the end, 12.6 million more than Plan A) and produces the strongest ratios (17.5%, 4.34x), but it is more than the rating requires, it makes the company a larger issuer, and it dilutes EPS growth to 4.6%. It buys protection against a downgrade of two notches that nobody is expecting.

Choice. Plan A, supplemented by a hybrid as a contingency, is the most defensible. It protects the rating with the least change to the dividend, and uses the hybrid only if the rate case or the market goes against the company.

Solution to Question 7: The rate case and regulatory lag

Equity required to hold 16.0% FFO to reported debt ($ millions).

Authorised equity ratio allowed ROE	9.0%	9.6%	10.0%	10.25%
48%	1,365.9	1,098.7	n/a	808.3
50%	1,266.6	988.3	801.7	684.6
52%	1,167.3	877.5	n/a	560.9

Pre-tax allowed returns on rate base at a 50% equity ratio: 8.321% at 9.0%, 8.716% at 9.6%, 8.979% at 10.0% and 9.143% at 10.25%. At 9.6%, the pre-tax return is 8.559% at 48% equity and 8.872% at 52%. **Each 0.1 point of allowed ROE is worth about $46 to $47 million of equity requirement; each 2 points of authorised equity ratio about $110 million.** The range of outcomes from the staff's likely 9.0% on 48% to the company's 10.25% on 52% spans equity needs of $1.37 billion to $0.56 billion. The financing plan must therefore be flexible.

If the order is at the staff position (9.0%, 48%), the equity requirement is $1,365.9 million, which is $377.6 million more than the base case. If the Commission orders 9.6% but cuts the equity ratio to 45% (imputed), the requirement is $1,264.3 million.

Partly forward-looking test year. With a weight of 0.5 on the current year-end rate base the equity needed falls to **$794.9 million**; with a weight of 1.0 it falls to **$599.6 million**, against $988.3 million. At the base plan equity, FFO to debt would be 16.5% to 16.6% (weight 0.5) and 17.0% to 17.1% (weight 1.0), and the earned ROE would rise to about 9.6% to 9.8% and 10.0% to 10.3%. Regulatory lag therefore costs roughly $190 million to $390 million of equity over the plan: the company should treat the test-year question as important as the ROE.

Keystone's adjusted debt. Holding the adjusted FFO to debt (debt plus $310 million) at 16.0% requires **$1,224.4 million** of equity, $236.1 million more than the reported-basis plan; holding it at 15.0% requires $809.7 million. In the base plan the adjusted ratio is 15.2% to 15.4% in 2027 to 2031, which is only 0.2 to 0.4 points above the threshold. This is the key sensitivity of the case: the cushion that appears comfortable on the reported basis is thin on the basis that the agency uses.

Solution to Question 8: The cost of a downgrade

Direct cost. A 45 basis point widening of the spread on new debt raises annual interest by 0.45% × cumulative new debt. In the all-debt plan, cumulative new debt in 2031 is $3,571.7 million, so the annual cost is **$16.1 million before tax, $12.2 million after tax**. If about $550 million of legacy debt is refinanced each year, a further $2,750 million is repriced by 2031, adding $12.4 million before tax. The total is $28.5 million pre-tax or **$21.7 million after tax**.

Cost of avoiding it. The $988.3 million of equity in Plan A replaces debt costing 5.3% before tax, so net income is lower by 988.3 × 5.3% × 0.76 = **$39.8 million** a year by 2031, plus $24.7 million of one-time issuance costs.

Break-even spread. The equity cost equals the downgrade cost when the spread widening is 39.8 / 0.76 / (3,571.7 + 2,750) = **0.83%, or 83 basis points**. The direct cost of a one-notch downgrade is therefore smaller than the earnings cost of the equity needed to avoid it.

Why protect the rating anyway? (i) Regulated debt costs are generally recovered through rates, so the economic burden of a spread increase falls mainly on customers, after a lag, which is a political cost that the company would like to avoid. (ii) The cost of the downgrade is not linear: one notch below BBB- takes the company out of the investment-grade universe, and Meridian would be one adverse rate order from that edge. (iii) A BBB- company has less access to commercial paper, collateral thresholds under swaps and power contracts are tighter, and refinancing in a crisis is harder. (iv) Equity is not a pure cost: the equity portion of rate base earns the allowed ROE, and equity issued at $42.00 against book value of $27.50 per share is accretive to book value. A reasonable answer is to **protect the rating at the margin but not over-insure it**: choose a target (16% reported) that leaves a modest cushion over the adjusted threshold, and use the hybrid or slower dividend growth as contingent tools.

Solution to Question 9: Execution

Choice of instrument. On the $988.3 million programme, the cost of issuance is 2.5% for an ATM ($24.7 million), 3.0% for forward sales ($29.6 million), and, for a block, 4.0% of proceeds plus a price about 4% below the market: the effective loss is $1 - 0.96 \times 0.96 = 7.8\%$, or **$77.5 million** if the whole amount were raised in blocks. The ATM is the cheapest and best suited to a steady, back-loaded programme; forward sales are useful to fix the price for a known need (for example the 2028 requirement) and to align settlement with the date of the order; a block is justified only for a sudden need and is the form that large holders have said they dislike. A mixture is appropriate: ATM as the main channel, with a forward for part of the 2028 and 2029 needs, and the dividend reinvestment plan as a small supplement.

Issue-price sensitivity. Using the equity amounts of Plan A (which are set by the ratio target and so do not change with price):

Issue price ($)	Shares issued (m)	2031 shares (m)	2031 EPS ($)	Change in EPS vs $42
36.00	27.45	207.5	3.11	−1.9%
42.00	23.53	203.5	3.17	
48.00	20.59	200.6	3.22	+1.4%

A 14% lower price raises the share count by 4.0 million and cuts 2031 EPS by about 1.9%. The EPS effect of the issue price is small relative to the effect of the rate case outcome (a 0.65 point change in allowed ROE changes 2031 EPS by about 7.6%, from $3.17 to $3.41). This is a key lesson: the financing decisions are second-order for EPS compared with regulatory decisions, but they are first-order for the credit rating.

Interest rates. If new debt costs 6.3% rather than 5.3%, FFO to debt in the Plan A model falls from a flat 16.0% to 15.96%, 15.89%, 15.83%, 15.78% and 15.73% over 2027 to 2031. Restoring a 16.0% ratio would require $1,094.6 million of equity, $106.3 million more. The 100 basis point increase therefore matters less to the ratio than the adjusted-debt issue, but it is not negligible, and part of the 2027 issuance should be hedged.

Solution to Question 10: Recommendation

The recommendation is in the board memorandum at the end of this section. The structure of a good answer is: (i) state the plan, with amounts by year; (ii) state the requested ROE and equity ratio and say why; (iii) state the dividend policy; (iv) identify the conditions under which the plan changes. The best answers rank the risks: the Keystone adjustment first, the rate case second, the share price third.

Sensitivity discussion

The model is most sensitive to the allowed return and the test year, then to the credit-agency definition, and least to the issue price. In rough terms: **0.1 point of allowed ROE is worth $46 to $47 million of equity; 2 points of authorised equity ratio about $110 million; the adjusted-debt definition about $236 million (at a 16% target); moving to a forward-looking test year worth $190 million to $390 million; and a 100 basis point change in the cost of new debt about $106 million.** The sum of the adverse cases (staff outcome, adjusted debt basis and higher rates) could push the equity requirement above $1.6 billion, while the favourable cases (company ask and a forward-looking year) could bring it well below $0.6 billion. Because the rate case outcome will be known in early 2028, the plan should be organised so that most of the equity is issued after the order, which in the model is true of 2028 onwards. The 2027 requirement ($45 million) is small, which is the key reason a 2027 financing can be done without taking a position on the outcome of the case.

Common student errors

1. **Using the market capital structure for the regulatory return** and the authorised structure for the market WACC; the two serve different purposes.

2. **Forgetting to gross up the equity return for tax** when computing the pre-tax allowed return.
3. **Using end-of-year rate base for EBIT in the same year**, thereby removing regulatory lag.
4. **Treating equity issuance as value destroying** because it dilutes EPS, without noting that equity issued above book value is accretive to book value and earns the allowed return.
5. **Ignoring the interest on the year's own new debt** or counting it for a full year when only half a year applies.
6. **Using reported debt only**, thereby missing the agency's adjustment.
7. **Setting the payout to 55% without computing the DPS in the first year**, which is a cut.
8. **Concluding that an earned ROE above the allowed ROE is a success**, when it reflects leverage above the authorised structure and invites a hypothetical-structure adjustment.
9. **Forgetting issuance costs** (net proceeds are 97.5% of gross for the ATM).
10. **Computing the downgrade cost on the full debt balance immediately**, rather than on new issuance and refinancing as it occurs.

Alternative defensible answers

Accept the downgrade risk. Question 8 shows that the direct cost of a one-notch downgrade is below the earnings cost of the equity that avoids it. A board with a strong view that the Commission will treat debt costs as pass-through could choose a target of 15.0% and issue only $0.5 billion to $0.8 billion of equity. This answer requires a clear view of the non-linear risks (the next notch is junk) and an acceptance of the dividend and the plan being stressed by a bad order.

Use hybrids more heavily. If management believes that Keystone's 50% equity credit will persist, then a larger hybrid programme would minimise dilution. The risk is concentration of exposure to a single agency's methodology and the heavier fixed charges.

Slow the programme. If the rate case is unfavourable, a deferral of the 2030 and 2031 capital expenditure (the Other category rises sharply in those years) would relieve the funding need by hundreds of millions of dollars at a cost to the growth in rate base.

Seek a minority investor in a subsidiary. A utility could sell a minority interest in the transmission business to an infrastructure fund, raising equity-like capital without issuing common stock. The case does not present this option, but it is common in practice and could be raised in discussion.

Board recommendation memorandum

To: Finance Committee, Meridian Power Company **From:** Samuel Okonkwo, Chief Financial Officer **Date:** 8 December 2026 **Re:** Financing plan for the 2027 to 2031 capital programme and rate case position

Recommendation. Approve (1) a financing plan for 2027 to 2031 built on **$988 million of common equity and $2.57 billion of senior notes**, with the equity issued mainly through an at-the-market programme and forward sales; (2) a rate case request of **10.0% ROE on a 50% equity ratio**, together with requests for a partly forward-looking test year and a renewable tracker; and (3) **maintenance of the 62% payout ratio** and no dividend cut during the plan period.

Why. The $7.5 billion capital programme creates value if the return is earned on time (about $255 million to $395 million, or $1.42 to $2.19 per share, at the market WACC) but not if recovery is delayed by a year. Financing it with debt alone would push FFO to debt from 16.8% to 14.6% by 2029 and 13.8% by 2031, and debt to EBITDA to 5.2x, which would trigger a Keystone downgrade. Equity of the amounts below holds FFO to debt at 16.0% each year.

$ millions	2027	2028	2029	2030	2031	Total
Common equity (gross)	45	276	270	227	171	988
Senior notes	624	491	537	481	442	2,574
Year-end FFO / debt (%)	16.0	16.0	16.0	16.0	16.0	

Cost to shareholders. Issuance of 23.5 million shares (13.1% of the shares outstanding, at $42.00) lowers 2031 EPS by about 6% against an all-debt plan. EPS and DPS both grow at about 5.1% a year, and book value per share is not diluted.

Risks. (a) *Keystone's adjusted debt* leaves only 0.2 to 0.4 points of cushion above the 15% threshold on the agency's basis; we will brief the agency before filing and keep a $400 million hybrid programme ready for 2028 to 2029 if needed. (b) *The rate case*: at the staff's likely outcome (9.0% on 48%) the equity requirement rises to $1.37 billion; most of the issuance falls after the order, so the plan can be adjusted. (c) *The share price*: at $36.00 the programme issues 27.5 million shares and 2031 EPS is $3.11. (d) *Interest rates*: a 100 basis point rise in new debt cost would add about $106 million to the equity requirement; we propose hedging part of the 2027 issuance.

Not recommended. A dividend cut to 55% (the DPS would fall to $1.46 in 2027); an all-debt plan; and a 50% actual equity ratio, which exceeds what the rating requires and would raise the equity programme to $1.5 billion.

Next steps. Authorise the ATM programme and a forward-sale agreement; file the rate case on 15 March 2027; brief Keystone in January; and report back to this committee after the order.

Chapter 9: Case: Quantra Systems, Dividend or Buyback?

Quantra Systems, its executives, its shareholders, its competitors and every number in this case are fictional and illustrative. Any resemblance to real companies or people is coincidental. The tax rates are stylized so that the arithmetic is transparent; they are not a statement of any actual tax code. This chapter applies the payout theory of Volume 1, Chapter 16 (dividends, repurchases, signaling and clienteles), the capital structure analysis of Volume 1, Chapter 15, and the market-efficiency and behavioral material of Volume 1, Chapter 12.

The Situation

On Monday, 9 November 2026, Priya Raman-Duval, Chief Financial Officer of Quantra Systems, opened a calendar entry that had been sitting on her screen for three weeks: “Board Capital Committee, Wednesday 9 December, 10:00. Item 1: Use of excess cash.” The committee would decide how to return a large pile of idle cash to shareholders, in what form, and on what timetable. Whatever it chose would be announced alongside fourth-quarter results on 28 January 2027, three days before the window opened in which shareholders could nominate directors for the spring annual meeting.

Quantra Systems is a mid-cap company, headquartered in a mid-sized inland city, that sells analytics software bundled with specialized edge-computing appliances to manufacturers, utilities and logistics operators. Roughly 58 percent of FY2026 revenue is recurring subscription and support; the remainder is appliance sales and implementation services, which are cyclical because customers buy hardware in refresh waves every four to five years. The company was founded in 1998 by Edmund Quantrell, a former equipment engineer, who still chairs the board emeritus and whose family trust holds 14 percent of the shares. The chief executive since 2019 is Dr. Mirela Okonkwo-Hart, a former head of product, who is respected internally for a disciplined product roadmap and criticized externally for an unwillingness to “do something” with the balance sheet.

The balance sheet is the problem, or the opportunity. At the end of FY2024 Quantra held $1.64 billion of cash and short-term investments; at the end of FY2025, $2.05 billion; the FY2026 forecast is $2.60 billion. The company earns about 4.8 percent pre-tax on that cash, far below the 7.5 percent earnings yield the market currently assigns to the equity (EPS of $3.60 on a share price of $48.00). It pays a regular quarterly dividend of $0.20 per share and buys back roughly $200 million of stock per year, which retires about 4 million shares a year against roughly 1.2 million shares issued under employee equity plans (the diluted share count has drifted down from 206.0 million to 200.0 million in two years). Net of those two programs, cash builds by more than half a billion dollars each year.

The people

- **Priya Raman-Duval, CFO.** Eight years at Quantra after a career in corporate banking. Believes in a dividend policy that “management can defend in a downturn.” She has to present the recommendation.
- **Dr. Mirela Okonkwo-Hart, CEO.** Prefers repurchases because she believes the shares are worth well above the market price: her internal DCF puts intrinsic value at about $58 per share. She also wants to keep dry powder for acquisitions.
- **Harold Weiss-Banerjee, Chair of the Capital Committee.** A retired industrial finance executive and the board’s most skeptical voice about buybacks: “Show me why paying $55 for something we say is worth $58 is a good use of money, and show me why it is better than giving the cash back evenly.”
- **Edmund Quantrell, Chair Emeritus.** His family trust has a tax basis of $1.50 per share. He dislikes dividends (“a dividend is the government’s cut of my cash flow”) but also dislikes selling, because the trust’s long-term plan is to hold

the stock until estate planning events.

- **Tomas Eklund, Treasurer.** Responsible for liquidity, the rating agency relationship and the execution plan, whichever instrument is chosen.
- **Meridian Capital Partners, activist fund.** On 14 October 2026 Meridian disclosed a 6.0 percent position (12.0 million shares, bought during the second and third quarters at an average of $41 per share). Its letter demands a $2.5 billion buyback within six months, funded by the cash plus $1.0 billion of new debt, and two board seats. It has said it is prepared to nominate directors at the 2027 annual meeting.

The decision

The Capital Committee has framed four candidate actions and asked the CFO to evaluate them on a common basis, with a recommendation that can be defended to long-term holders, to Meridian, to the rating agencies and to the employees whose bonuses depend on EPS:

1. **Special dividend.** A one-time distribution of $7.50 per share, $1,500 million in total, with no change to the regular dividend.
2. **Open-market repurchase.** A $1,500 million authorization executed over about five months at an expected average price of $52.80 (a 10 percent premium to today's price, reflecting the price pressure of the program and the possibility of a positive announcement effect).
3. **Tender offer.** A modified Dutch auction for $1,500 million at prices between $52 and $58, completed in the first quarter of 2027.
4. **Meridian's proposal.** A $2,500 million tender offer at $56, funded with $1,500 million of cash and $1,000 million of new ten-year senior notes at 6.4 percent.

Underneath those four choices sit the questions that Volume 1, Chapter 16 teaches us to ask. Is the payout a pure financial transaction, or does it carry information? Who among Quantra's shareholders is made better off after tax, and who is made worse off? What does EPS accretion really measure? And how much of the cash is truly "excess" once acquisitions, contingencies, offshore cash and the credit rating are taken seriously?

The deadline

The committee meets on 9 December, with materials due to directors on 2 December. Because the fiscal-year close on 31 December begins the blackout, open-market purchases cannot start before the 28 January earnings release, and a tender launched before then would have to be filed in December. Ms. Raman-Duval has about three weeks to produce a recommendation.

Industry Context

Quantra operates in the industrial analytics and edge-computing segment, which sits between enterprise software and industrial equipment. Three structural features of the industry shape the payout decision.

Cyclicality with a recurring core. Subscription and support revenue is sticky, with gross retention in the low nineties, and carries gross margins above 80 percent. Appliance and services revenue is lumpy: customers defer refresh cycles when capital budgets tighten, and a pause in the cycle can shave 10 to 15 percent off total revenue in a single year while operating costs, particularly research and development, are largely fixed. This asymmetry is why Ms. Raman-Duval insists on testing every payout commitment against a downside year, not merely a base case. It also explains why Quantra's EBIT margin has ranged from roughly 9 percent in a trough year to above 15 percent in a peak year.

Low capital intensity, high cash accumulation. Capital expenditure runs at about 4.7 percent of revenue, and free cash flow exceeds net income by a comfortable margin, so cash builds quickly unless it is deployed. Small analytics acquisitions have been the industry's usual answer, but integration risk has left many peers with idle cash and activist shareholders.

Peer payout practice has diverged. Of the five listed peers shown later, one pays no dividend and relies on repurchases, two combine a low dividend with buybacks, and two pay high dividends with little repurchase. Investors sort into clienteles: growth funds that prefer buybacks, income funds and retail holders that prefer dividends, and event-driven investors who want lumpy returns of capital.

Regulatory and tax environment (stylized). For the purposes of this case, the jurisdiction taxes qualified dividends received by taxable individuals at 24 percent and long-term capital gains at 20 percent. Tax-exempt institutions and employee retirement plans pay neither tax. Foreign institutions suffer a 15 percent withholding tax on dividends and are not taxed on capital gains. Mutual funds and ETFs are tax-transparent, and their end-investors, a mix of taxable and tax-deferred accounts, face blended effective rates of 12 percent on dividends and 8 percent on gains. The corporate tax rate is 21 percent. The jurisdiction also levies a 1 percent excise tax on the value of net share repurchases, payable by the company, and a special-purpose "offshore cash" regime under which cash moved from foreign subsidiaries to the parent is subject to a 5 percent leakage cost. Dividends and repurchases are both permitted from retained earnings, and the company has ample distributable reserves.

Financial Data

All figures in US$ millions except per-share data. Fiscal years end 31 December. FY2026E combines nine months of actual results with a three-month forecast. Quantra's diluted share count and basic share count are treated as identical at 200.0 million at the date of the case.

Exhibit 1: Income statement

US$ millions	FY2024A	FY2025A	FY2026E
Revenue	5,120	5,580	6,000
Cost of revenue	1,741	1,925	2,100
Gross profit	3,379	3,655	3,900
Research and development	1,020	1,110	1,200
Selling, general and administrative	1,545	1,665	1,780
Amortization of acquired intangibles	55	60	60
Operating income (EBIT)	759	820	860
Interest income	88	104	112
Interest expense	58	60	60
Pre-tax income	789	864	912
Income tax	166	181	192
Net income	623	683	720
Diluted shares (millions)	206.0	203.0	200.0
Diluted EPS (US$)	3.02	3.36	3.60
Dividends per share (US$)	0.72	0.76	0.80
Depreciation and amortization (included above)	270	285	300
EBITDA	1,029	1,105	1,160
Gross margin	66.0%	65.5%	65.0%
EBIT margin	14.8%	14.7%	14.3%

Exhibit 2: Balance sheet extract

US$ millions	31 Dec 2025A	31 Dec 2026E
Cash and short-term investments	2,050	2,600
Accounts receivable	1,010	1,090
Inventory and other current assets	260	280
Property, plant and equipment, net	1,150	1,170
Goodwill and intangibles	1,780	1,720
Other assets	410	440
Total assets	**6,660**	**7,300**
Accounts payable and accrued liabilities	720	770
Deferred revenue	980	1,060
Senior notes (face value)	1,200	1,200
Other long-term liabilities	330	270
Total liabilities	**3,230**	**3,300**
Shareholders' equity	**3,430**	**4,000**
Total liabilities and equity	6,660	7,300

The equity roll-forward for FY2026E is: opening equity 3,430, plus net income 720, plus stock-based compensation 210, less dividends 160, less repurchases 200, equals 4,000.

Exhibit 3: Cash flow and distributions

US$ millions	FY2024A	FY2025A	FY2026E
Net income	623	683	720
Depreciation and amortization	270	285	300
Stock-based compensation	180	195	210
Increase in net working capital	(30)	(38)	(40)
Cash flow from operations	1,043	1,125	1,190
Capital expenditure	(240)	(260)	(280)
Free cash flow (after interest and tax)	**803**	**865**	**910**
Dividends paid	(148)	(154)	(160)
Share repurchases	(150)	(200)	(200)
Acquisitions and other	(50)	(101)	0
Net change in cash	455	410	550
Ending cash and short-term investments	1,640	2,050	2,600

Exhibit 4: Management forecast, status quo (before any excess-cash distribution)

US$ millions	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Revenue	6,000	6,420	6,837	7,248	7,646
Revenue growth	7.5%	7.0%	6.5%	6.0%	5.5%
EBIT	860	950	1,039	1,123	1,200
EBIT margin	14.3%	14.8%	15.2%	15.5%	15.7%
EBITDA	1,160	1,270	1,379	1,483	1,580
Net income	720.0	791.7	862.1	928.5	989.4
EPS on 200.0m shares (US$)	3.60	3.96	4.31	4.64	4.95
Capital expenditure	280	300	320	340	360
Free cash flow before distributions	910.0	986.7	1,067.1	1,138.5	1,209.4

Forecast conventions. Net interest is held at the FY2026E level (interest income 112, interest expense 60) and the tax rate at 21 percent, so the forecast isolates operating growth. The share count is held at 200.0 million, which strips out both the continuing $200 million annual program and employee issuance; scenarios in this case are compared on that common base. Stock-based compensation is assumed to rise from 210 to 250. Free cash flow equals net income plus D&A plus stock-based compensation less capital expenditure less the increase in working capital.

Exhibit 5: Market data and valuation snapshot (2 November 2026)

Item	Value
Share price	$48.00
52-week range	$39.80 to $57.10
Shares outstanding	200.0 million
Equity market capitalization	$9,600 million
Net debt (debt 1,200 less cash 2,600)	-$1,400 million (net cash)
Enterprise value	$8,200 million
P/E on FY2026E EPS	13.3x
EV/EBITDA on FY2026E	7.1x
Regular dividend yield ($0.80)	1.67%
Buyback yield ($200m / $9,600m)	2.08%
Equity beta (5-year monthly)	1.15
Average daily trading volume	1.9 million shares ($91 million)
Management's intrinsic value estimate	about $58 per share
Sell-side target prices (12 analysts)	low $44, median $54, high $63
Sell-side ratings	5 buy, 6 hold, 1 sell
Analysts' preferred capital return	7 buyback, 3 special dividend, 2 neutral

Exhibit 6: Quarter-end share price history (US$)

Date	Dec 2024	Mar 2025	Jun 2025	Sep 2025	Dec 2025	Mar 2026	Jun 2026	Sep 2026	2 Nov 2026
Price	41.20	44.60	49.80	55.40	52.10	50.30	46.80	47.30	48.00

Exhibit 7: Peer group (all fictional; trailing data as of 2 November 2026)

Company	P/E (x)	EV/EBITDA (x)	Dividend yield	Buyback yield	Net debt/EBITDA (x)	Payout approach
Helix Data Systems	17.2	10.4	1.9%	2.8%	0.6	Steady dividend plus continuous buyback
Corvane Software	19.8	12.9	0.0%	4.1%	(0.8)	Buybacks only
Ostrel Analytics	15.1	9.2	2.6%	1.2%	1.1	Dividend-led, opportunistic buyback
Pembrook Digital	14.6	8.8	3.1%	0.5%	1.4	High dividend, token buyback
Tallis Networks	12.4	7.6	3.8%	0.0%	0.9	High dividend only
Peer median	**15.1**	**9.2**	**2.6%**	**1.2%**	**0.9**	
Quantra Systems	13.3	7.1	1.67%	2.08%	(1.2)	Low dividend plus steady buyback

Exhibit 8: Shareholder register by clientele

Clientele	% of shares	Shares (millions)	Tax on dividends	Tax on capital gains	Average basis per share
Founders' family trust	14%	28.0	24%	20%	$1.50
Retail direct, taxable	15%	30.0	24%	20%	$36.00
Tax-exempt institutions and employee plans	32%	64.0	0%	0%	not applicable
Mutual funds and ETFs (blended end-investor rates)	24%	48.0	12%	8%	$40.00
Foreign institutions	9%	18.0	15% (withholding)	0%	$38.00
Meridian Capital Partners (activist)	6%	12.0	24%	20%	$41.00
Total	**100%**	**200.0**			

Exhibit 9: Liquidity position and the Treasurer's excess-cash build (31 December 2026E)

Item	US$ millions	Comment
Cash and short-term investments	2,600	Of which 1,700 held in the United States and 900 held in foreign subsidiaries
Less: minimum operating cash	(600)	10% of revenue; 400 domestic and 200 foreign
Less: Verdana Analytics acquisition	(300)	Signed, closes Q1 2027, paid in cash from the parent
Less: contingency buffer	(200)	100 domestic and 100 foreign
Indicated excess cash	**1,500**	Treasurer's figure, before offshore leakage and the March 2027 maturity

Exhibit 10: Debt and rating metrics

Instrument	Face value	Coupon	Maturity	Annual interest
Senior notes A	$400m	4.6%	March 2027	$18.4m
Senior notes B	$800m	5.2%	June 2031	$41.6m
Total	**$1,200m**	**5.0% average**		**$60.0m**

The notes carry a covenant of net debt to EBITDA of 3.0x or less and EBIT to interest expense of 4.0x or more. The company is rated BBB+/Baa1; the agencies have publicly indicated that gross debt to EBITDA above 2.5x would put the rating under review. The Treasurer assumes the $400 million March 2027 maturity is refinanced at 6.0 percent, but no commitment has been signed. New ten-year senior notes could be issued today at 6.4 percent.

Complications and Exhibits

The following facts surfaced during the Treasurer's review and the investor roadshow. Some are firm; some are ambiguous, and students are expected to say which.

Exhibit 11: Pre-launch tender offer indications

Quantra’s financial adviser polled holders confidentially (without disclosing price ranges to anyone with inside information) to estimate how many shares would be offered at each price in a modified Dutch auction. The adviser describes the numbers as “indicative, not binding, and probably optimistic at the low end.” Tendering of at least the first 12.0 million shares is believed to be Meridian.

Price (US$)	Cumulative shares offered at or below this price (millions)
52	12
53	19
54	26
55	31
56	37
57	43
58	49

Under the proposed terms, the company would purchase $1,500 million of shares at a single uniform clearing price, defined as the lowest price at which the offer is fully subscribed, with pro-ration among shares tendered at or below the clearing price. The Edmund Quantrell family trust has said informally that it will not tender at any price in the range.

Exhibit 12: The annual bonus plan

The FY2027 annual bonus for the CEO, CFO and eight other officers is based on diluted EPS. The CEO’s target bonus is $2.4 million; the plan is linear between points and capped at the maximum.

Diluted EPS, FY2027	Payout as percent of target
Below $3.80	0%
$3.80 (threshold)	25%
$4.00 (target)	100%
$4.20 or above (maximum)	200%

A three-year performance share plan is based on compound annual growth in diluted EPS. The Compensation Committee has not yet addressed whether capital-return actions are to be neutralized.

Exhibit 13: Meridian Capital Partners’ letter of 14 October 2026 (summary)

Meridian argues that Quantra is under-levered, that the share price is depressed because the market assigns zero credit to future capital allocation, and that a $2.5 billion repurchase funded by $1.5 billion of cash and $1.0 billion of ten-year debt at 6.4 percent would raise EPS by roughly 10 percent. It asks for two board seats and an independent capital-allocation review. It states that it will accept a smaller program if the board commits to a multi-year leverage target of at least 1.5x gross debt to EBITDA. Meridian’s own average cost is $41 per share, and it acquired its stake after the January 2026 low.

Further complications

1. **Offshore cash.** Of the $2,600 million of cash, $900 million sits in foreign subsidiaries. The domestic requirement for a $1,500 million distribution is the distribution itself plus the $300 million Verdana payment, the $400 million domestic operating minimum and the $100 million domestic buffer: $2,300 million in total against $1,700 million of domestic cash. About $600 million must therefore be repatriated, at a 5 percent leakage cost. The Treasurer’s excess-cash figure does not reflect this $30 million cost.

2. **The March 2027 maturity.** If the $400 million of 4.6 percent notes are repaid from cash instead of refinanced, gross debt and cash both fall by $400 million, net debt is unchanged, but the Treasurer's "excess" would be recomputed as $1,100 million on a gross basis. Some directors read the Treasurer's 1,500 as a net figure; others as gross. The treatment of this item is ambiguous in the committee papers.
3. **Verdana Analytics.** The $300 million acquisition has a purchase price adjustment clause that could add up to $80 million if Verdana meets earn-out targets in 2027 and 2028. The earn-out is not in the Treasurer's table.
4. **Litigation.** A patent claim (the Brandt matter) has a reasonably possible loss range of $60 million to $150 million, unreserved. The contingency buffer of $200 million is meant to cover it among other things, but the General Counsel advises that a loss could arrive in the same quarter as a cyclical downturn.
5. **Excise tax on repurchases.** The 1 percent excise tax applies to the open-market program and the tender (net of any issuance by the company during the year). It does not apply to dividends. The Treasurer's EPS analysis ignores it. Employees' share issuance of about 1.2 million shares a year (worth roughly $58 million) would offset a small portion of the base.
6. **Price pressure and execution.** Average daily volume is 1.9 million shares. Limiting the company to 15 percent of daily volume implies the open-market program would take about 100 trading days, so the assumed 10 percent premium is a judgment, not a measurement.
7. **Dividend history.** The company has raised its dividend every year since 2012, by 4 to 6 percent annually, and never cut it. A "dividend growth" investor mandate that two large fund holders follow requires uninterrupted increases.
8. **Basis data.** The register's average tax bases come from a custodian survey and are rough. In particular, the 14 percent held by the founders' trust has a basis so low that almost the whole proceeds of any sale would be gain; the trust also has a stated intention to hold until an estate-planning event, at which point (under the stylized code) gains on shares held at death are not taxed.
9. **Disclosure.** An authorization to repurchase shares is not an obligation. In the last three years Quantra completed 90 percent of its announced authorization on average, but the committee papers show that two of its peers completed under half of theirs.

Assignment Questions

Answer all ten questions. Use the stylized tax rates and the data in Exhibits 1 to 13. Where the case is ambiguous, state your assumption and show how the answer changes under the main alternative.

Question 1: Excess cash and capacity. (a) Reconstruct the Treasurer's excess-cash figure and adjust it for the offshore leakage, the March 2027 maturity and the Verdana earn-out. (b) Test the post-distribution balance sheet against the covenants and the rating guardrail, for FY2026E and for the FY2027E base case. (c) Estimate the cumulative cash available for distributions to shareholders (the regular dividend and all repurchases) over FY2027E and FY2028E, first with no new debt and then with gross debt held at or below 1.0x EBITDA at the end of FY2028E. Ignore changes in interest income.

Question 2: The Modigliani-Miller benchmark. Assume no taxes and no information effects. (a) For a holder of 1,000 shares, show that the special dividend of $7.50 per share and a $1,500 million repurchase at $48 leave wealth unchanged. (b) Calculate the ex-dividend price and the post-repurchase price in each case. (c) Explain what each assumption of the benchmark leaves out.

Question 3: EPS accretion. (a) Calculate pro forma FY2026E EPS under the special dividend and under repurchases at $48.00, $52.80 and $54.00, with foregone interest of 4.8 percent pre-tax and a 21 percent tax rate. (b) Derive the break-even repurchase price at which accretion is zero. (c) Show why a constant P/E assumption overstates the price effect. (d) Evaluate Meridian's debt-financed $2,500 million tender at $56 on EPS and credit metrics.

Question 4: Value transfer. Using Exhibit 5 and the data above, (a) derive the per-share value to continuing holders after a $1,500 million repurchase as a function of intrinsic value V and purchase price P. (b) Compute the grid for V of $44, $48, $52, $58 and $64 and P of $48.00, $52.80, $54.00 and $55.00. (c) Compute the probability-weighted gain for continuing holders under the committee's three scenarios (25 percent at $44, 45 percent at $52, 30 percent at $64).

Question 5: Taxes and clienteles. (a) For each clientele in Exhibit 8, compute after-tax cash per share held under the $7.50 special dividend and under a pro rata tender at $54. (b) Compute the aggregate tax paid by shareholders in each case. (c) Compute the ex-dividend price drop predicted if the marginal investor is a taxable individual. (d) Discuss which holders are hurt by each choice, and how the founders' basis and the estate-planning intention affect the comparison.

Question 6: Signaling and dividend policy. (a) Using the Lintner model with a 35 percent target payout on FY2027E EPS and an adjustment speed of 0.30, find the dividend implied for next year. (b) Compare what a special dividend, a regular dividend increase and a repurchase each signal, and which is costly to mimic. (c) Using a prior that 28.6 percent of firms in Quantra's position are undervalued, compute the post-announcement price for mimic probabilities of 0, 0.1, 0.3, 0.6 and 1.0.

Question 7: Tender offer design. (a) From Exhibit 11, find the clearing price and the pro-ration factor in the modified Dutch auction. (b) Compute post-offer share counts, Meridian's position and the founders' ownership percentage under the tender and under open-market buying at $52.80. (c) Compare open-market, fixed-price tender and Dutch auction on cost, certainty, speed and the effect on activism.

Question 8: Compensation and agency. (a) Compute FY2027E pro forma EPS under each of the special dividend, the open-market program and the tender, and the resulting CEO bonus under Exhibit 12. (b) Explain the incentive problem. (c) Propose a neutralization method and calculate neutralized EPS.

Question 9: Stress test and policy flexibility. (a) Build FY2027E downside and severe cases and test free cash flow coverage of the dividend and buyback commitments in a hybrid program. (b) Test Meridian's debt-financed program against the covenants and the rating guardrail in the same cases. (c) Design a flexible payout framework with a base dividend and variable repurchases.

Question 10: Recommendation. Recommend a payout structure, size and timetable. Address (a) the response to Meridian, (b) the credit rating, (c) how the structure is communicated to the clienteles in Exhibit 8, and (d) what you would want to know that the case does not say. Draft a one-page board memorandum.

Teaching Note and Full Solutions

Learning objectives

After working through this case, students should be able to:

1. Distinguish genuinely excess cash from idle cash, using a cash-need schedule that includes operating minimums, committed acquisitions, contingencies and repatriation frictions.
2. State the Modigliani-Miller payout irrelevance proposition and identify which assumption each of taxes, information, agency and transaction costs relaxes (Volume 1, Chapter 16).
3. Compute EPS accretion, derive the break-even repurchase price, and explain why accretion is an accounting outcome and not a value outcome.

4. Quantify the value transfer between selling and continuing shareholders, and analyze investor clienteles by after-tax cash.
5. Calibrate a regular dividend with the Lintner model, compare signals by the cost of mimicry, and design a tender offer.
6. Recognize the agency problem in EPS-based pay, stress-test a leveraged proposal against covenants, and write a board recommendation that is robust to uncertainty about intrinsic value.

Suggested class plan (120 minutes)

Minutes	Segment	Activity
0 to 10	Opening poll	“Dividend, buyback or neither?” Record the vote on the board; revisit at the end.
10 to 25	Excess cash (Question 1)	Cold-call a student to reconstruct the Treasurer’s number; surface the offshore, maturity and earn-out adjustments.
25 to 40	Benchmark and EPS (Questions 2 and 3)	One student at the board derives the break-even price; the class discusses why a 7 percent accretion is not a 7 percent gain.
40 to 55	Value transfer (Question 4)	Two students take the roles of the CEO (V = $58) and the sell-side skeptic (V = $44) and argue the same transaction.
55 to 60	Break	
60 to 75	Taxes and clienteles (Question 5)	Break the room into five clientele groups, each computing its own after-tax proceeds; reconvene to tally the winners and losers.
75 to 90	Signaling and tender design (Questions 6 and 7)	Run the Bayesian update; construct the Dutch auction clearing price live.
90 to 105	Incentives and stress test (Questions 8 and 9)	Compute the bonus consequences; test Meridian’s proposal in the downside case.
105 to 120	Recommendation (Question 10)	Three teams present 90-second recommendations; the instructor reveals the sample memorandum; repeat the poll.

Instructors with a 75-minute session can drop Question 6(c), Question 7(c) and Question 8(c), and assign them as written work.

Solution to Question 1: Excess cash and capacity

(a) Reconstructing and adjusting the Treasurer’s figure. The Treasurer’s figure follows from Exhibit 9:

```
Cash and short-term investments            2,600
less minimum operating cash (10% x 6,000)   (600)
less Verdana acquisition                    (300)
less contingency buffer                     (200)
Indicated excess cash                       1,500
```

Three adjustments are needed.

Offshore leakage. A $1,500 million distribution must come from the parent. The parent’s domestic need is 1,500 (distribution) + 300 (Verdana) + 400 (domestic operating minimum) + 100 (domestic buffer) = 2,300. Domestic cash is 1,700, so 2,300 less 1,700 = 600 must be repatriated, at 5 percent leakage = 30.

Verdana earn-out. The contingent consideration of up to 80 is a real claim on cash within 24 months and belongs in the cash-need schedule.

Circularity. Because the leakage depends on the distribution D, the exact solution is found from the identity 2,600 = D + 300 + 80 + 600 + 200 + 0.05 x (D less 820), where the repatriation need is D + 880 - 1,700 = D - 820:

```
2,600 = D + 1,180 + 0.05D - 41
1.05 D = 1,461
D = 1,391.4
```

The first-order shortcut, 1,500 - 30 - 80 = 1,390, is accurate to within $2 million. The adjusted excess cash is therefore about **$1,390 million** on the refinancing reading.

March 2027 maturity. If the $400 million notes are repaid from cash, the gross excess falls to 1,100 before the other adjustments (about 990 after leakage and the earn-out). This is a choice among claimants, not a computational difference: $400 million goes either to shareholders (debt refinanced) or to bondholders.

Reading	Distribution to shareholders	Gross debt after	Cash after Verdana	Net debt after
A: refinance the notes	1,390	1,200	880	320
B: repay the notes	990	800	880	(80)

Reading B leaves net cash; reading A leaves a small net debt (0.28x EBITDA). Given a 6.0 percent refinancing cost against 4.8 percent earned on cash, reading A costs about 0.012 x 400 x 0.79 = $4 million a year after tax in carry. The committee should adopt reading A and record that the figure is net of the March 2027 refinancing.

(b) Covenants and rating guardrail. After distributing 1,390 and paying leakage of 30, cash at the 31 December 2026 balance date is 2,600 - 1,390 - 30 = 1,180, falling to 880 after the Verdana payment in Q1 2027.

Test	FY2026E basis	FY2027E base	Limit
Gross debt / EBITDA	1,200 / 1,160 = 1.03x	1,200 / 1,270 = 0.94x	2.5x (rating review)
Net debt / EBITDA (after Verdana)	320 / 1,160 = 0.28x	320 / 1,270 = 0.25x	3.0x (covenant)
EBIT / interest expense	860 / 60 = 14.3x	950 / 60 = 15.8x	4.0x (covenant)
Gross debt capacity at 2.5x	2,900 (headroom 1,700)	3,175 (headroom 1,975)	

All tests pass comfortably in the base case; the base case is not the binding case, and Question 9 repeats the test in the downside.

(c) Cumulative cash available over FY2027E and FY2028E. Start with cash of 2,600 and add free cash flow before distributions of 986.7 for FY2027E and 1,067.1 for FY2028E (Exhibit 4). Subtract Verdana (300) and the earn-out (80). The ending floor at 31 December 2028 is 10 percent of FY2028E revenue (683.7) plus the 200 buffer = 883.7.

```
2,600.0 + 986.7 + 1,067.1 - 300 - 80 - 883.7 = 3,390.1
```

Quantra can therefore distribute about **$3,390 million** over two years with no new debt, before leakage and ignoring interest lost on distributed cash. Allowing gross debt of 1.0x FY2028E EBITDA (1,379) adds 179 of new debt for **$3,569 million**. Net of the regular dividend and continuing program (2 x 360 = 720), incremental capacity is about $2,670 million. The question is timing and form, not whether capacity exists.

Solution to Question 2: The Modigliani-Miller benchmark

(a) Wealth equivalence for 1,000 shares. Before any action, equity value is 200.0 x 48 = 9,600, and 1,000 shares are worth $48,000.

Special dividend. The company pays 7.50 per share, $1,500 million in total. The ex-dividend equity value is 9,600 - 1,500 = 8,100, so the ex-dividend price is 8,100 / 200 = $40.50. The holder has 1,000 x 7.50 = $7,500 in cash and 1,000 x 40.50 = $40,500 in stock: **$48,000**.

Repurchase at $48. The company buys 1,500 / 48 = 31.25 million shares, leaving 168.75 million. Equity value is 8,100 and the price remains 8,100 / 168.75 = $48.00. The company bought 31.25 / 200 = 15.625 percent of each holder's shares if the repurchase is pro rata. The holder sells 156.25 shares for $7,500 and keeps 843.75 shares worth 843.75 x 48 = $40,500: **$48,000**. A holder who does not tender holds 1,000 shares worth $48,000 and owns a larger fraction of the company (1,000 / 168.75 million versus 1,000 / 200 million), but the same wealth.

(b) Prices. Ex-dividend price: $40.50 (a fall of exactly $7.50). Post-repurchase price: $48.00 (unchanged). The distinction is crucial: the *per-share price drop* is a feature of the dividend, not a loss of value. A student who reads "the price falls 15.6 percent" as a loss has confused price with wealth.

(c) What the benchmark omits. Taxes (Question 5); information, because a repurchase away from intrinsic value transfers wealth (Question 4) and may signal (Question 6); agency, because EPS-linked pay biases managers toward repurchases (Question 8); and transaction costs, which support clienteles.

Solution to Question 3: EPS accretion

(a) Pro forma FY2026E EPS. Baseline: net income 720, 200.0 million shares, EPS $3.60. Using $1,500 million of cash costs foregone interest of 1,500 x 4.8% = $72.0 million pre-tax, or 72.0 x (1 - 0.21) = **$56.88 million after tax**. Pro forma net income is 720 - 56.88 = 663.12 in every option.

Option	Shares repurchased (m)	Shares after (m)	Net income	EPS (US$)	Accretion
Special dividend $7.50	0	200.00	663.12	3.32	-7.9%
Repurchase at $48.00	31.250	168.75	663.12	3.93	+9.2%
Repurchase at $52.80	28.409	171.59	663.12	3.86	+7.3%
Repurchase at $54.00	27.778	172.22	663.12	3.85	+6.9%
Repurchase at $52.80 with 1% excise tax	28.409	171.59	662.55	3.86	+7.3%

For the excise-tax row the extra cash cost is 15 (1 percent of 1,500), which costs a further 15 x 4.8% x 0.79 = 0.57 of net income. Excise tax is therefore a real cash cost of $15 million but a negligible EPS effect (about 0.1 percent of EPS). The dividend is dilutive to EPS because the lost interest reduces earnings but the share count is unchanged. Arithmetic example: 663.12 / 171.59 = 3.8645; 3.8645 / 3.60 - 1 = 7.35 percent.

(b) Break-even price. Let c be the after-tax yield on cash (0.048 x 0.79 = 0.03792), E net income, N shares, and X dollars used. Pro forma EPS exceeds current EPS when

```
(E - cX) / (N - X/P) > E / N
N(E - cX) > E(N - X/P)
-N c X > -E X / P
E / (N P) > c
```

so accretion requires that the repurchase **earnings yield** EPS / P exceed the after-tax cash yield, and the break-even price is P* = EPS / c = 3.60 / 0.03792 = **$94.94**. At the current price the earnings yield is 3.60 / 48.00 = 7.50 percent, nearly double the 3.79 percent after-tax yield on cash, so every price below $94.94 is accretive. The break-even tells us the test is nearly meaningless: Quantra could repurchase at $90 and call it accretive. Accretion measures only the spread between the earnings yield and the after-tax yield on cash; it says nothing about whether $90 is a good price.

(c) The constant P/E fallacy. A naive analyst multiplies the old P/E of 48.00 / 3.60 = 13.33 by the new EPS of 3.8645 to forecast a price of $51.53, a 7.4 percent gain. The error: the old P/E blended the market's valuation of operations and of cash, with cash valued at about $1 per dollar but earning only 3.79 percent after tax (a P/E of 26 on cash earnings). Removing low-yielding cash *lowers* the blended multiple. The value-consistent calculation is: equity value after the repurchase is 9,600 - 1,500 = 8,100; shares are 171.59; price = 8,100 / 171.59 = **$47.21** (if purchased at $52.80), P/E = 47.21 / 3.8645 = **12.2x**. If purchased at $48, the price is exactly $48.00 and the P/E is 48.00 / 3.9296 = 12.2x. A buyback re-rates the stock only if it changes the market's view of cash flows, risk or governance, not by arithmetic.

(d) Meridian's debt-financed tender. Meridian proposes $2,500 million at $56: shares repurchased = 2,500 / 56 = 44.64 million, leaving 155.36 million.

```
Lost interest income on 1,500 of cash, after tax:   1,500 x 4.8% x 0.79 = 56.88
Interest on 1,000 new notes, after tax:             1,000 x 6.4% x 0.79 = 50.56
Pro forma net income: 720 - 56.88 - 50.56 = 612.56
EPS = 612.56 / 155.36 = 3.94, accretion +9.5%
```

Meridian's "roughly 10 percent" is right. But compare with a cash-only tender of $1,500 million at $56 (26.79 million shares, EPS 663.12 / 173.21 = 3.83, +6.3 percent): the extra $1,000 million of debt adds only 3.2 percentage points of accretion. The spread is thin: the incremental earnings yield is 3.60 / 56 = 6.43 percent against an after-tax debt cost of 6.4 x 0.79 = 5.06 percent, a margin of 1.4 points on $1,000 million, about $14 million of earnings a year, bought with permanent exposure to the downside. Credit metrics: gross debt 2,200 / FY2026E EBITDA 1,160 = 1.90x (1.73x on FY2027E); net debt (2,200 - 1,100) / 1,160 = 0.95x; EBIT / interest expense = 860 / 124 = 6.9x. These are acceptable in the base case, but the guardrail at 2.5x gross has a cushion of only 0.6x, which is eliminated by a 24 percent fall in EBITDA. Question 9 shows that the downside case breaches.

Solution to Question 4: Value transfer

(a) Derivation. Let V be intrinsic value per share, N = 200.0 million, X = $1,500 million, P the price paid, and n = X / P shares retired. Intrinsic equity value before is V x N. After paying X in cash for n shares, it is V x N - X, spread over N - n shares:

```
V' = (V x N - X) / (N - n)
Gain per continuing share = V' - V = n (V - P) / (N - n)
```

The gain is positive only when P < V. The total transfer to continuing holders is n(V - P), and sellers lose exactly that amount relative to holding. A repurchase is a zero-sum bet between sellers and stayers.

(b) Grid. Values of V' (percentage change from V in parentheses):

V	P = $48.00	P = $52.80	P = $54.00	P = $55.00
$44	43.26 (-1.7%)	42.54 (-3.3%)	42.39 (-3.7%)	42.26 (-4.0%)
$48	48.00 (0.0%)	47.21 (-1.7%)	47.03 (-2.0%)	46.89 (-2.3%)
$52	52.74 (+1.4%)	51.87 (-0.2%)	51.68 (-0.6%)	51.53 (-0.9%)
$58	59.85 (+3.2%)	58.86 (+1.5%)	58.65 (+1.1%)	58.47 (+0.8%)
$64	66.96 (+4.6%)	65.85 (+2.9%)	65.61 (+2.5%)	65.42 (+2.2%)

Worked cell (V = $58, P = $52.80): n = 28.409; V’ = (11,600 - 1,500) / 171.591 = 58.86; gain = 28.409 x 5.20 / 171.591 = $0.86.

The break-even is P = V. At the CEO’s own estimate of $58, a repurchase at $55 creates only $0.47 per continuing share (n(V - P) = 27.27 x 3 = $81.8 million, divided by 172.73 million shares). A one percent change in the purchase price moves value by about as much as the whole gain.

(c) Probability-weighted. With weights 25/45/30 on V = 44, 52, 64, expected V = 0.25 x 44 + 0.45 x 52 + 0.30 x 64 = $53.60. At P = $52.80 the expected V’ = 0.25 x 42.54 + 0.45 x 51.87 + 0.30 x 65.85 = $53.73, a gain of $0.13 (0.2 percent). At P = $55.00 expected V’ = $53.38, a loss of $0.22. The expected value to stayers is effectively zero, so the case for repurchase cannot rest on a bet that the market is wrong. It must rest on tax efficiency, flexibility, signaling or a lack of better uses for the cash.

Solution to Question 5: Taxes and clienteles

(a) After-tax cash per share held. A pro rata tender at $54 buys 1,500 / 54 = 27.78 million shares, or 13.889 percent of each holder’s position; proceeds are 0.13889 x 54 = $7.50 per share held, matching the dividend. Tax on the tender is the capital gains rate times (54 - basis) x 0.13889.

Clientele	Dividend tax	After-tax dividend	Tender tax	After-tax tender
Founders’ trust	1.800	5.700	1.458	6.042
Retail, taxable	1.800	5.700	0.500	7.000
Tax-exempt and employee plans	0.000	7.500	0.000	7.500
Mutual funds and ETFs	0.900	6.600	0.156	7.344
Foreign institutions	1.125	6.375	0.000	7.500
Meridian	1.800	5.700	0.361	7.139

Sample calculations: founders, dividend 0.24 x 7.50 = 1.80; tender 0.20 x (54 - 1.50) x 0.13889 = 0.20 x 7.292 = 1.458. Retail, tender 0.20 x (54 - 36) x 0.13889 = 0.20 x 2.50 = 0.50.

(b) Aggregate tax. Dividend: 28.0 x 1.80 + 30.0 x 1.80 + 48.0 x 0.90 + 18.0 x 1.125 + 12.0 x 1.80 = 50.4 + 54.0 + 43.2 + 20.25 + 21.6 = **$189.45 million** (12.6 percent of the distribution). Tender: 28.0 x 1.458 + 30.0 x 0.50 + 48.0 x 0.1556 + 12.0 x 0.361 = 40.83 + 15.0 + 7.47 + 4.33 = **$67.63 million** (4.5 percent). The difference of $121.8 million is a deadweight transfer to the tax authority under the dividend, borne entirely by taxable holders and by foreign holders through withholding. In a real tender, high-basis, tax-exempt and foreign holders would tender disproportionately and the founders would not, so the actual tax on the tender would be lower still.

(c) Ex-dividend price drop. If the marginal investor is a taxable individual, a price drop equals the dividend scaled by (1 - 0.24) / (1 - 0.20) = 0.95, so the price falls $7.125 to about $40.875, not $7.50 to $40.50. The stock falls less than the dividend because holders value a dollar of price at $0.80 after tax but a dollar of dividend at $0.76.

(d) Winners and losers. The founders (basis $1.50) are hurt most by a dividend, $1.80 of tax per share held. If gains are forgiven at death, a repurchase costs the family nothing, so they strongly prefer it, and it raises their stake from 14.0 to 16.2 percent. Tax-exempt holders are indifferent; foreign holders prefer the repurchase to avoid withholding. Income-seeking retail holders save $1.30 of tax per share held under a tender, far more than the roughly $0.01 cost of selling 0.156 share. A dividend-growth mandate values an uninterrupted regular dividend, which should not be sacrificed.

Solution to Question 6: Signaling and dividend policy

(a) Lintner. FY2027E EPS is $3.96 (Exhibit 4). Target dividend = 0.35 x 3.96 = $1.386. New dividend = old + speed x (target - old) = 0.80 + 0.30 x (1.386 - 0.80) = **$0.976**, or about $0.244 a quarter. Rounding to $0.25 a quarter ($1.00 a year) is a 25 percent increase and a payout of 1.00 / 3.96 = 25.3 percent, up from 22.2 percent today (0.80 / 3.60). The increase is far above the 4 to 6 percent pace of recent years and is itself a statement.

(b) What each action signals.

Action	What it says	Cost of mimicry	Credibility
Special dividend	Cash is temporarily abundant	Low: any firm with cash can pay it	Weak signal; no commitment
Regular dividend increase	Earnings are durable	High: cuts are punished (dividend-growth mandates, ratchets)	Strong
Open-market authorization	Shares are cheap	Moderate: an overvalued firm loses (V = 44, P = 52.80 loses 3.3 percent for stayers); but it is not an obligation	Moderate
Tender offer	Shares are cheap, at a premium	Higher: binding, and priced above market (V = 44, P = 55 loses 4.0 percent)	Strong

The tender separates a firm worth $58 (+0.8 percent for stayers) from one worth $44 (-4.0 percent), a gap of 4.8 points, about the premium itself. Costly signaling works only where a bad type finds mimicry costly.

(c) Bayesian update. The prior that Quantra is worth $58 (and not $44) must be consistent with a price of $48: p x 58 + (1 - p) x 44 = 48 gives p = 4/14 = 28.6 percent. If an overvalued firm mimics with probability q, the posterior is p’ = p / (p + (1 - p) q), and the price is 44 + 14 p’.

Mimic probability q	Posterior p’	Price (US$)	Change vs $48
0.0	100.0%	58.00	+20.8%
0.1	80.0%	55.20	+15.0%
0.3	57.1%	52.00	+8.3%
0.6	40.0%	49.60	+3.3%
1.0	28.6%	48.00	0.0%

The more costly it is for weak firms to mimic, the greater the price reaction. This is why the form of commitment (fixed-price tender over open-market authorization) matters more than the amount.

Solution to Question 7: Tender offer design

(a) Clearing price. For each price p, the company can afford 1,500 / p million shares. The offer clears at the lowest price at which cumulative shares offered covers that quantity.

Price	Shares offered (m)	Shares affordable (m)	Fully subscribed?
$52	12	28.85	No
$53	19	28.30	No
$54	26	27.78	No
$55	31	27.27	Yes

The clearing price is **$55**; the company buys 1,500 / 55 = 27.273 million shares. Pro-ration = 27.273 / 31 = **87.98 percent**: every holder who offered at or below $55 sells 87.98 percent of its tendered shares at $55.

(b) Ownership. After the tender: 200 - 27.273 = 172.727 million shares. Meridian, which tendered its 12.0 million at $52, sells 12.0 x 0.8798 = 10.557 million for $580.6 million (a gain of about $147.8 million over its $41 basis) and retains 1.443 million shares, or 0.84 percent. The founders (not tendering) hold 28.0 / 172.727 = **16.21 percent**. Under open-market buying at $52.80 (28.409 million shares, 171.591 million remaining), the founders hold 28.0 / 171.591 = 16.32 percent and Meridian, which is not forced to sell, holds 12.0 / 171.591 = 6.99 percent.

(c) Comparison.

Feature	Open market	Fixed-price tender	Dutch auction
Premium paid (illustrative)	4.80 x 28.41 = $136 million	6.00 x 27.78 = $167 million at $54	7.00 x 27.27 = $191 million at $55
Certainty of size	Low (non-binding)	High	High
Speed	About 100 trading days	About 20 business days	About 20 business days
Price discovery	None	None	Reveals holders' reservation prices
Effect on activism	Meridian stays and stays vocal	Meridian can exit	Meridian can exit

The tender pays a higher premium for certainty and speed, and it gives Meridian a profitable exit that reduces the pressure of a proxy contest. In expected value terms the open market is cheaper, but it is the weaker signal and leaves the activist in place.

Solution to Question 8: Compensation and agency

(a) FY2027E pro forma EPS and bonus. FY2027E status quo net income is 791.7 (Exhibit 4). Each $1,500 million action costs 56.88 of after-tax interest; assume execution at the start of the year.

Action	Net income	Shares (m)	EPS (US$)	Payout vs target	CEO bonus
Status quo	791.7	200.00	3.96	85%	$2.04m
Special dividend	734.8	200.00	3.67	0%	$0.00
Open market at $52.80	734.8	171.59	4.28	200% (capped)	$4.80m
Tender at $55	734.8	172.73	4.25	200% (capped)	$4.80m

Payout for status quo: 25 + (3.96 - 3.80) / 0.20 x 75 = 85 percent. If the open-market program runs evenly from February to June, the average weighted shares are 200 - 28.41 x (8.5 / 12) = 179.88 and lost interest is 56.88 x 0.7083 = 40.29, so EPS = 751.4 / 179.88 = $4.18 and the payout is about 189 percent, or $4.53 million. The conclusion survives.

(b) The problem. Two actions with identical shareholder value (Question 2) differ by $4.8 million in executive pay. The plan rewards EPS created by shrinking the share count, which Question 3(b) showed is unrelated to value.

(c) Neutralization. Add back the after-tax interest on cash used and restore repurchased shares at the weighted-average timing, so EPS is measured on the pre-distribution capital base: (734.8 + 56.88) / (171.59 + 28.41) = 791.7 / 200 = **$3.96** in every option. Alternatives: pay on free cash flow per share with a capital charge, or on return on invested capital (which is unaffected by distributions). The Compensation Committee should adopt neutralization before the board announces the decision, or the choice will look self-interested.

Solution to Question 9: Stress test and policy flexibility

(a) Cases. Hybrid program: $1,200 million tender at $55 (21.82 million shares retired, 178.18 million remaining) plus a regular dividend of $1.00 per share ($178.2 million). Lost interest on $1,200 million is 45.50 after tax. Operating cases: base (revenue 6,420, margin 14.8 percent); downside (revenue 5,520, a fall of 8 percent from FY2026E, margin 9 percent, working capital release of 30); severe (revenue 5,100, margin 6 percent, release of 60).

Case	EBIT	Net income	FCF	EBITDA	FCF / dividend	FCF / (dividend + $200m buyback)
Base	950.2	746.2	961.2	1,270.2	5.4x	2.5x
Downside	496.8	388.0	678.0	816.8	3.8x	1.8x
Severe	306.0	237.3	557.3	626.0	3.1x	1.5x

Base net income is (950.2 + 112 - 60) x 0.79 - 45.50 = 746.2; free cash flow adds 320 of D&A and 220 of stock-based compensation, and deducts capex of 280 and the working capital effect (an increase of 45 in the base case, releases of 30 and 60 in the stress cases). The dividend is covered at least three times in every case, so the hybrid is robust and leaves room for a smaller flexible buyback.

(b) Meridian's program. Gross debt 2,200, interest expense 60 + 64 = 124.

Case	Gross debt / EBITDA (limit 2.5x guardrail)	EBIT / interest (covenant 4.0x)	Net debt / EBITDA (covenant 3.0x)
Base	2,200 / 1,270.2 = 1.73x	950.2 / 124 = 7.7x	0.9x
Downside	2,200 / 816.8 = **2.69x (breach)**	496.8 / 124 = 4.01x (0.2 percent headroom)	1.3x
Severe	2,200 / 626.0 = **3.51x (breach)**	306.0 / 124 = **2.47x (covenant breach)**	1.8x

(Net debt uses cash of 1,100 for simplicity.) Meridian's proposal leaves no margin for a refresh pause: it triggers a rating review in the downside and a covenant default in the severe case. The current structure (1,200 of debt) shows gross leverage of 1.5x and 1.9x and interest cover of 8.3x and 5.1x in the same cases.

(c) Flexible framework. (i) A regular dividend at about 25 percent of normalized EPS, covered at least three times by severe-case FCF. (ii) Repurchases as the residual: about 60 percent of FCF after dividends and committed acquisitions (FY2027E base: (961 - 178) x 60 percent, about $470 million), reduced when the price exceeds 90 percent of the intrinsic range. (iii) A gross leverage range of 0.8x to 1.5x tested at downside EBITDA. (iv) Special dividends only when cash exceeds target for two successive years.

Solution to Question 10: Recommendation

The recommended structure (the hybrid, run as a package): (1) a **$1,200 million modified Dutch auction tender offer** at $52 to $58, launched immediately after the 28 January earnings release; (2) a **regular dividend increase to $0.25 per quarter**; (3) retention of the balance, about $190 million, as a flexible reserve for opportunistic repurchases below $52 and for the Verdana earn-out; (4) a committed **gross leverage range of 0.8x to 1.5x**; and (5) neutralized EPS in the bonus plan.

Why this form. The tender is tax-efficient (Question 5), a credible signal (Question 6), fast and certain, and gives Meridian an exit. The $1,200 million size leaves a reserve of about $190 million of the Treasurer's $1,500 million after the $80 million earn-out, the $12 million excise tax and about $16 million of repatriation leakage (1,200 x 1.01 + 300 + 400 + 100 - 1,700 = 312 repatriated, at 5 percent). At V = $58 and P = $55, stayers gain (58 x 200 - 1,200) / 178.18 = $58.37, or 0.6 percent; at V = $44

they lose 3.1 percent. The package is robust to the main uncertainty because the dividend increase and the leverage range deliver benefits that do not depend on valuation. FY2027E pro forma EPS is 746.2 / 178.18 = $4.19, 5.8 percent above the $3.96 status quo.

Sensitivity discussion

EPS accretion varies with the repurchase price and the after-tax yield on cash (percent accretion for $1,500 million):

Pre-tax cash yield	P = $44	P = $48	P = $52.80	P = $56	P = $60	P = $64
2.0%	16.6	14.6	12.7	11.7	10.5	9.5
3.0%	14.6	12.7	10.8	9.8	8.6	7.7
4.8%	11.0	9.2	7.3	6.3	5.3	4.3
6.0%	8.6	6.8	5.0	4.1	3.0	2.1

Accretion stays positive across the grid, confirming Question 3(b). *Value* is driven by P minus V (Question 4). The $122 million *tax* advantage of the tender shrinks if founders tender. *Debt cost and cyclicality* decide whether Meridian's proposal is feasible, since the downside breaches at 2.69x.

Common student errors

1. Calling the dividend "dilutive" and the buyback "accretive" as if EPS were value, or applying the old P/E to the new EPS.
2. Ignoring foregone interest, so that accretion appears to be 15.6 percent (the share count reduction) rather than 7.3 percent.
3. Forgetting that the repurchase price premium is paid by stayers, and sizing the program at the market price.
4. Taking Treasurer's excess cash at face value without the offshore, earn-out and maturity adjustments.
5. Reading the Dutch auction as a pay-as-bid auction: all accepted shares get the uniform clearing price.
6. Testing leverage only in the base case. ### Alternative defensible answers

A pure buyback ($1,500 million). Defensible if the committee holds to V = $58 and accepts a thinner dividend story; weaknesses are the premium, the commitment problem and the bonus problem.

A special dividend. Defensible if the committee believes the shares are fairly priced; it avoids value-transfer risk, but costs $122 million more tax and the founders oppose it.

Meridian's proposal. Defensible only for a far less cyclical company. A staged version with a 1.5x ceiling (debt of 1,740) gives downside leverage of 1,740 / 816.8 = 2.13x and interest cover of 496.8 / 94.6 = 5.3x.

Hold the cash for acquisitions. Defensible only with a specific pipeline; cash earns 3.79 percent after tax against a 7.5 percent earnings yield.

Board-style recommendation memorandum

To: Capital Committee of the Board of Directors, Quantra Systems **From:** Priya Raman-Duval, Chief Financial Officer **Date:** 2 December 2026 **Subject:** Return of excess cash; recommendation for decision on 9 December

Recommendation. Approve a $1,200 million modified Dutch auction tender offer ($52 to $58, launching after the 28 January release), raise the regular dividend 25 percent to $0.25 per quarter, retain about $190 million as a flexible repurchase reserve, adopt a through-the-cycle gross leverage range of 0.8x to 1.5x, and neutralize capital-return effects in the FY2027 bonus plan.

Cash available. The Treasurer's $1,500 million excess is $1,390 million after the offshore leakage and the Verdana earn-out, on the basis that the March 2027 notes are refinanced. We recommend deploying $1,200 million now.

Why a tender and not a special dividend. For our shareholders, a pro rata tender costs about $68 million in tax against $189 million for a special dividend, a saving of $122 million. A tender is a binding, premium-priced commitment and a stronger signal than an open-market authorization. It is not a bet on any one view of value: stayers gain 0.6 percent if the CEO is right and lose 3.1 percent if the skeptics are, which is why the regular dividend and leverage range, not the repurchase, carry the case.

What EPS says and does not say. The package raises FY2027E EPS to $4.19 from $3.96, but accretion is arithmetic: the break-even repurchase price is $94.94. We ask the committee not to treat it as evidence of value.

Meridian. We recommend engaging constructively. The package returns $1,200 million within weeks, gives Meridian a liquid exit at a gain of about $14 per share, and sets a leverage range with a ceiling of 1.5x. We do not recommend the $2,500 million debt-financed program: it breaches the 2.5x rating guardrail (2.69x) and is at the edge of the interest cover covenant (4.01x) in our downside case, for a gain in EPS of only 3.2 points beyond the cash-funded version.

Risks. The clearing price could reach $58 (stayers' gain at V = $58 is then nil); the Brandt litigation could use $150 million of the buffer; a deep refresh pause tests the dividend, which remains covered 3.1 times in our severe case.

Decision requested. Approve the package, authorize the Treasurer to refinance the March 2027 notes, and ask the Compensation Committee to adopt neutralized EPS before announcement.

Chapter 10: Case: Andara Foods, an Emerging-Market Bond Issue and FX Risk

Andara Foods, the Republic of Orvania, the Orvanian sol (ORS), the banks, the investors and every number in this case are fictional and illustrative. Exchange rates, interest rates, spreads and tax rules are invented and stylized so that the arithmetic is transparent; they describe no real country or market. This chapter applies the bond and debt-market material of Volume 1, Chapters 4 and 14, the foreign-exchange and derivatives material of Volume 1, Chapters 19 and 20, and the cost-of-capital and capital structure material of Volume 1, Chapters 11 and 15.

The Situation

On Monday, 16 November 2026, Anneliese Dragomir-Sato, Chief Financial Officer of Andara Foods S.A., received a two-line message from the chairman: “The board wants one structure to vote on on 14 December. Not a menu.” The CFO had spent six weeks assembling exactly such a menu. Her team had three term sheets, two conflicting economist forecasts, a hedging quotation that was valid for only ten days, and a growing suspicion that the cheapest-looking bond in the pile was not the cheapest.

Andara Foods is a packaged-food and edible-oil company headquartered in the port city of Calvera, in the Republic of Orvania, a middle-income economy of about 60 million people with a floating exchange rate, an inflation-targeting central bank and a history of sharp currency adjustments. The company was founded in 1971 by the Ventresca family, which still owns 58 percent of the shares; the remaining 42 percent trades on the Orvanian Stock Exchange. Andara sells rice, pasta, sauces and frozen products to Orvanian supermarkets and traditional retailers (55 percent of revenue), refined edible oils to households, bakeries and exporters (30 percent) and snacks and frozen foods (15 percent). In FY2026E revenue is ORS 36,000 million, roughly US$1.8 billion at today’s rate of ORS 20.00 per US dollar. About 22 percent of revenue is exported, priced and invoiced in US dollars, mainly refined oil sold to neighboring countries.

The project and the funding need

The board has approved in principle the Port Calvera complex, a crushing, refining and packaging plant that would raise oilseed processing capacity by 60 percent and create an export platform. The capital cost is ORS 4,000 million (US$200 million), spent mostly in FY2027 (ORS 3,200 million) and FY2028 (ORS 800 million). Management expects the plant to add ORS 1,100 million of EBITDA by FY2031, roughly half of it in US dollars. In addition, the company wants to refinance ORS 1,000 million (US$50 million) of floating-rate bank debt that matures within eighteen months and carries a 15.5 percent rate. Together this is a funding need of ORS 5,000 million, or US$250 million at the current exchange rate.

The people

- **Anneliese Dragomir-Sato, CFO.** Joined from a regional bank. She will present the recommendation, and she is personally uneasy about leaving US-dollar debt unhedged: “Our costs are in dollars and our customers pay in sols.”
- **Rafael Ibarrena-Quist, CEO.** A food engineer who wants the plant on schedule and the financing settled before the presidential campaign season. He is attracted by the Eurobond’s lower coupon: “Nine and a quarter versus fifteen and a half. Explain the difference to me in one sentence.”
- **Don Aurelio Ventresca, Chairman.** Representing the family. He remembers the devaluation of 2023, when a US-dollar import loan nearly cost the family control, and refuses to approve an instrument whose worst case he cannot describe.

- **Kofi Marchetti-Adeyemi, Treasurer.** Responsible for relationships with banks, rating agencies and swap counterparties, and for executing whatever the board approves.
- **Dr. Thandeka Voss-Okonjo, independent director, Chair of the Audit and Risk Committee.** A former central-bank economist. She wants the board paper to say, in numbers, what Andara is betting on.
- **Northbridge Securities (fictional).** Lead bookrunner for both the local and the international bond. Its banker has promised a Eurobond coupon of about 9.25 percent, a local-currency coupon of about 15.5 percent, and a swap quote that "is better than you think."

The decision

The board must choose one financing structure and the covenant package that goes with it. The three structures on the table are:

- **Option A: local-currency bond.** An ORS 5,000 million, five-year bullet bond at a 15.50 percent coupon, sold to Orvanian pension funds, insurers and banks. (A seven-year tenor is possible at a coupon of 16.25 percent.)
- **Option B: US-dollar Eurobond, unhedged.** A US$250 million, seven-year bullet at 9.25 percent, sold to international investors, with proceeds converted into sols at the spot rate.
- **Option C: US-dollar Eurobond with a cross-currency swap.** The same Eurobond, with a swap that converts US$175 million (70 percent) of the exposure into fixed-rate sols at 17.60 percent. Counterparty lines limit the swap to US$175 million.

Students are also asked to design Option D, a combination that might beat all three, and to specify the covenants.

The deadline

The board meets on 14 December 2026. The investor roadshow would run from 25 to 29 January 2027, and pricing is targeted for 4 February 2027, ahead of the campaign season for the October 2027 presidential election. The dealer's swap quotation expires on 26 November, and a delay beyond mid-February would push the plant's first milestone payment (ORS 1,200 million, due 15 March 2027) onto the existing bank lines.

Industry Context

Orvania's food-processing industry has three features that shape the financing choice.

A structural mismatch between revenue and cost currencies. Processors sell mostly in local currency to price-sensitive consumers, while their raw materials (oilseeds, wheat, packaging resin and a share of energy and freight) are priced off global markets in US dollars. When the sol weakens, imported costs rise immediately, but retail prices adjust slowly, partly because the government has twice in the past decade threatened price caps on staples. Domestic competitors with fewer imported inputs have the opposite exposure. As a result, a processor's operating profit is typically *short* the US dollar even before any dollar borrowing.

A shallow local bond market, a deep sovereign curve. Orvania has a liquid sovereign curve, but its corporate bond market is young: the largest ever corporate issue is ORS 2,500 million, most investors want maturities of five years or less, and a bond larger than ORS 3,000 million must be priced wide to clear. Local investors are mostly domestic pension funds, insurers and banks who hold to maturity and demand a premium over sovereign yields for credit and illiquidity. International investors, by contrast, buy Orvanian corporate Eurobonds as a diversified emerging-market allocation; they accept lower coupons in dollars because they do not bear the local-currency risk, but they price the sovereign, transfer-and-convertibility and political risk of Orvania into the spread.

A thin, expensive hedging market. Cross-currency swaps into sols are available from a handful of international dealers and from the two largest local banks, but tenors beyond five years are rare, counterparty credit lines are limited, and the pricing reflects the large interest-rate gap between the sol (about 12 percent on seven-year government paper) and the dollar (about 4.2 percent). Under covered interest parity this gap is the price of the hedge: it is not a fee that a clever negotiator can remove.

Currency history. The sol has depreciated against the dollar in every year of the past five, from ORS 14.00 at the end of 2021 to ORS 20.00 today, an average of about 7.4 percent a year. The pace has been uneven: a six-week sell-off in the spring of 2023, triggered by a commodity-price shock and a rating downgrade, took the sol down 17 percent before it partially recovered. During that episode the central bank imposed temporary limits on transfers of foreign currency abroad, and some companies could not service dollar debt on schedule even though they held sols.

Covenant norms. International bond covenants for emerging-market corporates are typically “incurrence” tests (limits on new debt, liens and restricted payments that apply only when the company takes an action) rather than “maintenance” tests. Local bank loans and some local bonds, in contrast, use maintenance tests measured every six months. The lead banks have proposed a hybrid package for Andara, reproduced later in the case, whose definitions will matter as much as its levels.

Financial Data

All amounts in Orvanian sols (ORS) millions unless stated. Fiscal years end 31 December. FY2026E combines ten months of actual results with a two-month forecast.

Exhibit 1: Income statement

ORS millions	FY2024A	FY2025A	FY2026E
Revenue	28,800	32,400	36,000
Cost of goods sold (excluding depreciation)	19,830	22,130	24,480
Gross profit before depreciation	8,970	10,270	11,520
Selling, distribution and administrative expenses	5,800	6,540	7,380
EBITDA	3,170	3,730	4,140
Depreciation and amortization	900	990	1,080
EBIT	2,270	2,740	3,060
Interest expense	880	930	960
Interest income	150	165	180
Foreign-exchange loss on US-dollar-linked items	220	310	0
Pre-tax income	1,320	1,665	2,280
Income tax (25%)	330	416	570
Net income	990	1,249	1,710
EBITDA margin	11.0%	11.5%	11.5%
Dividends (30% payout)	300	375	510

Exhibit 2: Balance sheet extract

ORS millions	31 Dec 2025A	31 Dec 2026E
Cash and equivalents	2,400	2,600
Receivables	3,600	4,000
Inventories	4,300	4,700
Property, plant and equipment, net	11,200	11,620
Other assets	1,300	1,400
Total assets	**22,800**	**24,320**
Payables	3,500	3,900
Accrued and other current liabilities	1,100	1,200
Bank debt (floating rate, average 15.5%)	6,400	6,000
Other long-term liabilities	1,100	1,320
Total liabilities	**12,100**	**12,420**
Shareholders' equity	**10,700**	**11,900**
Total liabilities and equity	22,800	24,320

Equity roll-forward: 10,700 + net income 1,710 - dividends 510 = 11,900. Pro forma metrics before the new financing: gross debt / EBITDA = 6,000 / 4,140 = 1.45x; net debt / EBITDA = (6,000 - 2,600) / 4,140 = 0.82x; EBITDA / interest expense = 4,140 / 960 = 4.3x.

Exhibit 3: Management forecast (budget exchange rate path, depreciation of 5 percent a year)

ORS millions	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E	FY2031E
Budget rate (ORS per US$, year-end)	20.00	21.00	22.05	23.15	24.31	25.53
Revenue	36,000	40,320	44,352	49,230	54,153	59,030
Revenue growth	11.1%	12.0%	10.0%	11.0%	10.0%	9.0%
EBITDA	4,140	4,677	5,234	6,105	6,986	7,733
EBITDA margin	11.5%	11.6%	11.8%	12.4%	12.9%	13.1%
of which Port Calvera plant	0	0	150	600	900	1,100
Capital expenditure	1,500	4,300	2,500	1,600	1,700	1,800
of which Port Calvera plant	0	3,200	800	0	0	0
US-dollar share of revenue	22%	22%	23%	25%	27%	28%
Net US-dollar operating position (US$ millions)	(154)	(154)	(128)	(71)	(1)	52

The net US-dollar operating position is US-dollar revenue less US-dollar-linked costs; a negative number means the company is short the dollar, so a depreciation of the sol hurts operating profit.

Exhibit 4: Exchange-rate and macro history

Item	End 2021	End 2022	End 2023	End 2024	End 2025	2 Nov 2026
ORS per US$	14.00	15.10	16.60	17.80	18.70	20.00
Change in the year		7.9%	9.9%	7.2%	5.1%	7.0% (ten months)
Orvanian inflation		7.9%	10.5%	9.2%	8.8%	8.5% (2026E)
US inflation		4.5%	3.8%	3.2%	2.9%	3.0% (2026E)

The simple average of the five annual changes is 7.4 percent. The spring 2023 sell-off took the sol down 17 percent in six weeks before a partial recovery; the year-end figure above shows only the net effect.

Exhibit 5: Interest-rate curves and parity data (2 November 2026)

Tenor	ORS government yield	US Treasury yield	ORS swap rate	US-dollar swap rate
1 year	12.5%	4.6%		
3 years	12.3%	4.3%		
5 years	12.1%	4.2%		
7 years	12.0%	4.2%	12.4%	4.3%

For covered-parity calculations the team uses flat 12.0 percent (sol) and 4.2 percent (dollar) zero rates. Orvania’s sovereign 7-year US-dollar bond yields 7.6 percent. Andara’s indicated Eurobond spread over the sovereign is 165 basis points, giving the 9.25 percent coupon; the corresponding spread of Andara’s local bonds over the 5-year ORS government yield is 340 basis points for ORS 5,000 million and 265 basis points for a smaller ORS 3,000 million issue.

Exhibit 6: Exchange-rate views (annual depreciation of the sol)

Source	View
Northbridge Securities economists	6.0% a year
Calvera Merchant Bank economists	4.0% a year
Purchasing-power parity (inflation 8.5% versus 3.0%)	to be computed
Covered interest parity (forward market)	to be computed
Five-year history	7.4% a year

Exhibit 7: Funding alternatives (indicative terms)

Feature	A: local bond	B/C: Eurobond
Size	ORS 5,000 million (a record; largest to date is ORS 2,500 million)	US$250 million (ORS 5,000 million at 20.00)
Tenor	5 years bullet (7 years at 16.25%)	7 years bullet
Coupon, annual pay	15.50% (14.75% for an ORS 3,000 million issue)	9.25% in US dollars (9.75% for a US$150 million issue)
Issuance cost	0.8% of face	1.6% of face (legal, rating, listing, underwriting)
Investors	Orvanian pensions, insurers, banks	International emerging-market funds
Covenants	Maintenance test, net debt / EBITDA of 4.0x or less	Hybrid package (Exhibit 9)
Rating	local scale AA-; global B+	B+ (stable)

Exhibit 8: Peer funding structures (all fictional; 2026)

Company	US-dollar share of revenue	US-dollar share of debt	Share of dollar debt hedged	Net debt / EBITDA	FX loss in 2023 as % of net income
Sunvale Foods	30%	62%	20%	2.1x	38%
Tamarind Agro	12%	18%	100%	1.5x	4%
Kestrel Provisions	5%	0%	not applicable	1.2x	0%
Lumbra Dairy	18%	35%	50%	2.4x	15%
Orvis Grain	25%	80%	10%	3.3x	120%
Andara Foods (before the new financing)	22%	0%	not applicable	0.8x	25% (FY2025 loss)

Exhibit 9: Covenant package proposed by the lead managers for the Eurobond

Covenant	Draft term
Maintenance leverage	Net debt / EBITDA of 3.5x or less, tested semi-annually; net debt translated at the spot rate on the test date; EBITDA as reported for the last twelve months
Maintenance interest cover	EBITDA / consolidated interest expense of 2.5x or more; interest expense includes net swap payments
Incurrence of debt	No new debt if pro forma net debt / EBITDA exceeds 3.0x, subject to a general basket of ORS 1,500 million
Definition of debt	Includes the net mark-to-market termination value of derivative contracts
Restricted payments	Dividends not to exceed 50% of net income; blocked if net debt / EBITDA exceeds 2.5x
Other	Negative pledge; limit on asset sales; change-of-control put at 101; cross-acceleration threshold of US$25 million; quarterly reporting

Exhibit 10: Exchange-rate scenarios (ORS per US$, year-end) and committee probabilities

Scenario	Probability	Year 1	Year 2	Year 3	Year 5	Year 7
S1: stability (2% a year)	10%	20.40	20.81	21.22	22.08	22.97
S2: parity-driven (5.3% a year)	25%	21.06	22.18	23.35	25.89	28.71
S3: history repeats (7.0% a year)	25%	21.40	22.90	24.50	28.05	32.12
S4: forward-implied (7.49% a year)	15%	21.50	23.11	24.84	28.70	33.16
S5: shock (35% devaluation in year 2, then 5%)	15%	21.00	28.35	29.77	32.82	36.18
S6: appreciation (2% a year)	10%	19.60	19.21	18.82	18.08	17.36

Exhibit 11: Operating exposure and swap quotation

Item	Value
US-dollar revenue (22% of FY2026E revenue)	ORS 7,920 million (US$396 million at 20.00)
US-dollar-linked costs (imported seeds, wheat, resin, energy, freight)	ORS 11,000 million (US$550 million)
Net US-dollar operating position	short ORS 3,080 million (US$154 million)
EBITDA effect of a 10% depreciation of the sol, after pricing pass-through	-ORS 190 million in the year of the move; -ORS 95 million a year thereafter
Swap line: Dealer A	US$100 million, 7 years
Swap line: Dealer B	US$75 million, 7 years
Swap structure	Initial exchange at 20.00; Andara pays fixed ORS 17.60% on the ORS notional and receives the Eurobond's 9.25% in dollars; final re-exchange at 20.00
Swap fee	0.40% of ORS notional, paid at inception
Collateral	Two-way credit support annex; threshold US$10 million
Quotation valid until	26 November 2026

Complications and Exhibits

The following facts emerged late in the Treasurer's review. Some are firm; others are ambiguous, and students should say which.

1. **Execution risk on the local bond.** The bookrunner puts the probability that an ORS 5,000 million local bond clears at 15.50 percent at about 70 percent. In the other 30 percent the book supports only ORS 3,000 million at 14.75 percent, and the remaining ORS 2,000 million would have to be raised from banks at a floating rate that is today 16.5 percent. A record-sized deal also uses up the local market's capacity for Andara for at least a year.

2. **Withholding tax on the Eurobond.** Orvania levies a 15 percent withholding tax on interest paid directly to non-residents. The 9.25 percent coupon assumes issuance through a treaty-eligible financing vehicle in a third country, which reduces the withholding rate to zero but takes about four months to establish. If the vehicle is not ready by February, investors will require a gross-up so that they receive 9.25 percent net, and the company must pay 9.25 / 0.85 = 10.88 percent.

3. **Tax treatment of foreign-exchange results.** Interest paid is deductible at the 25 percent rate in the amount of sols actually paid, so a dollar coupon is deductible at the exchange rate of the payment date. Unrealized translation losses on dollar debt are not deductible. The realized loss on repaying principal is deductible in the year of repayment, but the tax authority has contested similar deductions in the past, and the company's tax advisers rate the deduction "probable but not certain." Swap payments are deductible as paid.

4. **Convertibility and transfer risk.** In the 2023 sell-off the central bank limited transfers abroad for fourteen weeks. A company with sols but no dollars could not service dollar debt. The dealers' swap documentation allows termination if currency is no longer freely convertible. Neither the Eurobond nor the swap eliminates this risk, and Dr. Voss-Okonjo has asked how the structure behaves if transfers are blocked for a quarter.

5. **Swap line limit.** Dealers A and B together limit the swap to US$175 million. Any additional line would require cash collateral of US$50 million, which would sit idle at about 4 percent while the company borrows at 15 percent or more.

6. **Existing bank covenants.** Andara's existing bank loans contain a maintenance covenant of net debt to EBITDA of 3.0x or less, tested at spot exchange rates, with cross-default to any other debt. The banks have not agreed to amend this covenant.

7. **Rating agency.** The B+ rating is on stable outlook. The agency has said it would consider a downgrade if gross debt to EBITDA exceeds 3.0x for two consecutive quarters, and it adds a stress to unhedged dollar debt by converting it at a rate 30 percent weaker than spot.
8. **The plant's dollar earnings arrive late.** Port Calvera's EBITDA, and its dollar exports in particular, arrive from FY2029. The first two coupons on any dollar bond (FY2027 and FY2028) are paid when Andara's net operating position is still short dollars by US$154 million and US$128 million.
9. **Election risk.** Pricing in February is ahead of the October 2027 presidential election. The central bank governor has hinted that capital controls "remain an option" if outflows accelerate. The probabilities in Exhibit 10 are the committee's judgments, not market-implied.
10. **Call features.** The local bond is callable at par from year three; the Eurobond is non-callable for three years and then callable at a make-whole premium. The option value of these features is not priced in the quoted coupons and is ambiguous.

Assignment Questions

Answer all ten questions. Where the case is ambiguous, state the assumption and show how the answer changes under the main alternative. Compute all-in costs as the internal rate of return, in sols, of the net cash flows after issuance costs.

Question 1: Funding need and the true cost of the coupons. (a) Prepare sources and uses and the pro forma leverage at close. (b) Compute the all-in sol cost, as an IRR, of Option A (five-year and seven-year), Option B if the sol is unchanged at 20.00, and a hypothetical fully hedged Eurobond using the 17.60 percent swap. (c) Explain why the 9.25 percent and 15.50 percent coupons are not comparable.

Question 2: Parity and expectations. (a) Compute the covered-interest-parity forward rates for one to seven years and the implied annual depreciation. (b) Compute purchasing-power-parity depreciation. (c) Compute the break-even annual depreciation at which the unhedged Eurobond costs the same as Option A, five-year and seven-year. (d) Evaluate the views in Exhibit 6.

Question 3: Scenario costs and the hedge ratio. (a) For each scenario in Exhibit 10, compute the all-in sol cost of Option A, Option B and Option C (70 percent hedged). (b) Compute the probability-weighted cost and standard deviation. (c) Construct the table of hedge ratios from 0 to 100 percent and identify what the hedge costs per unit of risk removed.

Question 4: Operating exposure and the natural hedge. (a) Compute the company's net dollar position and the effect on EBITDA of a 20 percent devaluation. (b) Show that dollar debt is a wrong-way exposure for Andara. (c) Describe how the Port Calvera plant changes the picture by year, and conclude how much unhedged dollar debt the company can responsibly carry.

Question 5: The cross-currency swap. (a) Verify the 17.60 percent quote against covered interest parity. (b) Compute the annual cash cost of the swap relative to the dollar coupon at today's rate. (c) Value the US$175 million swap at the end of year 2 in scenarios S5 and S6, using the remaining five years, and discuss collateral and counterparty risk. (d) Explain what happens if convertibility is suspended.

Question 6: Credit metrics and covenant testing. (a) Compute gross debt to EBITDA, net debt to EBITDA and EBITDA to interest in years 1 to 3 for Options A, B and C and the combination in Question 9, in the base case. (b) Repeat for a stress case: scenario S5 with demand falling 10 percent in year 2 and 5 percent in year 3, floating bank rates up 400 basis points from year 2, and the EBITDA sensitivity of Exhibit 11. (c) Identify which covenant binds.

Question 7: Covenant design. (a) Critique each element of Exhibit 9. (b) Propose a revised covenant package and justify each level using your answer to Question 6. (c) Explain how currency definitions should be handled.

Question 8: Taxes and issuance frictions. (a) Compute the all-in cost of the unhedged Eurobond if the withholding gross-up applies. (b) Compute after-tax costs of Options A and C and of the unhedged Eurobond in scenario S3, with and without deductibility of the realized foreign-exchange loss on principal. (c) Discuss the implications for the choice.

Question 9: Tenor, refinancing risk and a combined structure. (a) Compute the all-in cost of refinancing a five-year bullet at year 5 for the last two years if rates are 0, 200 and 400 basis points higher. (b) Design and evaluate a combined Option D with ORS 2,000 million of local bond at 14.75 percent and a US$150 million Eurobond at 9.75 percent that is 60 percent hedged at a swap rate of 18.14 percent. (c) Compare Option D with Options A and C.

Question 10: Recommendation. Recommend one structure, a hedge ratio and a covenant package. Address the CEO's question ("explain the difference in one sentence"), the Chairman's request to describe the worst case, and the Audit Chair's question on convertibility. Draft a one-page board memorandum.

Teaching Note and Full Solutions

Learning objectives

After working through this case, students should be able to:

1. Compare bonds in different currencies by converting each to a common-currency all-in cost (IRR after fees), and explain why headline coupons are not comparable.
2. Use covered interest parity, purchasing-power parity and history to frame the exchange-rate bet implicit in an unhedged foreign-currency bond, and compute the break-even depreciation (Volume 1, Chapters 19 and 20).
3. Analyze a company's operating currency exposure and recognize when foreign-currency debt is a wrong-way exposure.
4. Price and value a cross-currency swap, interpret the hedge as the carry given up, and manage collateral, counterparty and convertibility risk.
5. Stress-test credit metrics and covenants under devaluation, and identify which covenant binds.
6. Design covenants whose definitions survive currency moves, and write a board recommendation that states plainly what the company is betting on.

Suggested class plan (120 minutes)

Minutes	Segment	Activity
0 to 10	Opening poll	"Which bond would you issue: 9.25 percent or 15.5 percent?" Record the vote; revisit at the end.
10 to 25	Cost of the coupons (Questions 1 and 2)	Compute IRRs and break-even depreciation live; ask the class who in the room thinks the sol will depreciate less than 5.6 percent a year.
25 to 45	Scenarios and hedge ratio (Question 3)	Split the class into three teams (hedge 0, 70 and 100 percent); each defends its choice against the scenario table.
45 to 60	Operating exposure (Question 4)	The instructor draws the net dollar position on the board; students find the wrong-way risk.
60 to 65	Break	
65 to 80	Swap mechanics (Question 5)	Walk through the quote, carry and mark-to-market; pose the convertibility question.
80 to 100	Credit and covenants (Questions 6 and 7)	Teams test the covenants in the stress case; one team rewrites Exhibit 9.
100 to 120	Frictions and recommendation (Questions 8 to 10)	Three teams give 90-second recommendations; reveal the sample memorandum; repeat the poll.

Instructors with a 75-minute session can assign Questions 8 and 9 as written work.

Solution to Question 1: Funding need and the true cost of the coupons

(a) Sources and uses and pro forma leverage.

ORS millions	Option A	Option B	Option C (70% hedged)
Bond proceeds at face	5,000	5,000	5,000
Use: Port Calvera plant	4,000	4,000	4,000
Use: refinance bank debt	1,000	1,000	1,000
Fees paid from cash	40	80	94
Gross debt after refinancing (at 20.00)	10,000	10,000	10,000

Fees: A is 0.8 percent of 5,000; B is 1.6 percent of 5,000; C adds 0.4 percent of the ORS 3,500 million swap notional (14). Debt after: existing bank debt 6,000 less 1,000 repaid, plus 5,000 new, equals 10,000. Pro forma metrics on FY2026E EBITDA of 4,140: gross debt / EBITDA = 10,000 / 4,140 = **2.42x** (from 1.45x); net debt / EBITDA = (10,000 - 2,600) / 4,140 = **1.79x**. On FY2027E EBITDA of 4,677: 2.14x gross and 1.58x net. Interest cover at close (FY2026E EBITDA, today's rate):

Option	Annual interest (ORS m)	EBITDA / interest
A	775 (bank) + 775 (bond) = 1,550	2.67x
B	775 + 23.125 x 20 = 1,237.5	3.35x
C (70% hedged)	775 + 616 + 138.75 = 1,529.75	2.71x
Fully hedged	775 + 880 = 1,655	2.50x

Note that the cheapest-looking option, B, has the highest interest cover at close. The Eurobond's lower coupon is what makes it attractive to the CEO and to lenders applying a simple cover test.

(b) All-in cost as IRR in sols.

Option A, five years. Net proceeds are 5,000 x (1 - 0.008) = 4,960. Flows: 775 a year for years 1 to 4, and 775 + 5,000 = 5,775 in year 5. The rate r solving 4,960 = 775 x annuity(r, 5) + 5,000 / (1 + r)^5 is **15.74 percent**. *Seven years at 16.25 percent:* coupons of 812.5 and the same fee give **16.45 percent**.

Option B at an unchanged exchange rate. Net proceeds are US$250 million x 0.984 x 20.00 = 4,920. Coupons are US$23.125 million x 20.00 = 462.5 a year, and 5,000 of principal in year 7. The IRR is **9.57 percent**.

Fully hedged Eurobond. Proceeds of 4,920 less a swap fee of 0.4 percent x 5,000 = 20 give 4,900. Swap payments are 17.60 percent x 5,000 = 880 a year and 5,000 at maturity. The IRR is **18.13 percent**.

Structure	Headline coupon	All-in cost in sols
A, five-year local	15.50%	15.74%
A, seven-year local	16.25%	16.45%
B, unhedged, sol unchanged	9.25% in dollars	9.57%
Eurobond fully hedged	9.25% plus swap	18.13%

(c) Why the coupons are not comparable. A coupon is the price of borrowing a particular currency. The 9.25 percent buys dollars; the sol cost of those dollars depends on how many sols must later be exchanged for them. Converting the dollar coupon into sols requires an exchange-rate forecast, or a hedge that fixes it. The answer to the CEO is therefore one sentence: *the 9.25 percent and the 15.5 percent are prices of different currencies, and the six-point gap is the market's estimate of how much the sol will lose against the dollar; if it loses more than about 5.6 percent a year, the dollar bond costs more.*

Solution to Question 2: Parity and expectations

(a) Covered interest parity. With sol and dollar zero rates of 12.0 and 4.2 percent, the t-year forward rate is F(t) = 20.00 x (1.12 / 1.042)^t. The annual ratio is 1.12 / 1.042 = 1.07486, an implied depreciation of **7.49 percent a year**.

Year	1	2	3	4	5	6	7
Forward rate	21.50	23.11	24.84	26.70	28.69	30.84	33.15
Cumulative depreciation	7.5%	15.6%	24.2%	33.5%	43.5%	54.2%	65.7%

(b) Purchasing-power parity. Depreciation equals the inflation ratio: 1.085 / 1.030 - 1 = **5.34 percent a year**; the seven-year PPP rate is 20.00 x 1.0534^7 = 28.8.

(c) Break-even depreciation. The unhedged Eurobond costs (1 + 0.0957)(1 + d) - 1 when the sol depreciates at d a year. Setting this equal to Option A's all-in cost:

```
Versus A, five-year:  (1 + 0.1574) / (1 + 0.0957) - 1 = 5.63%
Versus A, seven-year: (1 + 0.1645) / (1 + 0.0957) - 1 = 6.28%
```

(The fee-free approximations 1.155 / 1.0925 - 1 = 5.72 percent and 6.41 percent are close.) These figures were confirmed by solving the IRR equation numerically with the depreciation path as the unknown.

(d) Evaluating the views.

View	Depreciation d	Eurobond B all-in cost (approx.)	Versus A five-year (15.74%)	Versus A seven-year (16.45%)
Calvera Merchant Bank	4.0%	13.95%	B cheaper	B cheaper
PPP	5.34%	15.42%	B cheaper by 0.3	B cheaper
Northbridge	6.0%	16.14%	B dearer by 0.4	B cheaper
Five-year history	7.4%	17.68%	B dearer by 1.9	B dearer
Forward market	7.49%	17.78%	B dearer by 2.0	B dearer

The unhedged bond wins only if the sol depreciates by less than about 5.6 percent a year; the inflation differential alone, 5.3 percent, leaves a margin of 0.3 points; history and the forward market both say the margin is negative. The Eurobond's apparent advantage rests on a forecast that is more optimistic than the market's own price. A forward-market view is the only view an investor can trade against, and an unhedged borrower is a net seller of the sol at the forward rate; the hedge market charges the full 7.49 percent for that.

Solution to Question 3: Scenario costs and the hedge ratio

(a) All-in cost by scenario. Each cash-flow set follows Exhibit 10. Option A is a fixed-rate sol bond and costs 15.74 percent whatever the exchange rate. Option C hedges 70 percent via the swap (ORS notional 3,500 million at 17.60 percent) and leaves 30 percent (US$75 million) exposed.

Scenario	Probability	A (five-year local)	B (unhedged)	C (70% hedged)
S1: 2% a year	10%	15.74%	11.77%	16.15%
S2: 5.3% a year	25%	15.74%	15.38%	17.23%
S3: 7.0% a year	25%	15.74%	17.24%	17.83%
S4: 7.49% a year	15%	15.74%	17.78%	18.01%
S5: 35% shock	15%	15.74%	20.08%	18.80%
S6: -2% a year	10%	15.74%	7.38%	15.00%

(b) Expected cost and risk. Probability-weighted costs are: A 15.74 percent (standard deviation zero); B **15.75 percent** (standard deviation **3.55 points**); C **17.40 percent** (standard deviation **1.07 points**). The committee's probabilities make the unhedged Eurobond exactly as costly in expectation as the local bond, but with a cost ranging from 7.4 to 20.1 percent. The worst case for B is S5, 4.3 points above A; the best case is S6, 8.4 points below A.

Sample calculation, B in S5: the sol is at 21.00 after one year and 28.35 after two, then rises 5 percent a year to 36.18 at year 7. Coupons in sols are 23.125 x 21.00 = 485.6 in year 1, 655.6 in year 2 and so on; principal is 250 x 36.18 = 9,045. Against net proceeds of 4,920, the IRR is 20.08 percent.

(c) Hedge ratio frontier.

Hedge ratio	Expected cost	Standard deviation	Best case	Worst case
0% (B)	15.75%	3.55	7.38%	20.08%
30%	16.45%	2.48	10.72%	19.57%
50%	16.92%	1.77	12.88%	19.20%
70% (C)	17.40%	1.07	15.00%	18.80%
100%	18.13%	0.00	18.13%	18.13%

Each percentage point of hedge ratio costs about 0.024 points of expected cost and removes about 0.036 points of standard deviation: about **0.67 points of expected cost per point of risk removed**, roughly constant. The hedge is not insurance priced below its expected payoff; it exchanges expected cost for certainty at the forward rate. The most important point is that A, with a standard deviation of zero and an expected cost of 15.74 percent, dominates every row of the frontier; its weaknesses are not FX, but tenor, size and execution (Question 9).

Solution to Question 4: Operating exposure and the natural hedge

(a) Net position and sensitivity. US-dollar revenue is 22 percent x 36,000 = ORS 7,920 million; US-dollar-linked costs are ORS 11,000 million. The net position is 7,920 - 11,000 = **-ORS 3,080 million, short US$154 million** at 20.00. A 20 percent devaluation raises the gross cost of the net short by 0.20 x 3,080 = ORS 616 million. Management's estimate (Exhibit 11) is that after pricing pass-through EBITDA falls by 190 x 2 = **ORS 380 million** in the year of the move (8.1 percent of FY2027E EBITDA of 4,677) and by 95 x 2 = **ORS 190 million** a year thereafter (3.6 percent of FY2028E EBITDA of 5,234). The implied pass-through is 38 percent in the first year (1 - 190 / 308) and 69 percent in steady state (1 - 95 / 308).

(b) Wrong-way exposure. A natural hedge exists when debt and operating cash flows move in opposite directions. Here they move together:

Effect of a 20% devaluation	Local bond (A)	Unhedged Eurobond (B)
EBITDA, year of move	-380	-380
Principal liability	none	+ORS 1,000 (250 x 20 x 0.20)
Annual coupon in sols	none	+ORS 92.5 (23.125 x 20 x 0.20)
Combined net dollar short	US$154 million	US$154 + 250 = **US$404 million**

Borrowing US$250 million unhedged makes Andara's dollar short 2.6 times larger. A dollar loan is a hedge only for a company that is long dollars. The peer table in Exhibit 8 corroborates this: Orvis Grain, with 80 percent of its debt in unhedged dollars, lost 120 percent of net income in the 2023 sell-off.

(c) The plant changes the picture slowly. The net dollar operating position (Exhibit 3) is -154, -154, -128, -71, -1 and +52 (US$ millions) for FY2026 to FY2031. The cumulative operating shortfall over FY2027 to FY2030 is 154 + 128 + 71 + 1 = US$354 million. The company turns long only in FY2031, and by then its net long position of US$52 million would cover the US$23.1 million coupon more than twice but only a fifth of the US$250 million principal. The first two coupons (FY2027, FY2028) are paid while the company is short US$154 million and US$128 million. A reasonable guideline is that unhedged dollar debt should not exceed the net dollar position the company expects to hold when the principal falls due: about US$50 million. A US$250 million bond with the maximum 70 percent swap leaves US$75 million unhedged, above that limit, which argues for a smaller dollar tranche or a higher hedge ratio.

Solution to Question 5: The cross-currency swap

(a) Checking the quote against covered interest parity. A hedged dollar borrower pays approximately the dollar coupon compounded with the forward premium: (1 + 0.0925)(1 + 0.0749) - 1 = **17.43 percent**. The dealer's quote of 17.60 percent therefore embeds a margin of about **17 basis points** for credit, funding and liquidity. The upfront fee of 0.40 percent adds roughly another 7 basis points a year over seven years. The swap is fairly priced against parity; its cost is the parity itself.

(b) Annual carry on US$175 million.

```
Notional in sols: 175 x 20.00 = 3,500
Sol payment:  3,500 x 17.60% = 616.0 a year
Dollar coupon received (to pay the bond): 175 x 9.25% = 16.1875, equal to 323.75 sols at 20.00
Extra cost relative to an unhedged coupon at today's rate = 616.0 - 323.75 = 292.25 sols (US$14.6 million), or
8.35% of notional a year
```

This 8.35-point gap is the hedge cost. It compares with the forward-implied depreciation cost of 1.0925 x 7.49 percent = 8.18 points plus the dealer margin of about 0.17. The hedge costs roughly what the forward market says depreciation will cost; it is a bet on nothing, which is why hedging is rarely cheaper than bearing the risk unless the sol depreciates faster than the forward.

(c) Mark-to-market at the end of year 2. With five years remaining, the dollar leg is worth (at the 4.2 percent dollar rate) 16.1875 x 4.426 + 175 / 1.042^5 = 71.65 + 142.47 = US$214.12 million. The sol leg is worth, at 12.0 percent, 616 x 3.6048 + 3,500 / 1.12^5 = 2,220.6 + 1,985.9 = ORS 4,206.5 million.

Scenario	Sol per dollar at end of year 2	Dollar leg in sols	Sol leg	Swap value to Andara
S5: shock	28.35	6,070.4	4,206.5	+ORS 1,863.9 million (+US$65.7 million)
S6: appreciation	19.21	4,113.3	4,206.5	-ORS 93.2 million (-US$4.9 million)

In S5 the swap gains ORS 1,864 million, more than the ORS 1,461 million (175 x 8.35) by which the hedged bond principal has grown, because it also covers the higher cost of the remaining coupons. Under the credit support annex with a US$10 million threshold, the dealers would post collateral of about US$55.7 million to Andara. That is a source of liquidity, but also a concentration of exposure to two dealers. In S6 the value to Andara is negative but within the threshold, so no collateral is posted.

(d) Convertibility. A swap hedges value, not transferability. If transfers abroad are blocked for a quarter, Andara may hold sols but be unable to buy dollars to pay a Eurobond coupon; the swap's termination clause on inconvertibility may allow dealers to close it out at a time when the company can least manage a settlement. The sound mitigation is structural: retain export proceeds offshore in a dollar collection account (US$396 million a year of exports, against US$23 million of coupons) and hold a debt-service reserve of at least twelve months of coupons offshore. The local bond has no such exposure, which is a point in its favor that the Audit Chair should record.

Solution to Question 6: Credit metrics and covenant testing

(a) Base case. Assumptions: existing bank debt of 5,000 at 15.5 percent (interest 775); cash held at 2,600; dollar debt translated at the budget rate (21.00, 22.05, 23.15); EBITDA of 4,677, 5,234 and 6,105 for years 1 to 3. Option D is the combination of Question 9 (ORS 2,000 million local at 14.75 percent plus US$150 million Eurobond at 9.75 percent, 60 percent hedged at 18.14 percent).

(b) Stress case. The dollar rises to 21.00, 28.35 and 29.77 (S5); EBITDA is reduced from the base by 190 x (S / budget - 1) / 0.10 in year 2 and by half of that in year 3, where S / budget is 28.35 / 22.05 = 1.2857 (a deviation of 28.6 percent), then multiplied by 0.90 and 0.95 for demand. Year 2: (5,234 - 543) x 0.90 = **4,222**. Year 3: (6,105 - 271) x 0.95 = **5,542**. The floating bank loan reprices to 19.5 percent from year 2.

Worked example, Option B, year 2 in the stress case: debt = 5,000 + 250 x 28.35 = 12,087.5; net debt = 9,487.5; interest = 5,000 x 0.195 + 23.125 x 28.35 = 975 + 655.6 = 1,630.6. Gross debt / EBITDA = 12,087.5 / 4,222 = 2.86x; net debt / EBITDA = 2.25x; EBITDA / interest = 2.59x.

Gross debt / EBITDA	Base Y1	Base Y2	Base Y3	Stress Y1	Stress Y2	Stress Y3
A (local)	2.14	1.91	1.64	2.14	2.37	1.80
B (unhedged)	2.19	2.01	1.77	2.19	**2.86**	2.25
C (70% hedged)	2.15	1.94	1.68	2.15	2.52	1.94
D (combination)	2.15	1.93	1.67	2.15	2.49	1.91

Net debt / EBITDA	Base Y1	Base Y2	Base Y3	Stress Y1	Stress Y2	Stress Y3
A (local)	1.58	1.41	1.21	1.58	1.75	1.34
B (unhedged)	1.64	1.51	1.34	1.64	2.25	1.78
C (70% hedged)	1.60	1.44	1.25	1.60	1.90	1.47
D (combination)	1.60	1.44	1.24	1.60	1.87	1.44

EBITDA / interest	Base Y1	Base Y2	Base Y3	Stress Y1	Stress Y2	Stress Y3
A (local)	3.02	3.38	3.94	3.02	**2.41**	3.17
B (unhedged)	3.71	4.07	4.66	3.71	2.59	3.33
C (70% hedged)	3.04	3.39	3.93	3.04	**2.36**	3.08
D (combination)	3.08	3.43	3.99	3.08	**2.40**	3.13
Fully hedged Eurobond	2.83	3.16	3.69	2.83	**2.28**	2.99

A fully hedged Eurobond has the same debt in sols as Option A, so its gross and net leverage are identical to A's.

(c) Which covenant binds. Leverage never approaches the 3.5x maintenance level: the worst net figure is 2.25x (B, stress year 2), and the existing bank covenant of 3.0x at spot passes as well (25 percent headroom). The **interest-cover covenant of 2.5x is the binding test**: in the stress year it is breached by every structure except B, which passes only because its cash interest is low. The 3.0x gross-leverage rating guardrail is closest for B (2.86x, 4.7 percent headroom); with the agency's own stress of dollar debt at 30 percent weaker than spot, B's gross leverage at close would be (5,000 + 250 x 26) / 4,140 = 2.78x, versus 2.42x at spot. The surprising lesson is that hedging reduces covenant headroom in good times and bad: hedged structures carry higher cash interest (a 17.60 percent coupon versus 9.25 percent), so the structure that is safest against devaluation is the one that breaches a cover test first.

Solution to Question 7: Covenant design

(a) Critique of Exhibit 9.

Element	Problem	Fix
Net debt at spot on the test date	Devaluation raises dollar debt in sols instantly, so the covenant measures the exchange rate, not credit quality; the same shock that cuts EBITDA also inflates debt	Translate at a constant (budget) rate, or hedged debt at the swap rate and unhedged debt at spot
Interest cover of 2.5x	At a 15 to 17 percent cost of sol debt, cover is structurally low: a stable Andara with 2.1x gross leverage already has cover near 3.0x. The test is breached in the stress year by every structure except the unhedged bond	Lower to 2.0x, define on cash interest including net swap payments, add an equity cure
Debt includes derivative mark-to-market	In an appreciation the swap value to Andara turns negative and inflates debt; the test is asymmetric and creates a perverse incentive not to hedge	Exclude mark-to-market except termination amounts due and unpaid
EBITDA as reported	Unrealized translation losses of the sort in FY2025 (ORS 310 million) flow through earnings	Exclude unrealized foreign-exchange gains and losses from EBITDA
Maintenance tests in a Eurobond	International investors usually accept incurrence tests only; a technical default on a maintenance test hands lenders leverage that does not match the risk	Keep one maintenance test (leverage) and make the rest incurrence-based
Cross-acceleration of US$25 million	Andara's ORS bank covenants at 3.0x at spot could trigger a cross-default through a definitional mismatch	Align definitions across all documents; amend the bank covenant to the same constant-currency basis
No transfer-risk protection	A convertibility event can cause a payment default with no credit deterioration	Offshore export-proceeds account and a twelve-month debt-service reserve

(b) Proposed package.

Covenant	Proposed term	Test result
Maintenance net debt / EBITDA	3.5x or less at constant currency	Worst case 2.25x (B, stress Y2): EBITDA could fall a further 36 percent before breach
Maintenance interest cover	2.0x or more, cash basis, two equity cures permitted	Worst stress-year cover 2.28x (fully hedged), 2.36x (C), 2.40x (D), 2.41x (A), 2.59x (B)
Incurrence net debt / EBITDA	3.0x or less pro forma	Stress-year net leverage of 1.75x to 2.25x leaves ample headroom
Restricted payments	Dividends up to 50 percent of net income; blocked above 2.5x	Current payout is 30 percent
Reserves and structure	Offshore collection account; reserve of US$23 million	Addresses convertibility
Definitions	Exclude unrealized FX and derivative MTM	Removes artificial breaches

Headroom at the 2.0x interest cover test, as the fraction of EBITDA that can still fall in the stress year (1 - 2.0 / cover): A 17.0 percent, B 22.8 percent, C 15.3 percent, D 16.7 percent, fully hedged 12.3 percent. A fully hedged structure with 12 percent headroom is thin, which is another reason to keep equity-cure rights.

(c) Currency definitions. Three options. *Spot* is volatile and confuses currency and credit. A *budget (constant) rate* is stable but lets lenders ignore real devaluation. A *hedged-rate* translation, with swapped debt at the swap rate and unhedged debt at spot, is the most faithful to economic exposure. Illustration, Option B in stress year 2: net debt / EBITDA is 2.25x at spot (28.35), 1.87x at the budget rate (22.05: debt 5,000 + 250 x 22.05 = 10,512.5, net 7,912.5, divided by 4,222), and 2.25x on a hedged basis

(nothing is hedged). For Option C on a hedged basis: debt = 5,000 + 3,500 + 75 x 28.35 = 10,626, net 8,026, so 1.90x. The right choice depends on who bears the risk the covenant protects; a constant-rate test with a spot "long-stop" at a higher level (for example 4.5x) is a common compromise.

Solution to Question 8: Taxes and issuance frictions

(a) Withholding gross-up. Without the vehicle, the coupon becomes 9.25 / 0.85 = 10.88 percent. The cash flows of Option B with a flat exchange rate give an all-in cost of **11.22 percent** (against 9.57 percent), an extra 165 basis points. The break-even depreciation versus A falls to (1.1574 / 1.1122) - 1 = **4.06 percent**, below the inflation-differential rate of 5.3 percent. Without the vehicle, the Eurobond is not attractive even to a forecaster who believes PPP. The vehicle (about four months) is therefore a gating item for any dollar option.

(b) After-tax costs. The tax rate is 25 percent. Interest in sols is deductible as paid. Swap payments are deductible. For the unhedged bond, the realized FX loss on principal is deductible in year 7 (probable but not certain).

Structure	Pre-tax cost	After-tax cost	After-tax cost if principal FX loss not deductible
A, five-year local	15.74%	11.85%	11.85%
Fully hedged Eurobond	18.13%	13.66%	13.66%
B unhedged in S3 (7.0% a year)	17.24%	13.53%	14.74%
B unhedged in S5 (shock)	20.08%	15.88%	17.32%

For example, A: coupon of 775 after tax is 581.25; net proceeds 4,960; the IRR of those flows is 11.85 percent. Non-deductibility of the principal loss raises the cost of the unhedged bond by 1.2 points in S3 and 1.4 points in S5. The asymmetry matters: the tax authority shares part of a gain and none, or only part, of an unrealized loss.

(c) Implications. All the frictions cut against the dollar bond: a possible 165-point withholding penalty, uncertain deductibility, and a four-month lead time. Each one is invisible in the headline coupon.

Solution to Question 9: Tenor, refinancing risk and a combined structure

(a) Refinancing a five-year bullet. Issue ORS 5,000 million at 15.50 percent for five years, then refinance at year 5 for two more years at 15.50 percent plus a spread shock s, paying a 0.8 percent fee on the refinancing. The all-in cost over seven years is:

Rate shock at refinancing	All-in cost over seven years
0 bp	15.79%
+200 bp	16.17%
+400 bp	16.54%

The seven-year local bond costs 16.45 percent, so locking tenor is cheaper than rolling only if the refinancing shock exceeds about 350 basis points. Refinancing risk is real but modest in expectation: a 400-point shock on ORS 3,000 million raises annual interest by ORS 120 million.

(b) Option D. ORS 2,000 million local five-year at 14.75 percent plus US$150 million Eurobond at 9.75 percent, 60 percent hedged (US$90 million, ORS 1,800 million at 18.14 percent; the quote scales with the coupon: 17.60 + 2 x 0.27 = 18.14). The swap line used is US$90 million, within the US$175 million limit. Unhedged exposure is US$60 million, close to the guideline of Question 4.

Scenario	S1	S2	S3	S4	S5	S6
D all-in cost	15.67%	16.63%	17.17%	17.33%	18.04%	14.66%

Probability-weighted cost 16.79 percent; standard deviation 0.95 points; credit metrics in Question 6 (stress year 2: gross 2.49x, net 1.87x, cover 2.40x).

(c) Comparison.

Option	Expected cost	Standard deviation	Worst case	Stress Y2 cover	Stress Y2 gross leverage
A, five-year local	15.74%	0.00	15.74%	2.41x	2.37x
A2: ORS 3,000 million 5-year at 14.75% plus ORS 2,000 million 7-year at 16.25%	15.65%	0.00	15.65%	2.42x	2.37x
B, unhedged	15.75%	3.55	20.08%	2.59x	2.86x
C, 70% hedged	17.40%	1.07	18.80%	2.36x	2.52x
D, combination	16.79%	0.95	18.04%	2.40x	2.49x
Fully hedged Eurobond	18.13%	0.00	18.13%	2.28x	2.37x

Splitting the local issue (A2) is cheaper than the single ORS 5,000 million bond (15.65 versus 15.74 percent) because the first ORS 3,000 million prices at 14.75 percent, and it also reduces the size of the record issue and adds a seven-year piece; the assumption that the 16.25 percent quote applies to a smaller tranche should be confirmed with the bookrunner. Option D costs 1.14 points more than A2 (about ORS 57 million a year on ORS 5,000 million) and Option C costs 1.75 points more (ORS 87.5 million a year). The price of a dollar tranche is therefore about 1 to 2 points of annual cost, paid for diversification and tenor.

Solution to Question 10: Recommendation

Recommended structure. (1) Issue **ORS 5,000 million in the local market in two tranches**: ORS 3,000 million for five years at 14.75 percent and ORS 2,000 million for seven years at 16.25 percent, all-in cost **15.65 percent**, with no dollar debt. (2) Prepare, but do not execute, a dollar financing: complete the treaty-eligible financing vehicle and establish hedging lines so that a hedged Eurobond can be issued from FY2029, when the net dollar operating position approaches neutral (-US$1 million in FY2030, +US$52 million in FY2031), sized to the net long position plus a hedged remainder. (3) Adopt the covenant package of Question 7 for the local bond and the bank amendments. (4) **Fallback:** if the local book is below ORS 3,500 million, switch to Option D.

Why. Every comparison favors the local bond: it is the cheapest in expected cost, ties with the unhedged bond on expected cost while carrying no dispersion, has the best stress metrics apart from the unhedged bond's cash cover, carries no convertibility or withholding risk, and does not add to an operating dollar short. The Eurobond's headline coupon is lower, but the all-in cost depends on a depreciation view that history and the forward market contradict. Its advantage is tenor and diversification, which D delivers at a premium of about 1.1 points.

Sensitivity discussion

Break-even depreciation (versus the unhedged Eurobond) depends on the local all-in cost and the dollar all-in cost:

Local all-in cost	Dollar 9.57%	Dollar 10.57% (+100 bp)	Dollar 11.22% (withholding)
14.99% (ORS 3,000 million)	4.95%	4.00%	3.39%
15.74% (ORS 5,000 million)	5.63%	4.68%	4.06%
16.45% (seven-year)	6.28%	5.32%	4.70%
17.00%	6.78%	5.82%	5.20%

Each 100 basis points on either rate moves the break-even by about 0.9 points. The hedged cost moves one for one with the swap rate: 16.6, 17.6 and 18.6 percent give all-in costs of 17.11, 18.13 and 19.14 percent. *Probabilities:* shifting 10 points of probability from S6 to S5 raises the expected cost of the unhedged bond from 15.75 to 17.02 percent; a more benign set of weights (20, 30, 20, 10, 5, 15) lowers it to 14.31 percent. The recommendation therefore holds for a wide range of probabilities, but the unhedged bond wins in a benign set; this is what the committee is betting on.

Common student errors

1. Comparing 9.25 and 15.50 percent as if they were in one currency.
2. Computing the hedge cost as the swap fee, rather than the interest differential.
3. Treating the forward rate as a forecast with a premium rather than the price of the hedge.
4. Forgetting that dollar debt adds to an operating short.
5. Testing covenants in the base case only, or only on leverage and never on interest cover.
6. Assuming hedging makes covenants safer in every respect.
7. Ignoring withholding tax, deductibility and issuance costs.
8. Ignoring convertibility, and assuming a swap protects against a transfer block.

Alternative defensible answers

Option C (70 percent hedged Eurobond) is defensible if the board values investor diversification and the company wants to build an international franchise before the dollar earnings arrive; it costs 1.75 points more than A2. *Option D* is the natural fallback and arguably the best balance if the local market cannot absorb the size. *Option B* is defensible only with a documented view that depreciation will be below about 5.6 percent a year and a plan for the operating short; the evidence does not support it. *Hedging more with cash collateral* beyond the swap limit is defensible but expensive: US$50 million idle at about 4 percent while borrowing at 15 percent or more.

Board-style recommendation memorandum

To: Board of Directors, Andara Foods S.A. **From:** Anneliese Dragomir-Sato, Chief Financial Officer **Date:** 8 December 2026 **Subject:** Financing of Port Calvera and the refinancing of bank debt; recommendation for decision on 14 December

Recommendation. Issue ORS 5,000 million of sol-denominated bonds in two tranches (ORS 3,000 million for five years at 14.75 percent and ORS 2,000 million for seven years at 16.25 percent), adopt a revised covenant package, and defer any dollar issue until the plant's dollar earnings arrive in FY2029. If the local book is short, issue Option D (ORS 2,000 million local plus a US$150 million Eurobond, 60 percent hedged).

The CEO's question. The 9.25 percent coupon buys dollars and the 15.50 percent buys sols; the gap is the market's estimate of the sol's decline. The dollar bond costs less only if the sol falls by less than about 5.6 percent a year. History has been 7.4 percent, inflation parity implies 5.3 percent, and the forward market prices 7.5 percent.

The Chairman’s question: the worst case. An unhedged Eurobond in the shock scenario costs 20.1 percent, raises our sol debt by ORS 2,088 million within two years (US$250 million times a move of 8.35) while our EBITDA falls by up to 543 million. The recommended local bond has none of that exposure: its worst case is a refinancing of ORS 3,000 million in 2031 at up to 400 basis points higher, an additional ORS 120 million a year.

The Audit Chair’s question: convertibility. The local bond is unaffected by transfer restrictions. Any dollar issue will have an offshore export account and a twelve-month reserve of US$23 million.

Covenants. Constant-currency net leverage of 3.5x, cash interest cover of 2.0x with an equity cure, exclusion of unrealized FX and derivative mark-to-market, and alignment of the bank covenants. Under the stress case, the tightest cover is 2.42x, a margin of 17 percent against the 2.0x test and a breach of the lenders’ draft 2.5x.

Cost. All-in cost 15.65 percent; no dispersion. A hedged Eurobond would cost 1.1 to 1.75 points more (ORS 57 to 87 million a year).

Decision requested. Approve the local issue, the covenant package and a mandate to the Treasurer to prepare the dollar financing vehicle for FY2029.

Chapter 11: Case: Nimbus Robotics, a Series B Term Sheet Negotiation

All companies, people, funds, and figures in this case are fictional and illustrative. Any resemblance to real organizations is coincidental. The numbers are constructed for teaching and are not forecasts or statistics about any real market.

The Situation

On the morning of Tuesday, November 3, 2026, Priya Raman closed her laptop in the glass-walled conference room on the second floor of Nimbus Robotics' converted warehouse in Pittsburgh and looked at the two documents she had printed. One was a term sheet from Aldergate Ventures. The other was a term sheet from Brightwater Partners. Each offered to lead a $40 million Series B financing. Each would end the company's financing uncertainty for the next two years. And each would reshape who owned the company and who got paid first when it was eventually sold.

Priya is the cofounder and chief executive officer of Nimbus Robotics, Inc. She holds the chair of a four-person board. Her cofounder, Tomás Vidal, is the chief technology officer. The two founded the company in 2021, after Priya spent seven years running operations at a regional third-party logistics provider and Tomás spent six years in a university lab building grasping and manipulation systems. Their thesis was simple and, at the time, unfashionable: that warehouse picking robots would only reach mid-sized distributors if the customer did not have to buy them. Nimbus therefore sells its pick-and-place robot cells in two ways. A customer can buy the hardware outright, or, more commonly now, the customer can sign a three-to-five-year Robots-as-a-Service (RaaS) agreement and pay a monthly fee per deployed cell that bundles hardware, software, remote monitoring, and field service.

That choice of model explains the company's financing history. RaaS revenue is recurring and sticky, but Nimbus pays for each robot up front and collects the cost over years, so faster growth consumes more cash. By the end of September 2026, Nimbus had 41 customers, about 620 robot cells deployed, trailing twelve-month revenue of roughly $22 million, and $8.4 million in cash. Monthly net burn, including debt amortization, was near $1.2 million and rising. Management's model showed cash falling below the $3.0 million minimum-cash covenant in the company's venture debt facility in February 2027.

Nimbus was therefore not negotiating from a position of strength. It was, however, negotiating from a position of momentum. Revenue had grown 109 percent in fiscal 2025 and was on track to grow about 87 percent in fiscal 2026. Net revenue retention among RaaS customers was 128 percent. Two large customers had signed expansion orders in the third quarter. In September, Priya had allowed an introductory conversation with a growth-stage investor she had met at a logistics conference, and by mid-October two funds were competing.

The people

Priya Raman, 39, CEO, cofounder. Holds 3,200,000 common shares. Fully vested.

Tomás Vidal, 41, CTO, cofounder. Holds 2,600,000 common shares. Fully vested.

Dana Whitfield, partner at Crestline Ventures, which led the $13 million Series A in 2024. Holds a board seat. Crestline would like to put money into the Series B to maintain its position (its pro rata right) but has told Priya that it will not lead.

Marcus Oyelaran, managing director at Lantern Seed Fund, lead of the 2022 seed round. Lantern has a board observer, not a seat.

Erik Lindqvist, independent director, a former operating executive in industrial automation. Priya trusts his judgment on commercial questions and has asked him to review the legal terms with the company's counsel.

Nadia Brandt, Nimbus' vice president of finance, who built the model that underlies this case.

Aldergate Ventures ("Aldergate"), a growth-equity investor with a $900 million fund. Its partner, Julian Sato, has been warm, quick, and focused on the plan. Aldergate invests in industrial and logistics technology and usually leads with a clean, standard term sheet.

Brightwater Partners ("Brightwater"), a multi-stage fund with a related logistics operating affiliate. Its partner, Rosalind Teague, has offered a much higher headline valuation. She has also mentioned, in passing, that Brightwater's affiliate operates warehouses where Nimbus robots could be deployed, which could mean meaningful new customer volume.

The decision

The board will meet on Monday, November 9, to choose which investor to pursue. Brightwater's term sheet expires at 5:00 p.m. on Wednesday, November 11. Aldergate's expires on Friday, November 13. Both ask for a 45-day exclusivity period, a "no-shop" clause that would stop Nimbus from soliciting other investors once signed. Priya has told both firms that she will sign at most one. She wants to close by the middle of December 2026 so that the company has cash well before the February covenant date.

Priya has to decide four things. First, which term sheet, if either, to accept. Second, whether to counter, and if so what to ask for and what to give up. Third, how big the option pool should be. And fourth, how to explain the decision to a board containing two investors who prefer different outcomes, a seed fund worried about its price, and a cofounder, Tomás, who has asked the pointed question that opens the case discussion: "If Brightwater's number is 28 percent higher, why would we ever take the lower one?"

The case rests on one idea that students of Volume 1, Chapter 13 (raising equity) and Volume 1, Chapter 15 (capital structure) will recognize in a new setting: the headline valuation in a venture term sheet is only one input, and often not the most important one, into what each class of holder will actually receive. The price per share tells you how many shares the new investor gets. The liquidation preference, the participation right, the option pool, and the anti-dilution protection tell you what those shares are worth in each possible future.

Industry Context

Warehouse robotics and the shift to RaaS

Distribution centers face rising labor costs, high turnover, and customer expectations of faster fulfillment. For a decade, automation vendors served the largest operators with fixed, engineered systems such as conveyors, sortation lines, and shuttle storage, costing tens of millions of dollars per facility and taking a year or more to install. Mid-sized distributors, those operating between one and ten facilities, were largely underserved: their volumes did not justify a fixed system, and their capital budgets were tight.

Flexible robots changed the economics. A mobile or arm-based robot cell can be deployed in days, redeployed when the layout changes, and scaled one unit at a time. Improvements in machine vision and grasp-planning software made it practical to handle the long tail of items that previously required human hands. Vendors responded with subscription pricing that converts a capital purchase into an operating expense. For the customer, RaaS aligns payment with benefit and removes technology risk. For the vendor, it creates a stream of contracted revenue but moves the capital burden onto the vendor's balance sheet.

This is why robotics companies as a group raise large amounts of capital relative to their revenue and why financing structure matters so much. A robot cell that costs $48,000 to build and install and that earns $2,300 per month gross of service costs pays back in roughly two years. The company that funds that deployment with equity at a high price gives up less of the business than one that funds it at a low price; the company that funds it with debt on the robots themselves preserves ownership but takes on fixed obligations.

How exits are priced

Exits for companies like Nimbus have come in three forms: acquisition by a larger automation or logistics-technology vendor, acquisition by a financial sponsor once the company is profitable, or an initial public offering for the few that reach scale. Acquirers typically price on a multiple of trailing or forward revenue, adjusted for growth, gross margin, and the share of recurring revenue. The comparable-company and precedent-transaction tables in the next section are fictional and are used here only to anchor the exit scenarios that the board has asked the finance team to model. Students should not treat them as market data.

Financial Data

All dollar amounts are in thousands of U.S. dollars unless stated otherwise. Share counts are actual shares. Fiscal years end December 31. Forecasts are management’s plan case as prepared by Nadia Brandt and reviewed by the board in October 2026.

Exhibit 1: Income statement summary and plan

	FY2024A	FY2025A	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Robot sales (hardware)	4,100	7,300	11,000	15,000	20,000	25,000	30,000
RaaS subscription revenue	1,900	5,200	12,300	24,000	40,500	59,000	80,000
Services and software	800	1,700	3,200	6,000	9,500	14,000	20,000
Total revenue	**6,800**	**14,200**	**26,500**	**45,000**	**70,000**	**98,000**	**130,000**
Revenue growth	n/a	108.8%	86.6%	69.8%	55.6%	40.0%	32.7%
Gross profit	2,142	5,538	11,395	21,600	36,400	54,880	76,700
Gross margin	31.5%	39.0%	43.0%	48.0%	52.0%	56.0%	59.0%
Research and development	2,500	4,200	7,200	12,500	17,000	21,500	25,000
Sales and marketing	1,700	3,700	7,300	12,500	17,500	24,000	29,500
General and administrative	1,042	2,238	4,400	6,000	8,500	11,000	13,500
Total operating expenses	5,242	10,138	18,900	31,000	43,000	56,500	68,000
Operating income (loss)	**(3,100)**	**(4,600)**	**(7,505)**	**(9,400)**	**(6,600)**	**(1,620)**	**8,700**
Operating margin	(45.6%)	(32.4%)	(28.3%)	(20.9%)	(9.4%)	(1.7%)	6.7%

Operating expenses include stock-based compensation and depreciation. RaaS revenue rises from 28 percent of the total in FY2024 to 62 percent in FY2030 in the plan.

Exhibit 2: Operating metrics (as of September 30, 2026)

Metric	Value
Customers	41
Robot cells deployed	620
RaaS annualized recurring revenue (Q3 run-rate x 4)	$16,800
Net revenue retention, RaaS customers, trailing 12 months	128%
Gross logo churn, trailing 12 months	4%
Average build and installation cost per cell	$48
Average gross monthly fee per RaaS cell	$2.3
Contribution payback per cell (months, after service cost)	26
Fully loaded customer acquisition payback (months)	19
Backlog of signed but not yet deployed RaaS contracts	$11,400
Employees	118

The per-cell figures (cost and monthly fee) are in thousands of dollars. The contribution payback is longer than the simple cost-over-fee ratio of 21 months because field service and monitoring absorb part of the monthly fee.

Exhibit 3: Balance sheet

	Dec 31, 2025	Sep 30, 2026
Cash and equivalents	17,300	8,400
Accounts receivable	3,600	4,900
Inventory	4,100	5,600
Prepaid expenses and other	700	900
Total current assets	**25,700**	**19,800**
Property and equipment, net (including RaaS fleet)	7,400	8,900
Capitalized software and intangibles	1,200	1,100
Other assets	500	600
Total assets	**34,800**	**30,400**
Accounts payable	3,000	3,700
Accrued liabilities	2,000	2,600
Deferred revenue	3,600	4,900
Current portion of venture debt	2,000	2,000
Total current liabilities	**10,600**	**13,200**
Venture debt, noncurrent	14,000	12,000
Other long-term liabilities	1,000	1,100
Total liabilities	**25,600**	**26,300**
Convertible preferred stock (Seed and Series A, at cost)	16,000	16,000
Common stock and additional paid-in capital	3,200	3,600
Accumulated deficit	(10,000)	(15,500)
Total stockholders’ deficit (common)	**(6,800)**	**(11,900)**
Total liabilities, preferred, and deficit	**34,800**	**30,400**

Check: at September 30, 2026, total liabilities of 26,300 plus preferred stock of 16,000 plus the stockholders’ deficit of (11,900) equals 30,400, which equals total assets. The venture debt, from Harbor Lending Partners, carries an 11 percent interest rate, amortizes at $500 per quarter, requires minimum cash of $3,000 at all times, and includes a warrant for 60,000 common shares at $1.40 that is excluded from the cap table below for simplicity.

Exhibit 4: Cash flow forecast and funding gap (plan case, assuming $40,000 Series B closes mid-December 2026)

	Q4 2026E	FY2027E	FY2028E	FY2029E	FY2030E
Operating income (loss)	(2,500)	(9,400)	(6,600)	(1,620)	8,700
Add: depreciation and amortization	450	2,100	3,000	3,900	4,800
Add: stock-based compensation	150	900	1,200	1,600	2,000
Less: cash interest	(350)	(1,375)	(1,155)	(935)	(715)
Less: increase in working capital	(700)	(2,300)	(3,000)	(3,400)	(3,600)
Less: capital expenditure (fleet and equipment)	(1,000)	(3,600)	(4,600)	(5,200)	(5,800)
Free cash flow before financing	**(3,950)**	**(13,675)**	**(11,155)**	**(5,655)**	**5,385**
Venture debt repayment	(500)	(2,000)	(2,000)	(2,000)	(2,000)
Net Series B proceeds ($40,000 less $900 costs)	39,100				
Net change in cash	**34,650**	**(15,675)**	**(13,155)**	**(7,655)**	**3,385**
Ending cash	43,050	27,375	14,220	6,565	9,950

Beginning cash for the fourth quarter is 8,400. The Q4 2026 column reduces cash by 4,450 before the financing, so cash would be 3,950 on December 31, 2026 without the Series B, only 950 above the covenant minimum. The plan therefore funds itself with a thin cushion: the lowest year-end cash is 6,565 at the end of FY2029.

Exhibit 5: Capitalization before the Series B

Holder	Security	Shares	Price paid	Invested ($000)	Fully diluted %
Priya Raman	Common	3,200,000	n/a	n/a	27.59%
Tomás Vidal	Common	2,600,000	n/a	n/a	22.41%
Early employees (exercised)	Common	400,000	n/a	n/a	3.45%
Employee options outstanding	Options (avg. strike $1.40)	1,000,000	n/a	n/a	8.62%
Unallocated option pool	Reserved	600,000	n/a	n/a	5.17%
Lantern Seed Fund and others	Seed Preferred	1,200,000	$2.50	3,000	10.34%
Crestline Ventures and others	Series A Preferred	2,600,000	$5.00	13,000	22.41%
Total fully diluted		**11,600,000**		**16,000**	**100.00%**

Seed and Series A Preferred each carry a 1x non-participating liquidation preference and convert one-for-one into common. The Series A and Seed rank equally with each other (pari passu). The Series A price of $5.00 implied a post-money valuation of $58.0 million on 11.6 million fully diluted shares. Both existing series have broad-based weighted-average anti-dilution protection.

Exhibit 6: The two term sheets side by side

Term	Aldergate Ventures	Brightwater Partners
Amount	$40.0 million	$40.0 million
Pre-money valuation	$100.0 million	$128.0 million
Post-money valuation	$140.0 million	$168.0 million
Security	Series B Preferred	Series B Preferred
Liquidation preference	1.0x, non-participating	1.5x, participating
Participation cap	Not applicable	Total proceeds capped at 3.0x of invested amount ($120.0 million), after which the holder may convert
Seniority	Senior to Series A, Seed, and common	Senior to Series A, Seed, and common
Option pool	Unallocated pool equal to 10% of post-money fully diluted shares, created pre-money	Unallocated pool equal to 14% of post-money fully diluted shares, created pre-money
Anti-dilution	Broad-based weighted average	Full ratchet for 24 months, then broad-based weighted average
Board	Five seats: two common, one Crestline, one Aldergate, one independent	Five seats: two common, one Crestline, two Brightwater
Exclusivity	45 days	45 days
Other	Customary pro rata and information rights	Right of first offer on any sale; commercial introduction to affiliate's warehouses (non-binding)
Expires	Friday, November 13, 2026	Wednesday, November 11, 2026

Exhibit 7: Public comparable companies (fictional)

Company	NTM revenue growth	Gross margin	EV / NTM revenue
Conveyra Automation	22%	38%	3.1x
Palletix Systems	30%	44%	4.6x
Harbor Mechatronics	14%	33%	1.8x
Stackwell Logistics Tech	35%	52%	6.2x
Orbita Robotics	18%	41%	2.6x
Kestrel Motion	26%	47%	4.0x
Median	**24%**	**44%**	**3.55x**
Mean	**24%**	**43%**	**3.72x**

Exhibit 8: Precedent acquisitions (fictional)

Four precedent acquisitions of private robotics companies in 2024 to 2026 were struck at 2.2x to 5.2x trailing revenue, with a median of 3.35x.

Exhibit 9: The board's exit scenarios (for use in the negotiation analysis)

The finance team and Dana Whitfield agreed on five exit scenarios, each assumed to occur in 2030 or 2031, with the probabilities that the board assigned in October.

Scenario	Description	Exit equity value ($ millions)	Board probability
1. Distress	Asset sale or acquihire after a missed financing	45	15%
2. Underperform	Growth stalls near $50 million of revenue; sold at about 2.4x	120	25%
3. Delayed plan	Plan achieved about a year late; about $85 million of revenue at 3.5x	300	30%
4. Plan	FY2030 plan achieved; $130 million of revenue at 4.0x	520	22%
5. Breakout	Outperformance; about $160 million of revenue at 6.2x	1,000	8%

The probabilities sum to 100 percent. For simplicity, the exit values are net proceeds available to equity holders after any debt and transaction costs, and the exit analysis ignores future financing rounds, taxes, and the time value of money; the Teaching Note revisits those simplifications.

Complications and Exhibits

Exhibit 10: Hiring plan and option pool requirement (24 months after closing)

Role group	Planned hires	Average grant (options)	Options required
Senior leaders (vice president and above)	6	90,000	540,000
Senior engineers and managers	12	35,000	420,000
Other hires	20	12,000	240,000
Retention refresh for existing key employees	n/a	n/a	250,000
Total for 24 months	**38**		**1,450,000**

Nadia Brandt notes that the existing unallocated pool of 600,000 shares is available for use before any new pool is created, and that hiring is likely to run slightly behind the plan in the first year.

Exhibit 11: Counter-proposal ideas from Erik Lindqvist

Erik proposed two counters for the board to evaluate, each keeping $40.0 million as the round size.

Counter	Structure
C1: Aldergate counter	Aldergate's structure (1.0x non-participating, broad-based weighted average), but at a $112.0 million pre-money valuation and an unallocated pool of 8% of post-money fully diluted shares
C2: Brightwater counter	Brightwater at a $115.0 million pre-money valuation with 1.0x non-participating preference, no participation, a 12% post-money unallocated pool, broad-based weighted average anti-dilution, and a board of two common, one Crestline, one Brightwater, one independent

Exhibit 12: A hypothetical down round

To test anti-dilution, the board asked what would happen if, in 2028, the company needed to raise $20.0 million in a Series C at a price of $5.50 per share. This is lower than the Series B price under either term sheet and higher than the Series A price of $5.00, so only the Series B conversion price would adjust. Assume that for the weighted-average calculation, the pre-round fully diluted share count equals the post-Series-B fully diluted count (including the unallocated pool).

Exhibit 13: A plan-case Series C and a slippage case

If the plan is met, the board expects to raise a $30.0 million Series C in 2028 at a $260.0 million pre-money valuation, to fund fleet growth and international expansion, though the plan in Exhibit 4 does not strictly require it. In a slippage case, revenue in FY2027 to FY2029 is 20 percent below the plan, gross margin percentages and operating expenses are unchanged, working capital investment is 20 percent lower than plan and capital expenditure is 10 percent lower, and all other items match the plan.

Additional facts and open issues

Crestline's position. Under the Series A documents, issuing a new senior security requires Crestline's consent. Dana Whitfield has told Priya that Crestline will support whichever offer the board prefers but has noted that Brightwater's 1.5x participating preference reduces the proceeds of Series A in modest exits. Crestline also wants to exercise its pro rata right, which on a fully diluted basis is 22.41 percent of any new round; if it does, Aldergate or Brightwater will lead for a smaller dollar amount unless the round is upsized.

Lantern's position. Marcus Oyelaran says Lantern's $2.50 seed price looks excellent at either valuation; its concern is survival, and it prefers simpler terms.

Brightwater's secondary suggestion. Rosalind Teague mentioned by phone that Brightwater might buy up to $4.0 million of founder common at the Series B price. It is not in the term sheet and is not modeled; a sale would be taxable to the founders.

Brightwater's commercial introduction. The Brightwater affiliate operates warehouses that could become Nimbus customers. Rosalind has suggested that deployments could generate $6.0 million of RaaS annualized revenue by the end of 2028, but this is non-binding, not in the term sheet, and creates a related-party relationship that the board would have to monitor.

Diligence speed. Brightwater can close in four weeks; Aldergate needs six to seven for its customer-reference program. Priya's target close is mid-December.

Ambiguity in the pool language. Both term sheets require an "unallocated" pool equal to a stated percentage of post-money fully diluted shares. Aldergate's counsel confirmed that the 600,000 existing unallocated shares count toward the target. Brightwater's term sheet is silent. Students should use the interpretation that the existing unallocated shares count toward the target in both cases, and discuss the other interpretation in class.

Ambiguity in option data. Outstanding options (1,000,000) have an average strike of $1.40, though strikes range from $0.40 to $3.10 and about 35 percent are unvested. For the exit analysis, assume all vest and exercise at $1.40 if, and only if, the common per-share value exceeds $1.40, with exercise proceeds added to distributable value. Unallocated pool shares are cancelled at exit, and the lender's 60,000 warrants are ignored.

Venture debt. Harbor Lending must consent in a sale; management assumes it will, and Exhibit 9 values are net of debt.

The founders' personal position. Priya has said that at least $10 million for each founder in a modest sale would change her willingness to take risk. Tomás is more focused on keeping control of the product roadmap.

Assignment Questions

Answer each question with explicit calculations. Where a question refers to a term sheet, use the structure in Exhibit 6; where it refers to exit values, use the scenarios in Exhibit 9 and the assumptions in the Complications section. Use $1.40 as the common option strike and treat all 1,000,000 options as exercised when in the money.

Question 1. Price and capitalization. For each term sheet, compute the Series B price per share, the number of Series B shares issued, the size of the new unallocated option pool, the post-money fully diluted share count, and the post-money ownership percentage of each holder group. Show that price times post-money fully diluted shares equals the stated post-money valuation.

Question 2. The pool shuffle and effective valuation. For each term sheet, compute the effective pre-money valuation, defined as the Series B price times the 11,600,000 pre-round fully diluted shares. Explain the difference between the effective and headline pre-money values in terms of the option pool top-up, and state the cost to existing holders.

Question 3. The waterfall. Build the liquidation waterfall for each term sheet at exit values of $45 million, $100 million, $300 million, and $520 million. For each, state which series convert, the per-share value of common stock, and the proceeds of the Series B, Series A, Seed, Priya, Tomás, early employees, and option holders. Then compute combined founder proceeds at exit values of $60, $80, $120, $150, $200, $250, $400, $700, and $1,000 million.

Question 4. Breakeven. Find the exit value above which the founders do better under Brightwater's term sheet than under Aldergate's. Find the exit values at which the Series B converts to common under each. Explain the economics of the crossover.

Question 5. Expected value and risk. Using the board's probabilities, compute the expected combined founder proceeds and Priya's expected proceeds under each term sheet. Compute Priya's certainty equivalent under logarithmic utility of wealth, with utility defined as the natural log of proceeds plus $1.0 million. Repeat the expected-value calculation under a bull distribution (5%, 15%, 30%, 30%, 20% across the five scenarios) and a bear distribution (30%, 30%, 25%, 12%, 3%).

Question 6. The counters. Compute the capitalization and founder ownership under Counters C1 and C2 from Exhibit 11, and the founders' proceeds at the five scenario exit values and in expectation. Compare them with the two original term sheets. What pre-money valuation, with Aldergate's structure and a 10 percent pool, would make the founders' expected proceeds equal to those under Brightwater's term sheet? What pre-money would equalize proceeds in the plan scenario at $520 million?

Question 7. Option pool sizing. Using Exhibit 10, compute the pool percentage that would exactly cover 24 months of hiring under each term sheet. Compute the cost, in founder ownership points, of the 10 percent and 14 percent pools compared with the right-sized pool. Evaluate the alternative of a smaller pool with a commitment to top it up with the next round.

Question 8. Anti-dilution. Using Exhibit 12, compute the adjusted Series B conversion price and the additional conversion shares under broad-based weighted average and under full ratchet, for each term sheet. Compute the founders' ownership after the Series C under each case. Why does the full-ratchet protection matter more in the Brightwater structure than the pricing alone suggests?

Question 9. Runway and the next round. Using Exhibits 4 and 13, verify the plan-case ending cash, compute the slippage-case cash path, and determine when and by how much the company would have to raise additional capital. Compute the founders' ownership after the plan-case Series C under each term sheet, and discuss how the Brightwater board structure and right of first offer would affect the C round.

Question 10. Recommendation. Prepare a recommendation for the board: which term sheet to pursue, which terms to negotiate, in what order of priority, what the walk-away position is, and how to handle Crestline, Lantern, and Tomás. Support it with your numerical findings from Questions 1 to 9. Identify at least three risks to your recommendation and two pieces of information you would want before the November 9 meeting.

Teaching Note and Full Solutions

Learning objectives

After working through this case, students should be able to:

1. Translate a venture term sheet into a capitalization table, including the effect of an option pool created before the investment.
2. Distinguish headline, effective, and economic pre-money valuations and explain why they differ.
3. Build a liquidation waterfall with senior preferences, non-participating preferred with a conversion decision, and capped participating preferred.
4. Locate the crossover exit value at which two term sheets deliver equal outcomes to common holders, and interpret it.
5. Evaluate term sheets under uncertainty using probability-weighted outcomes and a simple risk-aversion adjustment, and test conclusions against alternative probability assumptions.
6. Apply broad-based weighted-average and full-ratchet anti-dilution formulas and trace their effect on founder ownership.
7. Size an option pool from a hiring plan rather than from a convention.
8. Connect financing terms to the operating plan and communicate a recommendation to a divided board.

Suggested class plan (90 minutes)

Time	Segment	Purpose
0:00 to 0:10	Cold open: Tomás's question. Poll the class on which term sheet to take, before any analysis.	Surface the headline-valuation bias
0:10 to 0:25	Q1 and Q2 at the board: build both cap tables, compute effective pre-money	Establish mechanics; reveal the pool shuffle
0:25 to 0:50	Q3 and Q4: waterfall in small groups; plot founder proceeds against exit value; find the crossover	The core of the case
0:50 to 1:05	Q5 and Q6: expected value, risk aversion, counters	Move from mechanics to decision
1:05 to 1:15	Q7, Q8, Q9: pool sizing, anti-dilution, runway (assign as pre-work or discuss briefly)	Show the second-order terms
1:15 to 1:30	Q10: students pitch recommendations; instructor reveals the sensitivity findings and repoll	Close the loop; compare to the opening poll

Instructors with 60 minutes can assign Questions 1, 2, 7, 8, and 9 as pre-work.

Solution to Question 1: price and capitalization

Step 1: the unchanging base. Exclude the existing unallocated pool, because the new pool target replaces it. The other holders are Priya 3,200,000; Tomás 2,600,000; early employees 400,000; outstanding options 1,000,000; Seed 1,200,000; and Series A 2,600,000. Their sum is 11,000,000 shares.

Step 2: the algebra of a pre-money pool. Let T be the post-money fully diluted share count, p the pool percentage of T, and s the Series B percentage of T. The identity is

```
T = 11,000,000 + p × T + s × T
T = 11,000,000 / (1 - p - s)
price = post-money valuation / T
```

The Series B percentage s equals new money divided by post-money valuation, because the investor’s ownership must equal the fraction of post-money value it contributes.

Aldergate. s = 40 / 140 = 28.5714 percent and p = 10 percent. Then 1 - 0.10 - 0.285714 = 0.614286, so T = 11,000,000 / 0.614286 = 17,906,977 shares. The price is $140,000,000 / 17,906,977 = $7.8182. The Series B receives 0.285714 × 17,906,977 = 5,116,279 shares (check: $40,000,000 / $7.8182 = 5,116,279). The pool is 0.10 × 17,906,977 = 1,790,698 shares, which is a top-up of 1,790,698 - 600,000 = 1,190,698 shares.

Brightwater. s = 40 / 168 = 23.8095 percent and p = 14 percent. Then 1 - 0.14 - 0.238095 = 0.621905, so T = 11,000,000 / 0.621905 = 17,687,596 shares. The price is $168,000,000 / 17,687,596 = $9.4982. The Series B receives 4,211,332 shares. The pool is 0.14 × 17,687,596 = 2,476,264 shares, a top-up of 1,876,264 shares.

Post-money capitalization:

Holder	Shares (Aldergate)	%	Shares (Brightwater)	%
Priya Raman	3,200,000	17.87%	3,200,000	18.09%
Tomás Vidal	2,600,000	14.52%	2,600,000	14.70%
Early employees	400,000	2.23%	400,000	2.26%
Options outstanding	1,000,000	5.58%	1,000,000	5.65%
Unallocated pool	1,790,698	10.00%	2,476,264	14.00%
Seed Preferred	1,200,000	6.70%	1,200,000	6.78%
Series A Preferred	2,600,000	14.52%	2,600,000	14.70%
Series B Preferred	5,116,279	28.57%	4,211,332	23.81%
Total	**17,906,977**	**100.00%**	**17,687,596**	**100.00%**

Check. For Aldergate, $7.8182 × 17,906,977 = $140.0 million. For Brightwater, $9.4982 × 17,687,596 = $168.0 million. The founders’ combined ownership falls from 50.00 percent to 32.39 percent under Aldergate and to 32.79 percent under Brightwater. The founders end up with almost the same ownership under both term sheets, even though Brightwater’s headline valuation is 28 percent higher. That near-equality is the first surprise of the case, and it comes entirely from the larger pool: Brightwater’s higher price is offset by the larger number of shares in the pool and the smaller dollar share of the company it must buy.

Solution to Question 2: the pool shuffle and effective valuation

The effective pre-money valuation values only the shares that existed before the round, including the existing unallocated pool of 600,000 shares, at the Series B price.

```
Aldergate: 7.8182 × 11,600,000 = $90.69 million
Brightwater: 9.4982 × 11,600,000 = $110.18 million
```

The same figures follow by subtracting the value of the pool top-up from the headline:

```
Aldergate: 100.00 - 1,190,698 × 7.8182 = 100.00 - 9.31 = $90.69 million
Brightwater: 128.00 - 1,876,264 × 9.4982 = 128.00 - 17.82 = $110.18 million
```

The headline premium of Brightwater over Aldergate is 28.0 percent. The effective premium is 110.18 / 90.69 - 1 = 21.5 percent, which also equals the ratio of the per-share prices, 9.4982 / 7.8182 - 1 = 21.5 percent. About a quarter of Brightwater's headline premium is consumed by its larger pool.

Cost to existing holders. The pool top-up costs existing holders $9.31 million under Aldergate and $17.82 million under Brightwater, in the sense that the new investor's price is set as if those shares were already outstanding. If we allocate that cost pro rata to the 11.6 million pre-round fully diluted shares, the founders, who hold 5.8 million of them (50 percent), bear $4.65 million and $8.91 million respectively.

Interpretation. If the company needs the options, someone must bear the dilution; a pre-money pool puts it all on existing holders, while a post-money pool shares it with the new investor.

The step-up from the Series A price of $5.00 is 1.56x for Aldergate and 1.90x for Brightwater. Both clear the seed and Series A investors' hurdles; neither triggers anti-dilution adjustments.

Solution to Question 3: the waterfall

Method. (1) Add option exercise proceeds (1,000,000 x $1.40 = $1.4 million) to the exit value if the common per-share price exceeds $1.40. (2) Pay the senior preference: $40.0 million (Aldergate) or 1.5 x $40.0 = $60.0 million (Brightwater). (3) Pay Series A ($13.0 million) and Seed ($3.0 million) preferences pari passu for any series that does not convert. (4) Distribute the remainder per share among common, options, converted preferred, and (Brightwater) the participating Series B on an as-converted basis, until the Series B reaches its $120.0 million total cap. A non-participating series converts if and only if the common per-share value exceeds its original price (Seed $2.50, Series A $5.00, Aldergate Series B $7.8182); we solve for the consistent set of decisions.

Exit of $45 million. Aldergate: the Series B takes its $40.0 million preference. The remaining $5.0 million goes to Series A and Seed pari passu in proportion to their preferences ($13.0 million and $3.0 million): Series A gets $5.0 × 13/16 = $4.06 million and Seed gets $0.94 million. Common receives nothing. Brightwater: the Series B preference is $60.0 million, which exceeds the exit value, so the Series B takes all $45.0 million. Everyone else receives nothing. Founders receive $0 under both. The difference falls on the Series A and Seed, who receive $5.0 million under Aldergate and nothing under Brightwater.

Exit of $100 million. Aldergate: the Series B takes $40.0 million. Test whether the Series A and Seed convert. If both convert, the amount to distribute is $100.0 + $1.4 (option proceeds) - $40.0 = $61.4 million, shared by 6,200,000 founder and employee shares, 1,000,000 options, 2,600,000 Series A shares, and 1,200,000 Seed shares, a total of 11,000,000 shares. The per-share value is $61.4 / 11.0 = $5.5818, which exceeds both $5.00 and $2.50, so both convert (and the Series B, whose conversion price is $7.8182, does not). Priya receives 3,200,000 × $5.5818 = $17.86 million. Tomás receives 2,600,000 × $5.5818 = $14.51 million. Early employees receive $2.23 million. Option holders receive 1,000,000 × ($5.5818 - $1.40) = $4.18 million. Series A receives $14.51 million; Seed receives $6.70 million; Series B receives $40.00 million. The total is $100.0 million. Brightwater: the Series B takes its $60.0 million preference first. Series A and Seed do not convert, because the amount left is small: they take $13.0 million and $3.0 million in preference. The remainder is $100.0 + $1.4 - $60.0 - $16.0 = $25.4 million, shared among common (6,200,000), options (1,000,000) and the participating Series B on an as-converted basis (4,211,332), a total of 11,411,332 shares: $2.2259 per share. Check that Seed would not prefer to convert: its $2.50 original price exceeds $2.2259. Priya receives 3,200,000 × $2.2259 = $7.12 million. Tomás receives $5.79 million. Early employees receive $0.89 million. Options receive 1,000,000 × ($2.2259 - $1.40) = $0.83 million. The Series B receives $60.0 + 4,211,332 × $2.2259 = $60.0 + $9.37 = $69.37 million. The total is $100.0 million. The Series B receives 69 percent of the proceeds with 23.8 percent of the shares.

Exit of $300 million. Aldergate: all series convert; the per-share value is $301.4 / 16,116,279 = $18.70. Priya receives $59.85 million, Tomás $48.62 million, options $17.30 million, Series A $48.62 million, Seed $22.44 million, and the Series B $95.68 million. Brightwater: the Series B takes its $60.0 million preference and participates, but participation is limited to $120.0 - $60.0 = $60.0 million, which binds once the per-share value exceeds $60.0 / 4.211332 = $14.25. With the cap binding, the Series B receives $120.0 million and the other 11,000,000 as-converted shares split $301.4 - $120.0 = $181.4 million, or $16.49 per share. The Series B would convert only if its as-converted value exceeded $120.0 million (a per-share value of $28.50); at $300 million that value is $83.4 million, so it does not. Priya receives $52.77 million and Tomás $42.88 million. Founders: $108.47 million (Aldergate) versus $95.65 million (Brightwater).

Exit of $520 million. Aldergate: all convert at $521.4 / 16,116,279 = $32.35 per share; Priya receives $103.53 million and Tomás $84.12 million. Brightwater: if the Series B converts, the per-share value is $521.4 / 15,211,332 = $34.28 and it receives $144.35 million, more than the $120.0 million cap, so it converts. Priya receives $109.69 million and Tomás $89.12 million. Founders: $187.64 million (Aldergate) versus $198.81 million (Brightwater).

Combined founder proceeds across exit values ($ millions):

Exit value	Aldergate: founders	Brightwater: founders	Aldergate: Priya	Aldergate: Tomás	Brightwater: Priya	Brightwater: Tomás
45	0.00	0.00	0.00	0.00	0.00	0.00
60	3.74	0.00	2.06	1.68	0.00	0.00
80	19.61	2.23	10.82	8.79	1.23	1.00
100	32.37	12.91	17.86	14.51	7.12	5.79
120	42.92	22.26	23.68	19.24	12.28	9.98
150	54.49	34.85	30.06	24.42	19.23	15.62
200	72.48	53.92	39.99	32.49	29.75	24.17
250	90.47	72.98	49.92	40.56	40.26	32.72
300	108.47	95.65	59.85	48.62	52.77	42.88
400	144.46	148.37	79.70	64.76	81.86	66.51
520	187.64	198.81	103.53	84.12	109.69	89.12
700	252.42	267.44	139.27	113.16	147.55	119.89
1,000	360.39	381.83	198.83	161.55	210.66	171.16

Under Aldergate the founders receive their first dollar once the exit exceeds the total preference stack of $40.0 + $13.0 + $3.0 = $56.0 million. Under Brightwater the stack is $60.0 + $13.0 + $3.0 = $76.0 million, so the founders receive nothing until the exit exceeds $76 million. The personal threshold that Priya mentioned, $10 million for each founder, is met by Priya at an exit of about $78 million and by Tomás (who holds fewer shares) at about $84 million under Aldergate. Under Brightwater the corresponding exit values are about $111 million for Priya and $120 million for Tomás. In the modest-sale zone of $80 to $120 million, the choice of term sheet therefore decides whether the founders reach their personal goal.

Conversion thresholds. Under Aldergate the Seed converts at an exit of about $73 million (price per common share of $2.50), the Series A at about $94 million ($5.00), and the Series B at about $125 million ($7.8182). Under Brightwater the Seed converts at about $103 million and the Series A at about $135 million. The Series B participation cap begins to bind at about $275 million and the Series B converts at about $432 million.

Implementation note. The reference solution was computed in Python by enumerating all eight conversion combinations at each exit value and selecting the one in which no series gains by changing its decision.

Solution to Question 4: breakeven

Where the term sheets cross. The founders do better under Brightwater only when the Series B's Brightwater payoff is small enough to offset its lower share count. The relevant region is the one in which Brightwater's Series B is capped at $120.0 million (exits between about $275 million and $432 million) and Aldergate's Series B has converted to common.

Aldergate region (all series converted). The founders hold 5,800,000 of 16,116,279 shares that participate at exit (the 11,000,000 other shares plus the 5,116,279 Series B shares; the unallocated pool is cancelled). Founder proceeds are

```
F_A(E) = 5,800,000 × (E + 1.4) / 16,116,279     (E in $ millions; 1.4 is option exercise cash)
```

Brightwater region (Series B capped at $120.0 million). The Series B receives $120.0 million, and the 11,000,000 other participating shares split the rest:

```
F_B(E) = 5,800,000 × (E + 1.4 - 120.0) / 11,000,000
```

Set the two equal and cancel the 5,800,000:

```
(E + 1.4) / 16.116279 = (E - 118.6) / 11.0
11.0 × (E + 1.4) = 16.116279 × (E - 118.6)
11.0 E + 15.4 = 16.116279 E - 1,911.4
5.116279 E = 1,926.8
E* = $376.6 million
```

At $376.6 million the founders receive $136.04 million under either term sheet (check: 5.8 × 378.0 / 16.116279 = 136.04). Remarkably, the Series B receives exactly $120.0 million under both at that point, so the other holders receive identical amounts. Above $376.6 million, Brightwater is better for the founders. The advantage is small: $3.9 million at $400 million, $11.2 million at $520 million, and $21.4 million at $1 billion.

Conversion thresholds. Aldergate's Series B converts at an exit of about $124.6 million (all converted: $7.8182 x 16,116,279 = $126.0 million less $1.4 million option cash). Brightwater's converts at about $432 million (per-share value $120.0 / 4,211,332 = $28.50; $28.50 x 15,211,332 = $433.5 million less $1.4 million).

Where Aldergate's advantage peaks. The founders' advantage under Aldergate is largest in the zone of modest outcomes. It is about $19.5 million at $100 million, peaks at about $21.0 million at an exit near $124.5 million (just where Aldergate's Series B converts), and is still $19.6 million at $150 million and $12.8 million at $300 million. The mid-range exits of $80 to $250 million cost the founders $17 to $21 million under Brightwater.

Economics of the crossover. Brightwater's structure is a bet by the founders that the exit will exceed about $377 million. Below that, the 1.5x senior preference plus participation transfers value from common holders to the Series B. The following table shows the return multiple of the Series B under each term sheet:

Exit value ($ millions)	Aldergate Series B multiple	Brightwater Series B multiple
45	1.00x	1.13x
120	1.00x	1.90x
300	2.39x	3.00x
520	4.14x	3.61x
1,000	7.95x	6.93x

Brightwater guarantees itself a return of at least 1.5x in any exit above $60 million and a return of 3.0x in any exit between $275 million and $432 million. Aldergate guarantees only its money back. At very high exits Brightwater gives up a bit more because its lower ownership (23.8 versus 28.6 percent) yields less than Aldergate's. A structure with such a pattern is,

economically, a downside protection product sold by the founders for the price of a higher headline valuation. The question is whether the headline price compensates for it, which Questions 5 and 6 answer.

Solution to Question 5: expected value and risk

Expected combined founder proceeds ($ millions):

Scenario	Probability	Aldergate	Brightwater
1. Distress ($45)	15%	0.00	0.00
2. Underperform ($120)	25%	42.92	22.26
3. Delayed plan ($300)	30%	108.47	95.65
4. Plan ($520)	22%	187.64	198.81
5. Breakout ($1,000)	8%	360.39	381.83
Expected value		**113.38**	**108.54**

```
Aldergate: 0.25 × 42.92 + 0.30 × 108.47 + 0.22 × 187.64 + 0.08 × 360.39
         = 10.73 + 32.54 + 41.28 + 28.83 = 113.38
Brightwater: 0.25 × 22.26 + 0.30 × 95.65 + 0.22 × 198.81 + 0.08 × 381.83
         = 5.57 + 28.70 + 43.74 + 30.55 = 108.54
```

Aldergate's structure is worth $4.8 million more in expectation, in spite of the 28 percent lower headline. The probability-weighted exit value is $321.15 million in both cases (0.15 × 45 + 0.25 × 120 + 0.30 × 300 + 0.22 × 520 + 0.08 × 1,000); the term sheets simply divide it differently. Priya's expected proceeds are $62.56 million (Aldergate) and $59.89 million (Brightwater); Tomás's are $50.83 million and $48.66 million.

Certainty equivalent. Priya's proceeds by scenario are $0, $23.68, $59.85, $103.53, and $198.83 million under Aldergate, and $0, $12.28, $52.77, $109.69, and $210.66 million under Brightwater. With utility u(x) = ln(x + 1.0):

```
Aldergate: ln(1) = 0; ln(24.68) = 3.2060; ln(60.85) = 4.1083; ln(104.53) = 4.6495; ln(199.83) = 5.2975
  E[u] = 0.15 × 0 + 0.25 × 3.2060 + 0.30 × 4.1083 + 0.22 × 4.6495 + 0.08 × 5.2975 = 3.4807
  CE = exp(3.4807) - 1 = 32.48 - 1 = $31.48 million

Brightwater: ln(1) = 0; ln(13.28) = 2.5863; ln(53.77) = 3.9847; ln(110.69) = 4.7067; ln(211.66) = 5.3550
  E[u] = 0.25 × 2.5863 + 0.30 × 3.9847 + 0.22 × 4.7067 + 0.08 × 5.3550 = 3.3059
  CE = exp(3.3059) - 1 = 27.27 - 1 = $26.27 million
```

The certainty-equivalent gap ($5.2 million) is larger than the expected-value gap ($2.7 million for Priya alone), because Brightwater's structure takes value away in the low and middle scenarios, where an additional dollar is worth more to a risk-averse founder, and adds it in the high scenarios.

Alternative probabilities (combined founders, $ millions):

Distribution	Aldergate	Brightwater	Difference
Board (15, 25, 30, 22, 8)	113.38	108.54	Aldergate +4.84
Bull (5, 15, 30, 30, 20)	167.35	168.04	Brightwater +0.69
Bear (30, 30, 25, 12, 3)	73.32	65.90	Aldergate +7.42

Moving linearly from the board distribution toward the bull distribution, the two term sheets have equal expected value when about 87.5 percent of the way to the bull case. At that point the probability of a plan-or-better outcome (scenarios 4 and 5) is 47.5 percent, versus 30 percent in the board's view. In plain terms, Brightwater's term sheet is only better in expectation if the founders believe the chance of reaching the plan is nearly half.[222]

Solution to Question 6: the counters

Counter C1 (Aldergate structure, $112 million pre-money, 8 percent pool). s = 40 / 152 = 26.3158 percent, so T = 11,000,000 / (1 - 0.08 - 0.263158) = 16,746,795 and the price is $152,000,000 / 16,746,795 = $9.0764. Series B shares = 4,407,051; pool = 1,339,744. Founders hold 34.63 percent; effective pre-money is $9.0764 x 11,600,000 = $105.29 million.

Counter C2 (Brightwater, $115 million pre-money, non-participating, 12 percent pool). s = 40 / 155 = 25.8065 percent, so T = 11,000,000 / (1 - 0.12 - 0.258065) = 17,686,722 and the price is $8.7636. Series B shares = 4,564,315; pool = 2,122,407. Founders hold 32.79 percent; effective pre-money is $101.66 million.

Because both counters use a 1.0x non-participating senior Series B, founder proceeds follow the Aldergate pattern, with the Series B converting at its own conversion price. The comparison ($ millions, combined founders):

Structure	Founders' ownership	Scenario 2 ($120)	Scenario 3 ($300)	Scenario 4 ($520)	Scenario 5 ($1,000)	Expected value
Aldergate as offered	32.39%	42.92	108.47	187.64	360.39	113.38
Brightwater as offered	32.79%	22.26	95.65	198.81	381.83	108.54
C1	34.63%	42.92	113.46	196.28	376.98	118.11
C2	32.79%	42.92	112.32	194.30	373.17	117.02

At $120 million the founders receive $42.92 million in all three non-participating structures, because the Series B takes its $40.0 million preference and does not convert, so the price per share is irrelevant. C1 improves expected founder proceeds by $4.73 million over the Aldergate offer; C2 improves them by $3.64 million. C2, which would turn Brightwater's term sheet into a clean structure at a still-generous price, beats the original Brightwater term sheet by $8.48 million in expectation. Both counters beat both originals in expectation and at every exit below the $377 million crossover. Both counters are also worth less to the investors, so the likelihood of acceptance is below 100 percent.

Equivalent pre-money valuations. With Aldergate's structure and a 10 percent pool, founders' expected proceeds equal those under Brightwater's term sheet ($108.54 million) at a pre-money valuation of about **$87.8 million**. In other words, Brightwater's $128.0 million headline is worth, in the board's own scenarios, the same to the founders as $87.8 million of clean valuation, $12.2 million less than Aldergate's offer. To equalize founder proceeds in the plan scenario alone ($198.81 million at $520 million), the Aldergate-type pre-money would have to be about **$120.5 million**. In the delayed-plan scenario ($95.65 million at $300 million) it is about **$71.6 million**. The value of Brightwater's structure to the founders is therefore very sensitive to the scenario: it is a $20 million gift to Brightwater at moderate exits and a $10 million benefit to the founders at high exits.

Solution to Question 7: option pool sizing

Pool needed. From Exhibit 10, 24 months of hiring requires 1,450,000 options. The existing unallocated pool is 600,000, so 850,000 new options are required. Solving for the pool percentage that provides exactly 1,450,000 unallocated shares:

```
T = (11,000,000 + 1,450,000) / (1 - s)
Aldergate: T = 12,450,000 / 0.714286 = 17,430,000;  p = 1,450,000 / 17,430,000 = 8.32%
Brightwater: T = 12,450,000 / 0.761905 = 16,340,625; p = 1,450,000 / 16,340,625 = 8.87%
```

With the right-sized pool, the Series B price would be $140.0 million / 17,430,000 = $8.0321 under Aldergate and $168.0 million / 16,340,625 = $10.2811 under Brightwater, and the founders' ownership would be 33.28 percent and 35.49 percent respectively.

Cost of oversizing. Measured in ownership points at closing:

Term sheet	Pool as proposed	Founders' ownership	Right-sized pool	Founders' ownership	Difference
Aldergate	10%	32.39%	8.32%	33.28%	0.89 points
Brightwater	14%	32.79%	8.87%	35.49%	2.70 points

Brightwater's 14 percent pool is 5.1 points larger than needed. If Brightwater's pool were 10 percent, the founders' ownership would be 34.90 percent, 2.11 points higher than at 14 percent.

Economic cost if the pool is unused. Unallocated shares are cancelled at exit, so the true cost of an oversized pool is the lower Series B price, which gives the Series B more shares. At the $520 million plan exit, founders would receive $189.24 million under a right-sized Aldergate pool versus $187.64 million (a $1.60 million difference), and $203.09 million under a right-sized Brightwater pool versus $198.81 million ($4.28 million). Unused pool shares disappear at exit, but the price paid for them does not come back.

Smaller pool with a later top-up. A 6 percent pool under Aldergate's structure (pool 1,008,734 shares; price $8.3273; founders 34.50 percent) covers about 70 percent of the 24-month need, roughly 17 months of hiring, and gives founders 2.11 more points at closing. The risk is a pre-money pool top-up demanded by the Series C investor when the company's leverage may be weaker. A reasonable position is a pool sized to 12 to 18 months of hiring (about 7 to 8 percent), with the plan documented for the investor.

Solution to Question 8: anti-dilution

The hypothetical Series C is $20.0 million at $5.50, so C = $20,000,000 / $5.50 = 3,636,364 new shares. The broad-based weighted-average formula is

```
CP2 = CP1 × (A + B) / (A + C)
A = fully diluted shares before the issue; B = shares the new money would buy at CP1; C = shares actually issued
```

Aldergate (A = 17,906,977; CP1 = $7.8182). B = $20,000,000 / $7.8182 = 2,558,140. CP2 = 7.8182 × (17,906,977 + 2,558,140) / (17,906,977 + 3,636,364) = 7.8182 × 20,465,117 / 21,543,341 = 7.8182 × 0.94995 = **$7.4269**. The Series B converts into $40,000,000 / 7.4269 = 5,385,835 shares, an increase of 269,556 shares.

Brightwater (A = 17,687,596; CP1 = $9.4982). B = 2,105,666. CP2 = 9.4982 × 19,793,262 / 21,323,960 = **$8.8164**. The Series B converts into 4,537,013 shares, an increase of 325,681 shares. Under the weighted-average formula, this is mild.

Full ratchet (Brightwater's first 24 months). The conversion price resets to $5.50, so the Series B converts into $40,000,000 / 5.50 = 7,272,727 shares, an increase of **3,061,395 shares** (73 percent more than the 4,211,332 original shares). For comparison, a full ratchet applied to Aldergate's structure would add 2,156,448 shares.

Founders' ownership after the Series C (total shares equal the original total plus extra Series B shares plus 3,636,364):

Case	Total shares	Founders (5,800,000)	Series B %	Series C %
Aldergate, weighted average	21,812,897	26.59%	24.69%	16.67%
Brightwater, weighted average (hypothetical)	21,649,640	26.79%	20.96%	16.80%
Brightwater, full ratchet (as offered)	24,385,355	23.78%	29.82%	14.91%
Aldergate, full ratchet (hypothetical)	23,699,789	24.47%	30.69%	15.34%

Before the Series C, the founders held 32.39 percent under Aldergate and 32.79 percent under Brightwater. After a down round at $5.50, the founders are left with 26.59 percent under Aldergate and 23.78 percent under Brightwater. The full ratchet costs the founders 3.0 ownership points relative to weighted average, and it makes the Series C investor's stake smaller as well (14.9 versus 16.8 percent), which raises the price the Series C investor will demand. The protection matters more in the Brightwater structure than the price alone suggests because it is paired with a 1.5x participating preference and board seats: in the very scenario where the ratchet triggers, Brightwater would hold nearly 30 percent of the company, a 1.5x senior claim, and two of five board seats.

Solution to Question 9: runway and the next round

Plan case. FY2027 free cash flow before financing is -9,400 + 2,100 + 900 - 1,375 - 2,300 - 3,600 = -13,675; less the 2,000 debt repayment, cash falls from 43,050 to 27,375. FY2028: -11,155 and -2,000 give 14,220. FY2029: -5,655 and -2,000 give 6,565. FY2030: +5,385 and -2,000 give 9,950. The plan funds itself, with a low point of $6.6 million at the end of FY2029, only about 2.2 times the $3.0 million covenant minimum.

Slippage case (revenue 20 percent below plan in FY2027 to FY2029, gross margin and opex unchanged, working capital 20 percent lower and capex 10 percent lower than plan):

	FY2027E	FY2028E	FY2029E	FY2030E
Gross profit shortfall (20% of plan gross profit)	(4,320)	(7,280)	(10,976)	(15,340)
Operating loss	(13,720)	(13,880)	(12,596)	(6,640)
Free cash flow before financing	(17,175)	(17,375)	(15,431)	(8,655)
Debt repayment	(2,000)	(2,000)	(2,000)	(2,000)
Ending cash	23,875	4,500	(12,931)	(23,586)

The FY2030 column applies the same 20 percent shortfall to FY2030. The company would breach the minimum-cash covenant in the first weeks of 2029 and be out of cash by about the second quarter of 2029. Because a financing takes six to nine months to arrange, a slippage-case Series C must be launched by early 2028, when cash is still about $24 million and the evidence of slippage is only just appearing. The required amount is about $15.9 million to reach the end of FY2029 with the covenant cushion, and about $26.6 million to reach the end of FY2030. A $25 to $30 million Series C is therefore needed in either case.

Founders' ownership after a plan-case Series C ($30 million at $260 million pre-money). The Series C buys 30 / 290 = 10.34 percent of the company. Founders' ownership is 32.39 percent × (1 - 0.1034) = **29.04 percent** under Aldergate and 32.79 percent × (1 - 0.1034) = **29.40 percent** under Brightwater (assuming no pool top-up and no anti-dilution adjustment, since $260

million pre-money is a price well above the Series B price).

Governance and the C round. Brightwater's two seats plus Crestline's would form a three-vote majority of five that can decide financing without the founders, and its right of first offer deters competing bidders. In the slippage case the Series C would fall in the first half of 2028, inside Brightwater's 24-month ratchet window, at a price that may be below the Series B price. The affiliate's potential $6.0 million of ARR by the end of 2028 is non-binding and equals about 8 percent of FY2028 plan revenue, far less than the $21.0 million peak founder disadvantage of the Brightwater structure.

Solution to Question 10: recommendation

Pursue the Aldergate structure, using Brightwater's offer as leverage, and open with Counter C1. Priorities in order of value to common holders: (1) a 1.0x non-participating preference and no ratchet; (2) price, with an effective pre-money of about $98 million or more (a $105 million pre-money with an 8 percent pool); (3) a pool sized to the hiring plan; (4) a board with common and investor seats balanced and a jointly chosen independent; (5) no right of first offer. The walk-away position is Aldergate's original term sheet, whose expected founder value ($113.4 million) exceeds Brightwater's ($108.5 million). Brightwater can win only with C2 or better: non-participating, no ratchet, and a pre-money above $115 million.

Sensitivity discussion

Pre-money sensitivity (Aldergate structure, 10 percent pool).

Pre-money ($ millions)	Price	Founders' ownership	Founders' expected value ($ millions)	Founders at $520 million
80	$6.18	29.88%	104.77	173.10
90	$7.00	31.23%	109.51	180.93
100	$7.82	32.39%	113.38	187.64
110	$8.64	33.39%	116.57	193.46
120	$9.45	34.27%	119.35	198.55
130	$10.27	35.05%	121.81	203.05

Each $10 million of pre-money adds about $2.5 to $4 million of expected founder value, with diminishing returns. Brightwater's package is worth to the founders what Aldergate's structure would be worth at a pre-money of only $87.8 million (Question 6).

Pool sensitivity (Aldergate, $100 million pre-money). Pools of 6, 8, 10, 12, 14, and 16 percent give founders' ownership of 34.50, 33.44, 32.39, 31.34, 30.28, and 29.23 percent. Each additional point of pool costs the founders about 0.5 ownership points and $1.5 million of effective valuation. The Aldergate-versus-Brightwater decision reverses only if the chance of plan-or-better is about 47.5 percent or more (Question 5).

Common student errors

1. Comparing headline pre-money valuations and concluding that Brightwater is 28 percent better.
2. Applying the pool percentage to the pre-round shares instead of solving T = base / (1 - p - s) simultaneously.
3. Forgetting that the existing unallocated pool counts toward the target.
4. Treating the Series B as always converting; conversion is a choice that depends on exit value.
5. Applying the participation cap to the participation amount rather than to total proceeds ($120.0 million in total).
6. Omitting option strikes or exercise proceeds, or letting the unissued pool share in exit proceeds.
7. Averaging founder proceeds across exits instead of weighting by probability.

8. Applying the ratchet price to the wrong share base.
9. Declaring a winner without a negotiation plan.

Alternative defensible answers

Take Brightwater, with changes. Teams that value the affiliate's customer opportunity ($6.0 million of ARR, if real) can defend Brightwater, but only with the participation removed or the cap lowered, the ratchet removed, and the board rebalanced. At a 47.5 percent chance of plan-or-better the term sheets are equal, so a team that believes in the plan could accept a structured deal at a materially higher price.

Take Aldergate as offered. Speed matters, but Aldergate's six to seven weeks still closes by early January if signed this week; ask Harbor Lending for a short covenant waiver to cover the gap.

Raise less, or run a three-way process. A $30 million round cuts dilution but leaves only about $6.6 million of low-point cushion and would require cutting roughly $10 million from the plan or accepting an earlier Series C. A third bidder such as Corvane Industrial is hard under no-shop terms but might be feasible with a short extension.

Board-style recommendation memo

To: Board of Directors, Nimbus Robotics, Inc. **From:** Priya Raman, Chief Executive Officer, with Nadia Brandt, Vice President of Finance **Date:** November 4, 2026 **Re:** Series B financing: recommendation for the November 9 meeting

Recommendation. Authorize management to negotiate and sign Aldergate Ventures' term sheet, opening with Counter C1 ($112.0 million pre-money, 8 percent pool, otherwise Aldergate's terms), and to decline Brightwater Partners' term sheet as written. If Brightwater offers Counter C2 (1.0x non-participating, no full ratchet, $115.0 million pre-money, 12 percent pool, a five-seat board with one Brightwater seat), we will consider it on equal terms and ask for a binding commercial commitment from its affiliate. Aldergate's original terms are the walk-away position.

Why. Brightwater's headline pre-money is $128.0 million against Aldergate's $100.0 million, but after the option pool the effective valuations are $110.2 million and $90.7 million, and after the structure the picture reverses. Founders' combined proceeds are higher under Aldergate in every exit below about $377 million, by up to $21.0 million near $125 million. On the board's own scenarios, expected founder proceeds are $113.4 million under Aldergate and $108.5 million under Brightwater, and Priya's certainty equivalent is $31.5 million versus $26.3 million. Brightwater is better only if the chance of plan-or-better approaches 48 percent, against the board's 30 percent. In weak outcomes the Series A and Seed also fare worse under Brightwater (nothing at a $45 million exit versus $5.0 million).

Risks. (1) Aldergate may not move on price or pool; we would then accept its original term sheet. (2) Aldergate's six-to-seven-week diligence could slip past the February covenant date; we will ask Harbor Lending now for a 60-day waiver. (3) Our scenarios are judgments, and the affiliate's customer introductions could raise the plan probability; we would reconsider only against a binding commitment of at least $3.0 million of ARR. (4) The plan has little cushion; a $25 to $30 million Series C should be launched by early 2028, because the slippage case breaches the covenant in early 2029.

Information requested before November 9. Aldergate's flexibility on price and pool (a call between Priya and Julian Sato scheduled for November 5), Crestline's and Lantern's views, and Harbor Lending's willingness to waive the covenant.

Decision requested. Authority to sign Aldergate's term sheet on or before November 13 on terms no less favorable to common holders than the original, and to continue discussions with Brightwater only on the basis of Counter C2.

Part III — Cases in Transactions and Restructuring

Chapter 12: Case: Tri-County and Lakeshore, a Regional Bank Merger

All companies, people, and figures in this case are fictional and illustrative. Any resemblance to real institutions is coincidental. The numbers are constructed for teaching and are not forecasts or statistics about any real market.

The Situation

At 7:15 on the morning of Wednesday, November 4, 2026, Maria Delgado, chief executive officer of Tri-County Bancorp, sat at the head of the boardroom table on the fourteenth floor of the bank's headquarters and listened to her chief financial officer, Samuel Okafor, read out a number she did not like. "On our own base case," he said, "tangible book value per share drops about six and a half percent the day we close. We earn it back in four and a half years on the simple method. On the crossover method, it is seven."

Tri-County Bancorp ("Tri-County" or "TCB") is a $6.2 billion bank holding company serving three adjoining counties and the small cities around them. It lends to local businesses, commercial real estate investors, and homeowners, and it funds itself mainly with deposits gathered through 61 branches. It has been consistently profitable, well capitalized, and publicly listed for two decades. Its board has long said that it wants to be "the bank that the next bank has to beat," which, in practice, means scale: spreading the fixed cost of technology, compliance, and risk management across a larger balance sheet.

Lakeshore Financial Corporation ("Lakeshore" or "LFC") is a $2.1 billion bank holding company headquartered on the opposite side of the lake in the next county. It has 24 branches, a loyal deposit base, a good small-business franchise, and a chief executive, Walter Haines, who is 66 and has told friends that he would like to hand the bank to a strong partner rather than a financial buyer. Nine of Lakeshore's branches sit within a few miles of a Tri-County branch. Lakeshore's shares trade at a modest premium to book value and a discount to Tri-County's valuation multiples.

In late October, after a series of conversations between Maria and Walter, Tri-County's advisor, Fairhaven Partners, delivered a draft proposal: Tri-County would acquire Lakeshore for 0.75 Tri-County shares plus $5.00 in cash for each Lakeshore share. At Tri-County's current stock price of $21.50, that is worth $21.125 per Lakeshore share, a 25.7 percent premium to Lakeshore's $16.80 closing price, and a total of about $295.8 million.

The people

Maria Delgado, 55, CEO of Tri-County, who led two small acquisitions earlier in her tenure and wants this one to be the transformational one.

Samuel Okafor, 49, CFO of Tri-County, who built the merger model and is cautious about tangible book value dilution.

Helen Prescott, 62, chair of Tri-County's board, a retired industrial executive who asks one question of every deal: "Who pays for this if we are wrong?"

Raj Menon, Tri-County's chief risk officer, who has reviewed Lakeshore's loan files and is concerned about its investor commercial real estate concentration.

Walter Haines, 66, CEO of Lakeshore, a sympathetic seller who nonetheless owes fiduciary duties to his shareholders and has made clear that his board will entertain other offers.

Fairhaven Partners, Tri-County's investment banking advisor, which prepared the valuation and synergy analysis.

The decision

Tri-County's board meets on Thursday, November 12, 2026. Lakeshore's board has said it will respond to a formal proposal by Thursday, November 19. Maria must recommend whether the board should approve the proposal as drafted, approve it with changes, or decline to proceed. If the board approves, the companies expect to announce the transaction on Monday, November 23, and to close in the second quarter of 2027, subject to regulatory approvals and shareholder votes.

The decision turns on five questions that this case asks students to answer in the order in which a bank board would raise them. Is the exchange ratio fair, and what does the offer imply on standard bank valuation measures? What accounting does the acquisition create, and how does it affect earnings per share and tangible book value per share? Are the synergies large and certain enough to justify the premium? Will the combined company remain comfortably capitalized, and for how long are its capital ratios compressed? And what could go wrong, because banks that stumble in mergers usually stumble on credit quality, deposit retention, or systems conversion rather than on the arithmetic of the exchange ratio.

Industry Context

Why regional banks merge

Regional banks earn most of their income from the spread between what they charge on loans and securities and what they pay on deposits, measured by net interest margin, plus fee income from services such as treasury management, wealth management, and card processing. Their largest cost is people and branches, and a growing share is technology and regulatory compliance. Many of those costs are semi-fixed: a $2 billion bank needs a risk management function, a cybersecurity team, a core processing system, and a regulatory reporting department that are not proportionally smaller than those of a $6 billion bank. Combining two banks therefore allows overlapping functions to be consolidated, branches within a few miles of each other to be merged, and systems to be run once rather than twice. These cost savings, usually expressed as a percentage of the target's non-interest expense, are the primary source of value in a bank merger. Revenue synergies are real but harder to deliver, and many boards exclude them from the numbers they rely upon.

How bank mergers are valued

Banks are valued on earnings and on book value, and especially tangible book value, which removes goodwill and other intangibles. The standard measures are price to earnings, price to tangible book value, and, for the deposit franchise, the core deposit premium: the amount paid above tangible book value, divided by core deposits. Buyers and sellers compare the offer to precedent transactions and to the buyer's own multiples. Two measures dominate the buyer's analysis. The first is earnings per share accretion or dilution, usually shown on an operating basis that excludes one-time merger costs and on a GAAP basis that includes them. The second is tangible book value per share dilution and the number of years of earnings accretion required to recover it, the earnback period. Many bank boards use informal limits on both: for example, operating EPS accretion by the second year and an earnback of no more than five years.

Purchase accounting

In an acquisition, the buyer records the target's assets and liabilities at fair value. Loans acquired at a discount to par (reflecting both credit risk and interest rates that are below current market rates) are written down, and the discount is accreted into interest income over the life of the loans. Securities with unrealized losses are likewise marked down and accreted. The buyer records a core deposit intangible, an asset representing the value of the low-cost deposit base, and amortizes it. Deferred taxes are recorded on the differences between book and tax values. The difference between the purchase price and the fair value of net identifiable assets is goodwill, which is not amortized but is deducted when computing tangible book value and regulatory capital. Under the current expected credit loss framework, the buyer also establishes an allowance for loan losses on the

acquired loans through a charge to earnings at closing, the so-called Day 2 provision. Students who have worked through Volume 1, Chapter 22 (mergers and acquisitions) will recognize the structure as the standard acquirer-versus-target value-creation framework, applied with bank-specific accounting.

Capital

A bank's ability to absorb losses and keep growing depends on its regulatory capital ratios. Common equity tier 1 (CET1) capital, essentially common equity less goodwill and most intangibles, divided by risk-weighted assets, is the most closely watched. Banks also report total capital (adding eligible subordinated debt and a limited amount of the allowance for loan losses), the tier 1 leverage ratio (capital divided by average total assets less deducted intangibles), and tangible common equity to tangible assets. Regulators set minimum levels and, above them, thresholds for a "well-capitalized" designation. In the fictional world of this case, those thresholds are 6.5 percent for CET1, 8.0 percent for tier 1 capital, 10.0 percent for total capital, and 5.0 percent for the leverage ratio, and Tri-County's board has adopted tighter internal limits that appear in the exhibits. A bank merger almost always reduces capital ratios on the closing date, because the buyer creates goodwill and intangibles, pays one-time costs, and takes on the target's risk-weighted assets, and then rebuilds them through retained earnings. Regulators approving a merger look closely at how fast the rebuilding happens.

Integration risk

The synergies on a spreadsheet are earned only if the banks are actually combined. The largest risks are customer and deposit attrition when branches close or the systems conversion goes poorly, departure of the target's lenders who carry client relationships, and unexpected credit losses in portfolios the buyer did not originate. These risks are asymmetric: merger accretion builds slowly and loss events arrive suddenly, which is why seasoned acquirers insist on conservative marks, prudent assumptions, and a structure that shares risk with the seller.

Financial Data

All dollar amounts are in millions of U.S. dollars unless stated otherwise, except per-share data. Share counts are in millions. Income statements are for the twelve months ended September 30, 2026 (LTM). The tax rate is 21 percent throughout.

Exhibit 1: Income statements (LTM)

	Tri-County	Lakeshore
Net interest income	212.0	71.0
Non-interest income	50.0	11.0
Total revenue	**262.0**	**82.0**
Non-interest expense	157.0	52.5
Pre-provision net revenue	**105.0**	**29.5**
Provision for credit losses	14.0	3.5
Pre-tax income	91.0	26.0
Income tax (21%)	19.1	5.5
Net income	**71.9**	**20.5**
Diluted shares (millions)	48.0	14.0
Earnings per share	**$1.50**	**$1.46**
Efficiency ratio (expense / revenue)	59.9%	64.0%
Return on average assets	1.18%	1.00%
Return on average equity	11.8%	9.8%

Exhibit 2: Balance sheets (September 30, 2026)

	Tri-County	Lakeshore
Cash and equivalents	300	110
Securities	1,300	420
Loans, net of allowance	4,300	1,450
Goodwill and other intangibles	80	15
Other assets	220	105
Total assets	**6,200**	**2,100**
Deposits	5,100	1,750
Borrowings	380	110
Other liabilities	100	25
Total liabilities	**5,580**	**1,885**
Common equity	**620**	**215**
Total liabilities and equity	**6,200**	**2,100**
Memo: gross loans	4,352	1,472
Memo: allowance for credit losses	52	22
Memo: core deposits	n/a	1,500

Exhibit 3: Regulatory capital and board policy limits

	Tri-County	Lakeshore	Tri-County board policy minimum
CET1 capital	520	190	
Risk-weighted assets	4,700	1,550	
CET1 ratio	**11.06%**	**12.26%**	9.50%
Total capital (CET1 plus allowance up to 1.25% of RWA; no subordinated debt)	572	209.4	
Total capital ratio	**12.17%**	**13.51%**	10.50%
Tier 1 leverage ratio (CET1 / total assets less intangibles)	**8.50%**	**9.11%**	7.00%
Tangible common equity / tangible assets	**8.82%**	**9.59%**	7.50%

CET1 capital is common equity less goodwill and intangibles and less other regulatory deductions of 20 for Tri-County and 10 for Lakeshore. The board’s merger guidelines also require operating EPS accretion by the end of year 2 and a tangible book value earnback of no more than five years on the simple method.

Exhibit 4: Market data and valuation (before announcement)

	Tri-County	Lakeshore
Share price	$21.50	$16.80
Shares (millions)	48.0	14.0
Market capitalization ($ millions)	1,032.0	235.2
Book value per share	$12.92	$15.36
Tangible book value per share	$11.25	$14.29
Dividend per share	$0.52	$0.40
Price / LTM earnings	14.4x	11.5x
Price / book value	1.66x	1.09x
Price / tangible book value	1.91x	1.18x

Exhibit 5: Proposed deal terms

Term	Proposal
Consideration per Lakeshore share	0.75 Tri-County shares plus $5.00 in cash
Exchange ratio	0.75 (fixed)
Value per share at $21.50	$21.125
New Tri-County shares issued	10.5 million
Cash consideration	$70.0 million
Total transaction value	$295.75 million
Funding of cash needs ($82.0 million: $70.0 million consideration plus $12.0 million deal fees)	$55.0 million subordinated debt at 6.75% (counts as tier 2 capital); $27.0 million holding company cash (opportunity cost 4.0%)
Governance	Two Lakeshore directors join the Tri-County board
Termination fee	3.5% of transaction value ($10.35 million)
Expected close	Second quarter 2027

Exhibit 6: Purchase accounting assumptions

Item	Assumption
Fair value discount on Lakeshore loans (on gross loans of 1,472)	38.0 (credit 20.0; interest rate 18.0)
Existing Lakeshore allowance eliminated	22.0
Day 2 allowance on acquired loans (1.5% of gross loans), expensed at close	22.08
Fair value write-down of Lakeshore securities	12.0
Core deposit intangible (2.0% of core deposits of 1,500)	30.0, amortized straight-line over 10 years
Accretion of loan and securities discounts	Straight-line over 8 years
Deferred taxes on fair value adjustments	21%
Risk-weighted assets of Lakeshore after marks	1,550 (no change)

Exhibit 7: Synergy and cost assumptions

Item	Assumption
Cost savings, run-rate	22% of Lakeshore non-interest expense of 52.5 = $11.55 million (Fairhaven's advisor case: 30% = $15.75 million)
Phase-in of cost savings	Year 1: 50%; Year 2: 85%; Year 3 and after: 100%
Revenue dis-synergy	$1.64 million per year pre-tax (2% of Lakeshore revenue)
Deposit attrition	3% of Lakeshore deposits ($52.5 million), replaced with borrowings at an incremental cost of 2.8%, or $1.47 million per year pre-tax
Deal fees	$12.0 million at close
Integration costs	$28.0 million pre-tax: $20.0 million in year 1 and $8.0 million in year 2
Revenue synergies	None assumed

Exhibit 8: Standalone forward projections (growth 4% per year)

	Year 1	Year 2	Year 3
Tri-County net income	74.78	77.77	80.88
Tri-County EPS (48.0 million shares)	$1.558	$1.620	$1.685
Lakeshore net income	21.32	22.17	23.06
Tri-County dividend per share	$0.520	$0.541	$0.562
Risk-weighted assets, combined (grows 4% per year from the closing amount)	6,500	6,760	7,030

Year 1 is the first twelve months after closing. Standalone tangible book value per share grows by EPS less dividends per share from the starting value of $11.25.

Exhibit 9: Lakeshore loan portfolio

Segment	Balance ($ millions)	% of gross loans
Investor commercial real estate	560	38.0%
Owner-occupied commercial real estate	220	14.9%
Commercial and industrial	330	22.4%
Residential mortgage	240	16.3%
Consumer	82	5.6%
Construction	40	2.7%
Total gross loans	**1,472**	**100.0%**

Non-performing loans are $16.0 million (1.09 percent of gross loans); the allowance covers 1.49 percent of gross loans. Within investor commercial real estate, about $162 million (11 percent of total gross loans) is office property.

Exhibit 10: Precedent regional bank transactions (fictional)

Acquirer / target	Price / tangible book	Price / earnings	Core deposit premium	Cost savings (% of target expense)	TBV dilution	Earnback (simple, years)
Ridgeway Bancorp / Millbrook Savings	1.52x	13.9x	6.8%	28%	5.8%	3.9
Summit Valley / Northgate Financial	1.35x	12.4x	5.1%	25%	4.7%	3.5
Cedarpoint / Bayfront Community	1.71x	15.2x	8.4%	32%	6.9%	4.6
Harbor Trust / Pinecrest Bank	1.28x	11.8x	4.2%	22%	3.9%	3.0
Median	**1.44x**	**13.2x**	**6.0%**	**26.5%**	**5.3%**	**3.7**

Complications and Exhibits

Exhibit 11: Market reaction scenarios for the Tri-County share price

Because the exchange ratio is fixed, the value Lakeshore holders receive depends on Tri-County's share price between announcement and closing. The board has asked for the outcomes at five prices: down 20 percent, down 10 percent, unchanged, up 10 percent, and up 20 percent from $21.50. It has also asked Fairhaven to price two alternatives: a floating ratio that delivers a fixed $16.125 of Tri-County stock per Lakeshore share (plus the $5.00 cash), and a collar that keeps the 0.75 ratio while the Tri-County price stays between $19.35 and $23.65 (within 10 percent of $21.50), floats the ratio to hold the value at the boundary when the price is below $19.35, and floats it to hold the value at the upper boundary when the price is above $23.65.

Additional facts and open issues

A rival. Walter Haines has told Maria, with some discomfort, that Lakeshore's advisor has been contacted by Harborview Bancshares, a larger regional bank, and that Harborview might propose an all-stock transaction worth about $22.50 per share. Harborview has not made a proposal. Under the draft agreement, Lakeshore would owe Tri-County a termination fee of 3.5 percent of transaction value if it accepted a superior proposal, and Tri-County would have a right to match.

Branch overlap and regulators. Nine Lakeshore branches are within a few miles of a Tri-County branch. Counsel believes regulators may require the sale of up to three branches holding about $210 million of deposits. This is not modeled in Exhibits 6 and 7; a sale would reduce deposits and net interest income and produce a gain on sale.

Credit quality. Raj Menon's team sampled Lakeshore's investor commercial real estate loans and found a few office-property loans with weakening debt service coverage and appraisals that were more than two years old. He considers the $20.0 million credit component of the loan mark light, but cannot yet say by how much. A portfolio of this size could produce unexpected provisions in the first two years.

Deposit attrition and branches. Lakeshore's deposit customers are loyal, but a branch consolidation and a core systems conversion (planned for the first quarter after closing) are the points at which banks historically lose deposits. The 3 percent attrition in Exhibit 7 is Fairhaven's estimate; Raj Menon would prefer to see 8 percent tested.

Core deposit estimate. The core deposit figure of $1,500 million and the 2.0 percent intangible are a valuation firm's preliminary estimates. The range for core deposits is $1,400 to $1,550 million, and the intangible could be 1.5 to 2.5 percent.

The synergy estimate. Fairhaven's advisor case uses 30 percent cost savings. Samuel Okafor's base case uses 22 percent, the low end of the precedent range, on the grounds that nine overlapping branches out of 24 is not a high-overlap combination and that Tri-County has already consolidated its own back office.

Retention. Three senior Lakeshore lenders, who carry about $300 million of relationships, have been asked to sign retention agreements costing about $4.0 million. These costs are inside the $28.0 million of integration costs.

Simplifications. The model accretes discounts straight-line, although accretion is front-loaded in practice. It amortizes the core deposit intangible straight-line, although many banks use an accelerated method. It treats all merger costs as tax deductible, which is optimistic for some deal fees. It keeps risk-weighted assets of Lakeshore unchanged after the marks, which is conservative. It holds Tri-County's share count constant (no buybacks) and ignores dividend changes after the merger.

Assignment Questions

Use Exhibits 1 to 10, the assumptions in Exhibit 11 and the Complications section, and a 21 percent tax rate. Show all calculations.

Question 1. Deal terms and implied multiples. Compute the value per Lakeshore share, the split between stock and cash, the premium to Lakeshore's market price, total transaction value, new shares issued, and Lakeshore holders' pro forma ownership. Compute the implied price to earnings, price to book, price to tangible book, and core deposit premium, and compare each with the precedent medians in Exhibit 10. State the exchange ratio that would deliver the same value in an all-stock transaction.

Question 2. Contribution analysis. Compute Lakeshore's contribution to the combined company on assets, net loans, deposits, tangible equity, LTM and year 1 net income, and market capitalization. Compare with Lakeshore holders' ownership on an actual basis and on an all-stock-equivalent basis. Is the exchange ratio fair on this analysis?

Question 3. Purchase accounting. Compute the fair value of Lakeshore's net identifiable assets, the goodwill created, the Day 2 provision, and the after-tax one-time costs at closing. Build the pro forma balance sheet at closing and show that it balances.

Question 4. Accretion and dilution. For years 1 to 3, compute pro forma operating net income, operating EPS, and EPS accretion or dilution against Tri-County standalone, and compute GAAP EPS accretion including one-time costs. How much of the year 2 accretion would remain without synergies, and without purchase accounting accretion?

Question 5. Synergy sensitivity. Compute the year 1 to 3 operating accretion and the earnback for synergies of 0, 10, 15, 22, 30, and 35 percent of Lakeshore's expense base. Find the level of run-rate cost savings at which year 2 GAAP EPS is neutral and at which the simple earnback is exactly five years. Compare with precedent.

Question 6. Tangible book value. Compute pro forma tangible common equity and tangible book value per share at closing, the dilution, and the earnback period by the simple method and the crossover method. Compare with the board's guideline and with the precedents.

Question 7. Capital. Compute pro forma CET1, total capital, tier 1 leverage, and tangible common equity ratios at closing. Compare with policy limits and with regulatory minimums. Project the CET1 ratio at the end of years 1 to 3, assuming a payout equal to the dividend per share in Exhibit 8 on all pro forma shares and 4 percent growth in risk-weighted assets. Evaluate an all-stock alternative.

Question 8. Fixed ratio, floating ratio, and collar. For the five price scenarios in Exhibit 11, compute the value per Lakeshore share, transaction value, shares issued, and year 2 operating accretion under the fixed ratio, the floating ratio, and the collar. Which structure best protects each party, and what does each do to capital?

Question 9. Risk and value. Compute year 1 to 3 accretion and the simple earnback under each of the following: synergies at half the base case, integration costs 50 percent higher, deposit attrition of 8 percent, extra provisions of $15 million in each of years 1 and 2, and all four combined. Then estimate the net present value of the synergies (after-tax, discounted at 10 percent, with a 2 percent growth rate after year 3), net of the integration costs, dis-synergies and fees, and compare it with the premium paid. At what price would the deal be NPV-neutral?

Question 10. Recommendation. Recommend whether the board should approve the proposal as drafted, approve it with changes, or decline. Specify the exchange ratio and structure you would propose, the conditions you would attach, the response to a Harborview approach, and the integration milestones you would ask the board to monitor. Support the recommendation with your numerical findings and identify the main risks.

Teaching Note and Full Solutions

Learning objectives

Students should be able to: (1) derive bank deal metrics (price to earnings, price to tangible book, core deposit premium) and compare them with precedents; (2) run a contribution analysis and judge an exchange ratio; (3) perform purchase accounting for a bank, including loan marks, the core deposit intangible, deferred taxes, goodwill, and the Day 2 provision; (4) build a pro forma accretion and dilution analysis on operating and GAAP bases; (5) compute tangible book value dilution and earnback by two methods; (6) project capital ratios through closing and rebuilding; (7) compare fixed, floating, and collared exchange ratios; (8) distinguish accretion from value creation; and (9) make a recommendation under integration risk.

Suggested class plan (90 minutes)

Time	Segment	Purpose
0:00 to 0:10	Opening poll: approve, change, or decline?	Record the initial view
0:10 to 0:25	Q1 and Q2: multiples and contribution	Is the price fair?
0:25 to 0:45	Q3 and Q4: purchase accounting and accretion (work in groups)	Mechanics
0:45 to 0:55	Q5 and Q6: synergy sensitivity and tangible book earnback	Where the board’s tests bind
0:55 to 1:05	Q7 and Q8: capital and structure	Closing-day constraints
1:05 to 1:20	Q9: risk scenarios and NPV (the central insight)	Accretion is not value
1:20 to 1:30	Q10: recommendation; repeat the poll	Compare with the opening view

Solution to Question 1: deal terms and implied multiples

```
Value per share = 0.75 × $21.50 + $5.00 = $16.125 + $5.00 = $21.125
Stock share of consideration = 16.125 / 21.125 = 76.3%; cash = 23.7%
Premium = 21.125 / 16.80 - 1 = 25.7%
Transaction value = 14.0 × 21.125 = $295.75 million
   stock: 14.0 × 0.75 = 10.5 million shares × $21.50 = $225.75 million
   cash: 14.0 × $5.00 = $70.0 million
Pro forma shares = 48.0 + 10.5 = 58.5 million; Lakeshore holders own 10.5 / 58.5 = 17.9%
```

Multiple	Calculation	Offer	Precedent median
Price / earnings	21.125 / 1.464	14.4x	13.2x
Price / book	21.125 / 15.357	1.38x	n/a
Price / tangible book	21.125 / 14.286	1.48x	1.44x
Core deposit premium	(295.75 - 200.0) / 1,500	6.4%	6.0%

Lakeshore's tangible equity is 215 - 15 = 200.0 and its LTM EPS is 20.5 / 14.0 = $1.464. The offer is slightly above the precedent median on every measure and rich on earnings (14.4x versus 13.2x), though within the precedent range. An all-stock equivalent would use an exchange ratio of 21.125 / 21.50 = **0.9826** Tri-County shares per Lakeshore share. Tri-County is buying Lakeshore at 1.48x tangible book with stock valued at 1.91x tangible book, which is favorable for tangible book value, and at 14.4x earnings with stock at 14.4x earnings, which is neutral for earnings.

Solution to Question 2: contribution analysis

Measure	Tri-County	Lakeshore	Lakeshore contribution
Total assets	6,200	2,100	25.3%
Net loans	4,300	1,450	25.2%
Deposits	5,100	1,750	25.5%
Tangible equity	540	200	27.0%
LTM net income	71.9	20.5	22.2%
Year 1 net income	74.78	21.32	22.2%
Market capitalization	1,032.0	235.2	18.6%

Lakeshore holders receive 17.9 percent of the pro forma shares plus $70.0 million of cash. Counting the cash as if it were paid in shares, the all-stock equivalent is 295.75 / 21.50 = 13.756 million shares, or 13.756 / (48.0 + 13.756) = **22.3 percent**, which matches Lakeshore's earnings contribution of 22.2 percent. Lakeshore contributes more of the balance sheet (25 to 27 percent) than the earnings (22 percent) because it is less profitable (ROE 9.8 versus 11.8 percent) and less efficient. The exchange ratio is fair to Lakeshore holders on earnings, and more than fair compared with its market value contribution of 18.6 percent. It is fair to Tri-County holders on earnings and on the balance-sheet measures, since Tri-County is paying less than the balance sheet contribution. The ownership percentage alone understates the price because of the cash.

Solution to Question 3: purchase accounting

Fair value of net identifiable assets:

```
Lakeshore tangible equity (215.0 - 15.0 existing goodwill)           200.00
Loan mark (fair value discount)                                      (38.00)
Eliminate existing allowance (loans now at fair value)                22.00
Securities write-down                                                (12.00)
Core deposit intangible (2.0% × 1,500)                                30.00
Deferred tax asset on loans/securities: (38 - 22 + 12) × 21% = 28 × 21%    5.88
Deferred tax liability on CDI: 30 × 21%                              (6.30)
Fair value of net identifiable assets                                201.58
Goodwill = 295.75 - 201.58                                            94.17
```

Day 2 provision and after-tax closing costs. The allowance on acquired loans is 1.5% × 1,472 = $22.08 million, expensed at closing. Deal fees are $12.0 million. After tax: (22.08 + 12.00) × (1 - 0.21) = 34.08 × 0.79 = **$26.92 million**.

Pro forma balance sheet at closing:

	Tri-County	Lakeshore	Adjustments	Pro forma
Cash	300.00	110.00	(27.00)	383.00
Securities	1,300.00	420.00	(12.00)	1,708.00
Loans, net	4,300.00	1,450.00	(38.08)	5,711.92
Goodwill and intangibles	80.00	15.00	109.17	204.17
Other assets	220.00	105.00	6.74	331.74
Total assets	**6,200.00**	**2,100.00**	**38.83**	**8,338.83**
Deposits	5,100.00	1,750.00	0.00	6,850.00
Borrowings	380.00	110.00	55.00	545.00
Other liabilities	100.00	25.00	0.00	125.00
Common equity	620.00	215.00	(16.17)	818.83
Total liabilities and equity	**6,200.00**	**2,100.00**	**38.83**	**8,338.83**

The adjustments: cash falls by $82.0 million paid out (70.0 consideration plus 12.0 fees) and rises by $55.0 million of subordinated debt, a net 27.0 from holding company cash. Net loans change by -38.00 (mark) + 22.00 (allowance eliminated) - 22.08 (new allowance) = -38.08. Goodwill and intangibles change by -15.00 + 94.17 + 30.00 = +109.17. Other assets change by the deferred tax asset 5.88, less the liability 6.30 netted, plus the tax benefit of the one-time charges, 34.08 × 21% = 7.16: a net +6.74. Equity changes by -215.00 + 225.75 (stock issued) - 26.92 = -16.17, which equals 620.00 + 225.75 - 26.92 = 818.83. Total assets (8,300 + 38.83 = 8,338.83) equal liabilities and equity (7,520.00 + 818.83), so the balance sheet balances.

Solution to Question 4: accretion and dilution

($ millions except per share)	Year 1	Year 2	Year 3
Tri-County standalone net income	74.78	77.77	80.88
Lakeshore standalone net income	21.32	22.17	23.06
Cost savings (50%, 85%, 100% of 11.55)	5.78	9.82	11.55
Accretion of loan and securities discounts ((38 + 12) / 8)	6.25	6.25	6.25
Core deposit intangible amortization	(3.00)	(3.00)	(3.00)
Interest on subordinated debt (55.0 × 6.75%)	(3.71)	(3.71)	(3.71)
Lost income on holding company cash (27.0 × 4.0%)	(1.08)	(1.08)	(1.08)
Deposit attrition cost (52.5 × 2.8%)	(1.47)	(1.47)	(1.47)
Revenue dis-synergy	(1.64)	(1.64)	(1.64)
Net pre-tax adjustments	1.12	5.17	6.90
After tax (× 0.79)	0.89	4.08	5.45
Pro forma operating net income	**96.98**	**104.02**	**109.39**
Pro forma shares (millions)	58.5	58.5	58.5
Pro forma operating EPS	**$1.658**	**$1.778**	**$1.870**
Tri-County standalone EPS	$1.558	$1.620	$1.685
Operating EPS accretion	**6.4%**	**9.8%**	**11.0%**
One-time costs after tax	(42.72)	(6.32)	0.00
GAAP net income	54.26	97.70	109.39
GAAP EPS	**$0.928**	**$1.670**	**$1.870**
GAAP EPS accretion	**(40.5%)**	**3.1%**	**11.0%**

The year 1 one-time costs are deal fees 12.00, integration 20.00, and the Day 2 provision 22.08, a total of 54.08 pre-tax and 54.08 × 0.79 = 42.72 after tax. In year 2, integration costs of 8.00 give 6.32 after tax. Standalone EPS is year 1 net income divided by 48.0 million shares: 74.78 / 48.0 = $1.558, and so on.

What drives the accretion? Without any cost savings, year 2 net income is 99.94 + (6.25 - 3.00 - 3.71 - 1.08 - 1.47 - 1.64) × 0.79 = 99.94 - 3.68 = 96.27, or EPS of $1.646 versus $1.620, so accretion is only 1.6 percent. Excluding purchase accounting accretion and intangible amortization but keeping the synergies, year 2 accretion is 7.0 percent (EPS $1.734). Excluding both synergies and purchase accounting effects, the deal is **1.1 percent dilutive** (NI 93.70, EPS $1.602 versus $1.620). In other

words the standalone earnings of Lakeshore, bought at 14.4 times with a financing cost, do not by themselves add to Tri-County's EPS; the accretion comes from synergies and from accretion of purchase accounting discounts, which are not a source of economic value.

Solution to Question 5: synergy sensitivity

Each row recomputes the Question 4 table with run-rate cost savings equal to the stated percentage of Lakeshore's $52.5 million expense base and the same 50, 85, and 100 percent phase-in.

Cost savings (% of expense; $ millions)	Op. accretion Y1	Y2	Y3	GAAP accretion Y2	Simple earnback (years)
0% ($0.0)	1.4%	1.6%	1.7%	(5.1%)	not recovered
10% ($5.25)	3.7%	5.3%	5.9%	(1.4%)	7.5
15% ($7.88)	4.8%	7.2%	8.0%	0.5%	5.9
22% ($11.55) base	6.4%	9.8%	11.0%	3.1%	4.6
30% ($15.75)	8.2%	12.7%	14.3%	6.1%	3.7
35% ($18.38)	9.4%	14.6%	16.4%	7.9%	3.3

Each point of cost savings (0.525 million pre-tax) changes year 2 operating net income by 0.525 × 0.85 × 0.79 = $0.35 million and year 2 accretion by about 0.35 / 58.5 / 1.62 = 0.37 percentage points. **Year 2 GAAP EPS is neutral at run-rate savings of about 13.7 percent ($7.2 million).** The **simple earnback is exactly five years at about 19.4 percent ($10.2 million)**; the base case of 22 percent clears the board's five-year guideline with a margin of only 2.6 percentage points of the expense base. By the crossover method, a five-year recovery would require 38 percent, above every precedent. The precedent range for savings is 22 to 32 percent, so the base case is at the low end, and the synergy case is conservative relative to the Fairhaven advisor case of 30 percent, but the tangible book value test is not tolerant of shortfalls: at 15 percent savings the earnback is 5.9 years.

Solution to Question 6: tangible book value

```
Standalone tangible equity = 620 - 80 = 540.0;  TBVPS = 540.0 / 48.0 = $11.25
Pro forma TCE at closing = 540.0 + 225.75 (stock issued) - 26.92 (after-tax closing costs)
                           - 94.17 (goodwill) - 30.00 (core deposit intangible) = 614.66
Pro forma TBVPS = 614.66 / 58.5 = $10.507
Dilution = 10.507 - 11.250 = -$0.743 per share = -6.6%
```

The dilution compares with a precedent median of 5.3 percent (range 3.9 to 6.9 percent). Note that the tangible equity at close omits the later integration costs, which reduce tangible book value as they are incurred.

Simple method. Cumulative operating EPS accretion per share (pro forma less standalone): year 1 $0.100; year 2 $0.158; year 3 $0.185; year 4 $0.189; year 5 $0.192. The cumulative total after four years is $0.631 and after five $0.823. The dilution of $0.743 is recovered at 4 + (0.743 - 0.631) / 0.192 = **4.6 years**, which meets the board's five-year guideline and is similar to the highest precedent (4.6 years) and above the median (3.7).

Crossover method. Each tangible book value per share path grows by EPS less dividends per share (and, for the pro forma path, GAAP income including one-time costs, and adding back the intangible amortization):

Year-end	Standalone TBVPS	Pro forma TBVPS	Difference
1	$12.29	$10.97	(1.32)
2	$13.37	$12.15	(1.22)
3	$14.49	$13.51	(0.99)
4	$15.66	$14.91	(0.75)
5	$16.87	$16.37	(0.50)
6	$18.13	$17.88	(0.25)
7	$19.45	$19.45	(0.00)
8	$20.81	$21.07	0.25

The paths cross at about **7.0 years**. The difference between the methods is instructive. The simple method asks how long the annual EPS pickup takes to repay the initial dilution, ignoring that the standalone bank is also compounding its tangible book. The crossover method recognizes that growth, and is slower because the accretion per share is small relative to the compounding base. A board that uses only the simple method sees a deal that passes; one that uses the crossover method sees a deal that does not.

Solution to Question 7: capital

```
Pro forma CET1 = 520.0 (Tri-County) + 225.75 (stock) - 26.92 (closing costs after tax)
                 - 94.17 (goodwill) - (30.00 - 6.30) (CDI net of deferred tax) = 600.96
Risk-weighted assets = 4,700 + 1,550 = 6,250
Total capital = 600.96 + 55.00 (subordinated debt) + 74.08 (allowance: 52.0 + 22.08, below the 1.25% × 6,250 =
78.1 cap) = 730.04
Tangible assets = 8,338.83 - 204.17 = 8,134.66
```

Ratio	Calculation	Pro forma	Standalone Tri-County	Policy	Regulatory well-capitalized
CET1	600.96 / 6,250	9.62%	11.06%	9.50%	6.50%
Total capital	730.04 / 6,250	11.68%	12.17%	10.50%	10.00%
Tier 1 leverage	600.96 / 8,134.66	7.39%	8.50%	7.00%	5.00%
TCE / tangible assets	614.66 / 8,134.66	7.56%	8.82%	7.50%	n/a

All ratios remain above policy and regulatory limits at closing, but the margins are thin: 12 basis points on CET1 and 6 on tangible common equity. A single after-tax charge of 0.0012 × 6,250 = $7.5 million (about $9.5 million pre-tax) would breach the CET1 policy minimum. A credit surprise of $15 million before tax would reduce CET1 by about 19 basis points (11.85 / 6,250).

Rebuild path. CET1 at the end of each year equals the prior CET1 plus GAAP net income, less dividends (0.52 growing 4 percent, times 58.5 million shares), plus the after-tax intangible amortization of 2.37 (the deduction falls as the intangible amortizes):

	Close	Year 1	Year 2	Year 3
GAAP net income		54.26	97.70	109.39
Dividends		(30.42)	(31.64)	(32.90)
CET1 capital	600.96	627.17	695.60	774.45
Risk-weighted assets	6,250	6,500	6,760	7,030
CET1 ratio	**9.62%**	**9.65%**	**10.29%**	**11.02%**

Year 1 is flat because the integration costs absorb the retained earnings; the ratio recovers to about 10.3 percent by the end of year 2 and to about 11 percent by year 3.

All-stock alternative. With no cash, no subordinated debt, and 13.756 million shares issued, CET1 at close is 520.0 + 295.75 - 26.92 - 94.17 - 23.70 = 670.96, or 10.74 percent, 112 basis points higher. Pro forma shares rise to 61.756 million, so year 2 operating accretion falls to 7.4 percent from 9.8 percent (4.4, 7.4, and 8.4 percent in years 1 to 3), because Tri-County's earnings yield on its own stock, 7.0 percent, exceeds the after-tax cost of the debt and cash it would otherwise use. But tangible book value per share at closing would be $11.09, a dilution of only 1.5 percent instead of 6.6 percent, because shares issued at 1.91 times tangible book finance the premium rather than debt. The all-stock deal buys capital strength and tangible book value and gives up some EPS; the cash component is what drives the tangible book dilution.

Solution to Question 8: fixed ratio, floating ratio, and collar

Under the fixed ratio, the shares issued are always 10.5 million, so the value to Lakeshore holders moves with Tri-County's price. Under the floating ratio, the value is fixed at $21.125 and the number of shares moves. The collar fixes the ratio inside the band of $19.35 to $23.65 and floats outside it.

Tri-County price	Fixed: value per share	Fixed: deal value	Floating: shares issued	Floating: Y2 accretion	Collar: value per share	Collar: shares issued	Collar: Y2 accretion
$17.20 (-20%)	$17.90	$250.6	13.13	5.0%	$19.51	11.81	7.3%
$19.35 (-10%)	$19.51	$273.2	11.67	7.6%	$19.51	10.50	9.8%
$21.50 (0%)	$21.13	$295.8	10.50	9.8%	$21.13	10.50	9.8%
$23.65 (+10%)	$22.74	$318.3	9.55	11.6%	$22.74	10.50	9.8%
$25.80 (+20%)	$24.35	$340.9	8.75	13.1%	$22.74	9.63	11.4%

Under the fixed ratio, Year 2 operating accretion stays at 9.8 percent in every case, because the share count does not change; what changes is the price Tri-County effectively pays (and the goodwill, though not CET1: a higher stock price raises goodwill and equity by the same amount). A fixed ratio protects Tri-County's EPS and shareholders from stock price declines while exposing Lakeshore holders to them; if Tri-County's price falls 20 percent the offer is worth only 6.5 percent more than Lakeshore's market price of $16.80 (17.90 / 16.80 - 1), which may invite a competing bid or a vote against. A floating ratio protects Lakeshore holders but makes Tri-County's EPS accretion vary from 5.0 to 13.1 percent. The collar shares the risk: Lakeshore receives at least $19.51 of value while Tri-County limits the extra shares issued in a decline to 1.31 million (11.81 versus 10.50) and gets the benefit of a bounded rise. For regulatory capital, none of these matters at closing, because a change in the stock price changes the stock issued and the goodwill equally.

Solution to Question 9: risk and value

Each scenario changes one assumption from the base case and recomputes the Question 4 table. Extra provisions are treated as operating costs.

Scenario	Op. accretion Y1	Y2	Y3	GAAP accretion Y1	Y2	Simple earnback (years)
Base	6.4%	9.8%	11.0%	(40.5%)	3.1%	4.6
Synergies at half (11%)	3.9%	5.7%	6.4%	(43.0%)	(1.0%)	7.1
Integration costs +50%	6.4%	9.8%	11.0%	(49.1%)	(0.3%)	4.6
Deposit attrition 8%	4.3%	7.7%	9.0%	(42.6%)	1.0%	5.5
Extra provisions $15 million in years 1 and 2	(6.6%)	(2.7%)	11.0%	(53.5%)	(9.4%)	6.6
All four combined	(11.2%)	(8.9%)	4.4%	(66.8%)	(18.9%)	more than 10

For attrition, the cost is 0.08 × 1,750 × 0.028 = $3.92 million pre-tax instead of 1.47. The extra provisions cost $15 million × 0.79 = $11.85 million after tax in each of years 1 and 2, or 0.20 per share. The combined downside, which is not extreme for a bank with a 38 percent investor commercial real estate concentration, produces EPS dilution for two years and pushes the earnback beyond a decade.

Net present value of the synergies. After-tax net synergies (cost savings less revenue dis-synergy and attrition cost, times 0.79):

```
Year 1: (5.775 - 1.64 - 1.47) × 0.79 = 2.105
Year 2: (9.8175 - 3.11) × 0.79 = 5.299
Year 3: (11.55 - 3.11) × 0.79 = 6.668, growing 2% per year thereafter

PV at 10% = 2.105 / 1.10 + 5.299 / 1.10² + [6.668 / (0.10 - 0.02)] / 1.10²
          = 1.914 + 4.379 + 83.35 / 1.21 = 1.914 + 4.379 + 68.88 = 75.17
After-tax one-time costs: fees 12.0 × 0.79 = 9.48 at close; integration 20.0 × 0.79 / 1.10 = 14.36; 8.0 × 0.79
/ 1.21 = 5.22; total 29.07
NPV of synergies net of costs = 75.17 - 29.07 = 46.10
Premium paid = 295.75 - 235.20 = 60.55
Value created for Tri-County = 46.10 - 60.55 = -14.45
```

With no growth in synergies, the NPV is 32.33 and the shortfall is $28.2 million. The deal is therefore accretive and still not NPV-positive at the proposed price: the premium is $60.6 million but the net value of the synergies is $46.1 million (some of which also accrues to the Lakeshore holders who will own 17.9 percent of the combined company). The accretion in the earnings table includes purchase accounting accretion of discounts, which is not a cash flow, and a measure of earnings per share that ignores the cost of the shares issued. The **NPV-neutral price** solves 235.20 + 46.10 = $281.3 million, or $20.09 per share, a 19.6 percent premium, which corresponds to an exchange ratio of (20.09 - 5.00) / 21.50 = 0.702.

Solution to Question 10: recommendation

Approve the transaction with changes: a lower exchange ratio of 0.70 (value $20.05 per share at $21.50, a 19.3 percent premium) with the $5.00 cash component, a collar of plus or minus 10 percent, and conditions on credit and synergies. At the revised terms the deal value is $280.7 million, goodwill is $79.1 million, 9.8 million shares are issued, pro forma shares are 57.8 million, and price to tangible book is 1.40x. Operating accretion is 7.7, 11.1, and 12.3 percent in years 1 to 3; tangible book value per share at closing is $10.63 (dilution 5.5 percent); the simple earnback is 3.5 years, the crossover earnback is 6.0 years; and the NPV of the synergies (46.1) is about equal to the premium (45.5). The memo below sets out the conditions.

Sensitivity discussion

Price. Each 0.01 change in the exchange ratio changes the offer by $0.215 per share and the transaction value by $3.0 million. The earnback, the NPV, and the capital margin all improve as the ratio falls:

Exchange ratio	Value per share	Premium	Deal value	Y2 accretion	TBV dilution	Simple earnback
0.70	$20.05	19.3%	$280.7	11.1%	5.5%	3.5
0.72	$20.48	21.9%	$286.7	10.5%	5.9%	3.9
0.75	$21.13	25.7%	$295.8	9.8%	6.6%	4.6

Synergies, credit, and deposits. Synergies are the dominant sensitivity: each point of cost savings is worth about 0.37 percentage points of year 2 accretion. Credit costs matter most for the capital margin: 12 basis points of CET1 cushion at closing equals only $7.5 million after tax. Deposit attrition is the quietest risk: five points of additional attrition (8 percent instead of 3 percent) cost about 2 percentage points of year 2 accretion.

Common student errors

1. Using the acquirer’s price for the target’s shares, or forgetting that the cash portion is not exchanged for stock.
2. Computing tangible book value without deducting the new core deposit intangible, or deducting it gross in the CET1 calculation instead of net of the deferred tax liability.
3. Treating the credit mark as an immediate hit to equity without recording the Day 2 allowance, or counting both the mark and the allowance without eliminating the target’s existing allowance.
4. Omitting the one-time costs from GAAP EPS, or including them in operating EPS.
5. Forgetting the financing cost of the cash consideration (subordinated debt interest and the lost income on holding company cash).
6. Computing earnback with year 1 GAAP EPS dilution rather than operating accretion, or confusing the simple and crossover methods.
7. Concluding from accretion that the deal creates value; accretion ignores the cost of equity and includes non-cash purchase accounting income.
8. Treating the fixed ratio as a fixed value, or the floating ratio as fixed shares.
9. Ignoring that a stock price change at closing changes goodwill but not CET1.

Alternative defensible answers

Approve as drafted. A board that values scale, deposit franchise, and the chance of losing the deal to Harborview may reasonably accept 0.75 plus $5.00, because the operating EPS test, the earnback test (4.6 years), and every policy ratio are met. It should do so only with the credit and integration conditions below and a realistic view that NPV is negative by about $14 million unless the synergies exceed the base case.

Decline. A board that weights the 38 percent investor real estate concentration and the thin capital margin (12 basis points on CET1) could defend declining, on the grounds that a combined downside leads to two years of dilution and a capital breach, and that the franchise could be acquired later from a weaker position.

All-stock. A board that prioritizes tangible book value (1.5 percent dilution) and capital (10.74 percent CET1) over EPS (7.4 percent year 2 accretion) could propose an all-stock deal at the 0.9826-equivalent ratio, or a lower ratio with more stock and less debt.

Board-style recommendation memo

To: Board of Directors, Tri-County Bancorp **From:** Maria Delgado, Chief Executive Officer, with Samuel Okafor, Chief Financial Officer **Date:** November 10, 2026 **Re:** Proposed acquisition of Lakeshore Financial Corporation: recommendation for the November 12 meeting

Recommendation. Approve a proposal to acquire Lakeshore at 0.70 Tri-County shares plus $5.00 in cash per share ($20.05 at $21.50), with a collar of plus or minus 10 percent, and authorize management to negotiate up to 0.72 (a value of $20.48) only if the conditions below are met. Do not match a Harborview proposal above $21.125 per share.

Why. At the proposed terms the transaction is 7.7 percent accretive to operating EPS in year 1 and 11.1 percent in year 2, dilutes tangible book value by 5.5 percent, recovers it in 3.5 years on the simple method, and leaves CET1 at about 9.6 percent at closing (above our 9.5 percent minimum; CET1 does not depend on the price because lower stock issued and lower goodwill offset). At the original terms, year 2 accretion is 9.8 percent, tangible book dilution is 6.6 percent, the simple earnback is 4.6 years, and the net present value of the synergies is $14 million less than the premium, because accretion partly reflects non-cash purchase accounting income. The 0.70 ratio makes the net present value roughly neutral. Excluding synergies and accounting accretion, the deal is slightly dilutive, so the case for it rests on cost savings.

Conditions. (1) A completed credit review of Lakeshore's investor commercial real estate portfolio before signing, and a price adjustment of $1.00 per share if the credit mark exceeds $30 million. (2) A signed plan to reach run-rate cost savings of at least 22 percent of Lakeshore's expense within 18 months. (3) The collar. (4) Retention agreements for the three lenders. (5) A capital plan that holds CET1 above 9.5 percent in each quarter, with the dividend maintained and buybacks suspended.

Risks. A combined downside (half the synergies, 50 percent higher integration costs, 8 percent deposit attrition, and $30 million of extra provisions) produces dilution for two years and an earnback beyond ten years. Regulators may require a sale of branches. Harborview may bid. The thin capital margin at closing means a $7.5 million after-tax surprise breaches our policy minimum.

Integration milestones for board monitoring. Deposit retention of at least 97 percent through conversion; core systems conversion on schedule in the first quarter after closing; cost savings run-rate of 50 percent by month 9 and 100 percent by month 18; classified loans and charge-offs of the acquired portfolio against the credit mark; and quarterly CET1 against the 9.5 percent minimum.

Decision requested. Authority to submit the proposal at 0.70 plus $5.00 with a collar, negotiate to a maximum of 0.72 on the conditions above, and walk away above $21.125 per share.

Chapter 13: Case: Brightwell Instruments, a Hostile Bid and Its Defense

Type: Case. Brightwell Instruments, Corvane Industrial Holdings, Altair Measurement Group, Larkspur Capital, Calloway Hart and every other company, fund, adviser and person named in this chapter are fictional and illustrative. All figures were constructed for teaching and bear no relation to any real company or transaction. Nothing in the chapter is legal advice; the discussion of directors' duties is deliberately general because the detail varies by jurisdiction and by the facts of each situation.

Prerequisites: Volume 1, Chapters 5 (equity valuation), 11 (WACC), 21 (enterprise valuation) and 22 (mergers and acquisitions); Volume 2 treatment of takeover defenses, deal structuring and valuation under uncertainty.

The Situation

On the morning of Monday, 2 November 2026, Corvane Industrial Holdings launched an unsolicited, all-cash tender offer for every outstanding share of Brightwell Instruments at $32.00 per share. The offer arrived three weeks after the Brightwell board had unanimously declined a private proposal at the same price, delivered by letter on 12 October. The tender is scheduled to expire at 5:00 p.m. on Tuesday, 1 December 2026, unless extended. The Brightwell board has scheduled a special meeting for Monday, 9 November 2026, at which it must adopt a formal recommendation to shareholders. Securities rules in the relevant market require that recommendation, or a statement that the board is unable to take a position, to be filed within ten business days of launch. Today is Wednesday, 4 November 2026.

Brightwell Instruments is a mid-cap designer and manufacturer of precision measurement instruments: laser interferometers, high-resolution torque and force sensors, and the calibration software and recurring service contracts that surround them. Henrik Sandoval, grandson of the 1958 founder, has been chief executive since 2014. Revenue for fiscal 2026, which ends on 31 December, is estimated at $1,200 million. Reported EBITDA is estimated at $240 million, a 20.0 percent margin. Roughly 31 percent of revenue comes from the Calibration Services division, a recurring-revenue business with 78 percent customer retention in its largest contracts; the remainder comes from instrument sales to aerospace, semiconductor equipment, medical device and energy customers. The company has 80.0 million basic shares outstanding, 3.0 million employee stock options with a weighted-average strike price of $18.00, and 1.0 million unvested restricted stock units that the plan documents say will vest in full on a change of control.

Until 2 October 2026, which analysts now call the "undisturbed date," Brightwell shares traded at $25.00. On that date Corvane voluntarily announced that it had accumulated 4.9 percent of the shares at an average cost of $24.10, a stake just under the 5 percent level at which disclosure would have been mandatory and well under the 15 percent trigger of the shareholder rights plan the Brightwell board adopted in 2022. The shares gained 9 percent on the disclosure, then moved to $29.80 on the private proposal leak, and closed at $32.70 on Tuesday, 3 November, the day after the tender launched. A price above the offer is the market's way of saying that it expects a higher bid, a rival, or both.

The people at the table are these.

- **Margaret Oyelaran**, independent chair of Brightwell and a former chief financial officer of a listed engineering group, known for deliberate, documented process.
- **Henrik Sandoval**, chief executive; he and his family trust own 6.5 percent of the shares. His change-of-control agreement is worth $9.6 million in cash plus accelerated equity, which makes him an interested party in the board's debate.

- **Priya Raman**, chief financial officer, who produced the management plan and privately doubts that its last two years are achievable.
- **Thomas Achterberg**, lead independent director and audit chair, who would chair a special committee.
- **Calloway Hart**, Brightwell's financial adviser, paid a fixed fee plus a success fee only if a transaction closes, a common structure and a potential source of bias.
- **Larkspur Capital**, a long-only fund that owns 9.8 percent, bought mostly at about $19, and has said it will "carefully scrutinize any decision to reject a premium of this size."
- **Rolf Vandermeer**, chief executive of Corvane, a diversified industrial group about three and a half times Brightwell's size by enterprise value.
- **Lena Koskinen**, chief executive of Altair Measurement Group, a larger listed instrument group active in adjacent rather than overlapping product categories, who telephoned Ms. Oyelaran on 30 October to say Altair "would be interested in a conversation about a combination on friendlier terms."

The decision is not simply "accept or reject." The board can recommend acceptance; reject and hold; reject and seek a higher price from Corvane; seek a white knight; or reject and put a stand-alone alternative in place, such as a leveraged recapitalization. It must also decide what to do about the rights plan, whether to open the books to Altair, whether to approach Corvane, and what to tell employees, customers and regulators.

You are an analyst at Calloway Hart. Ms. Oyelaran wants a written analysis, in order, of: what the bid is worth and what premium it carries; what Brightwell is worth on its own under different plans; what Corvane can afford to pay; what each defense does to value and to the bidder; how a white knight compares once certainty and timing are included; and what the board's process should look like.

Industry Context

Precision measurement is a quietly attractive corner of industrial technology. A laser interferometer that calibrates machine tools in an aerospace plant must be traceable to a recognized standard, supported by service engineers who can attend within a day, and accepted by quality auditors who have approved a specific vendor's process. Switching vendors means re-qualification, weeks of lost production and a documentary burden. That is the source of pricing power and of the recurring service and software revenue, and it is why peers trade at higher multiples than general industrials. Five features matter for the case.

Consolidation pressure. Customers are consolidating and want one vendor for sensors, software and service. Larger groups can bundle; smaller ones cannot. Independent mid-sized instrument makers are scarce assets, and scarcity supports control premiums. The six precedents in Table 13.8 carried a median premium of 32 percent and a median EV/EBITDA of about 12.1 times.

Cyclicality at the edges. Service and software revenue is stable, but instrument sales depend on aerospace, semiconductor-equipment and energy capital spending. Management argues the mix is shifting toward recurring revenue; the adviser agrees it is real but slow. A semiconductor-equipment slowdown in late 2025 cut instrument bookings by 11 percent in one quarter before they recovered.

Regulatory attention. Corvane has an interferometry business, acquired in 2019, that overlaps with part of Brightwell's laser line. Altair's portfolio is adjacent rather than overlapping, and its counsel expects a shorter review. These facts feed the certainty-adjusted comparison of offers.

Technology and capital intensity. Optical-comb and chip-scale sensing could erode premium interferometer positioning over a decade; Brightwell spends 7.5 percent of revenue on research and development. Capital expenditure runs at 4.5 percent of revenue against depreciation of 4.0 percent, and working capital absorbs about 15 percent of incremental revenue, so cash conversion from EBITDA is good.

The takeover environment. In the market where both companies are listed, boards have wide latitude to adopt a rights plan against an unsolicited offer, but must be able to show a reasonable basis for perceiving a threat, a proportionate response and no mere entrenchment. Shareholders can replace directors, and Brightwell's board is staggered into three classes of which one is elected each year. The case supplies the financial analysis any legal standard will depend upon; it does not set out the law.

Voting power now sits largely with institutions, passive funds and activists, who respond quickly to premium offers, and merger-arbitrage funds quickly set the price after a bid. Directors cannot count on a long quiet period, and the post-bid price is not an unbiased estimate of stand-alone value: it blends the probabilities of a deal, a bump, a counterbid and failure. Part of the assignment is to decode what $32.70 is saying.

Financial Data

All amounts are in US dollars. Millions unless stated; per-share amounts in dollars. Fiscal years end 31 December. FY2026E combines nine months of actuals with a three-month estimate; the valuation date used throughout the case is 31 December 2026, treated as "now" for simplicity. The case ignores stub-period adjustments. All data are illustrative.

Historical and current-year financials

Table 13.1: Brightwell Instruments, income statement summary (USD millions)

Line	FY2024A	FY2025A	FY2026E
Revenue	1,090.0	1,150.0	1,200.0
Gross profit	436.0	471.5	504.0
Gross margin	40.0%	41.0%	42.0%
EBITDA	207.0	224.0	240.0
EBITDA margin	19.0%	19.5%	20.0%
Depreciation and amortization	44.0	46.0	48.0
EBIT	163.0	178.0	192.0
Interest expense (incl. non-cash items)	38.0	37.0	36.0
Pre-tax income	125.0	141.0	156.0
Income tax (25%)	31.0	35.0	39.0
Net income	94.0	106.0	117.0
Basic shares (millions)	80.0	80.0	80.0
Basic EPS ($)	1.18	1.33	1.46
Dividend per share ($)	0.36	0.38	0.40

Cash interest on the $520 million of debt outstanding at the end of FY2026E is $33.8 million (6.5 percent). The additional $2.2 million in reported interest expense is non-cash amortization of financing costs and the interest component of pension cost. Use the $33.8 million cash figure when a cash interest burden is needed, and say so.

Table 13.2: Brightwell Instruments, balance sheet summary (USD millions)

Line	31 Dec 2025A	31 Dec 2026E
Cash and equivalents	95.0	120.0
Accounts receivable	180.0	190.0
Inventory	200.0	210.0
Other current assets	28.0	30.0
Property, plant and equipment, net	600.0	620.0
Goodwill and intangibles	480.0	480.0
Total assets	**1,583.0**	**1,650.0**
Accounts payable	115.0	120.0
Accrued liabilities	85.0	90.0
Debt: term loan and notes	540.0	520.0
Deferred tax and other long-term liabilities	108.0	110.0
Total liabilities	**848.0**	**840.0**
Shareholders’ equity	735.0	810.0
Total liabilities and equity	**1,583.0**	**1,650.0**
Net debt (debt less cash)	445.0	400.0

The debt is split into a $300 million term loan at a floating rate (currently 6.0 percent) with a change-of-control mandatory prepayment clause, and $220 million of senior notes at 7.2 percent fixed, callable at 103. Neither figure includes the unfunded pension obligation described under Exhibit C.

Market data and capitalization

Table 13.3: Brightwell share data and capitalization at the undisturbed price and at the offer

Item	Undisturbed (2 Oct 2026)	At $32.00 offer
Share price ($)	25.00	32.00
Basic shares (millions)	80.000	80.000
Options: 3.0m at $18.00 strike, net shares by treasury method	0.840	1.3125
RSUs (vest on change of control)	1.000	1.000
Fully diluted shares (millions)	81.840	82.3125
Equity value	2,046.0	2,634.0
Plus debt	520.0	520.0
Less cash	(120.0)	(120.0)
Enterprise value	2,446.0	3,034.0
EV / FY2026E EBITDA ($240.0m)	10.2x	12.6x
EV / FY2027E EBITDA, management plan ($272.2m)	9.0x	11.1x
EV / FY2027E EBITDA, board case ($260.8m)	9.4x	11.6x
Price / FY2026E EPS ($1.4625 unrounded)	17.1x	21.9x

Other price data: 30-day volume-weighted average price to 2 October, $24.60; 52-week high, $28.40; 52-week low, $19.80; closing price on 3 November, $32.70; average daily volume has risen from 0.4 million shares to 3.1 million shares since the bid.

Forecasts

The management plan was prepared in the ordinary course in September 2026, before Corvane’s approach became public. The “board case” is an alternative prepared by Calloway Hart with Ms. Raman after a review of the management plan’s largest assumptions; it lowers growth and margin expansion in every year. The downside case assumes a prolonged industrial slowdown. All three cases share the following drivers: depreciation and amortization 4.0 percent of revenue; capital expenditure 4.5 percent of revenue; investment in net working capital equal to 15 percent of the change in revenue; cash tax rate 25 percent on EBIT; no incremental debt. Free cash flow (FCF) in the tables is unlevered: NOPAT plus depreciation less capital expenditure less the investment in working capital.

Table 13.4: Management plan (USD millions)

Line	FY2027E	FY2028E	FY2029E	FY2030E	FY2031E
Revenue growth	8.0%	8.0%	7.0%	6.0%	5.0%
Revenue	1,296.0	1,399.7	1,497.7	1,587.5	1,666.9
EBITDA margin	21.0%	22.0%	22.5%	23.0%	23.0%
EBITDA	272.2	307.9	337.0	365.1	383.4
D&A	51.8	56.0	59.9	63.5	66.7
EBIT	220.3	251.9	277.1	301.6	316.7
NOPAT (EBIT less 25% tax)	165.2	189.0	207.8	226.2	237.5
Capital expenditure	58.3	63.0	67.4	71.4	75.0
Investment in NWC	14.4	15.6	14.7	13.5	11.9
Unlevered free cash flow	144.4	166.4	185.6	204.8	217.3

Table 13.5: Board case and downside case (USD millions)

Line	FY2027E	FY2028E	FY2029E	FY2030E	FY2031E
Board case revenue growth	6.0%	5.0%	5.0%	4.0%	4.0%
Revenue	1,272.0	1,335.6	1,402.4	1,458.5	1,516.8
EBITDA margin	20.5%	21.0%	21.5%	21.5%	21.5%
EBITDA	260.8	280.5	301.5	313.6	326.1
D&A	50.9	53.4	56.1	58.3	60.7
EBIT	209.9	227.1	245.4	255.2	265.4
NOPAT	157.4	170.3	184.1	191.4	199.1
Capital expenditure	57.2	60.1	63.1	65.6	68.3
Investment in NWC	10.8	9.5	10.0	8.4	8.8
Unlevered free cash flow	140.2	154.1	167.0	175.7	182.7
Downside revenue growth	3.0%	3.0%	3.0%	3.0%	3.0%
Revenue	1,236.0	1,273.1	1,311.3	1,350.6	1,391.1
EBITDA margin	19.5%	19.0%	19.0%	19.0%	19.0%
EBITDA	241.0	241.9	249.1	256.6	264.3
D&A	49.4	50.9	52.5	54.0	55.6
EBIT	191.6	191.0	196.7	202.6	208.7
NOPAT	143.7	143.2	147.5	151.9	156.5
Capital expenditure	55.6	57.3	59.0	60.8	62.6
Investment in NWC	5.4	5.6	5.7	5.9	6.1
Unlevered free cash flow	132.1	131.3	135.2	139.3	143.5

Discount rate and terminal assumptions. Calloway Hart’s weighted-average cost of capital for Brightwell is 8.5 percent, built from a 9.4 percent cost of equity (risk-free rate 4.2 percent, beta 0.91, equity risk premium 5.7 percent; see Exhibit F on how the beta was chosen), an after-tax cost of debt of 5.0 percent and a target debt-to-capital ratio of 20 percent: 0.80 x 9.4% + 0.20 x 5.0% = 8.52 percent, rounded to 8.5 percent. The terminal growth rate is 2.5 percent. Flows arrive at year-end. The board has asked for a probability weighting across the three cases of 30 percent management plan, 45 percent board case and 25 percent downside.

Peers, precedents and the bidders

Table 13.6: Trading peers (illustrative, at 2 October 2026 prices)

Company	EV / LTM EBITDA	P/E (LTM)	Revenue growth (2-yr CAGR)	EBITDA margin
Aldermere Sensors	11.2x	19.5x	6.0%	20.5%
Tessaro Metrology	9.4x	17.0x	5.0%	18.5%
Norvik Analytical	12.1x	22.0x	8.0%	21.5%
Parhelion Controls	10.3x	18.4x	5.5%	19.5%
Quillon Test Systems	8.7x	15.8x	4.0%	17.0%
Median	**10.3x**	**18.4x**	**5.5%**	**19.5%**
Mean	**10.3x**	**18.5x**	**5.7%**	**19.4%**
Brightwell (undisturbed)	10.2x	17.1x	4.9%	20.0%

Table 13.7: Corvane Industrial Holdings and the financing of the offer

Item	Value
Corvane share price ($)	48.00
Shares outstanding (millions)	150.0
Equity market value (USD millions)	7,200.0
Net debt (USD millions, assumed to carry 6.0% interest)	1,800.0
Enterprise value (USD millions)	9,000.0
EBITDA, LTM (USD millions)	1,050.0
EV / EBITDA	8.6x
Net income, expected FY2027 standalone (USD millions)	520.0
EPS ($) / P/E	3.47 / 13.8x
Net debt / EBITDA	1.7x
Committed bridge financing for the offer (USD millions)	2,700.0 at 7.0%
Corvane's estimated run-rate pre-tax cost synergies (USD millions)	45.0
One-time integration cost, pre-tax (USD millions)	60.0

Table 13.8: Precedent transactions in precision instruments (illustrative)

Target / acquirer	Premium to undisturbed price	EV / LTM EBITDA	Deal path
Dunmore / Harlow	34%	12.8x	Hostile, then agreed
Optek / Veritas Gauge	27%	11.4x	Friendly
Calder Labs / Strand	41%	13.9x	Hostile, then agreed after one raise
Pellam / Mirren	22%	10.6x	Friendly
Brandt Tools / Okano	36%	12.2x	Hostile, withdrawn
Lindqvist / Castor	30%	11.9x	Friendly
Median	**32%**	**12.1x**	
Mean	**31.7%**	**12.1x**	

Table 13.9: The two proposals on the table

Term	Corvane tender (launched 2 Nov)	Altair indication (30 Oct, by telephone)
Form of consideration	$32.00 cash	$20.00 cash plus 0.40 Altair shares
Altair share price at 3 Nov ($)	n/a	35.00
Headline value per share ($)	32.00	34.00
Premium to undisturbed	28.0%	36.0%
Conditions	50.1% minimum tender; rights plan redeemed; financing (committed); antitrust clearance; no due diligence	Due diligence; board recommendation; antitrust clearance; Altair shareholder approval not required
Estimated probability of completion	75%	90% (adjacent portfolios; see Exhibit E)
Expected time to close	11 months	7 months
Break fee if Brightwell accepts another offer	None offered	Altair requests 3% of equity value
Altair’s estimated run-rate synergies	n/a	$60 million pre-tax

Table 13.10: Ownership and rights plan terms

Holder	% of basic shares	Notes
Larkspur Capital	9.8%	Long-only; average cost about $19
Sandoval Family Trust	6.5%	Founder family; CEO is a trustee
Corvane Industrial Holdings	4.9%	3.92m shares at average cost $24.10
Passive index funds	21.0%	Vote according to published policies
Other institutions	38.3%	Includes three merger-arbitrage funds that began buying after the bid
Directors and officers (excluding family trust)	5.5%	Includes 1.2% held by independent directors
Retail and other	14.0%	

Rights plan: adopted 2022; one right per share; flip-in trigger at 15 percent beneficial ownership by an acquirer; exercise price $100 per right; on a trigger each right (except those held by the acquirer) entitles the holder to buy $200 of common stock for $100; the board may redeem rights for $0.001 per right at any time before the trigger; the board may alternatively exchange each unexercised right for one share of common stock after the trigger. A tender offer that is “qualifying” under the plan (all-cash, for all shares, open for at least 60 days, with a 75 percent minimum condition) is not exempted from the rights unless the board approves it.

Complications and Exhibits

These exhibits arrived in the 72 hours after the tender launched. Some are clean facts; some are ambiguous. Part of the exercise is deciding which belong in the analysis.

Exhibit A: Process history. The chief executives met at a conference on 14 September. On 2 October Corvane announced its 4.9 percent stake, and on 12 October it delivered a written proposal at $32.00. The board met on 15 October with Calloway Hart present but without a formal valuation, and declined by letter on 16 October. It formed no special committee and requested no written opinion. The tender launched on 2 November. One director has asked whether the October rejection was adequately informed.

Exhibit B: Executive change-of-control payments. Mr. Sandoval's agreement pays $9.6 million in cash if he leaves within 24 months of a change of control; Ms. Raman is entitled to $3.1 million; four other officers share $6.1 million. Unvested equity held by the six executives accelerates and is worth $11.2 million at $32.00. The total of $30.0 million is 1.14 percent of the equity value at the offer. Corvane has said privately that it expects to retain "most of the current management team," but its offer documents describe no arrangement for Mr. Sandoval.

Exhibit C: Pension. A closed defined-benefit plan has a projected-benefit-obligation deficit of $64 million pre-tax, or $48 million after tax at 25 percent. It is not in the $400 million of net debt in Table 13.2. Corvane's advisers treat it as debt-like; management's plan does not, and the forecast cash flows contain no special contributions.

Exhibit D: Calibration Services. The division has FY2026E revenue of $372 million and EBITDA of $96 million, so the instruments business has revenue of $828 million and EBITDA of $144 million. Oriel Capital has told Calloway Hart in confidence that it would pay 11.0 times EBITDA, or $1,056 million, subject to diligence. The estimated tax cost of a sale is $150 million. Corvane's public statements make clear that the services business is its main strategic attraction.

Exhibit E: Antitrust. Counsel estimates a 30 to 35 percent chance that a Corvane acquisition draws an in-depth review of the overlapping laser line, with a possible remedy of divesting about $40 million of revenue. Altair's portfolio is adjacent and the chance is much lower. The completion probabilities and timelines in Table 13.9 are Calloway Hart's judgments, not guarantees.

Exhibit F: Discount rate. Regression betas for Brightwell range from 0.80 to 1.12 depending on index and window; the peer-relevered average, preferred by Calloway Hart, is 0.91. Corvane's advisers may use a lower rate, which supports a higher price. The 8.5 percent WACC is a central estimate; the board should see the range.

Exhibit G: Third-quarter results. Unaudited Q3 revenue grew 5.2 percent and EBITDA margin was 20.4 percent, in line with plan. Bookings fell 3 percent year on year while backlog rose 6 percent. The data are not public, and Corvane's offer relies only on public information. Selective disclosure is not permitted.

Exhibit H: Shareholder dynamics. Larkspur's letter asks the board to form an independent committee, obtain a fairness opinion and "engage constructively." Three arbitrage funds with a combined 4.1 percent have built positions since the bid. Holders of 10 percent can call a special meeting; Larkspur and the family trust hold 16.3 percent together, and Corvane plus the arbitrage funds could reach 9.0 percent. The staggered board means the May 2027 meeting elects one class only. Corvane has indicated it may propose a short slate.

Exhibit I: An ambiguous signal. A trade publication reports that a "person close to Corvane" said its board has discussed a price "in the high 30s" as an upper limit. Corvane declines to comment. Treat it as, at most, one input to the probability of a raise.

Exhibit J: Measurement ambiguities. Defensible choices differ on: (1) counting the RSUs as fully dilutive (they vest on change of control, so the case says yes); (2) the treasury stock method at each price tested (the case does this) versus constant 82.3 million shares; (3) whether pension and non-cash interest belong in net debt; (4) terminal value from FCF growth (as in the case) versus an exit multiple; (5) valuing at 31 December 2026 versus 2 November 2026, a difference of about 1.4 percent of value at 8.5 percent.

Assignment Questions

Answer all ten questions. Show formulas, inputs and results; state any choice you make among the ambiguities in Exhibit J. Round per-share values to the cent and ratios to one decimal place unless told otherwise.

1. **Offer value and premium.** Using the treasury stock method, compute diluted shares, equity value and enterprise value at the $32.00 offer; EV/EBITDA on FY2026E and on FY2027E (management and board cases); the price/earnings multiple; and the premium to the undisturbed price, the 30-day VWAP and the 52-week high. Compare with Tables 13.6 and 13.8. What does the $32.70 price suggest?
2. **Stand-alone valuation.** Value Brightwell under the management, board and downside cases: enterprise value, equity value and value per diluted share (solve the share count iteratively). Compute the probability-weighted value, show the effect of treating the after-tax pension deficit as debt-like, and cross-check with the peer median multiple.
3. **Sensitivity.** For the board case, tabulate value per share for WACC from 7.5 to 9.5 percent in 0.5-point steps and terminal growth of 1.5, 2.0, 2.5 and 3.0 percent. Find the WACC and the growth rate (the other held at base) at which the board case equals $32.00. What share of enterprise value is terminal value, and what does it imply about reliability?
4. **The bidder's perspective.** Compute the present value of Corvane's synergies (after tax, 9 percent, 50 percent phase-in in year one and 100 percent after, one-time integration cost in year one, no growth after year two) and the maximum price Corvane could pay under each stand-alone view. What share of synergies does $32.00 hand to Brightwell's holders? Estimate Corvane's year-one and year-two EPS accretion or dilution if the bridge finances the offer, and its pro forma leverage.
5. **The rights plan.** Explain the economics of the flip-in right. Compute the dilution to Corvane if it crossed 15 percent and the board exchanged each non-Corvane right for one share, and the loss in its stake value. What does the plan do and not do for shareholders, and what must the board do to make the tender capable of completing?
6. **Alternative defenses.** (a) A leveraged recapitalization paying $12.00 a share with $1,000 million of new debt at 8.5 percent: leverage, interest cover, stub value (25 percent tax shield, 8 percent chance of distress costing 25 percent of enterprise value, options re-struck to $6.00) and total holder value. (b) A "Pac-Man" bid for 51 percent of Corvane at a 30 percent premium: cost and feasibility. (c) The Exhibit D sale with proceeds distributed: value per holder. Rank the three.
7. **The white knight.** Value Altair's proposal at Altair prices of $28, $30, $32, $35 and $38; compute equity value, EV/EBITDA and the 3 percent break fee at $34.00; compute certainty- and time-adjusted values for both proposals (9 percent, fallback $25.00); find the Altair price at which they are equal.
8. **Accept, reject or counter.** Build the decision tree for rejection (Corvane raises to $35.00 or walks away) and find the break-even raise probability using (a) a $26.50 fallback and (b) the probability-weighted DCF value. How should Exhibit I affect your estimate?
9. **Reading the market price.** In a three-state model (complete at $32.00, bump to $36.00, fail at a $26.00 fallback; 12 percent failure; six months at 6 percent), compute the implied probability of a bump and an arbitrageur's returns at $32.70. What does this mean for the board's bargaining position?
10. **Directors' duties, process and recommendation.** Describe in general terms the directors' duties, list the conflicts and how to manage them, specify the steps and information required before 9 November, and draft a recommendation and the actions to authorize.

Teaching Note and Full Solutions

Learning objectives

After working through this case students should be able to:

1. Convert a headline offer into diluted equity value, enterprise value, multiples and premiums, and judge them against peers and precedents.
2. Build a stand-alone DCF under several cases with the treasury stock method, and report a probability-weighted value with its sensitivities.
3. Value a bidder's synergies, compute its maximum price and the share of synergies a price transfers.
4. Explain a rights plan, quantify its dilution of a bidder and recognize its limits.
5. Evaluate a leveraged recapitalization, a counter-bid and an asset sale on value, deliverability and signaling.
6. Compare a white knight with a hostile bid on a certainty- and time-adjusted basis, and decode a post-bid price into implied probabilities.
7. Describe directors' duties in general terms, identify conflicts, and design a defensible process.

Suggested class plan (120 minutes)

Minutes	Segment	Activity
0-10	Opening	Cold-call: "You are Ms. Oyelaran. The bid is 28 percent above the market. Why not simply say yes?" Record first-reaction votes on the board.
10-30	Valuation (Questions 1-3)	Groups of four compute offer multiples and the three DCF values; reconvene and compare. Show the sensitivity grid.
30-50	Bidder economics (Questions 4-5)	Synergy value and maximum price; discuss the rights plan using the dilution figures.
50-70	Defenses and white knight (Questions 6-7)	Assign one defense to each of three groups; each group presents value, deliverability, signaling. Then compare the Altair indication.
70-85	Probabilities (Questions 8-9)	Build the decision tree on the board; decode the $32.70 price. Re-vote.
85-110	Board simulation (Question 10)	Students play the board, Larkspur, Corvane and Altair. The board must agree on a recommendation and a press release within 25 minutes.
110-120	Debrief	Compare the first vote with the final one; discuss process and what would change the decision.

With only 80 minutes, assign Questions 1, 2, 4 and 6 as preparation and teach Questions 7 to 10.

Solutions

Question 1: Offer value and premium

Step 1: diluted shares at $32.00. Options are in the money (strike $18.00 below $32.00). Treasury stock method: the options bring in 3.0m x $18.00 = $54.0m of proceeds, which repurchase $54.0m / $32.00 = 1.6875m shares at the offer price. Net new shares = 3.0 - 1.6875 = 1.3125m. (Equivalently 3.0 x (1 - 18/32) = 3.0 x 0.4375 = 1.3125.) RSUs vest on a change of control: 1.0m. Diluted shares = 80.0 + 1.3125 + 1.0 = 82.3125m.

Step 2: equity and enterprise value. Equity value = 82.3125 x $32.00 = $2,634.0m. Enterprise value = 2,634.0 + 520.0 - 120.0 = $3,034.0m.

Step 3: multiples.

```
EV / FY2026E EBITDA      = 3,034.0 / 240.0 = 12.64x
EV / FY2027E EBITDA (mgmt)  = 3,034.0 / 272.2 = 11.15x
EV / FY2027E EBITDA (board) = 3,034.0 / 260.8 = 11.63x
P/E (FY2026E)           = 32.00 / 1.4625  = 21.88x
```

Step 4: premiums.

```
To undisturbed $25.00: 32.00 / 25.00 - 1 = 28.0%
To 30-day VWAP $24.60:  32.00 / 24.60 - 1 = 30.1%
To 52-week high $28.40: 32.00 / 28.40 - 1 = 12.7%
```

Step 5: comparison. The premium of 28.0 percent is four points below the precedent median of 32 percent (and 3.7 points below the mean of 31.7 percent); applying the median premium to $25.00 gives $33.00, and the mean gives $32.92. The range of precedent premiums (22 to 41 percent) implies $30.50 to $35.25. But the EV/EBITDA multiple of 12.6x is above the precedent median of 12.1x and 2.3 turns above the peer median of 10.3x. Both statements are true. Brightwell traded at a higher multiple than peers before the bid, so a normal premium produces a higher multiple. The premium looks light; the multiple looks full. Quoting only one tells half the story.

Step 6: the $32.70 price. $32.70 is a 30.8 percent premium to undisturbed and $0.70 (2.2 percent) above the offer. The market is saying that the probability-weighted outcome is above $32.00, which requires a material probability of a higher bid. Question 9 quantifies it.

Question 2: Stand-alone valuation

Step 1: discount factors at 8.5 percent, year-end, for 2027 to 2031: 1/1.085 = 0.92166; 1/1.085^2 = 0.84946; 0.78291; 0.72157; 0.66505.

Step 2: present value of free cash flow (board case).

Year	FCF	Factor	PV
2027	140.25	0.92166	129.3
2028	154.07	0.84946	130.9
2029	167.03	0.78291	130.8
2030	175.72	0.72157	126.8
2031	182.75	0.66505	121.5
Sum			639.2

Step 3: terminal value. TV = FCF2031 x (1 + g) / (WACC - g) = 182.75 x 1.025 / 0.06 = 3,121.9. PV = 3,121.9 x 0.66505 = 2,076.2. The implied exit multiple is 3,121.9 / 326.1 = 9.6x FY2031 EBITDA, which is below the current peer median and therefore not aggressive. Enterprise value = 639.2 + 2,076.2 = 2,715.5 (11.3x FY2026E EBITDA). Equity value = 2,715.5 - 400.0 = 2,315.5.

Step 4: per share, solving for dilution. Start with 81.0m shares and compute price 2,315.5 / 81.0 = 28.59. At that price options add 3.0 x (1 - 18/28.59) = 1.11m, giving 82.11m shares and a price of 28.20. Iterate: at 28.20 options add 1.0851m, shares 82.085m, price 28.208. The iteration converges at **$28.21** on 82.086m diluted shares.

The same steps give the other cases.

Case	PV of FCF	PV of TV	EV	Equity	Diluted shares (m)	Value per share
Management	712.0	2,468.7	3,180.7	2,780.7	82.400	$33.75
Board	639.2	2,076.2	2,715.5	2,315.5	82.086	$28.21
Downside	535.1	1,630.0	2,165.1	1,765.1	81.506	$21.66

Terminal values are 77.6, 76.5 and 75.3 percent of enterprise value respectively.

Step 5: probability weighting. 0.30 x 33.75 + 0.45 x 28.21 + 0.25 x 21.66 = 10.125 + 12.695 + 5.415 = **$28.23** (using unrounded values, 28.231). Weighting enterprise values first gives the same per-share figure because the share count is nearly constant across cases: weighted EV is $2,717.4m, equity $2,317.4m.

Step 6: pension. Treating the $48m after-tax deficit as debt-like lowers each equity value by $48m: management $33.18, board $27.64, downside $21.08, weighted $27.66. The effect is about $0.57 a share, small but not trivial against a $32.00 bid, and Corvane's advisers will raise it.

Step 7: cross-check. The peer median of 10.3x on FY2026E EBITDA gives EV of $2,472m, equity of $2,072m and $25.31 a share; the peer range of 8.7x to 12.1x gives $20.74 to $30.45; the precedent median of 12.1x gives $30.45 (a control multiple, so it should exceed trading value). The board case at $28.21 exceeds the peer-median value because Brightwell's margin and growth are slightly above the peers'. A good answer notes that the peer multiple is a market view, while the DCF embeds plan optimism. The offer sits above the peer, weighted-DCF and precedent-multiple values and below the precedent-premium value ($33.00) and the management plan.

Question 3: Sensitivity

Board-case value per share (each cell re-solves enterprise value, equity and the diluted share count):

WACC	g = 1.5%	g = 2.0%	g = 2.5%	g = 3.0%
7.5%	29.34	31.81	34.77	38.39
8.0%	26.72	28.77	31.19	34.10
8.5%	24.47	26.20	**28.21**	30.59
9.0%	22.52	23.99	25.69	27.66
9.5%	20.82	22.08	23.52	25.19

For comparison, the management plan at 8.0 percent and 2.0 percent growth gives $34.41, and at 9.5 percent and 1.5 percent gives $24.97.

Break-even rates. With g = 2.5 percent, the board case equals $32.00 at a WACC of 7.88 percent (found by solving for the WACC at which the per-share value is 32.00; at 8.0 percent the value is $31.19 and at 7.5 percent it is $34.77). With WACC = 8.5 percent, the board case equals $32.00 at g = 3.26 percent, which is above any sensible long-run growth rate for a company in this sector. An adviser using a rate about 0.6 points below ours would find the board case worth the offer price, and Exhibit F's beta range makes that plausible.

Terminal value share. About 76 percent of enterprise value in the board case arrives after year five. The spread between the 7.5 and 9.5 percent rows is $11.25 per share at g = 2.5 percent, wider than the whole premium. The honest conclusion is that stand-alone valuation can bracket value at somewhere between the mid-$20s and the mid-$30s, and that the bid sits inside that range. Competition, which reveals what others will pay, has more evidential weight than any single DCF.

Question 4: The bidder's perspective

Synergy value (9 percent, after tax at 25 percent).

Item	Calculation	Value
Year 1 synergy (50% of $45m), after tax	22.5 x 0.75 = 16.875; / 1.09	15.5
Year 2 synergy, after tax	45 x 0.75 = 33.75; / 1.09^2	28.4
Perpetuity from year 3 (no growth)	33.75 / 0.09 = 375.0; / 1.09^2	315.6
Gross PV		359.5
Integration cost, after tax, year 1	60 x 0.75 = 45.0; / 1.09	(41.3)
Net synergy value		**318.2**

Per diluted share at $32.00: 318.2 / 82.3125 = $3.87.

Maximum price. Stand-alone equity value plus net synergies, divided by diluted shares:

View	Equity	+ synergies	Maximum price per share
Management plan	2,780.7	3,098.9	$37.53
Board case	2,315.5	2,633.7	$32.00
Downside	1,765.1	2,083.3	$25.44

On the board case, $32.00 is exactly Corvane's break-even price: 2,634.0 offer equity value less 2,315.5 stand-alone equity is a premium of $318.5m, which is 100.1 percent of the $318.2m synergy value. Corvane would be paying away all of its synergies (and a further $52.7m of fees if those are counted: 116.6 percent) to earn only its cost of capital. If Corvane's advisers believe the management plan, it can pay up to $37.53. The negotiating range therefore runs from $32 to about $37.50.

Accretion and dilution. Purchase price $2,634.0m plus fees of 2 percent ($52.7m) is funded with $2,686.7m of new debt at 7.0 percent, within the $2,700m bridge. After-tax interest = 2,686.7 x 0.07 x 0.75 = $141.05m.

	Year 1	Year 2
Corvane standalone net income	520.0	540.8 (+4%)
Brightwell net income, board case: (EBIT - 33.8) x 0.75	132.06	144.98
Less after-tax acquisition interest	(141.05)	(141.05)
Plus after-tax synergies	16.875	33.75
Pro forma net income	527.88	578.48
Pro forma EPS (150m shares)	$3.519	$3.857
Standalone EPS	$3.467	$3.605
Accretion	+1.5%	+7.0%

Without synergies year-one EPS is $3.4093, which is 1.7 percent dilutive. The $45m after-tax integration cost excluded from the table would reduce reported year-one EPS by $0.30 (about 8.7 percent), so reported year-one EPS would be $3.219, 7.1 percent below standalone. Accretion is therefore a product of synergies and growth, not of the purchase price alone: the target's earnings yield at the offer is 1/21.9 = 4.6 percent, below the 5.25 percent after-tax cost of the debt, so the deal is dilutive before synergies. EPS accretion is often necessary for a board but is not evidence of value creation.

Pro forma leverage. Net debt = 1,800 + 2,686.7 + 400 = $4,886.7m. EBITDA = 1,050 + 240 = $1,290m (1,312.5 with year-one synergies). Net debt/EBITDA = 3.79x (3.72x). Interest cover = 1,290 / (2,686.7 x 7% + 108 + 33.8) = 1,290 / 329.9 = 3.9x. Corvane can afford the deal without stretching.

Question 5: The rights plan

Economics. On a trigger each non-acquirer right allows the holder to pay $100 and receive $200 of stock at the then price. At $32.00 that is 6.25 shares per right. If every non-acquirer holder exercised, 68.0m rights (after Corvane reaches 12.0m shares, or 15 percent) would create 425m new shares, raising $6.8 billion. The threat, not the exercise, is the point.

Dilution by exchange. At the trigger Corvane holds 15% x 80.0 = 12.0m shares and the other holders 68.0m. An exchange of one share per non-Corvane right adds 68.0m shares: total 148.0m. Corvane's stake falls from 15.0 percent to 12.0/148.0 = **8.11 percent**. On the stand-alone board-case equity value of $2,315.5m, the stake falls in value from 0.15 x 2,315.5 = $347.3m to 0.0811 x 2,315.5 = $187.7m, a loss of **$159.6m**. To reach 50.1 percent after the exchange Corvane would need 74.1m shares instead of 40.1m, so it would have to buy 62.1m rather than 28.1m. The pill does not change Brightwell's total value; it makes unapproved accumulation prohibitively costly.

Limits. Corvane's toehold is 4.9 percent, well below 15 percent, and the tender crosses the threshold only on completion, so the pill is not yet triggered; leaving the rights in place prevents any tender from completing. A pill buys time and leverage; it cannot defeat a bid shareholders want, because shareholders can replace the board. Directors who keep a pill against a premium offer must explain the threat (inadequate price, coercive structure, information asymmetry), which is easier if the board is also seeking alternatives. To let the tender complete, the board would have to redeem the rights or exempt the offer, and should do so only if it concludes that $32.00 is a price shareholders should be allowed to accept.

Question 6: Alternative defenses

(a) Leveraged recapitalization. Dividend: 81.0m shares (basic plus RSUs) x $12.00 = $972m; fees 2.8 percent of $1,000m = $28m; total new debt $1,000m. Gross debt = 520 + 1,000 = 1,520 (6.33x FY2026E EBITDA); net debt = 1,400 (5.83x). Interest = 33.8 + 1,000 x 8.5% = $118.8m; EBITDA/interest = 240 / 118.8 = 2.02x.

Case, year 1	EBITDA	Net debt/EBITDA	EBITDA/interest	FCF after interest and tax
Management	272.2	5.14x	2.29x	55.3
Board	260.8	5.37x	2.19x	51.1
Downside	241.0	5.81x	2.03x	43.0

Stub value: enterprise value is the board case 2,715.5, plus the debt tax shield 0.25 x 1,000 = 250.0, minus expected distress cost 8% x 25% x 2,715.5 = 54.3, for 2,911.2. Less net debt 1,400 gives equity of 1,511.2. With options re-struck to a $6.00 exercise price (in the money) the per-share stub solves to **$18.20**, and the total value to a holder is 12.00 + 18.20 = **$30.20**, or $2.00 more than the stand-alone board case. That uplift is entirely the tax shield net of expected distress cost. At a distress probability of 15 percent the total is $29.64 and at 25 percent $28.83, close to the unlevered value. It delivers 38 percent of the offer in cash immediately, but headroom against any plausible covenant is thin (current EBITDA of 240 is only 1.0 percent above the 237.6 that gives 2.0x cover, and leverage starts at 5.8x), and the board would be exposing the company to a cyclical downturn to defend against a bidder.

(b) Pac-Man. Buying 51 percent of Corvane at a 30 percent premium costs 0.51 x 7,200 x 1.30 = $4,773.6m, which is 2.3 times Brightwell's whole equity value of $2,046m and more than four times its debt capacity at 6x EBITDA (6 x 240 - 400 = $1,040m). Infeasible.

(c) Sale of Calibration Services. Net proceeds 1,056 - 150 tax = $906m, or $11.19 on 81.0m shares. The remaining business has EBITDA of $144m; at the undisturbed multiple of 10.19x its EV is $1,467.6m, equity (after the unchanged $400m net debt) $1,067.6m, or $13.18 a share (options are out of the money at that price). Total **$24.37**, below the undisturbed price. The

division must trade at above 12.35x on the remainder to equal the board case $28.21. The sale destroys value through tax leakage and the loss of recurring revenue.

Ranking. On value: recap ($30.20), then Pac-Man (not computable), then asset sale ($24.37). On deliverability: asset sale and recap above Pac-Man. On signaling: the recap shows willingness to return capital; the asset sale signals a break-up; the Pac-Man signals desperation. None beats a higher offer, but the recap is a credible fallback if no bidder emerges.

Question 7: The white knight

Altair value by share price (value = $20.00 + 0.40 x price):

Altair price	$28	$30	$32	$35	$38
Value per Brightwell share	31.20	32.00	32.80	34.00	35.20

The offer equals Corvane's $32.00 at an Altair price of $30.00; each $1 move in Altair moves the offer $0.40. At $34.00 headline: diluted shares 80 + 3.0 x (1 - 18/34) + 1.0 = 82.4118m, equity $2,802.0m, EV $3,202.0m, 13.3x FY2026E EBITDA. The 3 percent break fee is $84.1m (about $1.02 per share, 2.6 percent of EV).

Certainty and time adjustment. Value = [p x offer + (1 - p) x fallback] / 1.09^t.

```
Corvane: [0.75 x 32.00 + 0.25 x 25.00] = 30.25;  1.09^(11/12) = 1.0822;  30.25 / 1.0822 = $27.95
Altair:  [0.90 x 34.00 + 0.10 x 25.00] = 33.10;  1.09^(7/12)  = 1.0516;  33.10 / 1.0516 = $31.48
```

Altair's proposal is worth $3.53 more on this basis. If Corvane raised to $36.00 its adjusted value would be (0.75 x 36 + 0.25 x 25) / 1.0822 = $30.72, still below Altair at $34.00. The two are equal when Altair's offer is [27.952 x 1.0516 - 2.5] / 0.9 = $29.88, i.e. an Altair share price of $24.70. The white knight is therefore robust to a 29 percent fall in Altair's shares. Cautions: Altair's proposal is an uncommitted telephone indication, the stock component transfers risk to Brightwell holders, and the 3 percent fee is a cost Corvane may price in.

Question 8: Accept, reject or counter

Value of rejecting = q x 35.00 + (1 - q) x fallback. Set equal to 32.00.

```
(a) fallback 26.50: q = (32.00 - 26.50) / (35.00 - 26.50) = 5.50 / 8.50 = 64.7%
(b) fallback 28.23: q = (32.00 - 28.23) / (35.00 - 28.23) = 3.77 / 6.77 = 55.7%
(c) fallback 25.00: q = 7.00 / 10.00 = 70.0%
```

q (raise)	EV of rejecting, fallback $26.50	fallback $28.23
30%	29.05	30.26
50%	30.75	31.62
60%	31.60	32.29
70%	32.45	32.97

The board should reject only if it believes a raise is more likely than roughly 56 to 65 percent, ignoring time value. A flat "no" clears a high bar; pairing rejection with a live alternative (Altair) improves both the raise probability and the fallback value. Exhibit I raises the estimate modestly if credible, but it is anonymous and should not carry the decision. The price at which Corvane would rationally stop is its management-plan break-even of $37.53.

Question 9: Reading the market price

States: complete at $32.00 (probability a), bump to $36.00 (b), fail with fallback $26.00 (c = 0.12). Six months at 6 percent: discount factor 1/1.06^0.5 = 0.97129. Price = 0.97129 x (32a + 36b) + 26 x 0.12 = 32.70, with a + b = 0.88.

```
32a + 36b = (32.70 - 3.12) / 0.97129 = 30.454
32 x 0.88 = 28.160;  b = (30.454 - 28.160) / 4 = 0.5736
a = 0.88 - 0.5736 = 0.3064
```

The market implies a **57.4 percent** probability of a bump, 30.6 percent that Corvane completes at $32.00 and 12 percent failure. For an arbitrageur buying at $32.70 the return is -2.1 percent (-4.2 percent annualized) if the deal closes at $32.00, +10.1 percent (+21.2 percent annualized) at a bump to $36.00, and -20.5 percent on failure. Buyers at this price are betting on a higher offer, not on completion at $32.00. For the board this means that shareholders will not tender at $32.00 when they can sell at $32.70 and that Corvane's tender is unlikely to reach the 50.1 percent minimum unless it raises. The board's bargaining position is strong, but only while the market believes a rival or a raise is possible; a flat rejection that forces Corvane to withdraw would collapse the price toward $26.

Question 10: Directors' duties, process and recommendation

Duties, in general terms (not legal advice). Directors owe duties of care (act on an informed basis), loyalty (act in the company's interests, not their own) and good faith. In a takeover, those duties are normally scrutinized more closely than ordinary business decisions because the board may have an interest in its own continuance. Directors should be able to show a reasonable basis for concluding that an offer is inadequate or threatening, that any defensive measure is proportionate, and that they have not used it to entrench themselves. In some jurisdictions the duty becomes one of securing the best value reasonably available once a sale of control is likely; in others directors may weigh the interests of employees, customers and other stakeholders insofar as they bear on the long-term success of the company. A board need not accept any offer or auction the company, but it must be informed and document its process.

Conflicts. (1) Executive payments of $30.0 million (Exhibit B) and uncertainty over post-deal roles: Mr. Sandoval and Ms. Raman should not vote or lead the process. (2) Calloway Hart's success fee, contingent on a completed deal: obtain a second adviser on a fixed fee for a fairness opinion. (3) The family trust's 6.5 percent, held by the chief executive: disclose and treat the trust as an interested holder. (4) Directors' own holdings (1.2 percent): disclose. (5) The October rejection was made without a formal valuation: the special committee should revisit it and record the reasoning.

Steps before 9 November. Form a committee of independent directors chaired by Mr. Achterberg; hire independent counsel and a second adviser; obtain management's plan, a challenge to it, and a written valuation range; sign a confidentiality and standstill agreement with Altair and offer diligence; ask Corvane whether it would pay above $35 and on what conditions; meet Larkspur and the three largest passive holders; prepare a recapitalization contingency and a disclosure plan for Q3 results; minute each decision.

Recommendation. Reject the tender as inadequate; keep the rights plan in place pending the process; authorize engagement with both Altair and Corvane; authorize a recapitalization as a contingency but not as an immediate action; and ask for the tender period to be extended. The memo follows.

Sensitivity discussion

The recommendation is most sensitive to four numbers. (1) WACC: a 0.6-point reduction takes the board case to $32.00; a 0.5-point reduction adds $3.54 to the management plan (8.0 percent, g = 2.5 percent gives $37.29). (2) The plan: management minus board case is $5.54 a share, 79 percent of the $7.00 premium over the undisturbed price. (3) The completion probability

for Corvane: at 75 percent the adjusted value is $27.95, at 100 percent it would be $32.00 / 1.0822 = $29.57, so antitrust risk alone costs about $1.62 a share. (4) Altair's share price, which moves the offer by $0.40 per dollar. The cheapest uncertainty to resolve, by asking, is Corvane's willingness to pay, and it dominates the decision tree.

Common student errors

- Using 80.0m or 81.0m shares at every price instead of the treasury stock method at each price.
- Computing the premium against the current $32.70 price or the 52-week high rather than the undisturbed price.
- Calling the board case "the" intrinsic value; the weighted $28.23 and the range matter.
- Omitting the pension, or double counting it by also deducting the non-cash interest.
- Using the cost of equity instead of WACC, or discounting the terminal value one year too few.
- Treating EPS accretion as proof the bid creates value for Corvane.
- Analyzing the pill as if it changed the total value of Brightwell; it changes who can control the outcome.
- Comparing the Altair and Corvane offers on headline price without time, probability and stock-price risk.
- Treating the post-bid price as the stand-alone value.
- Recommending "just say no" without a fallback or a documented process.

Alternative defensible answers

Three outcomes can be defended with the same numbers. **Accept**: if the board believes Corvane will not raise, Altair is not credible and the management plan is aspirational, then the $28.23 weighted value and $26.50 fallback make $32.00 fair, and the 56 to 65 percent raise threshold is too high. **Reject and counter**: the recommended course, relying on Altair, the market's pricing of a bump and the management plan's $33.75. **No recommendation**: the board states the facts and leaves the choice to shareholders, easier to defend if the board is split or conflicted. Grade the logic from numbers to choice, the treatment of conflicts, and whether the student states what evidence would change the answer.

Recommendation memo to the board

To: Board of Directors, Brightwell Instruments. **From:** Calloway Hart. **Date:** 6 November 2026 (for the meeting of 9 November). **Subject:** Response to the Corvane tender offer. *Illustrative.*

Recommendation. Do not recommend that shareholders tender at $32.00. Authorize management and the independent committee to pursue a higher price from Corvane and a competing proposal from Altair, and extend the process by at least three weeks.

Basis. (1) The offer is a 28.0 percent premium, below the 32 percent precedent median, at 12.6x EBITDA, above the 12.1x precedent median and 10.3x peer median. (2) Our probability-weighted stand-alone value is $28.23 (range $21.66 to $33.75); the offer exceeds it by $3.77 but is below the management plan's $33.75. (3) Corvane can afford more: its break-even is $32.00 on our case and $37.53 on the management plan; a $32.00 price transfers essentially all of its synergies. (4) Altair's indication is worth $34.00 headline and $31.48 after risk and time, against $27.95 for the Corvane bid; the two are not yet comparable as Altair's is not committed. (5) The market price of $32.70 implies a 57.4 percent chance of a bump; rejecting is justified if the true chance exceeds 56 to 65 percent, which is why the board needs Altair as a live alternative.

Actions. Maintain the rights plan and do not trigger an exchange; sign a standstill and open diligence with Altair; invite Corvane to submit a revised price above $35.00; ready a $12.00 recapitalization (value to a holder $30.20; 5.8x net leverage) for use only if no bidder emerges; do not pursue a counter-bid or asset sale; manage conflicts as set out in Question 10; brief Larkspur and the largest holders; and publish the Q3 data only with the recommendation and with counsel's advice.

Risks. Corvane withdraws and the price falls toward $26; Altair does not commit; shareholders tender anyway; a recap leaves little headroom in a downturn. **Reversal trigger.** If neither bidder improves by the end of the extended period, the board should reconsider whether $32.00 is the best value reasonably available.

Chapter 14: Case: Oakhaven Brands, a Consumer-Brand LBO

Type: Case. Oakhaven Brands, Ridgemont Partners, Northbridge Credit, Whitcombe & Rowe, Marlowe Foods and every other company, fund, adviser and person named here are fictional and illustrative. All figures were constructed for teaching and bear no relation to any real company, fund or transaction. Nothing here is investment advice.

Prerequisites: Volume 1, Chapter 14 (debt), Chapter 15 (capital structure), Chapter 21 (enterprise valuation) and Chapter 23 (LBOs, private equity and restructuring); Volume 2 treatment of leveraged finance and covenants.

The Situation

On Monday, 2 November 2026, Whitcombe & Rowe, the investment bank advising the family owners of Oakhaven Brands, sent final-round process letters to four bidders. Binding bids are due at 5:00 p.m. on Friday, 20 November 2026, with committed financing papers. Ridgemont Partners is one of the four. Its investment committee meets on Wednesday, 18 November 2026. Today is Wednesday, 4 November 2026, and the deal team has two weeks to settle on price, structure and a recommendation.

Oakhaven Brands makes premium pantry sauces, shelf-stable snacks and sparkling beverages, sold mainly through grocery chains, club and mass retailers, and a growing e-commerce channel. It was founded in 1971 by Arthur Pruitt, and his descendants own 78 percent of the equity; a regional investor owns the remaining 22 percent and wants liquidity. Chief executive Gwen Halloran, a twelve-year company veteran, and chief financial officer Ibrahim Nasser would roll $40 million into the new structure and receive a management incentive plan. FY2026 is on track to deliver revenue of $600 million and adjusted EBITDA of $120 million, a 20.0 percent margin. The business has $210 million of existing debt and $30 million of cash. Its three brands are described in Table 14.3.

The Ridgemont deal team is led by partner Naomi Castellanos and principal Daniel Ferreira. You are an associate on the team. Ridgemont Partners Fund VII has $5.2 billion of commitments and a concentration limit of 10 percent of commitments per investment, with any excess to be placed with co-investors. The investment committee's rules are these: a base-case gross IRR of at least 15 percent and a base-case multiple of at least 2.0x; a downside case that returns at least 1.0x of invested capital; and a "target" of 20 percent and 2.5x that the partners hope, but do not require, to reach. The committee will not underwrite an exit multiple above 11.0x EBITDA, whatever the entry price.

Whitcombe & Rowe has signaled that the sellers hope for 12.0 times adjusted EBITDA. Ms. Castellanos believes that Marlowe Foods, a listed strategic peer, is among the other bidders and that a strategic can pay more because it can capture cost synergies a sponsor cannot. Northbridge Credit, the lead arranger, has offered a committed package of a $480 million seven-year term loan B, $180 million of eight-year senior notes and a $75 million five-year revolver, which would be undrawn at closing. The terms are in Table 14.6, with the maintenance covenants in Table 14.7.

The questions for the team are the questions every sponsor asks. How should the purchase be financed, and how much equity does it need? Given a plan, how quickly does the debt come down, and how much cash is there for debt service? What do the investors earn if the plan is achieved, and where does that return come from? What if the exit multiple, or the operating plan, is worse? How close to the covenants does the structure run in a bad year? And what is the highest price at which Ridgemont can submit a bid and still meet its own rules? Ms. Castellanos wants a short memo for the committee and a complete model behind it. Assume the transaction closes on 31 December 2026, which is a simplification of the signing and closing timetable.

Industry Context

Branded consumer food and beverage is a favorite of private equity for reasons that deserve both respect and skepticism. Brands with loyal buyers can raise prices modestly each year, shelf placement is sticky once won, capital spending is light, working capital is moderate and cash flow conversion is high. Those features support leverage, since a business that converts about 80 percent of EBITDA to pre-interest cash can carry a heavy interest load.

The same industry has recognizable risks. Several are central to the case.

Input cost pass-through. Sugar, tomatoes, edible oils, glass and cartons are commodity-priced. Retailers resist price increases and the lag between a cost spike and a price action can compress margin for two to four quarters. Oakhaven passed through most of the 2022 inflation but only after a 190-basis-point margin trough.

Retailer power and private label. The top customer accounts for 21 percent of Oakhaven’s revenue, and the top five for 54 percent. Retailers use private-label alternatives to pressure branded suppliers, and a brand that loses distinction loses price. Brand investment in marketing is therefore a protective cost, not a discretionary one. Cutting it lifts EBITDA in the short term and erodes the franchise slowly.

Growth and category trends. Volume growth in the category is flat; growth comes from price, mix and new channels. Beverages are growing faster than pantry staples and have lower margins. E-commerce is attractive but carries higher fulfillment costs.

Leveraged finance conditions. Floating-rate term loans reprice with the benchmark rate. A business that carries 5.5 times leverage at an 8.25 percent term-loan rate has an interest bill that absorbs about half of EBITDA, so a rate shock, an EBITDA miss or both can bring coverage covenants into play quickly. Many tax regimes also limit the deductibility of net interest to a percentage of earnings, which raises the after-tax cost of debt for highly levered companies.

Exit routes. Sponsors exit through a sale to a strategic acquirer, a sale to another sponsor, or a public offering. Strategics have paid the highest multiples for scarce brands, but each route depends on conditions at the time. Exit multiples are the least controllable input in an LBO model, which is why committees insist on underwriting them conservatively.

The consolidation logic is also worth noting. Large food companies have been pruning slow-growth portfolios and buying smaller, faster-growing brands, which supports exit values. A sponsor that buys at a high multiple is, in effect, betting that this buying will continue. The assignment tests whether the numbers justify that bet at 11 times, at 12 times, or at neither.

Financial Data

All amounts are US dollars; millions unless stated. Fiscal years end 31 December. FY2026E combines nine months of actuals and a three-month estimate. Transaction close is assumed to be 31 December 2026, and the first projected year is FY2027. All data are illustrative.

Table 14.1: Oakhaven Brands, income statement summary (USD millions)

Line	FY2024A	FY2025A	FY2026E
Revenue	548.0	574.0	600.0
Gross profit	197.3	212.4	228.0
Gross margin	36.0%	37.0%	38.0%
Reported EBITDA	100.0	106.0	112.0
Add-backs (“restructuring and non-recurring”)	4.0	6.0	8.0
Adjusted EBITDA	**104.0**	**112.0**	**120.0**
Adjusted EBITDA margin	19.0%	19.5%	20.0%
Depreciation and amortization	19.2	20.1	21.0
Adjusted EBIT	84.8	91.9	99.0
Interest on existing debt	15.0	14.0	13.7
Adjusted pre-tax income	69.8	77.9	85.3
Tax at 25%	17.5	19.5	21.3
Adjusted net income	52.3	58.4	64.0
Capital expenditure	16.4	17.2	18.0

Table 14.2: Oakhaven Brands, balance sheet summary, before the transaction (USD millions)

Line	31 Dec 2025A	31 Dec 2026E
Cash	22.0	30.0
Accounts receivable	66.0	70.0
Inventory	80.0	85.0
Other current assets	9.0	10.0
Property, plant and equipment, net	158.0	160.0
Goodwill and brand intangibles	230.0	230.0
Total assets	**565.0**	**585.0**
Accounts payable	56.0	60.0
Accrued liabilities	42.0	45.0
Debt (term loan and notes)	225.0	210.0
Deferred tax	55.0	55.0
Other liabilities	19.0	20.0
Total liabilities	**397.0**	**390.0**
Shareholders’ equity	168.0	195.0
Net working capital (AR + inventory + other CA - AP - accruals)	57.0	60.0

Net working capital is 10 percent of revenue, and the model assumes that it stays at 10 percent of incremental revenue.

Table 14.3: Brand portfolio, FY2026E (USD millions)

Brand and category	Revenue	Growth (LTM)	EBITDA	Margin
Oakhaven Kitchen (pantry sauces)	270.0	3.0%	64.8	24.0%
Hearthstone (shelf-stable snacks)	180.0	4.0%	34.2	19.0%
Pine & Pepper (sparkling beverages)	150.0	8.0%	22.5	15.0%
Corporate costs			(1.5)	
Total	**600.0**	**4.5%**	**120.0**	**20.0%**

Channel mix: grocery 62 percent, club and mass 25 percent, e-commerce 8 percent, foodservice 5 percent. Largest customer 21 percent of revenue; top five 54 percent.

Operating cases

The three cases share these drivers: depreciation and amortization 3.5 percent of revenue; capital expenditure 3.0 percent of revenue; investment in net working capital 10 percent of the change in revenue; tax 25 percent of pre-tax income after interest and the amortization of financing fees. The “Base” case was prepared by Ridgemont with management and includes a $9 million procurement program that is the main source of margin expansion. “Upside” assumes stronger growth in beverages and faster procurement savings. “Downside” assumes a retailer destocking and commodity squeeze in FY2028.

Table 14.4: Operating case assumptions

Case and driver	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033
Base revenue growth	5.5%	5.5%	5.0%	5.0%	4.5%	4.0%	4.0%
Base EBITDA margin	21.0%	22.0%	22.5%	23.0%	23.0%	23.0%	23.0%
Upside revenue growth	7.0%	7.0%	6.0%	6.0%	5.0%	5.0%	5.0%
Upside EBITDA margin	21.0%	22.0%	23.0%	23.5%	24.0%	24.0%	24.0%
Downside revenue growth	2.0%	(3.0%)	1.0%	2.0%	2.0%	2.0%	2.0%
Downside EBITDA margin	19.5%	18.5%	18.5%	19.0%	19.0%	19.0%	19.0%

Table 14.5: Resulting revenue and EBITDA by case (USD millions)

Case and line	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033
Base revenue	633.0	667.8	701.2	736.3	769.4	800.2	832.2
Base EBITDA	132.9	146.9	157.8	169.3	177.0	184.0	191.4
Upside revenue	642.0	686.9	728.2	771.8	810.4	851.0	893.5
Upside EBITDA	134.8	151.1	167.5	181.4	194.5	204.2	214.4
Downside revenue	612.0	593.6	599.6	611.6	623.8	636.3	649.0
Downside EBITDA	119.3	109.8	110.9	116.2	118.5	120.9	123.3

Deal terms and financing

Table 14.6: Proposed purchase and financing terms

Item	Terms
Purchase price (enterprise value)	11.0x FY2026E adjusted EBITDA = $1,320.0m (the sellers hope for 12.0x)
Existing debt	$210.0m, repaid at closing; existing cash $30.0m used as a source
Term loan B	$480.0m (4.0x); seven years; benchmark 3.50% plus 4.75% = 8.25% floating; 1% annual amortization; all excess cash applied to prepayment (assumed)
Senior notes	$180.0m (1.5x); eight years; 10.00% fixed; bullet; non-call three years
Revolver	$75.0m; five years; undrawn at close; 7.75% when drawn; 0.50% fee on undrawn amount
Financing fees	2.5% of funded debt ($660.0m), capitalized and amortized over seven years straight line (tax deductible)
Advisory and transaction costs	1.5% of enterprise value
Minimum operating cash	$25.0m, left on the balance sheet at close and held thereafter
Management rollover	$40.0m invested alongside the sponsor in the same security
Management incentive plan (MIP)	8% of the total equity gain above invested equity, paid at exit
Exit costs	1.0% of exit enterprise value
Interest convention	Computed on opening balances (avoids circularity)

Table 14.7: Maintenance covenants proposed by Northbridge (tested at fiscal year-end)

Covenant	FY2027	FY2028	FY2029	FY2030	FY2031
Maximum net debt / EBITDA	6.25x	5.75x	5.25x	4.75x	4.25x
Minimum EBITDA / cash interest	1.75x	1.75x	2.00x	2.00x	2.25x

Net debt is total debt less cash. Cash interest includes the revolver commitment fee.

Peers and precedents

Table 14.8: Listed consumer-brand peers (illustrative)

Company	EV / LTM EBITDA	EBITDA margin	Revenue growth	Net debt / EBITDA
Marlowe Foods	12.8x	21.5%	4.5%	2.4x
Pennant Pantry Co.	11.4x	19.0%	3.5%	2.9x
Sundial Beverages	13.6x	24.0%	6.0%	1.8x
Corbel Confections	10.2x	18.0%	2.5%	3.4x
Hartwell Snacks	11.9x	20.5%	5.0%	2.6x
Median	**11.9x**	**20.5%**	**4.5%**	**2.6x**
Mean	**12.0x**	**20.6%**	**4.3%**	**2.6x**

Table 14.9: Precedent consumer-brand buyouts (illustrative)

Target / sponsor	Entry EV / EBITDA	Total leverage	Equity % of capitalization
Alder Pantry / Northgate	10.8x	5.25x	47%
Bluecrest Snacks / Tamarack	12.4x	5.75x	49%
Cobalt Beverages / Fenwick	11.6x	5.50x	51%
Dunmere Sauces / Aspen Hill	9.9x	5.00x	46%
Eastlake Brands / Granite Point	13.1x	6.00x	52%
Median	**11.6x**	**5.50x**	**49%**
Mean	**11.6x**	**5.50x**	**49%**

Complications and Exhibits

Exhibit A: The add-backs. Adjusted EBITDA of $120.0 million includes add-backs of $8.0 million for “restructuring and non-recurring items,” up from $4.0 million in FY2024 and $6.0 million in FY2025. Management says the items are plant consolidation costs that will end in FY2027. A skeptical lender’s quality-of-earnings review has so far accepted $5.0 million and challenged $3.0 million. At 11.0 times adjusted EBITDA the price is 11.8 times reported EBITDA of $112.0 million. The lenders’ leverage multiples are quoted on adjusted EBITDA.

Exhibit B: Procurement program. About 70 percent of the base-case margin expansion, roughly $9 million of savings, depends on renegotiating tomato-paste and packaging contracts that run to 2028. Management has signed letters of intent covering about a third of the volume. The base case also assumes that Oakhaven holds marketing spending at 6.5 percent of revenue.

Exhibit C: Customer concentration. The largest customer, a national grocery chain, has asked for a 1.5 percent price rollback in exchange for expanded shelf space for Hearthstone. The base case assumes no rollback. Accepting it would cost roughly $5 million of EBITDA a year before any volume benefit.

Exhibit D: The competing bid. Whitcombe & Rowe has told Ridgemont that another bidder has indicated 12.0 times. If Marlowe Foods is the bidder, a strategic acquirer might capture cost synergies of $12 to $15 million a year (procurement and overhead), worth perhaps one to one and a half turns of EBITDA at a capitalized multiple. A sponsor's synergies are limited to the operating plan.

Exhibit E: Interest and tax. The term loan floats with the benchmark rate; Northbridge offers a three-year swap that fixes the benchmark at 3.90 percent for half of the term loan. In the tax regime assumed here, net interest deductions are capped at 30 percent of EBITDA; disallowed interest is carried forward but the model should ignore the carryforward. The base case in Table 14.6 assumes full deductibility; Question 9 asks for the effect of the cap.

Exhibit F: Management terms. Ms. Halloran has asked for a larger incentive pool, 10 percent of the equity gain, and a higher rollover. A larger MIP reduces the sponsor's return; a larger rollover reduces the sponsor's equity cheque.

Exhibit G: Fund concentration. Fund VII's concentration limit is 10 percent of commitments, or $520.0 million. Three limited partners have asked about co-investment on a fee-free, carry-free basis. Co-investors would be treated like the sponsor on terms and returns.

Exhibit H: Measurement ambiguities. Defensible choices differ on whether to charge interest on opening or average balances (the case uses opening balances); whether transaction fees are capitalized or expensed (financing fees are capitalized and amortized; advisory costs are not deductible in the model); whether the MIP applies to total equity or to sponsor equity only; whether the minimum cash of $25.0 million is a use at closing; and whether exit proceeds are measured after exit costs. The case answers each of these as stated in Table 14.6, but a student who chooses differently and says so should get credit.

Assignment Questions

Answer all ten. State formulas and inputs. Use the Base case unless told otherwise, the terms in Table 14.6 and exit at the end of FY2031 at 11.0 times EBITDA, with sponsor returns measured after the MIP and exit costs. Round dollars to one decimal, multiples to two decimals, returns to one decimal place of a percentage.

1. **Entry valuation.** Compute the implied equity price paid to sellers at 11.0x and 12.0x, EV/revenue, EV/reported EBITDA and EV/FY2027E base EBITDA. Compare with the peers (Table 14.8) and the precedents (Table 14.9).
2. **Sources and uses.** Construct the sources and uses at 11.0x, solving for the sponsor equity cheque. Compute total, senior and net leverage, equity as a percentage of total sources, the sponsor's ownership, interest cover at closing, and the co-investment required by Exhibit G.
3. **Free cash flow.** For FY2027 to FY2031, build the base-case cash flow available for debt repayment: EBITDA less cash interest, cash taxes, capital expenditure and the investment in working capital. Show the tax calculation for FY2027.
4. **Debt schedule.** Build the year-by-year schedule for the term loan, notes and revolver with mandatory amortization and a full cash sweep. Report net leverage and interest cover each year and the percentage of the term loan repaid by FY2031.
5. **Returns.** Compute exit enterprise value, exit equity, the MIP, and the sponsor's proceeds, multiple of invested capital and IRR. Show the IRR before the MIP.
6. **Returns attribution.** Decompose the total equity gain into revenue growth, margin expansion, multiple change, debt paydown, and fee and exit-cost leakage. What share of value creation does each deliver?
7. **Exit scenarios.** Tabulate sponsor IRR and multiple for the three operating cases at exit multiples of 9.0x to 13.0x, and for the Base case by exit year (three to seven). What exit multiple does the Base case need for 15 percent and for 20 percent? At what exit multiple does the Downside case return 1.0x?

8. **Covenant headroom.** For Base and Downside, compute each year's leverage and coverage and the percentage by which EBITDA could fall before each covenant is breached. Find the uniform EBITDA shortfall to the Base case that causes the first breach, and the size of an equity cure in the Downside case with a further 10 percent EBITDA shortfall in FY2029.
9. **Financing sensitivities.** Compute the effect on Base IRR of (a) a 200-basis-point rise in the term-loan rate, (b) the 30 percent EBITDA interest cap of Exhibit E, (c) both together, and (d) total leverage of 4.5x to 6.5x at the same notes amount. Comment on what leverage does to the Downside.
10. **Price, structure and recommendation.** Find the maximum entry multiples consistent with a 15 percent, 18 percent and 20 percent base-case IRR (exit 11.0x). Evaluate the sellers' 12.0x. Recommend a bid, a walk-away price and the structure, and draft a short memo to the investment committee.

Teaching Note and Full Solutions

Learning objectives

After this case students should be able to:

1. Construct a sources-and-uses table and compute the leverage, coverage and equity statistics that lenders and sponsors quote.
2. Build an operating forecast into a free-cash-flow and debt schedule with mandatory amortization and a cash sweep, using opening-balance interest to avoid circularity.
3. Compute sponsor IRR and multiple of invested capital, including the effect of a management incentive plan and exit costs.
4. Attribute equity value creation to revenue growth, margin expansion, multiple change, deleveraging and leakage, and explain why attribution is not causation.
5. Build exit-scenario and exit-timing grids, and interpret the asymmetry between upside and downside.
6. Measure covenant headroom statically and dynamically, and size an equity cure.
7. Assess the effect of interest rates, interest-deductibility caps and leverage on returns and risk, and set a bid price against a committee's hurdle rules.

Suggested class plan (120 minutes)

Minutes	Segment	Activity
0-10	Opening	Poll: "Is 12.0x too much?" Record votes. Explain the committee's rules.
10-30	Structure (Questions 1-2)	Groups build sources and uses; compare equity percentages with precedents.
30-55	Model (Questions 3-4)	Walk through cash flow and the debt schedule on a shared spreadsheet; discuss interest on opening versus average balances.
55-75	Returns (Questions 5-6)	Compute returns and the attribution bridge; discuss why deleveraging is a third of value.
75-95	Scenarios and covenants (Questions 7-8)	Build the exit grids; examine the Downside covenant table.
95-110	Price (Questions 9-10)	Sensitivities and the maximum price; role-play the committee.
110-120	Debrief	Re-poll; compare with the first vote.

Solutions

Question 1: Entry valuation

Equity price to sellers = enterprise value less existing net debt (debt $210.0m less cash $30.0m = $180.0m).

```
At 11.0x:  EV = 11.0 x 120.0 = 1,320.0;  equity = 1,320.0 - 180.0 = 1,140.0
At 12.0x:  EV = 12.0 x 120.0 = 1,440.0;  equity = 1,440.0 - 180.0 = 1,260.0
```

Statistic	11.0x	12.0x
EV / FY2026E revenue	2.20x	2.40x
EV / reported EBITDA ($112.0m)	11.79x	12.86x
EV / FY2027E base EBITDA ($132.9m)	9.93x	10.83x
Equity price / FY2026E net income ($64.0m)	17.8x	19.7x

Comparison. The peer median is 11.9x (mean 12.0x), so 11.0x is 0.9 turns below the median and 12.0x is 0.1 above it. Oakhaven's growth (4.5 percent) and margin (20.0 percent) are close to the peer medians (4.5 and 20.5 percent), so on fundamentals 11.0x is not expensive. The precedent buyout median is 11.6x with a range of 9.9x to 13.1x; 11.0x is 0.6 below the median and 12.0x is 0.4 above. Two cautions apply. The peers are listed and carry 2.6x net debt, not 5.5x, so a buyer's equity must absorb more risk at the same multiple. And the add-backs mean that on reported EBITDA the price is 11.8x at the lower bid and 12.9x at the sellers' ask; the sellers' 12.0x on adjusted EBITDA is a precedent-high price on reported earnings.

Question 2: Sources and uses

Uses. Equity purchase price $1,140.0m; repayment of existing debt $210.0m; advisory and transaction costs 1.5% x 1,320.0 = $19.8m; financing fees 2.5% x 660.0 = $16.5m; cash to balance sheet $25.0m. **Total uses = $1,411.3m.**

Sources. Term loan B $480.0m; senior notes $180.0m; revolver $0; existing balance sheet cash $30.0m; management rollover $40.0m; sponsor equity is the balancing figure: 1,411.3 - 480.0 - 180.0 - 30.0 - 40.0 = **$681.3m**.

Sources	USD m	%	Uses	USD m	%
Term loan B	480.0	34.0%	Equity purchase price	1,140.0	80.8%
Senior notes	180.0	12.8%	Repay existing debt	210.0	14.9%
Revolver (undrawn)	0.0	0.0%	Advisory and transaction costs	19.8	1.4%
Existing cash	30.0	2.1%	Financing fees	16.5	1.2%
Management rollover	40.0	2.8%	Cash to balance sheet	25.0	1.8%
Sponsor equity	681.3	48.3%			
Total	**1,411.3**	**100.0%**	**Total**	**1,411.3**	**100.0%**

Statistics.

```
Total leverage     = 660.0 / 120.0           = 5.50x
Senior (TLB) lev.  = 480.0 / 120.0           = 4.00x
Net leverage       = (660.0 - 25.0) / 120.0  = 5.29x
Total equity       = 681.3 + 40.0            = 721.3  (51.1% of total sources; 52.2% of debt plus equity)
Sponsor ownership  = 681.3 / 721.3           = 94.45%; management 5.55%
Cash interest, year one on closing debt = 480.0 x 8.25% + 180.0 x 10.00% + 75.0 x 0.50%
                   = 39.60 + 18.00 + 0.375   = 57.975
Interest cover     = 120.0 / 57.975          = 2.07x
(EBITDA - capex) / interest = (120.0 - 18.0) / 57.975 = 1.76x
```

Equity at 51.1 percent of sources is above the precedent median of 49 percent because the 11.0x price exceeds the leverage that lenders will fund; leverage of 5.5x equals the precedent median. **Co-investment.** The concentration limit is 10% x 5,200 = $520.0m, so the Fund can write $520.0m of the $681.3m sponsor cheque, and co-investors must take **$161.3m** (23.7 percent of the sponsor equity). Student answers that use a different cash treatment should reconcile to the same sponsor-cheque logic.

Question 3: Free cash flow

Tax calculation, FY2027. Revenue = 600.0 x 1.055 = 633.0; EBITDA = 633.0 x 21.0% = 132.9. D&A = 3.5% x 633.0 = 22.2. Financing-fee amortization = 16.5 / 7 = 2.4. EBIT for tax = 132.9 - 22.2 - 2.4 = 108.4. Cash interest = 57.975 (opening balances). Pre-tax income = 108.4 - 58.0 = 50.4. Tax = 25% x 50.4 = 12.6.

Cash flow. Capital expenditure = 3.0% x 633.0 = 19.0. Working capital investment = 10% x (633.0 - 600.0) = 3.3. Cash flow before debt repayment = 132.9 - 58.0 - 12.6 - 19.0 - 3.3 = 40.1 (rounding).

Table 14.10: Base-case cash flow available for debt repayment (USD millions)

Line	FY2027	FY2028	FY2029	FY2030	FY2031
EBITDA	132.9	146.9	157.8	169.3	177.0
Cash interest (incl. commitment fee)	(58.0)	(54.7)	(50.4)	(45.2)	(39.0)
Cash taxes	(12.6)	(16.6)	(20.1)	(24.0)	(27.2)
Capital expenditure	(19.0)	(20.0)	(21.0)	(22.1)	(23.1)
Investment in working capital	(3.3)	(3.5)	(3.3)	(3.5)	(3.3)
Cash flow before debt repayment	**40.1**	**52.1**	**62.9**	**74.6**	**84.4**
As % of EBITDA	30.2%	35.5%	39.9%	44.1%	47.7%

Cumulative cash flow over five years is $314.0m. Figures may differ by $0.1m from sums of rounded entries. The conversion ratio rises from 30 to 48 percent of EBITDA because interest falls as debt is repaid; this is the engine of deleveraging. Interest absorbs 44 percent of FY2027 EBITDA.

Question 4: Debt schedule

Each year: cash available for repayment = opening cash (25.0) + cash flow before debt repayment - mandatory amortization - minimum cash (25.0); that amount (all of it) repays the term loan in the sweep. Mandatory amortization is 1% x 480.0 = 4.8. Interest on the term loan uses the opening balance: FY2027 480.0 x 8.25% = 39.6; FY2028 439.9 x 8.25% = 36.3; and so on.

The revolver is never drawn.

Table 14.11: Base-case debt schedule (USD millions)

Line	Close	FY2027	FY2028	FY2029	FY2030	FY2031
Term loan, opening		480.0	439.9	387.8	324.9	250.4
Mandatory amortization		(4.8)	(4.8)	(4.8)	(4.8)	(4.8)
Cash sweep		(35.3)	(47.3)	(58.1)	(69.8)	(79.6)
Term loan, closing	480.0	439.9	387.8	324.9	250.4	166.0
Senior notes	180.0	180.0	180.0	180.0	180.0	180.0
Revolver	0.0	0.0	0.0	0.0	0.0	0.0
Total debt	660.0	619.9	567.8	504.9	430.4	346.0
Cash	25.0	25.0	25.0	25.0	25.0	25.0
Net debt	635.0	594.9	542.8	479.9	405.4	321.0
Net debt / EBITDA	5.29x	4.48x	3.69x	3.04x	2.39x	1.81x
Total debt / EBITDA	5.50x	4.66x	3.86x	3.20x	2.54x	1.96x
EBITDA / cash interest	2.07x	2.29x	2.69x	3.13x	3.75x	4.53x

(The “Close” column uses FY2026E EBITDA of $120.0m.) By FY2031 the term loan is 65.4 percent repaid ((480.0 - 166.0) / 480.0) and total debt is down $314.0m, or 47.6 percent. The senior notes are non-call and are not touched, so the cost of the structure is partly the 10.00 percent coupon on $180.0m that cannot be prepaid.

Question 5: Returns

```
Exit EBITDA (FY2031)          = 177.0
Exit enterprise value         = 11.0 x 177.0          = 1,946.6
Exit costs (1.0%)             =                          (19.5)
Net debt at exit              = 346.0 - 25.0          = (321.0)
Exit equity value             =                          1,606.1
Equity gain                   = 1,606.1 - 721.3       =   884.8
MIP (8% of gain)              = 0.08 x 884.8          =    (70.8)
Equity to investors           = 1,606.1 - 70.8        = 1,535.3
Sponsor share (94.45%)        =                          1,450.2
Multiple of invested capital = 1,450.2 / 681.3       =  2.13x
IRR = 2.1285^(1/5) - 1                                  = 16.3%
```

Before the MIP the sponsor’s proceeds would be 94.45% x 1,606.1 = 1,517.0, a multiple of 2.23x and an IRR of 17.4 percent. The MIP therefore costs the sponsor 1.1 points of IRR. The base case clears the committee’s 15 percent and 2.0x floor but not the 20 percent and 2.5x target.

Question 6: Returns attribution

The total equity gain of $884.8m reconciles as follows. Entry equity of $721.3m equals entry enterprise value ($1,320.0m) less entry net debt ($635.0m) plus fees ($36.3m); exit equity equals exit EV less exit costs less exit net debt. Hence the gain = change in EV + reduction in net debt - entry fees - exit costs.

Source	Calculation	USD m	% of gain
Revenue growth	(769.4 - 600.0) x 20.0% margin x 11.0x	372.7	42.1%
Margin expansion	769.4 x (23.0% - 20.0%) x 11.0x	253.9	28.7%
Multiple change	(11.0x - 11.0x) x 177.0	0.0	0.0%
Debt paydown	635.0 - 321.0	314.0	35.5%
Entry fees	19.8 + 16.5	(36.3)	(4.1%)
Exit costs	1.0% x 1,946.6	(19.5)	(2.2%)
Total		**884.8**	**100.0%**

After the MIP of $70.8m, investors share $814.0m, of which the sponsor’s 94.45 percent is $768.9m (check: 1,450.2 - 681.3 = 768.9). Interpretation. About 71 percent of the gain comes from EBITDA growth, split between volume and price (42 percent) and margin (29 percent), and 36 percent from deleveraging; the exit multiple contributes nothing by construction. Margin expansion, which depends on the procurement program in Exhibit B, is the most fragile contributor. Attribution is a decomposition, not causation: deleveraging exists only because the business generates cash, and the cash depends on EBITDA. Students should also note that the “debt paydown” bar is partly funded by operating improvement and partly by the leverage itself, because interest on the borrowed money is the price of the leverage.

Question 7: Exit scenarios

For each case the model is rerun with that case’s EBITDA and cash flows; exit equity then follows the Question 5 method at the chosen multiple, year-five exit.

Table 14.12: Sponsor IRR and multiple of invested capital, exit at end of FY2031

Exit multiple	Downside	Base	Upside
9.0x	(3.9%) / 0.82x	11.0% / 1.68x	13.9% / 1.92x
10.0x	(0.4%) / 0.98x	13.8% / 1.91x	16.7% / 2.16x
11.0x	2.5% / 1.13x	16.3% / 2.13x	19.2% / 2.41x
12.0x	5.1% / 1.28x	18.7% / 2.35x	21.6% / 2.65x
13.0x	7.4% / 1.43x	20.8% / 2.58x	23.7% / 2.90x

Table 14.13: Exit timing at 11.0x (IRR / multiple)

Exit at end of	FY2029 (yr 3)	FY2030 (yr 4)	FY2031 (yr 5)	FY2032 (yr 6)	FY2033 (yr 7)
Downside	(2.6%) / 0.92x	1.2% / 1.05x	2.5% / 1.13x	3.4% / 1.22x	4.0% / 1.32x
Base	18.4% / 1.66x	17.6% / 1.92x	16.3% / 2.13x	15.3% / 2.35x	14.5% / 2.58x
Upside	21.7% / 1.80x	20.4% / 2.10x	19.2% / 2.41x	17.9% / 2.68x	16.9% / 2.98x

Break-evens (solved for the multiple). The Base case needs an exit multiple of 10.48x to earn 15 percent and 12.61x to earn 20 percent. The Downside case returns exactly 1.0x at an exit multiple of 10.12x; at the 11.0x the committee permits it returns 1.13x, so capital is preserved with a thin cushion of about 0.9 turns.

Interpretation. (1) A one-turn change in the exit multiple moves Base IRR by about 2.1 to 2.8 points; a switch from Base to Upside moves it by about 2.9 points at 11.0x, and from Base to Downside by 13.8 points. The distribution is asymmetric: the plan's operating risk is larger than the multiple risk. (2) IRR falls as the hold lengthens in every case because growth slows and the deleveraging engine flattens (interest savings shrink as a share of value), while the multiple rises; a sponsor that is scored on IRR is therefore tempted to exit early, which is the reverse of what a long-term brand owner would prefer. (3) A scenario weighting of 25 percent Downside, 55 percent Base and 20 percent Upside gives an expected multiple of 0.25 x 1.13 + 0.55 x 2.13 + 0.20 x 2.41 = 1.94x, which is below the 2.0x floor even though the Base case clears it. A committee that wants the floor to hold in expectation should say so; the rules in the case apply the floor to the Base case alone.

Question 8: Covenant headroom

Static headroom. For the leverage covenant: EBITDA can fall until net debt / EBITDA equals the covenant, so the cushion is 1 - (net debt / covenant) / EBITDA. For the coverage covenant: the cushion is 1 - (minimum cover x cash interest) / EBITDA. Both hold net debt and interest fixed, so they are static measures that ignore the feedback of a lower EBITDA on cash flow and debt reduction.

Table 14.14: Covenant tests and EBITDA cushion

Case, year	Net lev.	Covenant	Cushion	Cover	Minimum	Cushion
Base FY2027	4.48x	6.25x	28.4%	2.29x	1.75x	23.7%
Base FY2028	3.69x	5.75x	35.7%	2.69x	1.75x	34.9%
Base FY2029	3.04x	5.25x	42.1%	3.13x	2.00x	36.1%
Base FY2030	2.39x	4.75x	49.6%	3.75x	2.00x	46.6%
Base FY2031	1.81x	4.25x	57.3%	4.53x	2.25x	50.4%
Downside FY2027	5.05x	6.25x	19.2%	2.06x	1.75x	15.0%
Downside FY2028	5.21x	5.75x	9.4%	1.99x	1.75x	11.9%
Downside FY2029	4.88x	5.25x	7.1%	2.10x	2.00x	4.9%
Downside FY2030	4.35x	4.75x	8.5%	2.31x	2.00x	13.6%
Downside FY2031	3.93x	4.25x	7.6%	2.51x	2.25x	10.3%

Example (Downside FY2029): net debt 541.0; EBITDA 110.9; leverage 541.0 / 110.9 = 4.88x; cushion = 1 - (541.0 / 5.25) / 110.9 = 1 - 103.05 / 110.9 = 7.1 percent. Coverage 110.9 / 52.8 = 2.10x; cushion = 1 - (2.00 x 52.8) / 110.9 = 4.9 percent. The tightest point in the Downside is the FY2029 coverage test, because the covenant steps up from 1.75x to 2.00x just as EBITDA is still depressed from the FY2028 destocking.

Dynamic breaches. A uniform shortfall applied to every year of the Base EBITDA, with cash flow and debt paydown recomputed, produces the first breach (FY2027 coverage) at a shortfall of about 24 percent; at 25 percent the FY2027 cover is 1.72x versus the 1.75x minimum. Base IRR at shortfalls of 5, 10, 15 and 20 percent is 14.4, 12.5, 10.3 and 8.0 percent; the structure survives a bad year long before returns do. In the Downside case, a further uniform EBITDA shortfall of only 4 percent produces a breach in FY2029 (coverage). The thin Downside cushion of 5 to 9 percent is the true constraint on the structure.

Equity cure. Downside with an extra 10 percent shortfall: FY2029 EBITDA is 110.9 x 0.90 = 99.8, net debt 568.2, leverage 5.69x versus the 5.25x covenant, and coverage 1.84x versus 2.00x. To restore leverage the net debt must fall to 5.25 x 99.8 = 524.1, so the equity cure is 568.2 - 524.1 = **$44.1m**, 6.5 percent of the sponsor's $681.3m. Coverage requires EBITDA of 2.00 x 54.2 = 108.5, short by $8.7m; if the cure counts as deemed EBITDA, the combined cure is the larger of the two tests, and sponsors negotiate deemed-EBITDA cure rights in advance. Students should note that the cure is new money in a failing case, which reduces the multiple on a larger invested amount.

Question 9: Financing sensitivities

Table 14.15: Base-case sensitivities (exit FY2031 at 11.0x)

Scenario	Year-1 cash interest	Year-1 cover	Sponsor IRR	Multiple
Base	58.0	2.29x	16.3%	2.13x
(a) Term loan +200 bp (10.25%)	67.6	1.97x	15.8%	2.09x
(b) Interest cap at 30% of EBITDA	58.0	2.29x	16.2%	2.12x
(c) Both	67.6	1.97x	15.5%	2.06x

a. Extra interest in year one is 480.0 x 2.00% = 9.6, which cuts the sweep and the cover from 2.29x to 1.97x, 11 percent above the 1.75x minimum; Base IRR falls by 0.5 points. A swap on half of the term loan at a fixed benchmark of 3.90 percent would remove half of this exposure at a cost of 0.40 percent on $240m (about $1.0m a year). (b) Deductible interest in FY2027 is capped at 30% x 132.9 = 39.9, so 58.0 - 39.9 = 18.1 is disallowed and tax rises by 25% x 18.1 = 4.5 (from 12.6 to 17.1); the effect fades as interest falls and EBITDA rises, so the IRR cost is only 0.1 point, but it is largest when the Downside needs cash most. (c) The effects are close to additive.

Table 14.16: Leverage (notes fixed at 1.5x; term loan flexed)

Total leverage	Sponsor equity	Base IRR	Base multiple	Year-1 cover (Base)	Downside IRR	Downside covenants
4.5x	798.3	15.1%	2.02x	2.77x	3.1%	Pass
5.0x	739.8	15.7%	2.07x	2.51x	2.8%	Pass
5.5x	681.3	16.3%	2.13x	2.29x	2.5%	Pass
6.0x	622.8	17.1%	2.20x	2.11x	2.2%	Breach FY2028 onward
6.5x	564.3	17.9%	2.28x	1.96x	1.7%	Breach FY2028 onward

Each half-turn of leverage adds 0.6 to 0.8 points of Base IRR but costs 0.3 to 0.5 points of Downside IRR and, from 6.0x, breaches the covenants in the Downside. Leverage raises the expected return only if the downside can be survived; here the covenant structure, not the return calculation, sets the efficient limit at about 5.5x. This is the central lesson: leverage is a bet on the plan, and the risk grows faster than the reward.

Question 10: Price, structure and recommendation

Maximum entry multiples (Base IRR target, exit 11.0x, leverage fixed at $660.0m).

Target IRR	Maximum entry multiple	Enterprise value	Equity price to sellers
20%	10.12x	1,213.8	1,033.8
18%	10.57x	1,268.7	1,088.7
15%	11.36x	1,363.2	1,183.2

Table 14.17: Entry price sensitivity (IRR / multiple, Base case)

Entry multiple	EV	Sponsor equity	Exit at 11.0x	Exit at entry multiple
10.0x	1,200.0	559.5	20.5% / 2.54x	17.9% / 2.28x
10.5x	1,260.0	620.4	18.3% / 2.32x	17.0% / 2.20x
11.0x	1,320.0	681.3	16.3% / 2.13x	16.3% / 2.13x
11.5x	1,380.0	742.2	14.5% / 1.97x	15.7% / 2.07x
12.0x	1,440.0	803.1	12.9% / 1.83x	15.1% / 2.02x
13.0x	1,560.0	924.9	10.0% / 1.61x	14.2% / 1.95x

Evaluation of 12.0x. With the committee's exit cap of 11.0x, the sponsor earns 12.9 percent and 1.83x at 12.0x, below both floors, and the Downside multiple is 0.98x, which fails capital preservation. Only if the exit multiple is allowed to rise to 12.0x does the return reach 15.1 percent. Even the Upside case at 12.0x earns only 15.7 percent and 2.07x. The sellers' 12.0x is therefore outside Ridgemont's rules unless the team can show value that the Base case does not contain; Ridgemont cannot win at that price without breaking its own discipline or finding savings a strategic cannot.

Recommendation. Bid 11.0x ($1,320.0m enterprise value; $1,140.0m equity price). Set the walk-away at 11.3x ($1,356.0m; Base 15.2 percent and 2.03x; Downside 1.08x), just inside the 15 percent floor and with the downside still above 1.0x. Keep total leverage at 5.5x and negotiate: a coverage covenant of 1.75x held through FY2029 (so that the Downside cushion is at least 10 percent); deemed-EBITDA equity cure rights; a hedge on half of the term loan; the add-backs at the quality-of-earnings level; and a $161.3m co-investment (at 11.3x the sponsor cheque is $717.8m and the co-investment $197.8m). Where the bid must go above 11.3x to win, prefer contingent consideration tied to FY2027 EBITDA to a higher headline price.

Sensitivity discussion

The recommendation depends mainly on four inputs. (1) The exit multiple: 0.5 turns moves Base IRR about 1.2 points. (2) Margin: the procurement program contributes 29 percent of equity gain; a failure to reach it (margin held at 20 percent) is roughly the Downside's effect on value and the committee should require evidence for it. (3) Interest rates: 200 basis points costs 0.5 points of IRR and 0.3x of cover. (4) Price: each 0.5 turn costs about 1.6 to 2.0 points of IRR when the exit is fixed at 11.0x. Ask students to rank the four by the cost of reducing the uncertainty: the cheapest is the hedge, then the procurement diligence, then price discipline; the exit multiple is not controllable.

Common student errors

- Forgetting that existing net debt is deducted to get the equity price while existing debt is repaid as a use.
- Treating the fees as outside the sources and uses, or capitalizing advisory fees and amortizing them.
- Computing interest on the average balance in one year and the opening balance in another, producing inconsistent schedules.
- Sweeping cash above the minimum balance but leaving the revolver undefined; or sweeping before the mandatory amortization.
- Applying the MIP to sponsor equity only, or computing it on exit equity rather than on the gain.

- Computing IRR with the wrong number of periods (the close to the end of FY2031 is five years).
- Reading the attribution table as causal, or double counting fees.
- Calculating covenant headroom on total debt instead of net debt, or measuring it in turns rather than percentage of EBITDA.
- Ignoring that a covenant breach is a function of both EBITDA and the debt paydown that EBITDA funds.
- Treating the 12.0x ask as acceptable because “the strategic will pay it.”

Alternative defensible answers

Three answers can be defended with the same model. **Bid at or just below 11.0x and walk at 11.3x**: the recommended answer, which follows the rules as written. **Stretch to 11.5x to 12.0x**: defensible only if the committee accepts a lower floor in return for the Upside’s potential, or if the team believes the exit can be to a strategic at a higher multiple; the student should state that the exit assumption breaches the committee’s rule and quantify the cost (12.9 percent at 12.0x). **Lower leverage and a lower price**: a 5.0x structure at 10.5x gives 17.5 percent and 2.24x in the Base case, 1.24x in the Downside, and first-year net leverage of 4.0x, with more covenant room, and may suit a committee that weighs the downside more heavily. Credit the logic from the numbers to the choice and the explicit statement of what would change it.

Recommendation memo to the investment committee

To: Investment Committee, Ridgemont Partners Fund VII. **From:** Deal team. **Date:** 16 November 2026 (for the meeting of 18 November). **Subject:** Oakhaven Brands: final bid. *Illustrative.*

Recommendation. Approve a binding bid of $1,320.0m enterprise value (11.0x adjusted EBITDA), with authority to raise it to $1,356.0m (11.3x). Do not match the sellers’ 12.0x.

Structure. Sources of $1,411.3m: term loan $480.0m, notes $180.0m, management rollover $40.0m, existing cash $30.0m, and sponsor equity of $681.3m, of which $161.3m is to be placed with co-investors under the concentration limit. Total leverage 5.5x (net 5.29x), equity 51.1 percent.

Returns. Base case, exit FY2031 at 11.0x: 16.3 percent IRR and 2.13x after an 8 percent management incentive plan. Upside 19.2 percent and 2.41x; Downside 2.5 percent and 1.13x, which preserves capital. The 20 percent target is reached only at an exit of about 12.6x or at an entry price near 10.1x. About 71 percent of value comes from growth and margin, 36 percent from paying down debt.

Risks. The procurement program (about $9m) is not yet contracted beyond a third of volume; the largest customer has asked for a price rollback worth about $5m of EBITDA; add-backs of $8.0m are partly contested by the lenders; the Downside cushion on the FY2029 coverage covenant is 4.9 percent.

Conditions. Covenant coverage held at 1.75x through FY2029; deemed-EBITDA cure rights; a hedge on half the term loan; quality-of-earnings confirmation of at least $5.0m of add-backs; and a management incentive pool of no more than 8 percent. If the bid must exceed 11.3x to win, the team will return to the committee rather than extend.

Extension exercises with solutions

Exercise 1: Interest on average balances. Recompute the Base case charging the term-loan interest on the average of opening and closing balances, which creates a circularity (interest depends on the sweep, which depends on interest). *Solution.* Iterate each year until the closing balance stops changing. FY2027 cash interest falls to 56.3 (from 58.0), the sweep rises, and by FY2031 net debt is 309.9 against 321.0. Exit equity is higher and the sponsor IRR is 16.5 percent with a 2.14x multiple, versus 16.3 percent and 2.13x. The opening-balance convention is conservative by about 0.2 points; it is acceptable for a first model, but lenders’ models use averages or daily balances.

Exercise 2: Management's asks (Exhibit F). (a) Raise the MIP to 10 percent of the gain. (b) Raise the rollover to $80.0 million. *Solution.* (a) The MIP becomes 0.10 x 884.8 = 88.5; investors receive 1,606.1 - 88.5 = 1,517.6; the sponsor's 94.45 percent is 1,433.5, a multiple of 2.10x and an IRR of 16.0 percent: the extra two points of MIP cost 0.3 points of IRR. (b) With $80.0m rolled, sponsor equity is 1,411.3 - 480.0 - 180.0 - 30.0 - 80.0 = 641.3 and its ownership is 641.3 / 721.3 = 88.9 percent. Because both parties hold the same security the sponsor's multiple and IRR are unchanged at 2.13x and 16.3 percent, but the cheque falls by $40.0m and the co-investment needed falls to 641.3 - 520.0 = $121.3m. A higher rollover is worth giving because it aligns management without diluting returns, but it also concentrates management's wealth in a leveraged structure.

Exercise 3: What the strategic can pay. Suppose Marlowe Foods can capture $13.5 million of annual cost savings, taxed at 25 percent, capitalized at 8.5 percent. *Solution.* After-tax savings are 13.5 x 0.75 = 10.1; capitalized, 10.1 / 0.085 = $119.1m, or about 1.0 turn of the $120.0m EBITDA. If a strategic's own valuation of Oakhaven at 11.0x equals the sponsor's, it could pay up to roughly 12.0x and keep its return; that is exactly the sellers' ask, and it is why Ridgemont should not expect to win a price war. The sponsor's route to a better price is certainty, speed and management continuity, not a higher headline.

Chapter 15: Case: Rollan Industrial, a Distressed Manufacturer

Type: Case. Rollan Industrial Holdings, Halden Partners, Brannoch Drive Systems and every other company, lender, adviser and individual in this chapter are fictional and illustrative. All figures are invented for teaching and are in US dollars millions unless stated. Nothing here is a statement about any real business, court or statute; legal mechanics are simplified to the level a finance leader needs for decision-making and should be confirmed with counsel in practice.

The Situation

On the morning of Wednesday 4 November 2026, Marta Elkind, chief executive of Rollan Industrial Holdings, Inc. ("Rollan"), asked her chief financial officer to close the door of the conference room at the company's headquarters in Marlow Falls and to bring only one document: the latest thirteen-week cash flow forecast. Dev Raghunathan, the CFO, had produced it the previous evening. He slid it across the table without comment. The bottom row showed cash falling from $14.2 million to a negative number in the seventh week.

Rollan designs and manufactures industrial power-transmission products: heavy-duty gearboxes, hydraulic pumps and motors, and the precision castings that go into them. It operates three plants. The Dunmere plant assembles gearboxes; the Carrick plant makes hydraulic components; the Teague foundry produces castings for both and for outside customers. About 2,900 people work for the company, of whom roughly 1,650 are represented by a union under a collective agreement that runs through 2028. The business was bought in a leveraged buyout in 2021 by Halden Partners, a private equity sponsor, which owns 82 percent of the equity; management and a handful of co-investors hold the remaining 18 percent. Revenue peaked at $468.0 million in FY2024. For the twelve months to 30 September 2026 it was $412.0 million, and EBITDA had fallen by more than half, from $62.4 million to $29.7 million.

The capital structure that looked comfortable in 2021 is now unmanageable:

- a $75 million asset-based revolving credit facility (the "ABL") provided by Pelion Commercial Finance, with $38.0 million drawn;
- a first-lien term loan (the "Term Loan") with $178.5 million outstanding, agented by Calder Agency Bank for a syndicate of fourteen lenders;
- $120.0 million of 9.5 percent senior unsecured notes due 2028 (the "Notes");
- an underfunded defined-benefit pension plan, carried on the balance sheet at a $45.0 million deficit; and
- trade payables of $58.0 million and accrued liabilities of $31.0 million.

Total funded debt is $336.5 million, which is 11.3 times reported EBITDA and 8.9 times the "Adjusted EBITDA" of $38.0 million that management presents to lenders. Cash interest is about $32.3 million a year against EBITDA minus capital expenditure of $14.7 million. The company is not merely tight on liquidity; it cannot service its debt even if its sales stabilise today.

Three events define the next six weeks.

1. **Covenant default and forbearance.** In September Rollan breached the first-lien leverage covenant (maximum 4.25 times) and the fixed charge coverage covenant (minimum 1.10 times). The Term Loan lenders and the ABL lender signed a forbearance agreement on 16 October. It expires on **4 December 2026**, the date on which $4.5 million of quarterly Term

Loan interest is due. While the default continues, the ABL lender has reserved its rights and will not fund new draws, even though the borrowing base would support roughly $24 million of availability.

2. **The Notes coupon.** A $5.7 million semi-annual coupon is due on **15 December 2026**. Missing it starts a 30-day grace period; failing to pay by 14 January 2027 is an event of default under the Notes indenture.
3. **Pension funding.** A minimum required contribution of $6.0 million is due on **15 January 2027**. Missing it triggers a statutory lien in favour of the pension plan and a report to the plan's insurer.

The board has scheduled a special meeting for **Friday 20 November**. Management has been asked to bring one recommendation. The realistic choices are:

- **Option A, a consensual out-of-court restructuring.** Exchange part of the Term Loan and all of the Notes into equity and new debt, with $30 million of new super-senior money, all without a court filing.
- **Option B, a prepackaged Chapter 11 plan.** Solicit votes before filing, file around 7 December with a debtor-in-possession (DIP) financing commitment in hand, and emerge in about three months.
- **Option C, a Chapter 11 case built around a section 363 sale.** File with a stalking-horse bid from Brannoch Drive Systems, a strategic competitor, and run an auction.
- **Liquidation**, which nobody wants but which defines the floor against which every other option is tested.

The people in the room matter. Marta Elkind became CEO in 2023 after running operations; she is respected on the shop floor and is personally committed to keeping the plants open. Dev Raghunathan joined from an investment bank in 2022 and wrote the original credit agreement covenants into the financial model. Joseph Okonkwo-Hale chairs the board and represents Halden Partners, which wants to retain at least some equity and has hinted at new money if lenders cooperate. Fenwick Alder Advisors, the company's financial adviser, has built the forecasts in this chapter. An ad hoc group of five Term Loan lenders holds 58 percent of the loan and has hired its own advisers; an ad hoc group of noteholders led by Corvane Capital holds 46 percent of the Notes. Neither group trusts management's forecast.

Your task is to take the role of Fenwick Alder's engagement team. You must establish how much time Rollan has, what the business is worth, who gets what under each path, whether a deal can be done without a court, how a DIP financing should be sized and priced, and what the board should do on 20 November.

Industry Context

Rollan competes in the market for engineered power-transmission and fluid-power components. This is a long-cycle, capital-goods-adjacent industry. Customers are original equipment manufacturers (OEMs) in mining, agricultural machinery, material handling, wind-turbine auxiliary systems and construction equipment, plus an aftermarket channel that sells replacement gearboxes and pumps through distributors. Roughly 64 percent of Rollan's revenue is OEM and 36 percent aftermarket. The aftermarket carries gross margins about ten points higher than OEM sales and is the more stable stream.

Several features of the industry shape the restructuring analysis.

Operating leverage is high. Plants carry heavy fixed costs: skilled labour that is expensive to lay off, foundry furnaces that must run near capacity to be efficient, and machining centres with long depreciation lives. Between FY2024 and the twelve months to September 2026 revenue fell 12.0 percent while EBITDA fell 52.4 percent. The decremental margin (the fraction of lost revenue that fell through to EBITDA) was 58.4 percent. Cost take-out is possible, but it takes time and cash, and the company has been short of both.

Demand is cyclical and currently weak. Mining equipment orders softened as commodity prices fell; agricultural machinery demand has been depressed by farm incomes and high interest rates; OEM customers have been destocking. Industry analysts (illustrative) expect volumes to bottom in the first half of 2027 and recover gradually. The pace of recovery is the largest source of valuation uncertainty in this case.

Switching costs protect the franchise. A gearbox specified into a haul truck or a baler is qualified over twelve to twenty-four months. OEMs rarely change suppliers mid-programme. That gives a distressed supplier a degree of franchise protection, but only as long as customers believe the supplier will survive to deliver warranty parts for fifteen years. A court filing can therefore damage revenue even where it improves the balance sheet. This is the central trade-off between the out-of-court and court-supervised options.

Supplier concentration. Rollan buys specialty forgings from a few mills. Harlan Forge, a sole-source supplier, is owed $6.5 million and has threatened cash-in-advance terms; credit insurers have cut limits, so trade terms are shortening. That is the main driver of the rising vendor payments in the thirteen-week forecast.

Financial Data

The tables below are extracts prepared by Fenwick Alder from management accounts and lender reporting. All amounts are USD millions. Statements are extracts and do not reconcile line by line to a full set of accounts; where the case needs a figure that is not shown, it says so.

Table 1: Income statement extract

	FY2024A	FY2025A	LTM Sep-2026
Revenue	468.0	441.0	412.0
Cost of goods sold (excluding D&A)	346.0	336.9	321.3
Gross profit	**122.0**	**104.1**	**90.7**
Gross margin	26.1%	23.6%	22.0%
SG&A and other operating expense	59.6	60.0	61.0
EBITDA (reported)	**62.4**	**44.1**	**29.7**
EBITDA margin	13.3%	10.0%	7.2%
Depreciation and amortisation	24.5	24.2	24.0
EBIT	37.9	19.9	5.7
Interest expense	28.4	31.6	33.4
Pre-tax income	9.5	(11.7)	(27.7)
Income tax	2.4	0.0	0.8
Net income	**7.1**	**(11.7)**	**(28.5)**

LTM interest of $33.4 million comprises Term Loan interest of $17.5 million (about 9.8 percent on $178.5 million), ABL interest of $2.9 million, Notes interest of $11.4 million, amortisation of financing costs of $1.1 million and other items of $0.5 million. Cash interest is therefore about $32.3 million. Other LTM cash-flow items used in the covenant tests: capital expenditure $15.0 million, cash taxes $0.8 million, scheduled Term Loan amortisation $1.8 million. By management’s reporting convention, EBITDA in Tables 1 and 6 is stated after charging cash pension contributions ($6.0 million in the LTM period and $6.0 million a year in both forecast cases). The pension plan is therefore serviced inside operating cash flow in the valuations, and no separate pension claim appears in the reorganisation waterfalls when the plan is retained.

Table 2: Reconciliation of reported to Adjusted EBITDA (LTM Sep-2026)

Item	$ million	Comment
Reported EBITDA	29.7	Per Table 1
Add: restructuring and severance	2.9	Cash costs; two plant shifts consolidated
Add: pro forma cost savings	4.1	Annualised savings from actions not yet fully implemented
Add: inventory write-down	1.3	Described as “non-recurring”; similar write-downs occurred in FY2025
Adjusted EBITDA (credit agreement and management)	**38.0**	27.9% above reported

Table 3: Balance sheet extract

	31 Dec 2025	30 Sep 2026
Cash	22.5	14.2
Accounts receivable	66.5	61.0
Inventory	101.0	98.5
Other current assets	10.0	9.3
Current assets	**200.0**	**183.0**
Property, plant and equipment (net)	174.0	168.0
Goodwill and intangibles	74.0	74.0
Other non-current assets	13.0	12.0
Total assets	**461.0**	**437.0**
Accounts payable	62.0	58.0
Accrued liabilities	33.0	31.0
ABL revolver	30.0	38.0
Term Loan	180.0	178.5
Senior unsecured Notes	120.0	120.0
Pension deficit (accounting basis)	38.0	45.0
Other long-term liabilities (leases, environmental, other)	15.0	14.0
Total liabilities	**478.0**	**484.5**
Shareholders’ equity (deficit)	**(17.0)**	**(47.5)**

Book equity is negative. The movement in equity in nine months exceeds reported net income because the pension deficit increased and the change flows through other comprehensive income.

Table 4: Capital structure, terms and market prices (30 September 2026, prices as of 2 November 2026)

Instrument	Amount	Rate	Maturity	Security and ranking	Market price (per 100)
ABL revolver (commitment $75.0)	38.0	SOFR + 3.3% (about 7.6%)	2027	First lien on receivables, inventory and cash; second lien on fixed assets	100
Term Loan	178.5	SOFR + 5.5% (about 9.8%)	2028	First lien on PP&E and intangibles; second lien on current assets	62
Senior Notes	120.0	9.5% fixed	2028	Unsecured, guaranteed by subsidiaries	6.5
Pension deficit	45.0 (accounting) / 62.0 (termination basis)	n/a	n/a	Unsecured claim for any unfunded amount	n/a
Trade payables	58.0	none	30-45 days	Unsecured, subject to special treatment below	n/a
Sponsor and management equity	n/a	n/a	n/a	Residual	about 0

For quick reference: SOFR is taken as 4.3 percent. The market-implied value of the funded debt is $38.0 million for the ABL plus $110.7 million for the Term Loan (62 percent of $178.5 million) plus $7.8 million for the Notes, a total of $156.5 million. Accrued interest at a presumed filing date of 1 December 2026 would add $2.5 million to the Term Loan claim (making $181.0 million) and $5.2 million to the Notes claim (making $125.2 million).

Table 5: Thirteen-week cash flow forecast (direct method; weeks ending Friday, from 13 November 2026)

Opening cash is $14.2 million. Management’s operating minimum is $8.0 million.

Wk	W/E	Receipts	Payroll	Vendors	Opex and taxes	Capex	Debt service	Pension	Prof. fees	Net cash flow	Ending cash
1	Nov 13	7.6	3.4	3.9	0.9	0.3	0.0	0.0	0.5	(1.4)	12.8
2	Nov 20	7.5	0.0	3.9	0.9	0.3	0.0	0.0	0.5	1.9	14.7
3	Nov 27	7.5	3.4	4.0	0.9	0.3	0.0	0.0	0.5	(1.6)	13.1
4	Dec 4	7.4	0.0	4.1	0.9	0.3	4.8	0.0	0.5	(3.2)	9.9
5	Dec 11	7.3	3.4	4.3	0.9	0.3	0.0	0.0	0.5	(2.1)	7.8
6	Dec 18	7.2	0.0	4.4	2.1	0.3	5.7	0.0	0.5	(5.8)	2.0
7	Dec 25	7.1	3.4	4.5	0.9	0.3	0.0	0.0	0.5	(2.5)	(0.5)
8	Jan 1	6.9	0.0	4.6	0.9	0.3	0.3	0.0	0.5	0.3	(0.2)
9	Jan 8	6.8	3.4	4.6	2.1	0.3	0.0	0.0	0.5	(4.1)	(4.3)
10	Jan 15	6.8	0.0	4.5	0.9	0.3	0.0	6.0	0.5	(5.4)	(9.7)
11	Jan 22	7.0	3.4	4.5	0.9	0.3	0.0	0.0	0.5	(2.6)	(12.3)
12	Jan 29	7.1	0.0	4.4	0.9	0.3	0.3	0.0	0.5	0.7	(11.6)
13	Feb 5	7.2	3.4	4.4	0.9	0.3	0.0	0.0	0.5	(2.3)	(13.9)
Total		**93.4**	**23.8**	**56.1**	**14.1**	**3.9**	**11.1**	**6.0**	**6.5**	**(28.1)**	**(13.9)**

Notes to Table 5. Receipts decline from $7.6 million to $6.8 million a week because the destocking OEMs are pushing out orders and because customers are slowing payments to a supplier they fear may fail. Payroll of $3.4 million is paid every second week. Vendor payments rise from $3.9 million to $4.6 million a week as suppliers shorten terms from about 45 days toward cash on delivery; if terms had been held at their present level, vendor payments would have stayed at $3.9 million and cumulative disbursements would be $5.4 million lower. “Opex and taxes” includes $1.2 million of property and payroll tax payments in weeks 6 and 9. “Debt service” comprises Term Loan interest of $4.5 million (week 4, 4 December), the Notes coupon of $5.7 million (week 6, 15 December) and ABL interest of $0.3 million in each of weeks 4, 8 and 12. The pension contribution is the $6.0 million due on 15 January. Professional fees are $0.5 million a week for the pre-filing advisers.

Table 6: Forecast cases, 2027E-2031E

Management and the lenders’ advisers disagree about the shape of the recovery. Management’s case assumes the cost programme works and volumes recover; the lender case is a slower recovery with less margin improvement.

	2027E	2028E	2029E	2030E	2031E
Management case					
Revenue	405	424	445	465	484
EBITDA margin	8.0%	9.5%	10.5%	11.5%	12.0%
EBITDA	32.4	40.3	46.7	53.5	58.1
Capital expenditure	15	16	17	18	19
Lender case					
Revenue	395	408	420	432	445
EBITDA margin	7.0%	8.5%	9.5%	10.0%	10.5%
EBITDA	27.7	34.7	39.9	43.2	46.7
Capital expenditure	14	15	16	16	17
Depreciation and amortisation (both cases)	23	22	22	21	21

Valuation conventions agreed by the advisers: tax rate 25 percent on positive EBIT (existing net operating losses are ignored, which is conservative); net working capital equal to 20 percent of incremental revenue (so a falling revenue line releases cash); mid-year discounting; a weighted average cost of capital of 11.5 percent for a reorganised, smaller, cyclical industrial company; and a long-term growth rate of 2.0 percent. The valuation date is taken as the filing date, and 2027E is treated as the first full forecast year.

Table 7: Trading comparables and precedent transactions

Company (illustrative)	EV / NTM EBITDA	EBITDA margin	Net debt / EBITDA
Vantrel Gear	7.1x	14.5%	2.3x
Kessing Hydraulics	6.4x	12.8%	2.9x
Orlund Power Transmission	8.2x	16.1%	1.9x
Pelham Motion Systems	5.6x	10.9%	3.4x
Tarrow Drives	6.9x	13.7%	2.6x
Beacon Fluid Controls	6.2x	12.0%	3.0x
Median	**6.65x**	**13.25%**	**2.75x**
Rollan (at 2027E lender EBITDA, EV $205)	7.4x	7.0%	11.6x

Precedent transaction (illustrative)	EV / LTM EBITDA
Acquisition of a gearbox maker by a strategic buyer	5.4x
Acquisition of a pump manufacturer by a sponsor	5.8x
Acquisition of a hydraulic components business	6.1x
Acquisition of a castings and forgings group	6.5x
Acquisition of a drive-systems manufacturer by a strategic buyer	7.0x
Median	**6.1x**

Table 8: Liquidation analysis assumptions

Asset	Book value	Recovery rate	Estimated gross proceeds
Cash (at liquidation date, after further burn)	5.0	100%	5.0
Accounts receivable	61.0	75%	45.8
Inventory	98.5	35%	34.5
Other current assets	9.3	15%	1.4
Property, plant and equipment	168.0	38%	63.8
Goodwill and intangibles	74.0	5%	3.7
Other non-current assets	12.0	20%	2.4
Gross proceeds			**156.6**

Liquidation costs (trustee, brokers, auctioneers, wind-down payroll, environmental holdbacks) are 10 percent of gross proceeds. Administrative and priority claims in a liquidation total $10.0 million (accrued wages, WARN-type obligations, taxes). Current assets (cash, receivables, inventory, other current assets) are ABL collateral; fixed and intangible assets are Term Loan collateral. The Term Loan has a second lien on current assets behind the ABL.

Table 9: Proposed DIP financing term sheet (ad hoc Term Loan group, with ABL lender consent)

Term	Proposal
Structure	$40.0 million new-money senior secured superpriority term DIP; ABL continues under cash collateral and amended terms
Lenders	Members of the Term Loan ad hoc group, offered to all Term Loan lenders pro rata on request
Interest	SOFR + 8.0% (12.3 percent at assumed SOFR)
Fees	3.0 percent upfront, 1.0 percent exit fee on commitments
Roll-up	$40.0 million of prepetition Term Loan rolled into the DIP, dollar for dollar with new money, for lenders who participate
Milestones	Plan filed within 30 days of the petition date; confirmation within 90 days; effective date within 105 days
Conversion	DIP converts to exit term loan on emergence at the same rate
Case length assumed	Prepack: about 3.5 months. Sale case: about 5 months

Table 10: The strategic options as proposed by the parties

	Option A: out-of-court exchange	Option B: prepackaged Chapter 11	Option C: Chapter 11 with 363 sale
New money	$30.0m super-senior term loan from Term Loan lenders	$40.0m DIP	$30.0m DIP
Term Loan	$100.0m new first-lien loan; $78.5m exchanged for 88% of new equity	Takeback loan $45.0m plus 95% of new equity	Paid from sale proceeds
Notes	7% of new equity and warrants; covenants stripped by exit consent	5% of new equity shared with other unsecured creditors, as a negotiated gift	Share in residual (none expected)
Sponsor	5% of new equity	Nothing	Nothing
ABL	Amended and extended	Continues under cash collateral order; exit ABL	Repaid at closing
Consent threshold	All affected Term Loan lenders; 95% Notes participation condition	Two-thirds in amount and one-half in number of each voting class	Court approval of sale
Expected EV haircut for loss of confidence	1%	3%	Price set by auction
Time to completion	About 2.5 months	About 3.5 months	About 5 months

Complications and Exhibits

The following facts surfaced in the two weeks before the board meeting. Some sharpen the analysis; others are deliberately ambiguous. Students should say explicitly how each affects their answer, and when they choose to ignore one, why.

Exhibit A: The Adjusted EBITDA add-backs. The credit agreement allows add-backs for restructuring costs and pro forma savings, capped at 25 percent of reported EBITDA (that is $7.4 million at the current level). The inventory write-down is not obviously permitted. The lenders' advisers accept $2.9 million of restructuring costs but only half of the pro forma savings and none of the inventory write-down, arriving at $34.7 million. Management's figure of $38.0 million is therefore a negotiating position, not a fact. For covenant purposes the answer matters; for valuation, the relevant measure is the earnings the business will actually produce, which is what the forecast cases attempt to capture.

Exhibit B: Customer concentration and change-of-control clauses. Three OEM customers account for 31 percent of revenue. Two of them have contracts that allow termination on a "material adverse change in financial condition", and one requires a supplier to maintain a minimum net worth, which Rollan has already breached. In a free-fall filing, one of these customers has told the sales team it would begin qualifying a second source immediately. In a prepackaged filing with a short, well-explained timeline, the same customer has indicated it would "continue to monitor". A sale to Brannoch, which is a competitor of two of the three customers' other suppliers, might trigger the opposite concern. These statements are the origin of the EV haircuts in Table 10 and are, like all such haircuts, judgemental.

Exhibit C: Pension valuation basis. The $45.0 million deficit is measured under the accounting standard using a high-quality bond discount rate. If the plan were terminated, the insurer would compute the claim on a termination basis that is more conservative: $62.0 million. The plan is frozen to new entrants but covers 1,100 current and former employees. The union has made clear that termination would be treated as a breach of the collective agreement. Fenwick Alder's draft plan assumes the

pension is retained and funded at the minimum required level of $6.0 million a year for seven years, after which the schedule is reassessed, but the lenders' advisers argue that the pension claim should be allowed in the unsecured pool at $62.0 million and be rejected as a priority.

Exhibit D: Critical vendors and statutory priorities. Of the $58.0 million of accounts payable, $4.0 million represents goods delivered within 20 days of a filing and would be entitled to administrative priority. A further $18.0 million is owed to vendors that management designates as critical (including Harlan Forge, owed $6.5 million). Management proposes paying critical vendors in full, partly in cash during the case ($10.0 million) and partly through reinstated terms ($8.0 million). The remaining $36.0 million of trade debt is general unsecured. Other unsecured claims, principally lease rejection and contract damages, are estimated at $14.0 million.

Exhibit E: The sponsor's proposal. Halden Partners has offered to contribute $15 million of new equity in an out-of-court deal if, in return, it retains 15 percent of the new equity. The Term Loan group has countered that it would give the sponsor 5 percent and no more, and only if the sponsor also supports the deal publicly. The sponsor has not said what it would do in a Chapter 11, but its fund documents restrict further investment into a company that has filed.

Exhibit F: The Term Loan syndicate. The five lenders in the ad hoc group hold $103.5 million (58 percent). They have said they will support a deal in which they receive control of the equity. The other nine lenders hold $75.0 million. Two of those are collateralised loan obligation vehicles holding $25.0 million (14 percent) whose documents limit them to receiving "like-kind" securities and who have already said they cannot accept equity. The credit agreement allows the "Required Lenders" (more than 50 percent in amount) to waive covenants, but any reduction of principal or release of substantially all collateral requires every affected lender. A deal in which principal is exchanged for equity therefore needs unanimity out of court.

Exhibit G: The Brannoch bid. Brannoch Drive Systems has offered $185 million in cash for substantially all of Rollan's assets, assuming trade payables incurred in the ordinary course after filing and the collective agreement, but not the pension liability or prepetition unsecured debt. Brannoch has asked for a break-up fee of 3 percent of the price and expense reimbursement of up to $1.5 million if it is outbid. Management believes the likelihood of an overbid is 40 percent and that an overbid would clear at about $210 million.

Exhibit H: Environmental, tax and preference matters. The Teague foundry has historic soil contamination; a consultant estimates remediation at $3 million to $9 million, and $4.0 million is accrued within other long-term liabilities. The company has about $60 million of federal net operating losses, but a debt-for-equity exchange transferring more than half the equity may limit their use; their value is excluded from all valuations here. In the 90 days before a filing Rollan paid $9.2 million to unsecured creditors outside the ordinary course, of which counsel believes about half is recoverable; recoveries are excluded from the base waterfalls.

Exhibit I: Management incentives. The compensation committee proposes a $1.5 million key employee retention plan and a management incentive plan of 8 percent of new equity, granted after emergence. Both are likely to be challenged by an unsecured creditors' committee.

Ambiguities students must handle. Whether the Term Loan price of 62 says anything about enterprise value; whether the pension deficit is a financial or an operating liability; whether the sponsor's equity has option value that justifies a gift; and whether a prepackaged case can run to a three-month timetable before a creditors' committee has been formed.

Assignment Questions

Answer each question in turn and show every calculation. Where you make an assumption, state it. Unless told otherwise, use the data in Tables 1 to 10 and Exhibits A to I.

1. **Diagnosis.** Compute leverage, interest coverage and fixed charge coverage on both reported and Adjusted EBITDA, test the two covenants, and calculate the decremental margin between FY2024 and the LTM period. Is Rollan suffering from economic distress, financial distress, or both? What does the answer imply for the choice between reorganisation and liquidation?

2. **Liquidity runway.** Using Table 5, identify the week in which cash first falls below the $8.0 million operating minimum and the week in which cash first turns negative. How many weeks of runway does the company have? Compute the cumulative funding shortfall at week 13, and test whether the ABL availability of $24.0 million would have been sufficient had the lender been willing to fund. Quantify how much of the shortfall is attributable to tightening vendor terms.

3. **Valuation.** Compute the going-concern enterprise value using the discounted cash flow method for both forecast cases, prepare a sensitivity table across WACC and terminal growth, cross-check against the trading and transaction multiples, and recommend a valuation range and base case for use in the restructuring negotiations. Compare the DCF value with the market-implied value of the debt.

4. **Liquidation analysis and waterfall.** Compute net liquidation proceeds and the recovery for each class (ABL, Term Loan, Notes and other unsecured creditors). Then compute the absolute-priority waterfall in a going-concern reorganisation at enterprise values of $175 million, $205 million and $245 million. Identify the fulcrum security.

5. **Out-of-court restructuring.** Evaluate Option A. Compute the pro forma capital structure and the value received by the Term Loan lenders, the noteholders and the sponsor at the base-case value. Explain, using Exhibit F, why a consensual out-of-court deal is difficult and what a holdout can do. Compute the interest burden carried by 20 percent noteholder holdouts.

6. **DIP financing.** Size the DIP financing from a sources-and-uses analysis for a prepackaged case. Compute the all-in cost of the DIP in cash terms. Evaluate the roll-up: show the recoveries of participating and non-participating Term Loan lenders with and without the roll-up if 70 percent of the Term Loan participates.

7. **Plan value.** Compute the reorganised value under Option B, the waterfall to each class after the DIP, the takeback debt and the distribution of new equity, and the recovery percentages. Compute the leverage of the reorganised company and test whether the proposed exit capital structure is sustainable. Compute the enterprise value below which the plan would fail the best-interests-of-creditors test.

8. **Critical stakeholders.** Evaluate the treatment of the pension plan (retain versus terminate), the critical vendors and the sponsor. Compute the present value of the $6.0 million seven-year pension funding commitment at 6 percent, 7 percent and 8 percent and compare it with the recovery the pension plan would receive as an unsecured creditor.

9. **Choosing among the options.** Compute the Term Loan lenders' expected recovery under Options A, B and C. For Option A, assume a failure path in which the company files after a further delay, with an additional 5 percent loss of enterprise value and a DIP of $45.0 million. Compute the break-even probability of success of the out-of-court exchange. Weight the three enterprise value scenarios 25, 50 and 25 percent.

10. **Recommendation.** Write a one-page board recommendation for the 20 November meeting. State what the board should do, what it should say to each constituency, which triggers should cause a change of course, and what the next three weeks should look like.

Teaching Note and Full Solutions

Learning Objectives

After working through this case, students should be able to:

1. Diagnose whether a company is in economic distress, financial distress, or both, using leverage, coverage, decremental margin and the comparison of going-concern and liquidation values.
2. Build and interpret a thirteen-week direct cash flow forecast, identify the dates on which liquidity binds, and separate operating from financing-driven cash burn.
3. Value a distressed business with a DCF under conflicting forecasts, triangulate with multiples, and explain why market prices of debt do not translate directly into enterprise value.
4. Construct a liquidation analysis and an absolute-priority waterfall, identify the fulcrum security, and show how process costs and priming new money change who is in the money.
5. Explain why out-of-court restructurings fail (unanimity requirements, holdouts, cross-defaults) and how a prepackaged Chapter 11 plan overcomes them.
6. Size and price DIP financing, and evaluate a roll-up and its distributional effects among lenders.
7. Test a plan against the best-interests-of-creditors standard, assess feasibility of the exit capital structure, and compute recoveries by class.
8. Compare restructuring paths by probability-weighted recovery, break-even success probability and option value, and write a recommendation that deals with all stakeholders.

Suggested Class Plan (90 minutes)

Minutes	Segment	Questions	Teaching purpose
0-10	Cold open: the thirteen-week forecast on the screen. “It is 4 November. What do you tell the board?”	2	Establish urgency; separate operating from financing burn
10-25	Diagnosis and valuation: is this business worth saving?	1, 3	Economic versus financial distress; DCF sensitivity
25-40	Waterfall: who is in the money, who is the fulcrum?	4	Priority, collateral, liquidation floor
40-55	The out-of-court option and why it is hard	5	Holdouts, sacred rights, exit consents
55-70	DIP, roll-up and plan value	6, 7	Superpriority, cost of capital in distress, feasibility
70-80	Stakeholders and path choice: pension, vendors, sponsor, Brannoch	8, 9	Break-even probability, option value of the market check
80-90	Board vote: each team recommends and defends	10	Judgement under uncertainty

Instructors may split the class into five teams: management, the Term Loan ad hoc group, the noteholder group, the sponsor and the union and pension trustee. Each team prepares only its own recovery and negotiating position for the first sixty minutes, then the class convenes as the board. The tensions this creates among groups are more instructive than a single “right” answer.

Solutions

All results below were computed with Python and are rounded to one decimal place (percentages to one decimal place, multiples to two) unless stated. Small differences from hand calculation arise from rounding of intermediate values.

Question 1: Diagnosis

Step 1: leverage. Funded debt is ABL $38.0 + Term Loan $178.5 + Notes $120.0 = $336.5 million.

```
Total debt / reported EBITDA   = 336.5 / 29.7  = 11.33x
Total debt / Adjusted EBITDA   = 336.5 / 38.0  =  8.86x
Net debt / Adjusted EBITDA     = (336.5 - 14.2) / 38.0 = 322.3 / 38.0 = 8.48x
First-lien net debt            = 38.0 + 178.5 - 14.2 = 202.3
First-lien net / Adjusted      = 202.3 / 38.0  =  5.32x   (covenant: 4.25x)
First-lien net / reported      = 202.3 / 29.7  =  6.81x
First-lien net / lender-adj.   = 202.3 / 34.7  =  5.83x
```

Step 2: first-lien covenant test. At 4.25 times, the EBITDA needed to comply is 202.3 / 4.25 = $47.6 million. Even on management's $38.0 million, the shortfall is $9.6 million, or 20 percent. On reported EBITDA the shortfall is $17.9 million. The covenant is breached on every definition; there is no accounting argument that cures it.

Step 3: coverage. Cash interest is $32.3 million; scheduled amortisation is $1.8 million; capex $15.0 million; cash taxes $0.8 million.

```
EBITDA / cash interest              = 29.7 / 32.3 = 0.92x  (Adjusted: 38.0 / 32.3 = 1.18x)
(EBITDA - capex) / cash interest    = 14.7 / 32.3 = 0.46x
Fixed charge coverage (reported)    = (29.7 - 15.0 - 0.8) / (32.3 + 1.8) = 13.9 / 34.1 = 0.41x
Fixed charge coverage (Adjusted)    = (38.0 - 15.0 - 0.8) / 34.1 = 22.2 / 34.1 = 0.65x
Fixed charge coverage (lender-adj.) = (34.7 - 15.0 - 0.8) / 34.1 = 18.9 / 34.1 = 0.55x
```

The covenant minimum is 1.10 times. Even management's most generous figure is barely over half of it. Cash interest exceeds EBITDA less capex by $17.6 million a year.

Step 4: decremental margin.

```
Change in revenue FY2024 to LTM  = 412.0 - 468.0 = -56.0
Change in EBITDA                 = 29.7 - 62.4   = -32.7
Decremental margin               = 32.7 / 56.0   = 58.4%
```

By sub-period: from FY2024 to FY2025 revenue fell 27.0 and EBITDA fell 18.3 (62.4 - 44.1), a decremental margin of 67.8 percent; from FY2025 to the LTM period revenue fell 29.0 and EBITDA fell 14.4 (44.1 - 29.7), a decremental margin of 49.7 percent. The average over the whole period is 58.4 percent. This is the signature of a high fixed-cost business: roughly 58 cents of every lost revenue dollar disappeared from EBITDA.

Step 5: economic versus financial distress. The evidence points to both.

- *Economic distress* is real: gross margin has fallen from 26.1 to 22.0 percent, EBITDA has halved, and the company has lost pricing power with OEM customers. The business needs operational repair (cost take-out, footprint rationalisation) regardless of how it is financed.
- *Financial distress* is dominant: the business's going-concern value (about $205 million, Question 3) is well above its liquidation value (about $141 million net, Question 4), so it is worth more alive than dead. But the debt of $336.5 million is 164 percent of that value. At a sustainable 4.0 times leverage on 2028E lender-case EBITDA of $34.7 million the

business can carry about $138.8 million of debt; excess debt is therefore about $197.7 million. No amount of operating improvement within a reasonable period closes that gap.

Implication. Because going-concern value exceeds liquidation value and the problem is mainly the balance sheet, the right answer is reorganisation, not liquidation. Because the business also needs operating repair, the restructuring must come with new money and a credible operating plan, not merely a debt swap. A restructuring that fixes only the capital structure (Option A at 6.1 times 2027E EBITDA, Question 5) leaves the company exposed to a second filing.

Question 2: Liquidity runway

Step 1: find the breaches. Ending cash by week (Table 5): 12.8, 14.7, 13.1, 9.9, **7.8**, 2.0, **(0.5)**, (0.2), (4.3), (9.7), (12.3), (11.6), (13.9).

- Cash first falls below the $8.0 million minimum in **week 5** (week ending 11 December, cash $7.8 million).
- Cash first turns negative in **week 7** (week ending 25 December, cash negative $0.5 million), following the Notes coupon in week 6.
- The runway is therefore about **four to five weeks** to the operating minimum and about **six to seven weeks** to zero. Roughly four of those weeks are before the 4 December expiry of the forbearance agreement.

Step 2: decompose the burn. Cumulative net cash flow over 13 weeks is negative $28.1 million.

```
Operating cash flow (receipts - payroll - vendors - opex and taxes):
  93.4 - 23.8 - 56.1 - 14.1 = -0.6
Capex                         =  -3.9
Debt service                  = -11.1  (Term Loan 4.5 + Notes 5.7 + ABL 0.9)
Pension                       =  -6.0
Professional fees             =  -6.5
Total                         = -28.1
Non-operating burn            = 27.5 / 28.1 = 97.9% of the total
```

The core business is roughly cash-neutral over the quarter; the cash crisis is almost entirely created by debt service, the pension contribution and the advisers. That is further evidence of financial rather than operating distress, and it explains why a standstill on debt service would cure most of the immediate problem.

Step 3: would the ABL have been enough? To keep $8.0 million of minimum cash the company would need to draw (8.0 minus ending cash) in each week. The peak requirement is in week 13: 8.0 - (-13.9) = **$21.9 million**. Against availability of $24.0 million, headroom would be only **$2.1 million** at week 13. The ABL is, however, unavailable: the default blocks draws. Also, availability is not static; it moves with the borrowing base. Each $1 of lower eligible receivables cuts availability by about $0.85 (at an 85 percent advance rate). Receipts are falling, so a decline of just $2.5 million in eligible receivables would eliminate the headroom. Even in the best case, the ABL merely postpones the problem past February.

Step 4: the vendor-terms effect. If vendor payments had stayed at $3.9 million a week, the 13-week total would have been 13 x 3.9 = $50.7 million rather than $56.1 million, a difference of $5.4 million. Ending cash at week 13 would have been negative $8.5 million rather than negative $13.9 million. The funding requirement to restore the $8.0 million minimum would have been $16.5 million rather than $21.9 million. Tightening vendor terms thus explains 5.4 / 21.9 = **24.7 percent** of the peak shortfall. This is a classic distress spiral: the weaker the credit looks, the more cash it needs, which makes it look weaker. It is also the reason that a credible, early announcement of a financing solution is worth more than its direct cost.

Question 3: Valuation

Step 1: discounted cash flow. Free cash flow equals EBITDA minus cash taxes at 25 percent of positive EBIT, minus capex, minus the increase in net working capital (20 percent of the change in revenue). Mid-year discounting at 11.5 percent. The terminal value uses the 2031E free cash flow grown at 2.0 percent: TV = FCF2031 x (1 + g) / (WACC - g).

Lender case	2027E	2028E	2029E	2030E	2031E
Revenue	395.0	408.0	420.0	432.0	445.0
EBITDA	27.7	34.7	39.9	43.2	46.7
Less D&A	(23.0)	(22.0)	(22.0)	(21.0)	(21.0)
EBIT	4.7	12.7	17.9	22.2	25.7
Cash tax at 25%	(1.2)	(3.2)	(4.5)	(5.6)	(6.4)
Capex	(14.0)	(15.0)	(16.0)	(16.0)	(17.0)
Working capital cash effect (20% of revenue change)	3.4	(2.6)	(2.4)	(2.4)	(2.6)
Free cash flow	15.9	13.9	17.0	19.3	20.7
Discount factor (mid-year)	0.9470	0.8494	0.7618	0.6832	0.6127
PV of FCF	15.0	11.8	13.0	13.2	12.7

Sum PV FCF 65.7; terminal FCF 20.69; TV 222.2; PV TV 136.1; EV 201.8; DF4.5 0.6127

The management case is computed the same way: free cash flow of 16.4, 15.9, 19.3, 23.4 and 26.0 in 2027E to 2031E, PV of those flows $75.7 million, terminal FCF $26.01 million, terminal value $279.3 million and PV of terminal value $171.1 million.

Step 2: results. The lender case gives an enterprise value of **$201.8 million** (PV of 2027E-2031E free cash flow $65.7 million plus PV of terminal value $136.1 million; terminal value is 67.5 percent of the total). The management case gives **$246.8 million** ($75.7 million plus $171.1 million; terminal value 69.3 percent). The implied terminal value multiples are 4.76 times and 4.81 times terminal-year EBITDA, well below the trading median of 6.65 times, so the DCF is conservative in its exit assumption. That conservatism is deliberate: the capitalised terminal free cash flow reflects a business that still carries capex near depreciation and a cyclical margin, and the comps are healthier, less leveraged companies.

Step 3: sensitivity (enterprise value, $ million).

Lender case: WACC growth	1.0%	2.0%	3.0%
10.5%	207.5	225.6	248.5
11.5%	187.6	**201.8**	219.3
12.5%	171.2	182.6	196.3
13.5%	157.5	166.7	177.7

Management case: WACC growth	1.0%	2.0%	3.0%
10.5%	253.9	276.6	305.4
11.5%	229.0	**246.8**	268.8
12.5%	208.5	222.7	240.0
13.5%	191.3	202.9	216.7

A one-point rise in WACC reduces value by about 9 to 10 percent and a one-point fall in terminal growth by about 7 to 8 percent. The range from $157.5 million to $305.4 million is nearly a factor of two, which is why the parties disagree.

Step 4: multiples cross-check.

```
Trading comps, median 6.65x NTM EBITDA:
  x lender 2027E EBITDA  27.7 = 184.2
  x management 2027E     32.4 = 215.5
  Range 6.0x - 7.2x: lender 166.2 - 199.4 ; management 194.4 - 233.3
Precedent transactions, median 6.1x LTM EBITDA:
  x reported LTM 29.7    = 181.2     (range 5.4x - 7.0x: 160.4 - 207.9)
  x Adjusted LTM 38.0    = 231.8     (range 5.4x - 7.0x: 205.2 - 266.0)
```

The transaction multiples applied to Adjusted EBITDA overlap the management-case DCF; applied to reported EBITDA they sit below the lender-case DCF. The inconsistency reduces to a single question: whose EBITDA is right? The reasonable answer is that normalised EBITDA lies between reported ($29.7 million) and lender-adjusted ($34.7 million), because the restructuring costs are real cash costs but the savings from completed actions are real as well.

Step 5: recommended range. We recommend a range of **$175 million to $245 million** with a base case of **$205 million**. The base case is close to the lender-case DCF ($201.8 million) and the lender-case comps cross-check at the upper end ($199.4 million at 7.2 times), and it sits between the 75 percent lender / 25 percent management blend ($213.1 million) and the pure lender case. The low case ($175 million) approximates the lender-case DCF at a 12.5 to 13.5 percent WACC and the trading multiple at 6.3 times; the high case ($245 million) approximates the management-case DCF. A range is the honest answer; the board should regard the base case as the centre of a distribution, not as a point estimate.

Step 6: reconciliation with market prices. The market value of the funded debt is $156.5 million (Table 4); net of cash of $14.2 million, the market-implied enterprise value is $142.3 million, which is $63 million below the base case. This is not necessarily a market error. First, the debt prices embed the cost of getting through a restructuring: after $90.0 million of DIP, ABL and administrative claims that rank ahead of them (Question 7), the Term Loan lenders are in line for about $108.8 million at the base case, or 60.1 cents per dollar of claim, against a market price of 62. Second, some holders are forced sellers and the price includes some probability of a worse outcome. The right comparison is between the market price and the net recovery, not the gross EV; a student who revises the DCF down to $142 million has confused what the debt is worth with what the business is worth.

Question 4: Liquidation analysis and waterfall

Step 1: gross proceeds (Table 8).

```
Cash                     5.0 x 100% =  5.0
Accounts receivable     61.0 x  75% = 45.8
Inventory               98.5 x  35% = 34.5
Other current assets     9.3 x  15% =  1.4     ABL collateral pool  = 86.6
Property, plant, equipment 168.0 x 38% = 63.8
Goodwill and intangibles  74.0 x   5% =  3.7
Other non-current        12.0 x  20% =  2.4     Term Loan collateral = 69.9
Gross proceeds                              = 156.6
```

Step 2: costs and net proceeds. Costs are 10 percent of gross: $15.7 million. Allocating pro rata: net ABL collateral = 86.6 x 0.90 = $78.0 million; net Term Loan collateral = 69.9 x 0.90 = $62.9 million. Total net proceeds = **$140.9 million**.

Step 3: distribution.

```
ABL (first lien on current assets):  38.0 paid from 78.0   -> 100.0% recovery
Surplus ABL collateral to Term Loan (second lien)  = 78.0 - 38.0 = 40.0
Term Loan collateral                               = 62.9
Total available to Term Loan                       = 102.9
Less administrative and priority claims            = 10.0
Net to Term Loan                                   = 92.9
Term Loan claim (with accrued interest)            = 181.0
Term Loan recovery                                 = 92.9 / 181.0 = 51.3%
Term Loan deficiency claim                         = 88.1  (joins unsecured pool)
Notes, trade, pension, other unsecured             = 0.0%
```

In liquidation the ABL recovers in full, the Term Loan recovers 51.3 percent and everyone else recovers nothing. The sponsor is of course wiped out.

Step 4: going-concern waterfall at three values, simple absolute priority (no process costs). Claims: ABL $38.0 million; Term Loan $181.0 million; unsecured pool (Notes $125.2 million, trade $36.0 million, other $14.0 million = $175.2 million; the pension is assumed to be retained and so outside the pool; critical vendors are paid in the ordinary course).

Enterprise value	ABL	Term Loan paid	Term Loan recovery	Residual to unsecured	Unsecured recovery	Fulcrum
175	38.0	137.0	75.7%	0.0	0.0%	Term Loan
205	38.0	167.0	92.3%	0.0	0.0%	Term Loan
245	38.0	181.0	100.0%	26.0	14.8%	Unsecured pool (Notes)

On this simple view the Term Loan is the fulcrum security at $175 million and $205 million, and the Notes become the fulcrum only at $245 million, where they would recover about $18.6 million (14.8 percent of $125.2 million).

Step 5: why the simple waterfall is misleading. It ignores the cost of getting to the plan. After $40.0 million of DIP new money, $12.0 million of unpaid administrative claims and the effect of the filing on value (a 3 percent haircut, so that $205 million becomes $198.8 million), the senior claims total $90.0 million and the Term Loan recoveries fall to 44.1, 60.1 and 81.6 percent at the three values (Question 7). The Term Loan is then the fulcrum security in all three cases and the Notes are out of the money in all three. This is a central teaching point: process costs and priming new money move the fulcrum up the capital structure. Noteholders who bought at 6.5 are betting on either an unusually high value or a negotiated gift.

Question 5: Out-of-court restructuring

Step 1: pro forma capital structure under Option A. Value after a 1 percent loss of confidence: 205 x 0.99 = $202.9 million.

```
ABL (amended and extended)                  38.0
New super-senior term loan (new money)      30.0
New first-lien Term Loan                   100.0
Total debt                                 168.0
Equity value = 202.9 - 168.0               34.9   (17.2% of EV)
Debt / 2027E lender EBITDA = 168.0 / 27.7  6.06x    Debt / 2028E = 168.0 / 34.7 = 4.84x
Debt / EV                                  82.8%
```

Step 2: value to each constituency at the base case.

```
Term Loan lenders:  100.0 new loan + 88% x 34.9 = 100.0 + 30.7 = 130.8
                    recovery = 130.8 / 181.0 = 72.2%   (Question 9 uses 72.2%)
Noteholders:        7% x 34.9 = 2.4      recovery = 2.4 / 125.2 = 1.9%
Sponsor:            5% x 34.9 = 1.7
```

(The Term Loan lender's equity of $30.7 million uses the unrounded equity value of $34.95 million; rounding differences are less than $0.1 million.)

Step 3: sustainability. Annual cash interest under the proposal is 38.0 x 7.6% + 30.0 x 12.3% + 100.0 x 9.8% = 2.9 + 3.7 + 9.8 = **$16.4 million**, against lender-case 2027E free cash flow before interest of $15.9 million, a coverage of **0.97 times**. Out-of-court, the structure produces a company that still cannot pay its interest in the first year, so the exchange would probably need PIK features or a second round. In other words, the out-of-court deal improves recoveries mainly because it avoids the costs and haircuts of a court case, not because it solves the leverage problem.

Step 4: why consensual deals are hard (Exhibit F). Three obstacles:

1. **Unanimity for principal.** A reduction of Term Loan principal requires every affected lender. Two CLOs holding $25.0 million (14.0 percent) cannot take equity. One or both can block the deal by simply saying no, and the ad hoc group of 58 percent cannot overrule them. Any attempt to engineer around them (for example, an "uptier" transaction that primes non-participants) invites litigation, and the economics of the transaction are then hostage to the outcome of that litigation.
2. **Holdout dynamics among noteholders.** The exchange is a voluntary offer with a 95 percent participation condition. A noteholder who does not tender keeps a full-priced claim on a company whose other creditors have taken equity: the more others exchange, the better the holdout's credit. Exit consents can strip covenants but cannot reduce principal or interest. If 20 percent of the Notes ($24.0 million) hold out, the company continues to owe cash interest of 24.0 x 9.5% = **$2.3 million a year**, equal to 8.2 percent of lender-case 2027E EBITDA and 14.3 percent of 2027E free cash flow, and a $24.0 million claim remains due in 2028.
3. **Coordination and time.** The runway is about six weeks; an exchange needs an offering memorandum, consent solicitations and a 20-business-day tender. Failure carries the cost shown in Question 9.

In Chapter 11 a plan binds dissenting holders if each impaired class votes in favour by two-thirds in amount and a majority in number of those voting, or if the plan can be "crammed down" over a dissenting class. The CLOs can vote no and still be bound by the class vote. That is the reason a prepackaged Chapter 11 is the usual tool for a balance-sheet restructuring with fragmented holders.

Question 6: DIP financing

Step 1: sources and uses through emergence (prepackaged case, about 3.5 months).

Use	$ million
Operating cash deficit including capex and the pension contribution	15.0
Critical vendor cash payments	10.0
Professional fees paid during the case	8.0
Adequate protection payments (50% of Term Loan cash interest, ABL interest)	3.2
DIP fees and agent expenses	2.1
KEIP/KERP	1.5
Utility deposits and cash collateral	1.0
Total uses	**40.8**
Less: opening cash above the $8.0 million minimum (9.9 - 8.0)	(1.9)
Net funding need	**38.9**

A $40.0 million DIP covers the need with only $1.1 million (2.8 percent) of cushion. The cushion looks thin, but the ABL continues with availability of about $24.0 million under the cash collateral order, so total committed liquidity is 40.0 + 24.0 = $64.0 million, or 1.65 times the net need. Lenders would normally want a cushion of 10 to 15 percent, and the ABL availability supplies it, provided the borrowing base holds.

Step 2: cost of the DIP. Interest is SOFR 4.3% + 8.0% = 12.3 percent. With an average drawn balance of $30.0 million for 3.5 months:

```
Interest            = 30.0 x 12.3% x 3.5/12 = 1.08
Upfront fee         = 3.0% x 40.0           = 1.20
Exit fee            = 1.0% x 40.0           = 0.40
Total cash cost                             = 2.68
Annualised on average drawn balance
  = 2.68 / (30.0 x 3.5/12) = 2.68 / 8.75    = 30.6%
```

The headline 12.3 percent understates the true cost; the effective annualised cost on money actually drawn is about 31 percent because of the fees and the short life of the facility. Even so, the total cost of $2.7 million is small compared with the value at stake: a 3 percent EV haircut on $205 million is $6.2 million, and a free-fall filing without committed financing would risk a larger one (Exhibit B). The price of the DIP is the price of the option to avoid a disorderly case.

Step 3: the roll-up. The proposal rolls up $40.0 million of prepetition Term Loan dollar for dollar with new money, for participating lenders. A roll-up converts a prepetition claim that would recover only part of its value into a superpriority claim repaid in full. It does not change the total value available to the Term Loan class, but it moves value among its members. With 70 percent of the Term Loan ($126.7 million of $181.0 million) participating and the base-case value available to the class (before the roll-up) of $108.8 million:

```
Without roll-up:   every lender recovers 108.8 / 181.0 = 60.1%
With roll-up:      rolled 40.0 paid at 100%
                   remaining claims = 181.0 - 40.0 = 141.0 share 108.8 - 40.0 = 68.8
                   recovery on remaining claims = 68.8 / 141.0 = 48.8%
Participants (claim 126.7): (40.0 + 86.7 x 48.8%) / 126.7 = 82.3 / 126.7 = 65.0%
Non-participants (claim 54.3): 48.8%
Transfer: non-participants lose 60.1% - 48.8% = 11.3 points x 54.3 = 6.1
          participants gain 65.0% - 60.1% = 4.9 points x 126.7 = 6.1
```

The same pattern at the other two values:

Enterprise value	No roll-up (all lenders)	With roll-up: participants	With roll-up: non-participants
175	44.1%	50.9%	28.2%
205	60.1%	65.0%	48.8%
245	81.6%	83.8%	76.4%

Observations: a roll-up is a zero-sum transfer within the class; the damage to non-participants is largest in the downside; the transfer is smaller when the class is nearly fully repaid; and offering the roll-up to every Term Loan lender pro rata removes the transfer entirely. That is why Table 9 offers participation to all lenders “on request”, and why the CLOs, who cannot lend new money, will object. A court will ask whether the roll-up is necessary to obtain the financing and whether the terms are fair to non-participants.

Question 7: Plan value

Step 1: reorganised value and senior claims (base case). Option B assumes a 3 percent loss of value from the filing: 205 x 0.97 = **$198.8 million** (unrounded $198.85 million). Claims that rank ahead of the Term Loan are:

```
ABL (prepetition, continues as exit ABL)       38.0
DIP new-money term loan (converts to exit TL)  40.0
Administrative and priority claims at emergence 12.0
  (accrued professional fees 6.0; section 503(b)(9) goods 4.0; cure costs 2.0)
Total senior claims                            90.0
Value available to Term Loan and below = 198.8 - 90.0 = 108.8
```

Term Loan recovery before any gift is 108.8 / 181.0 = **60.1 percent**.

Step 2: the plan capital structure. The administrative claims are funded from the exit ABL. The Term Loan receives a $45.0 million takeback loan plus equity.

```
Exit ABL           = 38.0 + 12.0 = 50.0
Exit term loan (DIP conversion)  = 40.0
Takeback term loan to prepetition lenders = 45.0
Total exit debt                  = 135.0
Equity value = 198.8 - 135.0     =  63.8
Term Loan lenders: 45.0 + 95% x 63.8 = 45.0 + 60.7 = 105.7  ->  105.7 / 181.0 = 58.4%
Unsecured pool (Notes 125.2 + trade 36.0 + other 14.0 = 175.2):
                   5% x 63.8 = 3.2   ->  1.8% (Notes: 2.3 on 125.2 = 1.8%)
```

The plan therefore transfers $3.2 million of value from the Term Loan lenders to the unsecured creditors, which is the 5 percent "gift" that buys their votes. Under the absolute priority rule the notes should receive nothing; the gift is lawful only if it is part of a consensual plan, or is justified as a settlement of claims and challenges. Equity is allocated before dilution by the 8 percent management incentive plan.

Step 3: the three values.

Enterprise value (pre-haircut)	175	205	245
Reorganised value (x 0.97)	169.8	198.8	237.7
Senior claims	90.0	90.0	90.0
Value to Term Loan and below	79.8	108.8	147.7
Term Loan recovery before gift	44.1%	60.1%	81.6%
Equity value (after 135.0 debt)	34.8	63.8	102.7
Term Loan value (45.0 + 95% equity)	78.0	105.7	142.5
Term Loan recovery after gift	43.1%	58.4%	78.7%
Unsecured value (5% equity)	1.7	3.2	5.1
Unsecured recovery	1.0%	1.8%	2.9%

Step 4: feasibility. Exit debt of $135.0 million is 4.87 times 2027E lender-case EBITDA, 3.89 times 2028E and 4.17 times 2027E management-case EBITDA. Interest at the stated rates is 50.0 x 7.6% + 40.0 x 12.3% + 45.0 x 9.8% = 3.80 + 4.92 + 4.41 = **$13.1 million**.

```
EBITDA / interest, 2027E lender case          = 27.7 / 13.1 = 2.11x
(EBITDA - capex) / interest                   = 13.7 / 13.1 = 1.04x
FCF before interest / interest                = 15.9 / 13.1 = 1.21x
```

The structure is feasible but tight in the lender case in the first year; it improves as margins recover (2028E lender free cash flow before interest is $13.9 million, a coverage of 1.06 times, so the plan should allow a PIK election on the takeback loan for the first two years). Contrast Option A, where leverage was 6.1 times and interest coverage 0.97 times.

Step 5: best-interests test. Each holder in an impaired class who does not vote for the plan must receive at least what it would receive in a liquidation. Term Loan liquidation recovery is 51.3 percent (Question 4). Setting the plan's pre-gift Term Loan recovery equal to that: (EV x 0.97 - 90.0) / 181.0 = 0.513, so EV x 0.97 = 92.9 + 90.0 = 182.9 and **EV = $188.6 million**. Below that value, a dissenting Term Loan lender could argue that the plan fails the test. At $175 million, the Term Loan recovers 43.1 to 44.1 percent, below 51.3 percent: the low case fails the best-interests test. This is an important finding: the low end of the valuation range is not a feasible plan value at all, because lenders would rather liquidate (or sell) than take it. That leads directly to Question 9's option-value point.

Question 8: Critical stakeholders

Pension. Retaining the plan costs $6.0 million a year. The present value of seven years of $6.0 million:

```
at 6%: 6.0 x [1 - 1.06^-7] / 0.06 = 33.5
at 7%: 6.0 x [1 - 1.07^-7] / 0.07 = 32.3
at 8%: 6.0 x [1 - 1.08^-7] / 0.08 = 31.2
```

These are below the $45.0 million accounting deficit because the plan's assets are expected to earn a return, and below the $62.0 million termination-basis claim. If the plan were terminated, the insurer's claim of $62.0 million would join an unsecured pool of 175.2 + 62.0 = $237.2 million and would recover 62.0 x 3.19 / 237.2 = about **$0.8 million**, or 1.3 percent. Retention thus preserves roughly $32 million of present-value funding commitment (plus the associated benefits to 1,100 participants) at the price of about $6 million a year of cash, which is already inside the forecast EBITDA. Termination would not save the company anything near $62 million of cash: it would trigger a union grievance, risk strikes and customer concerns, and the claim itself recovers almost nothing. The correct recommendation is to retain the plan and negotiate a funding schedule.

Critical vendors. The $18.0 million owed to critical vendors is general unsecured debt that would recover about 1.8 percent. Paying 100 percent costs $10.0 million in cash during the case and $8.0 million through reinstated terms. The test is whether the cost is smaller than the value lost if supply stopped. Harlan Forge is sole-source, qualification of an alternative takes 9 to 12 months, and Table 5 already shows the company being squeezed. A week of lost forging deliveries stops the Dunmere line, which is part of about $7.9 million of weekly company revenue. Paying the critical vendors is clearly worth it, provided the payment is conditioned on customary trade terms for the period of the case and limited to those who are truly critical. Students often over-allocate: a vendor is critical only if it cannot be replaced at a lower cost than the claim.

Sponsor. Under absolute priority the equity gets nothing, because even at $245 million the Term Loan is not paid in full once costs are deducted. The sponsor's only legitimate levers are (a) new money, (b) the threat of delay and litigation and (c) the possibility that the lenders prefer an out-of-court deal that avoids cost. The $15 million of new equity requested for 15 percent of the equity in Exhibit E values the equity at $100 million post-money, but the base-case equity value after debt is $34.9 million (Option A) or $63.8 million (Option B). The sponsor is thus asking to buy at a price far above intrinsic value; the lenders' 5 percent offer for $15 million is not crazy either, because the sponsor receives 5 percent of $34.9 million, about $1.7 million, without investing. The right conclusion is that the sponsor should receive a small consensual gift (up to 5 percent) in return for a support agreement, and no more.

Question 9: Choosing among the options

Step 1: recoveries to the Term Loan lenders at each value. Using the formulas and values already derived:

Enterprise value	Option A success	Option A failure (files later)	Option B (prepack)	Option C: no overbid	Option C: overbid
175	57.8%	36.0%	43.1%	53.9%	63.3%
205	72.2%	50.5%	58.4%	53.9%	63.3%
245	91.5%	69.9%	78.7%	53.9%	63.3%

Notes. *Option A failure path:* value falls to 205 x 0.97 x 0.95 = $188.9 million, senior claims rise to 38.0 + 45.0 + 12.0 = $95.0 million, so the recovery is (188.9 - 95.0) / 181.0 = 51.9 percent before the 5 percent gift; after the gift and the equity-only residual it is 50.5 percent. *Option C:* price is market-determined: with no overbid the buyer pays $185.0 million, with an overbid $210.0 million. Costs are 4 percent of price; if overbid, the stalking horse is paid a break-up fee of 3 percent of $185.0 million ($5.55 million) plus $1.5 million of expenses. Senior claims are ABL 38.0, DIP 30.0 and administrative 12.0 = $80.0 million.

```
C, no overbid: 185.0 x 0.96 = 177.6 ; 177.6 - 80.0 = 97.6 ; 97.6 / 181.0 = 53.9%
C, overbid:    210.0 x 0.96 - (5.55 + 1.5) = 201.6 - 7.05 = 194.55 ; 194.55 - 80.0 = 114.55 ; / 181.0 = 63.3%
C, expected (60/40): 0.6 x 53.9 + 0.4 x 63.3 = 57.7%
```

Step 2: break-even probability for Option A at the base case. Let p be the probability that the exchange succeeds. Expected recovery = p x 72.2 + (1 - p) x 50.5. Setting this equal to Option B's 58.4:

```
p x 72.2 + (1 - p) x 50.5 = 58.4
p x (72.2 - 50.5) = 58.4 - 50.5
p = 0.0784 / 0.2171 = 36.1%   (36.4% using the rounded percentages)
```

Management's honest estimate of the probability of unanimous Term Loan consent plus a 95 percent noteholder tender is 25 to 35 percent given the CLO constraint. Expected recovery at p = 30 percent is 0.30 x 72.2 + 0.70 x 50.5 = **57.0 percent**, below Option B's 58.4 percent. Break-even is therefore above management's own estimate: the out-of-court path is not worth attempting alone.

Step 3: probability-weighted over values (25 / 50 / 25 percent).

```
Option B:  0.25 x 43.1 + 0.50 x 58.4 + 0.25 x 78.7 = 10.8 + 29.2 + 19.7 = 59.6%
Option A (p = 30%): 58.3% (see note)
Option C:  57.7% (price does not depend on the EV scenario)
```

(Note for Option A, with p = 30 percent: at the three values 0.3 x 57.8 + 0.7 x 36.0 = 42.5, 0.3 x 72.2 + 0.7 x 50.5 = 57.0 and 0.3 x 91.5 + 0.7 x 69.9 = 76.4; weighted 0.25 x 42.5 + 0.5 x 57.0 + 0.25 x 76.4 = 58.3.)

The expected recoveries are very close: 59.6, 58.3 and 57.7 percent. The ranking is therefore decided by variance, speed and optionality rather than by the mean. Option B has the widest spread (43.1 to 78.7 percent); Option C has the narrowest (53.9 to 63.3 percent), because the price is set by the market.

Step 4: value of the market check. If the lenders can run a sale as a floor, they would choose Option C's no-overbid recovery of 53.9 percent in the low case rather than Option B's 43.1 percent. The expected recovery from "prepack with a sale floor" is 0.25 x 53.9 + 0.50 x 58.4 + 0.25 x 78.7 = **62.3 percent**, which is 2.7 points (about $4.9 million on a $181.0 million claim) higher than the plain prepack. The stalking-horse bid is a put option on the business at $185 million, and it is worth about $4.9 million. This is why the recommendation uses it as leverage in the prepack negotiations rather than as the primary strategy.

Question 10: Recommendation

Board memorandum (answer to Question 10).

To: Board of Directors, Rollan Industrial Holdings, Inc. *From:* Fenwick Alder Advisors. *Date:* 20 November 2026. *Re:* Restructuring path.

Recommendation. Authorise management to pursue a prepackaged Chapter 11 (Option B) on a dual track: negotiate with the Term Loan ad hoc group and the noteholders for three weeks toward a consensual out-of-court exchange (Option A) while preparing the prepack, the $40.0 million DIP and a stalking-horse sale process with Brannoch as a floor. Set a drop-dead date of 4 December, the forbearance expiry, after which the company files on or about 7 December unless a fully documented, unanimous out-of-court agreement has been signed.

Why. (1) The company has about six weeks of runway and is below its $8.0 million minimum by 11 December; 97.9 percent of the 13-week burn is debt service, pension and advisers, so a restructuring cures the liquidity problem. (2) The business is worth more alive ($205 million base case) than in liquidation ($140.9 million net; Term Loan recovery 51.3 percent). (3) Out-of-court success needs every Term Loan lender and 95 percent of the Notes; break-even probability is 36.1 percent against our estimate of 25 to 35 percent, and a failed attempt costs about 8 points of Term Loan recovery. (4) The prepack gives expected Term Loan recovery of 59.6 percent against 58.3 percent for the exchange and 57.7 percent for a sale. (5) The Brannoch bid is a put worth about $4.9 million to the lenders: it removes the low-value outcome and should be kept alive.

What we say to each constituency. *Term Loan lenders:* you are the fulcrum; you receive the takeback loan and 95 percent of the equity; the DIP and roll-up are offered pro rata to all lenders. *Noteholders:* absolute priority gives nothing; a 5 percent gift is available for support. *Sponsor:* up to 5 percent for a support agreement; no new-money deal is needed. *Union and pension trustee:* the pension is retained and the collective agreement is assumed. *Customers and critical vendors:* a short case, committed financing and payment of critical vendors on normal terms.

Triggers to change course. (a) Cash below $8.0 million before the DIP is committed: file immediately. (b) A DIP commitment is not signed by 30 November: move to a sale case with Brannoch’s bid. (c) Customer terminations that cut lender-case revenue by more than 5 percent: reduce the plan value to the low case and prefer the sale floor. (d) The CLOs or a holdout group convert to a signed support agreement: accept Option A.

Next three weeks. Weeks 1 to 2 (to 4 December): term sheets exchanged with both ad hoc groups; DIP commitment letters signed; plan and disclosure statement drafted; customer and union communications; critical vendor list agreed. Week 3: solicitation of votes; decision on the filing by 4 December; first-day motions ready. Risks: the lender case may be right (EV $175 million, plan fails the best-interests test and a sale takes over); the DIP is thin without ABL availability; customer losses could exceed our 3 percent haircut.

Sensitivity Discussion

Enterprise value. Each $10 million of enterprise value moves Term Loan recovery by 5.4 points ($10.0 million x 0.97 / $181.0 million) in a prepack, because senior claims are fixed at $90.0 million and the Term Loan is the fulcrum. A value below $188.6 million fails the best-interests test. The 3 percent filing haircut costs the Term Loan about $6.2 million of value, or 3.4 points.

The probability of an out-of-court deal. Option A breaks even with the prepack at 36.1 percent. If the probability is 20 percent the exchange yields 54.9 percent; at 50 percent it yields 61.4 percent. A 10-point change in probability moves recovery by 2.2 points; but the failure path matters as much: if failing costs only 2 percent of EV rather than 5 percent, the failure recovery rises to 53.7 percent and the break-even probability falls to about 25 percent.

Roll-up and DIP. The roll-up’s transfer between lenders ($6.1 million at the base case) depends on who participates: with 100 percent participation it is zero. The DIP’s effective cost of about 31 percent is almost irrelevant to the decision compared with a 3 percent change in EV ($6.2 million).

Pension. Whether the deficit is treated as debt-like changes recoveries materially: deducting the full $45.0 million accounting deficit from the base-case equity would cut the equity value to $18.8 million and the Term Loan recovery from 58.4 percent to about 35 percent. This is a weakness in the treatment above, which treats contributions as an operating cost; a court-appointed valuation expert could choose the other convention. Students should flag this explicitly.

Common Student Errors

1. **Using the market price of the debt as the enterprise value** instead of comparing it to a net recovery after priming claims (Question 3).
2. **Ignoring that the ABL is blocked by the default.** Students count the $24.0 million of availability as liquidity and conclude there is no crisis (Question 2).
3. **Running a waterfall without process costs.** The simple waterfall puts the Notes in the money at $245 million; the realistic one does not (Question 4).
4. **Treating accrued interest as optional.** The claims at filing are $181.0 million and $125.2 million, not $178.5 million and $120.0 million.
5. **Forgetting unanimity for sacred rights.** Concluding that "50.1 percent of lenders can agree" a principal exchange (Question 5).
6. **Assuming a roll-up changes the class's total recovery** instead of how it is distributed (Question 6).
7. **Mixing the failure path of Option A with the success path** when computing expected values, or comparing percentages of different claim bases.
8. **Treating the pension deficit as a priority claim or as irrelevant,** rather than choosing a convention and defending it (Question 8).
9. **Recommending liquidation because book equity is negative.** Book equity is irrelevant to the reorganisation versus liquidation decision; going-concern value ($205 million) exceeds liquidation value ($140.9 million).
10. **Giving a point estimate of value** without a range.

Alternative Defensible Answers

- **Pure Option C (sale).** Some groups will prefer the sale for its certainty: the floor recovery of 53.9 percent, narrower spread and no plan-confirmation risk. This is defensible if the board is risk-averse about customer reaction or believes the lender case is the better forecast. It is less attractive because it leaves behind the pension and the unsecured claims and is likely to face union and customer concerns about the buyer's intentions.
- **Option A with a prepack fallback only.** A group may argue for starting with Option A and filing only if it fails. That is defensible if the cash runway is extended, for example if the ABL lender funds under a standstill, but the 13-week forecast leaves little time, and failure costs about 8 points.
- **A higher or lower base value.** A team that uses the management case ($246.8 million) will conclude that the Notes should receive more than a 5 percent gift and that the sponsor deserves more; a team using the lender case at a higher WACC will conclude that the Term Loan is impaired even in the sale. Both can be defended; what cannot be defended is not showing the sensitivity.
- **Pension as debt.** See the Sensitivity Discussion.
- **A different DIP.** Some will propose that the ABL lender provide the entire DIP in a traditional creeping roll-up, which reduces the dispute among Term Loan lenders but gives the ABL a greater role in the case.

Chapter 16: Case: Zephyr Components Buys into a Fictional Emerging Market

Type: Case. Zephyr Components, Inc., the Republic of Marenta, Tavira Drive Components, Orvane Group, the Marentan dinar and every other entity, person, regulator and market statistic in this chapter are fictional and illustrative. All numbers are invented for teaching. Legal and tax rules described are simplified, are not those of any real country, and must be confirmed with local advisers in practice. USD means US dollars; MRD means Marentan dinars. Amounts are millions unless stated.

The Situation

On 2 December 2026, Lukas Brandt, chief financial officer of Zephyr Components, Inc. ("Zephyr"), received an email from the head of corporate development, Priya Nandakumar, with a subject line he had learned to dread: "Tavira: seller wants an answer, and the number has moved." Zephyr is a US-based manufacturer of brake, transmission and driveline components for commercial vehicles, with revenue of $2.1 billion, EBITDA margin of 14 percent and investment-grade credit. For two years it had searched for a way into the fast-growing commercial-vehicle supply chain of the Republic of Marenta, a mid-sized emerging economy where truck and bus fleets are expanding and where Zephyr's imported products face a 17 percent tariff. Corporate development had now found a target.

Tavira Drive Components S.A. ("Tavira") is the leading local maker of drive shafts, clutches and axle components for Marenta's truck and bus assemblers. It is owned 100 percent by Orvane Group, a family-controlled conglomerate led by Teodor Kasrin, whose interests range from cement to retail banking. Tavira's revenue for 2026 is forecast at MRD 6,000 million (about $300 million) and EBITDA at MRD 960 million (a 16 percent margin). It has three plants near the port city of Alvara, 2,400 employees, long relationships with three assemblers and an aftermarket distribution network. It also has net debt of MRD 900 million.

Orvane has granted Zephyr exclusivity until **18 December 2026**. Its opening position is an enterprise value of **MRD 6,720 million, or 7.0 times 2026E EBITDA**, payable in cash, for 100 percent of Tavira. At the current exchange rate of MRD 20.00 per dollar, that is $336 million. Mr Kasrin has said he is also willing to discuss a joint venture in which Zephyr acquires 49 percent, but has not said at what price. The Zephyr board meets on **10 December** to decide what to offer, if anything.

The question is deceptively simple: what is Tavira worth to Zephyr, in dollars, and how should Zephyr be structured to hold it? Answering it requires solving several problems at once.

- **Country risk.** Marenta is rated B+ (illustrative). Its dollar sovereign bonds yield 4.6 percentage points above US Treasuries. The central bank has devalued the dinar sharply in the past. A good deal in dinars can be a poor deal in dollars.
- **Currency and inflation.** Marenta's inflation is 8.0 percent against 2.5 percent in the United States. Tavira's dinar cash flows grow nominally by 8 to 10 percent a year, but a dollar-based investor receives a currency that is steadily weakening. Forecasting in dinars and discounting at a dollar rate (or the reverse) is the most common error in cross-border valuation.
- **Repatriation.** Marenta taxes dividends at 12 percent, caps annual remittances at 60 percent of net income without special central-bank approval, and has an unpredictable queue for converting dinars to dollars. Free cash flow in Marenta is not the same as cash in Zephyr's hands.
- **Structure.** Should Zephyr buy 100 percent and take control, with the synergies that control brings, but with all of the country exposure? Or should it enter a 49 percent joint venture with a well-connected local partner, accepting less control in exchange for lower risk and an option to increase its stake?

- **Financing.** Should the acquisition debt be raised in dollars at 6.8 percent or in dinars at 14.0 percent?

The people involved: Imani Okafor-Reyes, Zephyr's chief executive, wants a clear recommendation and has told the board that Zephyr will not "buy a currency bet dressed up as a factory." Lukas Brandt owns the valuation. Priya Nandakumar wants the deal done; she has spent eighteen months on it. Tobias Mwangi, Zephyr's treasurer, will have to hedge whatever structure is chosen. Zephyr's general counsel, Ingrid Salazar, is worried about the central-bank approval process and about what happens if Marenta's government, which has nationalised assets twice in the past thirty years, takes an interest in a foreign-owned supplier to its military trucks.

Your task is to produce the valuation and structuring analysis that the CFO will present to the board on 10 December, and to recommend a price, a structure and a set of protective conditions.

Industry Context

The commercial-vehicle supply chain in emerging markets. Truck, bus and trailer fleets in emerging markets grow roughly in line with GDP and infrastructure spending, with a multiplier for fleet renewal as emission and safety standards tighten. Original equipment makers prefer local suppliers for heavy drivetrain components because of weight, logistics and tariff considerations; imported components face tariffs and long lead times. Local content rules, where they exist, create a protected position for established local suppliers such as Tavira.

Margin structure. Local suppliers earn mid-teen EBITDA margins, helped by lower labour costs but hurt by imported steel and bearings priced in dollars. Tavira imports about 30 percent of its input costs, which makes its margin sensitive to the exchange rate: when the dinar weakens, imported inputs cost more in dinars, and price increases to assemblers lag by two to three quarters.

Valuation conventions. Emerging-market industrial companies trade at lower multiples than their developed-market peers, a reflection of country risk, liquidity and governance discounts. Control transactions typically carry a premium of 20 to 30 percent to trading multiples. Local sellers often anchor on developed-market multiples or on "replacement cost", and buyers have to explain why a 5.5 times local multiple is not 9 times.

Cross-border deal practice. Buyers typically use some combination of: dollar discount rates with country risk premiums; scenario analysis for devaluation, capital controls and expropriation; political risk insurance; staged investment through joint ventures with call and put options; and royalty, management-fee and shareholder-loan structures to move cash out more efficiently than dividends alone. Each has costs and limits, all of which appear in this case.

Marenta in brief. Table 1 summarises the economy. Marenta has had three significant devaluations in the past twenty years and a currency that lost 38 percent against the dollar between 2019 and 2024. Its central bank holds foreign reserves covering 3.4 months of imports. The government is generally pro-investment but has used capital controls (a freeze on dividend remittances for 18 months) and renegotiated contracts with foreign investors in other sectors.

Financial Data

Table 1: Marenta macroeconomic and market data (illustrative, as of 30 November 2026)

Indicator	Value
Real GDP growth (forecast average)	2.5% to 3.0%
Inflation, latest 12 months (forecast 2027: 8.0%)	7.6%
Central bank policy rate	11.0%
Ten-year government bond yield (MRD)	12.0%
Sovereign rating	B+
Spread of Marenta's USD sovereign bond over US Treasury (5-year)	4.6%
US ten-year Treasury yield	4.3%
US inflation (long-run forecast)	2.5%
Spot exchange rate	MRD 20.00 per USD
One-year forward rate	MRD 21.48 per USD
Current account balance	(4.8%) of GDP
Foreign reserves	3.4 months of imports
External debt	62% of GDP
Corporate income tax rate	25%
Withholding tax on dividends to foreign shareholders	12%
Withholding tax on royalties and management fees	10%
Ratio of Marenta equity-market volatility to Marenta USD bond volatility	1.4

Table 2: Tavira income statement extract

MRD millions	2024A	2025A	2026E
Revenue	5,200	5,600	6,000
EBITDA	780	868	960
EBITDA margin	15.0%	15.5%	16.0%
Depreciation and amortisation	208	224	240
EBIT	572	644	720
Interest expense	118	138	154
Pre-tax income	454	506	566
Income tax (25%)	113.5	126.5	141.5
Net income	340.5	379.5	424.5
Average exchange rate (MRD per USD)	16.5	18.4	19.6
Revenue in USD millions	315.2	304.3	306.1
EBITDA in USD millions	47.3	47.2	49.0

In dinars Tavira has grown revenue by 15.4 percent in two years. In dollars its revenue has been flat. This difference is the central fact of the case.

Table 3: Tavira balance sheet extract (31 December 2026E, MRD millions)

Assets		Liabilities and equity	
Cash	200	Accounts payable	900
Accounts receivable	1,150	Accrued liabilities	260
Inventory	1,000	Short-term debt (14.0%)	300
Other current assets	110	Long-term debt (14.0%)	800
Property, plant and equipment	2,300	Other long-term liabilities	140
Intangibles	150	**Total liabilities**	**2,400**
Other non-current assets	90	**Shareholders' equity**	**2,600**
Total assets	**5,000**	**Total liabilities and equity**	**5,000**

Net debt is MRD 900 million (debt of 1,100 less cash of 200). Operating net working capital (receivables, inventory and other current assets less payables and accruals) is MRD 1,100 million, 18.3 percent of revenue; the forecast assumes incremental working capital of 16 percent of the change in revenue because of an inventory programme.

Table 4: Tavira forecast, 2027E-2033E (MRD millions; management case moderated by Zephyr)

MRD millions	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E
Inflation (MRD)	n/a	8.0%	7.0%	6.0%	5.5%	5.0%	5.0%	5.0%
Real growth	n/a	3.0%	3.0%	2.5%	2.5%	2.0%	2.0%	2.0%
Revenue	6,000.0	6,674.4	7,355.9	7,992.1	8,642.5	9,256.1	9,913.3	10,617.1
EBITDA margin	16.0%	16.0%	16.3%	16.6%	17.0%	17.3%	17.5%	17.5%
EBITDA	960.0	1,067.9	1,199.0	1,326.7	1,469.2	1,601.3	1,734.8	1,858.0
D&A (4.0% of revenue)	240.0	267.0	294.2	319.7	345.7	370.2	396.5	424.7
EBIT	720.0	800.9	904.8	1,007.0	1,123.5	1,231.1	1,338.3	1,433.3
Tax at 25%		200.2	226.2	251.8	280.9	307.8	334.6	358.3
Capex (5.5% of revenue)		367.1	404.6	439.6	475.3	509.1	545.2	583.9
Increase in NWC (16% of revenue change)		107.9	109.0	101.8	104.1	98.2	105.1	112.6
Free cash flow to firm		392.7	459.2	533.6	608.9	686.3	749.9	803.1

Notes to Table 4. Revenue growth is (1 + real growth) x (1 + inflation) - 1, so 2027E nominal growth is 11.2 percent (3.0 percent real). The tax rate is 25 percent of EBIT (the target is valued debt-free). Cash flows are shown at year-end for simplicity. Long-run assumptions beyond 2033E: inflation 5.0 percent, real growth 2.0 percent, so nominal terminal growth of 7.1 percent in dinars; US inflation 2.5 percent. Zephyr's own estimate of dollar-equivalent terminal growth is therefore 1.02 x 1.025 - 1 = 4.55 percent. Management's original forecast had EBITDA margins reaching 19 percent; Zephyr has capped them at 17.5 percent.

Table 5: Cost of capital inputs

Input	Value	Source or comment
US risk-free rate	4.3%	Ten-year Treasury
Mature-market equity risk premium	5.0%	Zephyr house view
Beta (unlevered peer drivetrain suppliers, relevered at target structure)	1.10	Peer regression
Zephyr home-market WACC	8.3%	Ke 9.8%, Kd 6.5% pre-tax, D/V 30%, tax 25%
Marenta sovereign USD spread	4.6%	Table 1
Equity-to-bond volatility ratio	1.4	Table 1
Rating-based country premium (alternative)	5.2%	Third-party table, illustrative
Exposure of Tavira's cash flows to Marenta (lambda)	1.0 for revenue; 0.7 if only domestic-demand share counted	Debated
Target capital structure	D/V 30%	Zephyr policy for acquisitions
Pre-tax cost of USD debt for a Marenta business	10.5%	Risk-free + sovereign spread + 1.6% corporate spread

Table 6: Comparable companies and transactions (illustrative)

Company	Market	EV / NTM EBITDA	EBITDA margin
Halvard Drivetrain	Marenta (listed)	5.1x	15.8%
Bellaro Brakes	Marenta (listed)	5.6x	14.2%
Sundar Auto Components	Neighbouring Tessaly	5.9x	16.5%
Nordal Castings	Neighbouring Tessaly	4.6x	12.9%
Perun Fasteners	Regional peer	6.4x	17.0%
Median emerging-market peers		**5.6x**	**15.8%**
Developed-market commercial-vehicle suppliers (median of six)		8.9x	14.8%
Precedent control transactions, emerging-market auto parts (median of four)		6.9x	

The seller's ask of 7.0 times is almost exactly the precedent control multiple. A buyer should still ask whether those precedents were struck in dinars or dollars, in which year, and with which buyer's discount rate.

Table 7: Repatriation rules and transaction costs

Mechanism	Rule
Dividends	12% withholding tax; remittances above 60% of current-year net income require central-bank approval, which is granted discretionarily and slowly
Conversion lag	Dinars paid as dividends convert into dollars about six months later, at the then-prevailing rate
Retained (trapped) cash	May be held in dinar deposits at 8.0 percent pre-tax; released in later years
Royalties	3% of net sales is the maximum rate the tax authority accepts as arm’s length; deductible locally; 10% withholding; no remittance cap; paid quarterly
Shareholder loans	Interest deductible; 10% withholding; principal repayment subject to central-bank registration
Sale of shares	No withholding tax; central-bank registration of the original investment guarantees conversion of sale proceeds
Transaction costs	2% of purchase price

Table 8: Country stress events (Zephyr’s risk team, illustrative)

Event	Share of stress events	Loss of value of the investment if the event occurs
Currency crisis (sharp devaluation; partly offset by pass-through)	60%	35%
Capital controls (remittance freeze of several years, with erosion of trapped cash)	25%	50%
Expropriation or forced sale at a discount	15%	100%

Zephyr’s risk team converts the sovereign spread into an annual probability that some stress event occurs by dividing the spread by an assumed loss given default on sovereign debt of 60 percent. A joint-venture structure with a well-connected local partner is judged to reduce losses from capital controls (to 30 percent, because the partner can net dividends against its own local-currency needs) and from expropriation (to 50 percent, because a partner with political relationships and a share of the equity is a less attractive target), but not those from currency crisis.

Table 9: The two structures on offer

	Structure 1: acquire 100%	Structure 2: 49% joint venture with Orvane
Price	EV MRD 6,720m (7.0x); equity MRD 5,820m; net debt MRD 900m	Opening: 6.6x EV for 49% (about MRD 3,105m, $155.2m); Zephyr to negotiate
Control	Full; consolidation; integration synergies	Joint control; board seats; vetoes on budget, dividends and related-party deals; Orvane keeps management
Synergies	Procurement and cross-selling: 0.6% of revenue in 2028, 1.2% from 2029 (pre-tax); one-off integration cost of MRD 150m in 2027	Zephyr assumed to capture half of its 49 percent share of the synergies
Call option	Not applicable	Zephyr may buy a further 26% at the end of 2029 at 7.0x trailing 2029 EBITDA (about MRD 2,415m, $106.1m at the expected 2029 rate)
Risk profile	Zephyr bears 100% of country risk	Country losses reduced as in Table 8; Zephyr exposed on 49% only
Local partner role	None	Licences, relationships with assemblers and officials, local-currency liquidity

Table 10: Financing quotes for $120 million of acquisition funding

Source	Rate	Comment
USD loan from Zephyr (guaranteed, placed at Tavira)	6.8% pre-tax	Principal repayable in USD in 2033
MRD loan from a Marentan bank	14.0% pre-tax	MRD 2,400m at spot; bullet in 2033
One-year forward hedge on MRD	MRD 21.48 per USD	Forward premium implied by the rate differential; hedging beyond two years unavailable

Complications and Exhibits

Exhibit A: Related-party sales. About 14 percent of Tavira's revenue comes from Alvara Coach, a bus assembler owned by Orvane Group, on 120-day payment terms and at prices that Zephyr's due diligence team believes are about 4 percent above what an arm's-length customer would pay. If Orvane remains an owner (in the joint venture) the arrangement is likely to continue; if Zephyr buys 100 percent, Alvara Coach will be free to renegotiate, and Orvane has hinted it could shift volume to a competitor. The forecast in Table 4 assumes the current terms continue. Reducing the margin on those sales to arm's-length levels would cut group EBITDA margin by about 0.6 percentage points.

Exhibit B: The military truck contract. Nine percent of revenue comes from a government contract for axle assemblies used in military trucks. The contract contains a clause allowing the ministry to terminate or re-tender if a foreign entity acquires "control" of a supplier. In the joint-venture structure, Orvane retains management control and the clause is arguably not triggered; in the 100 percent structure, it almost certainly is. The ministry has not indicated how it would exercise its right. This single contract also accounts for the general counsel's worry about nationalisation.

Exhibit C: Central-bank registration. Marenta's central bank registers foreign investments, and registered investments are guaranteed conversion of sale proceeds (Table 7). Registration takes about 90 days and requires evidence that the purchase price was converted through the official market. Dividend remittances above the 60 percent cap require discretionary approval, and in 2023 a different ministry froze remittances for 18 months.

Exhibit D: Competing exchange-rate views. Zephyr's economics team assumes purchasing-power parity: the dinar loses value at the inflation differential, which is 5.4 percent in 2027, falling to 2.4 percent a year. A broker's research note argues the dinar is overvalued, that reserves are falling, and that a 15 percent devaluation is likely in 2027. The forward rate implies a 7.4 percent annual depreciation, which is more than purchasing-power parity but less than the broker's view. The forward rate is a hedge price; it includes the interest-rate differential, which includes a risk premium.

Exhibit E: Political risk insurance quote. A multilateral-backed insurer has quoted cover against expropriation, political violence and currency inconvertibility on up to $300 million of investment for seven years, covering 90 percent of insured losses, for an annual premium of 1.0 percent of the insured amount. Currency devaluation is explicitly excluded, and capital controls are covered only if the central bank refuses conversion of registered dividends for more than 180 days.

Exhibit F: Tavira's management and employees. The chief executive of Tavira is Mr Kasrin's nephew and will leave if Zephyr buys 100 percent. Two plant managers are the real source of the customer relationships. The collective agreement covers 1,300 workers and requires a 24-month job guarantee on a change of control.

Exhibit G: Treaty status. A tax treaty between the United States and Marenta that would reduce the dividend withholding tax to 5 percent has been signed but not ratified. The valuation should use 12 percent, with 5 percent shown as an upside.

Exhibit H: Seller's financing offer. Orvane has offered, as an alternative to a cash price, to accept $200 million of the price in cash at closing and the remainder as a dollar-denominated vendor note, 5-year, at 6.0 percent, secured by Tavira's shares, which would reduce Zephyr's upfront cash but not the enterprise price. The case does not model this offer; students may comment on it.

Ambiguities students must handle.

- Whether the country risk premium should apply to the whole firm, to the equity only, or to the share of cash flows exposed to Marenta (the lambda debate).
- Whether the terminal value should use dinar inflation of 5 percent forever, even though the long-run exchange rate may depreciate by 2.4 percent a year.
- Whether to treat repatriation leakage as a cash-flow adjustment or a discount-rate adjustment.
- Whether synergies deserve the same country risk premium as Tavira's standalone cash flows.

Assignment Questions

Show every calculation. State assumptions. Unless told otherwise, use Tables 1 to 10 and Exhibits A to H. All values are in millions.

1. **Country risk premium and WACC.** Compute the country risk premium by (a) the sovereign spread alone, (b) the spread scaled by the ratio of equity to bond volatility, (c) a lambda-weighted premium (lambda 0.7), and (d) the rating-based figure. Compute the cost of equity, the after-tax cost of debt and the USD WACC under method (b). Compute Tavira's enterprise value under each method and discuss the range.
2. **Inflation, exchange rates and hedging.** Forecast the dinar-per-dollar rate from 2027 to 2033 under purchasing-power parity using the inflation path in Table 4. Compute the exchange rate implied by covered interest parity (ten-year MRD yield 12.0 percent, US 4.3 percent) for the same horizon. Compute the annual hedging cost relative to expected depreciation in 2027, and the effect on enterprise value of using the forward path rather than purchasing-power parity.

3. **Valuation in local currency.** Discount the dinar free cash flows at a time-varying dinar WACC consistent with the USD WACC from Question 1(b) and the inflation path. Compute the terminal value and the enterprise value in dinars and in dollars at the spot rate.
4. **Valuation in hard currency, and reconciliation.** Convert the same cash flows to dollars at the purchasing-power-parity rates and discount at the USD WACC. Show that the answer matches Question 3. Then compute the enterprise value under three common errors: (i) dinar cash flows discounted at the USD WACC and converted at spot; (ii) a constant long-run dinar WACC applied to all years; (iii) converted at a flat spot rate, with dollar terminal growth of 4.55 percent. Express each error as a percentage of the correct value.
5. **Sensitivity.** Prepare a table of enterprise value for USD WACC between 11.73 percent and 15.73 percent and long-run real growth between 1 percent and 3 percent; test margin changes of minus 3 and plus 1.5 percentage points; compute the share of value in the terminal value; and comment on which assumption matters most.
6. **Repatriation.** Compute, year by year for 2027 to 2033, dividends allowed under the 60 percent cap, the trapped cash balance, and the net dollars received after withholding tax and a six-month conversion lag. Decompose the leakage into withholding tax, lag and the cap. Then evaluate a 3 percent royalty structure and compute the value gain.
7. **Country risk as expected loss.** Convert the sovereign spread into an annual stress probability, compute the expected annual loss using Table 8, and compute the equivalent discount rate. Compare the resulting enterprise value to the country-risk-premium approach. Compute the value of the investment in each stress event conditional on a first event in 2029. Evaluate the political-risk-insurance quote and the joint-venture structure using the same machinery.
8. **Acquisition or joint venture.** Compute the value to Zephyr of the 100 percent acquisition (after repatriation leakage under the royalty structure and including synergies) and its net present value at the ask. Compute the break-even price. Compute the value of the 49 percent joint venture at the JV discount rate, the value of the call option on a further 26 percent using Black-Scholes (volatility 35 percent, risk-free rate 4.3 percent), and the NPV at 6.6 times. Compute the break-even multiple.
9. **Financing currency.** Compute the expected dollar cost of the dinar loan under purchasing-power parity and under the forward curve, compare it to the USD loan, and test debt service coverage in 2028 under a further 35 percent devaluation. Recommend a funding mix.
10. **Recommendation.** Write a board-style memorandum recommending a structure, a price range, protective conditions, and triggers to walk away.

Teaching Note and Full Solutions

Learning Objectives

After this case, students should be able to:

1. Estimate a country risk premium by several methods and build a hard-currency WACC for an emerging-market target.
2. Forecast exchange rates from inflation differentials (purchasing-power parity) and distinguish them from forward rates (covered interest parity).
3. Value a foreign business in local currency and in hard currency, and show the two are equivalent when inflation, exchange rates and discount rates are consistent.
4. Recognise and quantify common cross-border valuation errors.

5. Quantify repatriation leakage from withholding tax, conversion delay and remittance caps, and evaluate structures that reduce it.
6. Treat country risk as an expected loss, reconcile that with the premium approach, and evaluate political risk insurance.
7. Compare an acquisition with a joint venture including a call option, and compute break-even prices.
8. Choose the currency of acquisition financing and write a decision-ready recommendation.

Suggested Class Plan (90 minutes)

Minutes	Segment	Questions	Teaching purpose
0-10	Open with Table 2: “Revenue up 15 percent, or flat?”	2	Dinar growth versus dollar growth
10-25	Country risk premium and WACC; the range of answers	1	Methods and judgement
25-45	Local versus hard-currency valuation; the three errors	3, 4	Consistency principle
45-55	Sensitivity: what matters most?	5	Value drivers
55-65	Repatriation: free cash flow is not cash in the parent’s hands	6	Leakage and structure
65-75	Country risk as expected loss; insurance	7	Reconciling approaches
75-90	Acquisition versus JV, financing, board vote	8, 9, 10	Structure and decision

Solutions

Results were computed with Python and are rounded as shown. Enterprise values are in USD millions unless stated.

Question 1: Country risk premium and WACC

Step 1: premiums.

```
(a) Spread only:                    4.60%
(b) Spread x vol ratio:             4.60% x 1.4 = 6.44%
(c) Lambda-weighted (0.7 x (b)):    0.7 x 6.44% = 4.51%
(d) Rating-based:                   5.20%
```

Step 2: cost of equity, cost of debt and WACC under (b). Cost of equity = 4.3% + 1.10 x 5.0% + 6.44% = 4.3 + 5.5 + 6.44 = **16.24%**. After-tax cost of debt = 10.5% x (1 - 0.25) = **7.875%**. With D/V of 30 percent:

```
WACC = 0.70 x 16.24% + 0.30 x 7.875% = 11.368% + 2.3625% = 13.73%
```

Step 3: all methods.

Method	Premium	Cost of equity	WACC	Enterprise value ($m)	EV / 2026E EBITDA
(a) Spread only	4.60%	14.40%	12.44%	296.9	6.19x
(b) Spread x volatility ratio	6.44%	16.24%	13.73%	253.8	5.29x
(c) Lambda 0.7	4.51%	14.31%	12.38%	299.4	6.24x
(d) Rating-based	5.20%	15.00%	12.86%	281.3	5.86x
No premium (Zephyr home WACC 8.32%)	0.00%	9.80%	8.32%	633.9	13.2x

Interpretation. The country premium is the single most powerful assumption in the case: removing it more than doubles the value. The four methods give $253.8 million to $299.4 million, or 5.3 to 6.2 times EBITDA. All are below the seller's ask of $336 million (7.0 times). Method (b) is the most conservative; (c) is justified only if part of Tavira's exposure is not Marenta-specific, which is weak for a business that earns all its revenue locally (and the debt rate already includes the sovereign spread). The base case in the rest of the chapter is (b) at 13.73 percent, with 12.4 percent as the bottom of a defensible range.

Question 2: Inflation, exchange rates and hedging

Step 1: purchasing-power parity. S(t) = S(t-1) x (1 + inflation in Marenta) / (1 + US inflation of 2.5%). For 2027: 20.00 x 1.080 / 1.025 = 21.07. Continuing with the inflation path of Table 4:

Step 2: covered interest parity. Forward rate F(t) = S0 x [(1 + 12.0%) / (1 + 4.3%)]^t. For one year: 20.00 x 1.12 / 1.043 = 21.48 (a 7.38 percent depreciation).

Year	PPP rate	Covered-interest-parity rate	CIP vs PPP	MRD loss per year under PPP
2027	21.07	21.48	+1.9%	5.37%
2028	22.00	23.06	+4.8%	4.39%
2029	22.75	24.76	+8.9%	3.41%
2030	23.42	26.59	+13.6%	2.93%
2031	23.99	28.56	+19.1%	2.44%
2032	24.57	30.66	+24.8%	2.44%
2033	25.17	32.93	+30.8%	2.44%

Step 3: hedging cost and effect on value. In 2027, purchasing-power parity implies depreciation of 5.37 percent (21.07 / 20.00 - 1), while the forward implies 7.38 percent. Selling dinars forward at 21.48 rather than expecting 21.07 costs 21.48 / 21.07 - 1 = **1.9 percent** of the hedged amount in the first year, or 2.0 percentage points of carry (7.38 - 5.37). By 2033 the cumulative gap is 30.8 percent. Re-running the valuation with cash flows converted along the covered-interest-parity path gives an enterprise value of **$206.5 million** against $253.8 million, an 18.6 percent difference. The lesson is that a forward rate is a price for hedging, not a forecast; using it to value cash flows that are built on inflation assumptions mixes a risk-premium-laden interest differential with an inflation-based cash-flow forecast. A buyer should model exchange rates consistently with the inflation in the forecast and treat the difference as the cost of hedging (or of the risk premium Zephyr is being paid to bear).

Question 3: Valuation in local currency

Step 1: dinar WACC. The dinar discount rate consistent with a USD WACC of 13.73 percent is (1 + 13.73%) x (1 + inflation) / (1 + 2.5%) - 1, which varies by year with Marenta inflation: 2027: 1.1373 x 1.080 / 1.025 - 1 = 19.83 percent; 2031 and later: 1.1373 x 1.050 / 1.025 - 1 = 16.50 percent.

Step 2: present values. Cumulative discount factor = product of 1 / (1 + dinar WACC) for each year to date.

Year	FCFF (MRD)	Inflation	Local WACC	Cumulative discount factor	PV (MRD)
2027	392.7	8.0%	19.83%	0.8345	327.7
2028	459.2	7.0%	18.72%	0.7029	322.8
2029	533.6	6.0%	17.61%	0.5976	318.9
2030	608.9	5.5%	17.06%	0.5105	310.9
2031	686.3	5.0%	16.50%	0.4382	300.7
2032	749.9	5.0%	16.50%	0.3761	282.0
2033	803.1	5.0%	16.50%	0.3228	259.3
Sum of explicit PV					2,122.3

Step 3: terminal value. Extend 2033E by long-run nominal growth of (1.02 x 1.05) - 1 = 7.1 percent:

```
Revenue 2034  = 10,617.1 x 1.071 = 11,370.9
EBITDA (17.5%) = 1,989.9 ; D&A (4.0%) = 454.8 ; EBIT = 1,535.1
Tax (25%) = 383.8 ; Capex (5.5%) = 625.4 ; Increase in NWC = 0.16 x (11,370.9 - 10,617.1) = 120.6
FCFF 2034 = 1,989.9 - 383.8 - 625.4 - 120.6 = 860.1
TV (end-2033) = 860.1 / (16.50% - 7.10%) = 9,146.1
PV of TV = 9,146.1 x 0.3228 = 2,952.8
```

Step 4: enterprise value. EV = 2,122.3 + 2,952.8 = **MRD 5,075.0 million**. At spot of MRD 20.00 per dollar this is **$253.75 million**, or 5.29 times 2026E EBITDA. The terminal value is 58.2 percent of the total.

Question 4: Valuation in hard currency and reconciliation

Step 1: convert flows at the PPP rates (Question 2) and discount at 13.73 percent.

Year	PPP rate (MRD per USD)	FCFF (USD m)	Discount factor at 13.73%	PV (USD m)
2027	21.07	18.63	0.8793	16.38
2028	22.00	20.87	0.7731	16.14
2029	22.75	23.45	0.6798	15.94
2030	23.42	26.01	0.5977	15.54
2031	23.99	28.61	0.5256	15.04
2032	24.57	30.52	0.4621	14.10
2033	25.17	31.91	0.4063	12.96

The sum of the explicit PVs is $106.11 million. The terminal value in dollars is 9,146.1 / 25.17 = $363.36 million, discounted by 0.4063 to **$147.64 million**. Enterprise value = 106.11 + 147.64 = **$253.75 million**, identical to the local-currency result (5,075.0 / 20.00 = 253.75). The two methods are the same computation arranged differently: dividing each year’s dinar cash flow by the PPP rate and by (1.1373)^t is the same as discounting at the dinar WACC, because S(t) / S0 = product of (1 + inflation) / 1.025.

Step 2: three common errors.

```
(i)   Dinar FCFF discounted at USD WACC 13.73% (TV at 13.73% - 7.10%), divided by spot 20.00:   $386.6m
(+52.3%)
(ii)  Dinar FCFF discounted at a constant long-run dinar WACC of 16.50% in all years, / 20.00:   $268.9m
(+6.0%)
(iii) Dinar FCFF converted at a flat spot rate of 20.00, discounted at 13.73%, USD terminal
      growth 4.55%:                                                                              $313.4m
(+23.5%)
```

Error (i) mixes a nominal-dinar cash flow with a dollar discount rate and ignores depreciation: it overstates value by half and would turn an apparent loss at the ask into a gain ($386.6 million against a $336 million price). Error (iii) ignores depreciation of the currency in which the cash flows arrive. Error (ii) is subtle: a single long-run local rate understates the discount rate in the early years, when inflation is higher, and overstates value by 6 percent.

Question 5: Sensitivity

Step 1: WACC and growth (enterprise value, $ million).

USD WACC long-run real growth	1.0%	2.0%	3.0%
11.73%	302.5	327.4	360.7
12.73%	267.9	286.1	309.4
13.73%	240.2	**253.8**	270.7
14.73%	217.5	227.8	240.5
15.73%	198.5	206.6	216.3

Step 2: margin and exchange-rate shocks. A change in all EBITDA margins of minus 3.0 percentage points gives $218.7 million (-13.8 percent); minus 1.5 points gives $236.2 million (-6.9 percent); plus 1.5 points gives $271.3 million (+6.9 percent). A one-time devaluation of 35 percent in 2028 with 40 percent pass-through into dinar prices leaves 0.65 x (1 + 0.4 x 0.35 / 0.65) = 0.79 of the dollar value of subsequent flows, so the value is 0.79 x (253.8 - 16.4) + 16.4 = **$203.9 million (-19.6 percent)**, before any margin effect.

Step 3: ranking. A one-point rise in the discount rate cuts value by 10.2 percent; a one-point fall raises it by 12.7 percent. A one-point fall in real terminal growth cuts it by 5.3 percent. The exchange-rate path matters at least as much as any operating assumption (Question 2: -18.6 percent on the forward path). The terminal value is 58.2 percent of the total. The country-risk premium (embedded in the discount rate) and the currency path are the key variables, not the operating forecast.

Question 6: Repatriation

Step 1: dividends under the cap. Allowed dividend = 60 percent of net income (taken as NOPAT, because the target is valued debt-free). Trapped cash earns 8.0 percent in dinars. Net dollars = dividend x (1 - 12%) / (PPP rate six months after year-end); the conversion lag delays receipt by half a year, so the discounting is for t + 0.5 years at 13.73 percent.

Year	FCFF (MRD)	Net income (MRD)	Allowed dividend (60% of NI)	Dividend paid	Trapped balance, year-end	Net USD received	PV of net USD	PV if freely remitted
2027	392.7	600.7	360.4	360.4	32.3	14.7	12.1	16.4
2028	459.2	678.6	407.1	407.1	86.9	16.0	11.6	16.1
2029	533.6	755.3	453.2	453.2	174.3	17.3	11.0	15.9
2030	608.9	842.6	505.6	505.6	291.6	18.8	10.5	15.5
2031	686.3	923.3	554.0	554.0	447.2	20.1	9.9	15.0
2032	749.9	1003.7	602.2	602.2	630.6	21.3	9.2	14.1
2033	803.1	1075.0	645.0	645.0	839.2	22.3	8.5	13.0
Total							72.9	106.1

Free cash flow exceeds the allowed dividend in every year, so the cap binds throughout and the trapped balance grows each year to MRD 839.2 million (about $33.3 million at the 2033 rate) of cash is trapped. Assume it is released at the 2033 exit, with 12 percent withholding tax and a six-month lag: its present value is 839.2 x 0.88 / 25.17 x 0.3809 (the discount factor for 7.5 years), or **$11.2 million**. The terminal value ($147.6 million, assuming a share sale with no withholding tax and guaranteed conversion under Table 7) is unaffected.

Step 2: decomposition of the leakage.

```
Value if freely remitted (PV of 2027-2033 FCFF)                106.1
Less withholding tax: 12% x 106.1                               12.7
Less six-month conversion lag (WHT and PPP depreciation)         7.1
Less dividend cap net of PV of trapped cash (86.3 - 72.9 - 11.2)  2.2
Value of 2027-2033 distributions + trapped cash release         84.1  (72.9 + 11.2)
Leakage = 22.0 = 8.7% of the enterprise value of 253.8
Value to Zephyr under dividends-only structure = 253.8 - 22.0 = 231.7
```

Step 3: the royalty structure. A royalty of 3 percent of revenue (MRD 200.2 million in 2027) is deductible: it saves 25 percent of itself in dinar tax (MRD 50.0 million), so the net local cash cost is MRD 150.2 million; it is paid with 10 percent withholding and no cap. Per dinar of pre-tax profit, the dividend route delivers 0.75 x 0.88 = 0.66 after local tax and withholding, and the royalty route delivers 0.90 before US taxes (and about 0.79 after a US tax of 21 percent with credit for the 10 percent withholding; US taxes are outside the case). Re-running the model with a 3 percent royalty (net income reduced accordingly, same 60 percent cap on the remaining dividends):

```
PV of net dividends        54.5
PV of net royalties        40.4
PV of trapped cash release  3.0   (trapped balance falls to MRD 226.6m)
PV of terminal value      147.6
Value to Zephyr           245.5   (leakage 8.2 = 3.2% of EV)
Gain versus dividends-only 13.8  (+6.0% of enterprise value)
```

The royalty structure recovers about two-thirds of the leakage. It is limited by the arm’s-length ceiling of 3 percent and is exposed to the same transfer-pricing and political risks as dividends. A shareholder loan with deductible interest is a further channel. A 5 percent treaty rate (Exhibit G) would cut withholding leakage by roughly $7 million.

Question 7: Country risk as expected loss

Step 1: annual stress probability. p = spread / loss given default = 4.6% / 60% = **7.67 percent** a year.

Step 2: expected loss given an event (Table 8). L = 0.60 x 35% + 0.25 x 50% + 0.15 x 100% = 21.0 + 12.5 + 15.0 = **48.5 percent**. Expected annual loss of value = p x L = 7.67% x 48.5% = **3.72 percent**.

Step 3: equivalent discount rate. Start with the home-market WACC of 8.32 percent plus a systematic premium of 1.05 percent for bearing country risk that cannot be diversified away (a 1.5 percent addition to the cost of equity at 70 percent weight): 9.37 percent. Then the effective rate is (1.0937) / (1 - 0.0372) - 1 = **13.60 percent**. The enterprise value at 13.60 percent is **$257.7 million**, against $253.8 million from the country-risk-premium method, a difference of 1.5 percent. The two approaches reconcile: the 6.44 percent premium corresponds to an expected loss of about 3.8 percent a year (1 - 1.0937 / 1.1373) plus compensation for risk.

Step 4: what the expected value hides. The distribution of outcomes is wide. Conditional on a first stress event in 2029 (and no further events), the enterprise value under each event, computed at the rate of 9.37 percent (the value if nothing happens is $493.2 million, of which 2027 and 2028 flows contribute $34.5 million before the event and $458.7 million is at risk):

Outcome	Value ($ million)	Calculation
No stress event	493.2	No loss at 9.37%
Currency crisis in 2029	332.7	34.5 + (1 - 0.35) x 458.7
Capital controls in 2029	263.8	34.5 + (1 - 0.50) x 458.7
Expropriation in 2029	34.5	34.5 + 0

The probability of no stress event in seven years is (1 - 0.0767)^7 = 57.2 percent. A buyer paying $336 million is therefore betting that the country behaves (493.2) or only a currency crisis happens (332.7); in the other states it loses most of its investment.

Step 5: joint venture and insurance. In the joint venture, the loss given an event is 0.60 x 35% + 0.25 x 30% + 0.15 x 50% = 36.0 percent, so p x L = 2.76 percent and the effective rate is 1.0937 / 0.9724 - 1 = **12.48 percent**, which gives a whole-company enterprise value of **$295.6 million**, 14.7 percent above $257.7 million. Political risk insurance covering 90 percent of expropriation losses lowers L to 0.60 x 35% + 0.25 x 50% + 0.15 x 10% = 35.0 percent, an effective rate of 12.39 percent and an enterprise value of **$299.0 million**, a gain of $41.3 million. The premium of 1.0 percent on $300 million for seven years has a present value of 3.0 x [1 - 1.1373^-7] / 0.1373 = 3.0 x 4.324 = **$13.0 million**, so the net benefit is about $28 million. Insurance does not cover devaluation (the largest event by probability) and covers capital controls only after 180 days, so the real benefit is somewhat smaller, but at this price it is cheap protection against the tail.

Question 8: Acquisition or joint venture

Step 1: value of 100 percent.

```
Enterprise value (CRP method, 13.73%)                                  253.8
Less leakage under royalty structure (253.8 - 245.5)                    (8.2)
Value after repatriation                                                245.5
Plus synergies: PV 2027-2033 5.6 + terminal 17.6 = 23.2
  (after tax; cost savings 0.6% of revenue in 2028, 1.2% from 2029; MRD 150m one-off cost in 2027)
Value to Zephyr of 100%                                                 268.8
Price at ask: 7.0 x 960 / 20.00                                         336.0
Transaction costs 2%                                                      6.7
Total cost                                                              342.7
NPV at ask                                                              (74.0)
Break-even price (EV): 268.8 / 1.02 = 263.5 = 5.49x EBITDA
```

The seller's ask exceeds Zephyr's value by $74.0 million. Negotiating to 5.5 times (equal to the median emerging-market peer multiple, $264 million) breaks even; 5.0 times gives an NPV of plus $24.0 million.

Step 2: value of the 49 percent JV.

```
Whole-company EV at JV rate 12.48%                      295.6
49% share                                               144.8
Less leakage 3.3% (royalty structure)                   140.0
Plus 49% x 50% of synergies 23.2                          5.7
Value before option                                     145.7
Price: 6.6 x 960 x 49% / 20.00 = 155.2; with 2% costs   158.3
NPV before option                                       (12.6)
```

Step 3: the call option on 26 percent. The underlying is the present value of a 26 percent stake: V0 = 0.26 x 295.6 x (1 - 0.033) = **74.3**. The exercise price is 26 percent of 7.0 times 2029 EBITDA of MRD 1,326.7 million = MRD 2,414.6 million, or $106.1 million at the 2029 PPP rate of 22.75. Treating that dollar strike as fixed (a simplification, because the formula links the strike to EBITDA), with volatility 35 percent, risk-free rate 4.3 percent and T = 3 years:

```
d1 = [ln(74.3 / 106.1) + (0.043 + 0.5 x 0.35^2) x 3] / (0.35 x sqrt(3))
   = [-0.3566 + 0.3128] / 0.6062 = -0.0722
d2 = -0.0722 - 0.6062 = -0.6784
N(d1) = 0.4712 ; N(d2) = 0.2487
C = 74.3 x 0.4712 - 106.1 x e^(-0.043 x 3) x 0.2487 = 35.0 - 106.1 x 0.8790 x 0.2487 = 35.0 - 23.2 = 11.8
```

The option is worth about **$11.8 million**; at volatilities of 25 percent and 45 percent it is worth $6.8 million and $16.9 million.

Step 4: NPV and break-even. NPV of the JV including the option at 6.6 times = -12.6 + 11.8 = **-$0.8 million**. Break-even multiple = (145.7 + 11.8) / 1.02 / 0.49 x 20 / 960 = **6.57 times**. At 6.3 times the NPV is +$6.4 million; at 6.0 times, +$13.6 million; at 5.8 times, +$18.4 million.

Step 5: comparison. The 100 percent acquisition is worth less per dollar of price because Zephyr bears all the country risk (discount rate 13.73 percent against 12.48 percent) and pays a control premium; its synergy advantage ($23.2 million) is smaller than the extra country risk. A joint venture priced at or below about 6.3 times creates value and keeps an option on control. The military contract (Exhibit B) and the Alvara Coach terms (Exhibit A) point the same way: they are protected by Orvane remaining an owner.

Question 9: Financing currency

Step 1: expected dollar cost of the dinar loan. A dinar loan of MRD 2,400 million (equal to $120 million at spot) at 14.0 percent pays MRD 336 million a year and repays MRD 2,400 million in 2033. Converting each payment at the PPP rate and finding the dollar internal rate of return:

```
Dollar flows (PPP path): +120.0, -15.9, -15.3, -14.8, -14.3, -14.0, -13.7, -(13.3 + 95.4) = -108.7
IRR under PPP                          = 10.04%
IRR under covered-interest-parity path  =  6.16%
One-year hedged cost: 1.14 / (1.12 / 1.043) - 1 = 6.16%
USD loan                                =  6.80%
```

If the dinar depreciates as purchasing-power parity says, the dinar loan costs 10.0 percent in dollars, 3.2 points more than the dollar loan. It is cheaper than the dollar loan (6.2 percent against 6.8 percent) only if the dinar depreciates as fast as the forward curve implies, or if the loan can be hedged, which Table 10 says is not possible beyond two years.

Step 2: risk. The dollar loan creates a currency mismatch: dinar EBITDA services dollar debt. In 2028, interest on $120 million at 6.8 percent is $8.16 million, which is MRD 179.5 million at the PPP rate of 22.00 against EBITDA of MRD 1,199.0 million (coverage 6.7 times). After a further 35 percent devaluation the interest is MRD 242.3 million; if EBITDA also falls (volume down 8 percent, margin down 3 points: 7,355.9 x 0.92 x 13.3% = MRD 900.1 million), coverage is **3.7 times**, still comfortable. The principal of $120 million is MRD 3,020 million at the 2033 PPP rate (1.6 times 2033E EBITDA) or MRD 4,078 million after the shock (2.2 times). The dinar loan has coverage of only 3.2 times in 2027 (1,067.9 / 336.0) and 3.6 times in 2028 at 14.0 percent, but it has no currency mismatch.

Step 3: recommendation. Fund most of the debt in dollars, because it is cheaper in expectation and coverage remains adequate in the stress case; keep a modest dinar loan sized to Tavira's working capital (which is matched to dinar receivables and inventory). A 75 percent dollar, 25 percent dinar mix ($90 million and MRD 600 million) has a blended expected dollar pre-tax cost of 0.75 x 6.8% + 0.25 x 10.04% = 7.6 percent. Hedge the first twelve to twenty-four months of dividends and royalties with forwards, not the whole stream.

Question 10: Recommendation

Board memorandum. *To:* Board of Directors, Zephyr Components, Inc. *From:* Lukas Brandt, CFO. *Date:* 10 December 2026. *Re:* Tavira Drive Components.

Recommendation. Do not accept the 100 percent acquisition at 7.0 times. Counter with a **49 percent joint venture**, opening at 5.8 times 2026E EBITDA (about $136 million for the stake) and with a ceiling of **6.3 times** (about $151 million with costs), together with a call option on a further 26 percent. If Orvane insists on selling 100 percent, our ceiling is **5.5 times ($264 million)**. Walk away above these prices.

Why. (1) Tavira is worth $254 million to $299 million on a standalone, hard-currency basis (5.3 to 6.2 times); the ask of $336 million exceeds the top of that range, and even with synergies and an efficient royalty structure, 100 percent at the ask has an NPV of minus $74 million. At the lowest country premium we consider defensible (12.4 percent), the NPV is still about minus $24 million. (2) Dinar growth of 15 percent over two years is flat in dollars; the business grows in dollars only if it outpaces depreciation. (3) Country risk is the largest driver of value: the expected loss is 3.7 percent a year, there is a 43 percent chance of at least one stress event in seven years, and an expropriation outcome would leave us with almost nothing. (4) A joint venture reduces the value at risk (whole-company value rises 14.7 percent at the JV discount rate), keeps the military contract and the Alvara Coach terms intact, and gives us an option worth about $12 million on control. At 6.3 times it has an NPV of plus $6 million; it breaks even at 6.6 times.

Conditions. (a) Registration of the investment with the central bank before closing, and a written confirmation that registered dividends are eligible for conversion; (b) a written waiver from the ministry on the military contract; (c) a five-year supply agreement with Alvara Coach on current terms; (d) a 3 percent royalty and technical-assistance agreement approved by the tax authority; (e) political risk insurance on the investment, with the premium (about $1.4 million a year on $140 million) built into the price; (f) a minimum dividend policy and vetoes on related-party transactions; (g) funding mainly in dollars at the parent, with a modest dinar working-capital line; (h) forward hedges for twelve to twenty-four months of cash flows.

Triggers to walk away. The price exceeds 6.3 times for the JV or 5.5 times for 100 percent; Orvane refuses the central-bank registration condition or the Alvara Coach agreement; the sovereign spread rises above 7 percent or reserves fall below 2.5 months of imports before signing; the ministry signals that it will re-tender the military contract.

Risks. The dinar may fall faster than purchasing-power parity (a 35 percent devaluation in 2028 cuts value by about 20 percent); we depend on a local partner whose interests diverge from ours; the call option's price is fixed and may not be exercisable if the government objects; insurance does not cover devaluation.

Sensitivity Discussion

Price versus country premium. At the lowest country premium (12.38 percent, lambda 0.7), standalone enterprise value is $299.4 million; with a 3.2 percent leakage and $28.6 million of synergies the value is $318 million against a total cost at the ask of $342.7 million (NPV -$24 million). The ask breaks even only at a WACC of about 11.85 percent, which implies a cost of equity of 13.55 percent and a country premium of about 3.75 percent, below even the 4.6 percent sovereign spread. Justifying the ask requires believing that equity in Marenta is less risky than Marenta's own government bonds.

Exchange-rate path. Value falls 18.6 percent if the dinar follows the forward curve, and 19.6 percent after a 35 percent devaluation. The dinar loan, a royalty route or a JV do nothing about a real devaluation of the operating business; only price and cost pass-through and insurance help.

Joint venture parameters. The break-even multiple moves from 6.0 to 6.6 as option volatility rises from 25 to 45 percent (the option is worth $6.8 million to $16.9 million). If the call strike were 8.0 times rather than 7.0 times, the option would be worth less; if the partner's expropriation and controls loss reductions were smaller, the JV discount rate would rise toward 13 percent and break-even would fall.

Common Student Errors

1. **Discounting dinar cash flows at a dollar rate** (or the reverse), the 52 percent error in Question 4.
2. **Converting at the spot rate for every year** and ignoring depreciation.
3. **Using a forward rate as a forecast** and comparing it with an inflation-based cash-flow forecast.
4. **Adding a country premium and subtracting expected loss in cash flows** (double counting), or adding nothing at all.
5. **Applying the sovereign spread alone to equity**, ignoring that equity is riskier than the bond.
6. **Treating free cash flow as cash in the parent's hands**; ignoring withholding tax, the cap and the lag.
7. **Paying the control premium without pricing the synergies,** or pricing synergies at 100 percent with the standalone country premium and no integration cost.
8. **Forgetting the terminal-value growth consistency check:** dinar growth of 7.1 percent equals dollar growth of 4.55 percent only with depreciation of 2.4 percent.
9. **Reading the option value from a formula without noting the linked strike**; or ignoring the option altogether.
10. **Choosing the loan currency by headline rate** (14.0 versus 6.8 percent) without computing the expected dollar cost.

Alternative Defensible Answers

- **Buy 100 percent at a lower price.** A team that values the synergies and control highly (the military contract, integration, a platform for regional expansion) can defend a bid of up to $264 million for 100 percent, which is 5.5 times, with a seller note and insurance. It must explain why Orvane would accept a price that is 21 percent below its ask.
- **Walk away.** A risk-averse board could decide that no price compensates for the exposure; this is defensible if Zephyr's hurdle rate for Marenta is higher than ours and if alternative uses of $150 to $260 million in safer markets exist.
- **A different country premium.** Using the rating-based premium (5.2 percent, WACC 12.86 percent) gives $281 million and a higher ceiling; using lambda 0.7 gives $299 million. Both are defensible provided the team states why.
- **Cash-flow haircuts instead of a premium.** A team may value the base case at the home-market WACC and apply the scenario weights; it must reach about the same value (Question 7) and must avoid double counting.
- **Dinar-only valuation.** A team can complete the entire analysis in dinars if it uses a dinar discount rate consistent with the dollar rate, as in Question 3; it must still translate the result into dollars to compare with the price.

Chapter 17: Case: Sterling Allied, Break Up the Conglomerate?

Type: Case. Sterling Allied Industries, Halvard Precision, Kestrel Flow Systems, Marlowe Specialty Materials, Hearthstone Living, Redwater Partners, Larkin Brands, Brandt Weiss & Co. and every other company, fund, adviser and person named in this chapter are fictional and illustrative. All figures were constructed for teaching and bear no relation to any real company or transaction. Nothing in the chapter is tax, legal or investment advice; the tax rules described are a simplified, stylized regime built to resemble the general features of tax-free separation rules in several jurisdictions, and a real transaction would require specialist advice.

Prerequisites: Volume 1, Chapters 11 (WACC), 15 (capital structure), 21 (enterprise valuation) and 22 (mergers and acquisitions); Volume 2 treatment of corporate restructuring, divestitures and activist investors.

The Situation

On the evening of Tuesday, 3 November 2026, Victor Aldana, the chief executive of Sterling Allied Industries, read a nine-page letter from Redwater Partners for the third time. The letter had arrived that afternoon by courier and by email, addressed to the full board. It opened politely and closed bluntly. Redwater, which had quietly accumulated 6.2 percent of Sterling Allied's shares over fourteen months, argued that the company was "a collection of four good-to-excellent businesses held together by habit and a corporate center that no longer earns its keep." It demanded that the board announce, within sixty days, a plan to separate the company into four independent, listed companies within twelve months. If the board declined, Redwater would nominate four of the eleven directors at the annual meeting. The nomination window closed on Friday, 15 January 2027.

Sterling Allied is a diversified industrial company headquartered in Pittsburgh with fiscal 2026 revenue of roughly $9.5 billion. It was assembled over six decades by acquisition. Its four divisions have little in common except a shared name, a shared balance sheet and a shared head office.

Halvard Precision (Aerospace and Defense Components) makes machined and forged components, actuators and sealing systems for commercial and military aircraft. It is the crown jewel: high margin, high return on capital, growing at 7 percent a year on a long order backlog, and valued by the public market at the highest multiples in the group's peer sets.

Kestrel Flow Systems (Industrial Flow) manufactures pumps, valves and flow-control systems for water, energy and process industries. It is a steady, mid-cycle business with a large installed base and a profitable aftermarket service stream. Its returns are adequate and its growth is moderate.

Marlowe Specialty Materials produces engineered resins, coatings and specialty chemicals. It is cyclical, capital intensive and exposed to commodity feedstock prices. It earns healthy EBITDA margins but, because of the capital it ties up, returns that have been below its cost of capital for four consecutive years.

Hearthstone Living (Consumer Home Products) sells kitchen and bath fixtures, small appliances and home storage through home-improvement chains and a growing direct-to-consumer channel. It is the group's only consumer business, has a different customer base, a different seasonality, a different working-capital profile and, by common agreement inside the company, the weakest strategic fit.

Aldana, who has led Sterling Allied for six years, had not been surprised by the letter, only by its timing. The board had discussed "portfolio simplification" at its September retreat and had asked the finance team for a preliminary analysis. That work had been overtaken by events in the last three weeks. On 14 October, Colleen Odom, the chief executive of Larkin

Brands, a consumer products group, had telephoned Aldana and followed up with a non-binding proposal to acquire Hearthstone Living for $2.9 billion in cash. And the investment bank advising the board, Brandt Weiss & Co., had begun sounding out the market on whether a partial initial public offering of Halvard Precision could achieve a premium valuation.

So four distinct paths were suddenly on the table, each championed by a different constituency:

1. Redwater's path: a full four-way separation, presumably through a sequence of spin-offs and sales.
2. A spin-off of Hearthstone Living alone, tax-free to shareholders, favored by two of the independent directors and by a faction of senior executives who wanted to keep the industrial businesses together.
3. A cash sale of Hearthstone to Larkin Brands, favored by the chief financial officer, Priya Natarajan, who wanted the cash to reduce debt.
4. A carve-out initial public offering of a minority stake in Halvard Precision, favored by the investment bankers and by Halvard's own management.

Helen Marchetti, the lead independent director, had asked Aldana and Natarajan to present a joint recommendation, with full supporting analysis, at a special board meeting on Thursday, 19 November 2026. Marchetti's instruction was specific. She did not want a sermon on strategy. She wanted to know, per Sterling Allied share, what each path was worth after dis-synergies, taxes and costs; how confident the finance team was in that number; what happened if the market did not re-rate the stock the way Redwater said it would; and what the board should say to Redwater in the meantime.

Natarajan's team had prepared the data in this chapter. It is imperfect, as such data always is. You are asked to act as the finance team and advise the board.

The people

Victor Aldana, chief executive since 2020, a former head of the Kestrel division. He is skeptical that the discount is real and believes Redwater is "pricing in a world without friction." His own compensation is tied to relative total shareholder return over three years and to the group's return on invested capital.

Priya Natarajan, chief financial officer since 2022. She favors the sale of Hearthstone because it deleverages the company, but is uneasy about the tax cost and about the risk of a buyer walking away.

Helen Marchetti, lead independent director, a retired banker. She chairs the strategy committee and is the board's point of contact with Redwater.

Jonas Treadwell, founder of Redwater Partners. His letter cites the divisions' stand-alone peer multiples and an estimate that a separation would release "no less than 35 percent" of equity value. Redwater's holding period at similar companies has historically been two to three years.

The Sterling family trust holds 11 percent of the shares and has historically supported management. Its trustee has told Marchetti that the trust "would support any plan that raises the value of our shares and does not break the company in a way we cannot undo."

Colleen Odom, chief executive of Larkin Brands, the potential buyer of Hearthstone. Her proposal is non-binding and conditional on financing, due diligence and regulatory clearance. She has asked for an answer, and for exclusivity, within three weeks.

The decision

By 19 November the board must decide (a) which separation path, if any, to pursue and in what order, (b) whether to open exclusive talks with Larkin, (c) how to respond to Redwater, and (d) how to describe the plan to the market. A public announcement is tentatively planned for the first week of December, so that it precedes the January nomination deadline and gives shareholders time to assess it.

Industry Context

Diversified industrial groups have been out of fashion for a generation. The investment case for them rests on three ideas, each with a long academic and practical history: that a corporate center can allocate capital more efficiently than external markets, that a diversified group enjoys lower financing costs because its earnings are less volatile, and that shared services and purchasing yield cost savings. The case against rests on three mirror-image ideas: that headquarters allocates capital poorly and cross-subsidizes weak businesses with the cash flow of strong ones, that the market cannot see clearly into a complex group and so applies a "conglomerate discount," and that managers of a portfolio of businesses they only partly understand destroy value through unfocused strategy and bureaucratic reporting.

The empirical literature is mixed on how large the discount is and, more importantly, on why it exists. Some studies find that diversified firms trade at a discount to a portfolio of stand-alone peers; others show that much of the measured discount reflects the characteristics of firms that choose to diversify (they were already poor performers) rather than diversification itself; and still others find that the discount narrows or reverses in some periods. Students of Volume 1, Chapter 22 will recognize the same selection problem that complicates the measurement of acquisition premia. The practical lesson for a board is that a sum-of-the-parts valuation is a starting point for analysis, not a forecast of what shareholders will receive.

Separations in the industrial sector over the past several years have tended to follow a recognizable script. A diversified parent announces a plan to separate one or more businesses. The market reaction on announcement is on average positive but modest. The separated entities then trade on the multiples of focused peers, though usually with some initial pressure: spin-offs are often sold by index funds and by shareholders who do not want the new entity's stock, so the early trading of a newly spun company can be weak. Over the following year, operational improvements (focus, targeted incentives, cleaner capital allocation) are believed to contribute more to value than the multiple re-rating itself. A frequent disappointment is the stranded cost problem: a parent that sheds a division usually finds that a substantial share of the head-office cost allocated to that division does not leave with it.

Three structural routes exist for separating a business, and each has distinct economics:

- **A spin-off** distributes shares of the subsidiary to the parent's shareholders pro rata. It raises no cash for the parent beyond any pre-spin financing the subsidiary borrows and upstreams. Under the stylized tax regime in this case, a qualifying spin-off is tax-free at both the corporate and shareholder level, provided that the business meets an active-business test, that neither entity is acquired in a defined window after the spin, and that the spin is not a device for distributing earnings.
- **A sale** to a strategic or financial buyer raises cash and crystallizes value at a negotiated price that can include a share of the buyer's synergies. It triggers corporate-level tax on the gain over the tax basis, and it removes the business and its risks from the seller permanently.
- **A carve-out** (a minority initial public offering) sells a stake, typically 15 to 30 percent, to public investors while the parent retains control. It raises cash, creates a public price for the subsidiary and a currency for acquisitions and employee equity, and leaves the option of a later full spin-off or sale. It also leaves the parent holding a listed stake, with all the governance conflicts of controlling a partly owned subsidiary, and the stake may itself attract a holding-company discount.

Activist investors have become the principal catalyst for these transactions. A typical campaign begins with a stake of 1 to 10 percent, a published valuation of the parts, a request for board seats, and a threat of a proxy contest. Boards that respond with a credible, well-analyzed alternative generally fare better, in the sense of retaining control of the process, than boards that refuse to engage. But a credible alternative needs a rigorous counter-valuation, which is what this case asks you to build.

Finally, a note on the sectors involved. Aerospace components trade at premium multiples because of long program lives, aftermarket revenue and high barriers to entry. Industrial flow businesses trade at mid-range multiples with a premium for aftermarket mix. Specialty materials trade at a discount because of cyclicality and capital intensity. Branded consumer home products trade in between, with a premium for brands and direct-to-consumer exposure and a discount for retailer concentration. Diversified industrial groups that remain in the market trade at a discount to the weighted average of their parts’ sectors, which is the empirical fact on which Redwater’s thesis is built.

Financial Data

All figures are in millions of US dollars except per-share data. Fiscal years end 31 December; fiscal 2026 is an estimate built from nine months of actuals and a three-month forecast (“FY2026E”). Share data are diluted and, for simplicity, held constant at 320.0 million shares. The assumed corporate income tax rate in all calculations is 25 percent, and the company’s effective tax rate equals the statutory rate.

Table 1: Consolidated income statement

$ million	FY2024	FY2025	FY2026E
Revenue	9,050	9,260	9,500
EBITDA	1,325	1,382	1,440
Depreciation and amortization	(462)	(471)	(480)
EBIT	863	911	960
Net interest expense	(255)	(259)	(262)
Pre-tax income	608	652	698
Income tax at 25%	(152)	(163)	(174.5)
Net income	456	489	523.5
Diluted EPS ($)	1.43	1.53	1.64
Dividend per share ($)	0.60	0.62	0.64

Table 2: Consolidated balance sheet, 31 December 2026E

Assets	$ million	Liabilities and equity	$ million
Cash and equivalents	580	Accounts payable	1,060
Accounts receivable	1,350	Accrued liabilities	640
Inventories	1,420	Short-term debt	350
Other current assets	210	Long-term debt	4,550
Current assets	**3,560**	Pension liability (pre-tax)	520
Property, plant and equipment, net	3,650	Deferred tax liability	480
Goodwill	3,100	Other long-term liabilities	300
Other intangibles	780	**Total liabilities**	**7,900**
Other long-term assets	420	Shareholders' equity	3,610
Total assets	**11,510**	**Total liabilities and equity**	**11,510**

Gross debt is $4,900 million (short-term $350 million plus long-term $4,550 million) and net debt, after $580 million of cash, is $4,320 million, which is 3.0 times consolidated EBITDA. The pension plan is underfunded by $520 million before tax; contributions are tax-deductible, so the deficit is treated as a debt-like claim of $390 million after tax. Net operating assets (total assets less cash less non-interest-bearing liabilities) are $8,450 million, equal to equity plus net debt plus the pension deficit.

Table 3: Segment data, FY2026E

$ million	Halvard Precision	Kestrel Flow	Marlowe Specialty	Hearthstone Living	Segments	Corporate	Consolidated
Revenue	2,200	3,000	1,700	2,600	9,500	n/a	9,500
EBITDA	440	510	306	312	1,568	(128)	1,440
EBITDA margin	20.0%	17.0%	18.0%	12.0%	16.5%	n/a	15.2%
Depreciation and amortization	95	150	135	90	470	10	480
EBIT	345	360	171	222	1,098	(138)	960
Capital expenditure	120	155	140	95	510	10	520
Capital employed (net operating assets)	1,650	2,650	2,150	1,950	8,400	50	8,450
Segment cost of capital (WACC)	8.0%	8.5%	9.5%	8.0%	n/a	n/a	n/a
Tax basis of the business (for asset or share sale)	1,600	n/a	n/a	1,150	n/a	n/a	n/a

Corporate costs of $128 million cover the head office, shared IT, treasury, tax and legal, insurance, and group-level research. Management's allocation of those costs to the divisions on the basis of usage (not reported externally) is $30 million to Halvard, $36 million to Kestrel, $24 million to Marlowe and $38 million to Hearthstone. The corporate D&A of $10 million relates to head-office assets and is not allocated.

Table 4: Forecast EBITDA, FY2026E to FY2031E

$ million	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E	FY2031E	Growth p.a.
Halvard Precision	440.0	470.8	503.8	539.0	576.8	617.1	7.0%
Kestrel Flow	510.0	530.4	551.6	573.7	596.6	620.5	4.0%
Marlowe Specialty	306.0	315.2	324.6	334.4	344.4	354.7	3.0%
Hearthstone Living	312.0	319.8	327.8	336.0	344.4	353.0	2.5%
Segment total	1,568.0	1,636.2	1,707.8	1,783.1	1,862.2	1,945.4	4.4%
Corporate costs (grow 3% p.a.)	(128.0)	(131.8)	(135.8)	(139.9)	(144.1)	(148.4)	3.0%
Consolidated EBITDA	1,440.0	1,504.3	1,572.0	1,643.2	1,718.1	1,797.0	4.5%

Other forecast assumptions used in the Hearthstone cash-flow cross-check: revenue grows with EBITDA at 2.5 percent a year (constant margin), D&A and capex scale with revenue, net working capital is 12 percent of revenue, the terminal growth rate is 2.0 percent, and in the terminal year capex equals 106 percent of D&A.

Table 5: Market data and capitalization (close of 2 November 2026)

Item	Value
Share price	$21.30
Diluted shares outstanding	320.0 million
Equity market capitalization	$6,816 million
Net debt	$4,320 million
Pension deficit, after tax	$390 million
Enterprise value (market cap + net debt + after-tax pension deficit)	$11,526 million
EV / FY2026E consolidated EBITDA	8.0x
P/E, FY2026E	13.0x
Net debt / EBITDA	3.0x
EBITDA / net interest	5.5x
Dividend yield	3.0%
52-week trading range	$17.80 to $24.10
Credit ratings (agency A / agency B)	BBB- / Baa3, stable
Agency leverage trigger for downgrade to sub-investment grade	Net debt / EBITDA above 3.5x for two consecutive quarters
Weighted average cost of debt	5.35%

Shareholder base: institutional investors 71 percent (of which index funds about 24 percent), Sterling family trust 11 percent, Redwater Partners 6.2 percent, directors and management 1.3 percent, other 10.5 percent.

Table 6: Debt schedule

Instrument	Amount ($ million)	Maturity	Terms
Commercial paper and revolver	350	Rolling	Floating; backed by $1,000 million revolver
Term loan	700	2028	Floating; leverage covenant 4.0x
Senior notes	1,000	2029	5.0% fixed
Senior notes	1,200	2031	5.6% fixed; investors may put at 101 on change of control accompanied by a downgrade to sub-investment grade
Senior notes	1,000	2034	5.9% fixed
Senior notes	650	2036	6.1% fixed
Total	**4,900**		

Table 7: Trading comparables (EV / FY2026E EBITDA)

Peer sets were compiled by Brandt Weiss & Co. from fictional listed companies. The “selected” multiple used in the base-case sum of the parts is the peer median; the selected low-high range brackets the interquartile spread plus judgment.

Segment	Peer set (EV/EBITDA)	Median	Selected range
Aerospace components	Vantor Aerostructures 13.4x; Cobalt Flight Systems 12.5x; Altis Precision 11.8x; Rennick Defense Parts 13.0x; Orlen Avionics 12.1x; Tarn Aerospace 11.2x; Brannock Systems 13.8x	12.5x	11.0x to 14.0x
Industrial flow	Delphine Pumps 10.4x; Marrow Valve 9.8x; Calder Flow 9.5x; Ostrander Controls 9.1x; Hexham Process 8.7x; Pellam Fluid Power 10.1x; Quillon Water 9.2x	9.5x	8.5x to 10.5x
Specialty materials	Corvin Chemicals 8.9x; Ellery Polymers 8.0x; Tamsin Coatings 7.6x; Ridgeway Resins 7.2x; Sable Materials 8.4x; Westerly Specialty 7.9x; Norland Compounds 8.1x	8.0x	7.0x to 9.0x
Consumer home products	Aldous Housewares 9.3x; Brightwater Brands 8.5x; Coral Living 7.9x; Dunmore Home 8.8x; Easton Fixtures 8.1x; Fenwick Kitchens 9.0x; Garrity Storage 7.6x	8.5x	7.5x to 9.5x
Diversified industrial groups	Ashgrove Industries 7.9x; Pemberton Group 7.6x; Taurus Holdings 7.1x	7.6x	n/a

Table 8: Deal and transaction terms under discussion

Term	Sale of Hearthstone to Larkin	Spin-off of Hearthstone	Carve-out IPO of Halvard
Headline value	$2,900 million cash, enterprise value (9.3x FY2026E EBITDA)	Pro rata distribution, 1 SpinCo share for each 4 Sterling shares (80 million SpinCo shares)	25% of Halvard equity sold by Sterling (secondary)
Debt arrangements	Cash-free, debt-free	SpinCo borrows $700 million and pays it to Sterling as a pre-spin dividend, used to repay Sterling debt	Halvard raises debt of 2.5x EBITDA and upstreams the proceeds before the IPO
Transaction costs	$45 million advisory and legal	$150 million pre-tax (legal, IT separation, rebranding, financing)	3.5% underwriting and advisory fees; $60 million pre-tax separation costs
Tax	Taxable at 25% of gain over tax basis ($1,150 million)	Tax-free if the active-business and no-change-of-control tests are met	Taxable at 25% of gain over tax basis on shares sold (25% of $1,600 million)
Timing	Signing in about 8 weeks, closing about 6 months later	About 12 months	About 5 months
Conditions	Financing, diligence, antitrust clearance; reverse break fee of 3%	Tax ruling or opinion; Form 10 style registration; board and shareholder processes	Market conditions; usual IPO conditions
Internal probability of completing	70%	90%	85%
Typical IPO discount	n/a	n/a	10% to the fair value of the equity

Complications and Exhibits

The following facts emerged after the original analysis and should be incorporated in your answers where relevant.

Exhibit A: stand-alone costs and dis-synergies. The finance team's estimates of the costs of running each division as an independent company are in the table below. "Stand-alone corporate cost" replaces that division's share of the $128 million of group corporate costs. "Purchasing synergies lost" is the annual EBITDA the group currently earns through joint procurement of steel, resins, freight and packaging, which a division would lose if it left the group. These are run-rate pre-tax annual figures, assumed to be in place immediately after separation. For a spin-off or sale of Hearthstone alone, purchasing synergies lost are $14 million at Hearthstone plus $3 million each at Kestrel and Marlowe, a total of $20 million. For a carve-out of Halvard the loss is $2 million at Halvard.

$ million per year	Halvard	Kestrel	Marlowe	Hearthstone	Total
Allocated share of current corporate cost	30	36	24	38	128
Stand-alone corporate cost	55	62	48	58	223
Purchasing synergies lost on separation	2	10	12	14	38

Exhibit B: stranded costs. When Hearthstone leaves, $38 million of allocated head-office cost leaves the group's books on paper, but the finance team believes $6 million of it will remain permanently as stranded cost in the remaining group after two years of cost reduction. When Halvard is partly floated, the stranded amount is $4 million. Corporate costs of the remaining group are therefore $128 - 38 + 6 = $96 million after a Hearthstone separation and $128 - 30 + 4 = $102 million after a Halvard carve-out.

Exhibit C: separation costs. One-time costs are $480 million pre-tax for a four-way separation, $150 million for a Hearthstone spin-off (Table 8), $45 million of sale costs for the sale, and $60 million for the Halvard carve-out in addition to underwriting fees. Assume separation costs (other than underwriting and sale fees, which are netted against proceeds) are tax-deductible at 25 percent.

Exhibit D: debt capacity. The agencies' guidance and market evidence suggest that the sustainable leverage of each division as a stand-alone company, measured as net debt to its stand-alone EBITDA, is 2.5x for Halvard, 3.0x for Kestrel, 2.5x for Marlowe and 2.5x for Hearthstone. The diversified group at 3.0x is at the high end of its own investment-grade capacity because its earnings are diversified. Debt in excess of stand-alone capacity must be repaid, and reducing debt that is assumed to be permanent costs the 25 percent interest tax shield on the amount repaid (treat the lost tax shield as permanent debt: its value is the tax rate multiplied by the debt repaid).

Exhibit E: pension and indemnities. The $520 million pension deficit belongs to a legacy plan covering Marlowe and Kestrel employees. In every scenario except the four-way breakup it stays with the remaining group. In the four-way breakup it is carried as a debt-like claim of $390 million after tax, with no change in total value.

Exhibit F: the five-year active-business test. Hearthstone includes a kitchen-fixtures business, Brightline, acquired in a taxable purchase in March 2023 and now contributing about 25 percent of Hearthstone's EBITDA. Tax counsel advises that Brightline might not satisfy the five-year active-business requirement, but that the remainder of Hearthstone would. Counsel's view is that there is about a 10 percent chance that the spin-off would be treated as taxable at the corporate level. In that event Sterling would owe 25 percent of the excess of SpinCo's fair value over its $1,150 million tax basis.

Exhibit G: the shared plant. A plant in Ohio produces components for both Kestrel and Hearthstone. After any separation of Hearthstone, a ten-year supply agreement would apply, which the team has incorporated in the purchasing-synergy loss in Exhibit A. A buyer such as Larkin may be unwilling to take on that contract on identical terms.

Exhibit H: Redwater's presentation. Redwater's letter values each division at the median multiples in Table 7 and capitalizes the corporate costs, but ignores stand-alone costs, lost purchasing synergies, separation costs, debt capacity and taxes. It claims upside of at least 35 percent of the share price. It also states that the market applies a conglomerate discount of 28 percent. Check these claims.

Exhibit I: the diversified peers. Table 7 shows that the three listed diversified industrial peers trade at a median of 7.6x EBITDA, below Sterling Allied's 8.0x. Redwater's presentation does not mention this.

Exhibit J: possible delay and failure of the Larkin deal. If the Larkin sale does not complete (probability 30 percent), the board's best fallback would be a Hearthstone spin-off, delayed by about six months. The finance team estimates that a six-month delay, combined with the cost of an announced but failed sale process, would reduce the per-share value of the fallback by $0.40.

Exhibit K: the residual discount. The key uncertainty is how much of a conglomerate discount remains in the shares of the surviving company after a separation. For all single-division separation scenarios (spin, sale, carve-out) the finance team's base case assumes that the surviving, still-diversified group continues to trade at a discount to the sum of its parts of 12 percent of enterprise value, and that the new stand-alone company (the SpinCo or the newly listed Halvard) trades at its fair value with no

discount, other than the 10 percent IPO discount that unwinds shortly after listing. For the carve-out, the parent's retained stake in Halvard is assumed to trade at a holding-company discount of 10 percent to the value of the stake. In the four-way breakup the base case assumes no residual discount. Treat these as assumptions to be tested.

Exhibit L: operational detail on Marlowe. Marlowe's return on capital has been below its cost of capital for four years. Management has a plan to improve margins by 150 basis points over three years through plant closures costing $90 million. The plan is not in the forecasts of Table 4. Some directors privately say that the real value-creating move is to sell or close Marlowe rather than to spin Hearthstone.

Exhibit M: timing. The announcement window and the Redwater nomination deadline compress the timetable. A spin-off of Hearthstone would take about 12 months, so no spin-off could be completed before the 2027 annual meeting. A signed sale agreement or a launched IPO could be announced earlier.

Assignment Questions

1. **Segment economics.** For each division compute the EBITDA margin, EBIT, after-tax operating profit (NOPAT at 25 percent), return on capital employed, the spread over the division's cost of capital, and the ratio of imputed enterprise value (median multiple) to capital employed. Which divisions create value and which destroy it? What does this suggest about which businesses benefit from being inside the group?
2. **Sum of the parts.** Value each division at the median multiple and at the low and high ends of the selected ranges. Capitalize the corporate costs at the blended multiple. Bridge enterprise value to equity value and to a per-share value, and compute the conglomerate discount on both an enterprise-value and an equity-value basis. Test Redwater's claims (a 28 percent discount; at least 35 percent upside).
3. **Diagnosing the discount.** Compare Sterling Allied's multiple with the diversified peers. Discuss whether the discount is "real," how much of it a separation can capture, and the possible causes (capital allocation, complexity, investor clientele, leverage, return on capital). Perform a discounted cash flow cross-check of Hearthstone Living at WACC of 8.0, 8.5 and 9.0 percent to test the multiple used.
4. **The full breakup.** Estimate the value per share of Redwater's four-way breakup after stand-alone costs, lost purchasing synergies, one-time costs, stand-alone debt capacity and the loss of the interest tax shield. Test sensitivity to a residual 5 percent and 10 percent discount in the post-breakup values.
5. **Capital structure and credit.** Compute the leverage of the group after (a) a Hearthstone spin-off, (b) a sale of Hearthstone to Larkin, and (c) a Halvard carve-out. Evaluate rating, covenant and put-at-101 issues. What should the board do about the SpinCo debt allocation?
6. **Spin-off of Hearthstone.** Value SpinCo and RemainCo, compute the value per Sterling share after costs under the base case, and show how the answer depends on the residual discount that RemainCo suffers and on a possible discount for SpinCo. Incorporate the risk that the spin-off fails the tax test. What residual discount makes the board indifferent to doing nothing?
7. **Sale of Hearthstone to Larkin.** Compute after-tax proceeds, the value per Sterling share under the base case, and the sale price at which the sale is no better than the spin. Compute the probability-weighted value given the 30 percent chance of failure and the fallback in Exhibit J. Consider how the proceeds should be used.
8. **Carve-out of Halvard.** Compute the IPO proceeds, fees, tax and net cash to Sterling, and the value per Sterling share under the base case. Discuss what the carve-out does and does not do about the discount and what it implies for Redwater's argument.

9. **Ranking and alternatives.** Rank the four options on expected value per share, risk, speed, reversibility and strategic effect. Assess the argument that Marlowe, not Hearthstone, is the right division to exit. Consider combinations, such as a dual-track process.
10. **Recommendation and engagement with Redwater.** Prepare a recommendation for the 19 November board meeting, including how to respond to Redwater, what to announce, what milestones the board should set, and what evidence would change your recommendation.

Teaching Note and Full Solutions

Learning objectives

After working through this case, students should be able to:

1. Build a sum-of-the-parts valuation from segment data and trading comparables, including the treatment of corporate costs, net debt and debt-like claims, and distinguish enterprise-value from equity-value measures of a conglomerate discount.
2. Decompose a gross “break-up value” into the parts that are real (stand-alone costs, lost purchasing scale, lost debt capacity, one-time costs, taxes) and the part that is capturable through a change in how the market prices the shares.
3. Compare a spin-off, a sale and a carve-out on after-tax value per share, speed, reversibility, risk and strategic effect, and explain why the three routes differ in taxation and in who captures the synergies.
4. Analyze the capital-structure and credit consequences of a separation, including debt allocation, agency leverage thresholds and the effect of stranded costs.
5. Use probability weighting, sensitivity analysis and break-even analysis to convert a point estimate into a decision under uncertainty, and communicate that decision to a board.
6. Evaluate an activist’s presentation critically, identifying what it omits, and design a response that engages the shareholder without conceding the analysis.

Suggested class plan (100 minutes)

Minutes	Activity	Questions
0-10	Opening cold call: “Is Sterling Allied worth more in pieces?” Poll the room before any numbers; record the distribution of answers.	none
10-25	Segment economics and the sum of the parts. Small groups compute the SOTP on a single page; reconcile to $29.61. Discuss the corporate-cost capitalization.	1, 2
25-40	Diagnosing the discount: peers, DCF cross-check, the “real versus capturable” waterfall. Plenary.	3, 4
40-50	Capital structure and credit: why the spin pushes RemainCo toward the agency trigger. Short lecture, five minutes of discussion.	5
50-75	Three routes: split the class into spin, sale and carve-out teams; each presents a per-share value and its two biggest risks. Instructor records results on a common table.	6, 7, 8
75-90	Ranking, Marlowe alternative, dual-track design. Vote again; compare with the opening poll.	9
90-100	Board memo and response to Redwater: two students present the key paragraphs; the class critiques.	10

Board-room role-play works well for the last segment: assign Aldana, Natarajan, Marchetti, the family trustee and a Redwater representative, and ask each to state, in thirty seconds, what would make them vote against the recommendation.

Solutions

All calculations below were recomputed with Python (a verification printout appears at the end of the solutions). Figures are in millions of US dollars unless stated; per-share figures use 320.0 million shares.

Question 1: Segment economics

Formulas:

```
EBITDA margin = EBITDA / Revenue
EBIT          = EBITDA - D&A
NOPAT         = EBIT x (1 - 25%)
ROIC          = NOPAT / Capital employed
Spread        = ROIC - segment WACC
Imputed EV    = EBITDA x median multiple
EV / Capital  = Imputed EV / Capital employed
```

Halvard: margin 440 / 2,200 = 20.0 percent; EBIT 440 - 95 = 345; NOPAT 345 x 0.75 = 258.75; ROIC 258.75 / 1,650 = 15.7 percent; spread 15.7 - 8.0 = +7.7 points; imputed EV 440 x 12.5 = 5,500; EV / capital 5,500 / 1,650 = 3.33.

Kestrel: margin 510 / 3,000 = 17.0 percent; EBIT 360; NOPAT 270.00; ROIC 270 / 2,650 = 10.2 percent; spread 10.2 - 8.5 = +1.7 points; EV 510 x 9.5 = 4,845; EV / capital 1.83.

Marlowe: margin 306 / 1,700 = 18.0 percent; EBIT 171; NOPAT 128.25; ROIC 128.25 / 2,150 = 6.0 percent; spread 6.0 - 9.5 = -3.5 points; EV 306 x 8.0 = 2,448; EV / capital 1.14.

Hearthstone: margin 312 / 2,600 = 12.0 percent; EBIT 222; NOPAT 166.50; ROIC 166.5 / 1,950 = 8.5 percent; spread 8.5 - 8.0 = +0.5 points; EV 312 x 8.5 = 2,652; EV / capital 1.36.

Division	EBITDA margin	EBIT	NOPAT	ROIC	WACC	Spread	EV / capital
Halvard	20.0%	345	258.75	15.7%	8.0%	+7.7 pts	3.33
Kestrel	17.0%	360	270.00	10.2%	8.5%	+1.7 pts	1.83
Marlowe	18.0%	171	128.25	6.0%	9.5%	-3.5 pts	1.14
Hearthstone	12.0%	222	166.50	8.5%	8.0%	+0.5 pts	1.36

Economic profit (NOPAT minus WACC times capital) is: Halvard 258.75 - 0.08 x 1,650 = +126.75; Kestrel 270 - 0.085 x 2,650 = +44.75; Marlowe 128.25 - 0.095 x 2,150 = -76.00; Hearthstone 166.5 - 0.08 x 1,950 = +10.50. Group economic profit (before corporate costs) is therefore about +106.

Interpretation. Halvard creates far more value than its share of capital would suggest; Marlowe destroys about $76 million a year. Hearthstone, which management regards as the weakest strategic fit, roughly earns its cost of capital and is therefore not destroying value; its problem is lack of strategic fit and growth rather than poor returns. A subtle point: EBITDA margin ranks Marlowe (18.0 percent) above Kestrel in appearance but its return on capital is the worst. Students who stop at margin will misidentify the problem division. The result suggests that the businesses that benefit from being inside the group are the lower-return, capital-hungry ones (Marlowe, partially Kestrel), while the high-return business (Halvard) is probably subsidizing them, which is the classic cross-subsidy that supports an activist’s case. Note that return on capital is computed on book capital that includes acquisition goodwill, so it understates the return on the incremental capital of Hearthstone and Marlowe and overstates their “history” cost; discuss this in class.

Question 2: Sum of the parts

At the median multiples: Halvard 5,500 + Kestrel 4,845 + Marlowe 2,448 + Hearthstone 2,652 = 15,445. Segment EBITDA is 1,568, so the blended multiple is 15,445 / 1,568 = 9.850x. Corporate costs of $128 million are capitalized at the blended multiple: 128 x 9.850 = 1,260.8. Enterprise value of the group = 15,445 - 1,260.8 = 14,184.2.

Bridge to equity: 14,184.2 - net debt 4,320 - after-tax pension deficit 390 = 9,474.2. Per share: 9,474.2 / 320 = $29.61. This is 29.61 / 21.30 - 1 = +39.0 percent above the current price.

Low and high cases. Low multiples (11.0, 8.5, 7.0, 7.5): 440 x 11.0 = 4,840; 510 x 8.5 = 4,335; 306 x 7.0 = 2,142; 312 x 7.5 = 2,340; total 13,657; blended 13,657 / 1,568 = 8.71x; corporate 128 x 8.71 = 1,114.9; EV 12,542.1; equity 7,832.1; $24.48 per share. High multiples (14.0, 10.5, 9.0, 9.5): 6,160 + 5,355 + 2,754 + 2,964 = 17,233; blended 10.99x; corporate 1,406.8; EV 15,826.2; equity 11,116.2; $34.74.

Case	Gross segment EV	Corporate costs capitalized	Group EV	Equity	Per share	vs $21.30
Low	13,657.0	(1,114.9)	12,542.1	7,832.1	$24.48	+14.9%
Median	15,445.0	(1,260.8)	14,184.2	9,474.2	$29.61	+39.0%
High	17,233.0	(1,406.8)	15,826.2	11,116.2	$34.74	+63.1%

Market-implied values. Market capitalization is 21.30 x 320 = 6,816; enterprise value 6,816 + 4,320 + 390 = 11,526, which is 8.0x consolidated EBITDA of 1,440. Conglomerate discount on an enterprise-value basis: 1 - 11,526 / 14,184.2 = 18.7 percent. On an equity-value basis: 1 - 6,816 / 9,474.2 = 28.1 percent. The equity measure is larger because leverage magnifies the percentage. Redwater's 28 percent is therefore an equity-basis figure; its 35 percent upside claim is consistent with our 39 percent gross figure. The claims are arithmetically right but they are the starting point of the analysis, not its conclusion, because the median-multiple SOTP assumes that the parts could be separated at no cost and still be valued as if they enjoyed the group's synergies, purchasing power, debt capacity and shared overhead. Note that the low-case SOTP value ($24.48) is still 14.9 percent above the price; the conclusion that the gross SOTP exceeds the market is robust to the choice of multiples, but the magnitude is not.

Question 3: Diagnosing the discount

Peers. The diversified industrial peers trade at a median of 7.6x EBITDA; Sterling Allied trades at 8.0x. Sterling is therefore not cheap relative to its true comparables; the discount is relative to a hypothetical portfolio of focused companies, and it is also embedded in the sector. A board that takes the diversified-peer multiple as the right benchmark could conclude that the market treats Sterling no worse than any other conglomerate. The relevant question is whether a separation turns each part into a focused company and moves it from the 7.6x set to the focused set. Sterling's implied multiple on the SOTP basis is 14,184.2 / 1,440 = 9.85x, against a market multiple of 8.0x.

Possible causes of the discount, and how much each can be addressed:

- Cross-subsidy and capital allocation: Marlowe's -3.5 point spread is the clearest evidence. Fix by operational improvement, exit or restructuring; separation of the other divisions does not address it.
- Complexity and information: four segments, a corporate center, a shared balance sheet. A spin-off or sale simplifies this and is capturable.
- Investor clientele: aerospace and consumer investors differ; a single stock does not fit anyone's mandate. Partly capturable.
- Leverage and the pension: 3.0x leverage plus a 520 pension deficit limit rerating. Only partly addressed by separation (and may worsen on a spin, see Question 5).
- Real synergies: shared overhead and purchasing power exist; the market is correct to reflect their value, so the portion of the discount corresponding to them is not an inefficiency (see Question 4).

DCF cross-check of Hearthstone. The base-case EBITDA of 312 grows at 2.5 percent; revenue, D&A and capex scale with it. Free cash flow in each year is:

```
FCF = EBIT x (1 - 0.25) + D&A - Capex - change in NWC
NWC change = 12% x Revenue x g / (1 + g)    (equal to 12% x 2,600 x 1.025^(t-1) x 0.025)
```

Year	EBITDA	EBIT	FCF
1	319.8	227.5	157.7
2	327.8	233.2	161.7
3	336.0	239.1	165.7
4	344.4	245.0	169.9
5	353.0	251.2	174.1

Year 1 in detail: EBIT 227.5; after tax 170.6; plus D&A 92.25; less capex 97.375; less NWC increase 0.12 x 2,600 x 0.025 = 7.8; FCF = 170.625 + 92.25 - 97.375 - 7.8 = 157.7. Terminal value: terminal-year (year 6) FCF is 178.9 with g = 2.0 percent and capex at 106 percent of D&A; terminal value at year 5 = 178.9 / (WACC - 0.02). At 8.0 percent: TV = 2,980.9; PV of TV = 2,980.9 / 1.08^5 = 2,028.8; PV of the five cash flows 659.6; EV = 2,688.3, or 8.62x FY2026E EBITDA. At 8.5 percent: EV 2,480.8 (7.95x). At 9.0 percent: EV 2,302.9 (7.38x).

WACC	PV of FCF (years 1-5)	PV of terminal value	EV	EV / EBITDA	Terminal value share
8.0%	659.6	2,028.8	2,688.3	8.62x	75.5%
8.5%	650.8	1,830.0	2,480.8	7.95x	73.8%
9.0%	642.3	1,660.6	2,302.9	7.38x	72.1%

At its own 8.0 percent WACC the DCF supports the 8.5x peer median (8.62x). A 50-basis-point rise in WACC takes the value down by about 8 percent, so the peer multiple is a reasonable but not conservative anchor for Hearthstone. Importantly, the DCF is on Hearthstone's EBITDA before any stand-alone costs; that is why the conclusion is that the comparable multiple applied to $312 million overstates what SpinCo would be worth once it carries its own overhead.

Question 4: The full breakup

Step 1: stand-alone EBITDA of each division = segment EBITDA - stand-alone corporate cost - purchasing synergies lost.

Division	Segment EBITDA	Stand-alone corporate	Purchasing lost	Stand-alone EBITDA	Multiple	Stand-alone EV
Halvard	440	55	2	383	12.5x	4,787.5
Kestrel	510	62	10	438	9.5x	4,161.0
Marlowe	306	48	12	246	8.0x	1,968.0
Hearthstone	312	58	14	240	8.5x	2,040.0
Total	1,568	223	38	1,307		12,956.5

The sum of the stand-alone enterprise values is 12,956.5, against 14,184.2 for the SOTP with group-level corporate costs: the difference of 1,227.7 (or $3.84 per share) is the capitalized value of $95 million of extra corporate cost plus $38 million of lost purchasing synergies, each capitalized at the division's own multiple. That is value that is real, not a discount. Equity before any other adjustment is 12,956.5 - 4,320 - 390 = 8,246.5, or $25.77, up 21.0 percent on the price.

Step 2: debt capacity. Stand-alone capacity = leverage target x stand-alone EBITDA: Halvard 2.5 x 383 = 957.5; Kestrel 3.0 x 438 = 1,314.0; Marlowe 2.5 x 246 = 615.0; Hearthstone 2.5 x 240 = 600.0; total 3,486.5. Net debt must fall from 4,320 to 3,486.5, a shortfall of 833.5. Losing the tax shield on that debt costs 0.25 x 833.5 = 208.4, or $0.65 per share.

Step 3: one-time costs: 480 x (1 - 0.25) = 360.0, or $1.13 per share.

Step 4: equity value = 12,956.5 - 4,320 - 390 - 360.0 - 208.4 = 7,678.1; per share $23.99; versus the price of $21.30, +12.6 percent ($2.69 per share). Waterfall from Redwater's number:

Step	Per share ($)	Cumulative ($)
Gross SOTP gain over market price (29.61 - 21.30)	8.31	8.31
Stand-alone costs and lost purchasing synergies	(3.84)	4.47
One-time separation costs after tax	(1.13)	3.35
Lost interest tax shield on debt capacity shortfall	(0.65)	2.69
Net gain from breakup, zero residual discount	2.69	2.69 (+12.6%)

Sensitivity to a residual discount on the post-breakup values (applied to total EV): 5 percent: EV 12,308.7; equity 12,308.7 - 4,320 - 390 - 360 - 208.4 = 7,030.3; $21.97 (+3.1 percent). 10 percent: EV 11,660.9; equity 6,382.5; $19.95 (-6.4 percent). The break-even residual discount is therefore about 6.7 percent. Multiples sensitivity: because stand-alone EBITDA is 1,307, a one-turn increase in every multiple adds about 1,307 (about $4.08 per share, holding debt unchanged) and a one-turn fall removes the same: $28.08 and $19.91 respectively.

Conclusion: Redwater's 39 percent shrinks to about 12.6 percent under favorable assumptions and disappears if the post-breakup companies trade at a 7 percent or greater discount to their peers. That is the central quantitative message of the case.

Question 5: Capital structure and credit

Starting point: net debt 4,320 / EBITDA 1,440 = 3.0x; with the pension deficit added, (4,320 + 520) / 1,440 = 3.36x. The agency trigger in Table 5 is net debt above 3.5x for two consecutive quarters.

a. Spin-off. SpinCo borrows 700 and pays it to Sterling, so group net debt falls to 3,620. RemainCo EBITDA is 1,154 (derived in Question 6). RemainCo leverage = 3,620 / 1,154 = 3.14x, up from 3.0x, because the EBITDA that leaves (240 after stand-alone costs) is a larger share of the group than the debt that leaves (700 is 16 percent of debt, while the EBITDA lost is 20 percent). Pension-adjusted: (3,620 + 520) / 1,154 = 3.59x, above the 3.5x level. SpinCo itself starts at 700 / 240 = 2.92x, above the 2.5x capacity in Exhibit D (capacity is 600). The two entities are both too levered; the spin transfers a leverage problem to each, and no allocation of the 4,320 solves it, because total capacity falls (RemainCo at 3.0x supports 3,462, SpinCo at 2.5x supports 600, a total of 4,062 against the group's 4,320 less the pension effect). Board options: suspend discretionary repurchases and slow the dividend; assign debt so that RemainCo is closest to 3.0x (a SpinCo debt of 858 would be needed, 3.6x for SpinCo and not financeable), or use a third source such as Marlowe plant-closure savings or asset sales.

b. Sale to Larkin. Net proceeds of 2,428.75 (Question 7) retire debt; net debt falls to 1,891.25. Leverage is 1,891.25 / 1,154 = 1.64x. There is headroom to re-lever to 2.5x (net debt of 2,885, extra capacity of 993.75) or to 3.0x (3,462). The sale is the only route that improves credit quality.

c. Carve-out. Halvard raises 957.5 (2.5x) and upstreams it; Sterling nets 723.7 of IPO proceeds; consolidated net debt = 4,320 - 723.7 = 3,596.3 on consolidated EBITDA of 1,409 (Halvard 383 + Kestrel, Marlowe and Hearthstone 1,128 less 102 of corporate cost), a leverage of 2.55x. Because Halvard's debt is consolidated, the group improves modestly; parent-only net debt is 4,320 - 957.5 - 723.7 = 2,638.8.

Put at 101. The 2031 notes can be put only on a change of control combined with a downgrade to sub-investment grade. A spin-off or a sale of part of the assets is not a change of control as normally defined, so the put is not triggered directly, but indentures also restrict the sale of substantially all assets and require legal review; and a downgrade driven by the spin would raise refinancing costs on the 2028 term loan and the commercial paper program. Illustration: a 75 basis point repricing on 4,900 of debt costs about 36.8 pre-tax a year.

Recommendation: do not set the SpinCo debt at 700 without agency pre-clearance; pre-clear a ratings view and target RemainCo net debt / EBITDA of 3.0x or below by the date of the spin.

Question 6: Spin-off of Hearthstone

SpinCo. EBITDA = 312 - 58 (stand-alone corporate) - 14 (purchasing synergies lost) = 240. EV = 240 x 8.5 = 2,040. Less debt of 700: equity 1,340. Per SpinCo share (80 million shares): $16.75; per Sterling share (1,340 / 320): $4.19.

RemainCo. Segment EBITDA of Halvard, Kestrel and Marlowe = 440 + 510 + 306 = 1,256. Corporate cost = 128 - 38 + 6 = 96. Lost purchasing synergies at Kestrel and Marlowe = 3 + 3 = 6. RemainCo EBITDA = 1,256 - 96 - 6 = 1,154. Segment EV = 5,500 + 4,845 + 2,448 = 12,793, blended 10.186x. The 102 of cost and lost synergies capitalized at 10.186 is 1,039.0, so RemainCo EV before any residual discount = 11,754.1 (10.19x on the 1,154 of EBITDA). With a 12 percent residual discount: 11,754.1 x 0.88 = 10,343.6. Net debt after the spin = 4,320 - 700 = 3,620. Equity = 10,343.6 - 3,620 - 390 = 6,333.6, or $19.79 per Sterling share.

Total value to a Sterling shareholder = 6,333.6 + 1,340.0 - 150 x 0.75 = 6,333.6 + 1,340.0 - 112.5 = 7,561.1, or $23.63 per share; +10.9 percent. Per share: RemainCo $19.79 + SpinCo $4.19 - costs $0.35 = $23.63.

Tax risk. If the spin fails the active-business test (10 percent), corporate tax = 0.25 x (2,040 - 1,150) = 222.5, or $0.70 per share; expected cost = 0.10 x 222.5 = 22.25 or $0.07 per share. The risk-adjusted value is $23.56. The expected cost is small but the tail matters: a taxable spin would cost $0.70 per share and might also create tax for shareholders, so counsel should consider restructuring (carving Brightline out before the spin, or retaining it).

Sensitivity grid: value per Sterling share by RemainCo residual discount (rows) and SpinCo discount (columns):

RemainCo residual discount	SpinCo 0%	SpinCo 5%	SpinCo 10%
8%	25.10	24.78	24.46
12%	23.63	23.31	22.99
15%	22.53	22.21	21.89
18%	21.42	21.11	20.79
22%	19.96	19.64	19.32

Break-even. The value equals the current market capitalization of 6,816 when the RemainCo residual discount is 18.3 percent (SpinCo at no discount) or 17.5 percent (SpinCo at 5 percent). Because today’s group trades at an implied 18.7 percent discount to its SOTP, the spin creates value only if it removes more than a small fraction of the existing discount. Each 1 percentage point of RemainCo discount is worth about 117.5 of equity (11,754.1 x 0.01) or $0.37 per share. That sensitivity makes it plain that the entire investment case rests on RemainCo’s rerating, an item that is largely out of management’s control.

Question 7: Sale of Hearthstone to Larkin

After-tax proceeds. Price 2,900 less costs 45 = 2,855. Gain over tax basis = 2,855 - 1,150 = 1,705. Tax = 0.25 x 1,705 = 426.25. Net cash = 2,855 - 426.25 = 2,428.75, or $7.59 per share. The price of 2,900 is 9.29x EBITDA of 312, a premium of 0.8x over the 8.5x median, reflecting Larkin’s synergies.

Value. RemainCo EV after a 12 percent residual discount is 10,343.6 (as in Question 6; the same RemainCo EBITDA of 1,154, because the same overhead and purchasing effects apply). Net debt = 4,320 - 2,428.75 = 1,891.25. Equity = 10,343.6 - 1,891.25 - 390 = 8,062.3, or $25.19 per share, +18.3 percent.

Why the sale beats the spin. Difference in equity = 8,062.3 - 7,561.1 = 501.2 ($1.57 per share):

Effect	$ million
Larkin pays 2,900 versus SpinCo EV of 2,040 (no stand-alone costs for Larkin)	+860.0
Corporate tax on the gain	(426.25)
Sale costs (45) in place of after-tax separation costs (112.5)	+67.5
Total	+501.25

Break-even price. Equity from the sale rises by 0.75 for each $1 of price (the after-tax fraction), so it equals the spin value of 7,561.1 at a price of 2,231.7, which is 7.15x EBITDA. The sale is better than the spin at any price above 7.15x, because the buyer absorbs the stand-alone costs that SpinCo would carry, and because the leverage and pension arrangements are identical. Price sensitivity: at 2,650 (8.5x) the value is $24.61; at 2,800 $24.96; at 2,900 $25.19; at 3,000 $25.43; at 3,100 $25.66.

Probability weighting. With a 70 percent chance of completion and a fallback of the spin at $23.63 - $0.40 = $23.23: expected value = 0.70 x 25.19 + 0.30 x 23.23 = 17.63 + 6.97 = $24.60 per share, +15.5 percent. The expected value of the dual-track strategy exceeds the spin alone by $0.97 per share. Even if Larkin’s probability of closing were as low as 30 percent, expected value would be 0.30 x 25.19 + 0.70 x 23.23 = $23.82, still above a pure spin (the fallback’s $0.40 haircut is smaller than the sale’s advantage).

Uses of proceeds. The 2,428.75 of cash reduces net debt to 1.64x EBITDA. Management could retain it as debt reduction (credit-supportive), distribute it through a repurchase, or invest it in Halvard. Base-case value counts cash at par. Re-levering to 2.5x would add 993.75 of debt capacity and a tax shield of 0.25 x 993.75 = 248.4 ($0.78 per share) if used permanently, but a repurchase of 25 percent of the equity at the market price would be dilutive to value if the shares were undervalued and accretive if not, an issue of price rather than of the transaction. EPS: if only debt is repaid (interest saved 2,428.75 x 5.35 percent = 129.9), net income falls from 523.5 to 474.0 and EPS from $1.64 to $1.48, a 9.5 percent drop that the board must be ready to explain; value per share nevertheless rises.

Execution risks: buyer financing, antitrust clearance, the Ohio plant contract (Exhibit G), and the leak risk of a failed process. The reverse break fee of 3 percent (87) compensates only partly.

Question 8: Carve-out of Halvard

Stand-alone EBITDA = 440 - 55 - 2 = 383. EV = 383 x 12.5 = 4,787.5. Debt = 2.5 x 383 = 957.5. Equity = 3,830.0. Stake sold = 25 percent = 957.5 at fair value; at a 10 percent IPO discount, gross proceeds = 861.75. Fees 3.5 percent = 30.16. Tax basis of shares sold = 25 percent of 1,600 = 400. Gain = 861.75 - 30.16 - 400 = 431.59; tax = 107.90. Net cash = 861.75 - 30.16 - 107.90 = 723.69, or $2.26 per share.

Value of the parent after the IPO. Retained 75 percent at fair value = 2,872.5, less a 10 percent holding-company discount = 2,585.25. Remaining B, C and D divisions: EV at median multiples = 4,845 + 2,448 + 2,652 = 9,945 on EBITDA of 1,128 (blended 8.816x). Corporate cost = 128 - 30 + 4 = 102; capitalized = 102 x 8.816 = 899.3; EV = 9,045.7. With a 12 percent residual discount: 7,960.2. Parent net debt = 4,320 - 957.5 (Halvard debt upstreamed) - 723.69 = 2,638.8. Pension 390. Separation cost 60 x 0.75 = 45.

Equity = 2,585.25 + 7,960.2 - 2,638.8 - 390 - 45 = 7,471.7, or $23.35 per share, +9.6 percent.

Sensitivity: holding-company discount h and residual discount on the remaining group dr:

h / dr	Value per share	vs $21.30
5% / 10%	24.36	+14.4%
10% / 10%	23.91	+12.3%
10% / 12% (base)	23.35	+9.6%
15% / 15%	22.05	+3.5%
10% / 18%	21.65	+1.7%

What the carve-out does and does not do. It raises $724 million of net cash, establishes a public price for the most valuable asset, and gives Halvard its own currency; it also takes about $96 million of value (10 percent of 957.5) from the parent's shareholders in the IPO discount, though the discount reverses into the retained stake. It does nothing about the structure of the conglomerate: the parent still contains four divisions in effect, still holds a listed subsidiary at a holding-company discount, and has simply added minority shareholders, a governance conflict, and stand-alone costs. For Redwater, whose argument is that the discount arises from bundling, a carve-out is the weakest response; it creates a price for the best asset while leaving the rest as it was. It may be a useful first step toward a full spin of Halvard, but it should not be presented as the answer.

Question 9: Ranking and alternatives

Option	Base-case value per share	Gain vs $21.30	Risk-adjusted view	Time to complete	Reversible?
Sale of Hearthstone to Larkin	25.19	+18.3%	24.60 (probability-weighted, with spin fallback)	about 6 months	No
Full four-way breakup	23.99	+12.6%	21.97 at 5% residual; 19.95 at 10%	12 months or more	No
Spin-off of Hearthstone	23.63	+10.9%	23.56 after tax risk	about 12 months	No
Carve-out of 25% of Halvard	23.35	+9.6%	21.65 to 24.36 across h, dr	about 5 months	Partly

Ranking. On expected value, the sale ranks first, the breakup second at zero residual discount (but it is the most fragile to the residual discount), the spin third and the carve-out fourth. On risk-adjusted value, the sale with a spin fallback dominates: it has the highest expected value, the fastest timing and the best credit outcome. The breakup's value is entirely dependent on an assumption that no discount remains, and its breakeven residual is only 6.7 percent. The spin is a good fallback. The carve-out is a poor stand-alone response but a possible step later.

Strategic effect. The sale and spin both simplify the group and focus it on three industrial businesses (the strongest case for a focused RemainCo). The carve-out leaves complexity unchanged. The breakup eliminates the diversified structure but destroys the shared services and purchasing scale and increases leverage constraints.

Marlowe alternative. Marlowe earns 6.0 percent against 9.5 percent and destroys $76 million of economic profit a year. The restructuring plan in Exhibit L would add 150 basis points of margin on revenue of 1,700, or 25.5 of EBITDA; at the 8.0x peer multiple that is 204 of enterprise value, against after-tax plant-closure costs of 90 x 0.75 = 67.5, a net gain of 136.5 or $0.43 per share. That is modest compared with the Hearthstone transactions ($1.57 per share of difference between sale and spin, $3.89 of gain from the sale versus the price) and it does not address the strategic issue. A sale of Marlowe would be defensible only if the

tax basis and a buyer's price justify it, and the case supplies neither; students who recommend it must say what data they would need (tax basis, buyer synergies, cyclical timing). The strongest answer treats the Marlowe plan as a complementary action alongside the Hearthstone exit, not a substitute.

Combinations. A dual-track process (negotiate with Larkin while preparing the registration for a spin-off) captures the sale's higher price while retaining a credible fallback and negotiating leverage. Preparing the spin in parallel costs perhaps $20 to $30 million of preparatory work (not in the case) but reduces the 30 percent failure risk's cost. A staged plan (exit Hearthstone in 2027, revisit Halvard in 2028 when Halvard has a longer record as a stand-alone division, and set a ROIC target for Marlowe) is a natural sequence.

Question 10: Recommendation and engagement with Redwater

The recommendation memo appears at the end of this note. The key elements are summarized here for discussion.

1. Pursue the sale of Hearthstone to Larkin as the first track and prepare the spin-off as the second track, with a decision point at signing or at 31 January 2027.
2. Do not pursue a four-way breakup or a Halvard carve-out now.
3. Announce a credit-supportive plan: use the proceeds to reduce net debt to no more than 2.5x EBITDA and review capital returns thereafter.
4. Commit to a value-creation program at Marlowe with a ROIC milestone, and a reduction of corporate cost.
5. Engage Redwater in a private meeting with the lead independent director before announcement; share the analysis (not the Larkin terms); offer a board refreshment discussion for two new independent directors with portfolio-management experience, chosen with the family trust's input; do not concede the four-way breakup.
6. Define the evidence that would change the recommendation: Larkin's price below $2,230 million (7.15x), the existence of a bid for Halvard or Kestrel at a significant premium, evidence that peer conglomerates are rerating after separations, or a tax opinion that the spin would be taxable.

Verification printout (Python replica of the central results):

```
SOTP EV 15445.0, corp 1260.8, equity 9474.2, per share 29.61, premium 39.0%
EV discount 18.7%, equity discount 28.1%
Breakup equity 7678.1, per share 23.99, gain 12.6%
Spin equity 7561.1 per share 23.63; Sale net cash 2428.75 equity 8062.3 per share 25.19
Prob-weighted sale 24.60
Carve-out net cash 723.69 equity 7471.7 per share 23.35
```

Sensitivity discussion

Three parameters drive every answer. The first is the residual discount on whatever remains diversified. Each percentage point is worth about $0.37 per share in the spin and sale cases (11,754 x 1 percent / 320), so a 6-point error in the assumed 12 percent moves the value by $2.20, as much as the entire expected gain from the spin. Students should be pushed to ask what evidence supports 12 percent. The three listed diversified peers suggest that investors apply a discount that is not much lower than Sterling's own; a RemainCo that still holds three businesses is hardly a pure play. A skeptical analyst might prefer 15 to 18 percent, in which case the spin yields only +3 to +6 percent and the carve-out is near break-even.

The second is the multiple. Every one-turn change in the peer multiples moves the four-way breakup by roughly $4 per share, and the selected ranges span more than three turns for Halvard. Breakup value is the most multiple-sensitive of the options because it depends on all four multiples, while the sale depends on a negotiated price that is partly fixed.

The third is tax. The sale loses $1.33 per share to tax (426.25 / 320), while the spin loses nothing if the tests are met. The sale still wins because Larkin's price carries a synergy premium and avoids stand-alone costs; if the price fell below 7.15x EBITDA, the spin would win. Tax basis is the key hidden variable: had Hearthstone's basis been 500 rather than 1,150, the tax would be 0.25 x (2,855 - 500) = 588.75, an additional 162.5, which would reduce the sale's value by $0.51 per share and narrow its advantage over the spin from $1.57 to $1.06.

Two further sensitivities deserve a mention. Debt: the value of the lost tax shield depends on the 25 percent rate and on the assumption that debt is permanent; with a lower rate the cost of deleveraging in a breakup is smaller. Timing: a 12-month spin has a longer exposure to market and execution risk than a six-month sale, and any comparison should compare prices on a like-for-like timing basis rather than discounting only the options.

Common student errors

1. Applying the SOTP without capitalizing corporate costs, which overstates value by $1,261 million ($3.94 per share).
2. Using the enterprise-value discount where Redwater quotes an equity discount (or the reverse), and then failing to reconcile 18.7 percent with 28.1 percent.
3. Forgetting to subtract the after-tax pension deficit or treating it as already in net debt.
4. Applying the 8.5x multiple to Hearthstone's EBITDA of 312 for SpinCo, ignoring the $20 million of extra stand-alone cost and $14 million of lost purchasing benefit.
5. Treating the Larkin price of 2,900 as cash to shareholders without deducting 45 of costs and 426 of tax, or ignoring that the spin is tax-free.
6. Allocating debt to the spin by the share of revenue or EBITDA without checking RemainCo's leverage and the agency trigger.
7. Taking the carve-out proceeds at the IPO price without the IPO discount, fees and tax, or valuing the retained stake without any holding-company discount.
8. Concluding that the breakup is best because its headline value is high, without testing the residual discount.
9. Ranking by value per share alone, without probability weighting, timing or the fallback.
10. Recommending the sale of Marlowe without noting that the data to value it (tax basis, buyer synergies) are missing.
11. Discounting the options by the cost of equity for the time to completion while leaving the status-quo price undiscounted.

Alternative defensible answers

Spin first. A board that is skeptical of Larkin's financing or that wants to retain the option of a later merger of Hearthstone may prefer the spin; it is defensible if the price offered falls below 7.15x EBITDA or if Larkin demands conditions on the Ohio plant that erode the price. *Full breakup, staged.* If students argue that a focused Kestrel and Halvard would not need the head office and that the residual discount would be lower than 12 percent, a staged breakup beginning with Hearthstone and ending with Marlowe is defensible, provided the student shows that the 6.7 percent break-even residual discount is credible. *Hold and fix.* Relying on the Marlowe plan, a tighter capital-allocation policy and a lower corporate cost base may deliver more than a separation if the market is already pricing Sterling in line with diversified peers (8.0x versus 7.6x). The student must then explain how to defend against the proxy contest. *Sale of Marlowe.* Defensible with data. *Other structures.* A tax-efficient sale of Hearthstone through a merger with a listed consumer company in which Sterling shareholders retain a majority (a "Reverse Morris Trust" type of structure, which in this stylized regime would be tax-free if the tests are met), or a sale of Halvard to an aerospace buyer at a strategic premium, are options that the case does not price; strong answers identify them as questions for the banker.

Recommendation memo to the board

To: The Board of Directors, Sterling Allied Industries **From:** Victor Aldana, Chief Executive; Priya Natarajan, Chief Financial Officer **Date:** 19 November 2026 **Subject:** Portfolio options and response to Redwater Partners

Recommendation. We recommend that the board (1) authorize exclusive negotiations with Larkin Brands for the sale of Hearthstone Living at not less than $2,900 million enterprise value, with a price floor equal to 7.15x EBITDA ($2,230 million), a reverse break fee, and the supply agreement for the Ohio plant settled before signing; (2) authorize preparation of a tax-free spin-off of Hearthstone as a fallback, with a decision point at signing or 31 January 2027; (3) defer any four-way breakup and any Halvard carve-out; (4) approve the Marlowe improvement program with an ROIC milestone and a $20 million corporate cost reduction target; and (5) authorize the lead independent director to meet Redwater before the announcement.

Why. On our analysis, a sum of the parts at median peer multiples gives $29.61 per share, 39 percent above the market price. That value is not available to shareholders. After stand-alone costs ($95 million a year), lost purchasing benefits ($38 million), separation costs, and the loss of debt capacity, a four-way breakup would deliver at most $23.99 per share (+12.6 percent), and would deliver less than the current price if the post-breakup companies trade at a discount of 7 percent or more to peers. A Hearthstone spin-off is worth $23.63 (+10.9 percent) in our base case, but it depends on the remaining group's discount falling from an implied 19 percent to 12 percent, and it pushes leverage toward the agency trigger. The sale to Larkin is worth $25.19 (+18.3 percent) if completed, $24.60 on a probability-weighted basis with the spin as fallback, reduces leverage from 3.0x to 1.6x, and can be announced before the nomination window. Its principal costs are $426 million of tax and a 9.5 percent reduction in EPS before any use of the proceeds.

Risks. Buyer financing, antitrust and the Ohio contract; the residual discount on the three-division group; the tax test on the Brightline business (10 percent probability, $0.70 per share if it fails); execution risk at Marlowe; and the danger that an announced sale that fails weakens our position with Redwater.

What we would do differently if facts change. If Larkin's price falls below $2,230 million, or if financing is not committed, we would proceed with the spin. If an unsolicited bidder emerges for Halvard or Kestrel at a significant premium, we would revisit the portfolio plan as a whole. If peer separations show that residual discounts fall to under 8 percent, we would accelerate a fuller separation.

Response to Redwater. We recommend telling Redwater that the board takes its analysis seriously, agrees that Hearthstone does not belong in the portfolio and that the discount is a legitimate concern, and has a concrete plan with milestones; and that the board does not agree that a four-way breakup would deliver 35 percent, and is prepared to show the reconciliation. We recommend offering to discuss board refreshment and incentive design (relative total shareholder return and ROIC measured after a capital-allocation hurdle). We do not recommend sharing the identity or terms of the Larkin proposal.

Part IV — Modeling Companion

Chapter 18: Excel Modeling Standards and Building an Integrated Three-Statement and DCF Model

Type: Modeling. Meridian Fastener Co., the company modeled in this chapter, is fictional and illustrative; every number was constructed for teaching. All Python listings were executed and the printed output shown is the actual output. The Excel layouts and formula patterns describe the same model so that the spreadsheet and the Python replica can be reconciled cell by cell.

Prerequisites: Volume 1, Chapters 2 (financial statements), 6 (net present value), 11 (WACC) and 21 (enterprise valuation); Volume 3, Chapters 1 to 16, which supply the valuation questions the model is meant to answer.

Why modeling discipline matters

Every case in this volume ends in a number: a net present value, an equity value per share, a recommended bid. Behind each number sits a model, and behind most bad decisions sits a model that nobody could audit. A spreadsheet is the only tool in the finance function whose logic is written by one person, edited by dozens, trusted by executives and almost never tested. The result is familiar to anyone who has worked in a deal team: a balance sheet that balances because a plug was inserted at 2 a.m., a discount rate typed over a formula, a terminal value that is eighty percent of enterprise value and nobody has questioned it, a scenario switch that changes the income statement but not the cash flow.

This chapter has two purposes. The first is to give you a standard, a set of rules about structure, color, flow, checks and documentation, that makes a model readable and testable by a stranger. The second is to apply the standard in a fully worked integrated model: an income statement, balance sheet and cash flow statement that articulate (that is, are mechanically linked so that the balance sheet balances without a plug), a debt schedule with a revolver and a deliberate circular reference, three scenarios, error checks, a discounted cash flow valuation and sensitivity tables. The model is built twice, once as a specification for Excel (layout, formula patterns, settings) and once as a Python program that you can run, whose output serves as the answer key against which a spreadsheet build can be verified.

Why a Python replica of a spreadsheet model? Because an independent second implementation is the strongest test of a model that exists. If the Excel model and the Python program disagree in the third decimal place, one of them contains a mistake, and finding it teaches more than any checklist. Professional model auditors do the same thing, rebuilding key calculations independently. The replica also lets us run experiments (hundreds of scenarios, a bisection search for a break-even assumption, deliberate injection of errors to see whether the checks catch them) that would be tedious in a workbook.

By the end of the chapter you should be able to:

1. Lay out a workbook so that inputs, calculations and outputs are separate, flow is left to right and top to bottom, and every row contains one formula copied across.
2. Apply a color and formatting convention that makes inputs, formulas and links distinguishable at a glance.
3. Build a three-statement model whose balance sheet balances through the cash flow statement and a revolver, and not through a plug.
4. Handle circularity from interest on average balances in a controlled way, with a circuit breaker.
5. Build integrity checks that catch the errors that actually occur, and demonstrate that they do.

6. Construct a DCF from the statements, interpret the terminal value, and cross-check Gordon growth against an exit multiple.
7. Produce scenario and sensitivity outputs, including Excel data tables, and know their limits.

Part I: Modeling standards

Principles

Published modeling standards (the FAST standard is a widely cited example, and most banks and consultancies maintain internal ones) differ in detail but agree on a small number of principles. We adopt the following and refer to them by number later.

S1. Separate inputs, calculations and outputs. Every hard-coded number lives in an input area and appears once. Calculation sheets contain only formulas. Output sheets (summaries, charts, sensitivity tables) contain only links and formulas. The test: if you delete all the inputs, no calculation cell should contain a constant.

S2. One formula per row. Across the time axis, the formula in the first forecast column is copied unchanged to every other forecast column. If a row needs a different formula in one column (a first-year special case, for example), the model is restructured with a flag row instead, so that the formula is still the same everywhere. The test: select a row, press Ctrl+ (go to row differences) and see no differences.

S3. Consistent time axis. The same column means the same period on every sheet. In our layout, column E is the last actual year (FY2025A) and columns F to J are FY2026E to FY2030E on every sheet. A reference from one sheet to another then always points at the same column letter, which makes audit trivial.

S4. Flow from left to right and top to bottom. Inputs precede calculations, which precede outputs; a formula should mostly refer to cells above and to the left of it. The exceptions (the circular interest loop, for example) are explicit and documented.

S5. No hard-coded numbers in formulas. The formula `=F10*1.08` embeds an assumption in a place nobody will look. Write `=F10*(1+Scenarios!F8)`. The only constants allowed in formulas are structural: 0, 1, 12 for months, 0.5 for the mid-year convention only if documented.

S6. Units and signs are explicit and consistent. Every row has a unit label ($m, %, days, x). Revenues and cash inflows are positive and costs and cash outflows are negative on the statements, so that subtotals are simple sums. Percentages are stored as decimals and displayed as percentages.

S7. Checks everywhere, one master flag. Every important identity (balance sheet balances, cash ties, roll-forwards) has a check cell; a master check on the cover sheet aggregates them and is visible from every sheet.

S8. Simple formulas, short chains. If a formula needs more than about three nested functions or wraps beyond one line, split it into helper rows. Avoid volatile functions (OFFSET, INDIRECT, TODAY) that recalculate on every keystroke and hide dependencies. Prefer INDEX and MATCH (or XLOOKUP) to VLOOKUP with approximate matching. Avoid merged cells, which break copy-paste, and avoid links to external workbooks.

S9. Document as you build. A cover sheet states purpose, version, author, units, scenario conventions, color legend, and a change log. Every assumption carries a source or a rationale in an adjacent notes column.

S10. Protect against yourself. Data validation on switch cells (a list of scenario names or 1 to 3), conditional formatting on checks (green when OK, red when not), locked formula cells on shared versions, and versioned file names rather than overwriting.

Why these rules, in terms of failure modes

Each principle answers a failure that occurs often enough to have a name. *The overwritten formula* (a number typed over a formula in one column) is defeated by S2 and by the row-difference test. *The orphaned assumption* (the same growth rate typed in three places and updated in two) is defeated by S1. *The plug* (a row labelled "other" that forces assets to equal liabilities) is defeated by S7 and, more fundamentally, by building the cash flow statement from the changes in the balance sheet, so that there is nothing left to plug. *The misaligned column* (revenue for FY2027 in a column where the tax rate is FY2026) is defeated by S3. *The unit error* (thousands added to millions) is defeated by S6. *The invisible circularity* (a reference loop that was not intended and that Excel quietly resolved or reported only in the status bar) is defeated by an explicit circuit breaker and by the rule that iterative calculation is turned on only for a documented reason.

Color and formatting conventions

Color is the cheapest audit tool. The following convention is used in many practitioner models; the specific colors matter less than consistency, and the legend must appear on the cover sheet.

Element	Font color	Fill	Meaning
Hard-coded input (number typed by a person)	Blue	Light yellow, on key switches	Change me
Formula on the same sheet	Black	None	Do not overwrite
Link to another sheet in the same workbook	Green	None	Pulled from elsewhere
Link to an external workbook	Red	None	Fragile; avoid
Check cell (OK)	White on dark green via conditional format		Pass
Check cell (error)	White on red via conditional format		Fail
Section header	White on dark blue		Navigation
Subtotal row	Black, bold, top border		Sum of rows above
Assumption flagged for review	Black on orange		Open question

Number formats should be uniform: `#,##0.0;(#,##0.0);"-"` for currency amounts (negatives in parentheses, zeros as dashes so that blank-looking rows can be told from zero rows), `0.0%` for percentages, `0.0x` for multiples, and `0` for years, formatted as text-like labels (`"FY"0"E"`) so that the underlying value remains numeric. Freeze panes at the first forecast column and the header row. Group (outline) detailed schedules so that the statements can be read at one level of detail. Set a consistent column width (12 to 14 characters) so printed pages align.

The Excel shortcut worth building into habit: Ctrl+[jumps to the precedents of a cell and Ctrl+] to its dependents; Alt+M+P and Alt+M+D trace precedents and dependents with arrows; Ctrl+` toggles formula view; F5, Special, Constants highlights all hard-coded numbers on a sheet (on a calculation sheet there should be none); and Ctrl+Shift+Enter has been replaced in modern Excel by dynamic arrays, but legacy array formulas should be avoided on shared models.

Workbook structure

Sheet	Contents	Contains hard-codes?
Cover	Purpose, version, color legend, change log, master check, scenario selector display	Text only
Inputs	Opening balance sheet (FY2025A), common assumptions (tax rate, depreciation rate, working-capital days, interest rates, minimum cash, payout, switches), DCF inputs (risk-free rate, beta, equity premium, target leverage, terminal growth, exit multiple)	Yes, all in blue
Scenarios	Three blocks (Base, Upside, Downside) of year-by-year drivers (revenue growth, COGS %, SG&A %) plus a “Live” block selected by the scenario switch	Yes (blue), plus the live block (formulas)
Model	Income statement, schedules (PP&E, working capital, debt and interest), cash flow statement, balance sheet, in this order, one sheet so that the flow can be audited by scrolling	No
DCF	Unlevered free cash flow, discount factors, terminal value, enterprise-to-equity bridge, implied multiples	No
Sens	Data tables or direct-formula sensitivity grids	No
Checks	Each check in its own row with a pass/fail flag and the aggregate	No

The same columns are used throughout: column A is a narrow spacer, B holds row labels, C the unit, D a constant or note (for example a per-row source), E the last actual year and F to J the five forecast years. The first calculation row of every sheet carries the year in row 5 (`=E5+1` across), a period counter in row 6 (0 for FY2025A, `=E6+1` across) and a period-end date if needed.

Named ranges

Named ranges for scalar inputs (`Tax`, `DepRate`, `CapexPct`, `ARDays`, `InvDays`, `APDays`, `DaysInYr`, `TermRate`, `RevRate`, `CashRate`, `TermAmort`, `MinCash`, `RevLimit`, `Payout`, `CircSwitch`, `Scen`, `WACC`, `TermG`, `ExitMult`, `MidYear`) make formulas readable, `=-DepRate*E40` instead of `=-Inputs!$D$12*E40`. Do not name time series (they are identified by their row and column) and do not name ranges to hide structure. The convention in this chapter is that a name always refers to a single input cell on the Inputs sheet.

Part II: The integrated model

The company and the architecture

Meridian Fastener Co. makes industrial fasteners and sells through distributors. At 31 December 2025 (FY2025A) it has revenue of $800.0 million, net working capital of $169.0 million, net PP&E of $400.0 million, a $200.0 million term loan, no revolver borrowings, $60.0 million of cash and 50.0 million shares outstanding. Its opening balance sheet is deliberately simple, with no goodwill, leases, deferred taxes or minorities, so that every line in the model is traceable. Extensions are left to the exercises.

Opening balance sheet, FY2025A ($ million)			
Cash	60.0	Accounts payable	61.0
Accounts receivable	120.0	Revolver	0.0
Inventory	110.0	Term loan	200.0
Net PP&E	400.0	Shareholders’ equity	429.0
Total assets	**690.0**	**Total liabilities and equity**	**690.0**

The model’s logic follows a fixed order, and the order is the architecture:

1. Drivers (growth, margins) produce revenue, cost of goods sold, SG&A and EBITDA.
2. Schedules produce depreciation, capex and closing PP&E; and receivables, inventory and payables, hence net working capital and its change.
3. The debt schedule produces the mandatory term-loan repayment.
4. The income statement subtracts depreciation and interest and applies tax to give net income.
5. The cash flow statement starts from net income, adds back non-cash items and subtracts investment, repayment and dividends, giving the cash position before the revolver.
6. The revolver draws or repays so that cash does not fall below a minimum; closing cash follows.
7. The balance sheet is assembled from the closing balances; equity rolls forward by net income less dividends.
8. A check compares total assets to total liabilities and equity.

The crucial discipline is step 7: nothing on the balance sheet is forced. Cash comes from the cash flow statement, PP&E from its schedule, working-capital items from the working-capital schedule, debt from the debt schedule and equity from its roll-forward. If the check shows zero, it is because the model is internally consistent, not because something was adjusted.

Assumptions

Assumption	Value	Excel name	Note
Tax rate	25%	Tax	Applied to positive pre-tax income only
Depreciation rate	10% of opening net PP&E	DepRate	Declining-balance simplification
Capex	6.5% of revenue	CapexPct	
Receivable days	55	ARDays	On revenue, 365-day year
Inventory days	80	InvDays	On cost of goods sold
Payable days	45	APDays	On cost of goods sold
Term loan rate	6.5%	TermRate	On average balance
Revolver rate	7.0%	RevRate	On average balance
Interest on cash	2.0%	CashRate	On average balance
Term loan amortization	$30.0 million a year	TermAmort	Mandatory
Minimum cash	$30.0 million	MinCash	Revolver draws to maintain
Revolver limit	$100.0 million	RevLimit	A check, not a constraint
Dividend payout	50% of net income	Payout	Paid in the year earned
Circularity switch	1	CircSwitch	1 = average balances, 0 = opening balances
Shares outstanding	50.0 million	Shares	

Scenario drivers, one value per year from FY2026E to FY2030E:

Driver	Scenario	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Revenue growth	Base	8.0%	7.0%	6.0%	5.0%	4.0%
	Upside	10.0%	9.0%	8.0%	7.0%	6.0%
	Downside	3.0%	1.0%	0.0%	1.0%	1.0%
COGS % revenue	Base	64.0%	63.8%	63.6%	63.4%	63.2%
	Upside	63.6%	63.2%	62.8%	62.4%	62.0%
	Downside	65.2%	65.8%	66.4%	66.4%	66.4%
SG&A % revenue	Base	19.0%	18.9%	18.8%	18.7%	18.6%
	Upside	18.9%	18.7%	18.5%	18.3%	18.1%
	Downside	19.4%	19.6%	19.8%	19.8%	19.8%

The full model in Python

The program below implements steps 1 to 8 exactly as the spreadsheet will. Save it as `model_core.py`; later listings import from it. The innermost loop reproduces what Excel does when iterative calculation is enabled: it recalculates the circular block repeatedly, starting from the opening balances, until the change between passes is negligible.

```
# model_core.py -- integrated three-statement model, Meridian Fastener Co. (illustrative)
# $ millions unless stated. Valuation date = 31 Dec 2025 (end of FY2025A).

YEARS = [2026, 2027, 2028, 2029, 2030]
N = len(YEARS)

# ---- Opening balance sheet (FY2025A) -------------------------------------
OPEN = dict(revenue=800.0, cash=60.0, ar=120.0, inv=110.0, ppe=400.0,
            ap=61.0, revolver=0.0, term=200.0, equity=429.0)

# ---- Assumptions that do not change by scenario --------------------------
COMMON = dict(tax=0.25, dep_rate=0.10, capex_pct=0.065,
              ar_days=55, inv_days=80, ap_days=45, days=365,
              term_rate=0.065, rev_rate=0.070, cash_rate=0.020,
              term_amort=30.0, min_cash=30.0, revolver_limit=100.0,
              payout=0.50, circ=1, shares=50.0)

# ---- Scenario-specific assumptions (one list per driver, one value per year)
SCEN = {
    "Base":     dict(g=[.080, .070, .060, .050, .040],
                     cogs=[.640, .638, .636, .634, .632],
                     sga=[.190, .189, .188, .187, .186]),
    "Upside":   dict(g=[.100, .090, .080, .070, .060],
                     cogs=[.636, .632, .628, .624, .620],
                     sga=[.189, .187, .185, .183, .181]),
    "Downside": dict(g=[.030, .010, .000, .010, .010],
                     cogs=[.652, .658, .664, .664, .664],
                     sga=[.194, .196, .198, .198, .198]),
}

def build(scn="Base", bug=None, **override):
    """Return dict of lists (one entry per forecast year)."""
    a = {**COMMON, **override}
    s = SCEN[scn]
    keys = ["revenue", "cogs", "gross", "sga", "ebitda", "dep", "ebit",
            "int_term", "int_rev", "int_inc", "ebt", "taxes", "ni",
            "ar", "inv", "ap", "nwc", "dnwc", "capex", "ppe",
            "cash", "revolver", "term", "equity", "div",
            "cfo", "cfi", "cff", "netchg", "iters", "assets", "liabeq", "check"]
    r = {k: [0.0] * N for k in keys}
    p = dict(OPEN)                      # prior-year balances
    p["nwc"] = p["ar"] + p["inv"] - p["ap"]
    for i in range(N):
        rev = p["revenue"] * (1 + s["g"][i])
        cogs = rev * s["cogs"][i]
        sga = rev * s["sga"][i]
        ebitda = rev - cogs - sga
        dep = a["dep_rate"] * p["ppe"]
        ebit = ebitda - dep
        capex = a["capex_pct"] * rev
        ppe = p["ppe"] + capex - dep
        ar = rev * a["ar_days"] / a["days"]
        inv = cogs * a["inv_days"] / a["days"]
        ap = cogs * a["ap_days"] / a["days"]
        nwc = ar + inv - ap
        dnwc = nwc - p["nwc"]
        term = p["term"] - a["term_amort"]
        # --- circular block: iterate exactly as Excel's iterative calc does
```

```
        revolver, cash = p["revolver"], p["cash"]          # starting guess
        for it in range(1, 201):
            avg = (lambda o, c: (o + c) / 2) if a["circ"] else (lambda o, c: o)
            int_term = a["term_rate"] * avg(p["term"], term)
            int_rev = a["rev_rate"] * avg(p["revolver"], revolver)
            int_inc = a["cash_rate"] * avg(p["cash"], cash)
            ebt = ebit - int_term - int_rev + int_inc
            taxes = max(0.0, ebt) * a["tax"]
            ni = ebt - taxes
            div = max(0.0, a["payout"] * ni)
            cfo = ni + (0.0 if bug == "no_dep_addback" else dep) - dnwc
            cfi = -capex
            cash_pre = p["cash"] + cfo + cfi - a["term_amort"] - div
            draw = max(-p["revolver"], a["min_cash"] - cash_pre)
            new_rev, new_cash = p["revolver"] + draw, cash_pre + draw
            done = abs(new_rev - revolver) + abs(new_cash - cash) < 1e-9
            revolver, cash = new_rev, new_cash
            if done:
                break
        cff = -a["term_amort"] - div + draw
        equity = p["equity"] + ni - (0.0 if bug == "div_not_in_equity" else div)
        assets = cash + ar + inv + ppe
        liabeq = ap + revolver + term + equity
        vals = dict(revenue=rev, cogs=cogs, gross=rev - cogs, sga=sga,
                    ebitda=ebitda, dep=dep, ebit=ebit, int_term=int_term,
                    int_rev=int_rev, int_inc=int_inc, ebt=ebt, taxes=taxes,
                    ni=ni, ar=ar, inv=inv, ap=ap, nwc=nwc, dnwc=dnwc,
                    capex=capex, ppe=ppe, cash=cash, revolver=revolver,
                    term=term, equity=equity, div=div, cfo=cfo, cfi=cfi,
                    cff=cff, netchg=cfo + cfi + cff, iters=it,
                    assets=assets, liabeq=liabeq, check=assets - liabeq)
        for k, v in vals.items():
            r[k][i] = v
        p = dict(revenue=rev, cash=cash, ar=ar, inv=inv, ppe=ppe, ap=ap,
                 revolver=revolver, term=term, equity=equity, nwc=nwc)
    return r
```

A few implementation notes. The dictionary `r` holds one list per line item, so `r["ni"][2]` is FY2028E net income. The `bug` argument exists only to inject errors for the testing section later. The `override` keyword arguments let any common assumption be changed in one call, which is how the exercises run. The convergence test sums the absolute change in the revolver and in cash; the loop usually stops after six or seven passes.

Excel layout: the Model sheet

The Model sheet is built with the following row map. Each formula shown is for column F (FY2026E) and is copied unchanged across to column J (S2). Column E holds FY2025A values, linked in green from the Inputs sheet. Costs are negative on the income statement and cash outflows are negative on the cash flow statement (S6).

Row	Line item	Formula in F	Python key
10	Revenue	`=E10*(1+Scenarios!F30)`	revenue
11	Cost of goods sold	`=-F10*Scenarios!F31`	cogs
12	Gross profit	`=F10+F11`	gross
13	SG&A	`=-F10*Scenarios!F32`	sga
14	EBITDA	`=F12+F13`	ebitda
15	Depreciation	`=-DepRate*E36`	dep
16	EBIT	`=F14+F15`	ebit
17	Interest, term loan	`=-TermRate*IF(CircSwitch=1,AVERAGE(E41,F41),E41)`	int_term
18	Interest, revolver	`=-RevRate*IF(CircSwitch=1,AVERAGE(E45,F45),E45)`	int_rev
19	Interest income	`=CashRate*IF(CircSwitch=1,AVERAGE(E70,F70),E70)`	int_inc
20	Pre-tax income	`=SUM(F16:F19)`	ebt
21	Income tax	`=-MAX(0,F20)*Tax`	taxes
22	Net income	`=F20+F21`	ni
26	Accounts receivable	`=F10*ARDays/DaysInYr`	ar
27	Inventory	`=-F11*InvDays/DaysInYr`	inv
28	Accounts payable	`=-F11*APDays/DaysInYr`	ap
29	Net working capital	`=F26+F27-F28`	nwc
30	Increase in NWC	`=F29-E29`	dnwc
33	PP&E, opening	`=E36`	
34	Capex	`=CapexPct*F10`	capex
35	Depreciation	`=F15`	
36	PP&E, closing	`=SUM(F33:F35)`	ppe
39	Term loan, opening	`=E41`	
40	Repayment	`=-MIN(TermAmort,F39)`	
41	Term loan, closing	`=F39+F40`	term
43	Revolver, opening	`=E45`	
44	Draw / (repayment)	`=F67`	
45	Revolver, closing	`=F43+F44`	revolver
57	Net income	`=F22`	
58	Depreciation	`=-F15`	
59	(Increase) in NWC	`=-F30`	
60	Cash from operations	`=SUM(F57:F59)`	cfo
61	Capex	`=-F34`	
62	Cash from investing	`=F61`	cfi
63	Term loan repayment	`=F40`	
64	Dividends	`=-MAX(0,Payout*F22)`	div
65	Cash flow before revolver	`=F60+F62+F63+F64`	
66	Cash before revolver	`=E70+F65`	
67	Revolver draw / (repayment)	`=MAX(-E77,MinCash-F66)`	

Row	Line item	Formula in F	Python key
68	Net change in cash	`=F65+F67`	netchg
69	Closing cash	`=E70+F68`	cash
70	Balance sheet: cash	`=F69`	cash
71	Accounts receivable	`=F26`	
72	Inventory	`=F27`	
73	Net PP&E	`=F36`	
74	Total assets	`=SUM(F70:F73)`	assets
76	Accounts payable	`=F28`	
77	Revolver	`=F45`	
78	Term loan	`=F41`	
79	Shareholders' equity	`=E79+F22+F64`	equity
80	Total liabilities and equity	`=SUM(F76:F79)`	liabeq
81	Balance check	`=ROUND(F74-F80,6)`	check

Study three of these formulas. The revolver formula in row 67, `=MAX(-E77, MinCash-F66)`, says: the change in the revolver is whichever is larger of (a) minus the opening revolver balance (you cannot repay more than you owe) and (b) the amount needed to bring cash before the revolver up to the minimum. If cash before the revolver is above the minimum, term (b) is negative and the revolver is repaid, but never below zero; if it is below, term (b) is positive and the revolver is drawn. One formula handles draw and repayment, which is why it needs no IF statement. The equity formula in row 79 adds net income and the (negative) dividend, so nothing else touches equity. And the interest formulas use the circuit breaker discussed below.

Row 19's reference to `F70` (closing cash) together with row 18's reference to `F45` creates the circular chain: interest depends on closing balances, which depend on cash flow, which depends on net income, which depends on interest.

The assembled statements for the base case follow. The program that produced them is below; it uses a small helper module for formatting.

```
# report.py -- print statements as fixed-width tables
from model_core import *

def row(label, vals, fmt="{:>9.1f}", opening=None):
    cells = ([fmt.format(opening)] if opening is not None else [" " * 9]) + [fmt.format(0.0 if abs(v) < 5e-5
else v) for v in vals]
    return f"{label:<28}" + "".join(cells)

def header():
    return f"{'$ millions':<28}" + "".join(f"{y:>9}" for y in ["FY2025A"] + [f"FY{y}E" for y in YEARS])

def income_statement(r):
    L = [header()]
    for lab, k, sign in [("Revenue", "revenue", 1), ("Cost of goods sold", "cogs", -1),
                         ("Gross profit", "gross", 1), ("SG&A", "sga", -1),
                         ("EBITDA", "ebitda", 1), ("Depreciation", "dep", -1),
                         ("EBIT", "ebit", 1), ("Interest, term loan", "int_term", -1),
                         ("Interest, revolver", "int_rev", -1), ("Interest income", "int_inc", 1),
                         ("Pre-tax income", "ebt", 1), ("Income tax", "taxes", -1),
                         ("Net income", "ni", 1)]:
        op = OPEN["revenue"] if k == "revenue" else None
        L.append(row(lab, [sign * v for v in r[k]], opening=op))
    return "\n".join(L)

def balance_sheet(r):
    L = [header()]
    o = OPEN
    for lab, k, op in [("Cash", "cash", o["cash"]), ("Accounts receivable", "ar", o["ar"]),
                       ("Inventory", "inv", o["inv"]), ("Net PP&E", "ppe", o["ppe"]),
                       ("Total assets", "assets", o["cash"] + o["ar"] + o["inv"] + o["ppe"]),
                       ("Accounts payable", "ap", o["ap"]), ("Revolver", "revolver", o["revolver"]),
                       ("Term loan", "term", o["term"]), ("Shareholders' equity", "equity", o["equity"]),
                       ("Total liabilities + equity", "liabeq", o["ap"] + o["revolver"] + o["term"] +
o["equity"]),
                       ("Balance check", "check", 0.0)]:
        L.append(row(lab, r[k], opening=op, fmt="{:>9.1f}" if k != "check" else "{:>9.4f}"))
    return "\n".join(L)

def cash_flow(r):
    L = [header()]
    rows = [("Net income", r["ni"]), ("Depreciation", r["dep"]),
            ("(Increase) in net working capital", [-x for x in r["dnwc"]]),
            ("Cash from operations", r["cfo"]), ("Capital expenditure", r["cfi"]),
            ("Term loan repayment", [-COMMON["term_amort"]] * N),
            ("Dividends", [-x for x in r["div"]]),
            ("Revolver draw / (repayment)", [r["revolver"][i] - (OPEN["revolver"] if i == 0 else r["revolver"]
[i - 1]) for i in range(N)]),
            ("Net change in cash", r["netchg"]),
            ("Closing cash", r["cash"])]
    for lab, v in rows:
        L.append(row(lab, v))
    return "\n".join(L)
```

```
from report import *
r = build("Base")
print("INCOME STATEMENT"); print(income_statement(r))
print(); print("BALANCE SHEET"); print(balance_sheet(r))
print(); print("CASH FLOW STATEMENT"); print(cash_flow(r))
print(); print("Iterations to converge each year:", r["iters"])
```

```
INCOME STATEMENT
$ millions                    FY2025A  FY2026E  FY2027E  FY2028E  FY2029E  FY2030E
Revenue                        800.0    864.0    924.5    979.9   1028.9   1070.1
Cost of goods sold                     -553.0   -589.8   -623.2   -652.4   -676.3
Gross profit                            311.0    334.7    356.7    376.6    393.8
SG&A                                   -164.2   -174.7   -184.2   -192.4   -199.0
EBITDA                                  146.9    159.9    172.5    184.2    194.8
Depreciation                            -40.0    -41.6    -43.5    -45.5    -47.6
EBIT                                    106.9    118.3    129.0    138.7    147.1
Interest, term loan                     -12.0    -10.1     -8.1     -6.2     -4.2
Interest, revolver                        0.0     -0.5     -1.6     -2.7     -3.4
Interest income                           1.0      0.7      0.6      0.6      0.6
Pre-tax income                           95.8    108.4    119.8    130.5    140.1
Income tax                              -24.0    -27.1    -30.0    -32.6    -35.0
Net income                               71.9     81.3     89.9     97.8    105.1

BALANCE SHEET
$ millions                    FY2025A  FY2026E  FY2027E  FY2028E  FY2029E  FY2030E
Cash                            60.0     35.6     30.0     30.0     30.0     30.0
Accounts receivable            120.0    130.2    139.3    147.7    155.0    161.2
Inventory                      110.0    121.2    129.3    136.6    143.0    148.2
Net PP&E                       400.0    416.2    434.6    454.9    476.3    498.2
Total assets                   690.0    703.1    733.2    769.1    804.3    837.7
Accounts payable                61.0     68.2     72.7     76.8     80.4     83.4
Revolver                         0.0      0.0     14.9     31.8     44.4     52.3
Term loan                      200.0    170.0    140.0    110.0     80.0     50.0
Shareholders' equity           429.0    464.9    505.6    550.5    599.4    652.0
Total liabilities + equity     690.0    703.1    733.2    769.1    804.3    837.7
Balance check                 0.0000   0.0000   0.0000   0.0000   0.0000   0.0000

CASH FLOW STATEMENT
$ millions                    FY2025A  FY2026E  FY2027E  FY2028E  FY2029E  FY2030E
Net income                              71.9     81.3     89.9     97.8    105.1
Depreciation                            40.0     41.6     43.5     45.5     47.6
(Increase) in net working capital            -14.2    -12.6    -11.6    -10.2     -8.5
Cash from operations                    97.6    110.3    121.8    133.2    144.2
Capital expenditure                    -56.2    -60.1    -63.7    -66.9    -69.6
Term loan repayment                    -30.0    -30.0    -30.0    -30.0    -30.0
Dividends                              -35.9    -40.6    -44.9    -48.9    -52.5
Revolver draw / (repayment)              0.0     14.9     16.9     12.6      7.9
Net change in cash                     -24.4     -5.6      0.0      0.0      0.0
Closing cash                            35.6     30.0     30.0     30.0     30.0

Iterations to converge each year: [6, 7, 7, 7, 7]
```

Reading the output

Verify FY2026E by hand, as you should whenever you build a model, before trusting any later year. Revenue is 800.0 x 1.08 = 864.0. Cost of goods sold is 864.0 x 0.640 = 553.0 (553.0 to one decimal; the exact figure is 552.96) and SG&A is 864.0 x 0.190 = 164.2, so EBITDA is 146.9, a margin of 17.0 percent. Depreciation is 10 percent of opening net PP&E: 0.10 x 400.0 =

40.0, giving EBIT of 106.9. Term loan interest is 6.5 percent of the average of 200.0 and 170.0: 0.065 x 185.0 = 12.0. Interest income is 2 percent of the average of 60.0 and 35.6, which is 1.0. Pre-tax income is 95.8, tax is 24.0 and net income is 71.9.

Working capital: receivables are 864.0 x 55 / 365 = 130.2; inventory 553.0 x 80 / 365 = 121.2; payables 553.0 x 45 / 365 = 68.2. Net working capital is 130.2 + 121.2 - 68.2 = 183.2, up 14.2 from 169.0. Cash from operations is therefore 71.9 + 40.0 - 14.2 = 97.6. Capex is 6.5 percent of revenue, 56.2. After the 30.0 term loan repayment and dividends of 35.9 (half of net income), cash falls by 24.4, from 60.0 to 35.6, above the 30.0 minimum, so no revolver is needed in FY2026E.

From FY2027E the revolver is needed: the company is profitable and growing, yet distributes half its earnings and repays 30.0 of the term loan a year while investing in growth, so it borrows on the revolver (14.9 in FY2027E, rising to 52.3 by FY2030E) to meet the amortization. Total debt falls from 200.0 to 102.3 over the five years, but the mix shifts toward the more expensive revolver. That is a real finding that the model produces and that a three-line “free cash flow minus dividends” calculation would miss. The balance check row is zero in every year, and the cash flow statement’s closing cash equals the balance sheet’s cash by construction.

Part III: Integrity checks and circularity

What to check, and how

A check is a formula that returns pass or fail, written so that it fails when the model is wrong in a way that matters. The balance sheet check is necessary and not sufficient: a model can balance and still be wrong (a mislinked growth rate does not unbalance anything). A good suite therefore tests identities from several directions.

#	Check	What it catches	Excel pattern (column F, 0 = pass)
1	Assets equal liabilities plus equity	Unlinked or mislinked balance sheet items; sign errors	`=IF(ABS(Model!F81)<Tol,0,1)`
2	Cash flow closing cash equals balance sheet cash	Cash flow statement built independently of the balance sheet	`=IF(ABS(Model!E70+Model!F68-Model!F70)<Tol,0,1)`
3	Equity roll-forward	Dividends or net income omitted from equity	`=IF(ABS(Model!E79+Model!F22+Model!F64-Model!F79)<Tol,0,1)`
4	PP&E roll-forward	Capex or depreciation mislinked	`=IF(ABS(Model!E73+Model!F34+Model!F15-Model!F73)<Tol,0,1)`
5	Cash at least the minimum, unless the revolver is at its limit	Revolver logic failure	`=IF(OR(Model!F70>=MinCash-Tol,Model!F77>=RevLimit-Tol),0,1)`
6	Revolver within its limit	Liquidity shortfall (a business finding, not a modeling error)	`=IF(Model!F77<=RevLimit+Tol,0,1)`
7	No negative debt balances	Over-repayment	`=IF(MIN(Model!F41,Model!F77)>=-Tol,0,1)`
8	Circularity converged	Stale iteration values	`=IF(ABS(Model!F17+TermRate*IF(CircSwitch=1,AVERAGE(Model!E41,Model!F41),Model!E41))<Tol,0,1)`
9	Subtotals recompute	Overwritten subtotals	`=IF(ABS(Model!F12-(Model!F10+Model!F11))<Tol,0,1)`
10	No error values anywhere	`#REF!`, `#DIV/0!` propagating	`=IF(SUMPRODUCT(--ISERROR(Model!F10:F81))>0,1,0)`

The tolerance `Tol` is a named input, 0.000001. The Checks sheet puts each check in its own row, with the five forecast years across; a column at the right sums each row; a cell at the top, `=IF(SUM(F5:J14)=0,"OK","ERROR")`, is the master flag, and the Cover sheet links to it so that it is visible everywhere. Conditional formatting turns the flag green or red. Two refinements: check 6 is a business flag (the model is right, the company is short of liquidity) and deserves a different color from the pure integrity checks; and each check should have a "last tested" note.

The Python equivalent follows. It takes a model result and returns a dictionary of named pass/fail results and a master flag.

```
# checks.py -- integrity checks that feed one master flag
from model_core import *

TOL = 1e-6

def run_checks(r, a=COMMON):
    out = {}
    out["1 Balance sheet balances"] = max(abs(x) for x in r["check"]) < TOL
    # cash flow statement ties to balance sheet cash
    prev = OPEN["cash"]; ok = True
    for i in range(N):
        ok &= abs(prev + r["netchg"][i] - r["cash"][i]) < TOL
        prev = r["cash"][i]
    out["2 Cash flow ties to balance sheet cash"] = ok
    # equity roll-forward
    prev = OPEN["equity"]; ok = True
    for i in range(N):
        ok &= abs(prev + r["ni"][i] - r["div"][i] - r["equity"][i]) < TOL
        prev = r["equity"][i]
    out["3 Equity roll-forward"] = ok
    # PP&E roll-forward
    prev = OPEN["ppe"]; ok = True
    for i in range(N):
        ok &= abs(prev + r["capex"][i] - r["dep"][i] - r["ppe"][i]) < TOL
        prev = r["ppe"][i]
    out["4 PP&E roll-forward"] = ok
    out["5 Cash never below minimum"] = min(r["cash"]) >= a["min_cash"] - TOL or \
        all(r["cash"][i] >= a["min_cash"] - TOL or r["revolver"][i] >= a["revolver_limit"] - TOL for i in
range(N))
    out["6 Revolver within limit"] = max(r["revolver"]) <= a["revolver_limit"] + TOL
    out["7 No negative debt balances"] = min(min(r["revolver"]), min(r["term"])) >= -TOL
    out["8 Circularity converged"] = max(r["iters"]) < 200
    out["9 Gross profit = revenue - COGS"] = all(abs(r["gross"][i] - (r["revenue"][i] - r["cogs"][i])) < TOL
for i in range(N))
    return out

def master(r, a=COMMON):
    c = run_checks(r, a)
    return all(c.values()), c
```

A test suite is only trustworthy if it has been shown to fail. The next listing runs the checks on the correct model, then deliberately injects two classic errors using the `bug` argument, one at a time: forgetting to add depreciation back in the cash flow statement, and forgetting to deduct dividends from equity. It also cuts the revolver limit to $40 million to show a business flag.

```
from checks import *
ok, c = master(build("Base"))
print("BASE CASE: master check =", "OK" if ok else "ERROR")
for k, v in c.items(): print(f"  {k:<42}{'pass' if v else 'FAIL'}")
for bug in ["no_dep_addback", "div_not_in_equity"]:
    r = build("Base", bug=bug)
    ok, c = master(r)
    print(f"\nINJECTED BUG '{bug}': master check =", "OK" if ok else "ERROR")
    print("  balance check by year:", [round(x, 2) for x in r["check"]])
    print("  failing tests:", [k for k, v in c.items() if not v])
r = build("Base", revolver_limit=40.0)
ok, c = master(r, {**COMMON, "revolver_limit": 40.0})
print("\nRevolver limit cut to 40: master check =", "OK" if ok else "ERROR", "| failing:", [k for k, v in
c.items() if not v])
print("  revolver balances:", [round(x, 1) for x in r["revolver"]])
```

```
BASE CASE: master check = OK
  1 Balance sheet balances                   pass
  2 Cash flow ties to balance sheet cash     pass
  3 Equity roll-forward                      pass
  4 PP&E roll-forward                        pass
  5 Cash never below minimum                 pass
  6 Revolver within limit                    pass
  7 No negative debt balances                pass
  8 Circularity converged                    pass
  9 Gross profit = revenue - COGS            pass

INJECTED BUG 'no_dep_addback': master check = ERROR
  balance check by year: [-40.0, -81.62, -125.08, -170.57, -218.19]
  failing tests: ['1 Balance sheet balances', '6 Revolver within limit']

INJECTED BUG 'div_not_in_equity': master check = ERROR
  balance check by year: [-35.93, -76.57, -121.51, -170.43, -222.98]
  failing tests: ['1 Balance sheet balances', '3 Equity roll-forward']

Revolver limit cut to 40: master check = ERROR | failing: ['6 Revolver within limit']
  revolver balances: [0.0, 14.9, 31.8, 44.4, 52.3]
```

Three lessons. First, only the balance check caught the missing depreciation add-back: the cash flow statement still tied to the balance sheet because both were wrong in the same way. The balance sheet was out by exactly 40.0 in FY2026E, the amount of the omitted depreciation, and the gap compounds (81.62 after two years), which is itself diagnostic: an error equal to a line item points to that line item. Second, the second bug was caught by two separate checks (balance and equity roll-forward), and the failing test names tell you where to look: a failure in 3 but not in 2 suggests the problem is in equity, not in cash. Third, cutting the revolver limit does not unbalance anything, but check 6 flags that the plan is infeasible with the committed facility. A model that never says “no” is not being tested.

An extra practice used by professional auditors is the “stress test”: set growth to 0 percent, or set capex to zero, or set the tax rate to 100 percent, and confirm that the checks still pass and that the result is what a calculation by hand predicts. Errors hide in the corners.

Circular references

The circularity in this model is not an accident. Interest expense on the revolver depends on the average of the opening and closing revolver balances; the closing balance depends on the cash flow; the cash flow depends on net income; and net income depends on interest expense. The same loop exists through interest income on cash and through dividends, which depend on net

income. Using average balances is more accurate than using opening balances (a company that borrows during the year pays interest on the borrowing for part of it) and is the standard choice in lending and valuation models. The cost is a loop that Excel will refuse to calculate unless iterative calculation is enabled.

Excel settings. Enable iterative calculation under File, Options, Formulas. Set maximum iterations to 100 and maximum change to 0.000001 (Excel's default of 0.001 is coarse). This is an application-level setting that is stored with the first workbook opened in a session, so a model that depends on it should say so prominently on the Cover sheet, and check 8 above detects a stale loop.

The circuit breaker. If an error value (such as `#DIV/0!` or `#VALUE!`) enters a circular loop it feeds itself and never clears, even after the cause is fixed. The remedy is a switch: the formulas in rows 17 to 19 use `IF(CircSwitch=1, AVERAGE(opening, closing), opening)`, so that setting the switch to 0 breaks the loop, flushes the error, and allows the switch to be turned back on. The Python code uses the same switch (`circ`). Do not wrap the whole loop in `IFERROR(..., 0)`: that hides the symptom and can leave the model calculating nonsense.

Why the loop converges. Each pass is a contraction. If the revolver balance is changed by a small amount x, interest changes by (rate / 2) x, after-tax income by (rate / 2)(1 - tax) x, and, since half of any change in income is paid out as dividends, cash before the revolver by (rate / 2)(1 - tax)(1 - payout) x, which is the change in the next pass. The factor is 0.07 / 2 x 0.75 x 0.50 = 0.0131, so each pass reduces the error by a factor of about 76, and a tolerance of 10 to the minus 9 is reached in six or seven passes. The next listing prints the iteration for FY2027E, starting from the opening balances (as Excel does) and shows the ratio of successive changes. It then compares the model with the loop on and off.

```
from model_core import *

r = build("Base")
i = 1                                          # FY2027: first year the revolver is drawn
p = dict(cash=r["cash"][i-1], revolver=r["revolver"][i-1], term=r["term"][i-1])
term = r["term"][i]
ebit, dnwc, capex = r["ebit"][i], r["dnwc"][i], r["capex"][i]
a = COMMON
revolver, cash = p["revolver"], p["cash"]     # starting guess = opening balances
print(f"{'iter':>4}{'interest':>10}{'net income':>12}{'revolver':>10}{'cash':>8}{'change':>12}")
for it in range(1, 9):
    int_term = a["term_rate"] * (p["term"] + term) / 2
    int_rev = a["rev_rate"] * (p["revolver"] + revolver) / 2
    int_inc = a["cash_rate"] * (p["cash"] + cash) / 2
    ebt = ebit - int_term - int_rev + int_inc
    ni = ebt * (1 - a["tax"])
    div = a["payout"] * ni
    cash_pre = p["cash"] + ni + r["dep"][i] - dnwc - capex - a["term_amort"] - div
    draw = max(-p["revolver"], a["min_cash"] - cash_pre)
    new_rev = p["revolver"] + draw
    change = abs(new_rev - revolver)
    revolver, cash = new_rev, cash_pre + draw
    print(f"{it:>4}{int_term + int_rev - int_inc:>10.5f}{ni:>12.5f}{revolver:>10.5f}{cash:>8.2f}
{change:>12.7f}")

print()
on, off = build("Base", circ=1), build("Base", circ=0)
print(f"{'':<28}{'circ ON':>10}{'circ OFF':>10}{'difference':>12}")
for lab, k in [("FY2027 net interest cost", None), ("FY2030 revolver", "revolver"), ("FY2030 net income",
"ni"), ("FY2030 equity", "equity")]:
    if k is None:
        f = lambda x: x["int_term"][1] + x["int_rev"][1] - x["int_inc"][1]
        a_, b_ = f(on), f(off)
    else:
        a_, b_ = on[k][-1], off[k][-1]
    print(f"{lab:<28}{a_:>10.3f}{b_:>10.3f}{a_ - b_:>12.3f}")
```

```
iter  interest  net income  revolver    cash      change
   1   9.36393    81.71633  14.71120   30.00  14.7112008
   2   9.93436    81.28851  14.92511   30.00   0.2139105
   3   9.94184    81.28290  14.92792   30.00   0.0028076
   4   9.94194    81.28282  14.92796   30.00   0.0000368
   5   9.94194    81.28282  14.92796   30.00   0.0000005
   6   9.94194    81.28282  14.92796   30.00   0.0000000
   7   9.94194    81.28282  14.92796   30.00   0.0000000
   8   9.94194    81.28282  14.92796   30.00   0.0000000

                             circ ON  circ OFF  difference
FY2027 net interest cost       9.942    10.344      -0.402
FY2030 revolver               52.315    53.393      -1.078
FY2030 net income            105.091   104.525       0.566
FY2030 equity                651.978   650.900       1.078
```

The successive changes (14.71, 0.2139, 0.0028, 0.000037) shrink by a factor of 69 to 76 per pass, matching the theoretical 1 / 0.0131 = 76. The comparison shows the cost of the shortcut: using opening balances overstates FY2027E net interest cost by 0.402 (4.0 percent) mainly because it charges interest on the full opening term loan even though 30.0 of it is repaid during the

year, and it puts FY2030E equity out by 1.078. Small in a base case; larger when debt is a bigger share of the capital structure, as in leveraged buyouts (Volume 1, Chapter 23).

Alternatives to iteration, for completeness: (i) calculate interest on opening balances (no loop, but less accurate); (ii) solve the loop algebraically (possible for a single revolver but cumbersome with several tranches); (iii) a macro that copies the interest values and pastes them as constants until they stop changing (robust but requires macros and leaves stale values if someone forgets to run it). For models that will be handed to others, a switch and clear documentation are usually the best compromise.

Part IV: The discounted cash flow valuation

From statements to unlevered free cash flow

A DCF values the operations of the business, independent of how they are financed (Volume 1, Chapter 21). The cash flow to be discounted is therefore unlevered free cash flow:

```
UFCF = EBIT x (1 - tax rate) + Depreciation - Capex - Increase in net working capital
```

Notice what is absent: interest, interest income, dividends, the revolver, the term loan. Financing enters the valuation in two places only: the discount rate (through the weighted average cost of capital, which reflects the interest tax shield) and the bridge from enterprise value to equity value (through net debt). A valuation that changes when you change the dividend payout is wrong; Exercise 2 demonstrates that this model passes that test.

Excel layout, DCF sheet (columns F to J, as always).

Row	Line	Formula in F
10	EBIT	`=Model!F16`
11	Tax on EBIT	`=-F10*Tax`
12	NOPAT	`=F10+F11`
13	Add: depreciation	`=-Model!F15`
14	Less: capex	`=Model!F61`
15	Less: increase in NWC	`=-Model!F30`
16	Unlevered free cash flow	`=SUM(F12:F15)`
17	Discount period (years)	`=F$6-IF(MidYear=1,0.5,0)`
18	Discount factor	`=1/(1+WACC)^F17`
19	Present value of UFCF	`=F16*F18`

WACC inputs sit on the Inputs sheet: risk-free rate 4.0 percent, equity beta 1.10, equity risk premium 5.0 percent, pre-tax cost of debt 6.5 percent, target debt-to-value 25 percent. The cost of equity is `=Rf+Beta*ERP` = 9.5 percent and WACC is `=(1-DV)*Ke+DV*Kd*(1-Tax)` = 0.75 x 9.5 percent + 0.25 x 6.5 percent x 0.75 = 8.34 percent (Volume 1, Chapter 11). Note that the WACC uses a target capital structure, not the model's own forecast leverage, which falls over the period.

Terminal value and the bridge:

Item	Formula
Terminal value, Gordon growth	`=J16*(1+TermG)/(WACC-TermG)`
PV of terminal value	`=TV/(1+WACC)^J17`
Terminal value, exit multiple (cross-check)	`=ExitMult*Model!J14`
PV of exit-multiple terminal value	`=TVexit/(1+WACC)^J6`
Enterprise value	`=SUM(F19:J19)+PVTV`
Less: net debt at valuation date	`=Inputs!TermLoan+Inputs!Revolver-Inputs!Cash`
Equity value	`=EV-NetDebt`
Value per share	`=Equity/Shares`
Implied terminal EV/EBITDA	`=TV/Model!J14`

Do not use Excel's NPV function with the mid-year convention (it assumes end-of-period flows and equally spaced periods starting one period out) and be careful with XNPV (it discounts on actual dates using a 365-day year). `SUMPRODUCT` over the discount-factor row, or the explicit PV row above, is transparent and auditable.

Two conventions need explaining. The **mid-year convention** discounts each year's cash flow at t - 0.5 years, on the logic that cash arrives through the year and not on the last day. It raises value by roughly half a year of discounting (about WACC / 2, here 4 percent). With a Gordon growth terminal value computed on mid-year flows, the terminal value is also discounted at N - 0.5 (the formula above uses the same J17 period); an exit-multiple terminal value, which represents a sale price at the end of year N, is discounted at N. The **valuation date** is 31 December 2025, so the net debt used is the opening balance sheet's: 200.0 + 0.0 - 60.0 = 140.0.

The DCF in Python

```
# dcf.py -- WACC and DCF layered on model_core
from model_core import *

def wacc(rf=0.04, erp=0.05, beta=1.10, kd=0.065, tax=0.25, dv=0.25):
    ke = rf + beta * erp
    return (1 - dv) * ke + dv * kd * (1 - tax), ke

def ufcf(r, tax=COMMON["tax"]):
    return [r["ebit"][i] * (1 - tax) + r["dep"][i] - r["capex"][i] - r["dnwc"][i]
            for i in range(N)]

def dcf(r, w, g, midyear=True, exit_mult=None, tax=COMMON["tax"],
        net_debt=OPEN["term"] + OPEN["revolver"] - OPEN["cash"], shares=COMMON["shares"]):
    f = ufcf(r, tax)
    t = [i + 1 - (0.5 if midyear else 0.0) for i in range(N)]
    pv = [f[i] / (1 + w) ** t[i] for i in range(N)]
    if exit_mult is None:                       # Gordon growth on final-year FCF
        tv = f[-1] * (1 + g) / (w - g)
        pv_tv = tv / (1 + w) ** (N - (0.5 if midyear else 0.0))
    else:                                       # exit multiple on final-year EBITDA
        tv = exit_mult * r["ebitda"][-1]
        pv_tv = tv / (1 + w) ** N
    ev = sum(pv) + pv_tv
    eq = ev - net_debt
    return dict(fcf=f, t=t, pv=pv, tv=tv, pv_tv=pv_tv, ev=ev, equity=eq,
                price=eq / shares, tv_share=pv_tv / ev,
                implied_mult=tv / r["ebitda"][-1])
```

```
from dcf import *
r = build("Base")
w, ke = wacc()
print(f"Cost of equity {ke:.4%}; after-tax cost of debt {0.065*(1-0.25):.4%}; WACC {w:.4%}")
d = dcf(r, w, 0.025, midyear=True)
print(f"\n{'':<22}" + "".join(f"{'FY'+str(y)+'E':>10}" for y in YEARS))
print(f"{'EBIT':<22}" + "".join(f"{x:>10.1f}" for x in r["ebit"]))
print(f"{'NOPAT (EBIT x 0.75)':<22}" + "".join(f"{x*0.75:>10.1f}" for x in r["ebit"]))
print(f"{'+ Depreciation':<22}" + "".join(f"{x:>10.1f}" for x in r["dep"]))
print(f"{'- Capex':<22}" + "".join(f"{-x:>10.1f}" for x in r["capex"]))
print(f"{'- Increase in NWC':<22}" + "".join(f"{-x:>10.1f}" for x in r["dnwc"]))
print(f"{'Unlevered FCF':<22}" + "".join(f"{x:>10.2f}" for x in d["fcf"]))
print(f"{'Discount period (yrs)':<22}" + "".join(f"{x:>10.1f}" for x in d["t"]))
print(f"{'Discount factor':<22}" + "".join(f"{1/(1+w)**x:>10.4f}" for x in d["t"]))
print(f"{'PV of FCF':<22}" + "".join(f"{x:>10.2f}" for x in d["pv"]))
print(f"\nSum of PV of FCF        {sum(d['pv']):>10.2f}")
print(f"Terminal value (Gordon) {d['tv']:>10.2f}   implied EV/EBITDA {d['implied_mult']:.2f}x")
print(f"PV of terminal value    {d['pv_tv']:>10.2f}   ({d['tv_share']:.1%} of EV)")
print(f"Enterprise value        {d['ev']:>10.2f}")
print(f"Less net debt (200 - 60){140:>10.2f}")
print(f"Equity value            {d['equity']:>10.2f}")
print(f"Value per share ($)     {d['price']:>10.2f}")
print()
for mid in (True, False):
    x = dcf(r, w, 0.025, midyear=mid)
    print(f"{'Mid-year' if mid else 'End-year'} convention: EV {x['ev']:.1f}, price {x['price']:.2f}")
m = dcf(r, w, 0.025, exit_mult=9.0)
f5 = d["fcf"][-1]
g_impl = (m["tv"] * w - f5) / (m["tv"] + f5)
print(f"Exit multiple 9.0x: TV {m['tv']:.1f}, EV {m['ev']:.1f}, price {m['price']:.2f}, implied perpetual
growth {g_impl:.2%}")
```

```
Cost of equity 9.5000%; after-tax cost of debt 4.8750%; WACC 8.3438%

                       FY2026E   FY2027E   FY2028E   FY2029E   FY2030E
EBIT                     106.9     118.3     129.0     138.7     147.1
NOPAT (EBIT x 0.75)       80.2      88.7      96.8     104.0     110.3
+ Depreciation            40.0      41.6      43.5      45.5      47.6
- Capex                  -56.2     -60.1     -63.7     -66.9     -69.6
- Increase in NWC        -14.2     -12.6     -11.6     -10.2      -8.5
Unlevered FCF            49.78     57.62     64.96     72.45     79.92
Discount period (yrs)      0.5       1.5       2.5       3.5       4.5
Discount factor          0.9607    0.8867    0.8184    0.7554    0.6972
PV of FCF                47.83     51.09     53.17     54.73     55.72

Sum of PV of FCF          262.54
Terminal value (Gordon)  1401.81   implied EV/EBITDA 7.20x
PV of terminal value      977.40   (78.8% of EV)
Enterprise value         1239.94
Less net debt (200 - 60)  140.00
Equity value             1099.94
Value per share ($)        22.00

Mid-year convention: EV 1239.9, price 22.00
End-year convention: EV 1191.2, price 21.02
Exit multiple 9.0x: TV 1752.8, EV 1436.7, price 25.93, implied perpetual growth 3.62%
```

Interpreting the valuation

The base case value is $22.00 per share on an enterprise value of $1,239.9 million, which is 8.4 times FY2026E EBITDA of 146.9 and 6.4 times FY2030E EBITDA. Several features deserve scrutiny, because they are the questions an investment committee would ask.

Terminal value dominates. The present value of the terminal value is 977.4, or 78.8 percent of enterprise value, so the five forecast years explain only about a fifth of the value. This is normal for a growing business with a short explicit forecast, but it means the answer is mainly a statement about the terminal growth rate and WACC. The sensitivity tables in Part V quantify this.

The implied exit multiple. The Gordon growth terminal value of 1,401.8 equals 7.20 times FY2030E EBITDA. If peers trade at 9.0 times, the exit-multiple method gives a terminal value of 1,752.8 and a value of $25.93, $3.93 per share higher. The two methods answer different questions: Gordon growth asks what the cash flows are worth if growth fades to 2.5 percent, while the multiple asks what a buyer would pay. When they diverge, one of the assumptions (growth, WACC, the multiple) is out of line. Here, a 9.0 times exit multiple implies perpetual growth of 3.62 percent on end-year timing (3.43 percent on the mid-year Gordon convention used in the model, see Exercise 3), which is above any reasonable long-run nominal growth rate for a mature fastener business. The sensible reading is that the 9.0 times multiple would be paid for a business with better returns on incremental capital, or in a more favorable credit market, than Meridian's.

Terminal cash flow quality. The model's terminal cash flow is simply FY2030E UFCF grown at 2.5 percent. But FY2030E reflects 4.0 percent growth, so its working-capital investment (8.5) and capex (69.6, versus depreciation of 47.6) support faster growth than the terminal period assumes. A more careful method normalizes the terminal year with the value-driver formula: reinvestment equals growth divided by the return on new invested capital (RONIC). Exercise 6 does this: if new investment only earns the cost of capital (RONIC equal to WACC), the value falls to $21.35; if it earns 12 percent, it rises to $23.82. The as-built $22.00 lies between, which is reassuring but not a substitute for stating the assumption.

Mid-year versus end-year. The end-year convention gives $21.02 against $22.00. The difference, 4.7 percent, is mechanical and should be a visible switch, not a buried assumption.

Part V: Scenarios, data tables and sensitivity

Scenario switch

The scenario selector is one input cell, `Scen`, restricted by data validation to 1, 2 or 3. On the Scenarios sheet, each driver has three input rows (Base, Upside, Downside) and one live row that feeds the Model sheet:

Row	Content	Formula in F
8 to 10	Base: growth, COGS %, SG&A %	Inputs (blue)
14 to 16	Upside: growth, COGS %, SG&A %	Inputs (blue)
20 to 22	Downside: growth, COGS %, SG&A %	Inputs (blue)
30	Live revenue growth	`=CHOOSE(Scen,F8,F14,F20)`
31	Live COGS %	`=CHOOSE(Scen,F9,F15,F21)`
32	Live SG&A %	`=CHOOSE(Scen,F10,F16,F22)`

`CHOOSE` is transparent and fast. `INDEX(F8:F20, ...)` with computed offsets or `OFFSET` are more compact but harder to audit (S8). The key discipline is that every scenario-dependent driver goes through a live row; a scenario that changes revenue but not margins, because one driver was hard-wired, is a common error. The Python equivalent is the `SCEN` dictionary and the `scn` argument.

Data tables

A one- or two-variable **data table** (Data, What-If Analysis, Data Table) recalculates the whole workbook for each combination of input values and records one output. For the two-way table of value per share against WACC and terminal growth: put the output reference (`=DCF!price`) in the top-left corner cell, the growth values across the top row, the WACC values down the left column, select the block, and specify the row input cell (`TermG`) and the column input cell (`WACC`). Excel fills the interior with `{=TABLE(TermG,WACC)}`.

Three limitations matter. The input cells must be on the same sheet as the table (so, in practice, put the table on the sheet holding the inputs, or put copies of the inputs on the Sens sheet and make the model read them). Data tables slow a workbook sharply and interact badly with iterative calculation, since every cell of the table triggers a full iterative recalculation; set Calculation to "Automatic except data tables" and press F9 when you want the table refreshed. And a table that varies WACC must vary the cell the DCF actually reads: if WACC is a formula (as above), you must overwrite it with an input-driven override, or the table will vary nothing.

For these reasons many practitioners prefer **direct-formula grids** for DCF sensitivity. Because UFCF does not depend on WACC or growth, value per share can be written as one formula in each grid cell, using the UFCF and discount-period rows already computed:

```
=( SUMPRODUCT(DCF!$F$16:$J$16 / (1+$B20)^DCF!$F$17:$J$17)
   + DCF!$J$16*(1+C$19)/($B20-C$19) / (1+$B20)^DCF!$J$17
   - NetDebt ) / Shares
```

with WACC values in column B and growth values in row 19. This recalculates instantly, has no iterative-calculation problem and can be audited like any other formula. It cannot, of course, vary operating assumptions that change UFCF itself; for those (a margin sensitivity, for example) a data table, or repeated runs of a macro, is required. The Python replica handles both uniformly because it simply calls the model again.

The listing below produces the two sensitivity grids, the scenario summary, a probability-weighted value, and a one-at-a-time sensitivity (sometimes called a tornado analysis) around the base case.

```
import copy
from dcf import *
from checks import master
import model_core

w0, _ = wacc()
base = build("Base")

# ---- 1. Two-way data table: value per share, WACC x terminal growth --------
waccs = [w0 - 0.01, w0 - 0.005, w0, w0 + 0.005, w0 + 0.01]
gs = [0.015, 0.020, 0.025, 0.030, 0.035]
print("Value per share ($): rows = WACC, columns = terminal growth")
print(f"{'WACC':>8}" + "".join(f"{g:>9.1%}" for g in gs))
for w in waccs:
    print(f"{w:>8.2%}" + "".join(f"{dcf(base, w, g)['price']:>9.2f}" for g in gs))

print("\nValue per share ($): rows = WACC, columns = exit EV/EBITDA multiple")
ms = [7.0, 8.0, 9.0, 10.0, 11.0]
print(f"{'WACC':>8}" + "".join(f"{m:>9.1f}x" for m in ms))
for w in waccs:
    print(f"{w:>8.2%}" + "".join(f"{dcf(base, w, 0.025, exit_mult=m)['price']:>10.2f}" for m in ms))

# ---- 2. Scenario summary ---------------------------------------------------
print("\nScenario summary (WACC 8.34%, g 2.5%, mid-year)")
hdr = f"{'':<26}" + "".join(f"{s:>11}" for s in SCEN)
print(hdr)
res = {s: build(s) for s in SCEN}
dd = {s: dcf(res[s], w0, 0.025) for s in SCEN}
def line(lab, f, fmt="{:>11.1f}"):
    print(f"{lab:<26}" + "".join(fmt.format(f(s)) for s in SCEN))
line("FY2030 revenue", lambda s: res[s]["revenue"][-1])
line("FY2030 EBITDA", lambda s: res[s]["ebitda"][-1])
line("FY2030 EBITDA margin %", lambda s: 100 * res[s]["ebitda"][-1] / res[s]["revenue"][-1])
line("FY2030 net income", lambda s: res[s]["ni"][-1])
line("Peak revolver", lambda s: max(res[s]["revolver"]))
line("Enterprise value", lambda s: dd[s]["ev"])
line("Value per share ($)", lambda s: dd[s]["price"], "{:>11.2f}")
line("Implied TV EV/EBITDA (x)", lambda s: dd[s]["implied_mult"], "{:>11.2f}")
print(f"{'Master check':<26}" + "".join(f"{('OK' if master(res[s])[0] else 'ERROR'):>11}" for s in SCEN))
pw = {"Base": 0.5, "Upside": 0.25, "Downside": 0.25}
ev_pw = sum(pw[s] * dd[s]["price"] for s in SCEN)
print(f"\nProbability-weighted value per share (25/50/25): {ev_pw:.2f}")

# ---- 3. One-at-a-time shocks around the base case (tornado) ---------------
def price_with(mod=None, w=w0, g=0.025):
    saved = copy.deepcopy(model_core.SCEN)
    try:
        if mod: mod(model_core.SCEN["Base"])
        return dcf(build("Base"), w, g)["price"]
    finally:
        model_core.SCEN.clear(); model_core.SCEN.update(saved)
b = price_with()
shocks = [("COGS % of revenue +1 pt, every year", lambda s: s.update(cogs=[x + 0.01 for x in s["cogs"]])),
          ("Revenue growth -1 pt, every year",   lambda s: s.update(g=[x - 0.01 for x in s["g"]])),
          ("SG&A % of revenue +1 pt",             lambda s: s.update(sga=[x + 0.01 for x in s["sga"]]))]
print(f"\nBase value per share {b:.2f}")
for lab, m in shocks:
    v = price_with(m); print(f"  {lab:<38}{v:>7.2f}  ({v - b:+.2f})")
```

```
v = price_with(w=w0 + 0.01); print(f"  {'WACC +1 pt':<38}{v:>7.2f}  ({v - b:+.2f})")
v = price_with(g=0.015);     print(f"  {'Terminal growth -1 pt':<38}{v:>7.2f}  ({v - b:+.2f})")
```

```
Value per share ($): rows = WACC, columns = terminal growth
    WACC     1.5%     2.0%     2.5%     3.0%     3.5%
   7.34%    22.76    24.76    27.17    30.13    33.87
   7.84%    20.72    22.38    24.34    26.71    29.63
   8.34%    18.98    20.37    22.00    23.93    26.26
   8.84%    17.48    18.66    20.03    21.63    23.53
   9.34%    16.17    17.18    18.34    19.69    21.27

Value per share ($): rows = WACC, columns = exit EV/EBITDA multiple
    WACC      7.0x      8.0x      9.0x     10.0x     11.0x
   7.34%     21.71     24.44     27.18     29.91     32.64
   7.84%     21.21     23.88     26.55     29.22     31.89
   8.34%     20.72     23.32     25.93     28.54     31.15
   8.84%     20.24     22.79     25.34     27.89     30.44
   9.34%     19.77     22.26     24.76     27.25     29.74

Scenario summary (WACC 8.34%, g 2.5%, mid-year)
                                Base     Upside   Downside
FY2030 revenue                1070.1     1175.0      849.0
FY2030 EBITDA                  194.8      233.8      117.2
FY2030 EBITDA margin %          18.2       19.9       13.8
FY2030 net income              105.1      133.6       48.6
Peak revolver                   52.3       49.2       65.1
Enterprise value              1239.9     1489.9      693.3
Value per share ($)             22.00      27.00      11.07
Implied TV EV/EBITDA (x)         7.20       7.32       6.30
Master check                       OK         OK         OK

Probability-weighted value per share (25/50/25): 20.52

Base value per share 22.00
  COGS % of revenue +1 pt, every year     19.41  (-2.59)
  Revenue growth -1 pt, every year        21.71  (-0.29)
  SG&A % of revenue +1 pt                 19.44  (-2.56)
  WACC +1 pt                              18.34  (-3.65)
  Terminal growth -1 pt                   18.98  (-3.02)
```

What the sensitivities say

WACC and growth. Value per share ranges from $16.17 (WACC 9.34 percent, growth 1.5 percent) to $33.87 (7.34 percent, 3.5 percent), a factor of two. A 50 basis point change in WACC moves value by about $2.0 to $2.8 around the base case and a 50 basis point change in terminal growth by about $1.6 to $1.9. Because the two interact (the growth effect is larger when WACC is lower), a table is the right display and a single number is not.

Exit multiple. At the base WACC, the value ranges from $20.72 at 7.0 times to $31.15 at 11.0 times, about $2.60 per turn. The base-case Gordon value of $22.00 corresponds to an exit multiple of roughly 7.5 times on the exit-multiple method's timing (the Gordon terminal value itself is 7.2 times): one reason to be cautious about quoting a DCF value alongside a peer multiple without reconciling them.

Scenarios. Value per share is $22.00 in the base case, $27.00 in the upside and $11.07 in the downside. The downside value is half the base value although revenue in FY2030E is only 21 percent lower (849.0 against 1,070.1), because the margin squeeze (EBITDA margin 13.8 percent against 18.2 percent) flows entirely to profit and the lower growth releases less value. Operating

leverage, in other words, is doing the work. The weights of 25, 50 and 25 percent give a probability-weighted value of $20.52, below the base case, because the downside is further from the base than the upside. A model whose "base case" is also its expected case is rare.

Drivers. One percentage point on the COGS ratio is worth $2.59 per share and one on SG&A $2.56, very similar and close to the effect of a 1 point rise in WACC ($3.65) or a 1 point fall in terminal growth ($3.02). The surprise is revenue growth: cutting growth by 1 point in every year costs only $0.29 per share. The reason is economic: with a return on new capital close to the cost of capital, growth consumes about as much cash (capex and working capital) as it creates, so more growth adds little value. This is the intuition of Volume 1, Chapter 21 and of the value-driver formula in Exercise 6, and it is a useful reminder that a model that shows revenue growth as its main lever may be disguising what actually matters.

Part VI: Exercises with solutions

Each exercise can be done in the Python model by changing one argument, and in Excel by changing one input; the Excel result must match the Python result. The output of the program that solves Exercises 1 to 5 is shown after the exercises.

Exercise 1 (liquidity policy). The treasurer wants a minimum cash balance of $50 million instead of $30 million. What happens to the revolver? Does the model still pass its checks?

Exercise 2 (financing neutrality). Cut the payout ratio from 50 to 30 percent. What happens to the FY2030E revolver and equity? What happens to the DCF value?

Exercise 3 (goal seek). What terminal growth rate makes the Gordon-growth value per share equal to the value obtained from a 9.0 times exit multiple? Why does it differ from the 3.62 percent computed in Part IV?

Exercise 4 (working capital). Receivable days rise from 55 to 65 as the company extends terms to win share. What is the effect on cash, on the peak revolver and on value per share?

Exercise 5 (tax linkage). The tax rate falls from 25 to 21 percent. What is the new value per share? What mistake would a modeler make if the tax rate in the DCF were typed as a constant?

Exercise 6 (normalized terminal year). Replace the terminal cash flow with a value-driver version, FCF = NOPAT x (1 - g / RONIC), where NOPAT is FY2030E NOPAT grown at g. Compute value per share with RONIC equal to WACC and with RONIC of 12 percent.

Exercise 7 (Excel only). Make the revolver limit a hard constraint rather than a check: rewrite row 67. What happens to the balance sheet when the constraint binds, and what must you add to keep the model balanced?

```
from dcf import *
from checks import master
w0, _ = wacc()
base = build("Base"); p0 = dcf(base, w0, 0.025)["price"]

# Exercise 1: raise minimum cash from 30 to 50
r1 = build("Base", min_cash=50.0)
print("Ex1 min cash 50: closing cash", [round(x, 1) for x in r1["cash"]], "| revolver", [round(x, 1) for x in
r1["revolver"]],
      "| master", "OK" if master(r1, {**COMMON, "min_cash": 50.0})[0] else "ERROR")
print("    value per share", round(dcf(r1, w0, 0.025)["price"], 2), "(unchanged: financing does not enter
UFCF)")

# Exercise 2: payout 30% instead of 50%
r2 = build("Base", payout=0.30)
print("Ex2 payout 30%: FY2030 revolver", round(r2["revolver"][-1], 1), "equity", round(r2["equity"][-1], 1),
      "| value per share", round(dcf(r2, w0, 0.025)["price"], 2))

# Exercise 3: goal seek - terminal growth that equates Gordon value with the 9.0x exit-multiple value
target = dcf(base, w0, 0.025, exit_mult=9.0)["price"]
lo, hi = 0.0, 0.06
for _ in range(60):
    mid = (lo + hi) / 2
    if dcf(base, w0, mid)["price"] < target: lo = mid
    else: hi = mid
print(f"Ex3 target price {target:.2f}; goal-seek terminal growth = {mid:.4%}")

# Exercise 4: receivable days from 55 to 65
r4 = build("Base", ar_days=65)
print("Ex4 AR days 65: FY2026 cash", round(r4["cash"][0], 1), "peak revolver", round(max(r4["revolver"]), 1),
      "value per share", round(dcf(r4, w0, 0.025)["price"], 2), f"({dcf(r4, w0, 0.025)['price'] - p0:+.2f})")

# Exercise 5: tax rate sensitivity (must change both the statements and the DCF)
r5 = build("Base", tax=0.21)
print("Ex5 tax 21%: FY2030 net income", round(r5["ni"][-1], 1), "value per share", round(dcf(r5, w0, 0.025,
tax=0.21)["price"], 2))
```

```
Ex1 min cash 50: closing cash [50.0, 50.0, 50.0, 50.0, 50.0] | revolver [14.6, 35.4, 52.6, 65.7, 74.0] |
master OK
    value per share 22.0 (unchanged: financing does not enter UFCF)
Ex2 payout 30%: FY2030 revolver 0.0 equity 746.5 | value per share 22.0
Ex3 target price 25.93; goal-seek terminal growth = 3.4348%
Ex4 AR days 65: FY2026 cash 30.0 peak revolver 84.8 value per share 21.18 (-0.82)
Ex5 tax 21%: FY2030 net income 111.3 value per share 23.85
```

```
from dcf import *
w, _ = wacc(); g = 0.025
r = build("Base"); d = dcf(r, w, g)
# Normalized terminal-year free cash flow: reinvestment = g / RONIC, with RONIC = WACC (growth adds no value)
nopat_T = r["ebit"][-1] * (1 + g) * (1 - COMMON["tax"])
for ronic, lab in [(w, "RONIC = WACC"), (0.12, "RONIC = 12%")]:
    f_T = nopat_T * (1 - g / ronic)
    tv = f_T / (w - g)
    pv_tv = tv / (1 + w) ** (N - 0.5)
    ev = sum(d["pv"]) + pv_tv
    print(f"{lab:<14} terminal FCF {f_T:7.2f}  TV {tv:8.1f}  EV {ev:8.1f}  price {(ev - 140) / 50:6.2f}
implied TV/EBITDA {tv / r['ebitda'][-1]:.2f}x")
print(f"{'Model as built':<14} terminal FCF {d['fcf'][-1] * (1 + g):7.2f}  TV {d['tv']:8.1f}  EV
{d['ev']:8.1f}  price {d['price']:6.2f}  implied TV/EBITDA {d['implied_mult']:.2f}x")
```

```
RONIC = WACC   terminal FCF   79.22  TV   1355.6  EV   1207.7  price  21.35  implied TV/EBITDA 6.96x
RONIC = 12%    terminal FCF   89.54  TV   1532.3  EV   1330.9  price  23.82  implied TV/EBITDA 7.87x
Model as built terminal FCF   81.92  TV   1401.8  EV   1239.9  price  22.00  implied TV/EBITDA 7.20x
```

Solutions

Solution 1. Raising the minimum cash to 50.0 means the company must borrow earlier and more: the revolver is 14.6 in FY2026E (previously nothing), 35.4 in FY2027E and 74.0 by FY2030E, against 52.3. All checks pass, including the limit check (74.0 is below 100.0), but headroom is thinner: 26.0 undrawn at the end against 47.7 before. The value per share is unchanged at $22.00, since financing policy does not enter unlevered free cash flow. The extra 20.0 of cash is financed by revolver debt costing 7 percent while earning 2 percent, a negative carry of about 1.0 a year before tax; the valuation attributes that cost to the financing choice, not to operations. (In a more refined DCF, excess cash is deducted in net debt and valued at par.)

Solution 2. With a 30 percent payout the company retains more, so the revolver is not needed (FY2030E revolver 0.0) and equity at FY2030E is 746.5 against 652.0. Value per share is unchanged at $22.00 for the same reason as in Exercise 1. If your spreadsheet shows a different value, the DCF is picking up a financing item, most likely interest, through a mislinked row; this is the quickest test of the DCF's construction.

Solution 3. The target price is the exit-multiple value of $25.93. Bisection on the terminal growth rate gives 3.4348 percent. The earlier figure of 3.62 percent solves the end-of-year Gordon relation TV = FCF x (1 + g) / (WACC - g) against the exit-multiple terminal value discounted at N years; in the model the Gordon terminal value is discounted at N - 0.5 under the mid-year convention, which makes each dollar of terminal value worth more, so a lower growth rate reaches the same present value. The lesson is that an "implied growth rate" is only meaningful relative to a timing convention.

Solution 4. Receivables rise by 864.0 x 10 / 365 = 23.7 in FY2026E alone. FY2026E cash falls to the 30.0 minimum (a revolver draw is now needed in the first year), the peak revolver rises from 52.3 to 84.8 (still within the 100.0 limit, with headroom of 15.2) and value per share falls by $0.82 to $21.18. The sign of the effect on value is clear; the size depends on the timing and permanence of the change. A one-off 10 day increase is a permanent investment of about 24 growing with sales.

Solution 5. With a 21 percent tax rate FY2030E net income rises from 105.1 to 111.3 and value per share from $22.00 to $23.85. The statement and the DCF must both use the tax input: in the Python call the DCF receives `tax=0.21` as an argument. If the DCF had a typed 25 percent (a violation of S5), the net income would have risen but the valuation would have been unchanged, and the model would be internally inconsistent without any check noticing. The remedy is the named range `Tax` used everywhere, and a test: change the rate and confirm both NOPAT and net income move.

Solution 6. With RONIC equal to WACC (8.34 percent), terminal reinvestment is g / RONIC = 0.025 / 0.0834 = 30.0 percent of NOPAT, so terminal FCF is 79.22 (NOPAT of 113.1 less reinvestment of 33.9), terminal value 1,355.6 and value per share $21.35, an implied exit multiple of 6.96 times. With RONIC of 12 percent, reinvestment is 20.8 percent of NOPAT, terminal FCF 89.54, terminal value 1,532.3 and value $23.82, a multiple of 7.87 times. The as-built terminal FCF of 81.92 corresponds to a RONIC of about 9 percent: slightly above WACC. A reviewer would accept the as-built number if the business is believed to earn modest excess returns and would ask for the normalized method if the terminal value were a large part of a deal price, which it generally is.

Solution 7. A hard limit changes row 67 to `=MIN(MAX(-E77,MinCash-F66),RevLimit-E77)`: the draw cannot take the revolver above its limit. When the constraint binds, cash falls below the minimum, and, if cash before the revolver is negative, closing cash becomes negative: the balance sheet still balances (cash is simply negative, which is an implicit overdraft) but the model is describing something that cannot happen. The modeler must therefore add either a “funding shortfall” row (a plug of last resort, clearly labelled, with its own check that it is zero) or an explicit equity issuance or dividend cut. The shortfall row is preferable because it makes the infeasibility visible instead of curing it. Check 5 (cash at least the minimum unless the revolver is at its limit) is written to be compatible with this design.

Chapter summary

A good model is an argument that survives audit. It separates inputs from formulas, uses one formula per row on a consistent time axis, marks inputs in blue and formulas in black, and checks itself from several directions through a master flag that has been proven to fail when the model is broken. Its balance sheet balances because the cash flow statement is built from changes in the balance sheet, never through a plug. Circularity from average-balance interest is handled with iterative calculation and a circuit breaker, and converges rapidly because each pass shrinks the error by about 76-fold in this model. The DCF discounts unlevered cash flow, is insensitive to financing choices, and reports its terminal value share and implied multiples so that a reader can judge its credibility. Sensitivities are produced by data tables or direct-formula grids, and scenarios through a single switch feeding live driver rows. For Meridian Fastener Co. these tools produced a base value of $22.00 per share, a range of $11.07 to $27.00 across scenarios and a probability-weighted $20.52, with margins and the discount rate mattering far more than revenue growth.

The Python replica in this chapter is the answer key for a spreadsheet build. When your Excel model reproduces FY2026E net income of 71.9, FY2030E revolver of 52.3, enterprise value of 1,239.9 and value per share of $22.00 to the displayed precision, with the master check showing OK, you will have built the model correctly, and you will have the tools to extend it: to leases and deferred taxes, to multiple debt tranches, to a monthly timeline, or to the transaction structures of Chapters 13 to 17.

Chapter 19: Python for Corporate Finance: NPV, IRR, Bonds and Monte Carlo

Why Learn to Code in a Finance Course

Volumes 1 and 2 taught the ideas: time value of money (Volume 1, Chapter 3), bond pricing (Volume 1, Chapter 4), net present value (Volume 1, Chapter 6), capital budgeting (Volume 1, Chapter 7), and uncertainty and real options (Volume 1, Chapter 8). Every one of those chapters could be worked with a financial calculator and, for the harder cases, with a spreadsheet. This chapter shows how to do the same work in Python, and, more importantly, why a finance leader would want to.

The case for Python is not that it replaces Excel. Excel remains the lingua franca of corporate finance, and Chapter 20 of this volume uses Excel formula patterns side by side with Python for the leveraged buyout and merger models. The case is that code has three properties a spreadsheet struggles to match.

1. **Repeatability.** A script that loads data, computes a valuation and prints a table will give the same answer every time and can be rerun on next quarter's numbers in seconds. There are no hidden cells and no manual steps.
2. **Scale.** A Monte Carlo analysis with one hundred thousand trials, each of which is a full eight-year cash flow forecast, is a single line of vectorised code. In a spreadsheet it requires a data table, patience and a macro.
3. **Auditability.** A reviewer can read a function, run it against a case with a known answer, and test it. Code can be version-controlled, so you can see exactly what changed between last year's model and this year's.

The audience is the MBA student or finance leader who will manage analysts who code, review models, and occasionally want a quick answer without waiting. We teach a small, practical subset of the language and build a reusable toolkit: time-value functions, bond analytics and a project simulation engine.

Every listing in this chapter was executed, and every block labelled "Output" is the genuine printed result, pasted without editing. Where we describe a chart, we also executed the plotting code, and we report what the data in the chart shows, because a book cannot carry a live figure.

All companies and projects in this chapter are fictional and illustrative. Numbers are chosen to teach technique, not to describe any real business or market.

Environment Setup

Installing Python and the libraries

You need three things: a Python interpreter, a handful of numerical libraries, and somewhere to write code. For the interpreter, any recent version of Python 3 will do. This chapter was run on Python 3.13, but nothing in it depends on features newer than Python 3.8. For the libraries, the standard scientific stack is enough:

- **numpy** provides the fast array, the basic data structure for numerical work.
- **pandas** provides the DataFrame, a table with labelled rows and columns, which is the natural structure for a financial statement or a debt schedule.
- **scipy** provides numerical routines. We use only its root finder (`brentq`) and the normal distribution.
- **matplotlib** draws charts. It is optional for the calculations and used only in one section.

From a command line, the sequence is the same on Windows, macOS and Linux. Create an isolated environment so that this project's libraries do not collide with other projects, activate it, and install the packages:

```
python -m venv finance-env
source finance-env/bin/activate        (Windows: finance-env\Scripts\activate)
pip install numpy pandas scipy matplotlib
```

For writing code, a plain editor and the command line is the most transparent; Jupyter notebooks suit exploration and teaching. What matters is saving code in files so the analysis can be rerun.

Verify, seed, and set display options

The first listing does three things that every serious analysis script should do. It records the versions of the libraries, so a future reader knows what produced the numbers. It creates a single seeded random number generator, so that any simulation is reproducible. And it sets the display format so that tables print consistently.

```
import sys, platform
import numpy as np
import pandas as pd
import scipy

print("Python  :", platform.python_version())
print("numpy   :", np.__version__)
print("pandas  :", pd.__version__)
print("scipy   :", scipy.__version__)

# Reproducible random numbers: a single seeded generator used throughout the chapter
rng = np.random.default_rng(seed=2026)
print("First three uniform draws:", np.round(rng.random(3), 4))

# Display options so that tables print compactly and identically each run
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
pd.set_option("display.width", 100)
```

Output:

```
Python  : 3.13.18
numpy   : 2.5.3
pandas  : 3.0.5
scipy   : 1.18.1
First three uniform draws: [0.1789 0.6399 0.4673]
```

Modern numpy recommends creating a `Generator` object and passing it around rather than relying on hidden global state. Reproducibility matters: when a board member asks why the probability of a loss was 19.2 percent in the draft and 19.4 percent in the final, "we changed the random seed" is an embarrassing answer. Fix the seed, record it, and report simulation error.

numpy Basics: Cash Flows as Arrays

A cash flow stream is naturally a one-dimensional array: element zero is today, element one is the end of year one, and so on. A discount factor is `1 / (1 + r) ** t`. The key idea in numpy is **vectorisation**: an operation written once applies to every element of the array, with no loop. This is both shorter and much faster than a Python `for` loop.

The listing below discounts a five-year project, then uses **broadcasting** to compute the NPV at seven different discount rates in one expression. Broadcasting is numpy's rule for combining arrays of different shapes: a column of rates with shape (7, 1) and a row of times with shape (1, 5) combine to a grid with shape (7, 5), one row per rate and one column per year. The expression `rates[:, None]` is the standard way to turn a flat array into a column.

```
import numpy as np

# A five-year project: outlay today, then four years of inflows (USD thousands)
cash_flows = np.array([-1000.0, 250.0, 300.0, 350.0, 400.0])
years = np.arange(len(cash_flows))
r = 0.10

discount_factors = 1.0 / (1.0 + r) ** years
present_values = cash_flows * discount_factors

print("years            :", years)
print("discount factors :", np.round(discount_factors, 4))
print("present values   :", np.round(present_values, 2))
print("NPV at 10%       :", round(present_values.sum(), 2))

# Vectorisation: NPV at many discount rates with one broadcast operation
rates = np.arange(0.00, 0.31, 0.05)                    # shape (7,)
grid = cash_flows[None, :] / (1 + rates[:, None]) ** years[None, :]   # shape (7, 5)
npv_curve = grid.sum(axis=1)
for rate, v in zip(rates, npv_curve):
    print(f"rate {rate:4.0%}  NPV {v:9.2f}")

print("shape of grid:", grid.shape)
print("cumulative undiscounted:", np.cumsum(cash_flows))
```

Output:

```
years            : [0 1 2 3 4]
discount factors : [1.     0.9091 0.8264 0.7513 0.683 ]
present values   : [-1000.     227.27   247.93   262.96   273.21]
NPV at 10%       : 11.37
rate   0%  NPV    300.00
rate   5%  NPV    141.63
rate  10%  NPV     11.37
rate  15%  NPV    -96.93
rate  20%  NPV   -187.89
rate  25%  NPV   -264.96
rate  30%  NPV   -330.82
shape of grid: (7, 5)
cumulative undiscounted: [-1000.  -750.  -450.  -100.   300.]
```

Read the output carefully, because it contains the whole of capital budgeting in miniature. The undiscounted cash flows sum to 300, so a naive observer would call this a handsome project: pay 1,000, get 1,300 back. The NPV at 10 percent is only 11.37, positive but thin. At 15 percent it is negative 96.93. The point where the curve crosses zero is the internal rate of return, which lies between 10 and 15 percent. We will compute it exactly shortly; it is 10.48 percent.

The cumulative line shows the project still 100 short after year three, so simple payback is 3.25 years, computed in the next section.

pandas Basics: The Project as a Table

An array has no labels. A financial model does: years across or down, line items with names. The pandas DataFrame supplies the labels and a rich set of operations on columns. The mental model is a spreadsheet in which each column is a vector and a formula applies to the whole column.

The next listing builds a five-year project from operating assumptions. It computes EBITDA, depreciation, EBIT, tax, free cash flow, the discount factor, the present value and the cumulative present value. The pattern `df["ebitda"] = df["revenue"] - df["opex"]` is the pandas equivalent of filling a spreadsheet formula down a column.

```
import pandas as pd

pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
pd.set_option("display.width", 100)

df = pd.DataFrame({
    "year":    [0, 1, 2, 3, 4, 5],
    "revenue": [0.0, 1200.0, 1500.0, 1800.0, 1900.0, 1900.0],
    "opex":    [0.0, 800.0, 950.0, 1100.0, 1150.0, 1150.0],
    "capex":   [1500.0, 100.0, 100.0, 50.0, 50.0, 50.0],
}).set_index("year")

tax_rate = 0.25
df["ebitda"] = df["revenue"] - df["opex"]
df["depreciation"] = [0.0] + [300.0] * 5
df["ebit"] = df["ebitda"] - df["depreciation"]
df["tax"] = (df["ebit"] * tax_rate).clip(lower=0)
df["fcf"] = df["ebitda"] - df["tax"] - df["capex"]
df.loc[5, "fcf"] += 200.0      # salvage / working capital recovery in the final year

r = 0.09
df["df"] = (1 + r) ** -df.index.to_series()
df["pv"] = df["fcf"] * df["df"]
df["cum_pv"] = df["pv"].cumsum()

print(df[["revenue", "ebitda", "tax", "capex", "fcf", "df", "pv", "cum_pv"]])
print()
print("NPV =", round(df["pv"].sum(), 2))
```

Output:

```
      revenue  ebitda    tax    capex       fcf   df        pv    cum_pv
year
0        0.00    0.00   0.00 1,500.00 -1,500.00 1.00 -1,500.00 -1,500.00
1    1,200.00  400.00  25.00   100.00    275.00 0.92    252.29 -1,247.71
2    1,500.00  550.00  62.50   100.00    387.50 0.84    326.15   -921.56
3    1,800.00  700.00 100.00    50.00    550.00 0.77    424.70   -496.85
4    1,900.00  750.00 112.50    50.00    587.50 0.71    416.20    -80.65
5    1,900.00  750.00 112.50    50.00    787.50 0.65    511.82    431.17

NPV = 431.17
```

Several details are worth studying.

- **The index is the year.** Setting `year` as the index means `df.loc[5, "fcf"]` addresses year five directly, and `df.index.to_series()` lets us use the years in arithmetic. We adjust the final-year free cash flow by adding 200 of salvage

and working-capital recovery with `df.loc[5, "fcf"] += 200.0`. In Excel, this is the cell that someone types over a formula and forgets about. In code, the adjustment is a visible, commented line.

- **Tax uses `.clip(lower=0)`.** In year one, EBIT is 100 (EBITDA of 400 less depreciation of 300), so tax is 25. Had EBIT been negative the clip would have set tax to zero, which is the pandas equivalent of the numpy `maximum` above.
- **Free cash flow here** is EBITDA minus tax minus capital expenditure, a simplification that ignores working capital changes other than the terminal recovery.
- **The cumulative PV column tells the discounted payback story.** It turns positive during year five: after year four the project is still 80.65 short on a present-value basis, and year five's present value of 511.82 more than covers it. The NPV is 431.17 at a 9 percent discount rate. The first-year outlay is 1,500, so the project has a profitability index (Volume 1, Chapter 7) of about 1.29: each dollar invested creates 29 cents of value at the margin. A DataFrame also makes sensitivity analysis easy: wrap the calculation in a function of the discount rate and call it repeatedly.

The Excel NPV trap

A trap lurks in the spreadsheet function `NPV`. Excel's `=NPV(rate, values)` assumes the first value occurs at the end of period one, not today. To value a stream whose first element is the time-zero outlay, you write `=B5 + NPV(rate, C5:G5)`, or equivalently `=NPV(rate, B5:G5) * (1 + rate)`. Many an analyst has understated an NPV by discounting the initial outlay one year. Our Python `npv` puts element zero at time zero, so remember the two conventions differ when you reconcile.

Time-Value-of-Money Functions

Volume 1, Chapter 3 derived the formulas. Here we implement them as a small library and then test it. We write our own rather than import a ready-made package for two reasons: writing them makes the sign conventions and the timing conventions explicit, and the code is short enough to read in full.

Sign conventions and timing

We follow the convention used by Excel's financial functions and by most calculators: **cash received is positive, cash paid is negative**. So if you deposit 5,000 a year, the payment is -5,000 and the future value comes back positive. The functions take an optional `when` argument, 0 for payments at the end of each period (an ordinary annuity) and 1 for payments at the start (an annuity due). The relationships are the standard ones:

```
FV = -[ PV·(1+r)ⁿ + PMT·(1 + r·when)·((1+r)ⁿ - 1)/r ]
PV = -[ FV + PMT·(1 + r·when)·((1+r)ⁿ - 1)/r ] / (1+r)ⁿ
PMT = -[ PV·(1+r)ⁿ + FV ]·r / [ (1 + r·when)·((1+r)ⁿ - 1) ]
NPER = ln[(-FV + z)/(PV + z)] / ln(1+r),  where z = PMT·(1 + r·when)/r
```

The rate cannot be isolated algebraically, so `rate` and `irr` find it numerically. We use Brent's method (`scipy.optimize.brentq`), which requires a bracket, an interval over which the function changes sign, and is guaranteed to converge when one exists. This is more robust than Newton's method, which can wander off when the starting guess is poor.

The library

Save the following as `tvm.py`. Every later listing in this chapter that begins with `from tvm import *` relies on it.

```
"""Small time-value-of-money library used throughout this chapter."""
import numpy as np
from scipy.optimize import brentq
from datetime import date

def fv(rate, nper, pmt, pv=0.0, when=0):
    """Future value. Sign convention: cash paid out is negative. when=1 for annuity-due."""
    if rate == 0:
        return -(pv + pmt * nper)
    g = (1 + rate) ** nper
    return -(pv * g + pmt * (1 + rate * when) * (g - 1) / rate)

def pv(rate, nper, pmt, fv_=0.0, when=0):
    if rate == 0:
        return -(fv_ + pmt * nper)
    g = (1 + rate) ** nper
    return -(fv_ + pmt * (1 + rate * when) * (g - 1) / rate) / g

def pmt(rate, nper, pv_, fv_=0.0, when=0):
    if rate == 0:
        return -(pv_ + fv_) / nper
    g = (1 + rate) ** nper
    return -(pv_ * g + fv_) * rate / ((1 + rate * when) * (g - 1))

def nper(rate, pmt_, pv_, fv_=0.0, when=0):
    if rate == 0:
        return -(pv_ + fv_) / pmt_
    z = pmt_ * (1 + rate * when) / rate
    return np.log((-fv_ + z) / (pv_ + z)) / np.log(1 + rate)

def npv(rate, flows):
    flows = np.asarray(flows, dtype=float)
    return float(np.sum(flows / (1 + rate) ** np.arange(len(flows))))

def irr(flows, lo=-0.99, hi=10.0):
    flows = np.asarray(flows, dtype=float)
    f = lambda r: npv(r, flows)
    if f(lo) * f(hi) > 0:
        return float("nan")
    return brentq(f, lo, hi, xtol=1e-12)

def rate(nper_, pmt_, pv_, fv_=0.0, when=0):
    f = lambda r: fv_ - fv(r, nper_, pmt_, pv_, when)
    return brentq(f, -0.99, 10.0, xtol=1e-12)

def xnpv(rate_, flows, dates):
    d0 = dates[0]
    return sum(cf / (1 + rate_) ** ((d - d0).days / 365.0) for cf, d in zip(flows, dates))

def xirr(flows, dates):
    return brentq(lambda r: xnpv(r, flows, dates), -0.99, 10.0, xtol=1e-12)

def mirr(flows, finance_rate, reinvest_rate):
    flows = np.asarray(flows, dtype=float)
    n = len(flows) - 1
    pos = np.where(flows > 0, flows, 0.0)
    neg = np.where(flows < 0, flows, 0.0)
    fv_pos = np.sum(pos * (1 + reinvest_rate) ** (n - np.arange(len(flows))))
```

```
    pv_neg = np.sum(neg / (1 + finance_rate) ** np.arange(len(flows)))
    return (fv_pos / -pv_neg) ** (1 / n) - 1

def payback(flows):
    cum = np.cumsum(flows)
    for t in range(1, len(flows)):
        if cum[t] >= 0 > cum[t - 1]:
            return t - 1 + (-cum[t - 1]) / flows[t]
    return float("nan")

def discounted_payback(flows, r):
    flows = np.asarray(flows, dtype=float)
    return payback(flows / (1 + r) ** np.arange(len(flows)))

def ear(nominal, m):
    return (1 + nominal / m) ** m - 1
```

The functions `npv` and `irr` take flows whose first element is at time zero, and `irr` returns `nan` rather than an obscure error when there is no sign change over the bracket. `xnpv` uses actual days divided by 365, the convention of Excel's `XNPV`, so you can reconcile to a spreadsheet. `mirr` compounds positive flows forward at the reinvestment rate and discounts negative flows back at the financing rate. `payback` interpolates linearly within the year in which the cumulative flow turns positive.

Putting the functions to work

The next listing exercises each function on a problem that has a known answer or an easy sanity check. Read the comments, predict the answer, then compare with the output.

```
from tvm import *

# 1. Future value of saving 5,000 a year for 20 years at 7%
print("FV of 5,000/yr, 20y, 7%      :", round(fv(0.07, 20, -5000), 2))
# 2. Present value of a 30-year annuity of 12,000 at 6%
print("PV of 12,000/yr, 30y, 6%     :", round(pv(0.06, 30, -12000), 2))
# 3. Monthly payment on a 400,000 mortgage, 6.0% APR, 30 years
m = pmt(0.06 / 12, 360, 400000)
print("Mortgage payment (monthly)   :", round(m, 2))
# 4. How many months to repay it if we pay 3,000 a month?
print("Months to repay at 3,000     :", round(nper(0.06 / 12, -3000, 400000), 2))
# 5. Implied rate: pay 1,000 today, receive 1,610.51 in 5 years
print("Implied annual rate          :", round(rate(5, 0, -1000, 1610.51), 6))
# 6. Effective annual rate of 12% compounded monthly, and continuously
print("EAR 12% monthly              :", round(ear(0.12, 12), 6))
print("EAR 12% continuous           :", round(np.exp(0.12) - 1, 6))

# Project metrics
cf = [-1000, 250, 300, 350, 400]
print()
print("NPV @10%                     :", round(npv(0.10, cf), 2))
print("IRR                          :", round(irr(cf), 6))
print("Simple payback (yrs)         :", round(payback(cf), 3))
print("Discounted payback @10%      :", round(discounted_payback(cf, 0.10), 3))
print("MIRR (fin 10%, reinvest 6%)  :", round(mirr(cf, 0.10, 0.06), 6))

# Non-conventional flows: two sign changes -> two IRRs
odd = [-100, 330, -242]
print()
print("NPV at 5%, 10%, 20%          :", [round(npv(r, odd), 3) for r in (0.05, 0.10, 0.20)])
print("IRR search on [-0.99,10]     :", irr(odd))
print("IRR (low bracket 0, hi 0.15) :", round(irr(odd, 0.0, 0.15), 6))
print("IRR (low bracket 0.15, hi 2) :", round(irr(odd, 0.15, 2.0), 6))

# Irregular dates
from datetime import date
dates = [date(2026, 1, 15), date(2026, 7, 1), date(2027, 3, 31), date(2028, 1, 15)]
flows = [-10000, 2500, 4000, 6000]
print()
print("XNPV @8%                     :", round(xnpv(0.08, flows, dates), 2))
print("XIRR                         :", round(xirr(flows, dates), 6))
```

Output:

```
FV of 5,000/yr, 20y, 7%      : 204977.46
PV of 12,000/yr, 30y, 6%     : 165177.97
Mortgage payment (monthly)   : -2398.2
Months to repay at 3,000     : 220.27
Implied annual rate          : 0.1
EAR 12% monthly              : 0.126825
EAR 12% continuous           : 0.127497

NPV @10%                     : 11.37
IRR                          : 0.104845
Simple payback (yrs)         : 3.25
Discounted payback @10%      : 3.958
MIRR (fin 10%, reinvest 6%)  : 0.088889

NPV at 5%, 10%, 20%          : [-5.215, 0.0, 6.944]
IRR search on [-0.99,10]     : nan
IRR (low bracket 0, hi 0.15) : 0.1
IRR (low bracket 0.15, hi 2) : 1.2

XNPV @8%                     : 1203.13
XIRR                         : 0.171702
```

Walk through the results.

Annuities and the mortgage. Saving 5,000 a year for 20 years at 7 percent accumulates to 204,977.46, of which only 100,000 is deposits. A 400,000 loan at 6 percent APR over 360 months needs a monthly payment of 2,398.20 (negative, because it is a payment); note that we pass the monthly rate, 0.06/12, with the number of months, since passing an annual rate with a monthly period count is a classic error. At 3,000 a month the loan is repaid in 220.27 months, about 18.4 years.

Rates. A 1,000 investment that grows to 1,610.51 in five years has an implied rate of 10 percent. A 12 percent nominal rate compounded monthly is 12.6825 percent effective, and continuous compounding gives 12.7497 percent.

Project metrics. The same project we met in the numpy section has NPV 11.37 at 10 percent, an IRR of 10.4845 percent, a simple payback of 3.25 years and a discounted payback of 3.958 years. The MIRR, computed with a 10 percent financing rate and a 6 percent reinvestment rate, is 8.8889 percent, below the IRR. This illustrates the critique of IRR made in Volume 1, Chapter 6: the IRR implicitly assumes intermediate inflows are reinvested at the IRR itself. If the firm's actual reinvestment opportunity is only 6 percent, the realised return is lower. MIRR is below the 10 percent hurdle, which agrees with how thin the NPV is when the reinvestment assumption is made less generous.

Non-conventional cash flows. The stream (-100, +330, -242) has two sign changes and so two IRRs: with $x = 1 + r$ the NPV is zero when $100x^2 - 330x + 242 = 0$, which factors as $100(x - 1.1)(x - 2.2)$, so the IRRs are 10 and 120 percent. Our wide-bracket search returns `nan` because the NPV is negative at both ends, so there is no sign change for Brent's method to start from; brackets straddling each root recover both. When a pattern has more than one sign change, examine the NPV profile directly.

Irregular dates. For flows not on annual anniversaries, `xnpv` and `xirr` use actual dates; the four flows in the listing give an XIRR of 17.17 percent.

Amortization schedules

A loan schedule is the canonical DataFrame exercise: each row depends on the previous row's balance. We cannot vectorise a recursion of this kind directly, so we use a simple loop that appends rows and then convert the result into a DataFrame for summarising. (Chapter 20 uses the same approach for debt schedules with a cash sweep.)

```
import pandas as pd
from tvm import pmt

principal = 250000.0
apr = 0.06
n = 12   # first year only, for display
payment = -pmt(apr / 12, 360, principal)

rows, bal = [], principal
for k in range(1, 361):
    interest = bal * apr / 12
    princ = payment - interest
    bal -= princ
    rows.append((k, payment, interest, princ, bal))
sched = pd.DataFrame(rows, columns=["month", "payment", "interest", "principal",
"balance"]).set_index("month")
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
print(sched.head(3))
print(sched.loc[[120, 240, 360]])
print()
print("Total interest over life :", round(sched["interest"].sum(), 2))
print("Interest in year 1       :", round(sched.loc[1:12, "interest"].sum(), 2))
print("Interest in year 30      :", round(sched.loc[349:360, "interest"].sum(), 2))
print("Balance after 5 years    :", round(sched.loc[60, "balance"], 2))
```

Output:

```
       payment  interest  principal    balance
month
1     1,498.88  1,250.00     248.88 249,751.12
2     1,498.88  1,248.76     250.12 249,501.00
3     1,498.88  1,247.51     251.37 249,249.63
       payment  interest  principal    balance
month
120   1,498.88  1,048.32     450.55 209,214.31
240   1,498.88    679.14     819.73 135,008.97
360   1,498.88      7.46   1,491.42      -0.00

Total interest over life : 289595.47
Interest in year 1       : 14916.49
Interest in year 30      : 571.17
Balance after 5 years    : 232635.89
```

The schedule shows the familiar front-loaded interest pattern. In month one, 1,250.00 of the 1,498.88 payment is interest; only 248.88 reduces principal. By month 120, interest has fallen to 1,048.32, and in the final month interest is only 7.46. Over the life of the 250,000 loan, the borrower pays 289,595.47 in interest, more than the amount borrowed. Interest in year one is 14,916.49 and in year 30 only 571.17. After five years the balance is still 232,635.89, so only 17,364.11 of principal (about 6.9 percent) has been repaid, a fact that surprises many first-time borrowers. The final balance of -0.00 is floating-point residue and confirms the schedule closes to zero.

Bond Pricing, Yield, Duration and Convexity

Volume 1, Chapter 4 priced bonds as the present value of coupons and principal. The bond library is the same idea in code, with the addition of the sensitivity measures that a treasurer or portfolio manager uses to manage interest-rate risk.

The formulas

For a bond with face value F, annual coupon rate c, m payments per year and a yield y (quoted as an annual rate compounded m times a year), the cash flow at time t_k = k/m is C = F·c/m, with the final flow also including F. Then:

```
Price                = Σ CF_k / (1 + y/m)^k
Macaulay duration    = Σ t_k · PV(CF_k) / Price                (in years)
Modified duration    = Macaulay / (1 + y/m)
Convexity            = Σ PV(CF_k) · k(k+1) / [ Price · m² · (1 + y/m)² ]
Price change ≈ -ModDur · Δy + ½ · Convexity · (Δy)²
```

Macaulay duration is the present-value-weighted average time to receipt of cash. Modified duration is the percentage price change for a small change in yield (with the sign reversed). Convexity captures the curvature of the price-yield relationship; it is always positive for a plain coupon bond, which is good for the bondholder: prices rise more when yields fall than they drop when yields rise by the same amount.

The bond library

Save the following as `bonds.py`.

```
"""Bond analytics: price, yield to maturity, duration and convexity."""
import numpy as np
from scipy.optimize import brentq

def cashflows(face, coupon_rate, years, freq=2):
    n = int(round(years * freq))
    t = np.arange(1, n + 1) / freq
    cf = np.full(n, face * coupon_rate / freq)
    cf[-1] += face
    return t, cf

def price(face, coupon_rate, years, ytm, freq=2):
    t, cf = cashflows(face, coupon_rate, years, freq)
    return float(np.sum(cf / (1 + ytm / freq) ** (t * freq)))

def ytm_from_price(face, coupon_rate, years, px, freq=2):
    return brentq(lambda y: price(face, coupon_rate, years, y, freq) - px, -0.05, 1.0, xtol=1e-12)

def macaulay(face, coupon_rate, years, ytm, freq=2):
    t, cf = cashflows(face, coupon_rate, years, freq)
    pvs = cf / (1 + ytm / freq) ** (t * freq)
    return float(np.sum(t * pvs) / np.sum(pvs))

def modified(face, coupon_rate, years, ytm, freq=2):
    return macaulay(face, coupon_rate, years, ytm, freq) / (1 + ytm / freq)

def convexity(face, coupon_rate, years, ytm, freq=2):
    t, cf = cashflows(face, coupon_rate, years, freq)
    k = t * freq
    pvs = cf / (1 + ytm / freq) ** k
    return float(np.sum(pvs * k * (k + 1)) / (np.sum(pvs) * freq ** 2 * (1 + ytm / freq) ** 2))
```

The `cashflows` function builds the schedule once and every other function reuses it, so adding day-count conventions later means changing one function.

Price, yield and risk

```
from bonds import *

face, c, T = 1000.0, 0.05, 10
for y in (0.03, 0.04, 0.05, 0.06, 0.07):
    print(f"YTM {y:5.1%}  price {price(face, c, T, y):9.2f}")

p = price(face, c, T, 0.06)
print("\nImplied YTM from price", round(p, 2), "->", round(ytm_from_price(face, c, T, p), 6))

mac = macaulay(face, c, T, 0.06)
mod = modified(face, c, T, 0.06)
cvx = convexity(face, c, T, 0.06)
print(f"Macaulay duration : {mac:.4f} years")
print(f"Modified duration : {mod:.4f}")
print(f"Convexity         : {cvx:.4f}")

# How good is the duration (and duration+convexity) approximation?
print("\n shock   actual%   duration%   dur+cvx%")
for dy in (-0.02, -0.01, 0.01, 0.02):
    actual = price(face, c, T, 0.06 + dy) / p - 1
    d1 = -mod * dy
    d2 = -mod * dy + 0.5 * cvx * dy ** 2
    print(f"{dy:+.2%}  {actual:9.4%}  {d1:9.4%}  {d2:9.4%}")

# Duration across maturities and coupons at a 6% yield
print("\nMaturity  zero-coupon  5% coupon  8% coupon  (modified duration)")
for T_ in (2, 5, 10, 20, 30):
    row = [modified(face, cc, T_, 0.06) for cc in (0.0, 0.05, 0.08)]
    print(f"{T_:>5}y   " + "   ".join(f"{x:9.3f}" for x in row))
```

Output:

```
YTM  3.0%  price   1171.69
YTM  4.0%  price   1081.76
YTM  5.0%  price   1000.00
YTM  6.0%  price    925.61
YTM  7.0%  price    857.88

Implied YTM from price 925.61 -> 0.06
Macaulay duration : 7.8950 years
Modified duration : 7.6650
Convexity         : 71.7854

 shock   actual%   duration%   dur+cvx%
-2.00%   16.8693%   15.3301%   16.7658%
-1.00%    8.0366%    7.6650%    8.0240%
+1.00%   -7.3180%   -7.6650%   -7.3061%
+2.00%  -13.9872%  -15.3301%  -13.8944%

Maturity  zero-coupon  5% coupon  8% coupon  (modified duration)
    2y       1.942       1.871       1.835
    5y       4.854       4.341       4.130
   10y       9.709       7.665       7.074
   20y      19.417      12.011      10.905
   30y      29.126      14.340      13.160
```

Study the first block. A bond with a 5 percent coupon and ten years to maturity is priced at exactly 1,000.00 when the yield is 5 percent, our sanity check. At a 3 percent yield it is worth 1,171.69, at 7 percent only 857.88. The price-yield relationship is convex: moving the yield down by two points adds 171.69, while moving it up by two points subtracts only 142.12.

At a 6 percent yield the price is 925.61 and `ytm_from_price` recovers 6 percent exactly, confirming that price and yield functions are inverses of each other. The Macaulay duration is 7.895 years: on a present-value-weighted basis, you wait nearly eight years to receive your money, even though the bond matures in ten, because coupons arrive along the way. The modified duration is 7.665, which means that a one percentage point increase in yield reduces the price by about 7.7 percent. The convexity is 71.79.

The shock table compares the exact price change with the two approximations. For a one-point move, the duration-only estimate is off by about 0.35 percentage points (it says -7.665 percent when the true change is -7.318 percent), while adding the convexity term reduces the error to about 0.012 points (-7.306 percent). For a two-point move the duration estimate is off by 1.34 points but the duration-plus-convexity estimate is within 0.1 points. Duration alone always overstates losses from rising yields and understates gains from falling yields; convexity corrects both errors.

The last table shows how duration behaves. A zero-coupon bond’s modified duration is almost exactly its maturity divided by (1 + y/2): 9.709 for a ten-year, 29.126 for a thirty-year. Coupons shorten duration, and the effect is greater for higher coupons and for longer maturities. Notice how a 30-year bond with a 5 percent coupon has a modified duration of 14.34, less than half the maturity: duration grows far more slowly than maturity at long horizons. This is why a long bond with a high coupon is a much safer instrument than a long zero.

A bond portfolio

Yields differ by maturity, so a spot curve gives the right discount rate for each cash flow. The listing bootstraps spot rates from par yields (s_n = [(1+y_n)/(1 - y_n·Σ 1/(1+s_k)^k)]^(1/n) - 1), derives one-year forwards, prices a 5-year 6 percent bond off the curve, and then computes the duration of a small bond book.

```
import numpy as np
import pandas as pd
from bonds import price, modified

# Bootstrapping a spot curve from par yields (annual-pay par bonds, face 100)
par = pd.Series({1: 0.030, 2: 0.034, 3: 0.037, 4: 0.039, 5: 0.041})
spot = {}
for n, y in par.items():
    annuity = sum(1 / (1 + spot[k]) ** k for k in range(1, n))
    spot[n] = ((1 + y) / (1 - y * annuity)) ** (1 / n) - 1
spot = pd.Series(spot)
fwd = pd.Series({n: (1 + spot[n]) ** n / (1 + spot[n - 1]) ** (n - 1) - 1 if n > 1 else spot[1] for n in
spot.index})
curve = pd.DataFrame({"par": par, "spot": spot, "fwd_1y": fwd})
print((curve * 100).round(4))

# Price a 5-year 6% annual-pay bond off the spot curve, then compute its YTM
cf = np.array([6, 6, 6, 6, 106.0])
pv = sum(cf[i] / (1 + spot[i + 1]) ** (i + 1) for i in range(5))
print("\nPrice off spot curve:", round(pv, 4))
from scipy.optimize import brentq
y = brentq(lambda r: sum(cf[i] / (1 + r) ** (i + 1) for i in range(5)) - pv, 0, 1)
print("Yield to maturity   :", round(y * 100, 4), "%")

# Portfolio duration of a small bond book (market values in USD millions)
book = pd.DataFrame({
    "coupon": [0.03, 0.05, 0.07],
    "years":  [3, 7, 15],
    "ytm":    [0.040, 0.055, 0.062],
    "face_m": [50.0, 80.0, 40.0],
})
book["price"] = [price(100, r.coupon, r.years, r.ytm) for r in book.itertuples()]
book["mv_m"] = book["face_m"] * book["price"] / 100
book["moddur"] = [modified(100, r.coupon, r.years, r.ytm) for r in book.itertuples()]
w = book["mv_m"] / book["mv_m"].sum()
pd.set_option("display.float_format", lambda x: f"{x:,.3f}")
print()
print(book)
print("\nPortfolio market value (USD m):", round(book["mv_m"].sum(), 3))
print("Portfolio modified duration   :", round((w * book["moddur"]).sum(), 4))
print("Approx. loss for +50bp shift  : USD", round((w * book["moddur"]).sum() * 0.005 * book["mv_m"].sum(),
3), "m")
```

Output:

```
   par    spot  fwd_1y
1  3.0  3.0000  3.0000
2  3.4  3.4068  3.8153
3  3.7  3.7163  4.3380
4  3.9  3.9250  4.5538
5  4.1  4.1384  4.9964

Price off spot curve: 108.5047
Yield to maturity   : 4.0851 %

   coupon  years   ytm  face_m   price   mv_m  moddur
0   0.030      3 0.040  50.000  97.199 48.600   2.833
1   0.050      7 0.055  80.000  97.127 77.702   5.814
2   0.070     15 0.062  40.000 107.740 43.096   9.441

Portfolio market value (USD m): 169.397
Portfolio modified duration   : 5.8814
Approx. loss for +50bp shift  : USD 4.981 m
```

The upward-sloping par curve implies a higher spot curve (4.1384 percent at five years) and a still higher forward curve (4.9964 percent for year five). The bond priced off the spot curve is worth 108.5047 and yields 4.0851 percent.

Portfolio duration is the market-value-weighted average of component durations, exact for parallel shifts. Our 169.397 million book has a modified duration of 5.8814, and a parallel upward shift of 50 basis points would cost roughly 4.981 million.

Project Simulation: Monte Carlo NPV

Volume 1, Chapter 8 introduced scenario analysis, sensitivity analysis and simulation as ways of dealing with uncertainty in a capital budgeting forecast. A deterministic NPV gives one number. A simulation gives a distribution, and a distribution answers questions that a single number cannot: how likely is a loss, how bad is the downside, and which input drives the risk?

The method

A simulation proceeds in five steps: write the NPV as a function of its uncertain inputs; assign each input a distribution that reflects what you actually know; draw many sets of inputs and compute the NPV for each; summarise the distribution (mean, percentiles, probability of a loss); and report the simulation error.

The illustrative project

Our fictional company, Tidewater Components, is evaluating a new plant. The initial capital expenditure is 70 million dollars, depreciated straight-line over an eight-year life with no salvage value. Four inputs are uncertain:

Input	Distribution	Parameters	Rationale
Year-1 volume (thousand units)	Triangular	min 800, mode 1,000, max 1,300	Sales team has a most-likely figure and a skewed range
Price per unit (USD)	Normal	mean 50, std 6	Competitive pricing noise
Variable cost (share of price)	Uniform	52% to 68%	Input cost uncertainty with no preferred value
Annual volume growth	Normal	mean 3%, std 2%	Market growth

Fixed cash costs are 4.0 million a year, tax is 25 percent on positive EBIT, and the discount rate is 10 percent. All of these are treated as known, which is itself an assumption. We choose the triangular distribution for volume because the sales director can state a minimum, a most likely value and a maximum but not a mean and standard deviation. We choose the uniform

distribution for variable cost because the procurement team could give only a range. These choices matter for the answer, and an analyst should always document and defend them.

The code builds a two-dimensional array with one row per trial and one column per year, so that all 100,000 forecasts are computed simultaneously. Revenue in millions equals volume in thousands times price in dollars divided by 1,000.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(2026)
N = 100_000

# Uncertain inputs for a new-plant project (USD millions, illustrative)
units   = rng.triangular(left=800, mode=1000, right=1300, size=N)          # thousand units, year-1 volume
price   = rng.normal(loc=50.0, scale=6.0, size=N)                        # USD per unit
vc_pct  = rng.uniform(0.52, 0.68, size=N)                                # variable cost as share of price
growth  = rng.normal(0.03, 0.02, size=N)                                 # annual volume growth
capex   = 70.0
fixed   = 4.0                                                            # fixed cash costs, USD m/yr
tax     = 0.25
wacc    = 0.10
T       = 8
dep     = capex / T

years = np.arange(1, T + 1)
vol = units[:, None] * (1 + growth[:, None]) ** (years[None, :] - 1)     # (N, T) thousand units
revenue = vol * price[:, None] / 1000.0                                  # USD m
ebitda = revenue * (1 - vc_pct[:, None]) - fixed
ebit = ebitda - dep
fcf = ebitda - np.maximum(ebit, 0) * tax
pv = (fcf / (1 + wacc) ** years[None, :]).sum(axis=1)
npv = pv - capex

print("Simulated trials :", N)
print("Mean NPV         :", round(npv.mean(), 2))
print("Median NPV       :", round(np.median(npv), 2))
print("Std dev          :", round(npv.std(ddof=1), 2))
print("5th / 95th pct   :", np.round(np.percentile(npv, [5, 95]), 2))
print("P(NPV < 0)       :", round((npv < 0).mean(), 4))
print("Expected shortfall (mean NPV | worst 5%):", round(npv[npv <= np.percentile(npv, 5)].mean(), 2))
se = npv.std(ddof=1) / np.sqrt(N)
print("Std error of mean:", round(se, 4), " -> 95% CI half-width", round(1.96 * se, 3))

# Deterministic base case for comparison
base_fcf = np.array([ (1000*(1.03)**(t-1)*50/1000*(1-0.60) - fixed) for t in years])
base_fcf = base_fcf - np.maximum(base_fcf - dep, 0) * tax
print("\nBase-case NPV (point estimates):", round((base_fcf / 1.1 ** years).sum() - capex, 2))

# Text histogram
counts, edges = np.histogram(npv, bins=14)
print("\nHistogram of NPV (each # = 1% of trials)")
for c, lo, hi in zip(counts, edges[:-1], edges[1:]):
    print(f"{lo:7.1f} to {hi:7.1f} | {'#' * int(round(100 * c / N))}")
```

Output:

```
Simulated trials : 100000
Mean NPV         : 16.41
Median NPV       : 14.83
Std dev          : 18.56
5th / 95th pct   : [-11.18  49.43]
P(NPV < 0)       : 0.192
Expected shortfall (mean NPV | worst 5%): -16.48
Std error of mean: 0.0587  -> 95% CI half-width 0.115

Base-case NPV (point estimates): 13.32

Histogram of NPV (each # = 1% of trials)
  -46.6 to   -34.7 |
  -34.7 to   -22.8 |
  -22.8 to   -10.9 | #####
  -10.9 to     1.0 | ################
    1.0 to    13.0 | ########################
   13.0 to    24.9 | ########################
   24.9 to    36.8 | ################
   36.8 to    48.7 | #########
   48.7 to    60.6 | ####
   60.6 to    72.5 | #
   72.5 to    84.4 |
   84.4 to    96.4 |
   96.4 to   108.3 |
  108.3 to   120.2 |
```

Interpreting the results

The deterministic base case, using the central value for each input (volume of 1,000, price of 50, variable cost of 60 percent and growth of 3 percent), gives an NPV of 13.32 million. The simulation's mean NPV is 16.41 million, higher than the base case. This is not a bug. The triangular volume distribution has a longer right tail (the mode is 1,000 but the mean is 1,033), and the NPV rises by about 0.088 million for each extra thousand units, so the extra 33 thousand units of mean volume alone explain about 2.9 million of the gap. Smaller convexity effects, from the compounding of growth and the tax floor, add the rest. In general the expected value of a nonlinear function is not the function of the expected values, a result known as Jensen's inequality. The practical message: **a base case built on most-likely inputs is not the expected outcome**, and managers who approve projects on the base case are systematically misreading the risk-reward trade-off.

The median, 14.83, is below the mean, indicating right skew, which the histogram confirms (about a quarter of trials in each of the two central bins, a left tail of losses and a right tail past 70 million). The standard deviation is 18.56 million, larger than the mean. The 5th percentile is -11.18 million and the 95th is 49.43 million. The probability that the project destroys value is 19.2 percent, and the expected shortfall, the average NPV in the worst 5 percent of outcomes, is -16.48 million. A board that approves this project on a base-case NPV of 13 million is implicitly accepting roughly a one-in-five chance of destroying value; the right question is not "is the NPV positive?" but "can we live with a 1-in-20 chance of losing 11 million or more?"

Simulation error. With 100,000 trials the standard error of the mean is 0.0587, so the mean NPV is known to within about 0.115 million at 95 percent confidence, far smaller than the uncertainty in the inputs. The error falls with the square root of the number of trials, as the next listing shows.

Sensitivity (tornado) analysis and correlated inputs

A simulation tells you how wide the distribution is. A **tornado analysis** tells you why. We move each input in turn from a low to a high value while holding the others at base, and rank inputs by the swing in NPV. We then simulate correlated inputs, because in most businesses price and volume do not move independently: a firm that cuts its price generally sells more, so price and volume are negatively correlated across scenarios.

For the correlated case we use a standard trick. Draw two independent standard normals z1 and z2, form z2’ = ρ·z1 + √(1-ρ²)·z2, so that the pair (z1, z2’) has correlation ρ, and then transform each into the desired marginal distribution. Here the price is normal and the volume is lognormal, with a log-volatility of 15 percent.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(2026)
N = 100_000
capex, fixed, tax, wacc, T = 70.0, 4.0, 0.25, 0.10, 8
dep = capex / T
years = np.arange(1, T + 1)

def npv_model(units, price, vc_pct, growth):
    vol = np.atleast_1d(units)[:, None] * (1 + np.atleast_1d(growth)[:, None]) ** (years[None, :] - 1)
    revenue = vol * np.atleast_1d(price)[:, None] / 1000.0
    ebitda = revenue * (1 - np.atleast_1d(vc_pct)[:, None]) - fixed
    fcf = ebitda - np.maximum(ebitda - dep, 0) * tax
    return (fcf / (1 + wacc) ** years[None, :]).sum(axis=1) - capex

base = dict(units=1000.0, price=50.0, vc_pct=0.60, growth=0.03)
low  = dict(units=800.0,  price=38.0, vc_pct=0.68, growth=-0.01)
high = dict(units=1300.0, price=62.0, vc_pct=0.52, growth=0.07)
b = npv_model(**base)[0]
print(f"Base-case NPV: {b:.2f}")
rows = []
for k in base:
    lo = npv_model(**{**base, k: low[k]})[0]
    hi = npv_model(**{**base, k: high[k]})[0]
    rows.append((k, lo - b, hi - b, abs(hi - lo)))
tor = pd.DataFrame(rows, columns=["driver", "low_delta", "high_delta", "swing"]).sort_values("swing",
ascending=False)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
print(tor.to_string(index=False))

# Correlated inputs: price and volume move in opposite directions (rho = -0.5)
z = rng.standard_normal((N, 2))
rho = -0.5
z2 = rho * z[:, 0] + np.sqrt(1 - rho ** 2) * z[:, 1]
price = 50 + 6 * z[:, 0]
units = 1000 * np.exp(0.15 * z2 - 0.5 * 0.15 ** 2)
vc = rng.uniform(0.52, 0.68, N)
g = rng.normal(0.03, 0.02, N)
n_corr = npv_model(units, price, vc, g)
print()
print("Correlation(price, units) :", round(np.corrcoef(price, units)[0, 1], 3))
print("Mean NPV / Std            :", round(n_corr.mean(), 2), "/", round(n_corr.std(ddof=1), 2))
print("P(NPV<0)                  :", round((n_corr < 0).mean(), 4))
# Rank correlation of each input with NPV (which input matters most?)
inputs = pd.DataFrame({"units": units, "price": price, "vc_pct": vc, "growth": g})
rank = inputs.rank().corrwith(pd.Series(n_corr).rank())
print("\nSpearman rank correlation with NPV")
print(rank.round(3).to_string())
```

Output:

```
Base-case NPV: 13.32
driver  low_delta  high_delta  swing
 units     -17.53       26.30  43.83
 price     -21.04       21.04  42.07
vc_pct     -17.53       17.53  35.06
growth      -9.99       11.57  21.56

Correlation(price, units) : -0.499
Mean NPV / Std            : 12.78 / 16.71
P(NPV<0)                  : 0.235

Spearman rank correlation with NPV
units     0.44
price     0.23
vc_pct   -0.62
growth    0.30
```

The tornado table shows volume has the largest swing (43.83 million between the low case of 800 and the high case of 1,300 thousand units), followed closely by price (42.07), variable cost (35.06) and growth (21.56). Note that the swing for volume is asymmetric: the high case adds 26.30 million but the low case subtracts only 17.53, simply because the range we assumed (200 below the base, 300 above) is itself asymmetric. The swings for price and variable cost are symmetric, because they enter the model linearly and the low and high values are symmetric around the base. The ranking depends on the ranges chosen, which are judgments, not facts: a tornado is only as honest as its ranges.

With a price-volume correlation of -0.5 (the sample correlation is -0.499), the mean NPV is 12.78 million, the standard deviation 16.71 million and the probability of a loss 23.5 percent. Do not compare these directly with the first simulation, because we also changed volume from triangular to lognormal (with mean exactly 1,000), which lowers the mean. A clean test of correlation holds the marginals fixed and changes only ρ.

The rank correlations at the bottom answer a different question from the tornado: they reflect the distributions actually simulated rather than low and high cases chosen by the analyst. Variable cost has the largest rank correlation with NPV (-0.62), followed by volume (0.44), growth (0.30) and price (0.23). Price ranks low partly because the negative price-volume correlation makes its effect partly offset by volume. When tornado and rank correlations disagree, your low and high cases probably do not correspond to the distributions you assumed, and one of them needs revisiting.

How many trials are enough?

The next listing repeats the simulation twenty times at each of four sample sizes, using different seeds, and reports how much the estimated mean and loss probability vary from run to run. It then demonstrates **common random numbers**: when comparing two project designs, evaluate both on exactly the same random draws, so that sampling noise cancels. Design A is the plant we have been analysing. Design B is a more manual plant with a lower capital cost (58 million), lower fixed costs (3.0 million) but a variable cost that is six percentage points of price higher.

```
import numpy as np

def project_npv(units, price, vc, g, capex=70.0, fixed=4.0, tax=0.25, wacc=0.10, T=8):
    years = np.arange(1, T + 1)
    vol = units[:, None] * (1 + g[:, None]) ** (years[None, :] - 1)
    ebitda = vol * price[:, None] / 1000 * (1 - vc[:, None]) - fixed
    fcf = ebitda - np.maximum(ebitda - capex / T, 0) * tax
    return (fcf / (1 + wacc) ** years[None, :]).sum(axis=1) - capex

def draw(rng, n):
    return (rng.triangular(800, 1000, 1300, n), rng.normal(50, 6, n),
            rng.uniform(0.52, 0.68, n), rng.normal(0.03, 0.02, n))

print("Convergence of the Monte Carlo estimate (20 independent replications each)")
print("   N      mean of means   spread of means (std)   P(NPV<0) mean")
for n in (500, 2_000, 10_000, 50_000):
    means, probs = [], []
    for rep in range(20):
        rng = np.random.default_rng(1000 + rep)
        v = project_npv(*draw(rng, n))
        means.append(v.mean()); probs.append((v < 0).mean())
    print(f"{n:>7,}   {np.mean(means):12.3f}   {np.std(means, ddof=1):18.4f}   {np.mean(probs):12.4f}")

# Common random numbers: compare two designs using the same draws
rng = np.random.default_rng(99)
u, p, vc, g = draw(rng, 20_000)
a = project_npv(u, p, vc, g, capex=70.0)
b = project_npv(u, p, vc + 0.06, g, capex=58.0, fixed=3.0)       # cheaper, more manual plant: lower capex and
fixed cost, higher variable cost
d = b - a
print("\nMean NPV A / B          :", round(a.mean(), 2), "/", round(b.mean(), 2))
print("Mean of paired diff B-A :", round(d.mean(), 3), "  s.e.:", round(d.std(ddof=1) / np.sqrt(len(d)), 4))
print("P(B beats A)            :", round((d > 0).mean(), 4))
print("Std of NPV A / B        :", round(a.std(ddof=1), 2), "/", round(b.std(ddof=1), 2))
```

Output:

```
Convergence of the Monte Carlo estimate (20 independent replications each)
   N      mean of means   spread of means (std)   P(NPV<0) mean
    500         16.322               0.6539         0.1945
  2,000         16.436               0.4578         0.1943
 10,000         16.485               0.2237         0.1926
 50,000         16.456               0.0788         0.1935

Mean NPV A / B          : 16.61 / 16.98
Mean of paired diff B-A : 0.372   s.e.: 0.0162
P(B beats A)            : 0.5886
Std of NPV A / B        : 18.63 / 16.8
```

The spread of the estimated mean falls from 0.6539 with 500 trials to 0.0788 with 50,000, a ratio of 8.3, close to the theoretical ratio of √100 = 10 for a 100-fold increase in trials. The estimated probability of loss converges to 19.3 or 19.4 percent. For most capital budgeting purposes 10,000 trials gives a mean accurate to about 0.2 million and is sufficient; use more when you care about tail probabilities, because rare events need more samples to be observed at all.

The comparison of designs shows why common random numbers are powerful. Design A's mean NPV is 16.61 and design B's is 16.98: the difference is small relative to the standard deviation of each (18.63 and 16.80). Computed paired, draw by draw, the mean difference is 0.372 with a standard error of only 0.0162, so we can say with confidence that B is slightly better on average, by about 0.37 million. If we had used independent draws for the two designs, the standard error of the difference would be about 0.18 million (the quadrature of the two standard errors for 20,000 trials), more than ten times larger, and we could not have distinguished B from A at all. B wins in 58.9 percent of trials, and has a lower standard deviation. So B is both slightly better in expectation and less risky, but it is not a runaway winner, and other considerations such as strategic flexibility, a plant's ability to scale, and the quality of the product might decide the matter.

Plotting: Describing What the Charts Show

A chart is the quickest way to communicate a distribution or a profile. Because this book cannot reproduce images, the listing draws two panels, saves the figure, and prints facts about the plotted data. In your own work you would open the saved image or display it in a notebook.

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import os, tempfile

rng = np.random.default_rng(7)
rates = np.linspace(0, 0.30, 61)
cf = np.array([-1000, 250, 300, 350, 400.0])
npv = [(cf / (1 + r) ** np.arange(5)).sum() for r in rates]
sim = rng.normal(15, 40, 20000)

fig, ax = plt.subplots(1, 2, figsize=(10, 4))
ax[0].plot(rates * 100, npv, color="#1f4e79")
ax[0].axhline(0, color="grey", lw=0.8)
ax[0].set_xlabel("Discount rate (%)"); ax[0].set_ylabel("NPV (USD thousands)")
ax[0].set_title("NPV profile")
ax[1].hist(sim, bins=40, color="#1f4e79", alpha=0.85)
ax[1].axvline(0, color="red", lw=1)
ax[1].set_xlabel("NPV (USD millions)"); ax[1].set_title("Simulated NPV distribution")
fig.tight_layout()
path = os.path.join(tempfile.gettempdir(), "chapter19_figure.png")
fig.savefig(path, dpi=120)
print("Figure written:", os.path.basename(path), "|", os.path.getsize(path) > 10_000, "(size over 10 KB)")
# Where the NPV profile crosses zero (the IRR), read straight from the plotted data
i = np.where(np.diff(np.sign(npv)))[0][0]
print("NPV profile crosses zero between", round(rates[i] * 100, 1), "% and", round(rates[i + 1] * 100, 1),
"%")
print("Axes in figure:", len(fig.axes), "| lines in left panel:", len(ax[0].lines))
```

Output:

```
Figure written: chapter19_figure.png | True (size over 10 KB)
NPV profile crosses zero between 10.0 % and 10.5 %
Axes in figure: 2 | lines in left panel: 2
```

The left panel is an **NPV profile**: the discount rate on the horizontal axis from 0 to 30 percent and the NPV on the vertical axis. For our conventional project, the curve starts at 300 at a zero discount rate, and falls steadily, convex to the origin, crossing zero between 10.0 and 10.5 percent. That crossing is the IRR, and the figure makes the decision rule visible: accept if the cost of capital is to the left of the crossing, reject if it is to the right. If you drew two projects' profiles on the same axes, the crossing point of the two curves would be the **crossover rate** (the rate at which the two have the same NPV), which is the key to ranking mutually exclusive projects, as Exercise 2 below shows.

The right panel is a **histogram of simulated NPV** with a vertical red line at zero; the fraction of the histogram left of the line is the probability of a loss. In a presentation, annotate the mean and median, the loss probability, and the 5th and 95th percentiles.

Common Errors and How to Avoid Them

Error	Symptom	Prevention
Mixing annual rate with monthly periods	Payments or values that are off by a factor near 12	Convert both rate and periods to the same unit and name variables `monthly_rate`, `n_months`
Reporting IRR without checking sign changes	Multiple or no IRRs	Print the NPV profile; use NPV for decisions
Using the base case as the expected value	Systematic under or overstatement	Simulate; compare the mean with the base case
Unseeded simulations	Results change each run	Seed the generator and record the seed
Independent inputs that are really correlated	Risk understated	Model dependence explicitly; test several correlations

Exercises with Solutions

The exercises below were solved by running the listing that follows them. Try each by hand or on a calculator first.

Exercise 1: Balloon loan. A company borrows 1,000,000 at 7 percent per year, with annual payments calculated as if the loan amortised over 20 years, but the whole remaining balance is due as a balloon at the end of year 5. What is the annual payment and what is the balloon?

Exercise 2: Mutually exclusive projects. Project A has cash flows (-500, 50, 100, 250, 450) and project B has (-500, 350, 250, 100, 50). The cost of capital is 9 percent. Compute each NPV and IRR, find the crossover rate, and say which to choose at 5, 9 and 15 percent.

Exercise 3: Bond analytics. A 4.5 percent semiannual-pay bond with 8 years to maturity and face value 1,000 trades at 975. Compute its yield to maturity, modified duration and convexity. Estimate the new price if yields rise by 75 basis points using (a) the exact price formula and (b) duration plus convexity.

```
import numpy as np
from tvm import pmt, npv, irr
from bonds import price, ytm_from_price, modified, convexity

print("EXERCISE 1: balloon loan")
loan, rate_, n_am, n_term = 1_000_000, 0.07, 20, 5
pay = -pmt(rate_, n_am, loan)
bal = loan
for _ in range(n_term):
    bal = bal * (1 + rate_) - pay
print("Annual payment :", round(pay, 2))
print("Balloon at yr 5:", round(bal, 2))

print("\nEXERCISE 2: mutually exclusive projects")
A = [-500, 50, 100, 250, 450]
B = [-500, 350, 250, 100, 50]
for name, cf in (("A", A), ("B", B)):
    print(name, "NPV@9%:", round(npv(0.09, cf), 2), " IRR:", round(irr(cf) * 100, 3), "%")
diff = np.array(A) - np.array(B)
x = irr(diff, 0.0, 1.0)
print("Crossover rate (IRR of A-B):", round(x * 100, 3), "%")
for r in (0.05, 0.10, 0.15):
    print(f"  NPV A-B at {r:.0%}:", round(npv(r, diff), 2))

print("\nEXERCISE 3: bond analytics")
px = 975.0
y = ytm_from_price(1000, 0.045, 8, px)
print("YTM (semiannual BEY):", round(y * 100, 4), "%")
md = modified(1000, 0.045, 8, y)
cx = convexity(1000, 0.045, 8, y)
print("Modified duration   :", round(md, 4), " Convexity:", round(cx, 3))
new = price(1000, 0.045, 8, y + 0.0075)
print("Exact price at +75bp:", round(new, 3), "  change:", round(new / px - 1, 5))
print("Approx (dur+cvx)    :", round(-md * 0.0075 + 0.5 * cx * 0.0075 ** 2, 5))
```

Output:

```
EXERCISE 1: balloon loan
Annual payment : 94392.93
Balloon at yr 5: 859722.65

EXERCISE 2: mutually exclusive projects
A NPV@9%: 141.88  IRR: 17.884 %
B NPV@9%: 144.16  IRR: 26.785 %
Crossover rate (IRR of A-B): 8.741 %
  NPV A-B at 5%: 36.89
  NPV A-B at 10%: -10.79
  NPV A-B at 15%: -46.96

EXERCISE 3: bond analytics
YTM (semiannual BEY): 4.8812 %
Modified duration   : 6.6259  Convexity: 51.885
Exact price at +75bp: 927.941   change: -0.04827
Approx (dur+cvx)    : -0.04824
```

Solutions and commentary

Solution 1. The annual payment on the 20-year amortisation is 94,392.93. After five payments the balance is 859,722.65, which is the balloon. The borrower has paid 471,964.65 over five years and has repaid only about 140,277 of principal. The balloon is therefore 86 percent of the original loan, so the refinancing risk is dominant: the company's real exposure is to the interest rate and credit spread prevailing at year five.

Solution 2. Project A has an NPV of 141.88 at 9 percent and an IRR of 17.88 percent; project B has an NPV of 144.16 and an IRR of 26.79 percent. B ranks higher on IRR at every cost of capital, but the ranking by NPV depends on the rate. The crossover rate is 8.74 percent, the IRR of the difference stream A minus B. Below it, A's larger late cash flows dominate (at 5 percent, A leads by 36.89); above it, B's faster payback wins (at 10 percent B leads by 10.79; at 15 percent by 46.96). At 9 percent the two are almost identical, with B ahead by 2.28, because 9 percent is just above the crossover. The recommendation at 9 percent is B, but a decision that depends on a 0.26 point difference in the discount rate is fragile. Inspect the reinvestment assumption, the uncertainty in later flows for A, and any capital rationing, before committing.

Solution 3. The yield to maturity is 4.8812 percent. Modified duration is 6.6259 and convexity is 51.885. At a yield 75 basis points higher the exact price is 927.941, a decline of 4.827 percent; the duration-plus-convexity approximation gives 4.824 percent, accurate to three thousandths of a percentage point.

Chapter Summary

- Python gives repeatability, scale and auditability; fix the seed, record versions, and keep inputs separate from logic.
- A cash flow is an array; a model is a DataFrame; discounting is one vectorised line.
- A small library of tested TVM functions (fv, pv, pmt, nper, rate, npv, irr, xirr, mirr, payback) handles most corporate finance calculations. Always check sign conventions and timing conventions, particularly the difference between Excel's NPV and the NPV of a time-zero stream.
- IRR can fail or mislead with multiple sign changes; the NPV profile is the reliable diagnostic.
- Bond price, yield, duration and convexity come from one cash flow schedule; duration plus convexity approximates price changes well for moderate yield shifts.
- Monte Carlo simulation turns a single NPV into a distribution, revealing probabilities of loss, tail outcomes and the gap between base case and expected value.
- Use tornado analysis to find which inputs matter, correlation to represent dependence, replication to measure simulation error, and common random numbers to compare alternatives.

Chapter 20: Building LBO and Merger Models

Purpose and Scope

Volume 1, Chapter 22 explained why companies merge and how deals are valued, and Volume 1, Chapter 23 explained how private equity sponsors use leverage to buy and improve businesses. Volume 2 returned to both topics with more advanced valuation and structuring material. This chapter is where those ideas become working models. We build two of the most common models in corporate finance and investment banking from the ground up:

1. A **leveraged buyout (LBO) model**: sources and uses of funds, an operating forecast, a debt schedule with a cash sweep, exit returns and sensitivity tables.
2. An **accretion/dilution model** for an acquisition by a public company: financing structures, synergies, purchase accounting effects, pro forma earnings per share, break-even analysis and ownership.

For each model we show the **Excel layout and formula patterns** that practitioners use, then the same logic in **Python**, so you can see the model's structure without the clutter of cell references, and so you can run hundreds of scenarios quickly. All code was executed, and each block labelled "Output" is the actual printed result. The two companies, Northgate Industrial Services (the LBO target) and Meridian Software and Calloway Data (the acquirer and target in the merger model), are fictional and illustrative. All numbers were chosen to teach technique, not to describe any real company or deal.

A word of caution, repeated throughout the chapter: a model is a disciplined way of organising assumptions, not a source of truth. The most important outputs of any model are the sensitivity tables, which show how the answer depends on assumptions that nobody can know with confidence. A model that gives a single IRR to one decimal place should be treated with suspicion.

Part I: The LBO Model

How an LBO works, in one paragraph

A sponsor buys a company using a small amount of its own money (equity) and a large amount of borrowed money, secured on the company's assets and cash flows. The company then uses its own cash flow to pay interest and repay debt. At the end of a holding period, usually three to seven years, the sponsor sells the company. Because debt is repaid from the company's cash flow, the value of the equity grows even if the enterprise value does not. The return to the sponsor comes from three sources, which we separate in a value-creation bridge later in the chapter: growth in earnings, change in the valuation multiple, and the paydown of debt.

The Excel layout pattern

Experienced modellers lay out an LBO on a small number of sheets, each with a single purpose, with flows left to right across time. A typical structure is:

Sheet or block	Contents	Key design rule
Assumptions	Entry multiple, leverage, rates, fees, operating drivers, exit assumptions	Every hard-coded number lives here, in blue font; no number appears inside a formula
Sources and Uses	Debt tranches and equity versus purchase price and fees	Total sources must equal total uses; a check cell shows the difference
Operating model	Revenue, EBITDA, D&A, capex, working capital	Drivers are percentages or growth rates taken from Assumptions
Debt schedule	Opening balance, mandatory repayment, sweep, closing balance, interest	One block per tranche; interest links to the balance
Cash flow and credit statistics	Cash flow available for debt repayment, leverage and coverage ratios	Ratios are tested against covenants
Returns	Exit value, net debt, equity value, MOIC, IRR, bridge	Exit year selected by an index formula
Sensitivities	Data tables of IRR and MOIC	Reference the returns output cell

The formula patterns we use most often in the debt schedule are listed below. Row 10 holds the opening balance and column D onward holds the years; cell `$C$5` holds the original term loan.

Item	Excel formula pattern	Comment
Mandatory amortisation	`=MIN($C$5*$C$6, D10)`	A percentage of the original balance, never more than the opening balance
Cash available for sweep	`=D20 - D11`	Cash flow available for debt repayment less mandatory amortisation
Cash sweep	`=MAX(0, MIN($C$7*D12, D10 - D11))`	Sweep percentage times available cash, floored at zero, capped at what remains
Closing balance	`=D10 - D11 - D13`	Opening less mandatory less sweep
Interest on average balance	`=$C$8*AVERAGE(D10, D14)`	Creates a circular reference, since the closing balance depends on cash flow, which depends on interest
Interest on opening balance	`=$C$8*D10`	No circularity; slightly overstates interest when debt is being repaid
Circuit breaker	`=IF($C$9=1, AVERAGE(D10, D14), D10)`	A switch cell that breaks the loop when the model produces errors
Exit equity	`=INDEX(D30:H30, $C$10) - INDEX(D31:H31, $C$10)`	Exit year picks the column
IRR	`=IRR(C40:H40)` or `=(MOIC)^(1/N)-1`	The second form is valid when there are no interim distributions

Assumptions and the sources-and-uses table

Northgate Industrial Services is a fictional provider of maintenance and compliance services to industrial plants. It has last-twelve-month (LTM) revenue of 600 million dollars and EBITDA of 100 million. A sponsor proposes to buy it at 9.0 times LTM EBITDA, an enterprise value of 900 million, financed with a term loan B of 4.0 times EBITDA at 8.0 percent, senior notes of 1.5 times at 10.5 percent, and sponsor equity as the balancing figure. The sponsor also pays transaction fees of 1.5 percent of enterprise value, financing fees of 2.0 percent of debt raised, and leaves 10 million of cash on the balance sheet as a minimum operating balance.

The model code is in one module, `lbo.py`, so that every later listing can import it. Read it through once; the rest of this part explains each function in turn.

```
"""A compact LBO model: sources and uses, operating forecast, debt schedule with cash sweep, returns."""
import numpy as np
import pandas as pd

BASE = dict(
    ltm_ebitda=100.0, entry_multiple=9.0, exit_multiple=9.0, exit_year=5,
    tlb_x=4.0, notes_x=1.5,                          # debt as multiples of LTM EBITDA
    tlb_rate=0.080, notes_rate=0.105, tlb_amort=0.01, sweep=1.0,
    txn_fee_pct=0.015, fin_fee_pct=0.02, min_cash=10.0,
    revenue0=600.0, growth=[0.06, 0.06, 0.05, 0.05, 0.04],
    margin0=100.0 / 600.0, margin_step=0.005,       # EBITDA margin expands 50bp a year
    da_pct=0.03, capex_pct=0.035, nwc_pct=0.10, tax=0.25,
)

def sources_uses(p):
    ev = p["ltm_ebitda"] * p["entry_multiple"]
    tlb = p["ltm_ebitda"] * p["tlb_x"]
    notes = p["ltm_ebitda"] * p["notes_x"]
    debt = tlb + notes
    txn_fees = ev * p["txn_fee_pct"]
    fin_fees = debt * p["fin_fee_pct"]
    uses = ev + txn_fees + fin_fees + p["min_cash"]
    equity = uses - debt
    return dict(ev=ev, tlb=tlb, notes=notes, txn_fees=txn_fees, fin_fees=fin_fees,
                cash=p["min_cash"], uses=uses, equity=equity)

def operating_model(p):
    n = len(p["growth"])
    rev = p["revenue0"] * np.cumprod(1 + np.array(p["growth"]))
    prev = np.concatenate([[p["revenue0"]], rev[:-1]])
    margin = p["margin0"] + p["margin_step"] * np.arange(1, n + 1)
    ebitda = rev * margin
    da = rev * p["da_pct"]
    capex = rev * p["capex_pct"]
    d_nwc = (rev - prev) * p["nwc_pct"]
    df = pd.DataFrame({"revenue": rev, "ebitda": ebitda, "da": da, "ebit": ebitda - da,
                       "capex": capex, "d_nwc": d_nwc}, index=range(1, n + 1))
    return df

def debt_schedule(p, su, ops, average_interest=True, tol=1e-9):
    """Solve the circular interest/cash-sweep problem year by year by fixed-point iteration."""
    n = len(ops)
    tlb_open, notes_bal = su["tlb"], su["notes"]
    rows = []
    for t in range(1, n + 1):
        o = ops.loc[t]
        mand = min(p["tlb_amort"] * su["tlb"], tlb_open)
        tlb_close = tlb_open - mand                      # first guess: no sweep
        for it in range(100):
            tlb_basis = (tlb_open + tlb_close) / 2 if average_interest else tlb_open
            int_tlb = p["tlb_rate"] * tlb_basis
            int_notes = p["notes_rate"] * notes_bal
            interest = int_tlb + int_notes
            ebt = o.ebit - interest
            taxes = max(ebt, 0.0) * p["tax"]
            ni = ebt - taxes
            cfads = ni + o.da - o.capex - o.d_nwc        # cash flow available for debt repayment
            avail = cfads - mand
```

```
            sweep = max(0.0, min(p["sweep"] * avail, tlb_open - mand))
            new_close = tlb_open - mand - sweep
            if abs(new_close - tlb_close) < tol:
                break
            tlb_close = new_close
        rows.append(dict(year=t, tlb_open=tlb_open, mandatory=mand, sweep=sweep, tlb_close=tlb_close,
                         notes=notes_bal, interest=interest, taxes=taxes, net_income=ni,
                         cfads=cfads, iterations=it + 1))
        tlb_open = tlb_close
    sch = pd.DataFrame(rows).set_index("year")
    sch["cash_build"] = sch["cfads"] - sch["mandatory"] - sch["sweep"]
    sch["cash"] = su["cash"] + sch["cash_build"].cumsum()
    sch["total_debt"] = sch["tlb_close"] + sch["notes"]
    sch["net_debt"] = sch["total_debt"] - sch["cash"]
    sch["lev"] = sch["net_debt"] / ops["ebitda"]
    sch["cover"] = ops["ebitda"] / sch["interest"]
    return sch

def returns(p, su, ops, sch):
    t = p["exit_year"]
    exit_ev = ops.loc[t, "ebitda"] * p["exit_multiple"]
    exit_eq = exit_ev - sch.loc[t, "net_debt"]
    moic = exit_eq / su["equity"]
    irr = moic ** (1 / t) - 1
    return dict(exit_ev=exit_ev, exit_equity=exit_eq, moic=moic, irr=irr)

def run_lbo(**over):
    p = {**BASE, **over}
    su = sources_uses(p)
    ops = operating_model(p)
    sch = debt_schedule(p, su, ops, average_interest=p.get("average_interest", True))
    ret = returns(p, su, ops, sch)
    return p, su, ops, sch, ret
```

The first function, `sources_uses`, is the simplest and the one most often wrong in practice, because the fees are easy to forget. Total uses are the enterprise value, the two fees and the cash left on the balance sheet. Debt is fixed by the leverage multiples, so equity is the plug: `equity = uses - debt`.

```
import pandas as pd
from lbo import BASE, sources_uses

su = sources_uses(BASE)
pd.set_option("display.float_format", lambda x: f"{x:,.1f}")
sources = pd.DataFrame({"USD m": [su["tlb"], su["notes"], su["equity"]]},
                       index=["Term loan B (4.0x)", "Senior notes (1.5x)", "Sponsor equity"])
uses = pd.DataFrame({"USD m": [su["ev"], su["txn_fees"], su["fin_fees"], su["cash"]]},
                    index=["Purchase enterprise value (9.0x)", "Transaction fees", "Financing fees", "Cash to
balance sheet"])
sources.loc["Total sources"] = sources["USD m"].sum()
uses.loc["Total uses"] = uses["USD m"].sum()
print(sources, "\n")
print(uses, "\n")
print("Check sources - uses  :", round(sources.loc["Total sources", "USD m"] - uses.loc["Total uses", "USD
m"], 6))
print(f"Equity share of capital: {su['equity'] / su['uses']:.1%}")
print(f"Total debt / EBITDA    : {(su['tlb'] + su['notes']) / BASE['ltm_ebitda']:.1f}x")
```

Output:

```
                     USD m
Term loan B (4.0x)   400.0
Senior notes (1.5x)  150.0
Sponsor equity       384.5
Total sources        934.5

                                  USD m
Purchase enterprise value (9.0x)  900.0
Transaction fees                   13.5
Financing fees                     11.0
Cash to balance sheet              10.0
Total uses                        934.5

Check sources - uses  : 0.0
Equity share of capital: 41.1%
Total debt / EBITDA    : 5.5x
```

Sponsor equity is 384.5 million, 41.1 percent of the capital structure, against debt of 550 million, 5.5 times EBITDA. That level of leverage is aggressive but not unusual for a stable, cash-generative services business in a favourable credit market. Fees of 24.5 million (2.7 percent of enterprise value) are paid out of the sponsor's money and are a pure leakage: they add nothing to the value of the business. The check line confirms sources equal uses.

The operating forecast

The operating model translates a handful of assumptions into the free cash flow that will service the debt. Revenue grows from 600 million at 6, 6, 5, 5 and 4 percent. EBITDA margin starts at the LTM level of 16.7 percent and improves by 50 basis points a year as the sponsor's cost programme takes effect. Depreciation is 3.0 percent of revenue, capital expenditure 3.5 percent, working capital investment is 10 percent of the change in revenue, and the tax rate is 25 percent.

```
import pandas as pd
from lbo import BASE, operating_model

ops = operating_model(BASE)
ops["margin"] = ops["ebitda"] / ops["revenue"] * 100
pd.set_option("display.float_format", lambda x: f"{x:,.1f}")
pd.set_option("display.width", 110)
print(ops.round(1).T)
print()
print("EBITDA CAGR (LTM 100 -> year 5):", f"{(ops.loc[5, 'ebitda'] / 100) ** 0.2 - 1:.2%}")
```

Output:

```
            1     2     3     4     5
revenue 636.0 674.2 707.9 743.3 773.0
ebitda  109.2 119.1 128.6 138.7 148.2
da       19.1  20.2  21.2  22.3  23.2
ebit     90.1  98.9 107.4 116.4 125.0
capex    22.3  23.6  24.8  26.0  27.1
d_nwc     3.6   3.8   3.4   3.5   3.0
margin   17.2  17.7  18.2  18.7  19.2

EBITDA CAGR (LTM 100 -> year 5): 8.18%
```

By year five revenue is 773.0 million, EBITDA is 148.2 million (19.2 percent margin), and EBITDA has compounded at 8.18 percent a year from 100 million. The sponsor's equity story depends on this: about two-thirds of the EBITDA growth comes from revenue growth and one-third from margin expansion. In due diligence a sponsor would examine each of these, asking whether a 50 basis point annual margin gain is supported by identified cost actions, whether customers are contractually committed, and whether capex of 3.5 percent of revenue covers the maintenance the assets actually need. Note also that the forecast makes no allowance for a recession. We test downside cases in Exercise 6.

The debt schedule, the cash sweep and the circularity

The debt schedule is the heart of the model. In each year the company earns EBIT, pays interest and tax, adds back depreciation, and subtracts capital expenditure and working capital investment. What remains is **cash flow available for debt repayment** (the numerator of most credit analyses, and sometimes called cash flow available for debt service after interest). The term loan has a mandatory amortisation of 1 percent of the original balance a year. After that mandatory payment, a **cash sweep** requires the company to apply a specified percentage (here 100 percent) of the remaining cash to prepay the term loan. The senior notes are a bullet: no repayment until maturity, which is after our exit date.

The difficulty is a circularity. Interest on the term loan depends on the average balance during the year. The average balance depends on the closing balance. The closing balance depends on the sweep. The sweep depends on cash flow, which depends on net income, which depends on interest. Excel users resolve this by turning on iterative calculation and adding a circuit-breaker switch. In Python we solve it directly by **fixed-point iteration**: guess a closing balance, compute interest, tax, cash flow and the sweep, derive a new closing balance, and repeat until the change is smaller than a tolerance. This is the `for it in range(100)` loop in `debt_schedule`.

```
import pandas as pd
from lbo import run_lbo

p, su, ops, sch, ret = run_lbo()
pd.set_option("display.float_format", lambda x: f"{x:,.1f}")
pd.set_option("display.width", 120)
cols = ["tlb_open", "mandatory", "sweep", "tlb_close", "notes", "interest", "taxes", "net_income", "cfads"]
print(sch[cols].round(1).T)
print()
print(sch[["cash", "total_debt", "net_debt", "lev", "cover", "iterations"]].round(2).T)
print()
print("Cumulative TLB repaid (USD m):", round(su["tlb"] - sch.loc[5, "tlb_close"], 1))
print("Share of TLB repaid          :", f"{1 - sch.loc[5, 'tlb_close'] / su['tlb']:.1%}")
```

Output:

```
year           1     2     3     4     5
tlb_open   400.0 374.2 340.5 297.9 245.9
mandatory    4.0   4.0   4.0   4.0   4.0
sweep       21.8  29.7  38.6  48.0  58.2
tlb_close  374.2 340.5 297.9 245.9 183.8
notes      150.0 150.0 150.0 150.0 150.0
interest    46.7  44.3  41.3  37.5  32.9
taxes       10.8  13.6  16.5  19.7  23.0
net_income  32.5  40.9  49.6  59.2  69.0
cfads       25.8  33.7  42.6  52.0  62.2

year           1     2     3     4     5
cash        10.0  10.0  10.0  10.0  10.0
total_debt 524.2 490.5 447.9 395.9 333.8
net_debt   514.2 480.5 437.9 385.9 323.8
lev          4.7   4.0   3.4   2.8   2.2
cover        2.3   2.7   3.1   3.7   4.5
iterations   8.0   8.0   8.0   9.0   9.0

Cumulative TLB repaid (USD m): 216.2
Share of TLB repaid          : 54.1%
```

Read the first table down the columns. In year one the company generates 25.8 million of cash flow available for debt repayment after 46.7 million of interest and 10.8 million of tax. Of this, 4.0 million goes to mandatory amortisation and 21.8 million is swept, so the term loan falls from 400.0 to 374.2 million. As the loan shrinks and EBITDA grows, the sweep accelerates: by year five it is 58.2 million. Over the five years 216.2 million of term loan, 54.1 percent of the original balance, is repaid. The notes stay at 150 million throughout.

The second table shows the credit statistics. Net debt to EBITDA falls from 4.7 times at the end of year one to 2.2 times at the end of year five, partly through paying down debt and partly through the denominator growing. EBITDA to interest cover rises from 2.3 times to 4.5 times. A lender would test these ratios against covenants. A typical set requires maximum net leverage and minimum interest cover, with a margin of 25 to 35 percent of headroom against the plan in each year. Year one is the tightest: a leverage covenant set at 5.5 times would have headroom of only 0.8 turns. The cash balance stays at the 10 million minimum because all excess cash sweeps debt. The last row shows that each year’s solution took eight or nine passes of the loop.

Why the circularity matters (and how fast it converges)

Is the circularity worth the trouble? Many modellers avoid it by computing interest on the opening balance. The next listing compares the two approaches and then prints the fixed-point iteration for year one, so you can see the loop converge.

```
from lbo import run_lbo

print("Circularity: interest on average balance vs opening balance")
for avg in (True, False):
    p, su, ops, sch, ret = run_lbo(average_interest=avg)
    label = "average balance (circular)" if avg else "opening balance (no loop)  "
    print(f"{label}: year-1 interest {sch.loc[1, 'interest']:.3f}, year-5 TLB {sch.loc[5, 'tlb_close']:.2f}, "
          f"IRR {ret['irr']:.3%}, iterations yr1 {sch.loc[1, 'iterations']}")

# Show the fixed-point iteration converging for year 1
p, su, ops, sch, ret = run_lbo()
o = ops.loc[1]
mand = 0.01 * su["tlb"]
close = su["tlb"] - mand
print("\nYear-1 iteration trace (closing TLB):")
for k in range(1, 9):
    interest = 0.08 * (su["tlb"] + close) / 2 + 0.105 * su["notes"]
    ebt = o.ebit - interest
    ni = ebt - max(ebt, 0) * 0.25
    cfads = ni + o.da - o.capex - o.d_nwc
    new_close = su["tlb"] - mand - (cfads - mand)
    print(f"  pass {k}: interest {interest:8.4f}  closing TLB {new_close:9.4f}  change {new_close -
close:+.6f}")
    close = new_close
```

Output:

```
Circularity: interest on average balance vs opening balance
average balance (circular): year-1 interest 46.720, year-5 TLB 183.75, IRR 21.299%, iterations yr1 8
opening balance (no loop)  : year-1 interest 47.750, year-5 TLB 190.89, IRR 21.127%, iterations yr1 2

Year-1 iteration trace (closing TLB):
  pass 1: interest  47.5900  closing TLB  374.8975  change -21.102500
  pass 2: interest  46.7459  closing TLB  374.2644  change -0.633075
  pass 3: interest  46.7206  closing TLB  374.2454  change -0.018992
  pass 4: interest  46.7198  closing TLB  374.2449  change -0.000570
  pass 5: interest  46.7198  closing TLB  374.2448  change -0.000017
  pass 6: interest  46.7198  closing TLB  374.2448  change -0.000001
  pass 7: interest  46.7198  closing TLB  374.2448  change -0.000000
  pass 8: interest  46.7198  closing TLB  374.2448  change -0.000000
```

Using opening balances overstates year-one interest by 1.03 million (47.750 versus 46.720), leaves 190.89 million of term loan at the end of year five rather than 183.75, and understates the IRR by 17 basis points (21.127 versus 21.299 percent). The simplification is conservative and the difference is small here, but it grows with the sweep percentage and with the interest rate, and a lender’s model that differs from the sponsor’s by seven million dollars of debt at exit will cause arguments. The iteration trace shows why the loop is well behaved. Each pass changes the closing balance by only about one thirty-third of the previous change: the loop’s contraction factor is half the interest rate times one minus the tax rate, $0.08 \times 0.5 \times 0.75 = 0.03$. A loop converges when this factor is below one, which is always true for realistic rates, so the circularity is benign. The instability that analysts experience in Excel usually comes from an error value (such as a `#DIV/0!`) that enters the loop and cannot leave it. The circuit-breaker switch, which sets interest to the opening-balance version for one recalculation, flushes the error out.

Exit, returns and the value-creation bridge

At exit the sponsor sells the company at an assumed multiple of exit-year EBITDA, repays the remaining net debt, and keeps the rest. The money multiple (MOIC) is exit equity divided by equity invested. With no interim dividends, the IRR is `MOIC^(1/years) - 1`, the same as Excel's `IRR` applied to a stream of one outflow and one inflow. We assume the exit multiple equals the entry multiple, 9.0 times, which is the standard conservative convention: do not underwrite a return on multiple expansion.

The **value-creation bridge** decomposes the equity gain. Because enterprise value equals EBITDA times a multiple, and equity equals enterprise value minus net debt, the gain in equity value is exactly the sum of four terms: EBITDA growth valued at the entry multiple, the change in multiple applied to exit EBITDA, the reduction in net debt, and the negative of the fees paid at the outset. The listing computes all four and checks that they sum to the gain.

```
import pandas as pd
from lbo import run_lbo

p, su, ops, sch, ret = run_lbo()
t = p["exit_year"]
print(f"Exit EBITDA (year {t})      : {ops.loc[t, 'ebitda']:8.1f}")
print(f"Exit enterprise value      : {ret['exit_ev']:8.1f}   ({p['exit_multiple']:.1f}x)")
print(f"Less net debt at exit      : {sch.loc[t, 'net_debt']:8.1f}")
print(f"Exit equity value          : {ret['exit_equity']:8.1f}")
print(f"Sponsor equity invested    : {su['equity']:8.1f}")
print(f"MOIC                       : {ret['moic']:8.2f}x")
print(f"IRR                        : {ret['irr']:8.2%}")

# Value-creation bridge: equity gain = EBITDA growth + multiple change + net debt paydown - fees/leakage
entry_ebitda, exit_ebitda = p["ltm_ebitda"], ops.loc[t, "ebitda"]
ebitda_growth = (exit_ebitda - entry_ebitda) * p["entry_multiple"]
multiple_exp = (p["exit_multiple"] - p["entry_multiple"]) * exit_ebitda
entry_net_debt = su["tlb"] + su["notes"] - su["cash"]
deleveraging = entry_net_debt - sch.loc[t, "net_debt"]
fees = -(su["txn_fees"] + su["fin_fees"])
total = ebitda_growth + multiple_exp + deleveraging + fees
bridge = pd.Series({"EBITDA growth (at entry multiple)": ebitda_growth,
                    "Multiple expansion": multiple_exp,
                    "Net debt reduction": deleveraging,
                    "Transaction and financing fees": fees})
print()
for k, v in bridge.items():
    print(f"{k:34s}{v:9.1f}   {v / total:6.1%}")
print(f"{'Total equity gain':34s}{total:9.1f}")
print("Check vs exit equity - invested   :", round(ret["exit_equity"] - su["equity"] - total, 6))
```

Output:

```
Exit EBITDA (year 5)      :    148.2
Exit enterprise value     :   1333.4   (9.0x)
Less net debt at exit     :    323.8
Exit equity value         :   1009.7
Sponsor equity invested   :    384.5
MOIC                      :     2.63x
IRR                       :   21.30%

EBITDA growth (at entry multiple)     433.4    69.3%
Multiple expansion                      0.0     0.0%
Net debt reduction                    216.2    34.6%
Transaction and financing fees        -24.5    -3.9%
Total equity gain                     625.2
Check vs exit equity - invested   : -0.0
```

The sponsor invests 384.5 million and receives 1,009.7 million after five years, a multiple of 2.63 times and an IRR of 21.30 percent. The bridge shows where the 625.2 million gain comes from: 433.4 million (69.3 percent) from EBITDA growth, none from multiple expansion by construction, 216.2 million (34.6 percent) from debt reduction, and a 24.5 million drag from fees (3.9 percent). The check line confirms the decomposition is exact. Two lessons follow. First, in this deal operational improvement, not financial engineering, is the largest source of value, which is typical of the better sponsors' own accounts of their returns. Second, deleveraging is a substantial contributor, but note that it is itself funded by the cash flow that operations generate. If EBITDA does not grow, the paydown shrinks too, so the two sources are not independent.

Sensitivity tables

A single IRR is a point on a surface. The sponsor's investment committee wants to see the surface. In Excel the standard tool is the **data table** (What-If Analysis, Data Table). You build a grid with one assumption varying down the rows and another across the columns; the top-left corner cell contains a formula pointing at the output (here the IRR cell); and Excel fills the grid by substituting each pair of values into two designated input cells and recalculating. Two practical rules apply: the input cells must be on the same sheet as the table, and data tables slow recalculation, so many modellers set calculation to "automatic except data tables".

In Python, the equivalent is a loop over a function. Because we wrote `run_lbo` to accept overrides of any assumption, a sensitivity table is a double loop. The helper `table` below is general: give it two assumption names, two lists of values and an output metric.

```
import pandas as pd
from lbo import run_lbo

pd.set_option("display.width", 110)
def table(row_name, row_vals, col_name, col_vals, metric="irr"):
    out = pd.DataFrame(index=pd.Index(row_vals, name=row_name), columns=pd.Index(col_vals, name=col_name),
dtype=float)
    for r in row_vals:
        for c in col_vals:
            out.loc[r, c] = run_lbo(**{row_name: r, col_name: c})[4][metric]
    return out

t1 = table("entry_multiple", [8.0, 9.0, 10.0, 11.0], "exit_multiple", [8.0, 9.0, 10.0, 11.0])
print("IRR: entry multiple (rows) vs exit multiple (columns), 5-year exit")
print((t1 * 100).round(1))

t2 = table("tlb_x", [3.0, 4.0, 5.0], "exit_year", [3, 4, 5])
print("\nIRR: term loan leverage (rows, x EBITDA) vs exit year (columns)")
print((t2 * 100).round(1))

t3 = table("tlb_x", [3.0, 4.0, 5.0], "exit_multiple", [8.0, 9.0, 10.0], metric="moic")
print("\nMOIC: term loan leverage (rows) vs exit multiple (columns)")
print(t3.round(2))
```

Output:

```
IRR: entry multiple (rows) vs exit multiple (columns), 5-year exit
exit_multiple   8.0   9.0   10.0  11.0
entry_multiple
8.0             24.9  29.0  32.5  35.8
9.0             17.5  21.3  24.7  27.7
10.0            12.1  15.7  19.0  21.9
11.0             8.0  11.4  14.5  17.3

IRR: term loan leverage (rows, x EBITDA) vs exit year (columns)
exit_year     3     4     5
tlb_x
3.0        20.3  19.7  18.9
4.0        23.2  22.4  21.3
5.0        27.9  26.6  25.0

MOIC: term loan leverage (rows) vs exit multiple (columns)
exit_multiple  8.0   9.0   10.0
tlb_x
3.0            2.07  2.37  2.68
4.0            2.24  2.63  3.01
5.0            2.54  3.05  3.57
```

Read the first table along its diagonal. The base case, 9.0 times in and 9.0 times out, gives 21.3 percent. Paying 10 times but exiting at 9 reduces the IRR to 15.7 percent; paying 8 times and exiting at 9 raises it to 29.0 percent. Each turn of exit multiple is worth between 3.0 and 3.8 percentage points of IRR at the base entry price, while a turn of entry price is worth 5.6 points going from 9 to 10 times and 7.7 points going from 9 to 8 times. This asymmetry is the reason sponsors fight over price: entry price affects the equity cheque, which is the denominator of every return. The worst corner of the table, paying 11 times and exiting at 8 times, yields only 8.0 percent, below the sponsor’s cost of equity and likely below its fund hurdle rate.

The second table shows leverage and timing. More term loan raises the IRR: with 3.0 times of term loan the five-year IRR is 18.9 percent and with 5.0 times it is 25.0 percent, because the sponsor invests less equity for the same business. But this table cannot be read alone. Higher leverage raises the probability of default and the chance that covenants are breached in a downturn, and we have assumed the same interest rate at every leverage level; in practice lenders charge more for more risk. The shorter the holding period, the higher the IRR in this model (23.2 percent at three years versus 21.3 percent at five for 4.0 times), because the early years capture the steepest growth in EBITDA relative to the equity base and because we hold the exit multiple constant. In practice, a sponsor that exits early forgoes the later growth, so the IRR and the MOIC tell different stories, and sponsors care about both.

The third table is the MOIC. At 4.0 times leverage and a 9.0 times exit the multiple is 2.63. Paying down more debt or exiting at a higher multiple lifts it; at a 5.0 times term loan and a 10.0 times exit the sponsor makes 3.57 times its money, while at 3.0 times and 8.0 times it makes only 2.07.

Part II: The Accretion/Dilution Model

What the model answers

When a public company buys another company, its shareholders want to know what happens to earnings per share (EPS). **Accretion** means pro forma EPS exceeds the acquirer's standalone EPS; **dilution** means it is lower. EPS accretion is not a measure of value creation (Volume 1, Chapter 22 stressed this: a deal can be accretive and destroy value, or dilutive and create it), but it is watched closely by investors and boards, it determines how the deal will be reported, and it frames the negotiation over price, financing mix and synergies. So we model it, and then we look at what it does and does not tell us.

The mechanics

Pro forma net income equals the sum of the two companies' standalone net incomes, plus after-tax synergies, minus the after-tax financing cost of the acquisition, minus the after-tax amortisation of acquired intangible assets. Pro forma shares equal the acquirer's shares plus any new shares issued to pay for the target. The financing cost depends on the consideration mix: a cash purchase funded with borrowing adds interest expense, while a stock purchase adds shares. Excel's layout is a simple vertical build:

Row	Formula pattern	Comment
Offer price per share	`=Target_price*(1+Premium)`	Premium over the unaffected price
Equity purchase price	`=Offer_price*Target_shares`	Add net debt for enterprise value
New shares issued	`=Pct_stock*Equity_price/Acquirer_price`	Fixed exchange ratio assumed
New debt	`=(1-Pct_stock)*Equity_price + Fees`	Cash portion plus fees
After-tax interest	`=New_debt*Kd*(1-Tax)`	Foregone interest on cash can be used instead
After-tax amortisation	`=Intangibles_pct*Equity_price/Years*(1-Tax)`	From purchase price allocation
After-tax synergies	`=Synergies*Phase_in*(1-Tax)`	Phase-in by year
Pro forma NI	`=Acq_NI+Tgt_NI+Syn-Int-Amort`	
Pro forma EPS	`=Pro_forma_NI/(Acq_shares+New_shares)`	
Accretion	`=Pro_forma_EPS/Standalone_EPS-1`	

The fictional deal: Meridian Software, a public acquirer, has year-one net income of 540 million, growing 8 percent in year two, 250 million shares and a share price of 40 dollars (a market capitalisation of 10 billion dollars and a P/E multiple of 20 times year-zero earnings of 500 million). It proposes to acquire Calloway Data, which has 60 million shares at an unaffected price of 28 dollars, year-one net income of 99 million and year-two net income of 108.9 million (10 percent growth). The offer is a 30 percent premium. The base-case consideration is half stock, half cash, with the cash portion and the fees (2 percent of the equity price) funded by new debt at 6.5 percent pre-tax. Fifteen percent of the purchase price is allocated to identifiable intangible assets amortised over 10 years. Run-rate pre-tax synergies are 60 million, phased in at 50 percent in year one and 100 percent in year two. Integration costs are one-time and excluded from the adjusted EPS shown, which is the usual convention but which, as we discuss later, is a matter of judgement. Tax is 25 percent.

```
"""Accretion/dilution model for a public-company acquisition (all figures USD m except per-share)."""
import numpy as np
from scipy.optimize import brentq

DEAL = dict(
    acq_ni=[540.0, 583.2], acq_shares=250.0, acq_price=40.0,
    tgt_ni=[99.0, 108.9], tgt_shares=60.0, tgt_price=28.0,
    premium=0.30, pct_stock=0.50,
    kd=0.065, tax=0.25, fee_pct=0.02,
    intangibles_pct=0.15, amort_years=10,
    synergies=60.0, phase_in=[0.5, 1.0],
)

def accretion(**over):
    d = {**DEAL, **over}
    offer_px = d["tgt_price"] * (1 + d["premium"])
    equity_price = offer_px * d["tgt_shares"]
    fees = equity_price * d["fee_pct"]
    stock_issued = d["pct_stock"] * equity_price
    new_shares = stock_issued / d["acq_price"]
    new_debt = (1 - d["pct_stock"]) * equity_price + fees      # cash portion and fees borrowed
    interest_at = new_debt * d["kd"] * (1 - d["tax"])
    amort_at = d["intangibles_pct"] * equity_price / d["amort_years"] * (1 - d["tax"])
    out = dict(offer_px=offer_px, equity_price=equity_price, fees=fees, new_shares=new_shares,
               new_debt=new_debt, interest_at=interest_at, amort_at=amort_at, years=[])
    for y in range(2):
        syn_at = d["synergies"] * d["phase_in"][y] * (1 - d["tax"])
        ni = d["acq_ni"][y] + d["tgt_ni"][y] + syn_at - interest_at - amort_at
        shares = d["acq_shares"] + new_shares
        eps_pf = ni / shares
        eps_sa = d["acq_ni"][y] / d["acq_shares"]
        out["years"].append(dict(ni=ni, shares=shares, eps_pf=eps_pf, eps_sa=eps_sa,
                                 accretion=eps_pf / eps_sa - 1, syn_at=syn_at))
    return out

def breakeven_synergies(year=1, **over):
    """Pre-tax run-rate synergies at which pro forma EPS equals standalone EPS in the given year."""
    f = lambda s: accretion(**{**over, "synergies": s})["years"][year]["accretion"]
    return brentq(f, -500, 2000, xtol=1e-10)

def max_premium(year=1, **over):
    f = lambda pr: accretion(**{**over, "premium": pr})["years"][year]["accretion"]
    return brentq(f, -0.5, 3.0, xtol=1e-10)
```

Three simplifications deserve mention. The target's own debt is assumed to remain in place, so its interest is already in its net income and the equity purchase price is the relevant outlay. Debt raised is never repaid within the two-year window; in a real model the acquirer would apply cash flow to paydown. And the acquirer's share price is assumed unaffected by the announcement, which is rarely true and is tested in Exercise 5.

Base-case results

```
import pandas as pd
from ad import DEAL, accretion

a = accretion()
print(f"Offer price per target share : {a['offer_px']:.2f}  (premium {DEAL['premium']:.0%})")
print(f"Equity purchase price        : {a['equity_price']:,.1f}")
print(f"Deal P/E (on year-0 NI of 90): {a['equity_price'] / 90:.1f}x     Acquirer P/E: {DEAL['acq_price'] /
(500 / 250):.1f}x")
print(f"Earnings yield: target at offer {90 / a['equity_price']:.2%}, acquirer {500 / 250 /
DEAL['acq_price']:.2%}, after-tax debt {DEAL['kd'] * 0.75:.3%}")
print()

rows = []
for label, ps in (("All cash (debt)", 0.0), ("50% stock / 50% cash", 0.5), ("All stock", 1.0)):
    for syn_label, syn in (("no synergies", 0.0), ("with synergies", DEAL["synergies"])):
        r = accretion(pct_stock=ps, synergies=syn)
        rows.append((label, syn_label, r["new_shares"], r["new_debt"],
                     r["years"][0]["eps_pf"], r["years"][0]["accretion"] * 100,
                     r["years"][1]["eps_pf"], r["years"][1]["accretion"] * 100))
df = pd.DataFrame(rows, columns=["structure", "synergies", "new_shs", "new_debt", "EPS_y1", "acc_y1_%",
"EPS_y2", "acc_y2_%"])
pd.set_option("display.width", 130)
pd.set_option("display.float_format", lambda x: f"{x:,.3f}")
print(df.to_string(index=False))
print()
print("Standalone EPS: year 1", round(a["years"][0]["eps_sa"], 3), " year 2", round(a["years"][1]["eps_sa"],
3))

# Bridge from standalone to pro forma NI for the base 50/50 structure, year 2
y = a["years"][1]
print("\nYear-2 net income bridge (50/50, with synergies)")
print(f"  Acquirer standalone NI      {DEAL["acq_ni"][1]:9.2f}")
print(f"  + Target NI                 {DEAL["tgt_ni"][1]:9.2f}")
print(f"  + After-tax synergies       {y['syn_at']:9.2f}")
print(f"  - After-tax interest        {-a['interest_at']:9.2f}")
print(f"  - After-tax amortisation    {-a['amort_at']:9.2f}")
print(f"  = Pro forma NI              {y['ni']:9.2f}")
```

Output:

```
Offer price per target share : 36.40  (premium 30%)
Equity purchase price        : 2,184.0
Deal P/E (on year-0 NI of 90): 24.3x     Acquirer P/E: 20.0x
Earnings yield: target at offer 4.12%, acquirer 5.00%, after-tax debt 4.875%

           structure      synergies  new_shs  new_debt  EPS_y1  acc_y1_%  EPS_y2  acc_y2_%
     All cash (debt)   no synergies    0.000 2,227.680   2.023    -6.328   2.236    -4.161
     All cash (debt) with synergies    0.000 2,227.680   2.113    -2.161   2.416     3.555
50% stock / 50% cash   no synergies   27.300 1,135.680   2.016    -6.662   2.208    -5.367
50% stock / 50% cash with synergies   27.300 1,135.680   2.097    -2.905   2.370     1.589
           All stock   no synergies   54.600    43.680   2.010    -6.936   2.185    -6.357
           All stock with synergies   54.600    43.680   2.084    -3.516   2.332    -0.024

Standalone EPS: year 1 2.16  year 2 2.333

Year-2 net income bridge (50/50, with synergies)
  Acquirer standalone NI         583.20
  + Target NI                    108.90
  + After-tax synergies           45.00
  - After-tax interest           -55.36
  - After-tax amortisation       -24.57
  = Pro forma NI                 657.17
```

The first block explains the economics before the model does the arithmetic. At a 30 percent premium Meridian pays 36.40 per share, 2,184.0 million in total, which is 24.3 times the target's year-zero earnings of 90 million. Meridian trades at 20 times. A buyer paying 24.3 times for earnings with a 4.12 percent earnings yield, financed with stock whose earnings yield is 5.00 percent, issues "expensive" currency for "cheap" earnings: all-stock dilution is therefore expected before synergies. Debt financing at 6.5 percent costs 4.875 percent after tax, which is also more than the 4.12 percent the target earns on the price paid. So the deal needs synergies and growth to break even. The results confirm this.

The table gives six cases. Without synergies, every financing structure is dilutive in both years, between 4.2 and 6.9 percent. All-cash is the least dilutive in year two (-4.16 percent), all-stock the most (-6.36 percent), because Meridian's after-tax cost of debt (4.875 percent) is below its earnings yield (5.00 percent) and so debt is the cheaper currency on a pure earnings basis. With 60 million of run-rate synergies, year-two accretion is +3.56 percent for all cash, +1.59 percent for the 50/50 structure, and -0.02 percent (a break-even) for all stock. Year one is dilutive in every structure, because synergies are only half phased-in, ranging from -2.16 percent to -3.52 percent. Note the difference between the earnings-yield ranking (cash cheapest) and the ranking by risk: all-cash maximises EPS accretion but also maximises financial leverage, in this case 2,227.7 million of new debt on a company earning 540 million. A board would not stop at the EPS ranking.

The bridge at the bottom reconciles year-two net income for the 50/50 structure: 583.20 plus 108.90 plus 45.00 of after-tax synergies, minus 55.36 of after-tax interest and 24.57 of after-tax amortisation, gives 657.17. Dividing by 277.3 million shares gives pro forma EPS of 2.370, 1.59 percent above the standalone 2.333.

Break-even analysis and sensitivities

Practitioners rarely accept a single accretion number. They ask: how much synergy do we need? How high a premium can we pay? The two break-even functions use the same root finder as the IRR calculation in Chapter 19. The `breakeven_synergies` function finds the pre-tax run-rate synergies at which pro forma EPS equals standalone EPS, and `max_premium` finds the largest premium that leaves EPS unchanged.

```
import pandas as pd
from ad import accretion, breakeven_synergies, max_premium

print("Break-even pre-tax run-rate synergies (USD m) [synergy phase-in: 50% yr 1, 100% yr 2]")
for label, ps in (("All cash", 0.0), ("50/50", 0.5), ("All stock", 1.0)):
    print(f"  {label:10s} year 1: {breakeven_synergies(0, pct_stock=ps):7.1f}   year 2:
{breakeven_synergies(1, pct_stock=ps):7.1f}")
print()
print("Maximum premium with zero year-2 dilution, base synergies of 60:")
for label, ps in (("All cash", 0.0), ("50/50", 0.5), ("All stock", 1.0)):
    print(f"  {label:10s} {max_premium(1, pct_stock=ps):6.1%}")

pd.set_option("display.width", 120)
prem = [0.10, 0.20, 0.30, 0.40, 0.50]
pcts = [0.0, 0.25, 0.50, 0.75, 1.0]
grid = pd.DataFrame({f"{int(ps * 100)}% stock": [accretion(premium=pr, pct_stock=ps)["years"][1]["accretion"]
* 100 for pr in prem]
                     for ps in pcts}, index=pd.Index([f"{int(p * 100)}%" for p in prem], name="premium"))
print("\nYear-2 EPS accretion (%): premium (rows) vs share of stock consideration (columns)")
print(grid.round(2))

syn = [0.0, 30.0, 60.0, 90.0, 120.0]
grid2 = pd.DataFrame({f"{int(ps * 100)}% stock": [accretion(synergies=s, pct_stock=ps)["years"][1]
["accretion"] * 100 for s in syn]
                      for ps in pcts}, index=pd.Index([int(s) for s in syn], name="synergies"))
print("\nYear-2 EPS accretion (%): run-rate pre-tax synergies (rows) vs share of stock (columns)")
print(grid2.round(2))
```

Output:

```
Break-even pre-tax run-rate synergies (USD m) [synergy phase-in: 50% yr 1, 100% yr 2]
  All cash   year 1:    91.1   year 2:    32.4
  50/50      year 1:   106.4   year 2:    46.3
  All stock  year 1:   121.7   year 2:    60.2

Maximum premium with zero year-2 dilution, base synergies of 60:
  All cash    50.2%
  50/50       39.3%
  All stock   29.9%

Year-2 EPS accretion (%): premium (rows) vs share of stock consideration (columns)
         0% stock  25% stock  50% stock  75% stock  100% stock
premium
10%          7.07       6.03       5.08       4.21        3.41
20%          5.31       4.27       3.32       2.46        1.67
30%          3.55       2.52       1.59       0.74       -0.02
40%          1.80       0.79      -0.12      -0.93       -1.67
50%          0.04      -0.93      -1.80      -2.57       -3.27

Year-2 EPS accretion (%): run-rate pre-tax synergies (rows) vs share of stock (columns)
           0% stock  25% stock  50% stock  75% stock  100% stock
synergies
0             -4.16      -4.80      -5.37      -5.89       -6.36
30            -0.30      -1.14      -1.89      -2.57       -3.19
60             3.55       2.52       1.59       0.74       -0.02
90             7.41       6.18       5.07       4.06        3.14
120           11.27       9.84       8.55       7.37        6.31
```

To avoid dilution in year one, with only half of the synergies realised, the run-rate synergies must be 91.1 million for an all-cash deal, 106.4 million for the 50/50 structure and 121.7 million for all stock. By year two these fall to 32.4, 46.3 and 60.2 million. Synergies of 60 million therefore cover the all-cash and 50/50 hurdles in year two with margin, and fall just short of the all-stock hurdle of 60.2. The maximum premiums tell the same story from the other direction: with 60 million of synergies the buyer could pay up to 50.2 percent in cash, 39.3 percent for the mixed structure, and only 29.9 percent for all stock before year-two EPS is diluted.

The first grid shows the premium and financing trade-off. Each ten points of premium costs about 1.7 to 1.8 points of year-two accretion in either an all-cash deal (7.07 to 5.31 between 10 and 20 percent) or an all-stock deal (3.41 to 1.67). Moving from all cash to all stock costs roughly 3.7 points of accretion at a 10 percent premium (7.07 versus 3.41). The cells where accretion turns negative form a diagonal band from the lower-left to the upper-right of the grid: at a 40 percent premium any structure with half or more stock is dilutive. The second grid shows how valuable synergies are: each 30 million of run-rate pre-tax synergies adds about 3.9 points of accretion in an all-cash deal and about 3.2 points in the all-stock deal, since the same dollars are spread over more shares. These are the numbers a negotiating team carries into the room.

Contribution analysis and ownership

In a stock-for-stock deal, the more important question for the target's shareholders is what fraction of the combined company they will own, versus what fraction of its earnings and value they bring. **Contribution analysis** compares those figures, and is a standard exhibit in fairness opinions.

```
import pandas as pd
from ad import DEAL, accretion

# Contribution analysis: who contributes what, versus who owns what in an all-stock deal
a_ni, t_ni = DEAL["acq_ni"][0], DEAL["tgt_ni"][0]
a_mv = DEAL["acq_shares"] * DEAL["acq_price"]
t_mv = DEAL["tgt_shares"] * DEAL["tgt_price"]
contrib = pd.DataFrame({"acquirer": [a_ni, a_mv], "target": [t_ni, t_mv]}, index=["Year-1 net income",
"Unaffected equity value"])
contrib["target %"] = contrib["target"] / (contrib["acquirer"] + contrib["target"]) * 100
pd.set_option("display.float_format", lambda x: f"{x:,.1f}")
print(contrib)

for pr in (0.0, 0.30):
    r = accretion(pct_stock=1.0, premium=pr, synergies=0.0)
    own = r["new_shares"] / r["years"][0]["shares"]
    print(f"All-stock, premium {pr:.0%}: new shares {r['new_shares']:.1f}m, target holders own {own:.1%}, "
          f"exchange ratio {r['offer_px'] / DEAL['acq_price']:.4f}")

# P/E-differential rule of thumb: an all-stock, no-synergy deal is accretive only if target P/E paid <
acquirer P/E
for pr in (0.0, 0.10, 0.20, 0.30):
    px = DEAL["tgt_price"] * (1 + pr)
    pe_paid = px / (90 / 60)
    r = accretion(pct_stock=1.0, premium=pr, synergies=0.0, intangibles_pct=0.0, fee_pct=0.0)
    print(f"premium {pr:4.0%}: P/E paid {pe_paid:5.2f}x vs acquirer 20.00x -> year-1 accretion {r['years'][0]
['accretion']:+.3%}")
```

Output:

```
                         acquirer  target  target %
Year-1 net income           540.0    99.0      15.5
Unaffected equity value  10,000.0 1,680.0      14.4
All-stock, premium 0%: new shares 42.0m, target holders own 14.4%, exchange ratio 0.7000
All-stock, premium 30%: new shares 54.6m, target holders own 17.9%, exchange ratio 0.9100
premium   0%: P/E paid 18.67x vs acquirer 20.00x -> year-1 accretion +1.313%
premium  10%: P/E paid 20.53x vs acquirer 20.00x -> year-1 accretion -0.124%
premium  20%: P/E paid 22.40x vs acquirer 20.00x -> year-1 accretion -1.520%
premium  30%: P/E paid 24.27x vs acquirer 20.00x -> year-1 accretion -2.878%
```

Calloway contributes 15.5 percent of year-one net income and holds 14.4 percent of the combined unaffected equity value. At a zero premium, an all-stock deal gives Calloway's shareholders 14.4 percent of the combined company, which exactly matches its share of market value, as expected. At a 30 percent premium, the exchange ratio is 0.9100 Meridian shares per Calloway share and Calloway's holders own 17.9 percent, more than its 15.5 percent earnings contribution. They are being paid for the synergies in advance.

The last block tests the P/E rule of thumb. In an all-stock deal with no synergies, no fees and no purchase accounting, the deal is accretive if the target's P/E at the offer price is lower than the acquirer's. At a zero premium Calloway's P/E of 18.67 times is below Meridian's 20 times and accretion is +1.31 percent; at a 10 percent premium the P/E paid, 20.53 times, is just over 20 and the deal is slightly dilutive (-0.12 percent). The rule is based on the P/E calculated on the same year's earnings, whereas our table divides the offer price by year-zero earnings and compares accretion computed on year-one earnings, in which Calloway's earnings grow faster than Meridian's (10 percent versus 8 percent). That is why the zero-accretion premium in the model is about 9.1 percent, rather than the 7.1 percent at which the offer P/E (on year-zero earnings) equals 20. The rule of thumb is useful for a first reaction in a meeting, but the model is more accurate.

What EPS accretion does not tell you

An accretive deal can destroy value, and a dilutive one can create it. The **P/E rule** says nothing about risk: Meridian can raise EPS by borrowing more, but this raises its cost of equity. Accretion depends on the accounting treatment of integration costs and intangibles, both of which management can influence through judgement. The share price reaction to an announcement depends on whether the premium paid exceeds the present value of the synergies (Volume 1, Chapter 22), a calculation that this model does not perform. So after the accretion analysis the board should always ask the value question: is the present value of the after-tax synergies, capitalised at an appropriate rate, greater than the premium paid? Here the premium is 30 percent of 1,680 million, or 504 million. After-tax run-rate synergies of 45 million capitalised at an assumed 9 percent are worth 500 million, with no integration costs or growth. The deal therefore transfers essentially all of the synergy value to Calloway's shareholders, which is the usual outcome in a competitive process. Meridian's shareholders gain only if the synergies are larger or the integration is better than the market expects.

Model Review: A Checklist

Whether you built the model or inherited it, review it systematically before it goes into a committee paper.

Check	What to look for
Sources equal uses	The check cell is exactly zero, not “approximately zero”
Signs and units	Money in one currency unit, rates annual, interest on the right balance
Hard-codes	No numbers inside formulas; no overwritten formula cells in a row
Balance checks	Cash reconciles to the debt schedule; closing balance of one year equals opening of the next
Circularity	Switch cell in place; model survives a flush
Bridge reconciles	Value-creation bridge terms sum exactly to the equity gain
Sensitivity centre	The base case sits in the centre of every table and matches the headline output
Reasonableness	Growth, margin, and leverage paths compared with history and peers
Alternative method	Recompute the IRR independently, for example with a second tool
Stress	Flat and downside operating cases run through the model, with covenant tests

The “alternative method” check is where code and spreadsheet reinforce each other: building the same model twice, in Excel and Python, and reconciling them to the last decimal catches an astonishing number of errors that neither would reveal on its own.

Exercises with Solutions

The code below solves six exercises. Attempt each before reading the answers.

Exercise 1. Replace the term loan and notes with a single term loan of 5.5 times EBITDA. Compare the sponsor’s equity, interest cover, exit leverage and IRR when the loan costs 8.0 percent and 9.5 percent. **Exercise 2.** Remove the cash sweep and compare the five-year outcome. **Exercise 3.** Find the exit multiple that gives a 20 percent IRR, and the maximum entry multiple the sponsor can pay and still earn 20 percent with a 9.0 times exit. **Exercise 4.** In the merger model, change the all-cash funding rate to 9 percent and find the effect on accretion and break-even synergies. **Exercise 5.** In the 50/50 structure, Meridian’s share price falls from 40 to 32 before the deal closes. What happens to accretion? **Exercise 6.** Run two downside cases for Northgate: no growth and no margin gain, and a year-two recession in which revenue falls 12 percent with no margin improvement.

```
from scipy.optimize import brentq
from lbo import run_lbo
from ad import accretion, breakeven_synergies, max_premium

print("EXERCISE 1: all senior debt (TLB 5.5x, no notes) vs base")
for label, kw in (("Base (4.0x TLB + 1.5x notes)", {}), ("5.5x TLB only at 8.0%", dict(tlb_x=5.5,
notes_x=0.0)), ("5.5x TLB only at 9.5%", dict(tlb_x=5.5, notes_x=0.0, tlb_rate=0.095))):
    p, su, ops, sch, ret = run_lbo(**kw)
    print(f"  {label:30s} equity {su['equity']:6.1f}  yr-1 interest {sch.loc[1, 'interest']:5.1f}  "
          f"yr-1 cover {sch.loc[1, 'cover']:.2f}x  yr-5 net lev {sch.loc[5, 'lev']:.2f}x  MOIC
{ret['moic']:.2f}x  IRR {ret['irr']:.2%}")

print("\nEXERCISE 2: no cash sweep (excess cash accumulates)")
for sw in (1.0, 0.0):
    p, su, ops, sch, ret = run_lbo(sweep=sw)
    print(f"  sweep {sw:.0%}: yr-5 TLB {sch.loc[5, 'tlb_close']:6.1f}  cash {sch.loc[5, 'cash']:6.1f}  net
debt {sch.loc[5, 'net_debt']:6.1f}  "
          f"cumulative interest {sch['interest'].sum():6.1f}  IRR {ret['irr']:.3%}")

print("\nEXERCISE 3: exit multiple needed for a 20% IRR, and entry multiple that delivers 20% at a 9.0x exit")
f1 = lambda m: run_lbo(exit_multiple=m)[4]["irr"] - 0.20
f2 = lambda m: run_lbo(entry_multiple=m)[4]["irr"] - 0.20
print(f"  Required exit multiple : {brentq(f1, 5, 15):.3f}x")
print(f"  Maximum entry multiple : {brentq(f2, 7, 14):.3f}x")

print("\nEXERCISE 4: accretion/dilution - all-cash deal funded at a 9% cost of debt")
for kd in (0.065, 0.09):
    r = accretion(pct_stock=0.0, kd=kd)
    print(f"  kd {kd:.1%}: yr-1 {r['years'][0]['accretion']:+.2%}  yr-2 {r['years'][1]['accretion']:+.2%}  "
          f"break-even synergies yr-2 {breakeven_synergies(1, pct_stock=0.0, kd=kd):.1f}")

print("\nEXERCISE 5: 50/50 deal, acquirer share price falls to 32 before announcement")
for px in (40.0, 32.0):
    r = accretion(acq_price=px)
    print(f"  acquirer price {px:.0f}: new shares {r['new_shares']:.1f}m  yr-1 {r['years'][0]
['accretion']:+.2%}  yr-2 {r['years'][1]['accretion']:+.2%}")

print("\nEXERCISE 6: downside cases for the LBO (9.0x exit)")
cases = {"Base plan": {},
         "Flat: no growth, no margin gain": dict(growth=[0.0] * 5, margin_step=0.0),
         "Recession in year 2": dict(growth=[0.06, -0.12, 0.02, 0.04, 0.04], margin_step=0.0)}
for label, kw in cases.items():
    p, su, ops, sch, ret = run_lbo(**kw)
    print(f"  {label:32s} yr-5 EBITDA {ops.loc[5, 'ebitda']:6.1f}  peak net lev {sch['lev'].max():.2f}x  min
cover {sch['cover'].min():.2f}x  "
          f"MOIC {ret['moic']:.2f}x  IRR {ret['irr']:.2%}")
```

Output:

```
EXERCISE 1: all senior debt (TLB 5.5x, no notes) vs base
  Base (4.0x TLB + 1.5x notes)   equity  384.5  yr-1 interest  46.7  yr-1 cover 2.34x  yr-5 net lev 2.19x
MOIC 2.63x  IRR 21.30%
  5.5x TLB only at 8.0%          equity  384.5  yr-1 interest  42.9  yr-1 cover 2.55x  yr-5 net lev 2.07x
MOIC 2.67x  IRR 21.69%
  5.5x TLB only at 9.5%          equity  384.5  yr-1 interest  51.2  yr-1 cover 2.13x  yr-5 net lev 2.28x
MOIC 2.59x  IRR 20.95%

EXERCISE 2: no cash sweep (excess cash accumulates)
  sweep 100%: yr-5 TLB  183.8  cash   10.0  net debt  323.8  cumulative interest  202.8  IRR 21.299%
  sweep 0%: yr-5 TLB  380.0  cash  182.3  net debt  347.7  cumulative interest  234.8  IRR 20.717%

EXERCISE 3: exit multiple needed for a 20% IRR, and entry multiple that delivers 20% at a 9.0x exit
  Required exit multiple : 8.643x
  Maximum entry multiple : 9.209x

EXERCISE 4: accretion/dilution - all-cash deal funded at a 9% cost of debt
  kd 6.5%: yr-1 -2.16%  yr-2 +3.55%  break-even synergies yr-2 32.4
  kd 9.0%: yr-1 -9.90%  yr-2 -3.61%  break-even synergies yr-2 88.1

EXERCISE 5: 50/50 deal, acquirer share price falls to 32 before announcement
  acquirer price 40: new shares 27.3m  yr-1 -2.91%  yr-2 +1.59%
  acquirer price 32: new shares 34.1m  yr-1 -5.24%  yr-2 -0.85%

EXERCISE 6: downside cases for the LBO (9.0x exit)
  Base plan                       yr-5 EBITDA  148.2  peak net lev 4.71x  min cover 2.34x  MOIC 2.63x  IRR
21.30%
  Flat: no growth, no margin gain  yr-5 EBITDA  100.0  peak net lev 5.17x  min cover 2.14x  MOIC 1.28x  IRR
5.07%
  Recession in year 2             yr-5 EBITDA  102.9  peak net lev 5.23x  min cover 2.08x  MOIC 1.34x  IRR
6.00%
```

Solutions and commentary

Solution 1. Equity is unchanged at 384.5 million, since total debt is still 5.5 times. At 8.0 percent the single tranche is cheaper than the blended 8.0/10.5 percent mix, so year-one interest falls from 46.7 to 42.9 million, cover improves from 2.34 to 2.55 times and the IRR rises from 21.30 to 21.69 percent. But no lender would offer 5.5 turns of senior debt at the same price as 4.0 turns. At 9.5 percent, a more realistic price, interest is 51.2 million, cover is 2.13 times and the IRR is 20.95 percent, lower than the base case. The lesson is that the benefit of simplifying a capital structure depends on the price of the senior debt, and a sponsor should treat the cheapest tranche as a limited resource.

Solution 2. With no sweep the term loan only amortises by the mandatory 1 percent, ending at 380.0 million, and cash builds to 182.3 million, so net debt at exit is 347.7 million instead of 323.8. Cumulative interest rises from 202.8 to 234.8 million, because interest is paid on gross debt while the cash earns nothing in our model, and the IRR falls from 21.30 to 20.72 percent. The cost of hoarding cash is the spread between the loan rate and the cash yield, here a full 8 percentage points. Sponsors generally prefer to use free cash to repay expensive debt, or to pay dividends, rather than let it accumulate.

Solution 3. The sponsor needs an exit multiple of 8.643 times to earn 20 percent, so the base case has a margin of safety of about 0.36 turns, a cushion of 4 percent of the exit price. The maximum entry multiple for a 20 percent IRR at a 9.0 times exit is 9.209 times, only 0.21 turns, about 21 million dollars of enterprise value, above the base offer. This sets the sponsor’s walk-away price and is the kind of number a deal team tracks during an auction. (The brackets in the solver, 7 to 14, exclude entry multiples that would make debt exceed enterprise value, where equity is negative and the IRR undefined.)

Solution 4. At 6.5 percent funding the all-cash deal is dilutive in year one (-2.16 percent) and accretive in year two (+3.55 percent), needing 32.4 million of run-rate synergies to break even in year two. At 9 percent funding it is dilutive in both years (-9.90 and -3.61 percent) and break-even synergies rise to 88.1 million, nearly three times as much. The debt cost determines whether cash is the cheap or the expensive currency, and a 2.5 point rise in the borrowing rate moves the answer from a "buy for cash" recommendation to a "do not".

Solution 5. When Meridian's share price falls to 32, the stock portion needs 34.1 million new shares instead of 27.3 million. Year-one dilution deepens from 2.91 to 5.24 percent and the year-two result flips from +1.59 to -0.85 percent. A fixed exchange ratio protects the target's shareholders by value only if the price is stable; with a fixed number of shares, the acquirer bears the risk through dilution. This is why deals often include collars that adjust the exchange ratio within a range of acquirer prices.

Solution 6. In the flat case, with no growth and no margin gain, EBITDA stays at 100 million, peak net leverage is 5.17 times, minimum interest cover is 2.14 times, and the sponsor earns 1.28 times its money, an IRR of 5.07 percent. In the recession case, EBITDA at year five is 102.9 million, leverage peaks at 5.23 times and cover troughs at 2.08 times, and the IRR is 6.00 percent. In neither case does the company run out of cash (it can service debt throughout), but the return falls from the base 21.30 percent to about a quarter of that. A covenant set at 5.5 times would survive but with little headroom, and a covenant set at 5.0 times would have been breached. This is the real risk in an LBO: not bankruptcy in the typical case, but a sharp return reduction and a negotiating position with lenders in a downturn.

Chapter Summary

- An LBO model has five parts: sources and uses, an operating forecast, a debt schedule with mandatory amortisation and a cash sweep, exit returns, and sensitivity tables. Each belongs on its own sheet or function.
- The circularity of interest on average balances is benign: the iteration converges quickly (a contraction factor of about 0.03 here), and the choice between average and opening balances moved the IRR by 17 basis points. Always provide a circuit breaker in Excel.
- The value-creation bridge, EBITDA growth plus multiple change plus net debt reduction less fees, sums exactly to the equity gain. In the Northgate case, EBITDA growth supplied 69 percent of the gain and debt paydown 35 percent.
- Entry price matters more than exit multiple: one extra turn paid (9.0 to 10.0 times) costs 5.6 points of IRR, while a turn of exit multiple is worth only 3.0 to 3.8 points.
- The accretion/dilution model combines standalone earnings, after-tax synergies, after-tax financing costs, amortisation of acquired intangibles and share count. Which currency is cheaper depends on the acquirer's earnings yield versus the after-tax cost of debt.
- Break-even synergies and the maximum premium are the numbers that the negotiating team needs; the sensitivity grids show the trade-offs between price, mix and synergies.
- Accretion is not value. Compare the premium with the present value of synergies, and consider risk.
- Build and review models systematically: checks, circuit breakers, independent recomputation and downside cases.

Chapter 21: Options, Risk and Hedging Models in Python

Why This Chapter Exists

Volume 1, Chapters 18 to 20 treated options, derivatives and FX risk management through formulas, payoff diagrams and hand-worked examples. Practitioners work in spreadsheets and short programs, and are expected to know why the numbers come out as they do.

This chapter is a modeling workshop. Every listing is complete, runnable Python 3 (NumPy, SciPy, pandas) and was executed while the chapter was written; the printed results are the actual output, and seeded random results are reproducible. Simulated data are synthetic, and the companies named are fictional and illustrative.

The chapter proceeds from binomial trees and Black-Scholes (Sections 21.2 to 21.4), through value at risk (21.5 to 21.7), to hedge ratios and an FX hedge comparison (21.8 and 21.9), and ends with backtesting, exercises and solutions (21.10 to 21.13). Each section pairs the code with an Excel formula pattern.

Learning Objectives

After working through this chapter you should be able to:

- Build a Cox-Ross-Rubinstein binomial tree and price European and American options on it.
- Implement Black-Scholes and Garman-Kohlhagen, compute the Greeks and verify them by finite differences.
- Compute VaR and expected shortfall by historical, variance-covariance and Monte Carlo methods, and value the risk of option positions by full revaluation.
- Estimate a minimum-variance hedge ratio and compare forward, money-market and option hedges of a currency receivable.
- Backtest a VaR model with the Kupiec test.

Setting Up

All listings assume the following imports are available. In a clean environment, `pip install numpy scipy pandas` is enough.

```
numpy >= 1.24    scipy >= 1.10    pandas >= 2.0
```

Listings 21.3 and 21.6 import the Black-Scholes function defined in Listing 21.2. To reproduce them, save the `bs` function from Listing 21.2 (everything above the first `print`) into a file called `bsmodel.py` in the same folder.

Conventions: S is the spot price, K the strike, r the continuously compounded risk-free rate, q the continuous dividend yield, sigma the annualised volatility and T the time to expiry in years. All rates are continuously compounded unless the text says otherwise. Greeks are reported in the units traders use: vega per one percentage point of volatility, theta per calendar day and rho per one percentage point of the interest rate.

Section 21.1: From Volume 1 to Code, a Quick Bridge

Volume 1, Chapter 18 presented the one-period binomial model with a replicating portfolio, and then the Black-Scholes formula as the limit of many periods. The key ideas you will need again here are stated in compact form.

```
One period:  Δ = (C_u − C_d)/(S(u − d));  C = e^(−rΔt)[p*C_u + (1−p*)C_d];  p* = (e^((r−q)Δt) − d)/(u − d)
Black-Scholes:  d1 = [ln(S/K) + (r−q+σ²/2)T]/(σ√T);  d2 = d1 − σ√T
                Call = S e^(−qT) N(d1) − K e^(−rT) N(d2)
```

An option price is not a forecast: the real-world probability of a rise plays no part, and risk-neutral valuation is a pricing device only. The simulations below use real-world or zero-drift assumptions when measuring risk, and risk-neutral ones only when pricing.

Section 21.2: Binomial Trees

The Model

A recombining binomial tree divides the life of the option into n equal steps of length Δt = T/n. In each step the underlying either multiplies by u or by d. The Cox-Ross-Rubinstein (CRR) parameterisation chooses the factors to match the volatility:

```
u = e^(σ·√Δt)        d = 1/u        p* = (e^((r−q)Δt) − d) / (u − d)
```

Because d = 1/u, an up move followed by a down move returns the price to where it started. This is what makes the tree recombine and keeps the number of terminal nodes at n + 1 rather than 2^n, which is the difference between a model that runs in milliseconds and one that never finishes.

Valuation proceeds backwards. At maturity the option is worth its payoff at each terminal node. At each earlier node the value is the discounted risk-neutral expectation of the two successor values. For an American option there is one extra line: at every node the value is the larger of the continuation value just computed and the immediate exercise value.

The Code

Listing 21.1 implements the tree with vectorised backward induction. Instead of building a two-dimensional array of nodes, it carries a single vector of option values and shortens it by one element each step. The terminal prices are indexed by j, the number of down moves, so the price at a terminal node is $S \cdot u^{(n-j)} \cdot d^{j}$.

Listing 21.1: CRR binomial tree for European and American options

```
import numpy as np
from math import exp, sqrt

def crr_price(S, K, r, q, sigma, T, n, kind="call", american=False):
    """Cox-Ross-Rubinstein binomial tree, vectorised backward induction."""
    dt = T / n
    u = exp(sigma * sqrt(dt))
    d = 1.0 / u
    p = (exp((r - q) * dt) - d) / (u - d)
    disc = exp(-r * dt)
    j = np.arange(n + 1)
    ST = S * u ** (n - j) * d ** j            # terminal prices, j = number of down moves
    V = np.maximum(ST - K, 0.0) if kind == "call" else np.maximum(K - ST, 0.0)
    for i in range(n - 1, -1, -1):
        V = disc * (p * V[:-1] + (1 - p) * V[1:])
        if american:
            Si = S * u ** (i - np.arange(i + 1)) * d ** np.arange(i + 1)
            ex = np.maximum(Si - K, 0.0) if kind == "call" else np.maximum(K - Si, 0.0)
            V = np.maximum(V, ex)
    return V[0]

S, K, r, q, sigma, T = 100.0, 100.0, 0.05, 0.0, 0.25, 1.0
print("Inputs: S=100 K=100 r=5% q=0% sigma=25% T=1y")
print(f"{'steps':>6} {'Euro call':>10} {'Euro put':>10} {'Amer put':>10}")
for n in (1, 2, 5, 10, 50, 100, 500, 2000):
    c = crr_price(S, K, r, q, sigma, T, n, "call")
    p = crr_price(S, K, r, q, sigma, T, n, "put")
    a = crr_price(S, K, r, q, sigma, T, n, "put", american=True)
    print(f"{n:>6} {c:>10.4f} {p:>10.4f} {a:>10.4f}")
```

Output:

```
Inputs: S=100 K=100 r=5% q=0% sigma=25% T=1y
 steps  Euro call   Euro put   Amer put
     1    14.5706     9.6935     9.6935
     2    11.2110     6.3339     7.4726
     5    12.7946     7.9176     8.3423
    10    12.0923     7.2152     7.8643
    50    12.2867     7.4096     7.9520
   100    12.3113     7.4343     7.9636
   500    12.3311     7.4540     7.9724
  2000    12.3348     7.4577     7.9740
```

Reading the Output

The one-step tree is crude: it values the European call at 14.57 against roughly 12.33 at convergence, because only two terminal prices are possible. The value converges with a zigzag (11.21 at two steps, 12.79 at five, 12.09 at ten), a known CRR feature caused by where the strike falls relative to the terminal nodes. By 500 steps the call is within 0.005 of the Black-Scholes value of Listing 21.2 (12.3360).

Two further observations. The American put exceeds the European put by about 0.52 at 2,000 steps (7.9740 against 7.4577), the early-exercise premium: a deep in-the-money put holder may prefer to receive the strike now and earn interest. At one step the two are equal because the only exercise date is expiry. And put-call parity holds inside the tree: 12.3348 − 7.4577 = 4.8771 = S − K·e^(−rT). Parity is the cheapest diagnostic for any pricing code. (An American call on a non-dividend share equals the European call; with a positive dividend yield it is worth more.)

Excel Pattern for the Binomial Tree

With named cells S, K, r, q, sigma, T and n, put dt = T/n, u = EXP(sigma*SQRT(dt)), d = 1/u, p = (EXP((r-q)*dt)-d)/(u-d) and disc = EXP(-r*dt) in a header block. The stock lattice cell in column i, row j (j ≤ i) is `= S*u^(i-j)*d^j`. Terminal values are `=MAX(stock-K,0)` (call) or `=MAX(K-stock,0)` (put). Backward induction is `= disc*(p*right_same_row + (1-p)*right_next_row)`, and for an American put `= MAX(that, K - stockcell)`. Anchor p and disc with `$` references, and add `=ABS(tree_price - BS_price)` as a convergence check. ## Section 21.3: Black-Scholes and the Greeks

From Formula to Function

The Black-Scholes formula is closed-form, so a function implementing it can be called inside loops that revalue a portfolio tens of thousands of times. Listing 21.2 returns the price and principal sensitivities in one call, computing d1 and d2 once.

The Greeks are partial derivatives of the option price:

Delta Δ = ∂V/∂S (call e^(−qT)N(d1)); Gamma Γ = ∂²V/∂S² = e^(−qT)n(d1)/(Sσ√T); Vega ν = ∂V/∂σ = S e^(−qT)n(d1)√T; Theta Θ = ∂V/∂t; Rho ρ = ∂V/∂r. Delta is the shares that offset a small move, gamma how fast delta changes, vega the exposure to volatility, theta the daily cost of time and rho the interest-rate exposure.

Listing 21.2: Black-Scholes-Merton price and Greeks with finite-difference verification

```
from math import log, sqrt, exp
from scipy.stats import norm

def bs(S, K, r, q, sigma, T, kind="call"):
    d1 = (log(S / K) + (r - q + 0.5 * sigma**2) * T) / (sigma * sqrt(T))
    d2 = d1 - sigma * sqrt(T)
    if kind == "call":
        price = S * exp(-q*T) * norm.cdf(d1) - K * exp(-r*T) * norm.cdf(d2)
        delta = exp(-q*T) * norm.cdf(d1)
        theta = (-S*exp(-q*T)*norm.pdf(d1)*sigma/(2*sqrt(T))
                 - r*K*exp(-r*T)*norm.cdf(d2) + q*S*exp(-q*T)*norm.cdf(d1))
        rho = K*T*exp(-r*T)*norm.cdf(d2)
    else:
        price = K * exp(-r*T) * norm.cdf(-d2) - S * exp(-q*T) * norm.cdf(-d1)
        delta = -exp(-q*T) * norm.cdf(-d1)
        theta = (-S*exp(-q*T)*norm.pdf(d1)*sigma/(2*sqrt(T))
                 + r*K*exp(-r*T)*norm.cdf(-d2) - q*S*exp(-q*T)*norm.cdf(-d1))
        rho = -K*T*exp(-r*T)*norm.cdf(-d2)
    gamma = exp(-q*T) * norm.pdf(d1) / (S * sigma * sqrt(T))
    vega = S * exp(-q*T) * norm.pdf(d1) * sqrt(T)
    return dict(price=price, delta=delta, gamma=gamma, vega=vega/100,
                theta=theta/365, rho=rho/100, d1=d1, d2=d2)

g = bs(100, 100, 0.05, 0.0, 0.25, 1.0, "call")
print("ATM call, S=K=100, r=5%, sigma=25%, T=1")
for k, v in g.items():
    print(f"  {k:>6}: {v: .5f}")
pg = bs(100, 100, 0.05, 0.0, 0.25, 1.0, "put")
print("ATM put")
for k, v in pg.items():
    print(f"  {k:>6}: {v: .5f}")
c, p = g["price"], pg["price"]
print(f"Put-call parity: C - P = {c-p:.5f}; S - K e^(-rT) = {100-100*exp(-0.05):.5f}")

print("\nCall Greeks across spot (K=100, T=1, sigma=25%, r=5%)")
print(f"{'S':>5} {'price':>8} {'delta':>7} {'gamma':>8} {'vega/1%':>8} {'theta/day':>10}")
for S in (70, 80, 90, 100, 110, 120, 130):
    x = bs(S, 100, 0.05, 0.0, 0.25, 1.0)
    print(f"{S:>5} {x['price']:>8.3f} {x['delta']:>7.4f} {x['gamma']:>8.5f} {x['vega']:>8.4f}
{x['theta']:>10.5f}")

# finite-difference check of the analytic Greeks
h = 0.01
up, dn = bs(100+h,100,0.05,0,0.25,1)['price'], bs(100-h,100,0.05,0,0.25,1)['price']
mid = g['price']
print("\nFinite-difference check")
print(f"  delta FD {(up-dn)/(2*h):.5f}  analytic {g['delta']:.5f}")
print(f"  gamma FD {(up-2*mid+dn)/h**2:.5f}  analytic {g['gamma']:.5f}")
vu, vd = bs(100,100,0.05,0,0.2501,1)['price'], bs(100,100,0.05,0,0.2499,1)['price']
print(f"  vega  FD {(vu-vd)/0.0002/100:.5f}  analytic {g['vega']:.5f}")
```

Output:

```
ATM call, S=K=100, r=5%, sigma=25%, T=1
   price:  12.33600
   delta:  0.62741
   gamma:  0.01514
    vega:  0.37842
   theta: -0.01986
     rho:  0.50405
      d1:  0.32500
      d2:  0.07500
ATM put
   price:  7.45894
   delta: -0.37259
   gamma:  0.01514
    vega:  0.37842
   theta: -0.00683
     rho: -0.44718
      d1:  0.32500
      d2:  0.07500
Put-call parity: C - P = 4.87706; S - K e^(-rT) = 4.87706

Call Greeks across spot (K=100, T=1, sigma=25%, r=5%)
    S    price   delta    gamma  vega/1%  theta/day
   70    1.077  0.1353  0.01243   0.1522   -0.00636
   80    3.142  0.2852  0.01698   0.2717   -0.01200
   90    6.870  0.4616  0.01765   0.3574   -0.01699
  100   12.336  0.6274  0.01514   0.3784   -0.01986
  110   19.305  0.7600  0.01130   0.3420   -0.02052
  120   27.406  0.8541  0.00763   0.2746   -0.01969
  130   36.278  0.9154  0.00477   0.2017   -0.01824

Finite-difference check
  delta FD 0.62741  analytic 0.62741
  gamma FD 0.01514  analytic 0.01514
  vega  FD 0.37842  analytic 0.37842
```

Interpreting the Numbers

The at-the-money one-year call is worth 12.336 on a spot of 100, which is 12.3% of the underlying: with 25% volatility and a year to run, the option is expensive because there is a great deal of time value. The put is worth 7.459. The difference of 4.877 is exactly S − K·e^(−rT), confirming put-call parity to five decimals.

Delta of the call is 0.6274, not 0.5, because d1 = 0.325 includes the drift term (r + σ²/2)·T; the put delta of −0.3726 is the call delta minus one.

Gamma of 0.01514 means that for a one-dollar rise in the share price the call's delta rises by about 0.015. The across-spot table shows the characteristic shape: gamma is largest near the strike (0.0177 at a spot of 90 and 0.0151 at 100 for this one-year option) and falls away in both directions. Vega per one point of volatility is 0.378, so if implied volatility rises from 25% to 26% the call gains about 0.38. Theta of −0.0199 per day means the option loses roughly two cents a day to time decay.

The finite-difference block bumps spot by 0.01 for delta and gamma and volatility by 0.0001 for vega; agreement to five decimals confirms both price and Greek formulas. Test every Greek against a bump.

Excel Pattern for Black-Scholes and Greeks

```
d1    =(LN(S/K)+(r-q+sigma^2/2)*T)/(sigma*SQRT(T))
d2    =d1-sigma*SQRT(T)
Call  =S*EXP(-q*T)*NORM.S.DIST(d1,TRUE)-K*EXP(-r*T)*NORM.S.DIST(d2,TRUE)
Put   =K*EXP(-r*T)*NORM.S.DIST(-d2,TRUE)-S*EXP(-q*T)*NORM.S.DIST(-d1,TRUE)
Delta (call)  =EXP(-q*T)*NORM.S.DIST(d1,TRUE)
Gamma         =EXP(-q*T)*NORM.S.DIST(d1,FALSE)/(S*sigma*SQRT(T))
Vega per 1%   =S*EXP(-q*T)*NORM.S.DIST(d1,FALSE)*SQRT(T)/100
Theta per day (call) =(-S*EXP(-q*T)*NORM.S.DIST(d1,FALSE)*sigma/(2*SQRT(T))
                       -r*K*EXP(-r*T)*NORM.S.DIST(d2,TRUE)
                       +q*S*EXP(-q*T)*NORM.S.DIST(d1,TRUE))/365
Rho per 1%    (call)  =K*T*EXP(-r*T)*NORM.S.DIST(d2,TRUE)/100
```

In `NORM.S.DIST(x, cumulative)`, TRUE returns N(x) and FALSE the density n(x); mixing them up is the most common spreadsheet slip. A data table with spot down the rows reproduces the across-spot table. Goal Seek on `=Call_formula - quoted_price` backs out an implied volatility (Exercise 21.2 does it by bisection).

Section 21.4: Delta Hedging by Simulation

What the Greeks Mean in Money

A seller of a call who wants no exposure to the share price holds Δ shares against each option sold, financing the purchase by borrowing. As the share moves, delta changes, and the seller trades to keep the hedge in line. In the idealised Black-Scholes world rebalancing is continuous and the hedged position earns exactly the risk-free rate, so the seller’s profit is zero in every state of the world. In practice rebalancing is discrete; Listing 21.3 measures the residual risk.

The experiment sells one call at the Black-Scholes price, buys delta shares financed at the risk-free rate, and rebalances n times to the new Black-Scholes delta. At expiry the hedger delivers on the option, if exercised, and counts what is left. Prices are simulated with a real-world drift of 8% a year, deliberately different from the risk-free rate of 5%, to show that the hedge does not depend on the real-world drift.

Listing 21.3: Delta-hedging a short call with discrete rebalancing

```
import numpy as np
from bsmodel import bs   # the function from Listing 21.2 saved as bsmodel.py
from scipy.stats import norm

def delta_hedge(n_rebal, n_paths=20000, S0=100, K=100, r=0.05, sigma=0.25, T=1.0, mu=0.08, seed=7):
    """Sell one call, delta hedge n_rebal times, return final P&L distribution (per option)."""
    rng = np.random.default_rng(seed)
    dt = T / n_rebal
    S = np.full(n_paths, float(S0))
    c0 = bs(S0, K, r, 0, sigma, T)['price']
    delta = np.array([bs(S0, K, r, 0, sigma, T)['delta']] * n_paths)
    cash = c0 - delta * S                      # premium received, buy delta shares on borrowing
    for i in range(1, n_rebal + 1):
        z = rng.standard_normal(n_paths)
        S = S * np.exp((mu - 0.5 * sigma**2) * dt + sigma * np.sqrt(dt) * z)
        cash = cash * np.exp(r * dt)
        tau = T - i * dt
        if i < n_rebal:
            d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * tau) / (sigma * np.sqrt(tau))
            new_delta = norm.cdf(d1)
            cash -= (new_delta - delta) * S
            delta = new_delta
    payoff = np.maximum(S - K, 0)
    pnl = cash + delta * S - payoff
    return pnl, c0

print(f"{'rebalances':>10} {'mean P&L':>9} {'st.dev':>8} {'5th pct':>8} {'95th pct':>9}")
for n in (1, 4, 12, 52, 252):
    pnl, c0 = delta_hedge(n)
    print(f"{n:>10} {pnl.mean():>9.4f} {pnl.std():>8.4f} {np.percentile(pnl,5):>8.4f}
{np.percentile(pnl,95):>9.4f}")
print(f"Premium received per option: {c0:.4f}")
```

Output:

```
rebalances  mean P&L   st.dev  5th pct  95th pct
         1    0.0271   7.6079 -14.4138    9.0389
         4    0.0169   4.0683  -6.9005    6.4028
        12    0.0061   2.3898  -3.9968    3.8135
        52    0.0018   1.1969  -1.9423    1.9373
       252    0.0012   0.5471  -0.8813    0.8884
Premium received per option: 12.3360
```

Interpretation

The mean profit is near zero at every rebalancing frequency: 0.027 with a single hedge (that is, no rebalancing after the initial position), and 0.001 with daily rebalancing. This is the central result. The seller collected 12.336 and, on average, spent exactly that much replicating the option, regardless of the 8% real-world drift. The small positive mean shrinks as rebalancing increases.

The standard deviation is what changes. With a single hedge the standard deviation of profit and loss per option is 7.61, which is more than 60% of the premium. Quarterly rebalancing (four times) cuts it to 4.07, monthly (12 times) to 2.39, weekly (52 times) to 1.20 and daily (252 times) to 0.55. Each time the number of rebalances is multiplied by about four, the standard

deviation falls by about a factor of two. This is the square-root law: hedging error scales with 1/√n. It also gives the practical cost-benefit rule. Moving from weekly to daily rebalancing multiplies trading by almost five and cuts risk by a little over half; whether that is worthwhile depends on transaction costs, which this simulation ignores.

The tail percentiles show the short-gamma asymmetry: with one hedge the 5th percentile is −14.41 against a 95th of +9.04; with daily rebalancing, −0.88 and +0.89.

Gamma is thus both a risk and a cost: a short-gamma hedger buys as the market rises and sells as it falls, and the premium (and the theta earned) compensates for that. Whether a hedged option book profits depends on realised versus implied volatility.

Section 21.5: Value at Risk by Historical Simulation

Definition and Purpose

Value at risk (VaR) answers a single question: over a stated horizon, what loss will be exceeded only with a stated small probability? A one-day 99% VaR of $300,000 says that on 99 days out of 100 the portfolio should lose less than $300,000, and on one day in 100 it should lose more. VaR says nothing about how much more. Expected shortfall (ES), also called conditional VaR, fills that gap: it is the average loss on the days when the loss exceeds VaR. Regulators and risk committees increasingly prefer ES because it is sensitive to the shape of the tail and, unlike VaR, is subadditive, so diversification can never appear to increase measured risk.

```
VaR(α) = −quantile of the P&L distribution at probability (1 − α)
ES(α)  = −E[ P&L | P&L ≤ −VaR(α) ]
```

Volume 1, Chapter 20 introduced VaR conceptually as one of the tools of financial risk management. This chapter implements it. Three methods are standard, and they differ in what they assume.

Historical simulation applies the past n days of factor changes to today’s portfolio and reads the quantile: it assumes no distribution but depends on a thin tail sample. The variance-covariance method assumes jointly normal changes and is fast and decomposable but understates fat tails and mishandles options. Monte Carlo simulates changes from an assumed model and revalues the portfolio, handling non-linear payoffs at the cost of model dependence and sampling noise.

The Data and Code

Listing 21.4 creates a synthetic history of 1,000 daily observations for three risk factors affecting a fictional company, Harbourline Foods plc: a commodity price (the main input cost), the euro-dollar exchange rate (export sales) and an equity index (a pension-fund asset that the company bears risk on). The returns are drawn from a multivariate Student-t distribution with five degrees of freedom, scaled to the stated daily volatilities and correlations. The t distribution has fat tails, which is what real financial returns exhibit and what makes the comparison among methods informative. The exposures are $8 million to the commodity, $5 million to the exchange rate and $3 million to the equity index, a total of $16 million.

Listing 21.4: Historical VaR and expected shortfall, and a comparison with the normal approximation

```
import numpy as np
import pandas as pd

# Fictional portfolio of Harbourline Foods plc: synthetic daily returns, 3 risk factors
rng = np.random.default_rng(2024)
n = 1000
vols = np.array([0.011, 0.007, 0.009])          # daily vols: commodity, FX, equity index
corr = np.array([[1.0, 0.2, 0.3], [0.2, 1.0, 0.1], [0.3, 0.1, 1.0]])
cov = np.outer(vols, vols) * corr
L = np.linalg.cholesky(cov)
# fat tails: Student-t(5) scaled to unit variance, then correlated
t = rng.standard_t(5, size=(n, 3)) / np.sqrt(5 / 3)
rets = pd.DataFrame(t @ L.T, columns=["Commodity", "EURUSD", "EquityIdx"])
print(rets.describe().loc[["mean", "std", "min", "max"]].round(5))
print("Excess kurtosis:", rets.kurt().round(2).to_dict())

pos = np.array([8.0, 5.0, 3.0])                 # exposures in $ millions
port = rets.values @ pos                        # daily P&L, $m
print(f"\nPortfolio exposure: ${pos.sum():.0f}m;  daily P&L st.dev ${port.std(ddof=1):.4f}m")

def hist_var_es(pnl, level):
    q = np.percentile(pnl, 100 * (1 - level))
    var = -q
    es = -pnl[pnl <= q].mean()
    return var, es

print(f"\n{'level':>6} {'hist VaR':>9} {'hist ES':>9} {'normal VaR':>11}")
from scipy.stats import norm
for lvl in (0.95, 0.99):
    v, e = hist_var_es(port, lvl)
    nv = -(port.mean() + norm.ppf(1 - lvl) * port.std(ddof=1))
    print(f"{lvl:>6.0%} {v:>9.4f} {e:>9.4f} {nv:>11.4f}")

# 10-day VaR by square-root-of-time scaling, and by bootstrapped 10-day sums
v1, _ = hist_var_es(port, 0.99)
boot = rng.choice(port, size=(20000, 10)).sum(axis=1)
v10b, _ = hist_var_es(boot, 0.99)
print(f"\n99% 1-day VaR {v1:.4f};  sqrt(10) scaled {v1*np.sqrt(10):.4f};  bootstrap 10-day {v10b:.4f}")
```

Output:

```
      Commodity   EURUSD  EquityIdx
mean    0.00026  0.00011    0.00043
std     0.01068  0.00644    0.00894
min    -0.05145 -0.03270   -0.03440
max     0.07280  0.02694    0.04884
Excess kurtosis: {'Commodity': 3.37, 'EURUSD': 1.67, 'EquityIdx': 2.03}

Portfolio exposure: $16m;  daily P&L st.dev $0.1081m

 level  hist VaR   hist ES  normal VaR
   95%    0.1650    0.2409      0.1738
   99%    0.3009    0.3603      0.2475

99% 1-day VaR 0.3009;  sqrt(10) scaled 0.9517;  bootstrap 10-day 0.7924
```

What the Output Shows

The factor series have excess kurtosis of 3.37, 1.67 and 2.03 (a normal has zero). The commodity series has a daily move of −5.1% against a standard deviation of 1.07%, about 4.8 standard deviations.

The portfolio's daily P&L standard deviation is $0.108 million, 0.68% of the $16 million exposure, lower than a weighted average of the three factor volatilities because of imperfect correlation.

At 95% historical and normal VaR are close (0.1650 against 0.1738). At 99% they diverge: historical 0.3009 against normal 0.2475, an understatement of 18%. The 99% expected shortfall is 0.3603, 20% above VaR. This is the pattern for fat-tailed data: the normal fits near the centre and fails in the tail.

Horizon Scaling

The last line of Listing 21.4 shows square-root-of-time scaling gives a ten-day 99% VaR of 0.9517, against 0.7924 from a bootstrapped ten-day sum. Summing ten days pulls a fat-tailed distribution toward normality, so scaling overstates here; with volatility clustering the opposite error arises. Test the scaling on the data in question.

Excel Pattern for Historical VaR

```
VaR 99%:  = -PERCENTILE.INC(C2:C1001, 0.01)
ES 99%:   = -AVERAGEIF(C2:C1001, "<=" & PERCENTILE.INC(C2:C1001, 0.01))
```

With 1,000 observations the 1% tail holds about ten days, so historical VaR rests on a handful of points; the more extreme the quantile, the noisier the estimate.

Section 21.6: Parametric and Monte Carlo VaR

The Variance-Covariance Method

If factor changes are jointly normal with covariance matrix Σ and exposures are the vector x, the portfolio P&L is normal with variance x'Σx. VaR is then the z-score times the portfolio standard deviation, and ES has a closed form:

```
σ_p = √(x' Σ x)
VaR(α) = z_α · σ_p                    (zero-mean assumption)
ES(α)  = σ_p · n(z_α) / (1 − α)
```

The covariance matrix is Σ_ij = ρ_ij·σ_i·σ_j. The closed form also decomposes risk exactly: the Euler component VaR of position i is x_i·(Σx)_i/σ_p·z_α, and components sum to total VaR.

Monte Carlo

Monte Carlo simulation draws random factor changes and revalues the portfolio. Multiplying independent normal draws by the Cholesky factor L (Σ = LL') gives correlated draws; for fat tails use Student-t draws divided by √(ν/(ν − 2)) so variance stays at one.

Listing 21.5: Variance-covariance and Monte Carlo VaR, convergence, fat-tailed draws and component VaR

```
import numpy as np
from scipy.stats import norm

rng = np.random.default_rng(11)
vols = np.array([0.011, 0.007, 0.009])
corr = np.array([[1.0, 0.2, 0.3], [0.2, 1.0, 0.1], [0.3, 0.1, 1.0]])
cov = np.outer(vols, vols) * corr
pos = np.array([8.0, 5.0, 3.0])

# Closed-form (variance-covariance) VaR
sigma_p = np.sqrt(pos @ cov @ pos)
print(f"Portfolio daily sigma: ${sigma_p:.5f}m")
for lvl in (0.95, 0.99):
    z = norm.ppf(lvl)
    print(f"  Delta-normal VaR {lvl:.0%}: ${z*sigma_p:.4f}m   ES: ${sigma_p*norm.pdf(z)/(1-lvl):.4f}m")

# Monte Carlo: Cholesky-correlated normals
def mc_var(n_sims, dist="normal", df=5, seed=1):
    r = np.random.default_rng(seed)
    L = np.linalg.cholesky(cov)
    if dist == "normal":
        z = r.standard_normal((n_sims, 3))
    else:
        z = r.standard_t(df, (n_sims, 3)) / np.sqrt(df / (df - 2))
    pnl = (z @ L.T) @ pos
    out = {}
    for lvl in (0.95, 0.99):
        q = np.percentile(pnl, 100 * (1 - lvl))
        out[lvl] = (-q, -pnl[pnl <= q].mean())
    return out

print("\nMonte Carlo convergence (normal)")
print(f"{'sims':>9} {'VaR95':>8} {'VaR99':>8} {'ES99':>8}")
for n in (1_000, 10_000, 100_000, 1_000_000):
    o = mc_var(n)
    print(f"{n:>9,} {o[0.95][0]:>8.4f} {o[0.99][0]:>8.4f} {o[0.99][1]:>8.4f}")

o = mc_var(1_000_000, "t")
print(f"\nStudent-t(5) factors, 1m sims: VaR95 {o[0.95][0]:.4f}  VaR99 {o[0.99][0]:.4f}  ES99 {o[0.99]
[1]:.4f}")

# Sampling error of a 99% VaR estimate across 200 repeated 10,000-path runs
est = [mc_var(10_000, seed=s)[0.99][0] for s in range(200)]
print(f"Std error of 99% VaR with 10,000 paths: {np.std(est):.4f} (mean {np.mean(est):.4f})")

# Component VaR (Euler allocation) under the normal model
mcov = cov @ pos / sigma_p
comp = pos * mcov * norm.ppf(0.99)
print("\nComponent 99% VaR ($m):", {k: round(float(v), 4) for k, v in zip(["Commodity","EURUSD","EquityIdx"],
comp)}, " sum:", round(float(comp.sum()), 4))
```

Output:

```
Portfolio daily sigma: $0.11200m
  Delta-normal VaR 95%: $0.1842m   ES: $0.2310m
  Delta-normal VaR 99%: $0.2606m   ES: $0.2985m

Monte Carlo convergence (normal)
     sims    VaR95    VaR99     ES99
    1,000   0.1732   0.2623   0.3206
   10,000   0.1840   0.2605   0.2969
  100,000   0.1839   0.2605   0.2990
1,000,000   0.1837   0.2600   0.2979

Student-t(5) factors, 1m sims: VaR95 0.1765  VaR99 0.2853  ES99 0.3700
Std error of 99% VaR with 10,000 paths: 0.0042 (mean 0.2602)

Component 99% VaR ($m): {'Commodity': 0.1884, 'EURUSD': 0.0402, 'EquityIdx': 0.0319}  sum: 0.2606
```

Interpretation

The closed-form portfolio sigma is $0.1120 million, a little above the sample estimate of 0.1081 in Listing 21.4, because Listing 21.4 used a finite sample while Listing 21.5 uses the true covariance parameters that generated it. The delta-normal VaR at 99% is $0.2606 million and ES is $0.2985 million.

The convergence table shows the estimates approaching the closed form: at 1,000 simulations ES99 is 0.3206, about 7% above the exact 0.2985, because tail averages rest on only ten draws; at one million, VaR99 is 0.2600 against 0.2606. Repeating a 10,000-path run 200 times gives a standard error of 0.0042 (1.6% of VaR), so quoting four significant figures from 10,000 draws claims precision that does not exist.

With fat-tailed draws and the same volatilities and correlations, the 99% VaR rises from 0.2600 to 0.2853 (+10%) and ES from 0.2979 to 0.3700 (+24%), while the 95% VaR falls from 0.1837 to 0.1765: a fat-tailed distribution has more mass in the centre and far tail and less in the shoulders, so a 95% report alone can understate tail risk.

Finally, component VaR shows that the commodity position, 50% of the exposure ($8 million of $16 million), contributes $0.1884 million of the $0.2606 million total, 72%. It is both larger and more volatile than the other two positions and is positively correlated with both. A risk committee wanting to reduce VaR cheaply should look at that position first, which is the purpose of Section 21.8.

Section 21.7: Risk of Option Positions, When Delta Is Not Enough

The Problem with Linear Approximations

The variance-covariance method treats every position as linear in its risk factors. That is accurate for a share, a bond held over a short period or a currency balance. It is poor for options, whose payoff is curved. The long-call position below shows the point. Pennine Retail (a fictional listed company) is trading at $50. A treasury holds 10,000 six-month at-the-money calls on it as part of an executive-compensation hedge, struck at $50, with volatility 30% and a 4% risk-free rate. Three VaR estimates are compared over a ten-day horizon at 99%:

1. Delta-normal, which replaces the option by Δ shares.
2. Delta-gamma, which adds a second-order term: $\Delta V \approx \Delta \cdot \Delta S + \tfrac{1}{2} \cdot \Gamma \cdot \Delta S^2$.
3. Full revaluation, which draws 50,000 values of the share price at the horizon and revalues the option exactly with Black-Scholes at the reduced maturity.

Listing 21.6: VaR of an option position by three methods

```
import numpy as np
from bsmodel import bs
from scipy.stats import norm

# Option book: long 10,000 calls on a fictional share (Pennine Retail, S=50), K=50, 6m, sigma 30%, r 4%
S0, K, r, sigma, T = 50.0, 50.0, 0.04, 0.30, 0.5
N = 10_000
base = bs(S0, K, r, 0, sigma, T)
V0 = N * base['price']
print(f"Call value {base['price']:.4f}; position value ${V0:,.0f}; delta {base['delta']:.4f}; gamma
{base['gamma']:.5f}")

daily_vol = 0.30 / np.sqrt(252)
hold = 10                                    # 10-day horizon
s_h = daily_vol * np.sqrt(hold)
z99 = norm.ppf(0.99)

# 1) Delta-normal
dn = N * base['delta'] * S0 * s_h * z99
# 2) Delta-gamma (analytic approx. of the 1st-order + 2nd-order terms at the quantile)
dS = -z99 * s_h * S0                          # loss-tail move (spot falls)
dg = -(N * (base['delta'] * dS + 0.5 * base['gamma'] * dS**2))
# 3) Full revaluation Monte Carlo
rng = np.random.default_rng(5)
z = rng.standard_normal(50_000)
ST = S0 * np.exp(-0.5 * s_h**2 + s_h * z)
V1 = N * np.array([bs(s, K, r, 0, sigma, T - hold/252)['price'] for s in ST])
pnl = V1 - V0
fr = -np.percentile(pnl, 1)
fres = -pnl[pnl <= np.percentile(pnl, 1)].mean()
print(f"\n10-day 99% VaR on long-call position")
print(f"  Delta-normal      : ${dn:,.0f}")
print(f"  Delta-gamma       : ${dg:,.0f}")
print(f"  Full revaluation  : ${fr:,.0f}   (ES ${fres:,.0f}; 50,000 paths)")
```

Output:

```
Call value 4.6952; position value $46,952; delta 0.5794; gamma 0.03687

10-day 99% VaR on long-call position
  Delta-normal      : $40,275
  Delta-gamma       : $31,369
  Full revaluation  : $31,036   (ES $33,402; 50,000 paths)
```

Why the Estimates Differ

The position is worth $46,952 with delta 0.5794. Delta-normal VaR of $40,275 is 86% of the position value, which overstates risk because a long call has positive gamma: as the share falls, delta falls too, and the option loses less than a linear model predicts. Delta-gamma ($31,369) captures the cushion and is within about 1% of full revaluation ($31,036), a gap of the order of Monte Carlo error and third-order terms. Expected shortfall is $33,402; the tail is truncated because a long option cannot lose more than its premium. For a short option the signs reverse and delta-normal VaR understates the true figure, which is why books with material optionality need full revaluation or delta-gamma-vega methods. The principle for corporates holding embedded options (collars, convertibles, price caps): simulate the payoff, not its linearisation. ## Section 21.8: Estimating Hedge Ratios

The Cross-Hedging Problem

A perfect hedge needs a traded instrument that moves one-for-one with the exposure, and often none exists: a haulage firm buys diesel at pump prices that follow, but are not identical to, a benchmark futures price. The hedger must choose how much to trade and accept residual basis risk.

Define the change in the spot price of the exposure over a hedging interval as ΔS and the change in the futures price as ΔF. A hedger with a position of one unit of the exposure who holds h units of futures against it has a hedged outcome of ΔS − h·ΔF. The variance of that outcome is minimised at:

```
h* = Cov(ΔS, ΔF) / Var(ΔF) = ρ · σ_S / σ_F
Hedge effectiveness = 1 − Var(ΔS − h*·ΔF) / Var(ΔS) = ρ²
```

Two facts follow. h* is the slope of an OLS regression of spot changes on futures changes, which also supplies a standard error; and the fraction of variance removed is the regression R², the squared correlation (0.9 removes 81%, 0.5 only 25%).

Volume 1, Chapter 20 showed hedge ratios computed from a handful of numbers. Here the ratio is estimated from three years of weekly data on a fictional haulage company, Calder Haulage Ltd, which burns 2.4 million litres of diesel a year, hedged with a heating-oil futures contract whose standard size is, for this illustration, 100,000 litres. The data are synthetic: futures returns have 3% weekly volatility, and diesel returns are generated as 0.82 times futures returns plus independent noise representing basis risk. The code does not know the true 0.82; it must recover it from data.

Listing 21.7: Minimum-variance hedge ratio by formula and by regression, with out-of-sample test

```
import numpy as np
import pandas as pd

# Fictional: Calder Haulage buys diesel; hedges with a heating-oil futures contract (cross hedge).
rng = np.random.default_rng(99)
n = 156                                          # 3 years of weekly data
fut = rng.normal(0.0004, 0.030, n)               # weekly % change in futures
spot = 0.0003 + 0.82 * fut + rng.normal(0, 0.014, n)  # diesel moves with basis noise
df = pd.DataFrame({"spot": spot, "fut": fut})
print(df.describe().loc[["mean", "std"]].round(5))

# (a) Minimum-variance hedge ratio by formula
rho = df.corr().iloc[0, 1]
h_formula = rho * df.spot.std() / df.fut.std()
# (b) The same number from OLS: spot_ret = a + h * fut_ret + e
X = np.column_stack([np.ones(n), df.fut])
beta, res, *_ = np.linalg.lstsq(X, df.spot.values, rcond=None)
resid = df.spot.values - X @ beta
se = np.sqrt(resid @ resid / (n - 2) / ((df.fut - df.fut.mean())**2).sum())
print(f"\ncorr(spot,fut) = {rho:.4f}")
print(f"h* (rho*sd_s/sd_f) = {h_formula:.4f};  OLS slope = {beta[1]:.4f} (s.e. {se:.4f}); R^2 = {rho**2:.4f}")

# Hedge effectiveness: variance reduction for exposure of 1 unit
hedged = df.spot - beta[1] * df.fut
naive = df.spot - 1.0 * df.fut                   # 1:1 hedge
print(f"\nVariance of weekly outcome per unit exposure:")
print(f"  unhedged        {df.spot.var():.6f}")
print(f"  1:1 hedge       {naive.var():.6f}   ({1-naive.var()/df.spot.var():.1%} reduction)")
print(f"  min-var h*      {hedged.var():.6f}   ({1-hedged.var()/df.spot.var():.1%} reduction)")

# Contract sizing: 2,400,000 litres of diesel; futures contract of 100,000 litres (illustrative)
exposure_l, contract_l = 2_400_000, 100_000
n_contracts = beta[1] * exposure_l / contract_l
print(f"\nContracts to sell/buy: h* x exposure / contract size = {n_contracts:.2f}  -> round to
{round(n_contracts)}")

# Tailing the hedge and out-of-sample stability: estimate on first 104 weeks, test on last 52
tr, te = df.iloc[:104], df.iloc[104:]
h_in = np.polyfit(tr.fut, tr.spot, 1)[0]
for name, h in (("h in-sample", h_in), ("1:1", 1.0), ("none", 0.0)):
    print(f"  out-of-sample variance, {name:>11} (h={h:.3f}): {(te.spot - h*te.fut).var():.6f}")
```

Output:

```
         spot      fut
mean  0.00207  0.00160
std   0.02621  0.02799

corr(spot,fut) = 0.8109
h* (rho*sd_s/sd_f) = 0.7592;  OLS slope = 0.7592 (s.e. 0.0442); R^2 = 0.6575

Variance of weekly outcome per unit exposure:
  unhedged        0.000687
  1:1 hedge       0.000281   (59.1% reduction)
  min-var h*      0.000235   (65.7% reduction)

Contracts to sell/buy: h* x exposure / contract size = 18.22  -> round to 18
  out-of-sample variance, h in-sample (h=0.776): 0.000291
  out-of-sample variance,         1:1 (h=1.000): 0.000359
  out-of-sample variance,        none (h=0.000): 0.000834
```

Interpretation

The correlation between diesel and futures changes is 0.8109. Multiplying by the ratio of standard deviations (0.02621/0.02799 = 0.9364) gives h* = 0.7592, and the OLS slope is the same number, as it must be. Its standard error is 0.0442, so a 95% confidence interval runs from about 0.67 to 0.85. The true value used to generate the data, 0.82, lies within this interval, comfortably less than two standard errors from the estimate. In practice the true ratio is unknown, which justifies caution about hedging to the second decimal.

The R^2 of 0.6575 means the optimal hedge removes about two-thirds of the weekly variance. The unhedged variance of the weekly diesel return is 0.000687. A naive one-for-one hedge brings it down to 0.000281, a 59.1% reduction. The optimal ratio cuts it to 0.000235, 65.7%. The one-for-one hedge over-hedges and adds futures noise the diesel price does not share. The remaining 34% of the variance is basis risk, which no choice of h can eliminate in a cross hedge.

Sizing the Position

Contracts = h* × 2,400,000 / 100,000 = 18.22, which rounds to 18 (75% of the physical exposure; the treasurer may prefer 19). In practice the ratio is also tailed for daily margining.

Out-of-Sample Stability

The final block estimates the hedge ratio on the first 104 weeks and measures the variance of the hedged outcome on the final 52 weeks, which the estimation never saw. The in-sample ratio is 0.776, different from the full-sample 0.7592 because it uses less data. Out-of-sample, the hedged weekly variance is 0.000291 against 0.000359 for the one-for-one hedge and 0.000834 for no hedge. So the estimated hedge still removes 65% of the variance in the test year (1 − 0.000291/0.000834), compared with 57% for the naive hedge. This is a realistic expectation: out-of-sample effectiveness is typically a little worse than in-sample but, if the relationship is stable, still clearly useful.

Cautions: estimate on returns, not price levels (levels give spuriously high R^2); match the estimation interval to the hedging horizon; and re-estimate and monitor effectiveness, since IFRS 9 hedge-effectiveness tests rest on closely related regressions.

Excel Pattern for Hedge Ratios

```
h* = SLOPE(s, f)     R² = RSQ(s, f)     s.e. = INDEX(LINEST(s, f, TRUE, TRUE), 2, 1)
Contracts = ROUND(SLOPE(s,f) * exposure / contract_size, 0)
```

Section 21.9: Comparing FX Hedges for a Currency Receivable

The Decision

Ostrava Precision AB (fictional) is a manufacturer whose reporting currency is the US dollar. It has invoiced a customer EUR 5,000,000, payable in six months. Spot is 1.1000 USD per EUR. Six-month interest rates are 4.5% in dollars and 3.0% in euros (continuously compounded), and euro-dollar implied volatility is 9%. The treasurer has three hedging alternatives, plus the option of doing nothing:

1. **Forward contract.** Sell EUR 5,000,000 forward at the six-month forward rate.
2. **Money-market hedge.** Borrow euros today equal to the present value of the receivable, convert at spot to dollars and invest dollars for six months. The euro receivable repays the euro loan.
3. **Currency option.** Buy a six-month euro put, dollar call, struck at 1.0900, paying the premium up front.
4. **Remain unhedged.**

The forward follows from covered interest parity, F = S·e^((r_USD − r_EUR)·T), and the option is priced with the Garman-Kohlhagen formula, which is Black-Scholes with the foreign rate in the role of the dividend yield (put = K·e^(−r_d·T)·N(−d2) − S·e^(−r_f·T)·N(−d1)).

Listing 21.8: Forward, money-market, option and unhedged outcomes for a EUR 5 million receivable

```
import numpy as np
from math import exp, log, sqrt
from scipy.stats import norm

# Fictional exporter, Ostrava Precision AB, receives EUR 5,000,000 in 6 months; reporting currency USD.
S0, rU, rE, T = 1.1000, 0.045, 0.030, 0.5          # spot USD per EUR, USD and EUR rates
sigma = 0.09
notional = 5_000_000
F = S0 * exp((rU - rE) * T)                          # forward by covered interest parity
print(f"Spot {S0:.4f}; 6m forward {F:.4f} (forward points {1e4*(F-S0):.1f} pips)")

def garman_kohlhagen_put(S, K, rd, rf, sig, T):
    d1 = (log(S / K) + (rd - rf + 0.5 * sig**2) * T) / (sig * sqrt(T))
    d2 = d1 - sig * sqrt(T)
    return K * exp(-rd * T) * norm.cdf(-d2) - S * exp(-rf * T) * norm.cdf(-d1)

K = 1.0900                                           # EUR put / USD call strike
prem_per_eur = garman_kohlhagen_put(S0, K, rU, rE, sigma, T)
prem_usd = prem_per_eur * notional
prem_fv = prem_usd * exp(rU * T)                     # premium financed to maturity
print(f"EUR put strike {K:.4f}: premium {prem_per_eur:.5f} USD/EUR = ${prem_usd:,.0f} (future value
${prem_fv:,.0f})")

# Money-market hedge: borrow EUR now, convert, invest USD
eur_borrow = notional / (1 + rE * T)
usd_now = eur_borrow * S0
usd_mm = usd_now * (1 + rU * T)
print(f"Money-market hedge: borrow EUR {eur_borrow:,.0f}, get USD {usd_now:,.0f}; USD in 6m {usd_mm:,.0f} ->
effective rate {usd_mm/notional:.4f}")
print(f"Forward hedge:      USD {F*notional:,.0f} at {F:.4f}")

# Simulate terminal spot under a real-world drift of 0 (spot martingale) and under risk-neutral drift
rng = np.random.default_rng(3)
z = rng.standard_normal(500_000)
def outcomes(drift):
    ST = S0 * np.exp((drift - 0.5 * sigma**2) * T + sigma * sqrt(T) * z)
    unhedged = ST * notional
    fwd = np.full_like(ST, F * notional)
    opt = np.maximum(ST, K) * notional - prem_fv
    return ST, unhedged, fwd, opt

print(f"\n{'Terminal spot':>14} {'Unhedged':>12} {'Forward':>12} {'Put option net':>15}")
for s in (0.95, 1.00, 1.05, 1.10, 1.15, 1.20, 1.25):
    print(f"{s:>14.4f} {s*notional:>12,.0f} {F*notional:>12,.0f} {max(s,K)*notional-prem_fv:>15,.0f}")

ST, un, fw, op = outcomes(0.0)
print("\nSimulated USD proceeds (zero-drift spot, 500,000 paths)")
print(f"{'strategy':>10} {'mean':>11} {'st.dev':>10} {'5th pct':>11} {'worst':>11}")
for name, x in (("unhedged", un), ("forward", fw), ("put option", op)):
    print(f"{name:>10} {x.mean():>11,.0f} {x.std():>10,.0f} {np.percentile(x,5):>11,.0f} {x.min():>11,.0f}")
be_spot = F + prem_fv / notional
print(f"\nOption net proceeds exceed the forward only if S_T > {be_spot:.4f} (forward rate + financed premium
per EUR)")
print(f"Option floor in USD: {K*notional - prem_fv:,.0f} ({(K*notional-prem_fv)/notional:.4f} per EUR);
forward {F:.4f}")
print(f"P(option net outcome > forward outcome) = {(op > fw).mean():.3f}")
```


Output:

```
Spot 1.1000; 6m forward 1.1083 (forward points 82.8 pips)
EUR put strike 1.0900: premium 0.01927 USD/EUR = $96,345 (future value $98,538)
Money-market hedge: borrow EUR 4,926,108, get USD 5,418,719; USD in 6m 5,540,640 -> effective rate 1.1081
Forward hedge:      USD 5,541,405 at 1.1083

 Terminal spot     Unhedged      Forward  Put option net
        0.9500    4,750,000    5,541,405       5,351,462
        1.0000    5,000,000    5,541,405       5,351,462
        1.0500    5,250,000    5,541,405       5,351,462
        1.1000    5,500,000    5,541,405       5,401,462
        1.1500    5,750,000    5,541,405       5,651,462
        1.2000    6,000,000    5,541,405       5,901,462
        1.2500    6,250,000    5,541,405       6,151,462

Simulated USD proceeds (zero-drift spot, 500,000 paths)
  strategy        mean     st.dev     5th pct       worst
  unhedged   5,500,254    350,422   4,943,763   3,992,361
   forward   5,541,405          0   5,541,405   5,541,405
put option   5,517,032    228,947   5,351,462   5,351,462

Option net proceeds exceed the forward only if S_T > 1.1280 (forward rate + financed premium per EUR)
Option floor in USD: 5,351,462 (1.0703 per EUR); forward 1.1083
P(option net outcome > forward outcome) = 0.335
```

Reading the Results

Forward and money market. The forward rate is 1.1083, 83 pips above spot: a premium on the lower-rate currency that follows from interest rate parity and is not a forecast. It locks in USD 5,541,405. The money-market hedge yields USD 5,540,640 (1.1081); the USD 765 gap arises only because Listing 21.8 mixes continuous compounding in the forward with simple interest in the money-market leg.

The option. The put struck at 1.0900 costs 0.01927 dollars per euro, or USD 96,345 for the whole notional, 1.75% of the notional value of USD 5.5 million. Because the premium is paid today and the proceeds arrive in six months, the comparison with the forward must be on a common date, so the premium is carried forward at the dollar rate to USD 98,538. The net floor on proceeds is the strike times the notional less this financed premium: USD 5,351,462, or 1.0703 per euro. If the euro falls to 0.95, the option protects the firm to a net 1.0703 while the unhedged firm receives 0.95. If the euro rises to 1.25, the option still allows participation, and net proceeds are USD 6,151,462 against the forward's USD 5,541,405. The option and the forward give the same net proceeds at a spot of about 1.1280, the forward rate plus the financed premium per euro (1.1083 + 0.0197). Below that spot the forward is better; above it the option is better. The option also always beats the unhedged position when the euro is weak and trails it by exactly the financed premium whenever the euro ends above the strike.

The distribution. The simulation draws 500,000 terminal spot rates from a lognormal distribution with zero drift in the spot rate (so that the expected spot at maturity is today's spot of 1.1000), and calculates the dollar proceeds from each strategy. The forward has no dispersion: USD 5,541,405 in every path. The unhedged position has a mean of USD 5,500,254 and a standard deviation of USD 350,422, a worst outcome in the sample of USD 3,992,361 and a 5th percentile of USD 4,943,763. The option strategy has a mean of USD 5,517,032 and a standard deviation of USD 228,947; its 5th percentile equals its floor of USD 5,351,462, which is also its worst outcome.

Two interpretations matter. First, the option is not free insurance. It beats the forward in only 33.5% of paths, because the premium is certain and the upside it preserves pays off only above 1.1280. If the objective is a certain budget rate, the forward dominates; if it is protection with participation, the option does the job at a price. Its standard deviation (USD 228,947) is all upside, so a downside measure such as the 5th percentile is the relevant statistic.

Second, the mean outcome depends on the assumed drift. With zero drift the forward (USD 5,541,405) beats the unhedged position (USD 5,500,254) and the option (USD 5,517,032) on average because it embeds an 83-pip premium the zero-drift assumption does not expect to be delivered; if expected spot equalled the forward, the forward and unhedged means would match. Expected-value comparisons are only as good as the exchange-rate forecast, so a hedging policy should be framed around risk reduction, as in Volume 1, Chapter 20. A partial hedge or a collar (Exercise 21.6) can be modelled the same way.

Section 21.10: Backtesting VaR and Model Risk

Why Backtest

A VaR number is a prediction: on 1% of days the loss should exceed it. An exception is a day on which it does; for a correct 99% model the number of exceptions in n days is binomial(n, 1%). The Kupiec proportion-of-failures test formalises this.

With x exceptions in n days and a nominal exception probability p, the likelihood-ratio statistic is:

```
LR = −2 · ln[ (1−p)^(n−x) · p^x ] + 2 · ln[ (1−x/n)^(n−x) · (x/n)^x ]
```

LR is approximately chi-square with one degree of freedom, so the 5% critical value is 3.841. Listing 21.9 simulates 1,500 days of P&L from a Student-t distribution with four degrees of freedom (a deliberately fat-tailed process with a standard deviation of $0.1 million), estimates 99% VaR each day from the previous 500 days by a normal method and by historical simulation, and counts exceptions over the remaining 1,000 days.

Listing 21.9: Rolling-window VaR backtest with the Kupiec test

```
import numpy as np
from scipy.stats import norm, chi2

# Backtest a 99% one-day VaR on 1,000 days of fat-tailed P&L (Student-t, df=4), VaR estimated two ways
rng = np.random.default_rng(31)
T_all, window = 1500, 500
pnl = rng.standard_t(4, T_all) / np.sqrt(2) * 0.1       # sd = 0.1 ($m)
exc_norm, exc_hist = [], []
for t in range(window, T_all):
    w = pnl[t - window:t]
    var_n = -(w.mean() + norm.ppf(0.01) * w.std(ddof=1))
    var_h = -np.percentile(w, 1)
    exc_norm.append(pnl[t] < -var_n)
    exc_hist.append(pnl[t] < -var_h)

def kupiec(x, n, p=0.01):
    phat = x / n
    ll0 = (n - x) * np.log(1 - p) + x * np.log(p)
    ll1 = (n - x) * np.log(1 - phat) + (x * np.log(phat) if x > 0 else 0)
    lr = -2 * (ll0 - ll1)
    return lr, 1 - chi2.cdf(lr, 1)

n = T_all - window
print(f"Test days: {n}; expected exceptions at 99%: {0.01*n:.1f}")
for name, e in (("normal (window)", exc_norm), ("historical (window)", exc_hist)):
    x = int(np.sum(e))
    lr, pv = kupiec(x, n)
    print(f"{name:>20}: exceptions {x:>3}  rate {x/n:.2%}  Kupiec LR {lr:5.2f}  p-value {pv:.3f}")
```

Output:

```
Test days: 1000; expected exceptions at 99%: 10.0
     normal (window): exceptions  14  rate 1.40%  Kupiec LR  1.44  p-value 0.231
 historical (window): exceptions  13  rate 1.30%  Kupiec LR  0.83  p-value 0.362
```

Interpretation

With 1,000 test days, 10 exceptions are expected. The normal method produces 14 (1.40%) and the historical method 13 (1.30%). Both overshoot, which is the expected direction because the true distribution has fatter tails than the normal and because a 500-day window gives noisy tail estimates. Yet neither exceeds the Kupiec critical value: the LR statistics are 1.44 and 0.83, with p-values of 0.231 and 0.362. The test does not reject either model.

That result teaches an important lesson about backtests: they have low power. With a 1% tail and 1,000 days there are so few expected exceptions that even a model that is wrong by 40% in its exception rate cannot be statistically distinguished from a correct one. The Basel traffic-light approach to market-risk capital uses a 250-day window and classifies models as green (four or fewer exceptions at 99%), amber (five to nine) or red (ten or more). That threshold system is a compromise between wanting to catch bad models and not penalising good ones, and it implicitly accepts that a backtest cannot give certainty. A prudent risk manager treats backtest results as evidence to be weighed alongside an understanding of how the model works, not as a pass-fail certificate.

Two companion tests are common: independence (exceptions should not cluster) and tests of exception size, using ES.

Section 21.11: Exercises with Solutions

The solutions below were computed by Listing 21.10, whose output is shown in full. Attempt each exercise before reading the solution.

Listing 21.10: Solutions to Exercises 21.1 to 21.8

```
# Exercises 21.1-21.4 solutions
import numpy as np
from math import exp, sqrt, log
from scipy.stats import norm
from bsmodel import bs
import importlib.util, sys

print("Exercise 21.1: one-step tree, S=40, u=1.25, d=0.80, r=4%, K=42 call")
S, u, d, r = 40, 1.25, 0.80, 0.04
p = (exp(r) - d) / (u - d)
Cu, Cd = max(S*u-42, 0), max(S*d-42, 0)
C = exp(-r) * (p*Cu + (1-p)*Cd)
delta = (Cu - Cd) / (S*(u-d))
B = exp(-r) * (u*Cd - d*Cu) / (u - d)
print(f"  p*={p:.4f}  C={C:.4f}  delta={delta:.4f}  borrowing={-B:.4f}  check delta*S+B={delta*S+B:.4f}")

print("\nExercise 21.2: implied volatility of a call quoted at 9.00 (S=K=100,r=5%,T=1) by bisection")
lo, hi = 0.01, 1.0
for _ in range(60):
    mid = 0.5*(lo+hi)
    if bs(100,100,0.05,0,mid,1)['price'] > 9.0: hi = mid
    else: lo = mid
print(f"  implied sigma = {mid:.5f}")

print("\nExercise 21.3: parametric VaR of a $20m position, daily vol 1.4%, 99%, 10 days")
v = 20 * 0.014 * norm.ppf(0.99) * sqrt(10)
print(f"  VaR = ${v:.4f}m")

print("\nExercise 21.4: min-variance hedge, sd_s=0.021, sd_f=0.026, rho=0.88, exposure 12m")
h = 0.88 * 0.021 / 0.026
var_red = 0.88**2
print(f"  h*={h:.4f}; hedge notional {h*12:.3f}m; variance reduction {var_red:.2%}")

print("\nExercise 21.5: GK put, S=1.25 (USD/GBP), K=1.22, rU=4.5%, rG=3.5%, sigma=8%, T=0.25, notional GBP
2m")
S,K,rd,rf,sg,T = 1.25,1.22,0.045,0.035,0.08,0.25
d1=(log(S/K)+(rd-rf+0.5*sg**2)*T)/(sg*sqrt(T)); d2=d1-sg*sqrt(T)
put=K*exp(-rd*T)*norm.cdf(-d2)-S*exp(-rf*T)*norm.cdf(-d1)
print(f"  premium {put:.5f} USD/GBP = ${put*2e6:,.0f};  forward {S*exp((rd-rf)*T):.5f}")

print("\nExercise 21.6: zero-cost collar for the Ostrava receivable. Buy EUR put at 1.09, sell EUR call at
strike Kc with equal premium")
from scipy.optimize import brentq
S0, rU, rE, T, sg = 1.10, 0.045, 0.030, 0.5, 0.09
def gk(S,K,rd,rf,sig,T,kind):
    d1=(log(S/K)+(rd-rf+0.5*sig**2)*T)/(sig*sqrt(T)); d2=d1-sig*sqrt(T)
    if kind=='put': return K*exp(-rd*T)*norm.cdf(-d2)-S*exp(-rf*T)*norm.cdf(-d1)
    return S*exp(-rf*T)*norm.cdf(d1)-K*exp(-rd*T)*norm.cdf(d2)
pp = gk(S0,1.09,rU,rE,sg,T,'put')
Kc = brentq(lambda k: gk(S0,k,rU,rE,sg,T,'call')-pp, 1.10, 1.40)
print(f"  put premium {pp:.5f}; call strike for zero cost Kc = {Kc:.4f}")
print(f"  Collar range: receive between 1.0900 and {Kc:.4f} per EUR -> USD {1.09*5e6:,.0f} to {Kc*5e6:,.0f}")

print("\nExercise 21.7: Kupiec test, 250 days, 6 exceptions at 99% VaR")
x, n, p = 6, 250, 0.01
phat = x/n
lr = -2*((n-x)*log(1-p)+x*log(p) - ((n-x)*log(1-phat)+x*log(phat)))
from scipy.stats import chi2
```

```
print(f"  expected {n*p:.1f}; LR = {lr:.4f}; p-value = {1-chi2.cdf(lr,1):.4f}; 5% critical value
{chi2.ppf(0.95,1):.3f}")

print("\nExercise 21.8: delta-hedge 100,000 calls (S=100,K=100,r=5%,sigma=25%,T=1): shares to hold; after S
rises to 102, shares to add")
b0 = bs(100,100,0.05,0,0.25,1); b1 = bs(102,100,0.05,0,0.25,1)
print(f"  initial delta {b0['delta']:.5f} -> hold {100000*b0['delta']:,.0f} shares (cost
${100000*b0['delta']*100:,.0f})")
print(f"  new delta {b1['delta']:.5f} -> add {100000*(b1['delta']-b0['delta']):,.0f} shares; gamma approx
{100000*b0['gamma']*2:,.0f}")
```

Output:

```
Exercise 21.1: one-step tree, S=40, u=1.25, d=0.80, r=4%, K=42 call
  p*=0.5351  C=4.1132  delta=0.4444  borrowing=13.6646  check delta*S+B=4.1132

Exercise 21.2: implied volatility of a call quoted at 9.00 (S=K=100,r=5%,T=1) by bisection
  implied sigma = 0.16109

Exercise 21.3: parametric VaR of a $20m position, daily vol 1.4%, 99%, 10 days
  VaR = $2.0598m

Exercise 21.4: min-variance hedge, sd_s=0.021, sd_f=0.026, rho=0.88, exposure 12m
  h*=0.7108; hedge notional 8.529m; variance reduction 77.44%

Exercise 21.5: GK put, S=1.25 (USD/GBP), K=1.22, rU=4.5%, rG=3.5%, sigma=8%, T=0.25, notional GBP 2m
  premium 0.00735 USD/GBP = $14,702;  forward 1.25313

Exercise 21.6: zero-cost collar for the Ostrava receivable. Buy EUR put at 1.09, sell EUR call at strike Kc
with equal premium
  put premium 0.01927; call strike for zero cost Kc = 1.1277
  Collar range: receive between 1.0900 and 1.1277 per EUR -> USD 5,450,000 to 5,638,665

Exercise 21.7: Kupiec test, 250 days, 6 exceptions at 99% VaR
  expected 2.5; LR = 3.5554; p-value = 0.0594; 5% critical value 3.841

Exercise 21.8: delta-hedge 100,000 calls (S=100,K=100,r=5%,sigma=25%,T=1): shares to hold; after S rises to
102, shares to add
  initial delta 0.62741 -> hold 62,741 shares (cost $6,274,095)
  new delta 0.65697 -> add 2,956 shares; gamma approx 3,027
```

Exercise 21.1: One-Step Binomial Option

A share trades at 40 and in one year will be either 50 (u = 1.25) or 32 (d = 0.80). The continuously compounded risk-free rate is 4%. Price a one-year European call struck at 42, and give the replicating portfolio.

Solution. p* = (e^0.04 − 0.80)/0.45 = 0.5351; the call pays 8 up and 0 down, so C = e^(−0.04) × 0.5351 × 8 = 4.1132. Δ = 8/(40 × 0.45) = 0.4444 shares, financed by borrowing 0.4444 × 40 − 4.1132 = 13.6646.

Exercise 21.2: Implied Volatility

A one-year at-the-money call (S = K = 100, r = 5%, no dividends) is quoted at 9.00. Find the implied volatility.

Solution. The price is increasing in sigma, so bisect between 1% and 100%; sixty halvings give sigma = 16.109%.

Exercise 21.3: Parametric VaR

A $20 million position has a daily volatility of 1.4%. Compute the ten-day 99% VaR under normality with zero mean.

Solution. Ten-day sigma = 20 × 0.014 × √10 = $0.8854 million; VaR = 2.3263 × 0.8854 = $2.0598 million, 10.3% of the position.

Exercise 21.4: Minimum-Variance Hedge

Spot and futures changes have standard deviations of 2.1% and 2.6% and a correlation of 0.88. The exposure is $12 million. Find the optimal hedge ratio, the hedge notional and the proportion of variance eliminated.

Solution. h* = 0.88 × 0.021/0.026 = 0.7108, so the futures notional is 0.7108 × 12 = $8.529 million. Effectiveness ρ^2 = 77.44%; hedged standard deviation falls from 2.1% to 0.997%.

Exercise 21.5: Currency Put on a Sterling Receivable

A US firm will receive GBP 2 million in three months. Spot is 1.2500 USD per GBP, the dollar rate is 4.5% and the sterling rate 3.5%, volatility is 8%, and the firm considers a put on sterling struck at 1.2200. Find the premium and the forward rate.

Solution. d1 = [ln(1.25/1.22) + (0.045 − 0.035 + 0.0032) × 0.25]/(0.08 × 0.5) = 0.6898 and d2 = 0.6498. The put is 1.22·e^(−0.01125)·N(−0.6498) − 1.25·e^(−0.00875)·N(−0.6898) = 0.00735 USD per GBP, or $14,702 on GBP 2 million (0.59% of notional). The forward is 1.25 × e^(0.0025) = 1.25313.

Exercise 21.6: Zero-Cost Collar

For the Ostrava receivable of Section 21.9, find the strike of a euro call, sold by the firm, that makes the collar zero-cost when combined with the 1.0900 put.

Solution. Solve for the call strike at which the Garman-Kohlhagen call price equals 0.01927: Kc = 1.1277. The firm is guaranteed between USD 5,450,000 (at 1.0900) and USD 5,638,665 (at 1.1277), pays no premium, and receives the market rate in between.

Exercise 21.7: Kupiec Test

A 99% VaR model produced 6 exceptions in 250 trading days. Test it.

Solution. Expected exceptions 2.5, observed 6 (2.4%). LR = 3.5554, p-value 0.0594, just short of the 3.841 critical value, so the model is not rejected at 5%; it is a Basel amber outcome and deserves scrutiny, but 250 days cannot settle the matter.

Exercise 21.8: Delta Hedging a Written Option Book

A bank has sold 100,000 one-year at-the-money calls (S = K = 100, r = 5%, sigma = 25%). How many shares should it hold, and how many must it add if the share price rises to 102?

Solution. Initial delta is 0.62741, so hold 62,741 shares ($6,274,095). After the rise delta is 0.65697, so buy 2,956 more. The gamma approximation (100,000 × 0.01514 × 2 = 3,027) is slightly high because gamma falls as the share rises. The bank buys after the price has risen: a short-gamma hedger buys high and sells low.

Section 21.12: Common Errors in Modeling

1. **Mixing real-world and risk-neutral drifts.** Price with the risk-neutral process; measure risk with the real-world (or zero-drift) one.
2. **Wrong units on the Greeks.** Vega per unit instead of per point is 100 times too large; theta per year instead of per day is 365 times too large.
3. **Confusing the `NORM.S.DIST` flag.** TRUE is the cumulative probability, FALSE the density.
4. **Too few simulations, too many digits.** Report a standard error; 10,000 draws give about 1.6% error on a 99% VaR.

5. **Delta-normal VaR for options.** It ignores curvature: it overstates risk for long options and understates it for short ones.
6. **Blind square-root-of-time scaling, regressing levels instead of changes, judging a hedge by expected value, comparing premiums and settlements on different dates, and treating a passed backtest as proof.**

Section 21.13: Chapter Summary

The binomial tree prices European and American options and converges to Black-Scholes. The Greeks, verified by finite differences, acquire financial meaning in the delta-hedging simulation, where residual risk falls with the square root of the number of rebalances. Fat tails make the normal approximation understate 99% VaR by 18% in our sample, and linear delta approximations misstate option risk by about a quarter. A hedge ratio is a regression slope whose effectiveness is the squared correlation: the 18-contract diesel hedge removed two-thirds of the variance. The FX comparison showed that a forward fixes the outcome while an option fixes a floor and keeps the upside for a premium. Chapter 22 turns to sixty graded problems on the valuation foundations of Volume 1, Chapters 1 to 12.

Key Formulas Collected

```
CRR tree:       u = e^(σ√Δt), d = 1/u, p* = (e^((r−q)Δt) − d)/(u − d)
Black-Scholes:  C = S e^(−qT) N(d1) − K e^(−rT) N(d2)
Put-call parity: C − P = S e^(−qT) − K e^(−rT)
Garman-Kohlhagen: replace q by the foreign rate r_f and S by the exchange rate
Parametric VaR: z_α · σ_p · √h          ES = σ_p · n(z_α)/(1 − α)
Min-variance hedge ratio: h* = ρ σ_S/σ_F ;  effectiveness = ρ²
Covered interest parity: F = S e^((r_d − r_f)T)
Kupiec LR = −2 ln[(1−p)^(n−x) p^x] + 2 ln[(1−x/n)^(n−x)(x/n)^x]
```

Part V — Practice and Instructor Materials

Chapter 22: Comprehensive Problem Sets with Full Solutions I: Foundations and Valuation

How to Use This Problem Set

This chapter contains sixty problems covering Volume 1, Chapters 1 to 12: the corporation, financial statements, time value of money, bonds, equity valuation, net present value, capital budgeting, uncertainty and real options, risk and return, the CAPM, cost of capital and market efficiency. The problems are graded in three levels, twenty at each.

Level	What it tests	Suggested time
Basic	A single concept applied directly; one or two formulas	5 to 8 minutes
Intermediate	Two or three linked steps, or a concept with an interpretation	10 to 15 minutes
Challenging	Multi-step modeling, a trap or a judgement call	20 to 30 minutes

All companies are fictional and all data are illustrative. Every numerical answer in this chapter was computed by a Python script using the formulas shown, and the figures in the solutions are taken from that output; they are rounded for display, so a hand calculation may differ in the last digit. Dollar amounts are in dollars unless stated ($m for millions, $k for thousands). Rates are annual and compounded annually unless stated.

Work each problem on paper first. Then read the solution for the method, not only the answer: each solution states the formula, substitutes the numbers, and ends with an interpretation sentence. The chapter-review notes at the head of each part restate the key formulas and the mistakes most frequently made. A companion problem set covers Volume 1, Chapters 13 to 24.

Problem Map

Part	Volume 1 chapter	Problems	Basic	Intermediate	Challenging
1	The Corporation and Its Objectives	22.1 to 22.4	1	1	2
2	Financial Statements and Ratios	22.5 to 22.10	3	2	1
3	Time Value of Money	22.11 to 22.16	2	1	3
4	Bonds and Interest Rates	22.17 to 22.22	2	2	2
5	Equity Valuation	22.23 to 22.28	2	3	1
6	Net Present Value	22.29 to 22.33	2	1	2
7	Capital Budgeting	22.34 to 22.39	2	2	2
8	Uncertainty and Real Options	22.40 to 22.44	1	1	3
9	Risk and Return	22.45 to 22.49	2	2	1
10	The Capital Asset Pricing Model	22.50 to 22.54	2	2	1
11	Cost of Capital	22.55 to 22.58	1	2	1
12	Market Efficiency and Behavioral Finance	22.59 to 22.60	0	1	1
Total		60	20	20	20

Part 1: The Corporation and Its Objectives (Volume 1, Chapter 1)

Review. Limited liability caps owners' losses at their investment; agency costs are the value lost when managers' incentives differ from owners'; equity behaves like a call option on firm assets, so shareholders can gain from risk at creditors' expense.

Typical errors. Treating the corporate form's tax treatment as a benefit, and forgetting that the manager bears only her ownership share of any cost.

Problem 22.1 (Basic)

A business earns $200,000 before tax. As a corporation it pays corporate tax of 25% and distributes all earnings as dividends taxed at 20%. As a sole proprietorship, the owner pays 35% personal tax. Compare the owner's after-tax cash and the total effective tax rate in each form (illustrative rates).

Solution. Corporation: $200,000 x (1 - 0.25) = $150,000; dividend tax 20% leaves $120,000, a total tax of 40.0%. Proprietorship: $200,000 x (1 - 0.35) = $130,000, a total tax of 35.0%. Double taxation costs the corporate owner $10,000 more in this illustration; the corporate form is justified by limited liability and access to capital, not by tax.

Problem 22.2 (Intermediate)

Firm Dunmore Tools (fictional) fails with assets of $150,000 and debts of $400,000. An investor put in $50,000 for a 25% stake. Compute her total loss (a) if the firm is a corporation and (b) if it is a general partnership in which her share of any shortfall is her ownership percentage.

Solution. Shortfall = $400,000 - $150,000 = $250,000. (a) Limited liability caps the loss at the $50,000 invested. (b) She loses $50,000 plus 25% x $250,000 = $62,500, a total of $112,500, 2.25 times her investment. Limited liability transfers the residual $250,000 shortfall risk to creditors, who price it into interest rates.

Problem 22.3 (Challenging)

A manager who owns 5% of the shares approves a corporate jet costing $2.0 million that she values at $1.2 million in personal benefit. (a) What is her net gain? (b) What is the value destroyed? (c) What ownership share would make her decline?

Solution. (a) She bears 5% x $2.0m = $0.10m of the cost and receives $1.20m, a net gain of $1.10m. (b) The firm spends $2.00m for a benefit worth $1.20m, destroying $0.80m. (c) She declines when her share of the cost exceeds her benefit: s x 2.0 > 1.2, so s > 60%. This is the agency problem of Volume 1, Chapter 1: with a small stake, managers capture the benefit and bear little of the cost, hence the use of incentive pay and monitoring.

Problem 22.4 (Challenging)

A firm has assets worth $100m (illustrative) and zero-coupon debt of $80m face due in one year; r = 5%. Treat equity as a call option on assets. Find equity and debt values when asset volatility is 30%, then when management shifts into riskier projects and volatility becomes 50%.

Solution. Equity = Black-Scholes call with S = 100, K = 80, r = 5%, T = 1. At 30%: E = $26.46m and debt = 100 - 26.46 = $73.54m. At 50%: E = $31.79m and debt = $68.21m. Equity gains $5.33m and creditors lose exactly that amount, though the asset value of $100m did not change. This is risk shifting: shareholders profit from variance at creditors' expense, which is why covenants exist. Riskless debt would be worth 80 x e^(-0.05) = $76.10m.

Part 2: Financial Statements and Ratios (Volume 1, Chapter 2)

Review. Current ratio = CA/CL; DuPont: ROE = margin x turnover x leverage; cash conversion cycle = DSO + DIO - DPO; EFN = increase in assets - increase in spontaneous liabilities - retained earnings.

Typical errors. Mixing year-end and average balances without saying so, and using sales-based days for inventory instead of COGS-based days.

Problem 22.5 (Basic)

From this extract of Pelham Components' balance sheet (fictional), compute the current ratio, the quick ratio and the cash ratio.

Current assets	$m	Current liabilities	$m
Cash	80	Payables and accruals	190
Receivables	160	Short-term debt	120
Inventory	240		
Other	40	Total	310

Solution. Current assets = 80 + 160 + 240 + 40 = 520. Current ratio = 520/310 = 1.68. Quick ratio = (80 + 160)/310 = 0.77 (inventory and other assets excluded). Cash ratio = 80/310 = 0.26. The firm covers short-term liabilities from liquid assets only 77% of the time, so it relies on turning inventory into cash.

Problem 22.6 (Basic)

Pelham Components reports revenue of $1,200m, cost of goods sold $780m, SG&A $240m, depreciation $60m and interest $30m; the tax rate is 25%. Compute gross, operating and net margins.

Solution. Gross profit = 1,200 - 780 = 420; margin 35.00%. EBIT = 420 - 240 - 60 = 120; margin 10.00%. Pre-tax income = 120 - 30 = 90; net income = 90 x 0.75 = 67.5; net margin 5.62%. Interest coverage = 120/30 = 4.0x.

Problem 22.7 (Basic)

A firm earns a return on equity of 15% and pays out 40% of earnings. Compute the retention ratio, the growth rate from retained earnings using beginning equity (g = ROE x b) and the sustainable growth rate using ending equity.

Solution. Retention b = 1 - 0.40 = 0.60. Using beginning equity, g = 0.15 x 0.60 = 9.00%. Using ending equity (ROE measured on average or ending equity), g = ROE x b/(1 - ROE x b) = 0.09/0.91 = 9.89%. The two differ because retained earnings also enlarge the base on which next year's return is earned; state which definition you use.

Problem 22.8 (Intermediate)

Pelham Components has net income $54m, revenue $1,200m, assets $900m and equity $360m. (a) Decompose ROE with the DuPont identity. (b) If assets are unchanged but equity were $600m (more equity, less debt), what happens to ROE?

Solution. (a) Net margin = 54/1,200 = 4.50%; asset turnover = 1,200/900 = 1.333; equity multiplier = 900/360 = 2.50. ROE = 0.0450 x 1.333 x 2.50 = 15.00% (check: 54/360 = 15.00%). (b) Equity multiplier becomes 900/600 = 1.50, so ROE = 9.00% (ignoring the interest saving). Lower leverage cuts ROE by 6.00 percentage points; ROE gains from leverage are bought with financial risk.

Problem 22.9 (Intermediate)

Using revenue $1,200m, COGS $780m, receivables $160m, inventory $240m and payables $130m (365-day year, year-end balances), compute days sales outstanding, days inventory, days payables and the cash conversion cycle. How much cash is released by cutting the cycle by 10 days?

Solution. DSO = 160/1,200 x 365 = 48.7. DIO = 240/780 x 365 = 112.3. DPO = 130/780 x 365 = 60.8. Cash conversion cycle = 48.7 + 112.3 - 60.8 = 100.1 days. Each day of the cycle ties up COGS/365 = $2.137m (valued at cost), so 10 days releases about $21.4m (about $32.9m if measured at sales value for receivables).

Problem 22.10 (Challenging)

Haldane Packaging (fictional) has sales of $1,000m, assets of $800m (all varying with sales) and spontaneous liabilities (payables, accruals) of $150m. Net margin is 8% and the payout ratio 50%. Sales will grow 20%. Compute the external financing needed (EFN) and the internal growth rate.

Solution. Next-year sales = $1,200m. Required asset increase = 800 x 0.20 = 160. Spontaneous liabilities increase by 150 x 0.20 = 30. Net income = 0.08 x 1,200 = 96; retained = 48. EFN = 160 - 30 - 48 = $82m. Internal growth rate (ROA-based): ROA = 0.08 x 1,000/800 = 10% on beginning assets; g = ROA x b/(1 - ROA x b) = 0.05/0.95 = 5.26% (ignoring spontaneous liabilities).

Part 3: Time Value of Money (Volume 1, Chapter 3)

Review. FV = PV(1+r)^n; annuity PV = C[1-(1+r)^-n]/r; growing annuity PV = C/(r-g)[1-((1+g)/(1+r))^n]; EAR = (1+APR/m)^m - 1.

Typical errors. Using the nominal rate with the wrong number of periods, and discounting nominal cash flows at a real rate.

Problem 22.11 (Basic)

What is $10,000 worth after 8 years at a 6% stated annual rate if interest is compounded (a) annually and (b) semiannually? What is the effective annual rate in (b)?

Solution. (a) FV = 10,000 x 1.06^8 = $15,938.48. (b) Semiannual rate 3%, 16 periods: FV = 10,000 x 1.03^16 = $16,047.06. EAR = 1.03^2 - 1 = 6.09%. More frequent compounding adds $108.58 over eight years.

Problem 22.12 (Basic)

What is the present value of $5,000 received at the end of each year for 12 years at 7%? What if the payments are made at the start of each year?

Solution. PV = 5,000 x [1 - 1.07^(-12)]/0.07 = 5,000 x 7.9427 = $39,713.43. For an annuity due, multiply by 1.07: $42,493.37, since every payment arrives one year sooner.

Problem 22.13 (Challenging)

A $250,000 mortgage has a 6% nominal annual rate compounded monthly and a 30-year term. Find (a) the monthly payment, (b) the balance after 5 years, (c) interest paid in year 1, and (d) the payoff time and interest saved if the borrower adds $300 a month.

Solution. (a) Payment = 250,000 x 0.005/[1 - 1.005^(-360)] = $1,498.88. (b) Balance after 60 payments = 250,000 x 1.005^60 - 1498.88 x (1.005^60 - 1)/0.005 = $232,635.89. (c) Year-1 payments total $17,986.52; principal repaid is 250,000 - 246,929.97 = $3,070.03, so interest is $14,916.49. (d) With $1,798.88 a month the loan is repaid after 239 months (19.9 years); total interest falls from $289,595 to $178,136, a saving of $111,459.

Problem 22.14 (Intermediate)

A contract pays $40,000 at the end of year 1, growing 3% a year. At an 8% discount rate find the value if it lasts (a) 20 years and (b) forever. What share of the perpetuity value comes from years 21 onward?

Solution. (a) PV = C/(r - g) x [1 - ((1+g)/(1+r))^n] = 40,000/0.05 x [1 - (1.03/1.08)^20] = 800,000 x 0.6125 = $490,002. (b) Perpetuity = 40,000/0.05 = $800,000. Years 21 onward are worth $309,998, 38.7% of the total: a growing stream’s value is back-loaded when r - g is small.

Problem 22.15 (Challenging)

A saver wants real income of $80,000 a year for 25 years, first received at the end of the year after retirement at 65. She saves a constant real amount at the end of each year for 35 years (from 31 to 65). At a real return of 3%, how much must she save each year? What is the required balance at 65?

Solution. Required balance at 65 = 80,000 x [1 - 1.03^(-25)]/0.03 = 80,000 x 17.4131 = $1,393,052. Annual saving S solves S x (1.03^35 - 1)/0.03 = 1,393,052; the factor is 60.4621, so S = $23,040 per year (28.8% of the target income). Working in real terms avoids forecasting inflation; the answer is in today’s dollars.

Problem 22.16 (Challenging)

A $24,000 car can be financed with 48 monthly payments of $560 starting next month. Find the monthly rate, the APR and the effective annual rate. If the dealer instead offers 0% for 48 months and a $1,500 cash rebate for paying cash, which is better at a 6% opportunity rate (compounded monthly)?

Solution. Solve 560 x [1 - (1+i)^(-48)]/i = 24,000: i = 0.4724% per month. APR = 12 x i = 5.67%; EAR = (1+i)^12 - 1 = 5.82%. Zero-percent alternative: 24,000/48 = $500 a month, PV at 0.5% = 500 x 42.5803 = $21,290; paying cash costs 24,000 - 1,500 = $22,500. The 0% financing costs $1,210 less in present value than paying cash, so take the 0% financing and invest the cash at 6%.

Part 4: Bonds and Interest Rates (Volume 1, Chapter 4)

Review. Price = sum of discounted coupons plus face; modified duration = Macaulay/(1+y); price change = -D_mod x dy + 0.5 x convexity x dy^2; forward rate from spot rates; immunization matches PV and duration.

Typical errors. Forgetting that a semiannual bond has 2n periods at y/2, and quoting yield as the coupon rate.

Problem 22.17 (Basic)

A $1,000 face bond pays a 5% coupon semiannually and matures in 10 years. Find its price at a yield to maturity of 6% (semiannual compounding) and state whether it trades at a premium or discount.

Solution. Coupon = $25; 20 periods at 3%. Price = 25 x [1 - 1.03^(-20)]/0.03 + 1,000/1.03^20 = 371.94 + 553.68 = $925.61. It trades at a discount because the coupon rate (5%) is below the yield (6%).

Problem 22.18 (Basic)

A 5-year, 6% annual-coupon bond with $1,000 face trades at $950. Find the current yield and the yield to maturity.

Solution. Current yield = 60/950 = 6.32%. Solve 950 = sum 60/(1+y)^t + 1,000/(1+y)^5: y = 7.23%. Check: price at 7.23% = $950.00. YTM exceeds the current yield because the bond also pulls up to par.

Problem 22.19 (Intermediate)

A 3-year bond pays a 6% annual coupon (face $1,000); yield 5%. Compute price, Macaulay and modified duration and convexity, and estimate the price change for a +50 basis point shift using duration alone and duration plus convexity. Compare with the exact change.

Solution. Price = $1,027.23. Macaulay duration = sum t x PV(CF)/P = 2.8358 years; modified = 2.8358/1.05 = 2.7007. Convexity = 10.091. Duration alone: -1.350%; with convexity: -1.338%. Exact repricing at 5.5%: $1,013.49, a change of -1.338%. Convexity corrects the linear estimate by 0.013%.

Problem 22.20 (Intermediate)

Spot rates (annual compounding) are 3.0% for 1 year, 3.5% for 2 years and 4.0% for 3 years. Price a 3-year 5% annual-coupon bond (face $1,000), find its yield to maturity, and compute the implied forward rate from year 2 to year 3.

Solution. Price = 50/1.03 + 50/1.035^2 + 1,050/1.04^3 = 48.54 + 46.68 + 933.45 = $1,028.67. YTM solves the single rate repricing the bond: 3.968%, close to the 3-year spot because most of the value comes at year 3. Forward 2y3y: $(1.04^{3}/1.035^{2}) - 1$ = 5.007%.

Problem 22.21 (Challenging)

A pension fund owes $1,000,000 in 5 years. The yield curve is flat at 5%. It can buy 3-year and 8-year zero-coupon bonds. Find the investment in each that immunizes the liability (match present value and duration), and explain what protects the fund against a parallel shift.

Solution. PV of liability = 1,000,000/1.05^5 = $783,526. A zero has duration equal to maturity. Weights solve w x 3 + (1-w) x 8 = 5: w = 0.60 in the 3-year, 0.40 in the 8-year. Investments: $470,116 and $313,410. Face amounts: 3-year $544,218, 8-year $463,050. Because asset and liability have equal PV and duration, and the asset side has greater convexity, a parallel shift leaves the surplus non-negative to first order and positive to second order. Rebalancing is needed as time passes.

Problem 22.22 (Challenging)

A 1-year zero-coupon bond of Orwell Textiles (fictional) with $1,000 face trades at $900. The risk-free rate is 4%. The probability of default is 8% with 40% recovery of face. Find the promised yield, expected payoff, expected yield and the default premium decomposition.

Solution. Promised yield = 1,000/900 - 1 = 11.11%. Expected payoff = 0.92 x 1,000 + 0.08 x 400 = $952; expected yield = 952/900 - 1 = 5.78%. Spread over risk-free: promised 7.11%; of this, expected loss accounts for 5.33% and the remaining 1.78% is compensation for bearing default risk (a risk premium) over the 4% risk-free rate.

Part 5: Equity Valuation (Volume 1, Chapter 5)

Review. P0 = D1/(r-g); g = ROE x b; two-stage values add a terminal value; EV = NOPAT(1 - g/ROIC)/(WACC - g); PVGO = P - EPS1/r.

Typical errors. Applying the Gordon formula when g is not below r, and growing the wrong dividend (D0 rather than D1).

Problem 22.23 (Basic)

Halden Utilities (fictional) will pay a dividend of $2.40 next year, expected to grow 4% a year forever. The required return is 10%. Find the value per share, the dividend yield and the implied price if the required return rises to 11%.

Solution. P0 = D1/(r - g) = 2.40/(0.10 - 0.04) = $40.00. Dividend yield = 2.40/40.00 = 6.00%, and capital gains yield = g = 4%, summing to r. At 11%: P0 = 2.40/0.07 = $34.29, a fall of 14.3% for one point on the required return, showing the high sensitivity of growth stocks to r - g.

Problem 22.24 (Basic)

A firm expects next-year earnings per share of $3.20, retains 60% of earnings and earns 12% on new equity investment. The required return is 10%. Find the growth rate, next dividend, price and forward P/E.

Solution. g = ROE x b = 0.12 x 0.60 = 7.20%. D1 = 3.20 x 0.40 = $1.28. P0 = 1.28/(0.10 - 0.072) = $45.71. Forward P/E = 45.71/3.20 = 14.29x. A no-growth policy would give 3.20/0.10 = $32.00, so the growth policy adds $13.71 per share because ROE 12% exceeds r = 10%.

Problem 22.25 (Intermediate)

Corvane Software (fictional) just paid a dividend of $1.50. Dividends will grow 15% a year for 4 years, then 5% forever. The required return is 11%. Find the value of the share.

Solution. Dividends years 1-4: $1.7250, $1.9837, $2.2813, $2.6235. PV of these = $6.56. Terminal value at year 4 = 2.6235 x 1.05/(0.11 - 0.05) = $45.91; PV = 45.91/1.11^4 = $30.24. Value = 6.56 + 30.24 = $36.80. The terminal value is 82% of the price, so the long-run growth assumption matters most.

Problem 22.26 (Intermediate)

A share trades at $70 and expects earnings of $5.00 next year. If the firm paid out all earnings with no growth, the required return of 10% would apply. Find the no-growth value and the present value of growth opportunities (PVGO), and the share of price attributable to growth.

Solution. No-growth value = EPS1/r = 5.00/0.10 = $50.00. PVGO = 70 - 50.00 = $20.00, or 28.6% of the price. The current P/E is 14.0x against a no-growth P/E of 10.0x. Roughly 29% of the price depends on investments not yet made, which is why growth stocks are more exposed to changes in expectations.

Problem 22.27 (Intermediate)

A firm has EBIT of $400m, a 25% tax rate, depreciation $100m, capital expenditure $130m and an increase in net working capital of $20m. Free cash flow will grow 3% a year; WACC is 9%. Net debt is $1,200m and there are 100m shares. Find FCFF, enterprise value, equity value and the value per share (FCFF this year is the base; the next flow is 1.03 times it).

Solution. NOPAT = 400 x 0.75 = 300. FCFF = 300 + 100 - 130 - 20 = 250. EV = 250 x 1.03/(0.09 - 0.03) = $4,291.7m. Equity = 4,291.7 - 1,200 = $3,091.7m; per share $30.92. Net debt is 28.0% of EV; a one-point rise in WACC to 10% would cut EV to $3,679m, equity to $2,479m.

Problem 22.28 (Challenging)

Use the value driver formula EV = NOPAT x (1 - g/ROIC)/(WACC - g) with next-year NOPAT of $300m and WACC of 9%. Compute enterprise value for (a) no growth, (b) 4% growth at ROIC 14%, and (c) 4% growth at ROIC 8%. Explain the result.

Solution.

Growth g	ROIC	Value
0%	14%	$3,333m
4%	14%	$4,286m
4%	8%	$3,000m

a. With g = 0 no reinvestment is needed: 300/0.09 = $3,333m. (b) Reinvestment rate = 4%/14% = 28.6%; FCF = 300 x 0.7143 = 214.3; EV = 214.3/0.05 = $4,286m, 28.6% more than (a). (c) Reinvestment is 4%/8% = 50%; FCF = 150; EV = $3,000m, 10.0% below (a). Growth creates value only when ROIC exceeds WACC; at ROIC 8% < 9% the firm would do better distributing the cash.

Part 6: Net Present Value (Volume 1, Chapter 6)

Review. NPV = sum CF_t/(1+r)^t; choose among mutually exclusive projects by NPV; IRR can be multiple or misleading; use equivalent annual cost for unequal lives; match real flows with real rates.

Typical errors. Ranking exclusive projects by IRR, and comparing PVs of costs for assets with different lives.

Problem 22.29 (Basic)

A project costs $1,000 now and returns $300, $400, $500 and $300 at the end of years 1 to 4. The cost of capital is 10%. Compute NPV, IRR, payback and discounted payback.

Solution. NPV = -1,000 + 300/1.1 + 400/1.1^2 + 500/1.1^3 + 300/1.1^4 = $183.87. IRR = 18.03%. Payback: cumulative inflows are 300, 700, 1,200, so payback = 2 + 300/500 = 2.60 years. Discounted flows are 272.7, 330.6, 375.7, 204.9; cumulative 272.7, 603.3, 979.0, 1183.9, so the discounted payback is 3.10 years. NPV is positive and IRR exceeds 10%: accept.

Problem 22.30 (Basic)

Mutually exclusive projects A and B each cost $500. A returns $200 a year for 4 years; B returns $1,000 in year 4 only. Compute NPV and IRR at 10%, say which to choose, and find the crossover rate.

Solution. A: NPV = -500 + 200 x 3.1699 = $133.97; IRR = 21.86%. B: NPV = -500 + 1,000/1.1^4 = $183.01; IRR = 18.92%. A has the higher IRR, but B has the higher NPV at 10% (B is bigger and later); with mutually exclusive projects choose by NPV, so choose B. The crossover is the IRR of the difference (B - A) = [0, -200, -200, -200, 800]: 15.09%. Below that rate B is better; above it A is better.

Problem 22.31 (Challenging)

A mine project has cash flows -100, +230, -132 at times 0, 1, 2 (the last is a restoration cost). Find the IRRs. Compute NPV at 5%, 15% and 25%, and the modified IRR with finance and reinvestment rates of 12%. Should the project be accepted at a 12% cost of capital?

Solution. Solving -100 + 230/(1+r) - 132/(1+r)^2 = 0 gives two IRRs: 10.0% and 20.0%. NPV(5%) = -0.68; NPV(15%) = 0.19; NPV(25%) = -0.48. NPV is positive only between the two roots, so the project has value at 12% (NPV = 0.13). MIRR: PV of outflows at 12% = 100 + 132/1.12^2 = 205.23; FV of the inflow at year 2 = 230 x 1.12 = 257.6; MIRR = (257.6/205.23)^(1/2) - 1 = 12.03%, which only just exceeds 12%: the project is marginally acceptable. Multiple IRRs arise whenever the sign of cash flow changes more than once.

Problem 22.32 (Intermediate)

Machine A costs $100 and has annual operating costs of $20 for a 3-year life. Machine B costs $150 and has operating costs of $12 for a 5-year life. Both provide the same service and will be replaced at the end of their lives. At 10%, which has the lower equivalent annual cost?

Solution. PV of costs A = 100 + 20 x 2.4869 = 149.74; EAC = 149.74/2.4869 = $60.21. B: PV = 150 + 12 x 3.7908 = 195.49; EAC = $51.57. B has the lower EAC by $8.64 a year, so choose B. Comparing PVs directly (149.74 against 195.49) would wrongly favour A because the lives differ.

Problem 22.33 (Challenging)

A project requires $1,000 now and yields real (constant-price) cash flows of $400 a year for 3 years. The nominal discount rate is 9% and expected inflation is 2.5%. Show that discounting nominal flows at the nominal rate and real flows at the real rate give the same NPV, and show the error from mixing the two.

Solution. Real rate = 1.09/1.025 - 1 = 6.341%. Real NPV = -1,000 + 400 x 2.6562 = $62.49. Nominal flows: 410.00, 420.25, 430.76; NPV at 9% = $62.49, identical. Error: discounting real flows at 9% gives $12.52, understating value by $49.97; discounting nominal flows at the real rate overstates it to $115.37. Match the units of flows and rates.

Part 7: Capital Budgeting (Volume 1, Chapter 7)

Review. Incremental after-tax cash flows: OCF = (savings)(1-t) + depreciation x t; add working capital and after-tax salvage; the tax effect of selling an old asset depends on book value; indivisible projects under capital rationing need enumeration.

Typical errors. Counting sunk costs, omitting working capital recovery, and ignoring the tax on salvage or the tax shield on a loss.

Problem 22.34 (Basic)

Equipment costs $600,000, is depreciated straight-line to zero over 5 years and saves $200,000 a year before tax. The tax rate is 25%. It will be sold for $50,000 at the end of year 5. Find the annual operating cash flow, the after-tax salvage value and the NPV at 10% (amounts in $000).

Solution. Depreciation = 600/5 = 120 a year. OCF = 200 x (1 - 0.25) + 120 x 0.25 = 150 + 30 = 180. After-tax salvage = 50 x (1 - 0.25) = 37.5 (book value is zero, so the whole price is a taxable gain). NPV = -600 + 180 x 3.7908 + 37.5/1.1^5 = $105.63k. The depreciation tax shield is worth 30 x 3.791 = $113.7k.

Problem 22.35 (Basic)

A product sells for $40 with variable cost of $25 a unit. Annual fixed costs are $120,000, including $40,000 of depreciation. Find the accounting break-even and cash break-even volumes, and the degree of operating leverage at 10,000 units.

Solution. Contribution margin = 40 - 25 = $15. Accounting break-even = 120,000/15 = 8,000 units. Cash break-even = (120,000 - 40,000)/15 = 5,333 units. At 10,000 units, EBIT = 10,000 x 15 - 120,000 = $30,000; DOL = contribution/EBIT = 5.0. A 10% rise in volume lifts EBIT by 50%. Neither break-even equals the NPV break-even, which would also cover the cost of capital.

Problem 22.36 (Intermediate)

A project needs $500k of equipment (straight-line over 6 years, no salvage) and $60k of net working capital, recovered at the end. It produces pre-tax cash savings of $180k a year for 6 years; tax rate is 30%; discount rate 10%. Compute the NPV and IRR.

Solution. Depreciation = 83.33; OCF = 180 x 0.70 + 83.33 x 0.30 = 151.00 a year. Flows: t0 = -560; t1-t5 = 151.00; t6 = 151.00 + 60 = 211.00. NPV = $131.51k; IRR = 17.35%. Working capital reduces NPV by 60 - 60/1.1^6 = $26.1k: it is a cost of tied-up funds, not an expense.

Problem 22.37 (Intermediate)

A project costs $300,000 (straight-line to zero over 5 years) with price $50, variable cost $30, 10,000 units a year, fixed cash costs $100,000, tax 25% and discount rate 12%. Compute base NPV and the NPV if (a) price falls 10%, (b) units fall 10%, and (c) variable cost rises 10%. Which input is most sensitive?

Solution. OCF = [(p - vc) x q - 100,000 - 60,000] x 0.75 + 60,000; annuity factor = 3.6048. Base: NPV = $24,430. (a) p = 45: NPV = -$110,749 (change -135,179). (b) q = 9,000: -$29,642 (change -54,072). (c) vc = 33: -$56,678 (change -81,107). Price is most sensitive, since a $5 price change applies to every unit with no offsetting cost, and the project turns NPV-negative after a price fall of about 1.8%.

Problem 22.38 (Challenging)

An old machine (book value $200k, 5 years of straight-line depreciation remaining, no final salvage) can be sold now for $90k. A new machine costs $500k, depreciated over 5 years to zero, and cuts operating costs by $140k a year before tax. Tax is 25%; the discount rate is 10%. Should the firm replace? Show the net initial outlay and incremental flows.

Solution. Loss on sale = 200 - 90 = $110k, tax saving = 0.25 x 110 = $27.5k. Net outlay = 500 - 90 - 27.5 = $382.5k. Incremental depreciation = 500/5 - 200/5 = 60. Incremental OCF = 140 x 0.75 + 60 x 0.25 = 120.0. NPV = -382.5 + 120.0 x 3.7908 = $72.39k; IRR = 17.16%. The NPV is positive, so replace. The decision survives a fall in savings to the break-even level of $$$114.5k a year.

Problem 22.39 (Challenging)

A firm has a $500k capital budget and these indivisible projects ($000):

Project	Outlay	NPV	PI
A	300	120	0.400
B	250	95	0.380
C	200	80	0.400
D	150	50	0.333
E	100	45	0.450

Choose the combination that maximizes total NPV. Does ranking by profitability index give the best answer?

Solution. Enumerate all combinations within the budget. The three best: {A,C} cost 500, NPV 200; {B,D,E} cost 500, NPV 190; {C,D,E} cost 450, NPV 175. Best: {A,C} with NPV 200 using 500 of the budget. Ranking by PI gives the order E, A, C, B, D. Taking projects in that order while money remains selects {E,A} (outlay 400, NPV 165), which is 35 below the optimum. PI ranking works only when projects are divisible or the budget is used up exactly; with indivisible projects, enumerate or use integer programming.

Part 8: Uncertainty and Real Options (Volume 1, Chapter 8)

Review. Decision trees use expected values with later decisions optimised; use risk-neutral probabilities p = (1+r-d)/(u-d) for tree-based options; real options include expand, abandon and defer; the Black-Scholes call values the right to invest.

Typical errors. Applying real-world probabilities in a risk-neutral tree, and treating static NPV as the whole story when flexibility exists.

Problem 22.40 (Basic)

A drug-development project (fictional) costs $400m now. With probability 40% it succeeds and has a present value of $900m; otherwise its value is $200m. Compute the expected value and NPV. What is the expected value if a 1-year pilot costing $30m can reveal the outcome before the full investment, which would then be made only if success is revealed (ignore discounting)?

Solution. Expected PV = 0.40 x 900 + 0.60 x 200 = 480; NPV = 480 - 400 = $80m, so invest without a pilot. In the failure state the full investment loses 200 - 400 = -200. With the pilot, invest only on success: expected value = 0.40 x (900 - 400) - 30 = $170m. The pilot adds $90m: it avoids the failure-state loss (0.60 x 200 = $120m) at a cost of $30m. This is a decision-tree (abandonment) argument: information has value when it can change the decision.

Problem 22.41 (Challenging)

A project costs $105m and has a present value of $100m, so NPV = -$5m. In one year its value will be either $150m or $75m (risk-free rate 5%). At that time the firm can expand by 50% (adding 50% of project value) for $40m. Value the expansion option and the project with the option.

Solution. Risk-neutral probability of the up state = (1.05 - 0.75)/(1.5 - 0.75) = 0.400. Expansion payoff: up = 0.5 x 150 - 40 = 35.0; down = max(0.5 x 75 - 40, 0) = 0.0 (do not expand). Option value = 0.400 x 35.0/1.05 = $13.33m. Project with option = -5 + 13.33 = $8.33m: accept, because the flexibility to expand turns a negative static NPV positive. Static NPV ignores the option to commit more only in the good state.

Problem 22.42 (Intermediate)

A project is worth $100m if continued. In one year it will be worth either $130m or $70m. The firm can abandon for a salvage value of $90m at that time. The risk-free rate is 5%. Value the abandonment option.

Solution. Risk-neutral probability up = (1.05 - 0.70)/(1.30 - 0.70) = 0.5833. Abandonment payoff = max(90 - V, 0): up 0, down 20. Option value = (0.5833 x 0 + 0.4167 x 20)/1.05 = $7.94m. The option establishes a floor of $90m on the project's value in a year; it is a European put with strike 90 on a $100m underlying and is worth 7.9% of project value.

Problem 22.43 (Challenging)

A firm can invest $95m in a project worth $100m today, or wait one year. In a year the project's value will be $140m or $70m (risk-free rate 6%; the project's value does not leak away while waiting). Compare investing now with waiting and state the cost of investing now.

Solution. Invest now: NPV = 100 - 95 = $5m. Wait: risk-neutral probability up = (1.06 - 0.70)/(1.40 - 0.70) = 0.5143. In the up state invest and get 140 - 95 = 45; in the down state 70 < 95, so do not invest (0). Value of waiting = 0.5143 x 45/1.06 = $21.83m. Waiting is worth $16.83m more, so the NPV rule 'invest if NPV > 0' is wrong when the option to wait is valuable; invest now only if cash flows lost by waiting exceed this premium.

Problem 22.44 (Challenging)

A firm holds the right to launch a project in 3 years by spending $110m. The present value of the project's cash flows is $100m, with volatility 35% and a 5% risk-free rate (no cash flow is lost by waiting). Value the option to launch and comment on the present NPV.

Solution. The static NPV is 100 - 110 = -$10m. The option is a Black-Scholes call with S = 100, K = 110, r = 5%, sigma = 35%, T = 3: d1 = [ln(100/110) + (0.05 + 0.06125) x 3]/(0.35 x sqrt(3)) = 0.3933; d2 = d1 - 0.6062 = -0.2129. Call = $25.94m. The right is worth $25.9m although NPV is negative, because the firm launches only if value exceeds the cost. Value rises with volatility: at 50% volatility the option is worth $35.3m.

Part 9: Risk and Return (Volume 1, Chapter 9)

Review. Portfolio variance = w1^2 s1^2 + w2^2 s2^2 + 2 w1 w2 rho s1 s2; minimum variance weight for asset 1 = (s2^2 - cov)/(s1^2 + s2^2 - 2 cov); tangency portfolio maximises the Sharpe ratio.

Typical errors. Averaging volatilities, and using correlation where covariance is needed.

Problem 22.45 (Basic)

Compute the expected return, variance and standard deviation of a stock with these scenario returns.

Scenario	Probability	Return
Recession	25%	-12%
Normal	50%	8%
Boom	25%	26%

Solution. E(R) = 0.25 x (-12%) + 0.50 x 8% + 0.25 x 26% = 7.50%. Deviations: -19.50%, 0.50%, 18.50%. Variance = 0.25 x (-0.195)^2 + 0.50 x (0.005)^2 + 0.25 x (0.185)^2 = 0.01808; standard deviation = 13.44%. Coefficient of variation = 1.79.

Problem 22.46 (Basic)

Asset X has expected return 9% and volatility 20%; asset Y has expected return 13% and volatility 30%. Correlation is 0.30. Find the expected return and standard deviation of a 60/40 portfolio, and compare with the weighted average of volatilities.

Solution. E(Rp) = 0.6 x 9% + 0.4 x 13% = 10.60%. Variance = (0.6 x 0.20)^2 + (0.4 x 0.30)^2 + 2 x 0.6 x 0.4 x 0.30 x 0.20 x 0.30 = 0.0144 + 0.0144 + 0.00864 = 0.03744. Standard deviation = 19.35%, versus a weighted average of 24.00%: diversification removes 4.65 percentage points of volatility.

Problem 22.47 (Intermediate)

For assets X and Y of the previous problem (volatilities 20% and 30%, correlation 0.30), find the minimum-variance portfolio and its volatility. Why is the lower-volatility asset not 100% of the portfolio?

Solution. w_X = (s_Y^2 - rho s_X s_Y)/(s_X^2 + s_Y^2 - 2 rho s_X s_Y) = (0.09 - 0.018)/(0.04 + 0.09 - 0.036) = 0.7660. So 76.6% X and 23.4% Y. Variance = 0.03485, volatility 18.67%, below X's 20% alone. Because correlation is below s_X/s_Y = 0.667, adding some of the riskier asset reduces risk.

Problem 22.48 (Intermediate)

Each of n stocks has volatility 40% and every pair has correlation 0.25. For an equally weighted portfolio compute volatility for n = 1, 5, 25 and 100, and the limit. How much risk is diversifiable?

Solution. Variance = s^2/n + (1 - 1/n) x rho x s^2.

Stocks (n)	Portfolio volatility
1	40.00%
5	25.30%
25	21.17%
100	20.30%
infinity	20.00%

The floor is s x sqrt(rho) = 0.40 x 0.5 = 20.00%. Single-stock variance of 0.16 splits into undiversifiable 0.0400 (25%) and diversifiable 0.1200 (75%). The first 5 stocks capture most of the benefit; the next 95 add little.

Problem 22.49 (Challenging)

Risky asset A has expected return 10% and volatility 18%; B has 15% and 28%; correlation 0.25; risk-free rate 4%. Find the tangency (maximum Sharpe ratio) portfolio, its Sharpe ratio, and the fraction of wealth an investor with risk aversion A = 4 puts in it.

Solution. Excess returns: A 6.00%, B 11.00%; covariance = 0.25 x 0.18 x 0.28 = 0.01260. w_A = [eA s_B^2 - eB cov]/[eA s_B^2 + eB s_A^2 - (eA + eB) cov] = 0.5416; w_B = 0.4584. Portfolio return = 12.29%, volatility = 17.95%, Sharpe = (12.29% - 4%)/17.95% = 0.462 (A alone: 0.333; B alone: 0.393). Optimal risky fraction y = (E - rf)/(A x s^2) = 0.643: 64.3% in the tangency portfolio, 35.7% in the risk-free asset.

Part 10: The Capital Asset Pricing Model (Volume 1, Chapter 10)

Review. E(R) = rf + beta(E(Rm) - rf); beta = Cov/Var; alpha = actual - required return; Hamada: beta_E = beta_A[1 + (1-t)D/E].

Typical errors. Regressing prices instead of returns, and relevering with the wrong tax or debt assumptions.

Problem 22.50 (Basic)

A stock has a beta of 1.3. The risk-free rate is 4% and the market risk premium is 6%. Find the required return. What if beta is 0.7?

Solution. Required return = rf + beta x MRP = 4% + 1.3 x 6% = 11.8%. With beta 0.7: 4% + 0.7 x 6% = 8.2%. Each 0.1 of beta adds 0.6 percentage points to the required return.

Problem 22.51 (Basic)

A portfolio holds 50% in a stock with beta 1.4, 30% with beta 0.9 and 20% with beta 0.5. Find portfolio beta and its required return with rf = 4% and MRP = 6%. Also find the beta of a stock whose covariance with the market is 0.045 when the market variance is 0.030.

Solution. Portfolio beta = 0.5 x 1.4 + 0.3 x 0.9 + 0.2 x 0.5 = 1.07. Required return = 4% + 1.07 x 6% = 10.42%. Stock beta = Cov(Ri, Rm)/Var(Rm) = 0.045/0.030 = 1.50.

Problem 22.52 (Intermediate)

Analysts forecast a 14% return for a stock with beta 1.2. The risk-free rate is 4% and the market risk premium is 6%. Compute the required return and alpha and say whether the stock is over- or underpriced. If the stock pays no dividend and is priced at $50 expecting to be worth $57 in a year, what price makes it fairly priced?

Solution. Required = 4% + 1.2 x 6% = 11.20%; alpha = 14% - 11.20% = 2.80%, so it plots above the security market line and is underpriced. Expected return from $50 to $57 is 14%, consistent with the forecast. Fair price = 57/(1 + 0.112) = $51.26, above $50: buying pressure should raise the price by $1.26.

Problem 22.53 (Intermediate)

Estimate the stock's beta, alpha and R-squared from these eight periods of returns (illustrative data).

Period	Stock	Market
1	12%	8%
2	-5%	-4%
3	20%	12%
4	8%	5%
5	-10%	-8%
6	15%	10%
7	3%	2%
8	9%	6%

Solution. Cov(stock, market) = 0.00692; Var(market) = 0.00476. Beta = 0.00692/0.00476 = 1.456. Alpha = mean(stock) - beta x mean(market) = 0.0650 - 1.456 x 0.0387 = 0.0086 per period. R-squared = 0.993: 99.3% of the stock's variance is market (systematic) risk and 0.7% is diversifiable. With only eight observations the estimate has wide error bands.

Problem 22.54 (Challenging)

Calder Bearings (fictional) has an equity beta of 1.4 at a debt/equity ratio of 0.5; the tax rate is 25%. Assume debt beta of zero. Find the asset (unlevered) beta, and the equity beta and cost of equity if the company moves to D/E = 1.0 (rf = 4%, MRP = 6%).

Solution. Asset beta = 1.4/(1 + 0.75 x 0.5) = 1.0182. Relevered equity beta = 1.0182 x (1 + 0.75 x 1.0) = 1.7818. Cost of equity now = 4% + 1.4 x 6% = 12.4%; at D/E = 1.0 it is 4% + 1.782 x 6% = 14.69%. Leverage raises the equity beta from 1.40 to 1.78 while business risk (asset beta) is unchanged.

Part 11: Cost of Capital (Volume 1, Chapter 11)

Review. WACC = (E/V)re + (D/V)rd(1-t) + (P/V)rp; pure-play betas unlevered and relevered give project-specific rates; under MM with taxes, re = ru + (ru-rd)(1-t)D/E.

Typical errors. Using book weights, using the coupon rate as cost of debt, and applying a company-wide WACC to projects of different risk.

Problem 22.55 (Basic)

A firm has equity of $600m (market) and debt of $400m. The cost of equity is 11%, pre-tax cost of debt 6% and the tax rate 25%. Compute WACC.

Solution. Weights: E/V = 0.60, D/V = 0.40. After-tax cost of debt = 6% x 0.75 = 4.5%. WACC = 0.60 x 11% + 0.40 x 4.5% = 8.40%. The debt tax shield lowers WACC by 0.40 x 6% x 0.25 = 0.60 percentage points.

Problem 22.56 (Intermediate)

A firm's capital structure by market value is $500m equity, $300m debt and $100m preferred stock. The costs are 12% (equity), 6.5% pre-tax (debt) and 8% (preferred); tax rate 30%. Compute WACC, and the NPV of a project costing $200m with perpetual after-tax flows of $17m if the project has the firm's average risk.

Solution. V = 900; weights: 0.5556, 0.3333, 0.1111. After-tax cost of debt = 6.5% x 0.70 = 4.550%. WACC = 0.5556 x 12% + 0.3333 x 4.550% + 0.1111 x 8% = 9.072%. NPV = 17/0.09072 - 200 = -$12.61m. The project's IRR is 17/200 = 8.50%, against a hurdle of 9.07%: reject.

Problem 22.57 (Intermediate)

A division of Calder Bearings is to be valued with a discount rate based on three listed pure-play comparables (tax 25%).

Comparable	Equity beta	Debt/equity
C1	1.30	0.40
C2	1.10	0.25
C3	1.45	0.50

The division will be financed at D/E = 0.30 with pre-tax debt cost 5.5%; rf = 4%, MRP = 6%. Find the division's WACC.

Solution. Unlevered betas = beta_E/(1 + 0.75 x D/E): 1.0000, 0.9263, 1.0545; average 0.9936. Relever at D/E = 0.30: 0.9936 x (1 + 0.75 x 0.30) = 1.2172. Cost of equity = 4% + 1.2172 x 6% = 11.30%. Weights: E/V = 1/1.3 = 0.7692, D/V = 0.2308. WACC = 0.7692 x 11.30% + 0.2308 x 4.125% = 9.647%. Using the parent's WACC for a division whose business risk differs would misprice its projects.

Problem 22.58 (Challenging)

A firm generates a perpetual after-tax free cash flow of $100m when all-equity financed; the unlevered cost of equity is 10%, the cost of debt is 5% and the tax rate 25%. Using the Modigliani-Miller tax-adjusted formulas, compute the cost of equity, WACC and firm value at D/V = 0%, 30% and 50%, and verify with the tax-shield approach at 30%.

Solution. Cost of equity re = ru + (ru - rd)(1 - t)D/E; WACC = (E/V) re + (D/V) rd (1 - t).

D/V	D/E	Cost of equity	WACC	Firm value
0%	0.000	10.00%	10.00%	$1,000.0m
30%	0.429	11.61%	9.25%	$1,081.1m
50%	1.000	13.75%	8.75%	$1,142.9m

Check at 30%: value = V_U + t x D = 1,000.0 + 0.25 x D where D = 0.30 x V. Solving V = 1,000 + 0.075 V gives V = 1,081.1, matching the table. WACC falls with leverage because debt is tax-advantaged, yet the cost of equity rises; this ignores distress costs, which Volume 1, Chapter 15 balances against the tax shield.

Part 12: Market Efficiency and Behavioral Finance (Volume 1, Chapter 12)

Review. Abnormal return = actual - expected from the market model; CAR is a sum of abnormal returns; the test statistic is CAR over its standard error; autocorrelation tests weak-form efficiency.

Typical errors. Reading a significant abnormal return as proof of inefficiency without considering the joint-hypothesis problem.

Problem 22.59 (Intermediate)

The market model for Fenwick Foods (fictional) is R = 0.0002 + 1.2 x Rm. Around an announcement (days -2 to +2) returns were as shown. Find daily abnormal returns, the cumulative abnormal return (CAR), and the t-statistic if daily abnormal return standard deviation is 1.5%.

Day	Stock	Market	Abnormal
-2	0.40%	0.20%	0.14%
-1	1.20%	0.50%	0.58%
0	-0.60%	-0.40%	-0.14%
1	2.50%	0.30%	2.12%
2	1.80%	0.60%	1.06%

Solution. Abnormal return = R - (0.0002 + 1.2 Rm). Values above (day 0 = announcement). CAR = sum = 3.76%. Standard error of CAR = 1.5% x sqrt(5) = 3.354%; t = 3.76%/3.354% = 1.12. The critical value is 1.96, so the CAR is not significant at 5%. Any such test is a joint test of market efficiency and the benchmark model used to define abnormal returns.

Problem 22.60 (Challenging)

Test weak-form efficiency using the lag-1 autocorrelation of these 12 monthly returns (illustrative data). Report the autocorrelation, the approximate standard error 1/sqrt(n) and your conclusion, and explain why a trading rule based on it might still not be profitable.

Month	Return
1	2%
2	-1%
3	3%
4	1%
5	-2%
6	4%
7	-3%
8	2%
9	1%
10	-1%
11	3%
12	-2%

Solution. Mean return = 0.583%. Autocorrelation = sum of d_t x d_(t-1) / sum d_t^2 = -0.720. Standard error is about 1/sqrt(12) = 0.289, so t = -2.49; the 5% bounds are +/- 0.566. The autocorrelation is outside the bounds, so the statistic is nominally significant; with only twelve observations the test has little power and a single pattern of this kind could easily be chance. Even a significant statistic need not be exploitable: after transaction costs, price impact and risk adjustment, small predictability may not pay, which is the practical content of semi-strong and weak-form efficiency.

Closing Notes for Instructors

The problems are designed so that Basic items can be used as quiz questions, Intermediate items as homework, and Challenging items as discussion or exam questions. Several problems pair deliberately: Problems on mutually exclusive projects and unequal lives warn against IRR and present-value comparisons; the real-option problems show that a negative static NPV is not a final verdict; and the cost-of-capital problems show how leverage changes the cost of equity without changing business risk. Ask students to state, for each Challenging problem, one assumption that would reverse the answer.

A suggested marking scheme awards half of the marks for correct setup (formula, inputs, units), a third for correct arithmetic and the remainder for interpretation. Because the solutions were generated by code, instructors can adapt any problem by changing the inputs in its statement and recomputing with the stated formulas; the structure of the solution does not change.

Chapter 23: Comprehensive Problem Sets with Full Solutions II: Financing, Derivatives and Transactions

All companies, funds and figures in this chapter are fictional and illustrative. Tax rates and market rates are stylized so that the arithmetic is transparent.

How to Use This Problem Set

This chapter contains 60 problems that cover Volume 1, Chapters 13 to 24: raising equity, debt, capital structure, payout policy, working capital, options, derivatives, risk management and foreign exchange, enterprise valuation, mergers and acquisitions, leveraged buyouts and private equity, and sustainability, governance and fintech. It is the second of two problem chapters; the first covers the earlier valuation and investment material. There are five problems per chapter in a fixed pattern: two basic (single-concept, one or two formulas), two intermediate (multi-step, some judgment) and one challenging (integrates several ideas or requires a modeling choice). Basic problems are numbered 1, 2, 6, 7, 11, 12 and so on; the pattern repeats every five problems, so problems 5, 10, 15, …, 60 are the challenging ones.

Every solution is shown step by step and every number was recomputed in Python and rounded only for display. When you check your own work, expect small differences in the last digit if you round intermediate steps. Work each problem on paper first, then compare; the most valuable learning comes from the steps where your approach differs from the printed one. Instructors will find the same problems mapped to learning outcomes in Chapter 24.

Problems and Solutions

Part A: Raising Equity (Volume 1, Chapter 13)

Key tools. Rights offering: ex-rights price = (N × P + n × S) / (N + n), where N is existing shares, n is new shares and S is the subscription price. Underpricing = first-day close / offer price − 1. Net proceeds = gross proceeds × (1 − gross spread) − other fees.

Problem 1 (Basic): Rights offering arithmetic

Question. Bayfield Tools (illustrative) has 10 million shares trading at $40. It announces a rights offering of one new share for every five held, at a subscription price of $32. Compute the number of new shares, the gross proceeds, the theoretical ex-rights price and the value of one right.

Solution.

1. New shares = 10,000,000 / 5 = 2,000,000.
2. Gross proceeds = 2,000,000 × $32 = $64.0 million.
3. Market value before = 10,000,000 × $40 = $400.0 million; after = $400.0m + $64.0m = $464.0m over 12,000,000 shares.
4. Ex-rights price = $38.667.
5. Value of a right = cum-rights price − ex-rights price = $40.00 − $38.667 = $1.333. A shareholder needs five rights per new share, so the right is also worth ($38.667 − $32) / 5 = $1.333, confirming consistency.

Answer. 2,000,000 new shares; $64.0m proceeds; ex-rights price $38.667; each right worth $1.333.

Problem 2 (Basic): IPO underpricing and total cost

Question. Calder Robotics (illustrative) sells 8 million shares at an IPO price of $25. The gross spread is 7%. The stock closes the first day at $31.50. Compute the underpricing, the money left on the table, the net proceeds and the issuer's total cost of going public as a percentage of the aftermarket value of the shares sold (ignore other fees).

Solution.

1. Underpricing = 31.50 / 25.00 − 1 = 26.0%.
2. Money left on the table = (31.50 − 25.00) × 8,000,000 = $52.0 million.
3. Gross proceeds = 8,000,000 × $25 = $200.0 million; spread = $14.0 million; net proceeds = $186.0 million.
4. Direct cost = $14.0m; indirect cost = $52.0m; total = $66.0m.
5. Aftermarket value of shares sold = 8,000,000 × 31.50 = $252.0m, so total cost = 26.19%.

Answer. Underpricing 26.0%; $52.0m left on the table; net proceeds $186.0m; total cost 26.19% of aftermarket value.

Problem 3 (Intermediate): Seasoned equity offering and announcement effect

Question. Harlow Energy (illustrative) has 50 million shares at $20. It announces a 10 million-share seasoned offering at $18.50 with 4% total fees. The stock falls 3% on the announcement as investors infer overvaluation. Assuming the market is otherwise efficient and the cash is held as cash (no value created), compute net proceeds, the post-issue share price, and the wealth change per old share relative to $20.

Solution.

1. Net proceeds = 10,000,000 × $18.50 × 0.96 = $177.60 million.
2. Post-announcement price = $20 × 0.97 = $19.40, so pre-issue equity value = 50,000,000 × 19.40 = $970.0m.
3. Add retained cash: post-issue equity = $970.0m + $177.60m = $1,147.60m.
4. Shares after = 60,000,000, so price = $19.127.
5. Change per old share = 19.127 − 20.00 = -$0.873 (-4.37%). Of this, -$0.60 is the announcement (information) effect and the remainder reflects dilution from issuing below the post-news price plus fees.

Answer. Net proceeds $177.60m; post-issue price $19.127; old holders lose $0.873 per share (4.37%).

Problem 4 (Intermediate): Greenshoe economics

Question. An underwriter buys 20 million shares from an issuer at $15 less a 7% spread and sells 23 million shares to investors at $15, creating a 3 million-share short position via a 15% over-allotment option. Compute the underwriter's gross profit if (a) the stock trades at $17 and the option is exercised, and (b) the stock trades at $14 and the underwriter covers the short in the market. Why does the greenshoe support the aftermarket price?

Solution.

1. Spread per share = $15 × 7% = $1.05; over-allotment = 20,000,000 × 15% = 3,000,000 shares; total sold = 23,000,000.
2. a. The underwriter exercises the option and buys 3,000,000 more shares at $13.95. Profit = 23,000,000 × $1.05 = $24.15 million.
3. b. Buying 3,000,000 shares at $14 to close the short earns ($15 − $14) × 3,000,000 = $3.00m on top of the spread on the base 20,000,000 shares ($21.00m). Profit = $24.00m.

Answer. (a) $24.15m; (b) $24.00m. The short-covering purchases are the price support.

Problem 5 (Challenging): IPO with primary and secondary components

Question. Before its IPO, Tessel Logistics (illustrative) has 50 million shares: founders 30 million (60%) and a venture fund 20 million (40%). In the IPO the company sells 10 million new shares (primary) and the venture fund sells 5 million of its shares (secondary) at $22 with a 6.5% gross spread. Compute (a) post-IPO ownership percentages and market capitalization, (b) net cash to the company and to the fund, (c) the price per share needed for the primary tranche alone to net $250 million, and (d) the number of primary shares needed at $22 for $250 million net.

Solution.

1. Post-IPO shares = 50,000,000 + 10,000,000 = 60,000,000. Founders 30m = 50.0%; fund 20m − 5m = 15m = 25.0%; public = 10m + 5m = 15m = 25.0%.
2. Market capitalization = 60,000,000 × $22 = $1,320.0 million.
3. Net to company = 10,000,000 × 22 × 0.935 = $205.70m; net to fund = 5,000,000 × 22 × 0.935 = $102.85m.
4. c. Price needed = 250,000,000 / (10,000,000 × 0.935) = $26.74.
5. d. Shares needed at $22 = 250,000,000 / (22 × 0.935) = 12,153,622 shares, which would raise public float to 27.6% of a 62.15m-share base and dilute existing holders further.

Answer. Founders 50.0%, fund 25.0%, public 25.0%; market cap $1,320.0m; net proceeds $205.70m (company) and $102.85m (fund); price $26.74 or 12.154 million shares.

Part B: Debt Financing (Volume 1, Chapter 14)

Key tools. Bond price = coupon × annuity factor + face × discount factor. Level loan payment = L × r / (1 − (1+r)^−n). After-tax cost of debt = y × (1 − t).

Problem 6 (Basic): Bond pricing and after-tax cost

Question. Marlow Foods (illustrative) issues a 10-year bond with a 6% annual coupon and $1,000 face value. Market yield for comparable risk is 7%. Compute the issue price, and the after-tax cost of debt at a 25% tax rate (using the yield).

Solution.

1. Annuity factor = (1 − 1.07^−10) / 0.07 = 7.0236.
2. PV of coupons = 60 × 7.0236 = $421.41.
3. PV of face = 1,000 / 1.07^10 = $508.35.
4. Price = $929.76, a discount of $70.24 because the coupon is below the market yield.
5. After-tax cost of debt = 7% × (1 − 0.25) = 5.25%.

Answer. Price $929.76; after-tax cost 5.25%.

Problem 7 (Basic): Amortizing term loan

Question. A $5,000,000 term loan has five equal annual payments at 8%. Compute the payment, the interest and principal in year 1, and the balance after year 2.

Solution.

1. Payment = 5,000,000 × 0.08 / (1 − 1.08^−5) = $1,252,282.27.

2. Year 1 interest = 5,000,000 × 8% = $400,000.00; principal = $852,282.27; balance = $4,147,717.73.
3. Year 2 interest = $331,817.42; principal = $920,464.85; balance = $3,227,252.87.
4. Check: the remaining balance equals the PV of the last three payments = 3,227,252.87.

Answer. Payment $1,252,282.27; year 1 interest $400,000.00, principal $852,282.27; balance after year 2 $3,227,252.87.

Problem 8 (Intermediate): Callable bond: yield to call versus yield to maturity

Question. A 10-year, 8% annual-coupon $1,000 bond is callable after year 5 at $1,050. The market yield to maturity is 6%. Compute the price at the 6% yield to maturity, then the yield to call if the bond trades at that price. Which yield should an investor treat as the more realistic return, and why?

Solution.

1. Price = 80 × annuity(6%, 10) + 1,000 / 1.06^10 = $588.81 + $558.39 = $1,147.20.
2. Yield to call solves 1,147.20 = 80 × annuity(y, 5) + 1,050 / (1+y)^5. Bisection gives y = 5.454%.
3. Because coupon (8%) exceeds the yield, the bond trades at a premium and the issuer has an incentive to call at $1,050; the yield to call of 5.45% is below the yield to maturity of 6.00%.

Answer. Price $1,147.20; yield to call 5.454% versus 6.000% to maturity; use yield to call (yield to worst).

Problem 9 (Intermediate): Bond refunding decision

Question. Dunmore Utilities (illustrative) has $100 million of 9% bonds with 15 years remaining, callable at 104. It can issue $100 million of 15-year 6.5% bonds with $2 million issuance costs. Tax rate is 25% and both call premium and issuance costs are treated as immediately deductible (a simplification). Compute the after-tax refunding outlay, the after-tax annual interest saving, and the NPV of refunding using the after-tax new borrowing rate.

Solution.

1. Call premium = $100m × 4% = $4.0m; add issuance cost $2.0m; after-tax outlay = $6.0m × 0.75 = $4.500m.
2. Annual saving = $100m × (9% − 6.5%) × 0.75 = $1.875m.
3. Discount rate = 6.5% × 0.75 = 4.875%; annuity factor for 15 years = 10.4679.
4. PV of savings = $19.627m; NPV = $19.627m − $4.500m = $15.127m.

Answer. NPV of refunding = $15.13 million (outlay $4.50m, PV of savings $19.63m).

Problem 10 (Challenging): Convertible bond decomposition and break-even

Question. A 5-year convertible has a $1,000 face value, a 3% annual coupon and a conversion ratio of 25 shares. The stock trades at $32 and pays a $0.80 annual dividend. Straight debt of equal risk yields 7%. The convertible trades at par ($1,000). Compute (a) the bond floor, (b) the conversion value, (c) the implied value of the embedded option, (d) the conversion premium per share and in percent, and (e) the break-even period based on income pickup.

Solution.

1. a. Floor = 30 × annuity(7%, 5) + 1,000 / 1.07^5 = $123.01 + $712.99 = $835.99.
2. b. Conversion value = 25 × $32 = $800.00.
3. c. Implied option value = price − floor = 1,000 − 835.99 = $164.01 per bond.
4. d. Conversion price = 1,000 / 25 = $40.00; premium per share = $40.00 − $32.00 = $8.00, or 25.0%.

5. e. Income pickup = coupon $30 − dividends on 25 shares ($20) = $10.00 per year. Break-even = premium paid (1,000 − 800 = $200 per bond) / $10 = 20.0 years, much longer than the 5-year life, so the investor is paying for optionality, not income.

Answer. Floor $835.99; conversion value $800.00; option value $164.01; premium 25.0%; break-even 20.0 years.

Part C: Capital Structure (Volume 1, Chapter 15)

Key tools. MM with taxes: V_L = V_U + t × D for permanent debt. rE = rU + (rU − rD) × (D/E) × (1 − t). WACC = (E/V) × rE + (D/V) × rD × (1 − t).

Problem 11 (Basic): Modigliani-Miller with taxes

Question. Eastgate Components (illustrative) is unlevered with value $500 million. It issues $200 million of permanent debt and uses the proceeds to repurchase shares. The tax rate is 25%. Compute the levered firm value, the equity value after the recapitalization, and the gain per dollar of debt.

Solution.

1. PV of tax shield = t × D = 0.25 × 200 = $50.0m.
2. V_L = 500 + 50 = $550.0m.
3. Equity = V_L − D = 550.0 − 200 = $350.0m.

Answer. V_L = $550.0m; equity = $350.0m; gain = $50m.

Problem 12 (Basic): Cost of equity and WACC under MM

Question. An unlevered cost of capital is 10%, the cost of debt is 5%, D/E = 0.5 and the tax rate is 25%. Compute the cost of equity and WACC, and compare WACC with the unlevered cost.

Solution.

1. rE = 10% + (10% − 5%) × 0.5 × 0.75 = 11.875%.
2. Weights: E/V = 1/1.5 = 66.67%, D/V = 33.33%.
3. WACC = 66.67% × 11.875% + 33.33% × 5% × 0.75 = 9.167%.

Answer. rE = 11.875%; WACC = 9.167%.

Problem 13 (Intermediate): Trade-off theory and optimal debt

Question. Rowan Pharma Distribution (illustrative) has unlevered value $800 million and a 25% tax rate. Distress would cost 25% of unlevered value ($200 million). The probability of distress at debt levels of $0, $100m, $200m, $300m and $400m is 0%, 2%, 6%, 15% and 30%. Find the value-maximizing debt level.

Solution.

1. Firm value = V_U + 0.25 × D − P(distress) × $200m. Computed values:

Debt ($m)	PV tax shield	P(distress)	Expected distress cost	Firm value
0	0.0	0%	0.0	800.0
100	25.0	2%	4.0	821.0
200	50.0	6%	12.0	838.0
300	75.0	15%	30.0	845.0
400	100.0	30%	60.0	840.0

2. The maximum is at debt of $300m with value $845.0m.
3. Marginal analysis: from $300m to $400m the extra tax shield of $25m is outweighed by the extra expected distress cost of $30m, so the optimum lies below $400m.
4. Debt/value at the optimum = 35.5%.

Answer. Optimal debt $300m; firm value $845.0m.

Problem 14 (Intermediate): EBIT-EPS indifference analysis

Question. Brightwater Marine (illustrative) has 50 million shares at $20 and needs $200 million. Plan A issues equity at $20; Plan B issues 6% debt. Tax rate is 25%. Find the EBIT at which EPS is identical, and compare EPS at EBIT of $60m and $100m.

Solution.

1. Plan A issues 200m / 20 = 10 million shares (60m total). Plan B has interest of 200 × 6% = $12m with 50m shares.
2. Indifference: EBIT × 0.75 / 60 = (EBIT − 12) × 0.75 / 50, so 50 × EBIT = 60 × EBIT − 720 and EBIT = 720 / 10 = $72.0m.
3. At EBIT = 60: EPS(A) = $0.750; EPS(B) = $0.720. Equity is better below the break-even.
4. At EBIT = 100: EPS(A) = $1.250; EPS(B) = $1.320. Debt is better above it.

Answer. Break-even EBIT = $72.0m; EPS at 60: A 0.750 vs B 0.720; at 100: A 1.250 vs B 1.320.

Problem 15 (Challenging): Adjusted present value with financing side effects

Question. A project costs $100 million and produces a perpetual unlevered after-tax cash flow of $12 million; the unlevered cost of capital is 10%. The firm will finance it with $60 million of permanent 6% debt (tax rate 30%, issuance costs 2% of the debt) plus $20 million of the debt's proceeds being replaced by a government-subsidized perpetual loan at 3% (the other $40 million is market-rate). Compute the APV. Treat the subsidized $20 million as part of the $60 million total and value the subsidy separately as the after-tax interest saving discounted at 6%.

Solution.

1. Base-case NPV = 12 / 0.10 − 100 = $20.0m.
2. Tax shield on permanent debt = 0.30 × 60 = $18.0m.
3. Issuance costs = 2% × 60 = $1.2m (negative).
4. Subsidy value = 20 × (6% − 3%) × (1 − 0.30) / 0.06 = $7.0m.
5. APV = 20.0 + 18.0 − 1.2 + 7.0 = $43.8m.

Answer. APV = $43.8 million.

Part D: Payout Policy (Volume 1, Chapter 16)

Key tools. Dividend per share = EPS × payout ratio. Sustainable growth g = ROE × retention. Gordon value P = D1 / (rE − g).

Problem 16 (Basic): Dividend mechanics

Question. Cobalt Retail (illustrative) has 100 million shares, EPS of $3.00 and a 40% payout ratio. The share price is $48. Compute dividend per share, total dividends, retained earnings, dividend yield and the theoretical ex-dividend price drop.

Solution.

1. DPS = 3.00 × 0.40 = $1.20.
2. Total dividends = 100,000,000 × 1.20 = $120.0m; retained earnings = (3.00 − 1.20) × 100m = $180.0m.
3. Dividend yield = 1.20 / 48 = 2.50%.
4. Ex-dividend price falls by about the dividend (no taxes): 48.00 − 1.20 = $46.80.

Answer. DPS $1.20; total $120.0m; retained $180.0m; yield 2.50%; ex-div price $46.80.

Problem 17 (Basic): Open-market repurchase and EPS

Question. A firm earns $400 million with 100 million shares trading at $50. It uses $200 million of excess cash to repurchase shares at $50. Assume lost interest on the cash is ignored. Compute the new share count, EPS before and after, and the post-buyback price if the P/E multiple is unchanged. Is any value created?

Solution.

1. Shares repurchased = 200,000,000 / 50 = 4,000,000; remaining = 96,000,000.
2. EPS before = 400/100 = $4.00; after = 400 / 96 = $4.1667 (4.17% accretion).
3. P/E before = 50 / 4.00 = 12.5x; with constant P/E the price would be 12.5 × 4.1667 = $52.08.

Answer. Shares 96.0m; EPS $4.00 to $4.1667; accretion is cosmetic since value per share remains $50.

Problem 18 (Intermediate): Debt-financed buyback

Question. Using the same $400 million of earnings and 100 million shares, now at a price of $60, the firm borrows $300 million at 6% and repurchases shares at $60 (tax rate 25%). Compute the new EPS, the earnings yield, the after-tax cost of debt and the break-even interest rate at which EPS is unchanged.

Solution.

1. Shares repurchased = 300,000,000 / 60 = 5,000,000; remaining 95,000,000.
2. After-tax interest = 300m × 6% × 0.75 = $13.50m; net income = $386.50m.
3. EPS = 386.50 / 95 = $4.0684 versus $4.00 before (1.71%).
4. Earnings yield = 4.00 / 60 = 6.667% exceeds after-tax debt cost of 4.500%, so EPS rises.
5. Break-even: r × 0.75 = 6.667%, so r = 8.889%. EPS accretion does not imply value creation, because the added financial risk raises the cost of equity and the P/E should fall.

Answer. EPS $4.0684 (+1.71%); break-even interest rate 8.889%.

Problem 19 (Intermediate): Dividend policy and growth valuation

Question. Next year's EPS will be $5.00, ROE on retained earnings is 15% and the cost of equity is 12%. Using the Gordon model, compute the value per share under payout ratios of 40% and 70%, and explain the result. Then compute the value at a payout of zero and state the limit of the model.

Solution.

1. At 40% payout: g = 15% × 0.60 = 9.0%; D1 = $2.00; P = 2.00 / (12% − 9.0%) = $66.67.
2. At 70% payout: g = 15% × 0.30 = 4.5%; D1 = $3.50; P = 3.50 / (12% − 4.5%) = $46.67.
3. At 0% payout: g = 15.0%, D1 = 0 and the Gordon formula yields zero, a failure of the model (growth of 15% exceeds the discount rate so the perpetuity does not converge). The perpetuity-growth model requires g < rE.
4. Value falls as the payout rises because ROE (15%) exceeds the cost of equity (12%): retained earnings earn more than shareholders could earn elsewhere, so retention creates value. If ROE equalled 12%, payout policy would be irrelevant.
5. Check at ROE = rE: P = 5.00 / 0.12 = $41.67 for any payout.

Answer. 40% payout: $66.67; 70% payout: $46.67; zero payout breaks the model since g > rE.

Problem 20 (Challenging): Dividend versus repurchase for a taxable investor

Question. A taxable investor holds 1,000 shares at $50 with a cost basis of $20. The firm distributes $2.00 per share either as a dividend (taxed at 30%) or through a pro rata repurchase at $50 in which the investor tenders a proportionate number of shares (capital gains taxed at 15% on the gain over basis). Compare after-tax wealth immediately after each choice, ignoring tax on unrealized gains.

Solution.

1. Dividend: cash = 1,000 × $2.00 = $2,000; tax = 30% = $600; after-tax cash = $1,400. Shares then trade at $48: 1,000 × 48 = $48,000. Wealth = $49,400.
2. Repurchase: investor tenders $2,000 / $50 = 40 shares and receives $2,000. Gain = 40 × (50 − 20) = $1,200; tax = 15% = $180; after-tax cash = $1,820.
3. Remaining 960 shares at $50 (no ex-dividend drop) = $48,000. Wealth = $49,820.
4. Difference = $420 in favor of the repurchase, equal to the tax saving: dividend tax $600 versus capital-gains tax $180.

Answer. Wealth after dividend $49,400; after repurchase $49,820; repurchase wins by $420.

Part E: Working Capital (Volume 1, Chapter 17)

Key tools. DIO = inventory / COGS × 365; DSO = receivables / sales × 365; DPO = payables / COGS × 365; CCC = DIO + DSO − DPO. Cost of forgoing a discount d/(1−d) × 365/(N − discount days). EOQ = √(2DS/H).

Problem 21 (Basic): Cash conversion cycle

Question. Pinecrest Hardware (illustrative) reports sales $730m, COGS $511m, inventory $70m, receivables $100m and payables $56m. Compute DIO, DSO, DPO and the cash conversion cycle, then the cash released if the CCC is cut by 10 days.

Solution.

1. DIO = 70 / 511 × 365 = 50.0 days.
2. DSO = 100 / 730 × 365 = 50.0 days.

3. DPO = 56 / 511 × 365 = 40.0 days.
4. CCC = 50.0 + 50.0 − 40.0 = 60.0 days.
5. Cutting 10 days of sales-based working capital releases about 10 × 730 / 365 = $20.0m (using sales as the base; on a COGS base, receivables and inventory days release different amounts).

Answer. DIO 50.0, DSO 50.0, DPO 40.0, CCC 60.0 days; cash released about $20.0m.

Problem 22 (Basic): Cost of trade credit

Question. Supplier terms are 2/10 net 30. Compute the annualized simple cost and the effective annual cost of forgoing the discount, and state whether a firm able to borrow at 12% should take the discount.

Solution.

1. Periodic cost = 0.02 / 0.98 = 2.0408% for 20 additional days of credit.
2. Simple annualized cost = 2.0408% × 365 / 20 = 37.24%.
3. Effective annual cost = (1.020408)^(365/20) − 1 = 44.59%.

Answer. Simple 37.24%; effective 44.59%; take the discount.

Problem 23 (Intermediate): Economic order quantity

Question. A distributor uses 24,000 units per year, ordering costs $150 per order and holding costs $6 per unit per year. Compute the EOQ, orders per year, total annual cost, and the cost penalty if it instead orders 2,000 units at a time.

Solution.

1. EOQ = √(2 × 24,000 × 150 / 6) = √1,200,000 = 1,095.4 units.
2. Orders per year = 24,000 / 1,095.4 = 21.91.
3. Total cost at EOQ = ordering 3,286.34 + holding 3,286.34 = $6,572.67.
4. At Q = 2,000: ordering = 12 × 150 = $1,800; holding = 1,000 × 6 = $6,000; total = $7,800. Penalty = $1,227.33 (18.7%).
5. At the EOQ, ordering and holding costs are equal, a useful check.

Answer. EOQ 1,095.4 units; cost $6,572.67; ordering 2,000 costs $1,227.33 more.

Problem 24 (Intermediate): Credit policy change

Question. Sales are $50m with variable costs of 70% of sales, DSO of 40 days and bad debts of 1% of sales. A looser credit policy would raise sales to $54m, DSO to 55 days and bad debts to 2% of sales. The cost of funds is 10%, and receivables are financed at variable cost. Evaluate the change (ignore taxes, and use a 365-day year).

Solution.

1. Incremental contribution = (54 − 50) × 30% = $1.20m.
2. Receivables: old = 50 × 40 / 365 = $5.479m; new = 54 × 55 / 365 = $8.137m. Incremental investment at variable cost = (8.137 − 5.479) × 0.70 = $1.860m.
3. Carrying cost = 10% × 1.860 = $0.186m.
4. Incremental bad debts = 54 × 2% − 50 × 1% = 1.08 − 0.50 = $0.58m.
5. Net benefit = 1.20 − 0.186 − 0.58 = $0.434m per year. Positive, so adopt, but the margin is thin: if bad debts reach 2.5%, the gain turns to $0.164m.

Answer. Net benefit $0.434m per year; adopt, with sensitivity to the bad-debt rate.

Problem 25 (Challenging): Miller-Orr cash management

Question. A firm has fixed transaction cost of $50 per transfer between securities and cash, daily cash-flow standard deviation of $1,200, an annual interest rate of 4% (daily rate 4%/365) and a minimum cash balance of $10,000. Compute the Miller-Orr spread, return point and upper limit, then the approximate average cash balance using (4 × return point − lower limit) / 3.

Solution.

1. Daily rate i = 0.04 / 365 = 0.00010959; variance $\sigma^2 = 1{,}200^2 = 1{,}440{,}000$.
2. Spread = 3 × (0.75 × 50 × 1,440,000 / 0.00010959)^(1/3) = $23,695.
3. Return point = 10,000 + spread/3 = $17,898; upper limit = 10,000 + spread = $33,695.
4. Average balance ≈ (4 × 17,898 − 10,000) / 3 = $20,531.

Answer. Spread $23,695; return point $17,898; upper limit $33,695; average cash about $20,531.

Part F: Options (Volume 1, Chapter 18)

Key tools. Put-call parity (continuous compounding): C − P = S − K × e^(−rT). One-period binomial: q = (1 + r − d) / (u − d), value = [q × Cu + (1 − q) × Cd] / (1 + r). Black-Scholes: C = S N(d1) − K e^(−rT) N(d2), d1 = [ln(S/K) + (r + σ²/2)T] / (σ√T), d2 = d1 − σ√T.

Problem 26 (Basic): Put-call parity

Question. A European call on a non-dividend-paying stock has S = $50, K = $50, T = 1 year, r = 5% (continuous) and price $6.00. Find the put price and the cost of a protective-put position.

Solution.

1. PV of strike = 50 × e^(−0.05) = $47.561.
2. P = C − S + PV(K) = 6.00 − 50.00 + 47.561 = $3.561.
3. A protective put (stock plus put) costs $53.561 and guarantees at least $50 at expiry.
4. Equivalent identity: stock + put = call + PV(K) = 6.00 + 47.561 = 53.561, confirming the result.

Answer. Put = $3.561.

Problem 27 (Basic): One-period binomial option

Question. A stock at $100 will rise to $120 or fall to $90 in one period. The risk-free rate is 4% per period. Price a call with strike $105 using risk-neutral probabilities and verify with a replicating portfolio.

Solution.

1. q = (1.04 − 0.90) / (1.20 − 0.90) = 0.4667.
2. Payoffs: Cu = 120 − 105 = 15.0; Cd = 0.
3. C = 0.4667 × 15 / 1.04 = $6.7308.
4. Replication: delta = 15 / 30 = 0.50 shares; borrowing = 0.5 × 90 / 1.04 = $43.269; cost = 50 − 43.269 = $6.7308, matching the risk-neutral price.

Answer. Call = $6.7308; delta = 0.50.

Problem 28 (Intermediate): Black-Scholes valuation

Question. S = \$80, K = \$75, r = 4% (continuous), σ = 30%, T = 0.5, no dividends. Compute d1, d2, call price, put price, call delta and verify put-call parity.

Solution.

1. d1 = [ln(80/75) + (0.04 + 0.045) × 0.5] / (0.30 × √0.5) = 0.5046.
2. d2 = 0.5046 − 0.2121 = 0.2925.
3. N(d1) = 0.6931; N(d2) = 0.6150.
4. Call = 80 × 0.6931 − 75 × e^(−0.02) × 0.6150 = \$10.232.
5. Put = 75 × e^(−0.02) × N(−d2) − 80 × N(−d1) = \$3.747.
6. Parity: C − P = 6.485; S − PV(K) = 6.485. Delta = N(d1) = 0.6931.

Answer. Call \$10.232; put \$3.747; delta 0.6931.

Problem 29 (Intermediate): American versus European put (two-period tree)

Question. S = \$50, u = 1.2, d = 1/1.2, risk-free rate 5% per period, two periods, strike \$52. Price the European and American puts and identify where early exercise is optimal.

Solution.

1. q = (1.05 − 0.8333) / (1.2 − 0.8333) = 0.5909. Terminal stock prices: uu = 72.00, ud = 50.00, dd = 34.72. Put payoffs: 0.00, 2.00, 17.28.
2. European: Pu = 0.7792, Pd = 7.8571; P0 = 3.4997.
3. American at node d (S = 41.67): continuation = 7.8571 versus immediate exercise = 10.33; exercise value is larger, so exercise early. Value = 10.3333.
4. At node u (S = 60.00): exercise = 0.00, continuation 0.7792, so hold.
5. American P0 = max(52 − 50, (q × 0.7792 + (1−q) × 10.3333) / 1.05) = \$4.4645.
6. Early exercise premium = 4.4645 − 3.4997 = \$0.9647.

Answer. European put \$3.4997; American put \$4.4645; early exercise optimal in the down node.

Problem 30 (Challenging): Warrant valuation with dilution

Question. A company has 10 million shares at \$30 and issues 2 million two-year warrants with strike \$35. Volatility is 35%, r = 4% (continuous) and there are no dividends. Value an ordinary call (no dilution), then the warrant using the dilution-adjusted approach W = N/(N+M) × C(S + M·W/N), solved iteratively.

Solution.

1. Ordinary Black-Scholes call at S = 30: \$5.0020.
2. Dilution adjustment: warrant holders' claim on the equity means the effective underlying value per share is (N × S + M × W) / N, and the payout is shared across N + M shares, so W = [N / (N + M)] × C((N S + M W) / N).
3. Iteration (start W = 5.0020) converges to W = \$4.5899.
4. Dilution discount = 8.24% relative to the plain call. Total warrant value = 2,000,000 × 4.5899 = \$9.180m.

Answer. Plain call \$5.0020; dilution-adjusted warrant \$4.5899; total \$9.180m.

Part G: Derivatives (Volume 1, Chapter 19)

Key tools. Forward price with continuous yield q: F = S × e^((r − q)T). Swap fixed rate = (1 − DF_n) / Σ DF_t. Minimum-variance hedge ratio h* = ρ × σS / σF.

Problem 31 (Basic): Forward price on a dividend-paying index

Question. An index is at 100, the continuously compounded risk-free rate is 5%, the dividend yield is 2% and the forward matures in six months. Compute the fair forward price. If the forward trades at 102.50, describe the arbitrage and its profit per unit at maturity.

Solution.

1. F = 100 × e^((0.05 − 0.02) × 0.5) = $101.511.
2. The quoted 102.50 exceeds fair value by 0.989, so the forward is overpriced.
3. Cash-and-carry: sell the forward at 102.50, buy e^(−0.02 × 0.5) units of the index (dividends reinvested grow this to exactly one unit at expiry) for $99.005 borrowed at 5%. At maturity repay $101.511, which equals the fair forward price, and deliver one unit against the 102.50 forward price.
4. Net profit per unit = 102.50 − 101.511 = $0.989, risk-free.

Answer. Fair forward 101.511; arbitrage profit 0.989 per unit.

Problem 32 (Basic): Interest-rate swap as a hedge

Question. A borrower pays floating SOFR + 1.50% on $80 million. It enters a swap to receive SOFR and pay a fixed 4.20%. SOFR over four annual periods is 3.50%, 4.00%, 4.80% and 5.50%. Compute the effective borrowing rate each year and the dollar interest in year 4 with and without the swap.

Solution.

1. Effective rate = (SOFR + 1.50%) − (SOFR − 4.20%) = 5.70% every year, regardless of SOFR. The table shows each component:

Year	SOFR	Unhedged rate	Swap net receipt	Hedged rate
1	3.50%	5.00%	-0.70%	5.70%
2	4.00%	5.50%	-0.20%	5.70%
3	4.80%	6.30%	0.60%	5.70%
4	5.50%	7.00%	1.30%	5.70%

2. Year 4 without swap: 80m × 7.00% = $5.60m. With swap: 80m × 5.70% = $4.56m.
3. Year 1: swap payment is negative (pays 4.20% vs receives 3.50%), so interest with swap is 80m × 5.70% = $4.56m versus unhedged $4.00m; the swap costs the borrower $0.56m in year 1 but saves $1.04m in year 4.

Answer. Effective rate is 5.70% in all years; year 4 interest $4.56m hedged vs $5.60m unhedged.

Problem 33 (Intermediate): Swap rate and value of an off-market swap

Question. Zero-coupon rates (annual compounding) for years 1 to 4 are 4.0%, 4.4%, 4.7% and 4.9%. Compute the discount factors, the par four-year swap rate and the value to the fixed-rate payer of an existing four-year swap on $100 million with a fixed rate of 4.00%.

Solution.

1. Discount factors: DF1 = 0.96154; DF2 = 0.91749; DF3 = 0.87128; DF4 = 0.82584;
2. Annuity (sum of DFs) = 3.57615.
3. Par swap rate = (1 − 0.82584) / 3.57615 = 4.8699%.
4. The existing swap pays 4.00%, below the market par rate, so the fixed payer holds an asset. Value = (par rate − 4.00%) × annuity × notional = (4.8699% − 4.0000%) × 3.57615 × \$100m = \$3.111m.
5. Equivalent view: value = PV of the fixed-rate advantage each year, discounted at the zero curve.

Answer. Par swap rate 4.8699%; swap value to fixed payer \$3.111m.

Problem 34 (Intermediate): Minimum-variance and beta hedges

Question. (a) A manager has a \$40 million exposure with spot-futures correlation 0.90, σS = 25% and σF = 22%. Find the minimum-variance hedge ratio and the variance reduction achieved. (b) An equity manager holds a \$60 million portfolio with beta 1.2 and wants beta 0.4 using index futures priced at 4,000 with a \$250 multiplier. How many contracts?

Solution.

1. a. h* = ρ σS / σF = 0.90 × 0.25 / 0.22 = 1.0227; futures notional to short = 1.0227 × \$40m = \$40.91m.
2. Hedge effectiveness = ρ^2 = 0.81, so variance is cut by 81%.
3. b. Contract value = 4,000 × 250 = \$1,000,000.
4. N = (0.4 − 1.2) × 60,000,000 / 1,000,000 = -48, i.e., sell 48 contracts.

Answer. (a) h* = 1.0227, variance reduced by 81%; (b) sell 48 contracts.

Problem 35 (Challenging): No-arbitrage band with frictions

Question. A commodity trades at \$100. One-year futures are quoted at 106. Borrowing costs 5%, lending earns 3.5%, storage costs \$2 payable at year-end, and round-trip transaction costs are \$0.50. Compute the cash-and-carry upper bound and the reverse cash-and-carry lower bound, and the arbitrage profit if futures are quoted at 106, 108 and 104.

Solution.

1. Upper bound (buy spot, store, short future) = 100 × 1.05 + 2 + 0.50 = \$107.50.
2. Lower bound (an owner sells spot, lends at 3.5%, saves storage, goes long future) = 100 × 1.035 + 2 − 0.50 = \$105.00.
3. At F = 106 the price lies inside the band [105.00, 107.50]: profit = \$0.00.
4. At F = 108: sell future, buy spot with borrowed funds; profit = 108 − 107.50 = \$0.50.
5. At F = 104: reverse cash-and-carry; profit = 105.00 − 104 = \$1.00 (only available to those who own or can short the commodity).

Answer. Band [105.00, 107.50]; profits 0.00, 0.50, 1.00.

Part H: Risk Management and Foreign Exchange (Volume 1, Chapter 20)

Key tools. Covered interest parity (direct quote, domestic per foreign): F = S × (1 + r_d) / (1 + r_f). Money-market hedge replicates the forward. Immunization matches the duration of assets and liabilities and the present values.

Problem 36 (Basic): Covered interest parity

Question. Spot is \$1.10 per euro, one-year USD interest is 5% and one-year euro interest is 3%. Compute the one-year forward rate and the forward points. Which currency trades at a forward premium?

Solution.

1. F = 1.10 × 1.05 / 1.03 = $1.1214 per euro.
2. Forward points = (1.1214 − 1.1000) × 10,000 = 213.6 pips.
3. The euro trades at a forward premium because its interest rate is lower than the dollar rate; interest differentials are offset by the forward discount or premium.

Answer. Forward $1.1214; +213.6 points; euro at a premium.

Problem 37 (Basic): Transaction exposure hedging comparison

Question. A US firm will receive €5 million in six months. The six-month forward is $1.12 per euro. A euro put (right to sell euros) with strike $1.11 costs $0.02 per euro (ignore interest on the premium). Compute dollar proceeds for spot outcomes of 1.05, 1.12 and 1.20.

Solution.

1. Proceeds: unhedged = 5m × spot; forward = 5m × 1.12; option = 5m × (max(spot, 1.11) − 0.02).

Spot at maturity	Unhedged ($m)	Forward ($m)	Put option net ($m)
1.05	5.250	5.600	5.450
1.12	5.600	5.600	5.500
1.20	6.000	5.600	5.900

2. The forward fixes $5.600m; the put sets a floor of 5m × (1.11 − 0.02) = $5.450m while retaining upside, at an upfront cost of $0.100m.
3. Break-even spot for the option versus the forward = 1.12 + 0.02 = $1.14; above this the option wins; below it the forward wins (down to 1.11).

Answer. Forward $5.600m in all cases; option floors at $5.450m and exceeds the forward only if spot is above $1.14.

Problem 38 (Intermediate): Money-market hedge of a foreign-currency payable

Question. A US importer owes £2 million in one year. Spot is $1.25 per pound, USD borrowing costs 5.5% and GBP deposits earn 4.0%. Design a money-market hedge, compute its cost in dollars at maturity, compare it with a bank forward quote of $1.2750, and state the CIP forward rate.

Solution.

1. Pounds to deposit today = 2,000,000 / 1.04 = £1,923,077.
2. Dollars needed today = 1,923,077 × 1.25 = $2,403,846, borrowed at 5.5%.
3. Dollar repayment at maturity = 2,403,846 × 1.055 = $2,536,058, an effective rate of $1.2680 per pound.
4. CIP forward = 1.25 × 1.055 / 1.04 = $1.2680, equal to the money-market rate.

Answer. Hedged cost $2,536,058 (1.2680 per £); forward at 1.2750 is $13,942 worse.

Problem 39 (Intermediate): Duration immunization

Question. A pension fund owes $10 million in exactly five years. Available assets are a 3-year zero-coupon bond and an 8-year zero-coupon bond, both yielding 6% (annual compounding). Build a portfolio with present value equal to the liability and Macaulay duration of 5 years, and test a parallel yield shift of +/−1%.

Solution.

1. PV of liability = 10,000,000 / 1.06^5 = $7,472,582.
2. Duration condition: 3 w + 8 (1 − w) = 5 gives w(3-year) = 0.60 and w(8-year) = 0.40.
3. Invest $4,483,549 in the 3-year zero and $2,989,033 in the 8-year zero.
4. Revalue after an instantaneous parallel yield shift:

Yield change	Asset value	Liability value	Surplus
-1%	7,837,380	7,835,262	2,119
0%	7,472,582	7,472,582	0
1%	7,131,742	7,129,862	1,880

Answer. Invest 60% in the 3-year and 40% in the 8-year zero; surplus stays non-negative for plus or minus 1%.

Problem 40 (Challenging): Parametric VaR and risk decomposition

Question. A $100 million portfolio holds 60% in Asset X (annual volatility 20%) and 40% in Asset Y (10%); correlation is 0.30. Compute annualized portfolio volatility, the 10-day 99% VaR (z = 2.3263, 250 trading days, zero mean), the undiversified VaR, and each asset's percentage contribution to portfolio risk.

Solution.

1. Variance = $(0.6 \times 0.20)^2 + (0.4 \times 0.10)^2 + 2 \times 0.3 \times 0.6 \times 0.4 \times 0.20 \times 0.10 = 0.018880$; volatility = 13.740%.
2. 10-day volatility = 13.740% × √(10/250) = 2.748%.
3. VaR(99%, 10-day) = 2.3263 × 2.748% × $100m = $6.39m.
4. Undiversified VaR (sum of stand-alone) = 2.3263 × √0.04 × (0.12 + 0.04) × $100m = $7.44m; diversification benefit = $1.05m.
5. Risk contribution of X = w × cov(X, P) / σP^2 = 83.9%; of Y = 16.1% (sum = 100.0%). X holds 60% of capital but 83.9% of risk.

Answer. Volatility 13.74%; 10-day 99% VaR $6.39m; X contributes 83.9% of risk.

Part I: Enterprise Valuation (Volume 1, Chapter 21)

Key tools. FCFF = EBIT × (1 − t) + D&A − capex − ΔNWC. Terminal value (Gordon) = FCFF_(n+1) / (WACC − g). Equity value = EV − net debt.

Problem 41 (Basic): Free cash flow to the firm

Question. A company has EBIT of $150 million, a tax rate of 25%, depreciation of $40m, capital expenditure of $60m and an increase in net working capital of $10m. Compute NOPAT and FCFF.

Solution.

1. NOPAT = 150 × 0.75 = $112.5m.
2. FCFF = 112.5 + 40 − 60 − 10 = $82.5m.
3. Reinvestment = capex − D&A + ΔNWC = 30 million, a reinvestment rate of 26.7%.

Answer. NOPAT $112.5m; FCFF $82.5m.

Problem 42 (Basic): Perpetuity-growth enterprise value

Question. Treat the FCFF of $82.5 million from Problem 41 as next year's cash flow, with WACC 9% and perpetual growth 3%. Net debt is $400m and there are 50 million shares. Compute EV, equity value, per-share value and the EV/FCFF multiple.

Solution.

1. EV = 82.5 / (0.09 − 0.03) = $1,375.0m.
2. Equity = 1,375.0 − 400 = $975.0m.
3. Per share = 975.0 / 50 = $19.50.
4. EV/FCFF = 1 / (WACC − g) = 16.67x. Raising g by 0.5 point to 3.5% would lift EV to $1,500.0m: high sensitivity.

Answer. EV $1,375.0m; equity $975.0m; $19.50 per share.

Problem 43 (Intermediate): Five-year DCF with terminal value

Question. Forecast FCFF is 100, 110, 121, 130 and 138 ($m) for years 1 to 5. WACC is 9% and terminal growth 3%. Compute the explicit-period PV, terminal value, PV of terminal value, EV and the share of EV from the terminal value.

Solution.

1. PV of explicit cash flows: 91.74 + 92.58 + 93.43 + 92.10 + 89.69 = $459.55m.
2. Terminal value at year 5 = 138 × 1.03 / 0.06 = $2,369.00m.
3. PV of terminal value = 2,369.00 / 1.09^5 = $1,539.69m.
4. EV = 459.55 + 1,539.69 = $1,999.24m.
5. Terminal value share = 77.0%. When this exceeds about three-quarters of value, the valuation rests on terminal assumptions; check implied multiple TV / FCFF5 = 17.17x.

Answer. EV $1,999.24m; terminal value is 77.0% of EV.

Problem 44 (Intermediate): Valuation by comparable multiples

Question. Five comparable companies trade at EV/EBITDA of 9.0x, 10.5x, 8.5x, 11.0x and 9.5x. The target has EBITDA of $200m, net debt of $450m and 40 million shares. Compute the mean and median multiples, the implied EV and per-share value range using the median, and the full range.

Solution.

1. Mean = 9.70x; median = 9.5x; low 8.5x; high 11.0x.
2. EV at median = 200 × 9.5 = $1,900m; equity = 1,450m; per share = $36.25.
3. EV range = $1,700m to $2,200m; per share $31.25 to $43.75.
4. Leverage amplifies the range in the equity value: a 29% spread in EV (8.5x to 11.0x) becomes a wider spread in equity because net debt is fixed.

Answer. Median value $36.25 per share; range $31.25 to $43.75.

Problem 45 (Challenging): Value-driver formula and growth

Question. Next year's NOPAT is $100 million and WACC is 9%. Compute enterprise value using EV = NOPAT × (1 − g/ROIC) / (WACC − g) for (a) g = 4%, ROIC = 15%; (b) g = 4%, ROIC = 9%; (c) g = 0; (d) g = 4%, ROIC = 20%. Interpret.

Solution.

1. a. Reinvestment rate = 4% / 15% = 26.67%; FCF = 73.33; EV = 73.33 / 0.05 = $1,466.67m.
2. b. ROIC = WACC: reinvestment rate = 44.44%; EV = 55.56 / 0.05 = $1,111.11m.
3. c. No growth: EV = 100 / 0.09 = $1,111.11m.
4. d. ROIC 20%: reinvestment 20%; EV = $1,600.00m.
5. Results: (b) equals (c): when ROIC = WACC growth adds no value. Growth value in (a) = 355.6m; in (d) = 488.9m. Growth creates value only when ROIC exceeds WACC.

Answer. (a) $1,466.7m; (b) $1,111.1m; (c) $1,111.1m; (d) $1,600.0m; only ROIC above WACC makes growth valuable.

Part J: Mergers and Acquisitions (Volume 1, Chapter 22)

Key tools. NPV to acquirer = synergies − premium (cash deal). Premium = offer price − standalone price. In stock deals, post-merger value is shared pro rata, so target holders share in synergies and in the risk of overpayment.

Problem 46 (Basic): Value created in a cash acquisition

Question. An acquirer pays $360 million cash for a target with standalone value $300 million. The PV of synergies is $100 million. Compute the premium, the NPV to the acquirer's shareholders, the gain to target shareholders and the share of synergies captured by each side.

Solution.

1. Premium = 360 − 300 = $60m (20%).
2. Acquirer NPV = synergies − premium = 100 − 60 = $40m.
3. Target gain = premium = $60m.
4. Shares of synergies: target 60%, acquirer 40%.
5. Maximum premium the acquirer can pay without destroying value is $100m (a 33.3% premium).

Answer. Premium $60m; acquirer NPV $40m; target gain $60m.

Problem 47 (Basic): Exchange ratio and EPS dilution in a stock deal

Question. An acquirer (100m shares, price $60, EPS $4.00) offers all stock for a target (50m shares, price $40, net income $120m) at a 30% premium. Compute the offer price, exchange ratio, new shares issued, pro forma EPS (no synergies) and the P/E ratios of each firm.

Solution.

1. Offer price = 40 × 1.30 = $52.00.
2. Exchange ratio = 52.00 / 60 = 0.8667 acquirer shares per target share.
3. New shares = 50,000,000 × 0.8667 = 43.333 million; total = 143.333m.
4. Pro forma net income = 400 + 120 = $520m; EPS = 520 / 143.333 = $3.6279, dilution of 9.30%.
5. Acquirer P/E = 60 / 4 = 15.0x; target P/E at market = 2,000 / 120 = 16.67x (at the offer price 21.67x). Paying a higher P/E than one's own with stock dilutes EPS.

Answer. Exchange ratio 0.8667; EPS $3.6279 vs $4.00 (-9.30%).

Problem 48 (Intermediate): Value sharing in a stock-for-stock merger

Question. Acquirer A has 100 million shares at $60; target B has 40 million shares at $40. Synergies are worth $300 million. A offers 0.80 of its shares for each B share. Compute post-merger price, gains to each side, and the break-even synergy at which A's holders are no worse off.

Solution.

1. Standalone values: A = $6,000m; B = $1,600m. Combined with synergies = $7,900m.
2. New shares issued = 40 × 0.80 = 32m; total = 132m; A holders own 75.76%.
3. Post-merger price = 7,900 / 132 = $59.848.
4. Gain to A holders = (59.848 − 60) × 100 = −$15.2m; gain to B holders = 59.848 × 32 − 1,600 = $315.2m. Sum = $300.0m = synergies.
5. Break-even: A holders need 75.76% × (8,000 + S) = 6,000, so S = $320m. The 0.80 ratio embeds a premium of 0.8 × 60 / 40 − 1 = 20%, i.e., $320m on B's $1,600m; A's holders break even exactly when synergies equal the premium paid.

Answer. Price $59.848; A loses $15.2m, B gains $315.2m; break-even synergies $320m.

Problem 49 (Intermediate): Debt-financed acquisition accretion

Question. Using the data of Problem 47, the acquirer instead pays $52 per target share in cash, financed entirely by new 6% debt (tax 25%). Compute the debt raised, after-tax interest, pro forma EPS and the break-even interest rate. Compare with the stock deal of Problem 47.

Solution.

1. Debt = 50,000,000 × $52 = $2,600m.
2. After-tax interest = 2,600 × 6% × 0.75 = $117.0m.
3. Pro forma net income = 520 − 117.0 = $403.0m; shares unchanged at 100m; EPS = $4.030 versus $4.00 (0.75%).
4. Break-even interest: 120 = 2,600 × r × 0.75, r = 6.154% (the target's earnings yield at the offer, 4.615%, divided by 1 − t).

Answer. EPS $4.030 (+0.75%); break-even rate 6.154%.

Problem 50 (Challenging): Merger arbitrage and implied deal probability

Question. A target trades at $40 before a $52 cash bid. After the announcement it trades at $49.50. If the deal fails the stock is expected to fall back to $41. Completion, if it happens, is in six months. Compute the gross spread, annualized return if the deal closes, the implied probability of success ignoring time value, and the implied probability when discounting at 5% per year (continuous).

Solution.

1. Gross spread = 52 − 49.50 = $2.50, or 5.05%.
2. Annualized return if closed = (1 + 5.05%)^2 − 1 = 10.36%.
3. Implied probability (no discounting): 49.50 = p × 52 + (1 − p) × 41 gives p = (49.50 − 41) / (52 − 41) = 77.3%.
4. With discounting (factor 0.97531 for six months): 49.50 = [p × 52 + (1 − p) × 41] × 0.97531, so p = (50.7531 − 41) / 11 = 88.7%.
5. Downside if the deal fails = 49.50 − 41 = $8.50 versus upside $2.50: arbitrageurs bear a 3.4-to-1 downside-to-upside ratio, hence the high annualized spread is compensation for the risk of a break.

Answer. Spread 5.05%; annualized 10.36%; implied p = 77.3% (undiscounted), 88.7% (discounted).

Part K: Leveraged Buyouts and Private Equity (Volume 1, Chapter 23)

Key tools. Sources equal uses. Sponsor equity = total uses − debt. MOIC = exit equity / entry equity; IRR = MOIC^(1/n) − 1 for a single entry and exit.

Problem 51 (Basic): Sources and uses

Question. A sponsor buys a company with EBITDA of $125 million at 8.0x. Fees are 2% of the purchase price. Debt is 4.5x EBITDA. Compute total uses, debt, sponsor equity and the equity percentage.

Solution.

1. Purchase EV = 125 × 8.0 = $1,000.0m.
2. Fees = 2% × 1,000 = $20.0m; total uses = $1,020.0m.
3. Debt = 4.5 × 125 = $562.5m.
4. Sponsor equity = 1,020.0 − 562.5 = $457.5m = 44.9% of capitalization.
5. Leverage check: debt/EV = 56.2%; net debt/EBITDA = 4.5x.

Answer. Uses $1,020.0m; debt $562.5m; equity $457.5m (44.9%).

Problem 52 (Basic): MOIC and IRR

Question. The sponsor equity from Problem 51 is $457.5 million. After five years the equity is sold for $900 million. Compute MOIC and IRR, and the exit value needed for a 20% IRR.

Solution.

1. MOIC = 900 / 457.5 = 1.967x.
2. IRR = 1.967^(1/5) − 1 = 14.49%.
3. For a 20% IRR: required exit equity = 457.5 × 1.2^5 = $1,138.4m (MOIC 2.488x).
4. Shortfall relative to 20% target = $238.4m.

Answer. MOIC 1.967x; IRR 14.49%; 20% IRR needs $1,138.4m.

Problem 53 (Intermediate): Five-year LBO model

Question. Continue with the deal of Problem 51 (opening debt $562.5m, equity $457.5m). EBITDA grows 8% per year. Each year: pre-interest, pre-tax free cash flow is 55% of EBITDA; depreciation is 15% of EBITDA (for tax); interest is 9% of opening debt; tax is 25% of (EBITDA less depreciation less interest); all remaining cash repays debt. Exit at 8.0x EBITDA in year 5. Compute the debt schedule, exit equity, MOIC and IRR.

Solution.

1. Annual calculation (cash for paydown = 0.55 × EBITDA − interest − tax):

Year	EBITDA	Interest	Tax	Cash for debt paydown	Closing debt
1	135.0	50.6	16.0	7.6	554.9
2	145.8	49.9	18.5	11.8	543.2
3	157.5	48.9	21.2	16.5	526.7
4	170.1	47.4	24.3	21.8	504.8
5	183.7	45.4	27.7	27.9	476.9

2. Year-5 EBITDA = 125 × 1.08^5 = $183.67m; exit EV = 8.0 × 183.67 = $1,469.3m.
3. Exit equity = EV − closing debt = 1,469.3 − 476.9 = $992.4m.
4. MOIC = 992.4 / 457.5 = 2.169x; IRR = 16.75%.
5. Leverage at exit = 476.9 / 183.7 = 2.60x versus 4.50x at entry.

Answer. Exit equity $992.4m; MOIC 2.17x; IRR 16.8%.

Problem 54 (Intermediate): Value creation bridge

Question. For Problem 53, decompose the equity value creation into EBITDA growth, multiple expansion and debt paydown. Then repeat for an exit multiple of 9.0x. Entry fees reduce sponsor value.

Solution.

1. At 8.0x exit: EBITDA growth = (EBITDA5 − 125) × 8.0 = (183.67 − 125) × 8 = $469.3m.
2. Multiple change = 0 at 8.0x. Debt paydown = 562.5 − 476.9 = $85.6m. Entry fees = −$20.0m.
3. Sum = 469.3 + 85.6 − 20.0 = $534.9m, which equals exit equity 992.4 − entry equity 457.5 = $534.9m.
4. At 9.0x: multiple expansion = (9.0 − 8.0) × 183.67 = $183.7m; total value creation = $718.6m; exit equity = $1,176.1m.
5. MOIC at 9.0x = 2.571x; IRR = 20.78%. The bridge shows that deleveraging is 15% of operating plus financial value creation at 8.0x.

Answer. 8.0x exit: growth $469m, paydown $86m, fees -$20m; at 9.0x add $184m from multiple expansion.

Problem 55 (Challenging): Private equity fund waterfall

Question. A $500 million fund charges a 2% annual management fee on commitments for ten years (funded from LP contributions, so total LP contributions equal $500m of which $200m is fees and $300m is invested), with carried interest of 20% over an 8% compounded hurdle with 100% GP catch-up. Gross distributions are $900m at the end of year 5 for all capital (treat all contributions as made at time 0 for simplicity). Compute the distribution to LPs and GP, LP net MOIC and IRR. Then repeat with gross distributions of $700m.

Solution.

1. Fee drag: fees = 2% × 500 × 10 = $100m, so invested capital = $400m (a 20% drag on commitments).
2. Tier 1, return of contributed capital: $500m to LPs. Tier 2, preferred return = 500 × (1.08^5 − 1) = $234.66m to LPs.
3. Tier 3, catch-up: GP receives 100% until it holds 20% of profits: pref × 0.20 / 0.80 = $58.67m.
4. Tier 4, residual = 900 − 793.33 = $106.67m split 80/20: LP $85.34m, GP $21.33m.
5. Total GP carry = $80.00m = 20.0% of total profit of $400m. LP receives $820.00m: MOIC 1.640x, IRR = 10.40%.

6. With $700m: profit = $200m, below the hurdle amount of 234.66m, so GP carry = $0.00m and LPs receive $700m (MOIC 1.40x, IRR 6.96%).
7. Gross MOIC on invested capital = 900 / 400 = 2.25x versus LP net MOIC 1.64x: fees and carry take 0.61 turns.

Answer. GP carry $80.0m; LP net $820.0m, MOIC 1.64x, IRR 10.40%; at $700m no carry.

Part L: Sustainability, Governance and Fintech (Volume 1, Chapter 24)

Key tools. Greenium = conventional yield − green yield. Internal carbon price converts tonnes of CO2 to dollars in the cash flows. Cumulative voting: shares needed to elect n of N directors = S × n / (N + 1) + 1.

Problem 56 (Basic): Green bond economics

Question. A firm issues $500 million of 7-year green bonds at 4.85% versus 5.00% for a conventional bond of the same risk. Third-party verification costs $0.6m upfront and annual impact reporting costs $0.15m. Compute the annual interest saving (greenium), its present value at 4.85%, the PV of the added costs and the net benefit.

Solution.

1. Greenium = 5.00% − 4.85% = 15 basis points. Annual saving = $500m × 0.0015 = $0.750m.
2. PV of savings = 0.750 × annuity(4.85%, 7) = $4.363m.
3. PV of costs = 0.60 + 0.15 × 5.8180 = $1.473m.
4. Net benefit = 4.363 − 1.473 = $2.891m (pre-tax).

Answer. Saving $0.750m per year; net PV benefit $2.89m.

Problem 57 (Basic): Carbon price exposure

Question. A firm emits 2.0 million tonnes of CO2 a year. A carbon price of $60 per tonne is introduced, and the firm can pass through 40% of the cost to customers. Tax is 25% and there are 100 million shares. Compute gross cost, net cost after pass-through, after-tax cost and EPS impact.

Solution.

1. Gross cost = 2,000,000 × 60 = $120m.
2. Pass-through = 40% × 120 = $48m; net pre-tax cost = $72m.
3. After-tax cost = 72 × 0.75 = $54m.
4. EPS impact = 54 / 100 = $0.54 per share.
5. Every $10 of carbon price changes EPS by $0.09.

Answer. Net after-tax cost $54m; EPS reduced by $0.54.

Problem 58 (Intermediate): Internal carbon price in capital budgeting

Question. A project costs $200m and generates $45m a year for 8 years; WACC is 9%. It emits 0.3 million tonnes of CO2 a year. The firm applies an internal shadow carbon price of $50 per tonne in year 1 rising 5% a year; the cost is tax-deductible at 25%. Compute NPV with and without carbon pricing, the PV of carbon costs and the break-even initial carbon price.

Solution.

1. PV of cash flows = 45 × annuity(9%, 8) = $249.07m; NPV without carbon = $49.07m.

2. Year-1 carbon cost = 0.3 × 50 = $15.0m pre-tax ($11.25m after tax), growing 5% a year; PV of after-tax carbon costs = $72.71m.
3. NPV with carbon = 49.07 − 72.71 = -$23.64m.
4. Break-even initial price: NPV falls to zero when the price multiplier is 0.675 times $50, i.e. $33.74 per tonne.

Answer. NPV $49.07m without carbon, −$23.64m with; break-even carbon price $33.74 per tonne.

Problem 59 (Intermediate): Cumulative voting and dual-class control

Question. (a) A company has 1,000 shares outstanding and a nine-member board elected at once. Under cumulative voting, how many shares are needed to elect 1, 2 and 3 directors? What can a 30% holder guarantee? What is the requirement under straight voting? (b) A dual-class firm has 90 million Class A shares (1 vote) and 10 million Class B (10 votes). A founder holds all Class B and 5 million Class A shares. Compute her economic ownership and voting power.

Solution.

1. a. Shares needed = S × n / (N + 1) + 1 with N = 9, S = 1,000:

Directors to elect	Shares needed	Percent of shares
1	101	10.1%
2	201	20.1%
3	301	30.1%

2. A 30% holder (300 shares) can guarantee two seats (needs 201) but not three (needs 301). Under straight voting, a holder needs a majority (501 shares) to control every seat, and a 30% holder elects none; cumulative voting thus gives minorities representation.
3. b. Total votes = 90 × 1 + 10 × 10 = 190 million. Founder's votes = 5 + 100 = 105 million = 55.3% of votes.
4. Economic ownership = (5 + 10) / 100 = 15%. The wedge between voting power (55.3%) and cash-flow rights (15%) is the agency cost associated with dual-class structures, which sunset provisions try to limit.

Answer. (a) 101, 201 and 301 shares for 1, 2 and 3 seats; a 30% holder guarantees 2; (b) economic 15%, voting 55.3%.

Problem 60 (Challenging): Digital lender unit economics

Question. A marketplace lender spends $9 million to acquire 15,000 borrowers. The average loan is $12,000; the platform earns a 3.0% origination fee and a 0.8% annual servicing fee on an average outstanding balance of 55% of the loan over a 3-year life. Variable cost per borrower is $60. 40% of borrowers take a second loan after three years, and each later cohort repeats the same behavior (discount rate 12%). Compute CAC, first-loan contribution, LTV with repeat business, LTV/CAC, and the repeat rate needed for LTV/CAC = 1.

Solution.

1. CAC = 9,000,000 / 15,000 = $600.
2. Origination fee = 12,000 × 3% = $360; servicing = 12,000 × 0.8% × 0.55 × 3 = $158.40; revenue = $518.40; contribution = $458.40 after $60 variable cost.
3. First-loan LTV/CAC = 458.40 / 600 = 0.76x: the platform loses money on a one-off borrower.
4. Repeat factor per cycle = 0.40 / 1.12^3 = 0.2847; LTV = 458.40 / (1 − 0.2847) = $640.86; LTV/CAC = 1.07x.
5. Break-even (LTV = CAC): required discounted repeat factor x = 1 − 458.40 / 600 = 0.2360, so repeat probability = 0.2360 × 1.12^3 = 33.2%.

Answer. CAC $600; LTV $641; LTV/CAC 1.07x; break-even repeat rate 33.2%.

Chapter 24: Instructor Manual and Examination Bank

This chapter is for instructors. It describes how to design a one-semester course around Volume 1, Chapters 13 to 24, the cases and problem sets of this volume, and how to assess students fairly. All companies and figures are fictional and illustrative. Every numeric answer in the exams and question bank was recomputed in Python; the model answers show rounded values.

Purpose and Use

Volume 3 supplies the practice material that Volumes 1 and 2 deliberately omit. This chapter turns that material into a coherent course and supplies a ready-to-run assessment package: two midterm examinations, two final examinations (each with a 100-point marking scheme and full model answers) and a 60-question multiple-choice bank. Instructors may use Midterm 1 and Midterm 2 together with Final A, keeping Final B as the make-up or the following year’s form, or may mix questions across forms. All four exams have been balanced so that each is worth exactly 100 points.

Course Design

Learning outcomes

By the end of the course a student should be able to:

1. Price and structure equity and debt issues, and explain issuance costs, underpricing and signaling.
2. Choose and defend a capital structure using Modigliani-Miller, trade-off, pecking-order and adjusted-present-value reasoning.
3. Evaluate dividend and repurchase decisions, and manage working capital and short-term funding.
4. Value options (binomial and Black-Scholes), and use forwards, futures, swaps and options to hedge interest-rate, currency and commodity risk.
5. Build and critique an enterprise valuation (DCF and multiples), including the role of growth and ROIC.
6. Analyze mergers, leveraged buyouts and private-equity fund economics, and the implications of governance, sustainability and fintech business models.
7. Communicate a decision in a board-style memo that states assumptions, alternatives and risks.

Fourteen-week schedule

Week	Topic	Reading	Cases, problems and assessment
1	Equity financing: rights, IPOs, SEOs	Vol 1 Ch 13	Chapter 7 (family IPO case); Problems 1-5
2	Debt financing and bond markets	Vol 1 Ch 14	Problems 6-10
3	Capital structure theory and practice	Vol 1 Ch 15	Chapter 15 (distressed manufacturer); Problems 11-15
4	Payout policy	Vol 1 Ch 16	Chapter 9 (payout decision); Problems 16-20
5	Working capital and short-term finance	Vol 1 Ch 17	Problems 21-25; Midterm 1 review
6	Options and option valuation	Vol 1 Ch 18	Problems 26-30; **Midterm 1** (Weeks 1-3 content)
7	Derivatives: forwards, futures, swaps	Vol 1 Ch 19	Problems 31-35
8	Risk management and foreign exchange	Vol 1 Ch 20	Problems 36-40; **Midterm 2** (Weeks 4-6 content)
9	Enterprise valuation	Vol 1 Ch 21	Problems 41-45; Case memo 1 due
10	Mergers and acquisitions	Vol 1 Ch 22	Chapter 13 (hostile bid case); Problems 46-50
11	Leveraged buyouts, private equity and restructuring	Vol 1 Ch 23	Chapter 11 (Series B term sheet); Problems 51-55
12	Sustainability, governance and fintech	Vol 1 Ch 24	Problems 56-60; Case memo 2 due
13	Integration: cross-chapter cases	Vol 1 Ch 13-24	Mock final (Final Examination A, untimed)
14	Review and final examination	All	**Final Examination** (Final B as the make-up form)

Each weekly session of about three hours works well as: 30 minutes on the concept and a quick quiz drawn from the question bank, 90 minutes on the case or problems, and 45 minutes of debrief. Problems 1 to 60 of Chapter 23 are assigned as the weekly problem sets; the five problems per topic provide a graded ladder, and the answers are available to students afterward so that class time is spent on judgment rather than arithmetic.

Assessment weights

Component	Weight (%)
Class participation and case discussion	10
Two case memos (10 each)	20
Midterm Examination 1	15
Midterm Examination 2	15
Final Examination	40
Total	**100**

Exam coverage

Exam	Topics covered	Weight on computation	Weight on interpretation
Midterm 1	Rights offerings, IPO costs, bonds and duration, MM, EBIT-EPS	75%	25%
Midterm 2	Buybacks and dividends, working capital, Black-Scholes, executive options	80%	20%
Final A	FX and swap hedging, DCF, merger accretion, LBO	85%	15%
Final B	IPO and capital structure, futures hedging and VaR, value drivers and multiples, PE waterfall and carbon NPV	80%	20%

Teaching Approach

Cases. Use the case chapters of this volume in a cold-call format: one student frames the decision, another presents the numbers, a third challenges the assumptions. Reserve the final 20 minutes for the alternative defensible answers listed in each Teaching Note, so students learn that finance questions frequently have more than one acceptable recommendation if it is supported.

Problems. Students attempt each problem before reading the solution. Common errors, such as dividing by the wrong base for underpricing, forgetting the tax shield or mixing continuous and discrete compounding, should be collected and revisited in class.

Modeling. The modeling chapters are best taught with laptops open. Require students to submit their spreadsheets with a short assumptions page; grade for transparency and checks (balance, sources equal uses) as well as for the final number.

Calibration. Before the first exam, run a 20-minute practice test of 10 questions from the bank, so students learn the difficulty and format. Results guide the review sessions.

Grading Rubrics

Case memo rubric (100 points, scaled to 10% of the grade)

Criterion	Points	Excellent (90-100%)	Adequate (60-75%)	Weak (below 50%)
Problem definition and decision framing	15	States the decision, deadline and constraints precisely	Decision stated but constraints incomplete	Restates the case facts only
Quantitative analysis (correct, complete, assumptions stated)	30	All numbers reproduce; sensitivities included	Method correct with arithmetic slips	Missing or unexplained calculations
Use of theory and evidence from Volume 1	15	Frameworks selected and applied critically	Frameworks named but applied mechanically	Theory absent or misapplied
Judgment, risks and alternatives	20	Weighs two or more alternatives and quantifies downside	Mentions risks without quantification	One-sided argument
Recommendation clarity and board-ready writing	20	Clear recommendation, next steps and a one-page memo	Recommendation present but hedged	No clear recommendation

Class participation

Score each session 0 to 3: 0 absent or silent; 1 present, answers when asked; 2 contributes a correct analytical point; 3 advances the discussion by challenging an assumption or connecting two ideas. Average the scores and scale to 10 points. Quality matters more than frequency; do not reward repetition of the case facts.

Exam marking conventions

1. **Method before arithmetic.** In a numeric part, award about 60% of the points for correct method and set-up, 20% for arithmetic accuracy and 20% for units, labeling and interpretation.
2. **Error carried forward.** If a wrong answer from an earlier part is used correctly in a later part, award full method points for the later part. Do not penalize the same mistake twice.
3. **Rounding.** Accept answers within 1% of the model answer, or within the stated rounding rule, unless the question demands an exact value. Interpolated or iterative answers (yield, IRR) are accepted within 0.1 percentage point.
4. **Explanations.** Two-point explanatory items are awarded for a correct statement plus a reason; one point if only the statement is given.
5. **Alternative methods.** Any valid method earns full credit; for example, a replicating portfolio is as good as risk-neutral valuation.
6. **Moderation.** Two markers should independently grade five scripts per question and reconcile before grading the rest; investigate any question with a mean below 45% of the points as a possible setting or teaching problem.

Target distribution

Grade band	Typical share of students	Description
A (85+)	15-20%	Correct, complete, with insight beyond the formulas
B (70-84)	40-45%	Sound method; occasional numerical or interpretive gaps
C (55-69)	25-30%	Correct tools, inconsistent execution
Below 55	10% or less	Fundamental gaps; offer remediation

Academic integrity and accommodation

Use different forms (A and B) in alternating rows, or change the input numbers using the model answers as a guide. Closed-book exams with a calculator are recommended; provide the formula list in the questions rather than allowing sheets of notes if exams are time pressed. Provide extra time and alternative formats according to institutional policy, and keep the make-up form (Final B) in reserve.

Examination Bank

The four exams follow. Each question shows part marks in square brackets; the marking scheme tables list the model answer and how to award the points. All amounts are in US dollars.

Midterm Examination 1 (Volume 1, Chapters 13 to 15)

Time allowed: 90 minutes. Closed book; a financial calculator is permitted. Show all steps; partial credit is awarded for correct method. Round money to the nearest cent and rates to two decimals of a percent unless told otherwise.

Question 1: Raising equity (25 points)

Aldwych Packaging (illustrative) has 40 million shares at $25 and needs $120 million.

- a. It sells shares through a rights offering at $20. How many shares are issued and in what ratio? [5]
- b. Compute the theoretical ex-rights price. [6]
- c. Compute the value of a right per old share. A holder has 2,000 shares: show that wealth is the same whether she exercises or sells her rights. [7]
- d. Alternatively, an underwriter sells shares to the public at $22 with a 5% gross spread. How many shares must be sold to net $120 million, and what is the dollar fee? [7]

Question 2: Debt (25 points)

A bond has an $1,000 face value, an 8-year life and a 5% annual coupon.

- a. Compute the price at a 6% yield. [5]
- b. Compute Macaulay and modified duration. [8]
- c. Estimate the price change for a +50 bp yield change using duration and compare with the exact change. [6]
- d. Describe how to find the yield if the price is $950, and state the yield. [6]

Question 3: Capital structure (25 points)

Brenton Foods (illustrative) has perpetual EBIT of $90 million, tax rate 25%, unlevered cost of capital 9%. It borrows $300 million of permanent debt at 5%.

- a. Compute the unlevered value. [4]
- b. Compute the levered value using MM with taxes. [4]
- c. Compute the cost of equity. [6]
- d. Compute WACC and verify it reproduces the levered value. [6]
- e. Explain two reasons why the MM tax result overstates the benefit of debt. [5]

Question 4: Financing choice (25 points)

Tanner Instruments (illustrative) has 40 million shares at $30 and needs $150 million. Plan A issues equity; Plan B issues 7% debt. Tax rate 30%.

- a. How many shares does Plan A issue and what is the annual interest under Plan B? [4]
- b. Find the EBIT at which EPS is the same under both plans. [7]
- c. Compare EPS at EBIT = $120 million and compute the degree of financial leverage under Plan B. [6]
- d. Discuss, with reference to pecking order and signaling, why managers may prefer debt to equity even above the break-even EBIT, and one reason the analysis is incomplete. [8]

Marking Scheme and Model Answers: Midterm Examination 1 (Volume 1, Chapters 13 to 15)

Question 1 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	New shares = 120m / 20 = 6.0m; ratio = 6/40 = 3 new for 20 held.	2 pts shares, 2 pts ratio, 1 pt units
(b)	6	(40 × 25 + 120) / 46 = $24.348.	3 pts market value after, 2 pts share count, 1 pt answer
(c)	7	Right = 25 − 24.348 = $0.652. Exercise: buy 300 shares for $6,000; wealth = 2,300 × 24.348 − 6,000 = $50,000. Sell: 2,000 × 24.348 + 2,000 × 0.652 = $50,000. Both equal 2,000 × 25 = $50,000.	2 pts right value, 2 pts each wealth calculation, 1 pt conclusion
(d)	7	Gross = 120 / 0.95 = $126.32m; shares = 5.742m; fee = $6.32m.	3 pts gross-up, 2 pts shares, 2 pts fee; deduct 2 if 120 × 5% used as fee

Question 2 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	50 × annuity(6%, 8) + 1,000 / 1.06^8 = 310.49 + 627.41 = $937.90.	3 pts annuity, 2 pts face
(b)	8	Macaulay = Σ t × PV(CF) / price = 6.740 years; modified = 6.740 / 1.06 = 6.359.	5 pts for PV-weighted timing, 3 pts for modified
(c)	6	Approx change = −6.359 × 0.005 × 937.90 = -$29.82. Exact price at 6.5% = $908.67, change -$29.23. The difference (0.59) is convexity.	2 pts approximation, 2 pts exact, 2 pts interpretation
(d)	6	Solve 950 = 50 × annuity(y, 8) + 1,000 / (1+y)^8 by trial or financial calculator: y = 5.80%.	3 pts setup, 3 pts value (accept 5.7% to 5.9%)

Question 3 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	4	V_U = 90 × 0.75 / 0.09 = $750m.	2 pts NOPAT, 2 pts capitalization
(b)	4	V_L = 750 + 0.25 × 300 = $825m; equity = $525m.	3 pts, 1 pt equity
(c)	6	rE = 9% + (9% − 5%) × (300/525) × 0.75 = 10.714%.	3 pts formula, 3 pts numbers
(d)	6	WACC = (525/825) × 10.714% + (300/825) × 5% × 0.75 = 8.182%. Check: 67.5 / 8.182% = $825.0m.	3 pts WACC, 3 pts check
(e)	5	Any two: financial distress costs, personal taxes reducing the debt advantage, agency costs of debt, loss of tax shield if EBIT falls below interest, limited debt capacity.	2.5 pts per reason with explanation

Question 4 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	4	Shares = 150/30 = 5m (45m total); interest = 150 × 7% = $10.5m.	2 pts each
(b)	7	0.7E/45 = 0.7(E − 10.5)/40 gives 40E = 45E − 472.5, E = $94.5m.	3 pts setup, 3 pts algebra, 1 pt answer
(c)	6	EPS(A) = $1.867; EPS(B) = $1.916; DFL = 120 / (120 − 10.5) = 1.096.	2 pts each EPS, 2 pts DFL
(d)	8	Pecking order: internal funds first, then debt, then equity because of information asymmetry. Equity issuance signals overvaluation and lowers the price; debt does not. Incomplete because EPS ignores risk, the effect on P/E and cost of equity, and distress costs.	3 pts pecking order, 3 pts signaling, 2 pts limitation

Total: 100 points.

Midterm Examination 2 (Volume 1, Chapters 16 to 18)

Time allowed: 90 minutes. Closed book; a financial calculator is permitted. Show all steps. Where a question says to ignore taxes, do so.

Question 1: Payout policy (25 points)

Corvus Devices (illustrative) has 120 million shares at $45, net income of $360 million and $180 million of excess cash.

- a. If it repurchases shares at $45, compute shares retired and EPS before and after (ignore lost interest). [6]
- b. If it pays the cash as a dividend instead, compute the dividend per share and the ex-dividend price. Why is pre-tax wealth the same? [6]
- c. An investor holds 1,000 shares with a basis of $30; dividends are taxed at 25% and gains at 15%. Compare after-tax wealth under the two payouts (the investor tenders proportionately). [8]
- d. Next year EPS will be $3.30, ROE is 14%, payout is 50% and the cost of equity is 11%. Value the share by the Gordon model. [5]

Question 2: Working capital (25 points)

Hallam Supplies (illustrative): sales $900m, COGS $630m, inventory $90m, receivables $100m, payables $63m (365-day year).

- a. Compute DIO, DSO, DPO and the cash conversion cycle. [7]
- b. A supplier offers 3/15 net 45. Compute the annualized simple cost of forgoing the discount. [5]
- c. Cutting DSO by 6 days releases how much cash, and what is it worth per year at an 8% cost of funds? [5]
- d. Annual demand is 18,000 units, order cost $200, holding cost $9 per unit per year. Compute the EOQ and the total annual ordering plus holding cost. [8]

Question 3: Option valuation (25 points)

A stock trades at $60. A European call has strike $62 and expires in 0.75 years; r = 5% (continuous) and volatility is 28%. No dividends.

- a. Compute d1 and d2. [5]
- b. Compute the call price. [6]
- c. Compute the put price using put-call parity. [5]
- d. A dealer sells 10,000 calls. How many shares does she hold to be delta-neutral? [5]
- e. If the stock rises to $61, estimate the call price using delta and compare with the exact value. [4]

Question 4: Executive options (25 points)

Meridian Labs (illustrative) grants 1 million at-the-money options (strike $40, stock $40) with a 5-year life; volatility 30%, r = 4% (continuous), dividend yield 1%. Options vest straight-line over 4 years. Tax rate 25%; 100 million shares.

- a. Value one option using Black-Scholes with a dividend yield. [8]
- b. Compute the grant-date value and the annual expense. [5]
- c. Compute the after-tax expense and EPS impact per year. [6]
- d. At an average price of $55 and strike $40, compute dilutive shares by the treasury stock method. [6]

Marking Scheme and Model Answers: Midterm Examination 2 (Volume 1, Chapters 16 to 18)

Question 1 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	6	Shares retired = 180m / 45 = 4.0m; EPS = 360/120 = $3.00 before, 360/116 = $3.1034 after.	2 pts shares, 2 pts each EPS
(b)	6	DPS = 180/120 = $1.50; ex-dividend price = $43.50. In a frictionless market a dollar of dividend equals a dollar less in price, and the buyback leaves value per share unchanged at $45, so each holder receives $1.50 in value either way.	2 pts DPS, 2 pts price, 2 pts argument
(c)	8	Dividend: cash 1,500 × 0.75 = 1,125 plus shares 1,000 × 43.50 = 43,500; wealth $44,625. Buyback: tender 33.33 shares for $1,500; gain = 33.33 × 15 = $500; tax $75; wealth = 1,500 − 75 + 966.67 × 45 = $44,925. Buyback higher by $300.	3 pts each wealth, 2 pts comparison
(d)	5	g = 0.14 × 0.5 = 7%; D1 = 3.30 × 0.5 = $1.65; P = 1.65 / (0.11 − 0.07) = $41.25.	2 pts g, 1 pt D1, 2 pts price

Question 2 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	7	DIO = 52.1; DSO = 40.6; DPO = 36.5; CCC = 56.2 days.	1.5 pts each ratio, 2.5 pts CCC
(b)	5	0.03/0.97 × 365/30 = 37.63%.	2 pts periodic cost, 3 pts annualization
(c)	5	900 × 6/365 = $14.79m; × 8% = $1.18m per year.	3 pts, 2 pts
(d)	8	EOQ = √(2 × 18,000 × 200 / 9) = 894.4 units; total cost = 4,025 + 4,025 = $8,050.	4 pts EOQ, 4 pts cost

Question 3 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	d1 = [ln(60/62) + (0.05 + 0.0392) × 0.75] / (0.28 × √0.75) = 0.1407; d2 = -0.1018.	3 pts d1, 2 pts d2
(b)	6	N(d1) = 0.5559; N(d2) = 0.4595; C = 60 × N(d1) − 62 e^(−0.0375) N(d2) = $5.919.	2 pts normal values, 4 pts price
(c)	5	P = C − S + K e^(−rT) = 5.919 − 60 + 59.718 = $5.637.	3 pts method, 2 pts value
(d)	5	Delta = N(d1) = 0.5559; hold 5,559 shares.	3 pts delta, 2 pts shares
(e)	4	Delta estimate: 5.919 + 0.5559 × 1 = 6.474; exact = 6.488. Difference 0.013 due to gamma.	2 pts each

Question 4 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	8	C = S e^(−qT) N(d1) − K e^(−rT) N(d2) with d1 = [ln(1) + (0.04 − 0.01 + 0.045) × 5] / (0.30 × √5) gives $12.171 per option.	4 pts d1, 2 pts d2, 2 pts price
(b)	5	Value = 1,000,000 × 12.171 = $12.171m; expense = $3.043m per year for 4 years.	3 pts total, 2 pts annual
(c)	6	After-tax = 3.043 × 0.75 = $2.282m; EPS impact = $0.0228.	3 pts tax, 3 pts EPS
(d)	6	Shares issued = 1m; shares bought back = 40m / 55 = 0.727m; dilutive shares = 0.273m.	3 pts proceeds, 3 pts net

Total: 100 points.

Final Examination A (Volume 1, Chapters 13 to 24)

Time allowed: 180 minutes. Closed book; a financial calculator is permitted. Answer all four questions. Each question is worth 25 points. Show all steps; partial credit is awarded for correct method.

Question 1: Foreign-exchange and interest-rate hedging (25 points)

A US importer owes EUR 4 million in nine months (0.75 year). Spot is $1.10 per euro; nine-month simple interest rates are 4.5% (USD) and 2.5% (EUR) per year.

- a. Compute the covered-parity forward rate. [4]
- b. Describe and cost a money-market hedge; compare with the forward. [6]
- c. The firm can instead buy a euro call with strike $1.12 for a premium of $0.015 per euro (financed at the USD rate). Compute the total dollar cost at spot rates of 1.05, 1.12 and 1.20. [7]
- d. The firm also pays SOFR + 2.00% on a $60 million loan and enters a swap to receive SOFR and pay 4.40% fixed. Compute the effective rate and the swap receipt when SOFR is 5.50%. [5]
- e. State two conditions under which the firm should leave the exposure unhedged. [3]

Question 2: Enterprise valuation (25 points)

Dunmore Systems (illustrative) forecasts FCFF of 80, 88, 96, 103 and 109 ($m) in years 1 to 5. WACC is 9.5%, terminal growth 3%, net debt $420m and 60 million shares.

- a. Compute the PV of the explicit cash flows. [6]
- b. Compute the terminal value, its PV, and enterprise value. [6]
- c. Compute the equity value, value per share and the share of EV from the terminal value. [5]
- d. Compute value per share for WACC of 8.5%, 9.5%, 10.5% and growth of 2%, 3%, 4% and comment. [8]

Question 3: Mergers and acquisitions (25 points)

Acquirer: 80 million shares at $50, EPS $3.50. Target: 30 million shares at $30, EPS $2.00. The acquirer offers all stock at a 35% premium. Tax rate is 25%.

- a. Compute the offer price, exchange ratio and new shares issued. [5]
- b. Compute pro forma EPS and dilution (no synergies). [7]
- c. Compute the after-tax and pre-tax synergies needed for EPS break-even. [6]
- d. If the PV of synergies is $400m, compute the post-merger share price and the gains to each group of shareholders. [7]

Question 4: Leveraged buyout (25 points)

A sponsor buys a firm with EBITDA of $80m at 9.0x. Fees are $15m. Debt is 5.0x EBITDA at 8%. EBITDA grows 6% a year. Each year pre-interest, pre-tax cash flow is 52% of EBITDA; depreciation is 15% of EBITDA; tax is 25% of (EBITDA − depreciation − interest); all cash repays debt. Exit is after five years.

- a. Prepare sources and uses and compute sponsor equity. [5]
- b. Build the debt schedule for years 1 to 5. [10]
- c. Compute exit equity, MOIC and IRR at a 9.0x exit multiple. [6]
- d. Repeat at an 8.0x exit and comment on the sensitivity. [4]

Marking Scheme and Model Answers: Final Examination A (Volume 1, Chapters 13 to 24)

Question 1 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	4	F = 1.10 × (1 + 0.045 × 0.75) / (1 + 0.025 × 0.75) = $1.1162.	2 pts formula, 2 pts value
(b)	6	Buy EUR 4m / 1.01875 = EUR 3,926,380 now for $4,319,018, borrow it at 4.5%; repay $4,464,785 in nine months = $1.1162 per euro, equal to the forward ($4,464,785).	3 pts steps, 2 pts cost, 1 pt comparison
(c)	7	Premium compounded = 0.015 × 1.03375 = $0.0155. Cost per euro = min(spot, 1.12) + 0.0155: spot 1.05 gives $1.0655 (total $4,262,025); spot 1.12 gives $1.1355 ($4,542,025); spot 1.20 gives $1.1355 ($4,542,025).	2 pts premium, 5 pts three outcomes
(d)	5	Effective = (SOFR + 2.00%) − (SOFR − 4.40%) = 6.40%. Receipt at SOFR 5.50% = (5.50% − 4.40%) × 60m = $0.66m.	3 pts, 2 pts
(e)	3	Any two: exposure naturally offset by receipts, immaterial size, the firm's cost of capital is low and cash flow volatility is not costly, or hedge cost exceeds expected benefit.	1.5 pts each

Question 2 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	6	PV = Σ FCFF_t / 1.095^t = $360.46m.	3 pts discounting method, 3 pts sum
(b)	6	TV = 109 × 1.03 / 0.065 = $1,727.23m; PV = $1,097.18m; EV = $1,457.64m.	3 pts TV, 1 pt PV, 2 pts EV
(c)	5	Equity = 1,457.64 − 420 = $1,037.64m; per share $17.29; terminal share = 75.3%.	2 pts, 2 pts, 1 pt
(d)	8	Per-share values: WACC 8.5%: g 2% = $18.13, g 3% = $21.80, g 4% = $27.10; WACC 9.5%: g 2% = $14.70, g 3% = $17.29, g 4% = $20.83; WACC 10.5%: g 2% = $12.08, g 3% = $13.99, g 4% = $16.49. Value is far more sensitive to WACC and g than to the explicit forecasts; the spread illustrates the risk in terminal value.	6 pts for the nine values (two-thirds point for each correct), 2 pts for comment

Question 3 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	Offer = 30 × 1.35 = $40.50; ratio = 40.50/50 = 0.810; new shares = 24.30m.	2 pts each for offer and ratio, 1 pt shares
(b)	7	NI = 280 + 60 = $340m; shares = 104.30m; EPS = $3.260; dilution = 6.86%.	3 pts NI and shares, 2 pts EPS, 2 pts dilution
(c)	6	Required NI = 3.50 × 104.30 = $365.05m; shortfall = $25.05m after tax; pre-tax = 25.05 / 0.75 = $33.40m.	3 pts shortfall, 3 pts gross-up
(d)	7	Values: A $4,000m, B $900m, synergies $400m; price = 5,300 / 104.30 = $46.980. A holders gain (46.980 − 50) × 80 = -$241.6m; B holders gain 46.980 × 24.30 − 900 = $241.6m; total $400m. A's holders lose despite $400m of synergies because the 35% premium (about $315m) plus B's pro rata share of synergies exceeds A's own share.	3 pts price, 2 pts each gain

Question 4 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	Uses = EV 720 + fees 15 = $735m; debt = $400m; equity = $335m (45.6%).	2 pts uses, 1 pt debt, 2 pts equity
(b)	10	Y1: EBITDA 84.8, interest 32.0, tax 10.0, paydown 2.1, debt 397.9; Y2: EBITDA 89.9, interest 31.8, tax 11.1, paydown 3.8, debt 394.2; Y3: EBITDA 95.3, interest 31.5, tax 12.4, paydown 5.6, debt 388.5; Y4: EBITDA 101.0, interest 31.1, tax 13.7, paydown 7.7, debt 380.8; Y5: EBITDA 107.1, interest 30.5, tax 15.1, paydown 10.1, debt 370.7.	2 pts per year (EBITDA, interest, tax, paydown, closing debt)
(c)	6	Exit EV = 9.0 × 107.06 = $963.5m; equity = $592.8m; MOIC = 1.770x; IRR = 12.09%.	2 pts EV, 1 pt equity, 1 pt MOIC, 2 pts IRR
(d)	4	At 8.0x equity = $485.8m; MOIC 1.450x; IRR 7.72%. One turn of exit multiple changes IRR by 4.4% (percentage points): returns are very sensitive to exit multiple.	2 pts numbers, 2 pts comment

Total: 100 points.

Final Examination B (Volume 1, Chapters 13 to 24)

Time allowed: 180 minutes. Closed book; a financial calculator is permitted. Answer all four questions. Each question is worth 25 points. Show all steps; partial credit is awarded for correct method.

Question 1: Initial public offering and capital structure (25 points)

Pendle Biosystems (illustrative) has 80 million shares, of which founders hold 56 million. It sells 20 million new shares at $18 with a 7% gross spread. The stock closes day one at $21.60. Tax rate is 25%.

- a. Compute first-day underpricing and the money left on the table. [5]
- b. Compute net proceeds to the company. [5]
- c. Compute post-IPO ownership of the founders and the public float percentage. [5]
- d. The company uses $150 million of proceeds to repay permanent debt. Using MM with taxes, find the change in firm value from the lost tax shield, and explain why value may nonetheless rise. [7]
- e. Name one purpose of a lock-up agreement. [3]

Question 2: Hedging and risk measurement (25 points)

A fund manages an $80 million equity portfolio with beta 1.1. It wants beta 0.5 using index futures at 5,000 with a $50 multiplier. Separately an airline has 6 million gallons of fuel exposure; futures contracts cover 42,000 gallons.

- a. How many futures contracts must the fund trade to reach beta 0.5? [8]
- b. The fuel spot-futures correlation is 0.85, $\sigma S = 35\%$ and $\sigma F = 32\%$. Compute the minimum-variance hedge ratio and the number of contracts. [8]
- c. The fund’s daily portfolio volatility is 1.2%. Compute the 1-day 99% VaR ($z = 2.3263$) before and after the beta hedge, assuming risk is all market risk and scales with beta. [9]

Question 3: Valuation cross-check (25 points)

Garnet Industrial (illustrative) has next-year NOPAT of $150m, WACC 8.5%, growth 3%, ROIC on new capital 12%, EBITDA $260m, net debt $600m, 70 million shares. Peer median EV/EBITDA is 8.5x.

- a. Compute EV using the value-driver formula. [7]
- b. Compute EV if ROIC equals WACC and explain the difference. [5]
- c. Compute EV from the multiple and the implied EV/EBITDA of the DCF value. [8]
- d. Blend the DCF and multiple equally and compute the value per share. [5]

Question 4: Private equity economics and carbon-adjusted investment (25 points)

i. A $400m fund has all capital contributed at time 0 and distributes $760m after six years; hurdle 8% compounded, carry 20%, full GP catch-up. (ii) A project costs $150m and yields $30m a year for 8 years (WACC 9%), emitting 0.2m tonnes a year with a shadow carbon price of $40 rising 4% a year (tax 25%).

- a. Compute the preferred return, GP carry and LP net multiple and IRR. [13]
- b. Compute the project NPV with and without carbon cost and comment. [12]

Marking Scheme and Model Answers: Final Examination B (Volume 1, Chapters 13 to 24)

Question 1 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	5	Underpricing = 21.60/18 − 1 = 20.0%; left on table = 3.60 × 20m = $72m.	2 pts, 3 pts
(b)	5	20m × 18 × 0.93 = $334.8m.	3 pts method, 2 pts value
(c)	5	Total shares = 100m; founders = 56/100 = 56% (from 70%); public float = 20% of shares.	2 pts, 2 pts, 1 pt
(d)	7	Lost shield = 0.25 × 150 = $37.5m, so V falls by $37.5m on that account. Offsetting benefits: lower expected distress costs and agency costs, and greater financial flexibility; the net effect is positive only if the distress cost saving exceeds $37.5m.	3 pts calculation, 2 pts sign, 2 pts offsetting effects
(e)	3	Lock-up prevents insiders selling shortly after the IPO, signaling confidence and preventing a price collapse from supply.	3 pts

Question 2 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	8	N = (0.5 − 1.1) × 80,000,000 / (5,000 × 50) = -192: sell 192 contracts.	4 pts formula, 2 pts contract value, 2 pts sign
(b)	8	h* = 0.85 × 0.35 / 0.32 = 0.9297; contracts = 0.9297 × 6,000,000 / 42,000 = 132.8, about 133 (long).	4 pts ratio, 4 pts contracts
(c)	9	Before: 2.3263 × 1.2% × 80m = $2.233m. After: × 0.5/1.1 = $1.015m.	4 pts before, 5 pts after

Question 3 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	7	EV = 150 × (1 − 0.03/0.12) / (0.085 − 0.03) = $2,045.5m.	3 pts reinvestment, 4 pts EV
(b)	5	EV = 150 × (1 − 0.03/0.085) / 0.055 = $1,764.7m = 150/0.085. Growth creates no value when ROIC = WACC; the difference of $280.7m is the value of growth at a 12% return.	3 pts EV, 2 pts explanation
(c)	8	EV = 260 × 8.5 = $2,210m; DCF implied multiple = 2,045.5/260 = 7.87x.	4 pts, 4 pts
(d)	5	Blend = $2,127.7m; equity = $1,527.7m; per share = $21.82.	2 pts blend, 1 pt equity, 2 pts per share

Question 4 (25 points)

Part	Pts	Model answer	Mark allocation
(a)	13	Pref = 400 × (1.08^6 − 1) = $234.75m. After capital and pref, remaining = 760 − 400 − 234.75 = $125.25m. Catch-up = 234.75 × 0.25 = $58.69m; residual 66.56 split 80/20. GP carry = $72.00m = 20.0% of profit; LP gets $688.00m: 1.720x, IRR 9.46%.	3 pts pref, 3 pts catch-up, 3 pts split, 4 pts LP multiple and IRR
(b)	12	PV of inflows = $166.04m; NPV = $16.04m. PV after-tax carbon cost = $37.58m; NPV = -$21.53m. Without carbon the project is accepted; including it NPV is lower but negative, so the decision reverses; test with higher prices.	4 pts base NPV, 5 pts carbon, 3 pts comment

Total: 100 points.

Multiple-Choice Question Bank

The bank has 60 questions, five for each of Volume 1, Chapters 13 to 24, in chapter order (questions 1-5 Chapter 13, 6-10 Chapter 14, and so on). Questions with calculations are keyed to values recomputed in Python. Key positions are distributed as follows (A: 20, B: 15, C: 14, D: 11). A 20-question paper drawn at random with a balanced chapter mix and a 30-minute limit is a suitable quiz.

Questions

1. A firm has 20 million shares at $30 and offers rights to buy one new share at $24 for every four held. The theoretical ex-rights price is:
(A) $27.00 (B) $28.80 (C) $29.50 (D) $30.00

2. An IPO is priced at $15 and closes day one at $18. Underpricing is:
(A) 20.0% (B) 16.7% (C) 3.0% (D) 83.3%

3. Which theory best explains why the share price typically falls when a firm announces a seasoned equity offering?
(A) Asymmetric information: managers issue when shares are overvalued (B) The Gordon growth model (C) Put-call parity (D) The tax shield of debt

4. The primary purpose of a greenshoe (over-allotment) option is to:
(A) Reduce the gross spread (B) Raise the offer price (C) Allow price stabilization after the offering (D) Transfer ownership from the issuer to the underwriter

5. Compared with a general cash offer, a rights offering mainly:
(A) Raises more cash per share (B) Eliminates the announcement effect (C) Guarantees a higher share price (D) Protects existing holders from ownership dilution and usually has lower flotation costs

6. A five-year zero-coupon bond with face value $1,000 and a 6% annual yield costs:
(A) $747.26 (B) $800.00 (C) $1,000.00 (D) $700.00

7. A bond with a coupon above its yield trades at a premium and is callable. Which statement is correct?
(A) Yield to call exceeds yield to maturity (B) Yield to call is below yield to maturity (C) They are equal (D) Yield to call cannot be computed

8. A firm borrows at 8% with a 30% tax rate. The after-tax cost of debt is:
(A) 8.0% (B) 2.4% (C) 5.6% (D) 10.4%

9. A covenant forbidding additional debt above a stated leverage ratio is called a:
(A) Positive covenant (B) Sinking fund (C) Call provision (D) Negative (restrictive) covenant

10. The Macaulay duration of a zero-coupon bond equals:
(A) Its time to maturity (B) Half its maturity (C) Its yield (D) Its coupon rate

11. In a world with no taxes and no frictions, Modigliani-Miller Proposition I states that firm value:
(A) Rises with debt (B) Is independent of capital structure (C) Falls with debt (D) Depends on dividend policy

12. An unlevered firm worth \$400m issues \$100m of permanent debt; the tax rate is 30%. Levered value is:
(A) \$400m (B) \$470m (C) \$430m (D) \$500m

13. With rU = 10%, rD = 6%, D/E = 1 and no taxes, the cost of equity is:
(A) 10% (B) 16% (C) 12% (D) 14%

14. Pecking-order theory predicts firms prefer financing in the order:
(A) Internal funds, debt, equity (B) Equity, debt, internal funds (C) Debt, equity, internal funds (D) Equity, internal funds, debt

15. Which is an indirect cost of financial distress?
(A) Legal fees in bankruptcy court (B) Lost customers and suppliers' tightening of credit (C) Interest tax shield (D) Underwriter fees

16. A firm pays a \$2 dividend. Ignoring taxes, the stock price on the ex-dividend date should fall by about:
(A) \$0 (B) \$1 (C) \$2 (D) \$4

17. A debt-financed buyback raises EPS when:
(A) The earnings yield exceeds the after-tax cost of debt (B) The P/E ratio is low and rates are high (C) The dividend yield is below the earnings yield (D) Shares trade below book

18. ROE = 12% and the retention ratio = 60%. Sustainable growth is:
(A) 4.8% (B) 12.0% (C) 20.0% (D) 7.2%

19. Why is a dividend cut often followed by a price decline?
(A) It signals lower expected future earnings (B) It increases the tax rate (C) It reduces shares outstanding (D) It always forces a buyback

20. Lintner's model says managers:
(A) Change dividends with each quarter's earnings (B) Smooth dividends and adjust gradually toward a target payout (C) Never pay dividends (D) Pay out all free cash flow

21. DIO = 45, DSO = 38 and DPO = 30 days. The cash conversion cycle is:
(A) 113 days (B) −37 days (C) 53 days (D) 83 days

22. Terms 2/10 net 40: the annualized simple cost of forgoing the discount is about:
(A) 24.8% (B) 2.0% (C) 12.2% (D) 74.5%

23. Annual demand 10,000, order cost \$100, holding cost \$4 per unit. The EOQ is about:
(A) 500 units (B) 1,000 units (C) 250 units (D) 707 units

24. A firm financing all permanent and seasonal current assets with long-term debt follows a:
(A) Aggressive policy (B) Conservative policy (C) Matching policy (D) Zero-working-capital policy

25. Lockbox systems aim to:
(A) Reduce collection float (B) Increase disbursement float (C) Lower inventory (D) Increase credit sales

26. For European options on a non-dividend stock, C − P equals:
(A) K − S (B) S + K (C) S − K e^(−rT) (D) S e^(−rT) − K

27. All else equal, an increase in volatility:
(A) Lowers call values (B) Lowers put values (C) Leaves option values unchanged (D) Raises both call and put values

28. The delta of a deep in-the-money call is close to:
(A) 1 (B) 0 (C) 0.5 (D) −1

29. A stock can go up 20% or down 10% and the risk-free rate is 5% per period. The risk-neutral probability of an up move is:
(A) 0.40 (B) 0.50 (C) 0.60 (D) 0.75

30. It is never optimal to exercise early a(n):
(A) American put on any stock (B) American call just before a large dividend (C) American call on a non-dividend-paying stock (D) Deep in-the-money American put

31. A non-dividend stock is $50, r = 4% continuous. The 6-month forward price is:
(A) $50.00 (B) $51.01 (C) $51.00 (D) $49.00

32. Futures differ from forwards mainly because futures are:
(A) Marked to market daily and exchange-traded (B) Always cash settled (C) Unregulated (D) Customized

33. A pay-fixed swap gains value when floating rates:
(A) Fall (B) Stay unchanged (C) Become negative (D) Rise

34. To cut a portfolio's beta from 1.2 to 0.8 with index futures, the manager:
(A) Sells index futures (B) Buys index futures (C) Buys calls (D) Shorts the risk-free asset

35. A market in which futures prices are below spot prices is in:
(A) Contango (B) Equilibrium (C) Backwardation (D) Arbitrage

36. Spot $1.20 per euro, USD rate 4%, EUR rate 2%. The one-year forward is about:
(A) $1.1769 (B) $1.2235 (C) $1.2000 (D) $1.2480

37. A firm whose long-run competitiveness is affected by currency swings faces:
(A) Economic exposure (B) Transaction exposure only (C) Translation exposure only (D) No exposure

38. Value at Risk measures:
(A) The maximum possible loss (B) The average loss (C) The expected return (D) A loss threshold exceeded with a given small probability over a horizon

39. To immunize a single liability, assets should have present value equal to the liability and:
(A) Maturity longer than the liability (B) Zero convexity (C) Duration equal to the liability's duration (D) Floating rates

40. Which is a natural hedge for a firm with euro revenues?
(A) Dollar-denominated debt (B) Euro-denominated debt (C) A larger dividend (D) A longer receivables period

41. EBIT 200, tax 25%, depreciation 30, capex 50, increase in NWC 10. FCFF is:
(A) 120 (B) 150 (C) 110 (D) 140

42. FCFF next year is 90, WACC 10%, growth 4%. Enterprise value is:
(A) 900 (B) 2,250 (C) 1,500 (D) 1,000

43. When terminal value represents most of enterprise value, the analyst should:
(A) Ignore terminal value (B) Test sensitivity to WACC and g and cross-check with multiples (C) Lower the tax rate (D) Extend the forecast with higher growth

44. Equity value from EV/EBITDA valuation equals:
(A) EV minus net debt (and other claims) (B) EV plus net debt (C) EV divided by shares (D) EBITDA minus interest

45. If ROIC equals WACC on new investment, growth:
(A) Adds maximum value (B) Destroys value (C) Lowers WACC (D) Adds no value

46. An acquirer pays a premium of $50m for synergies worth $80m. NPV to the acquirer's shareholders is:
(A) $80m (B) $50m (C) $30m (D) −$30m

47. A stock-financed acquisition tends to be EPS-dilutive when the:
(A) Target's P/E is below the acquirer's (B) Target's P/E exceeds the acquirer's P/E (C) Target is private (D) Deal is financed with cash

48. A poison pill is designed to:
(A) Make a hostile takeover prohibitively dilutive without board approval (B) Raise the target's cash (C) Lower debt (D) Replace the board

49. The regulatory concern most likely to block a horizontal merger is:
(A) Large goodwill (B) Tax shields (C) High leverage (D) Market power and reduced competition

50. The winner's curse in auctions implies that winning bidders:
(A) Always gain most synergies (B) Pay below value (C) Tend to overpay (D) Avoid premiums

51. Total uses are $600m and debt is 4.0x EBITDA of $100m. Sponsor equity is:
(A) $200m (B) $400m (C) $100m (D) $300m

52. A sponsor earns 2.5x MOIC over five years. IRR is about:
(A) 25.0% (B) 20.1% (C) 12.5% (D) 30.0%

53. Which is NOT a main source of LBO returns?
(A) EBITDA growth (B) Debt paydown (C) A lower tax rate on dividends (D) Multiple expansion

54. A 20% carry over an 8% hurdle with full catch-up means the GP:
(A) Receives 20% of committed capital (B) Receives 8% of profits (C) Receives nothing above the hurdle (D) Receives 20% of total profit once the hurdle is exceeded and fully caught up

55. A dividend recapitalization:
(A) Raises debt to pay a dividend to the sponsor (B) Repays debt with equity (C) Reduces risk (D) Is tax-free to all holders

56. Green bond yield 4.85% versus conventional 5.00%. The greenium is:
(A) 5 bp (B) 15 bp (C) 1.5 bp (D) 150 bp

57. An internal carbon price in capital budgeting is used to:
(A) Reduce WACC (B) Offset taxes (C) Reflect the cost of expected emissions in project cash flows (D) Replace market prices

58. Under cumulative voting with 600 shares and five directors elected, shares needed to guarantee one seat:
(A) 101 (B) 301 (C) 121 (D) 60

59. In a dual-class structure the founder’s voting power exceeds her economic stake because:
(A) She owns more cash flows (B) Her shares carry more votes per share (C) Voting is cumulative (D) She holds preferred stock

60. A fintech lender has LTV of $900 and CAC of $450. LTV/CAC is:
(A) 2.0x (B) 0.5x (C) 4.5x (D) 3.0x

Answer Key and Rationales

Q	Key	Rationale
1	B	(4 × 30 + 24) / 5 = 28.80.
2	A	18 / 15 − 1 = 20%; 16.7% divides by the wrong base.
3	A	Equity issuance signals overvaluation (Myers-Majluf).
4	C	The short position lets the underwriter buy back shares if the price falls.
5	D	Pre-emptive rights let holders maintain proportional ownership; the right's value offsets the discount.
6	A	1,000 / 1.06^5 = 747.26.
7	B	Shortening the horizon removes future premium-bond coupons and the call price is lower than the premium price, so the yield is lower.
8	C	8% × (1 − 0.30) = 5.6%.
9	D	It restricts management actions; it protects lenders from asset substitution and dilution of their claim.
10	A	There is a single cash flow at maturity.
11	B	Arbitrage by investors replicates leverage.
12	C	400 + 0.30 × 100 = 430.
13	D	10% + (10% − 6%) × 1 = 14%.
14	A	Adverse selection costs are lowest for internal funds and highest for equity.
15	B	Indirect costs arise before formal bankruptcy and are often larger than direct costs.
16	C	A dollar paid out reduces firm value by a dollar.
17	A	Each dollar repurchased retires earnings equal to the earnings yield but costs only the after-tax interest rate, so EPS rises when the yield is higher.
18	D	g = ROE × retention = 0.12 × 0.6 = 7.2%.
19	A	Dividends convey information because managers smooth payouts.
20	B	Dividends follow a partial-adjustment pattern.
21	C	45 + 38 − 30 = 53.
22	A	0.02/0.98 × 365/30 = 24.83%.
23	D	√(2 × 10,000 × 100 / 4) = 707.1.

Q	Key	Rationale
24	B	It uses long-term funds for temporary needs, which lowers risk but raises cost.
25	A	They speed receipt of customer payments.
26	C	Put-call parity.
27	D	Options have convex payoffs, so greater dispersion raises value.
28	A	It behaves like the stock.
29	B	(1.05 − 0.90) / (1.20 − 0.90) = 0.50.
30	C	The call's time value exceeds the interest saved by delaying exercise.
31	B	50 × e^0.02 = 51.01.
32	A	Daily settlement and a clearinghouse reduce credit risk.
33	D	The holder pays a fixed rate and receives a higher floating rate.
34	A	Short futures reduce market exposure.
35	C	Backwardation reflects a high convenience yield or short-term scarcity.
36	B	1.20 × 1.04 / 1.02 = 1.2235.
37	A	Economic exposure affects future operating cash flows.
38	D	VaR is a quantile, not a worst case.
39	C	Matching duration offsets first-order interest-rate effects.
40	B	Euro liabilities offset euro inflows.
41	A	150 + 30 − 50 − 10 = 120.
42	C	90 / 0.06 = 1,500.
43	B	Terminal assumptions dominate value.
44	A	Debt holders are paid before equity.
45	D	Reinvested capital earns exactly its cost.
46	C	Synergies − premium = 30.
47	B	The acquirer issues too many shares for each dollar of target earnings.
48	A	Rights become exercisable at a discount if a bidder crosses a threshold.
49	D	Antitrust authorities assess concentration.
50	C	The highest estimate is likely an overestimate.
51	A	600 − 400 = 200.
52	B	2.5^(1/5) − 1 = 20.1%.
53	C	Returns come from operations, deleveraging and the exit multiple.

Q	Key	Rationale
54	D	Catch-up restores the 80/20 split on all profits.
55	A	It returns capital early, raising IRR but also leverage.
56	B	5.00 − 4.85 = 0.15%.
57	C	It stress-tests projects against regulation and policy.
58	A	600 × 1 / (5 + 1) + 1 = 101.
59	B	The wedge between control and cash-flow rights is the agency cost.
60	A	900 / 450 = 2.0, below the common 3x guideline.